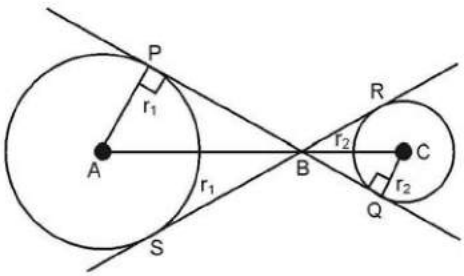
Sl. No.	*Position of two circles*	*Number of common tangents*
1.	One circle lies entirely inside the other circle	Zero
2.	Two circles touch internally	One
3.	Two circles intersect in two distinct points	Two
4.	Two circles touch externally	Three
5.	One circle lies entirely outside the other circle	Four

Direct common tangents and transverse common tangents

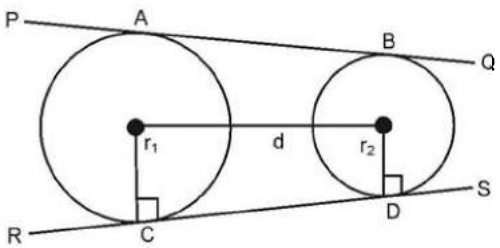
1. *Transverse common tangent* In the figure given below, PQ and RS are the transverse common tangents. Transverse common tangents intersect the line joining the centre of the two circles. They divide the line in the ratio $r_1 : r_2$.
$AB : BC = r_1 : r_2$

   Assume AC = Distance between centres = d



   $PQ^2 = RS^2 = d^2 - (r_1 + r_2)^2$

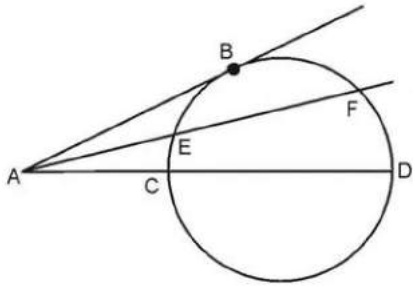
2. *Direct common tangent*



   In the figure given above, PQ and RS are direct common tangents.

   Points A and C are the point of tangency for the first circle and similarly points B and D are the point of tangency for the second circle. AB and CD are known as lengths of the direct common tangents and they will be same.
$CD^2 = AB^2 = d^2 - (r_1 - r_2)^2$

Secants

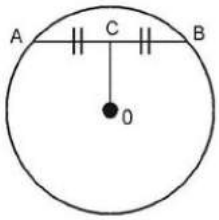


In the figure given above, AB is a tangent and ACD is a secants

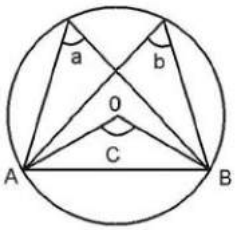
i. $AB^2 = AC \times AD$
ii. $AE \times AF = AC \times AD$

Important theorems related to circle

1. If C is the mid-point of AB, then OC is perpendicular to AB. And vice versa is also true.

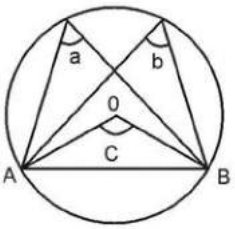


2. Angles in the same segment will be equal.



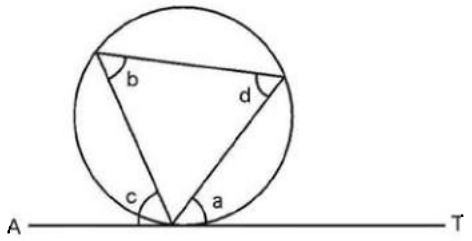
   In the figure given above, $a = b$.

3. Angle subtended by a chord at the centre is two times the angle subtended on the circle on the same side. In the figure given below, $2a = 2b = c$.



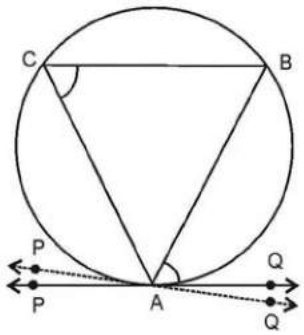
4. Angle subtended by a diameter of the circle is a right angle.

5. Alternate segment theorem



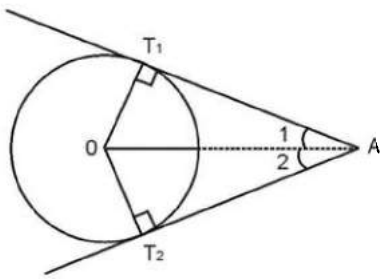
In the figure above, AT is the tangent. ∠a = Alternate segment ∠b ∠c = Alternate segment ∠d

6. Converse of alternate segment theorem If a line is drawn through an end point of a chord of a circle so that the angle formed by it with the chord is equal to the angle subtended by the chord in the alternate segment, then the line is a tangent to the circle.



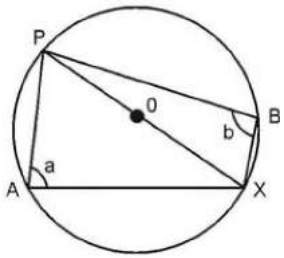
AB is a chord of a circle and a line PAQ such that ∠BAQ = ∠ACB, where C is any point in the alternate segment ACB, then PAQ is a tangent to the circle.

7. Tangent drawn to a circle from a point are same in length.
In the figure below, tangents are drawn to the circle from point A and AT_1 and AT_2 are the tangents.
i. $AT_1 = AT_2$ ii. ∠1 = ∠2
iii. $AT_1^2 + OT_1^2 = AT_2^2 + OT_2^2 = AO^2$



Cyclic Quadrilateral

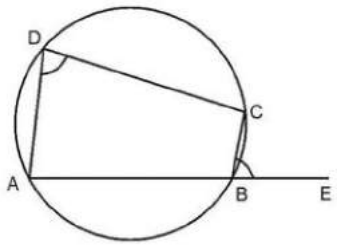
Consider the figure given below:



If we have $a + b = 180°$ and quadrilateral AXBP has all its vertices on a circle, then such a quadrilateral is called a Cyclic quadrilateral.

For a cyclic quadrilateral, the sum of the opposite angles of a quadrilateral in a circle is 180°.

It can also be seen that exterior ∠CBE = internal ∠ADC = 180° − ∠ABC.



Using Brahmagupta's formula to find out the area of a cyclic quadrilateral We know that A + B = p So, area of cyclic quadrilateral

$$= \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Where terms used are having their meaning.
[Cos 90° = 0]

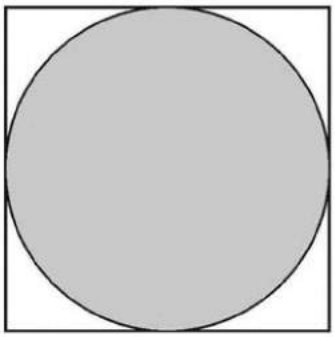
Circle Packing

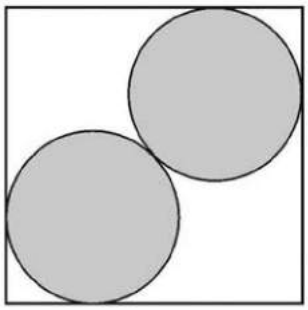
N circles have been packed inside a square of side length R unit and radius of the circle is to be calculated.

Case 1 When $n = 1$

Obviously in this case, diameter of the circle = side of the square.

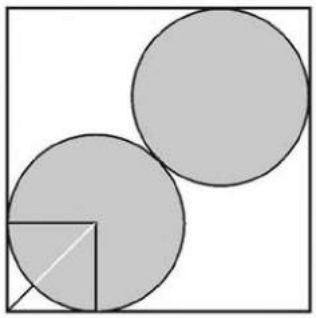
So, the radius of the circle $= \frac{1}{2}$ R unit





Case 2 When $n = 2$

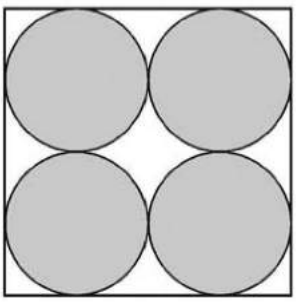
Consider this figure



This is a right-angled triangle with sides r, r and r $\sqrt{2}$. Hence the diagonal of the square = 4r + 2 r $\sqrt{2}$ = $\sqrt{2}$ R

So, $r = \frac{\sqrt{2}}{4+2\sqrt{2}}R$

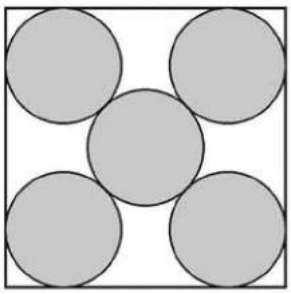
Case 3 When $n = 4$



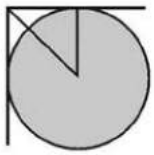
Obviously, in this case 2 × diameter of a circle = side of square.

So, the radius of a circle $= \frac{1}{4}$ R

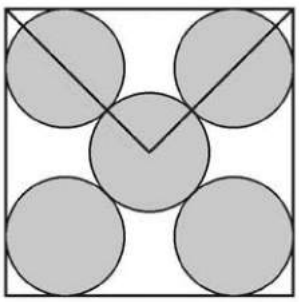
Case 4 When $n = 5$



Consider this figure:



If the radius of the circle is r, then the distance between the centre of the circle and the vertex of a square = r $\sqrt{2}$

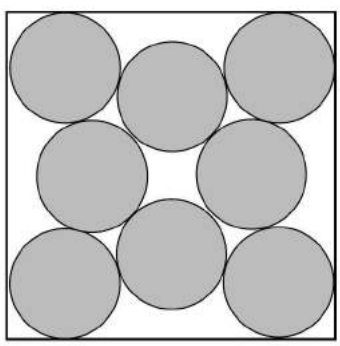


Now, consider this figure, the triangle formed here is a right-angled triangle:

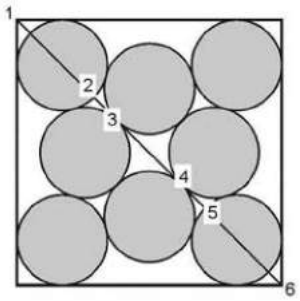
1/$\sqrt{2}$(1/2 of the diagonal of the square) = r + r + r $\sqrt{2}$

= r(2 + $\sqrt{2}$), so, $r = \frac{1}{\sqrt{2}\times(2+\sqrt{2})}R$

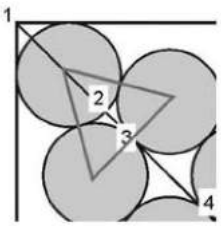
Case 5 When $n = 8$



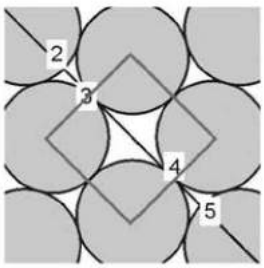
We will find out the value of r in 3 steps here:



Step 1 Find 1 - 2 = 5 - 6
Step 2 Find 2 - 3 = 4 - 5
Step 3 Find 3 - 4
Step 1 1 - 2 = 5 - 6 = r + r √2
Step 2 For 2 - 3
This is an equilateral triangle with side length = 2r. Height of this triangle r √3

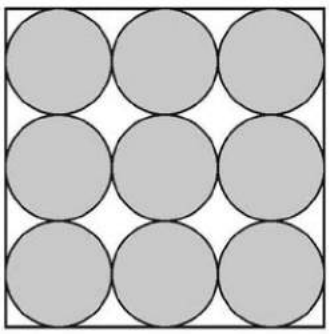


So, 2 - 3 = 4 - 5 = r√3 – r
Step 3



This is a square with side length 2 r and diagonal = 2r √2
So, 3 - 4 = 2r
Hence 1 – 6 = 2(r + r√2) + 2(r√3 – r) + 2r = R√2. Now r can be calculated.

Case 6 When *n* = 9

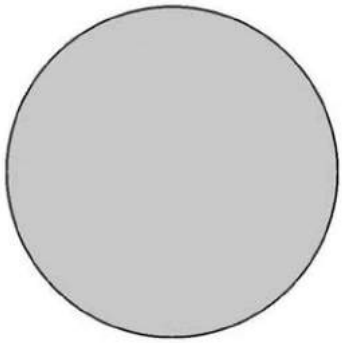


Radius of circle = r
So, the side of a square = 3 × diameter of a circle = 6r
So, 6r = R unit, r = R/6 unit

Circles Inside Circles

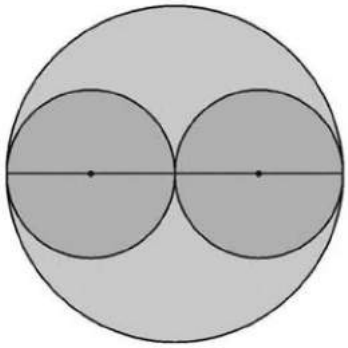
Radius of the outer circle R = 1 unit and n similar circles of 'r' radius have been inserted inside this outer circle.

N = 1



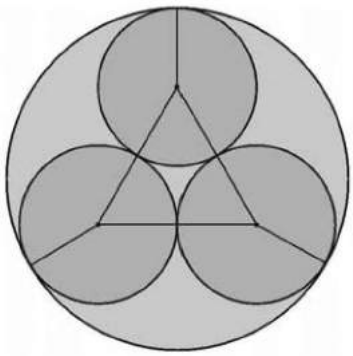
Obviously R = r

N = 2



Obviously, 4 r = 2 R Hence r = R/2

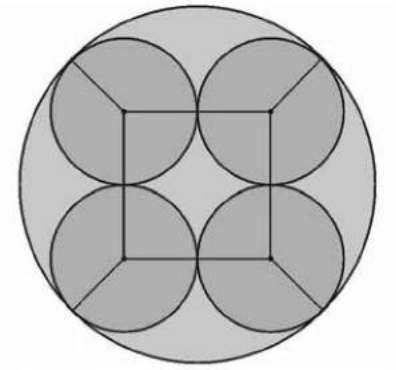
N = 3



The triangle formed inside will be an equilateral triangle of the side length 2 r. The centre of this circle will be incentre/centroid/circumcentre of this equilateral triangle. So, the distance between the centre of any smaller circle to the centre of the bigger circle = 2/3 median = x = 2/3 [√3/2 × a], where a = 2r.

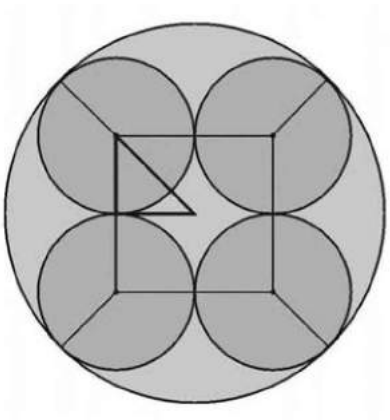
Now, r + x = R. Now put the values of x from the above condition to find the value of r.

N = 4



The square formed inside will be of the side length 2r. Now consider this figure,

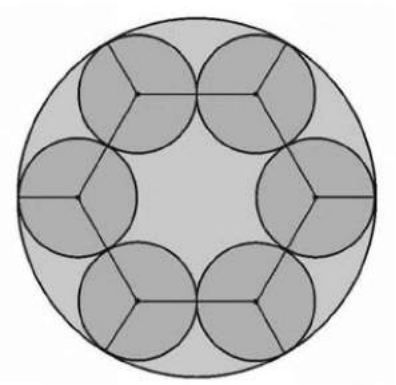
N = 5



The figure inside is a right angled triangle, with base and height being equal to r, and hypotenuse = r$\sqrt{2}$ = r + x. Hence x = r($\sqrt{2}$ – 1)

Now 2R = 4 r + 2 x. Hence r = 2(R – x)/4[Now put the value of x from the above condition.]

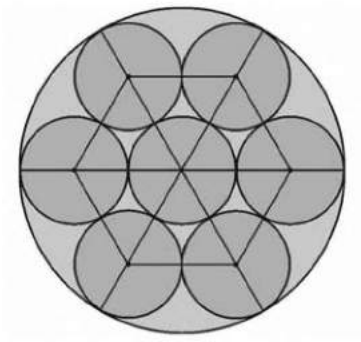
N = 6



Here we can put a similar circle inside all the six circles making it a 7 circle figure. Now, the situation is the same as that of a circle packing with N = 7

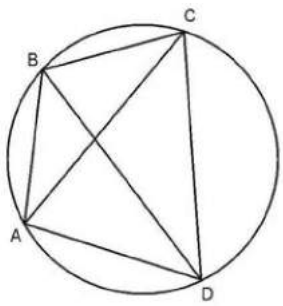
In this case, 6r = R, so, r = 1/6 R

N = 7



In this case, 6r = R, so, r = 1/6 R

Ptolemy's Theorem of Cyclic Quadrilateral



For a cyclic quadrilateral, the sum of the products of the two pair of the opposite sides equal the product of the diagonals.

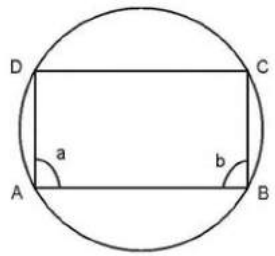
In the figure given above, AB × CD + BC × DA = AC × BD

In any cyclic quadrilateral of side lengths a, b, c and d, the length of the diagonals p and q will be equal to:

$$p = \sqrt{\frac{(ab+cd)(ac+bd)}{ad+bc}} \quad q = \sqrt{\frac{(ac+bd)(ad+bc)}{ab+dc}}$$

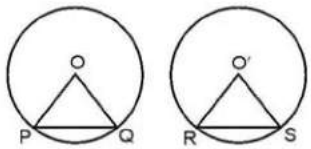
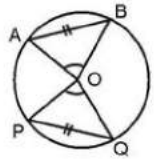
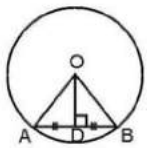
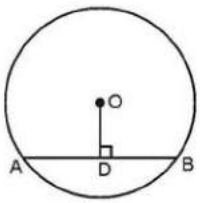
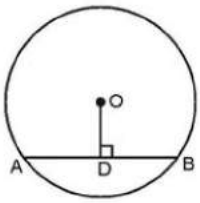
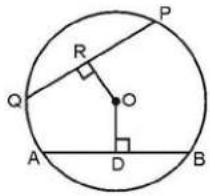
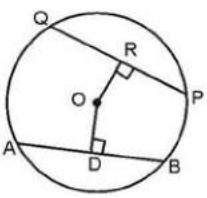
Example 11 Under some special conditions, it is given that a cyclic quadrilateral ABCD is a parallelogram. What kind of figure will ABCD be?

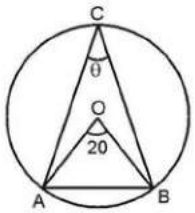
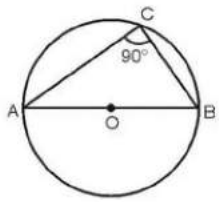
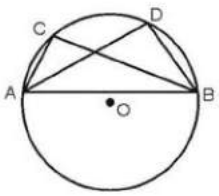
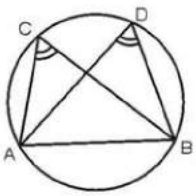
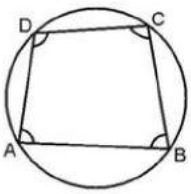
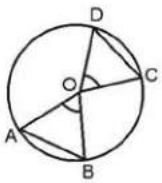
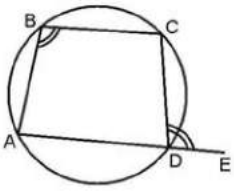
Solution ABCD is a parallelogram. Thus $a + b = 180°$. And ∠A + ∠C = 180°

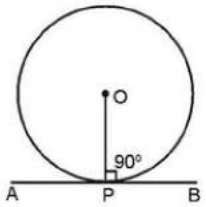
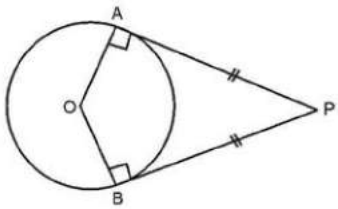
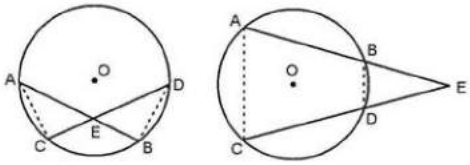
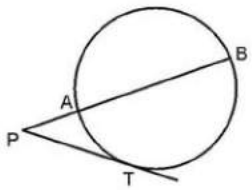
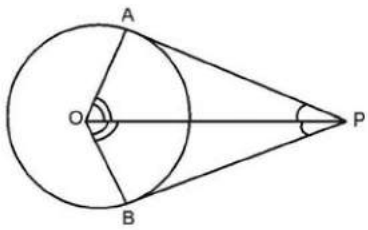
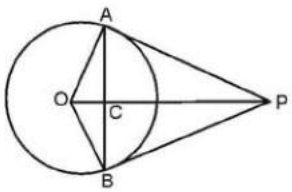
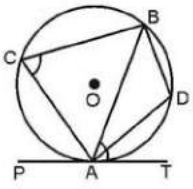


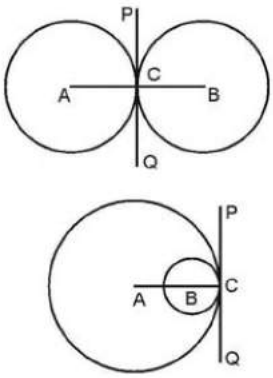
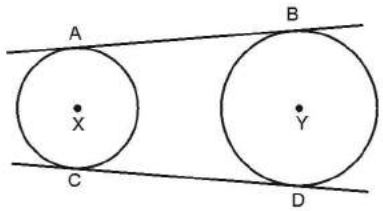
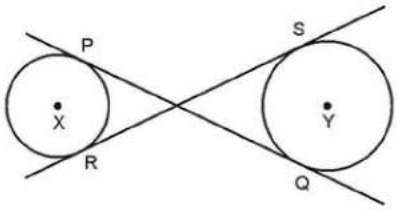
This means that a parallelogram drawn inside a circle is always a rectangle.

Summarizing the discussion regarding circle

Sl. No.	*Theorem/Property*	*Diagram*
1.	In a circle (or congruent circles) equal chords are made by equal arcs. {OP = OQ} = {O'R = O'S) PQ = RS and PQ = RS	
2.	Equal arcs (or chords) subtend equal angles at the centre PQ = AB (or PQ = AB) ∠POQ = ∠AOB	
3.	The perpendicular from the centre of a circle to a chord bisects the chord i.e., if OD ⊥ AB (OD is perpendicular to AB).	
4.	The line joining the centre of a circle to the mid-point of a chord is perpendicular to the chord. AD = DBOD ⊥ AB	
5.	Perpendicular bisector of a chord passes through the centre i.e., OD ⊥ AB and AD = DB ∴ O is the centre of the circle	
6.	Equal chords of a circle (or of congruent circles) are equidistant from the centre ∴ AB = PQ ∴ OD = OR	
7.	Chords of a circle (or of congruent circles) are equidistant from the centre ∴ OD = OR ∴ AB = PQ	

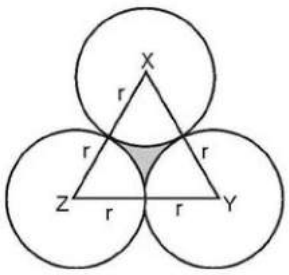
Sl. No.	*Theorem/Property*	*Diagram*
8.	The angle subtended by an arc (the degree measure of the arc) at the centre of a circle is twice the angle subtended by the arc at any point on the remaining part of the circle. m ∠AOB = 2m ∠ACB.	
9.	Angle in a semicircle is a right angle.	
10.	Angles in the same segment of a circle are equal i.e., ∠ACB = ∠ADB	
11.	If a line segment joining two points subtends equal angle at two other points lying on the same side of the line containing the segment, then the four points lie on the same circle. ∠ACB = ∠ADB ∴ Points A, C, D, B are concyclic i.e., lie on the circle	
12.	The sum of pair of opposite angles of a cyclic quadrilateral is 180°. ∠DAB + ∠BCD = 180° and ∠ABC + ∠CDA = 180° (Inverse of this theorem is also true)	
13.	Equal chords (or equal arcs) of a circle (or congruent circles) subtended equal angles at the centre. AB = CD (or AB = CD) ∠AOB = ∠COD (Inverse of this theorem is also true)	
14.	If a side of a cyclic quadrilateral is produced, then the exterior angle is equal to the interior opposite angle. m ∠CDE = m ∠ABC	

Sl. No.	*Theorem/Property*	*Diagram*
15.	A tangent at any point of a circle is perpendicular to the radius through the point of contact. (Inverse of this theorem is also true)	
16.	The lengths of two tangents drawn from an external point to a circle are equal i.e., AP = BP	
17.	If two chords AB and CD of a circle, intersect inside a circle (outside the circle when produced at a point E), then AE × BE = CE × DE	
18.	If PB be a secant which intersects the circle at A and B and PT be a tangent at T then PA × PB = $(PT)^2$	
19.	From an external point from which the tangents are drawn to the circle with centre O, then (a) they subtend equal angles at the centre (b) they are equally inclined to the line segment joining the centre of that point ∠AOP = ∠BOP and ∠APO = ∠BPO	
20.	If P is an external point from which the tangents to the circle with centre O touch it at A and B then OP is the perpendicular bisector of AB. OP ⊥ AB and AC = BC	
21.	If from the point of contact of a tangent, a chord is drawn then the angles which the chord makes with the tangent line are equal respectively to the angles formed in the corresponding alternate segments. In the adjoining diagram. ∠BAT = ∠BCA and ∠BAP = ∠BDA	

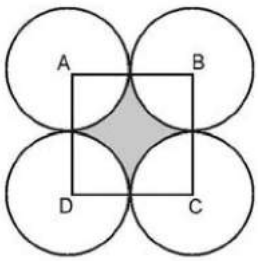
Sl. No.	Theorem/Property	Diagram
22.	The point of contact of two tangents lies on the straight line joining the two centres. (a) When two circles touch externally then the distance between their centres is equal to sum of their radii, i.e., AB = AC + BC (b) When two circles touch internally the distance between their centres is equal to the difference between their radii. i.e., AB = AC – BC	
23.	For the two circles with centre X and Y and radii r_1 and r_2. AB and CD are two Direct Common Tangents (DCT), then the length of DCT $=\sqrt{(\text{Distance between centres})^2-(r_1-r_2)^2}$	
24.	For the two circles with centre X and Y and radii r_1 and r_2 PQ and RS are two transverse common tangent, then length of TCT $=\sqrt{(\text{Distance between centres})^2-(r_1+r_2)^2}$	

Some important points

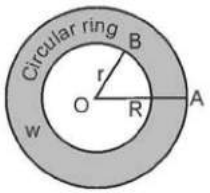
1. If three circles, each of radius r, are so kept that each circle touches the other two, then the area of the shaded region is $r^2\,[\sqrt{3}-(\pi/2)]$ which is approximately equal to $(4/25)\,r^2$.



2. If four circles, each of the radius r, are so kept that each circle touches the other two, then the area of the shaded region is $r^2\,(4-\pi)$ which is equal to $(6/7)r^2$.



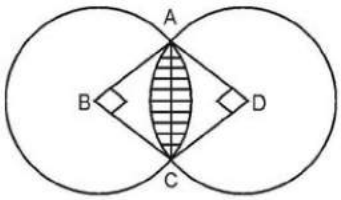
3. In the diagram given below, if the radius (OA) of the outer circle is R and the radius (OB) of the inner circle is r, then the width (w) of the ring is (R – r) and the area of the shaded region is $\pi(R^2-r^2)$ or $\pi(R-r)\,(R+r)$ or $\pi(w)\,(R+r)$



Example 12 Two identical circles intersect so that their centers, and the points at which they intersect, form a square of side 1 cm. The area in sq. cm of the portion that is common to the two circles, is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}-1$ (c) $\frac{\pi}{5}$ (d) $\sqrt{2}-1$

Solution



Shaded area = 2 × (area of sector ADC – area of ΔADC)

$$=2\times\left(\frac{\pi}{4}\times1^2-\frac{1}{2}\times1\times1\right)$$

$$=\frac{\pi}{2}-1$$

Hence, option (b)

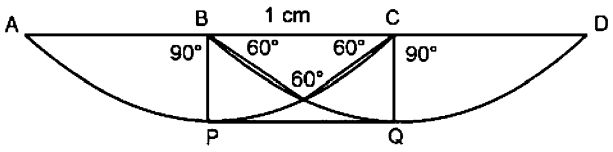
Example 13 Four points A, B, C and D lie on a straight line in the X – Y plane, such that AB = BC = CD, and the length of AB is 1 cm. An ant at A wants to reach a sugar particle at D. But there are insect repellents kept at points B and C. The ant would not go within one metre of any insect repellent. The minimum distance the ant must traverse to reach the sugar particle (in m) is

(a) $3/\sqrt{2}m$ (b) $1+\pi$

(c) $\frac{4\pi}{3}$ (d) 5

Solution

In the drawn figure, ant will go along APQD since it cannot be within a distance of 1 cm from the repellents kept at B and C.



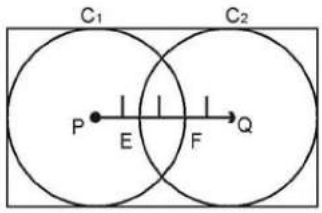
$$AP=\frac{90}{360}\times2\pi\times1=\frac{\pi}{2}$$

Also AP = QD $=\frac{\pi}{2}$

So the minimum distance = AP + PQ + QD

$$=\frac{\pi}{2}+1+\frac{\pi}{2}=1+\pi$$

Example 14 Two circles C_1 and C_2, having the same radius of 2 cm and centers at P and Q respectively, intersect each other such that the line of centers PQ intersects C_1 and C_2 at F and E respectively. EF = 1 cm. The whole assembly is enclosed in a rectangle of the minimum area. The perimeter of the rectangle is



Hence, breadth of the rectangle = 4 cm

And length = 7 cm

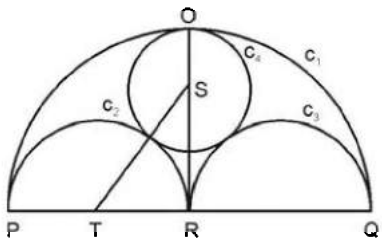
Perimeter = 2 × (7 + 4) = 22 cm

Example 15 Semi-circle C_1 is drawn with a line segment PQ as its diameter with centre at R. Semicircles C_2 and C_3 are drawn with PR and QR as diameters respectively, both C_2 and C_3 lying inside C_1. A full circle C_4 is drawn in such a way that it is tangent to all the three semicircles C_1, C_2 and C_3. C_4 lies inside C_1 and outside both C_2 and C_3. The radius of C_4 is

(a) $\frac{1}{3}PQ$ (b) $\frac{1}{6}PQ$

(c) $\frac{1}{\sqrt{2}}PQ$ (d) $\frac{1}{4}PQ$

Solution



Assume that the radius of C_4 = r and PQ = k.

Now, PR = k/2 = RQ = RO

⇒ RS = (k/2) – r

RT = k/4

ST = (k/4) + r

Applying Pythagoras theorem in triangle STR.

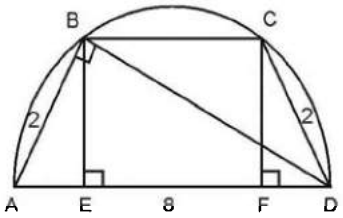
$$\left(\frac{k}{4}+r\right)^2=\left(\frac{k}{4}\right)^2+\left(\frac{k}{2}-r\right)^2$$

⇒ r = k/6 = PQ/6

Example 16 On a semicircle with diameter AD, chord BC is parallel to AD. Further each of the chords AB and CD has length 2 units, while AD = 8 units. What is the length of BC?

(a) 7.5 (b) 7
(c) 7.75 (d) 8

Solution



Finding area of DABD $\frac{1}{2}$ AB × BD $= \frac{1}{2}$ AD × BE

So, BE = $(\sqrt{15})/2$

Hence AE $= \frac{1}{2}$

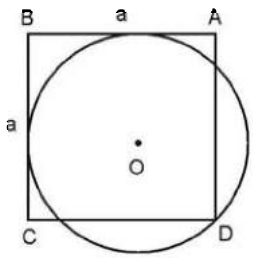
Now, BC = EF = $8 - \left(\frac{1}{2} + \frac{1}{2}\right) = 7$

Alternatively, this question can be solved by using Ptolemy's theorem also.

Example 17 The adjacent sides AB, BC of a square of side 'a' units are tangent to a circle. The vertex D of the square lies on the circumference of the circle. The radius of the circle could be:

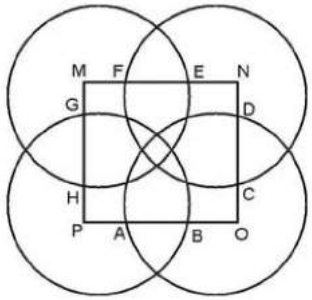
(a) $a(2 - \sqrt{2})$ (b) $a(\sqrt{2} - 1)$
(c) $a(\sqrt{2} + 1.5)$ (d) $a(\sqrt{2} + 1)$

Solution



OD = r, OB = $r\sqrt{2}$
Hence, $r + r\sqrt{2} = \sqrt{2}\,a$
Hence, $r = a(2 - \sqrt{2})$

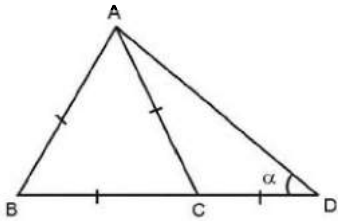
Example 18 M, N, O and P are centres of four intersecting circles each having a radius of 15 cm. If AB = 7 cm, CD = 5 cm, EF = 6 cm, GH = 8 cm, what is the perimeter of the quadrilateral MNOP?



(a) 84 cm (b) 45 cm
(c) 94 cm (d) 124 cm

Solution MN = 30 – 6 = 24 cm
NO = 30 – 5 = 25 cm
OP = 30 – 7 = 23 cm
PM = 30 – 8 = 22 cm
So, the perimeter = 24 + 25 + 23 + 22 = 94 cm

Example 19

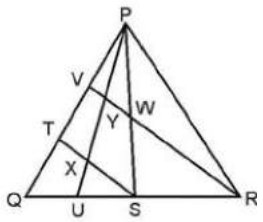


AB = BC = AC = CD. Find ∠a.

(a) 30° (b) 60°
(c) 15° (d) None of these

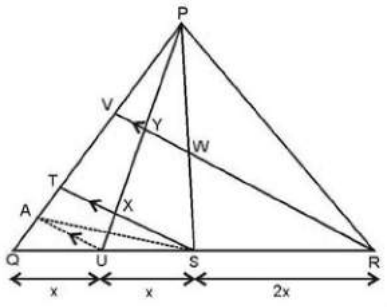
Solution ∠ACB = 60° ∠ACD = 180 – 60 = 120° ∠CAD = ∠CDA = 30° ∠α = 30°

Example 20 In the figure shown here QS = SR, QU = SU, PW = WS and ST||RV. What is the value of the $\frac{\text{Area of } \Delta PSX}{\text{Area of } \Delta PQR}$?



(a) $\frac{1}{5}$ (b) $\frac{1}{7}$
(c) $\frac{1}{6}$ (d) $\frac{1}{9}$

Solution



Draw a line form U such that it is parallel to ST (and hence RV, also) joining AS, to get ΔQSA.

Now, let us first find the area of ΔSUX.

PW = WS (W is the mid point of PS)

Consider the ΔPXS, SX||YW and PX and PS are transversals to those parallel lines, we must have

$$\frac{PY}{YX}=\frac{PW}{WS}=\frac{1}{1}=1$$

$\Rightarrow$ PY = YX

Similarly we also get

PV = TV (in ΔPTS; VW||ST) and QA = AT (in ΔQTS: AU||ST) and QT = VT (in ΔVQR; ST||VR)

Combining all these $QA = AT = \frac{1}{2}(VT) + \frac{1}{2}(VP)$

Now, in ΔRUY

$$\frac{TS}{SR}=\frac{x}{2x}=\frac{1}{2}=\frac{UX}{XY}\Rightarrow UX=\frac{1}{2}(XY)$$

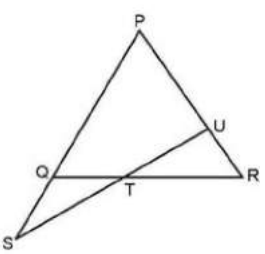
so, in ΔPUS, $UX=\frac{1}{2}XY=\frac{1}{2}PY\Rightarrow\frac{\text{Area }(\Delta SUX)}{\text{Area }(\Delta PUS)}=\frac{1}{5}$

$\Rightarrow$ Area $(\Delta SUX)=\frac{1}{5}$ Area (ΔPUS)

$=\frac{1}{5}\left(\frac{1}{4}\text{area }\Delta PQR\right)\Rightarrow$ Area $(\Delta SUX)=\frac{1}{2}(\Delta SUX)$, then

Area (ΔPSX) = Area (ΔPSU) – area (Δ SUX) $=\frac{1}{4}=\frac{1}{5}$ (ΔPQR)

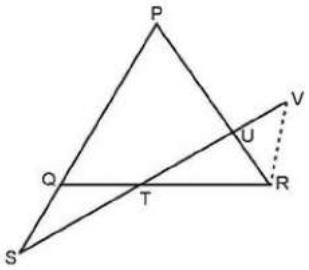
Example 21 Through T, the mid-point of the side QR of a ΔPR, a straight line is drawn to meet PQ produced to S and PR at U, so that PU = PS. If length of UR = 2 units then the length of QS is:



(a) $2\sqrt{2}$ Units (b) $\sqrt{2}$ Units

(c) 2 Units (d) Cannot be determined

Solution We have QT = TR and PU = PS and UR = 2 units



Draw RV||PS that meets SU extended at V.

Now, in ΔQST and ΔTVR

$\angle QST = \angle TVR$ (alternate angles as PS||VR) and $\angle QTS = \angle VTR$

QT = TR

$\therefore$ DQST and DTVR are congruent.

$\therefore$ QS = VR ... (1)

Now $\angle QST = \angle PUS = \angle VUR = \angle UVR$

In ΔUVR, $\angle VUR = \angle RVU$

or, RV = UR = 2 ...(2)

From (1) and (2)

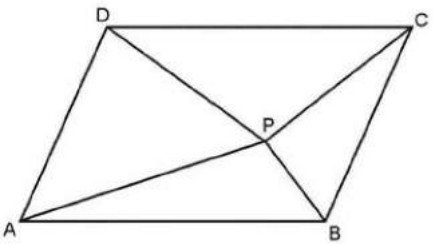
QS = VR = UR = 2 units

Example 22 ABCD is a parallelogram and P is any point within it. If the area of the Parallelogram ABCD is 20 units, then what is the sum of the areas of the DPAB and DPCD?

(a) 5 units (b) 10 units

(c) 12 units (d) Cannot be determined

Solution



Let AB = CD = a and x, y be the lengths of the perpendiculars from P on AB and CD respectively, then

Area of $(\Delta PAB + \Delta PCD) = \frac{1}{2}ax + \frac{1}{2}ay$

$=\frac{1}{2}a(x+y)=\frac{1}{2}ah$

Given area of the parallelogram = 20 units

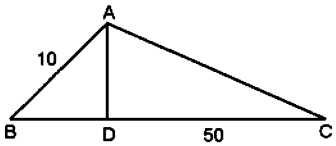
Area of $(\Delta PAB + \Delta PCD) = \frac{1}{2}\times 20 = 10$ units

Example 23 Nayantara bought a triangular piece of land of area 150 m^2. He took a piece of rope and measured the two sides of the plot and found the largest side to be 50 m and another side to be 10 m. What is the exact length of the third side?

(a) $40\sqrt{3}$ m (b) $30\sqrt{2}$ m

(c) $\sqrt{1560}$ m (d) 32 m

Solution



Let the triangle be ABC.

Now, area of ΔABC = (1/2) (BC) (AD) (Where D is a point on BC such that $AD \perp BC$)

Now, AD has to be equal to $\frac{\text{Area } \triangle 2}{BC}$

$$AD = \frac{(150)(2)}{50} \text{ m} = 6 \text{ m}$$

Now ΔBDA is a right-angled triangle

$\therefore BD^2 = AB^2 - AD^2 = 10^2 - 6^2 = 8^2 \text{ m}^2$

$\Rightarrow$ BD = 8 m and DC = BC – BD = 42 m

In a right-angled ΔADC, $AC^2 = AD^2 + DC^2 = (6)^2 + (42)^2$

$\Rightarrow AC = \sqrt{1800} = 30\sqrt{2}$

Alternatively, going through the options

$30\sqrt{2} = 42$

$40\sqrt{3} = 68$

$\sqrt{1560} = 39$

$\therefore$ Using the principle that the sum of any two sides of a triangle is greater than the third side. The two given sides are 50 and 10.

From the choices the above condition is satisfied only for choice (b).

In Choice (a) 10 + 50 < 68

In Choice (c) 39 + 10 < 50

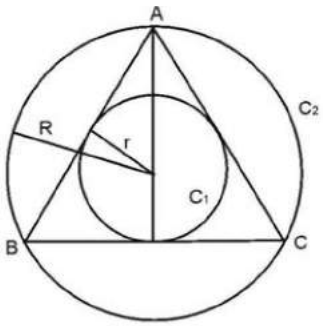
In Choice (d) 32 + 10 < 50

Example 24 C_1 and C_2 are two concentric circles with radii 5 cm and 9 cm respectively. If A, B and C are points on C_2 such that AB and AC are tangent to C_1 at how many points does BC intersect C_1?

(a) 0 (b) 1

(c) 2 (d) Cannot be determined

Solution

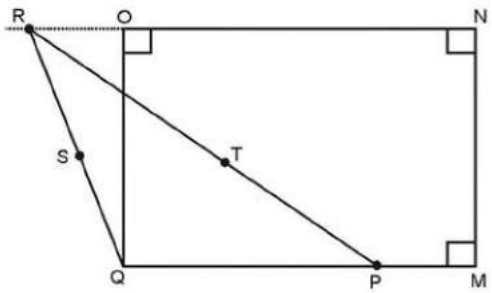


Consider two concentric circles with radii r and R (r < R). If $r = \frac{R}{2}$ and A, B and C are points on the outer circle such that AB and AC are tangent to the inner circle, then BC is also a tangent.

(In this case, C_1 and C_2 become the incircle and circumcircle of ABC respectively, which will be an equilateral triangle)

If $r < \frac{R}{2}$, BC does not intersect or touch the inner circle and if $r > \frac{R}{2}$ then BC intersects the inner circle at two points.

Example 25 A rectangle MNOQ is drawn and length 'NO' is extended to point R and a triangle QPR is drawn with QP $= \frac{2}{3}$ QM. Angle QRP = 45° and side QR $4\sqrt{7}$ cm, S and T are the midpoints of sides QR and PR respectively. If ST = 6 units, the area (in sq. cm) of the rectangle is



(a) 112 (b) 144

(c) 288 (d) 256

Solution Since the line joining the mid-points of two sides of a triangle is parallel and equal to half the third side, we have PQ = 2(ST), $\Rightarrow$ PQ = 12 cm

Since, PQ $= \frac{1}{5}$ QM

Now, $\angle ORP = 45°$

Draw $PV \perp ON$

In ΔRVP, $\tan 45° = \frac{PV}{VR} = \frac{PV}{OR+OV} = \frac{\text{breadth}}{12+OR}$

(Since OV = PQ = 12 cm) also, breadth^2 + OR^2 = (QR)^2

In ΔROQ, $\Rightarrow$ Breadth = x + 12 where OR = x

$\Rightarrow (x+12)^2 = \left(4\sqrt{17}\right)^2 - x^2$ (given that QR = $\left(4\sqrt{17}\right)$

$\Rightarrow 2x^2 + 24x - 128 = 0$

$\Rightarrow x^2 + 12x - 24 = 0$

$\Rightarrow x = 4$ or -16

Since $x > 0$, $x = 4$

Breadth = x + 12 i.e., 16 cm, and the area of MNOQ = 16 × 18 = 288 sq. cm.

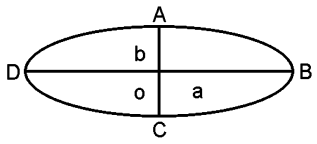
Example 26 ΔABC has sides AB, AC measuring 2001 and 1002 units respectively. How many such triangles are possible with all integral sides?

(a) 2001 (b) 1002 (c) 2003 (d) 1004

Solution Value of BC will lie in between 999 and 3003. Hence 999 < BC < 3003.

So, the total values possible for BC = 2003

Ellipse The path of a moving point which moves in such a way that its distance from a fixed point (focus) bears a constant ratio with its distance from a fixed line (directrix)



Given: OB = OD = a (Semi-major axis)

OA = OI = b (Semi = minor axis)

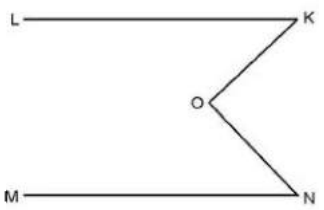
- Area = πab
- Perimeter = $\pi(a + b)$

Maths behind the formula (Area = πab) One way to see why the formula is true is to realize that the ellipse is just a unit circle that has been stretched by a factor 'a' in the x-direction and by factor 'b' in the y-direction. Hence the area of the ellipse is just $a \times b$ times the area of unit circle.

PRACTICE EXERCISES

WARM UP

Q.1. In the given figure, MN and KL are parallel lines.



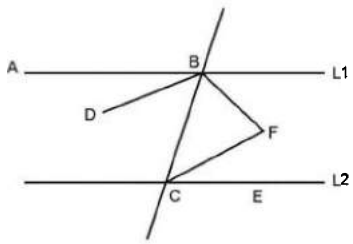
$\angle LKO = 70°$, $\angle KON = 100°$

Find $\angle MNO$.

(a) 20° (b) 30°

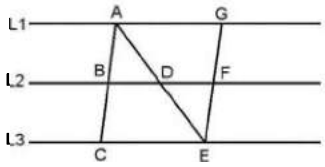
(c) 40° (d) 50°

Q.2. In the given figure, L1 and L2 are parallel lines. $\angle ABC = 80°$. If the lines BD and CF are parallel and $\angle DBC = 30°$, find $\angle FCE$.



(a) 30° (b) 45° (c) 50° (d) 60°

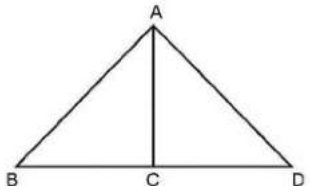
Q.3. The lines L1, L2 and L3 are parallel. If AB = 4 cm, BC = 6 cm, AD = 12 cm and FG = DE then find EF.



(a) 27cm (b) 36 cm
(c) 48 cm (d) None of these

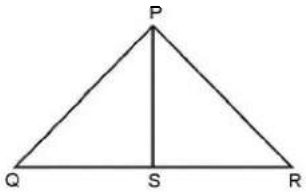
Q.4. The supplement of an angle is five times the angle. What is the measurement of the angle?
(a) 30° (b) 65° (c) 40° (d) 45°

Q.5. In the given figure, AB = AD. ∠ACB = 95° + ∠BAC and ∠BAD = 150°. Find ∠ACB.



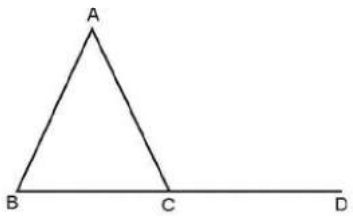
(a) 110° (b) 120° (c) 130° (d) 140°

Q.6. In the given figure, PS is the altitude drawn to the side QR of the triangle PQR. ∠PRS = ∠QPS − 20°. ∠PQS = 20°, ∠PRQ = 50°. What is the value of ∠QPR?



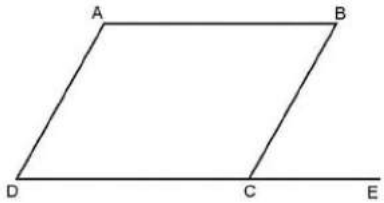
(a) 100° (b) 110° (c) 120° (d) 130°

Q.7. In ΔABC, ∠ACD = 100° and ∠ABC = 40°. Find ∠BAC



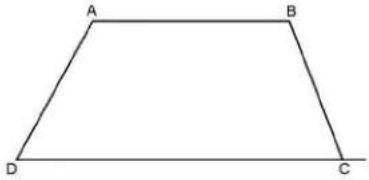
(a) 40° (b) 100° (c) 80° (d) 60°

Q.8. In the given figure, ABCD is a parallelogram. Find 2∠ABC – ADC if A = ∠135°.



(a) 50° (b) 60°
(c) 65° (d) 75°

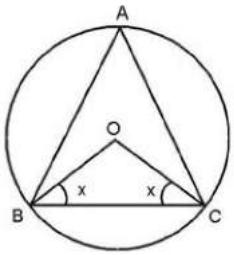
Q.9. ABCD is an isosceles trapezium with lines AB parallel to CD. If ∠DCB = 40°, ∠BAD equals



(a) 40° (b) 80° (c) 100° (d) 140°

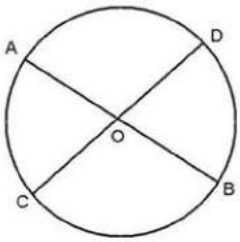
Q.10. If each interior angle of a regular polygon is 135°, find the number of diagonals.
(a) 54 (b) 48
(c) 20 (d) None of these

Q.11. In the given figure, ΔABC is circumscribed by a circle having its centre at O. If OB and OC are phe angle bisector of ∠B and ∠C respectively, what is the value of ∠BAC?



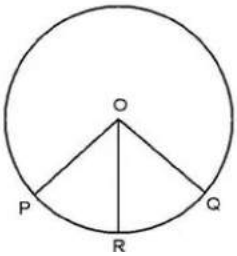
(a) 45° (b) 60° (c) 75° (d) 50°

Q.12. In the given figure, AB and CD are two chords of a circle intersecting at O. If AO = 4 cm and OB = 6 cm and OC = 3 cm. Find OD.



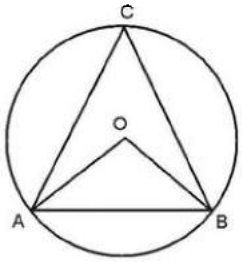
(a) 4 cm (b) 6 cm (c) 8 cm (d) 10 cm

Q.13. In the given figure, if 2 ∠POR = 3 ∠ROQ and ∠POQ = 100° then find ∠POR.



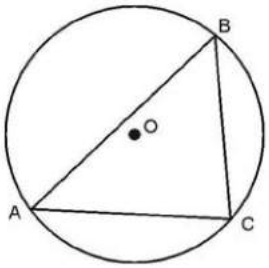
(a) 20° (b) 30° (c) 45° (d) 60°

Q.14. In the above figure, O is the centre of the circle, and ∠AOB = 120°. Find ∠ACB.



(a) 60° (b) 30°
(c) 45° (d) 50°

Q.15. In the given figure, O is the centre of the circle, ∠AOC = 160°. If AB = AC, then find ∠BAC.

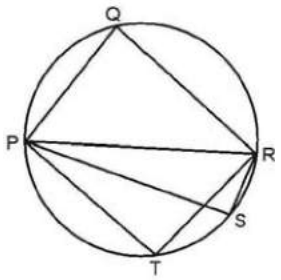


(a) 45° (b) 50°
(c) 60° (d) 75°

Q.16. PQRS is a cyclic quadrilateral. If PQR is an equilateral triangle, find ∠RSP.

(a) 120° (b) 60°
(c) 30° (d) 150°

Q.17. In the given figure, PT = TR, ∠PQR = 100°. Find ∠PRT.

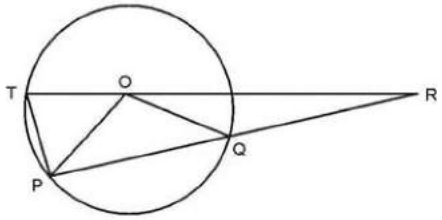


(a) 40° (b) 50°
(c) 60° (d) 70°

Q.18. In a triangle ABC, the incentre is at O. If ∠BOC = 100°, find ∠BAC.

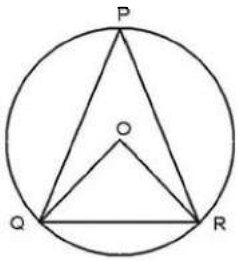
(a) 10° (b) 20°
(c) 30° (d) 40°

Q.19. In the given figure, O is the center of the circle, OQ = QR and ∠QRO = 15°. Find ∠OQP.



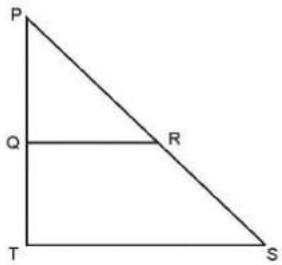
(a) 120° (b) 150° (c) 135° (d) 105°

Q.20. In the given figure, O is the centre of the circle and ∠OQP + ∠ORP = 70°. Find ∠ORQ.



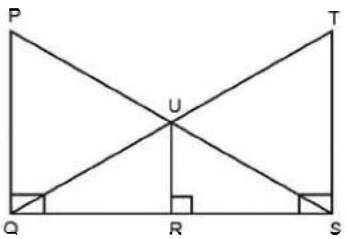
(a) 20° (b) 30°
(c) 40° (d) 50°

Q.21. In the given figure PQ × PS = PT × PR. If ∠PQR = ∠PST + 30° and ∠PTS = 100° then find ∠PRQ.



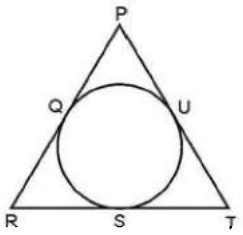
(a) 100° (b) 70° (c) 130° (d) 50°

Q.22. In the above figure, QR = 4 cm and RS = 12 cm. TS = 8 cm and QU is extended to T. Find PQ.



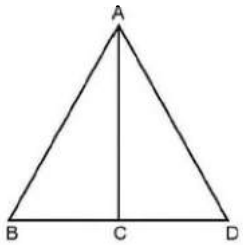
(a) 3 cm (b) 24 cm
(c) 21 cm (d) None of these

Q.23. In the given figure PR, PT and RT are tangents to the circle at Q, U and S respectively. PR = (RT+3) cm; PR = (PT+1) cm. If the perimeter of triangle RPT is 26 cm, what is value of QR + PT?



(a) 12 cm (b) 3 cm (c) 14 cm (d) 15 cm

Q.24. In the given figure, ∠CAB = 1/2 ∠ACD, ∠BAD = 85° and ∠ADC = 40°. Find ∠ACB.



(a) 45° (b) 55°
(c) 70° (d) 60°

Q.25. How many different regular polygons can be formed with the interior angle exceeding the exterior angle and the sum of the interior angles not exceeding 180°?
(a) 7 (b) 8
(c) 9 (d) None of these

FOUNDATION

Q.1. The ratio of the sides of ΔABC is 1:2:4. What is the ratio of the altitudes drawn onto these sides?
(a) 4:2:1 (b) 1:2:4
(c) 1:4:16 (d) None of these

Q.2. The largest angle of a triangle of sides 7 cm, 5 cm and 3 cm is
(a) 45° (b) 60° (c) 90° (d) 1200

Q.3. The three sides of a triangle measure 6 cm, 8 cm and 10 cm respectively. A rectangle equal in area to the triangle has a length of 8 cm. The perimeter of the rectangle is:
(a) 11 cm (b) 22 cm
(c) 16 cm (d) None of these

Q.4. X and Y are respectively two points on the sides DC and AD of parallelogram ABCD. The area of ΔABX is equal to
(a) $\frac{1}{3}\times$ area of ΔBYC
(b) Area of ΔBYC
(c) $\frac{1}{2}\times$ area of ΔBYC
(d) 2 × area of ΔBYC

Q.5. A rectangular enclosure 40 m × 36 m has a horse tethered to a corner with a rope of 14 m in length. What is the ratio of the respective areas it can graze, if it is outside the enclosure and if it is inside the enclosure?
(a) 2 (b) 2:3
(c) 1:4 (d) 3:1

Q.6. What is the area of the triangle having the sides 4, 6 and 12 units?
(a) 14.28 (b) 12.6
(c) 16.3 (d) Not possible

Q.7. Identical spherical balls are spread on a table top so as to form an equilateral triangle. How many balls are needed so that a side of the equilateral triangle contains n balls?
(a) $\frac{n(n+1)}{2}$ (b) $n^2 - 1$
(c) $n(n-1)$ (d) $n!$

Direction for questions 8 and 9: Read the passage below and solve the questions based on it.

The area of a square is equal to the area of a rectangle. Moreover, the perimeter of the square is also equal to the perimeter of the rectangle.

Q.8. The side of the square is equal to the:
(a) Length of the rectangle
(b) Breadth of the rectangle
(c) Cannot be determined
(d) Both [1] and [2]

Q.9. The length of the rectangle is equal to the:
(a) Breadth of the rectangle
(b) Side of the square
(c) Cannot be determined
(d) Both [1] and [2]

Q.10. A quadrilateral is inscribed in a circle. If an angle is inscribed in each of the segments outside the quadrilateral, then what is the sum of the four angles?
(a) 270° (b) 360°
(c) 540° (d) 720°

Q.11. In a triangle PQR, PQ = 6 cm, QR = 8 cm and PR = 10 cm. The length of the median bisecting the shortest side is (approx.):
(a) 10 cm (b) 8.5 cm
(c) 9 cm (d) None of these

Q.12. A square is inscribed in a circle which is inscribed in an equilateral triangle. If one side of the triangle is 'a', find the area of the square.
(a) $\frac{1^2}{3}$ (b) $\frac{a^2}{6}$
(c) $\frac{3a^2}{8}$ (d) $\frac{\pi a^2}{12}$

Q.13. A cyclic quadrilateral is such that two of its adjacent angles are divisible by 6 and 10 respectively. One of the remaining angles will necessarily be divisible by:
(a) 3 (b) 4
(c) 8 (d) None of these

Q.14. In the above question, if one of the remaining angles is necessarily divisible by 16, then its measure can be:
(a) 128° (b) 48°
(c) 132° (d) 112°

Q.15. The volume of spheres are proportional to the cubes of their radii. Two spheres of the same material weigh 3.6 kg and 2.7 kg and the radius of the smaller one is 2 cm. If the two were melted down and formed into a single sphere, what would be its radius?
(a) 4 cm (b) 4.3 cm
(c) 3 cm (d) 2.6 cm

Direction for questions 16 and 17: Read the passage below and solve the questions based on it.

$f(x)$ is the area of a square where, x is the side of a square. $g(x)$ is the perimeter of square where, x is the side of a square. $h(x, y)$ is the area of a rectangle where x is the length and y is the breadth. $i(x, y)$ is the perimeter of a rectangle where x is the length and y is the breadth.

Q.16. The value of $g[h\{f\{i\ (2,3)\}, 2\}$ is equal to
(a) 450 (b) 600
(c) 800 (d) 1000

Q.17. The value of $\frac{[f(3)-g(2)]}{h(4,2)-i(6,2)} \times \frac{g(6)}{f(1)}$ is equal to
(a) −3 (b) 2
(c) 1 (d) −4

Direction for questions 18 and 19: Read the passage below and solve the questions based on it.

A 10 m long piece of wire is cut into two pieces, one of which is bent into a circle and the other into the square enclosing it.

Q.18. The area of the square outside the circle is
(a) 1200 cm² (b) 2400 cm²
(c) 3600 cm² (d) 4200 cm²

Q.19. The ratio of the radius of the circle to the perimeter of the square is
(a) 8:1 (b) 1:8 (c) 4:1 (d) 1:4

Q.20. There are two regular polygons with the number of sides in the ratio of 4:5 and the interior angles in the ratio of 25:26. The number of sides in the first polygon are
(a) 8 (b) 10 (c) 12 (d) 15

Q.21. A square is inscribed in a semi circle of radius 10 cm. What is the area of the inscribed square? (Given that the side of the square is along the diameter of the semicircle.)
(a) 70 cm² (b) 50 cm²
(c) 25 cm² (d) 80 cm²

Q.22. ABCD is a parallelogram in which $\angle B = 70°$. Find the number of points X in the plane of the parallelogram such that it is equidistant from its vertices.
(a) zero (b) one (c) two (d) three

Q.23. There is a fan with 3 blades at 120° to each other whose central circular disc has an area of 3p cm² and a blade is $(20 - \sqrt{3})$ cm long. If the tips of the blades are joined so as to form an equilateral triangle, what will be its area?
(a) 900 cm² (b) $900\sqrt{3}$ cm²
(c) $(900 + 9x)$ cm² (d) $(3x + 300)$ cm²

Q.24. Two circles of an equal radii are drawn, without any overlap, in a semicircle of radius 2 cm. If these are the largest possible circles that the semicircle can accommodate, what is the radius (in cm) of each of the circles?
(a) 0.414 (b) 0.828
(c) 0.172 (d) 0.586

Q.25. The sides of a regular octagon are extended to form a star. Find the measure of the internal angle at each point of the star.
(a) 45° (b) 90° (c) 135° (d) 60°

Q.26. PQRS is trapezium, in which PQ is parallel to RS, and PQ = 3 (RS). The diagonal of the trapezium intersect each other at X, then the ratio of ΔPXQ and ΔRXS is

(a) 6:1 (b) 3:1 (c) 9:1 (d) 7:1

Direction for questions 27 and 28: Read the passage below and solve the questions based on it.

All the angles of a quadrilateral form a GP and its common ratio is a natural number. Exactly two of the angles are obtuse and none of the angles is a right angle.

Q.27. What is the measurement of the largest angle?
(a) 124° (b) 156°
(c) 192° (d) None of these

Q.28. What is the measurement of the smallest angle?
(a) 18° (b) 24°
(c) 36° (d) None of these

Q.29. Two circles with centres C1 and C2 and radii 6 cm and 8 cm respectively cut each other at right angles. Find the length of the common chord.
(a) 10 cm (b) 4.8 cm
(c) 9.6 cm (d) 5 cm

Q.30. Let ABCDEF be a regular hexagon. What is the ratio of the area of the triangle ACE to that of the hexagon ABCDEF?
(a) 1/3 (b) 1/2 (c) 2/3 (d) 5/6

Q.31. The sides of a triangle are in the ratio of $\frac{1}{2}:\frac{1}{3}:\frac{1}{4}$.
If the perimeter is 52 cm, then the length of the smallest side is
(a) 9 cm (b) 10 cm
(c) 11 cm (d) 12 cm

Q.32. A pond 100 m in diameter is surrounded by a circular grass walk-way 2 m wide. How many square metres of grass is the on the walk-way?
(a) 98 π (b) 100 π
(c) 204 π (d) 202 π

Q.33. A triangle and a parallelogram are constructed on the same base such that their areas are equal. If the altitude of the parallelogram is 100 m, then the altitude of the triangle is
(a) 100 m (b) 200 m
(c) $100\sqrt{2}$ m (d) $10\sqrt{2}$ m

Q.34. A rhombus OABC is drawn inside a circle, whose centre is at O, in such a way that the vertices A, B and C of the rhombus are on the circle. If the area of the rhombus is $32\sqrt{3}$ m², then the radius of the circle is
(a) 64 m (b) 8 m (c) 32 m (d) 46 m

Q.35. Let A be the area of a square inscribed in a circle of radius 'r' and let B be the area of a hexagon inscribed in the same circle. Then B/A equals:
(a) $2\sqrt{3}$ (b) $\frac{3\sqrt{3}}{4}$
(c) $\frac{2\sqrt{3}}{4}$ (d) None of these

Q.36. The dimensions of a rectangular box are in the ratio of 1:2:4 and the difference between the costs of covering it with the cloth and a sheet at the rate of Rs 20 and Rs 20.5 per sq m respectively is Rs 126. Find the dimensions of the box.
(a) 2 m, 6 m, 9 m (b) 6 m, 12 m, 24 m
(c) 1 m, 2 m, 4 m (d) None of these

Q.37. If the sides 50 m and 130 m of the triangular field meet at an angle of 72°, then find the area in which wheat is cultivated. (Sin 72° = 0.9510, Cos 72° = 0.309)
(a) 100 πm² (b) 125 πm²
(c) 160 πm² (d) None of these

Q.38. Four horses are tied on the four corners of a square field of 14 m length so that each horse can touch just the other two horses. They were able to graze in the area accessible to them for 11 days. For how many days is the ungrazed area sufficient for them?
(a) 3 days (b) 4 days
(c) 5 days (d) 2 days

Q.39. The ratio of the area of a square inscribed in a semicircle to that of the area of a square inscribed in the circle of the same radius is
(a) 2:1 (b) 2:3 (c) 2:5 (d) 2 : 7

Q.40. The area of a rectangle is given by $A = 8x^2 - 2x - 15$. If the length was larger than the breadth, then the length can be
(a) $(3x + 5)$ (b) $(5x + 3)$
(c) $(4x + 5)$ (d) $(2x - 3)$

Q.41. The ratio of the area of a square to that of the square drawn on the its the diagonal is
(a) 1:4 (b) 2:1 (c) 1:2 (d) 1:3

Q.42. An equilateral triangle had 7 cm long sides. Equal circles were drawn about the three vertices of the triangle so that each circle touched the other two. Calculate the area of space enclosed between these circles.
(a) 3.934 cm² (b) 1.967 cm²
(c) 3.9 cm² (d) 1.95 cm²

Q.43. What is the area of the triangle in which two of its medians 9 cm and 12 cm long intersect at the right angles?
(a) 72 (b) 60 (c) 56 (d) 48

Q.44. If one leg of an isosceles right-angled triangle is increased by 6 cm and that of the other leg decreased by 4 cm, then the area of the triangle decreases by 24 sq cm. Find the length of the leg of the original triangle.
(a) 36 cm (b) 30 cm
(c) 24 cm (d) None of these

Q.45. ABCD is a parallelogram in which ∠D = 120°. The bisector of ∠D bisects the side AB. If the length of

the bisector is 5 cm, then what is the perimeter of the parallelogram?
(a) 30 cm (b) 25 cm
(c) 40 cm (d) 15 cm

Q.46. Four horses are tethered at four corners of a square plot of side 14 m so that the adjacent horses can just reach one another. There is a small circular pond of area 20 m² at the centre. Find the ungrazed area.
(a) 22 m² (b) 42 m²(c) 84 m² (d) 168 m²

Q.47. Euclid has a triangle in mind. Its longest side has length 20 and another of its sides has length 10. Its area is 80. What is the exact length of its third side?
(a) $\sqrt{260}$ (b) $\sqrt{250}$
(c) $\sqrt{240}$ (d) $\sqrt{270}$

Q.48. AB is the hypotenuse in the right angled triangle ABC. N is the point inside the triangle which divides the triangle in three equal parts (ΔABN, ΔCAN, ΔBCN). What is the distance between the circumcentre of this triangle from this point N?
(a) $\frac{AB}{4}$ (b) $\frac{AB}{6}$
(c) $\frac{AB}{3}$ (d) $\frac{2}{1+\sqrt{3}}$

Q.49. Two sides of a triangle are 4 and 5. Then, for the area of the triangle, which one of the following bounds is the sharpest?
(a) <10. (b) ≤ 10. (c) ≤ 8. (d) > 5.

Q.50. ABC is a triangle and P is a point inside it such that ∠BPC = ∠CPA = ∠APB. Then P is:
(a) The point of intersection of medians
(b) The incentre
(c) The circumcentre
(d) None of these

MODERATE

Q.1. The angles of a triangle are in the ratio of 4:1:1. Then the ratio of the largest side to the perimeter is
(a) $\frac{2}{3}$ (b) $\frac{1}{2+\sqrt{3}}$
(c) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (d) $\frac{2}{1+\sqrt{3}}$

Q.2. In ΔABC, P and Q are mid-points of sides AB and BC respectively, right angled at B, then
(a) $AQ^2 + CP^2 = AC^2$
(b) $AQ^2 + CP^2 = \frac{4}{5} AC^2$
(c) $AQ^2 + CP^2 = \frac{3}{4} AC^2$
(d) $AQ2 + CP2 = \frac{5}{4} AC2$

Q.3. Three identical right angle cones with base radius r are placed on their bases so that each is touching the other two. The radius of the circle drawn through their vertices is:
(a) Smaller than r
(b) Equal to r
(c) Larger than r
(d) Depends on the height of the cones

Q.4. The line AB is 6 m, in length and is tangent to the inner of the two concentric circles at point C. It is known that the radii of the two circles are integers. The radius of the outer circle is-------, where A and B are points on the outer circle.
(a) 5 m (b) 4 m (c) 6 m (d) 3 m

Q.5. Under the usual 2-dimensional coordinate system the equation $|x| + |y| = 1$, where x and y are real numbers, represents
(a) A rhombus which is not a square
(b) A parallelogram which is not a rhombus
(c) A square whose sides are not parallel to the coordinate axes
(d) A square with sides parallel to the coordinate axis

Q.6. Two chords of lengths a and b of a circle subtend 60° and 90° angles at the centre respectively. Which of the following is correct?
(a) $b = \sqrt{2a}$ (b) $b = \sqrt{2b}$
(c) $a = 2b$ (d) $b = 2a$

Q.7. PQR is a right-angled triangle with ∠Q = 90°, S is the mid-point of PR, and QS $= \sqrt{117}$ cm. The sum of the length of sides PQ and QR is 30 cm. Area of ▽PQR is
(a) 216 cm² (b) 108 cm²
(c) 54 cm² (d) 162 cm²

Q.8. In a trapezium, the diagonals intersect at point O. The ratio of the length of one of the diagonals from one vertex of the trapezium to the point O to its entire length is 2:5. Find the ratio of its parallel sides (smaller side:larger side).
(a) 2 : 5 (b) 2 : 3 (c) 2:7 (d) 5 : 7

Q.9. A chord of length 32 cm is placed inside a circle of

radius 20 cm and a point whose distance from the centre of the circle is 13 cm, is marked on the chord. Calculate the lengths of the segment of the chord.
(a) 21 cm and 11 cm (b) 19 cm and 13 cm
(c) 16 cm each (d) 18 cm and 14 cm

Q.10. The sum of the lengths of the hypotenuse and one of the perpendicular sides of a right angled triangle is L. When the area of this triangle is maximum, the angle between these two sides is:
(a) 45° (b) 22.5°
(c) 60° (d) None of these

Q.11. In a triangle, the lengths of the two larger sides are 8 cm and 7 cm. The angles of the triangle are in an arithmetic progression. The length of the remaining side can be:
(a) 3 cm
(b) 5 cm
(c) Either 3 cm or 5 cm
(d) None of these

Q.12. The biggest possible regular hexagon H is cut out of an equilateral triangle X. The biggest possible equilateral triangle Y is cut out from the hexagon H. What is the ratio of the areas of the equilateral triangles X and Y?
(a) 5 : 1 (b) 6 : 1 (c) 8 : 1 (d) 3 : 1

Q.13. Nine parallel chords are drawn in a circle of diameter 10 cm. If the distance between any two of the adjacent chords is 1 cm, which of the following statements is always true?
(a) One of the chords is the diameter of the circle.
(b) At least two of the chords must be of an equal length.
(c) The difference between the lengths of any two adjacent chords on the same side of the diameter is greater than 1 cm.
(d) None of these

Direction for questions 14 and 15: Read the passage below and solve the questions based on it.

There are three equal circles of unit radii touching each other.

Q.14. Find the area of the triangle circumscribing the three circles.
(a) $\sqrt{3}(\sqrt{3}+1)^2$ (b) $(\sqrt{3}+1)^2$
(c) $\sqrt{3}(\sqrt{3}+1)$ (d) None of these

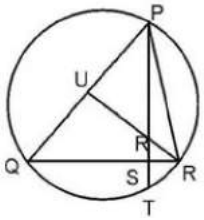
Q.15. What would be the area of the remaining portion if the same three circles are circumscribed by another circle?
(a) $\pi\left(\frac{2}{\sqrt{3}}+1\right)^2$ (b) $\pi\left(\frac{2}{3}+1\right)^2-3\pi$
(c) $\left(\frac{3}{\sqrt{2}}+1\right)^2\pi-3\pi$ (d) $\left(\frac{1}{\sqrt{3}}+1\right)^2\pi-3\pi$

Q.16. A circle passes through the vertex A of an equilateral triangle ABC and is tangent to BC at it mid-point. Find the ratio in which the circle divides each of the sides AB and AC.
(a) 1 : 1 (b) 3 : 2 (c) 3 : 1 (d) 2 : 1

Q.17. In any quadrilateral ABCD, the diagonal AC and BD intersect at a point X. If E, F, G and H are the mid-points of AX, BX, CX and DX respectively, then what is the ratio of (EF + FG + GH + GE) to (AD + DC + CB + BA)?
(a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) $\frac{3}{4}$ (d) Data insufficient

Q.18. In ∇PQR, the incircle touches the sides QR, RP and PQ at T, U and S respectively. If the radius of the incircle is 4 units and QT, RU and PS are consecutive integers, then what is the area of the triangle (in sq. units)?
(a) 42 (b) 84 (c) 64 (d) 96

Q.19. If altitude PS meets the circumcircle of ∇PQR at T and H is the orthocenter, then what is the length of the line segment HT? (Given that $\angle RUP = \angle PSQ = 90°$)



(a) 4 cm (b) 5 cm (c) 6 cm (d) 8 cm

Q.20. A vertical lamp-post OP stands at the centre O of a square ABCD. Let h and b denote the length OP and AB respectively. Suppose $\angle APB = 60°$ then the relationship between h and b can be expressed as
(a) $2b^2 = h^2$ (b) $2h^2 = b^2$
(c) $3b^2 = 2h^2$ (d) $3h^2 = 2b^2$

Q.21. Consider two different cloth-cutting processes. In the first one, *n* circular cloth pieces are cut from a square cloth piece of side a in the following steps: the original square of side a is divided into n smaller squares, not necessarily of the same size; then a circle of the maximum possible area is cut from each of the smaller squares. In the second process, only one circle of the maximum possible area is cut from the square of side a and the process ends there. The cloth pieces remaining after cutting the circles are scraped in both the process. The ratio of the total area of the scrap cloth generated

in the former to that in the latter is
(a) 1:1 (b) $\sqrt{2}:1$
(c) $\frac{n(4-x)}{4n-x}$ (d) $\frac{4n-x}{n(4-x)}$

Q.22. Let S_1 be a square of side a. Another square S_2 is formed by joining the mid-points of the sides of S_1. The same process is applied to S_2 to form yet another square S_3, and so on. If $A_1, A_2, A_3, \ldots$ be the areas and $P_1, P_2, P_3, \ldots$ be the perimeters of $S_1, S_2, S_3, \ldots$ respectively, then the ratio of $\frac{P_1+P_2+P_3+.....}{A_1+A_2+A_3+.....}$ equals.
(a) $\frac{(2+\sqrt{2})}{a}$ (b) $\frac{2(2-\sqrt{2})}{a}$
(c) $\frac{2(2+\sqrt{2})}{a}$ (d) $\frac{2(1+2\sqrt{2})}{a}$

Q.23. One of the angles of a parallelogram is of 150°. Altitudes are drawn from the vertex of this angle. If these altitudes measure 6 cm and 8 cm, then find the perimeter of the parallelogram.
(a) 28 cm (b) 42 cm
(c) 56 cm (d) 64 cm

Q.24. A square, whose side is 2 meters, has its corners cut away so as to form an octagon with all sides equal. Then, the length of the each side of the octagon, in meters is:
(a) $\frac{\sqrt{2}}{\sqrt{2}+1}$ (b) $\frac{2}{\sqrt{2}-1}$
(c) $\frac{2}{\sqrt{2}-1}$ (d) $\frac{\sqrt{2}}{\sqrt{2}+1}$

Q.25. A certain city has a circular wall around it and this wall has four gates pointing north, south, east and west. A house stands outside the city three km north of the north gate, and it can be seen from a point nine km east of the south gate. What is the diameter of the wall that surrounds the city?
(a) 6 km (b) 9 km
(c) 12 km (d) None of these

Q.26. The perimeter of a right-angled triangle is four times the shortest side. The ratio of the other two sides is
(a) 5:6 (b) 3:4
(c) 4:5 (d) 2:3

Q.27. A circle is inscribed in an equilateral triangle and a square is inscribed in the circle. The ratio of the area of the triangle to the area of the square is
(a) $\sqrt{3}:\sqrt{2}$ (b) $3\sqrt{3}:2$
(c) $3:\sqrt{2}$ (d) $\sqrt{2}:1$

Q.28. The sides of a triangle are given to be $x^2 + x + 1$, $2x + 1$ and $x^2 - 1$. Then the largest of the three angles of the triangle is
(a) 75° (b) $\frac{x}{x+1}\pi$
(c) 120° (d) 135°

Q.29. ABC is a triangle inscribed in a circle. AD and AE are straight lines drawn from the vertex A to the base BC parallel to the tangents at B and C respectively. If AB = 5 cm AC = 6 cm, and CE = 9 cm, then the length of BD (in cm) equals.
(a) 7.5 (b) 10.8 (c) 7.0 (d) 6.25

Q.30. In a triangle ABC, the medians AM and CN to the sides BC and AB respectively, intersect at the point O. Let P be the mid-point of AC and let MP intersect CN at Q. If the area of the triangle OMQ is s square units, the area of ABC is
(a) 16s (b) 18s (c) 12s (d) 24s

Q.31. ABCD is a cyclic quadrilateral and the points A, B and C form an equilateral triangle. What is the sum of the lengths of line segments DA and DC?
(a) DB (b) DB/2 (c) $\sqrt{2}$ DB (d) DB/$\sqrt{2}$

Q.32. PQRS is a trapezium with PQ and RS parallel, PQ = 6 cm, QR = 5 cm, RS = 3 cm and PS = 4 cm. The area of PQRS is:
(a) 27 cm² (b) 12 cm²
(c) 18 cm² (d) None of these

Q.33. The side AB of a rectangle ABCD is a tangent to a circle which passes through the points C and D. The centre of the circle does not lie within the rectangle ABCD. If the length of the rectangle is twice the breadth, then what is the radius of the circle (in terms of breadth of the rectangle)?
(a) Breadth $(2 - \sqrt{2})$ (b) Breadth
(c) 2 × Breadth (d) None of these

Q.34. Let C_1 and C_2 be the inscribed and circumscribed circles of a triangle with sides 3 cm, 4 cm and 5 cm. The $\frac{\text{area of } C_1}{\text{area of } C_2}$ equals.
(a) $\frac{16}{25}$ (b) $\frac{4}{25}$ (c) $\frac{9}{25}$ (d) $\frac{9}{16}$

Q.35. The sides of a triangle are given by: $\sqrt{b^2+c^2}, \sqrt{c^2+a^2}$ and $\sqrt{a^2+b^2}$ where a, b, c are positive. Then the area of the triangle equals.

(a) $\frac{1}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}$

(b) $\frac{1}{2}\sqrt{a^4+b^4+c^4}$

(c) $\frac{\sqrt{3}}{2}\sqrt{b^2c^2+c^2a^2+a^2b^2}$

(d) $\frac{\sqrt{3}}{2}(bc+ca+ab)$

Q.36. Let the bisector of the angle at C of a triangle ABC intersect the side AB in a point D. Then the geometric mean of CA and CB:
(a) Is less than CD
(b) Is equal to CD
(c) Is greater than CD
(d) Does not always satisfy any one of the foregoing property

Q.37. There is a pole at the centre of a stadium which is in the shape of an equilateral triangle. The angle subtended by any side of the stadium at the top of the tower is right-angled. What is the ratio of the height of the pole to the perimeter of the stadium?
(a) $\sqrt{6}$ (b) $\frac{\sqrt{6}}{18}$
(c) $\frac{\sqrt{6}}{3}$ (d) $\frac{\sqrt{6}}{4}$

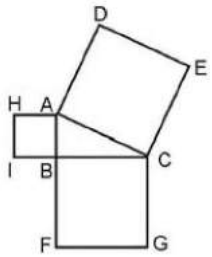
Q.38. If in a triangle ABC with a, b, c denoting sides opposite to angles A, B and C respectively, a = 2 b and A = 3 B, then the triangle
(a) Is isosceles
(b) Is right-angled but not isosceles
(c) Is right-angled and isosceles
(d) Not necessarily be any of the above types

Q.39. The length of the common chord of two circles of radii 15 cm and 20 cm, whose centres are 25 cm apart, is (in cm)
(a) 24 (b) 25 (c) 15 (d) 20

Q.40. PQRS is a trapezium with PQ and RS parallel. PQ = 6 cm, QR = 5 cm, RS = 3 cm, PS = 4 cm. The area of PQRS is
(a) 27 cm^2 (b) 12 cm^2
(c) 18 cm^2 (d) Cannot be determined

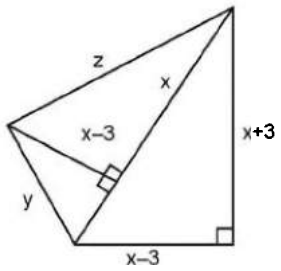
Q.41. If the number of square inches in the area of a square is equal to the number of inches in its circumference, then the diagonal of the square is equal to
(a) 4 (b) $4\sqrt{2}$ (c) $3\sqrt{2}$ (d) $\sqrt{2}$

Q.42. All the three quadrilaterals ADEC, ABIH and BCGF are squares and ABC = 90°. If the area of ADEC = x^2 and the area of AHIB = y^2 ($x^2 > y^2$), then the area of BCGF is:



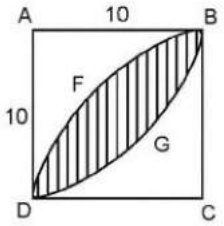
(a) $(x + y)(x - y)$ (b) $(x + y)^2$
(c) $(x - y)^2$ (d) None of these

Q.43. Based on the figure below, what is the value of x, if $y = 10$.



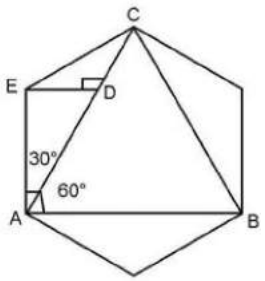
(a) 10 (b) 11
(c) 12 (d) None of these

Q.44. In the figure, ABCD is a square with side 10. BFD is an arc of a circle with centre C. BGD is an arc of a circle with centre A. What is the area of the shaded region?

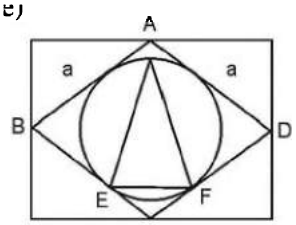


(a) $100\pi - 50$ (b) $100 - 25\pi$
(c) $50\pi - 100$ (d) $25\pi - 10$

Q.45. In the figure below, if the perimeter of ∇ABC is p, then the perimeter of the regular hexagon is
(a) $\frac{3p}{\sqrt{2}}$ (b) $\frac{\sqrt{2}p}{3}$
(c) $\frac{\sqrt{3}p}{2}$ (d) $\frac{2p}{\sqrt{3}}$



Q.46. What is the area of the inner equilateral triangle if the side of the outermost square is 'a'? (ABCD is a square)



(a) $\frac{3\sqrt{3}a^2}{32}$ (b) $\frac{3\sqrt{3}a^2}{64}$

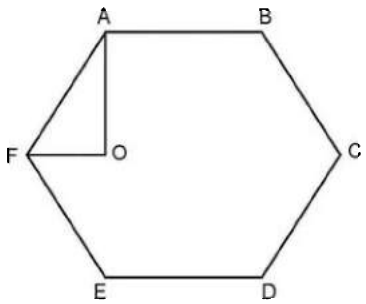
(c) $\frac{5\sqrt{3}a^2}{32}$ (d) $\frac{\sqrt{3}a^2}{16}$

Q.47. Each side of a given polygon is parallel to either the X or the Y axis. A corner of such a polygon is said to be convex if the internal angle is 90° or concave if the internal angle is 270°. If the number of convex corners in such a polygon is 25, then the number of concaves must be

(a) 20 (b) 0

(c) 21 (d) 22

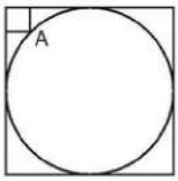
Q.48. In the figure below, ABCDEF is a regular hexagon and AOF = 90°. FO is parallel to ED. What is the ratio of the area of the triangle AOF to that of the hexagon ABCDEF?



(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{24}$ (d) $\frac{1}{18}$

Q.49. In the figure below, the rectangle at the corner measures 10 cm × 20 cm. The corner A of the rectangle is also a point on the circumference of the circle. What is the radius of the circle in cm?



(a) 10 cm (b) 40 cm

(c) 50 cm (d) None of these

Q.50. A piece of paper is in the shape of a right-angled triangle and is cut along a line that is parallel to the hypotenuse, leaving a smaller triangle. There was a 35% reduction in the length of the hypotenuse of the triangle. If the area of the original triangle was 34 square inches before the cut, what is the area (in square inches) of the smaller triangle?

(a) 16.665 (b) 16.565

(c) 15.465 (d) 14.365

ADVANCED

Q.1. A ball of diameter 15 cm is floating so that the top of the ball is 5 cm above the smooth surface (water) of the pond. What is the circumference in centimeters of the circle formed by the contact of the water surface with the ball?

(a) $10\sqrt{2}\pi$ (b) 50 p

(c) 10 p (d) $5\sqrt{2}\pi$

Q.2. Let S be as an arbitrary point on the side PQ of an acute-angled ▽PQR. Let T be the point of intersection of QR extended with the straight line PT drawn parallel to SR through P. Let U be the point of intersection of PR extended with the straight line QU drawn parallel to SR through Q. If PT = a and QU = b, then the length of SR is

(a) $\frac{a+b}{ab}$ (b) $\frac{a-b}{ab}$

(c) $\frac{ab}{a+b}$ (d) $\frac{ab}{a-b}$

Q.3. The municipal authorities of three cities A, B and C jointly constructed three straight roads connecting these cities. The area enclosed by three roads is $150\sqrt{39}$ km^2. The authorities of city B found the length of the road connecting it to the city A is 40 km and that to the city C is 50 km. What is the length of the road connecting the cities A and C, if it is the longest of the three roads?

(a) $20\sqrt{3}$ km (b) $30\sqrt{3}$ km

(c) $30\sqrt{6}$ km (d) $10\sqrt{55}$ km

Q.4. In an equilateral D, 3 coins of radii 1 unit each are kept in such a way that they touch each other and also the sides of the triangle. What is the area of the triangle (in sq. units)?

(a) $4 + 5\sqrt{2}$ (b) $6 + 4\sqrt{3}$

(c) $4 + 6\sqrt{3}$ (d) $3 + 8\sqrt{3}$

Q.5. In a scalene Δ, sum of all the sides can be at most 13 units. How many triangles are possible?

(a) 3 (b) 4 (c) 5 (d) 6

Q.6. Three circles of equal radii have been drawn inside an equilateral triangle, of side a, such that each circle touches the other two circles as well as two sides of the triangles. Then, the radius of each circle is

(a) $\frac{a}{2(\sqrt{3}+1)}$ (b) $\frac{a}{2(\sqrt{3}-1)}$

(c) $\frac{a}{\sqrt{3}-1}$ (d) $\frac{a}{\sqrt{3}-1}$

Q.7. Consider a circle with unit radius. There are seven adjacent sectors, $S_1, S_2, S_3, \ldots S_7$, in the circle such that their total area is $\frac{1}{8}$ th of the area of the circle. Further, the area of the jth sector is twice that of the (j – 1)th sector, for j = 2,... 7. What is the angle, in radians, subtended by the arc of S_1 at the centre of the circle?

(a) $\frac{\pi}{508}$ (b) $\frac{\pi}{2040}$

(c) $\frac{\pi}{1016}$ (d) $\frac{\pi}{1524}$

Q.8. In a triangle ABC, let C_1 be any point on the side AB other than A or B. Join CC_1. The line passing through A and parallel to CC_1 intersects the line BC extended at A_1. The line passing through B and parallel to CC_1 intersects the line AC extended at B_1. The lengths AA_1, BB_1, CC_1 are given to be p, q, r units respectively. Then

(a) $r = \frac{pq}{p+q}$ (b) $r = \frac{p+q}{4}$

(c) $r = \frac{\sqrt{pq}}{2}$ (d) None of these

Q.9. Two circles APQC and PBDQ intersect each other at the points P and Q and APB and CQD are two parallel straight lines. Then only one of the following statements is always true. Which one is it?

(a) ABDC is a cyclic quadrilateral
(b) AC is parallel to BD
(c) ABDC is a rectangle
(d) ΔACQ is a right angle

Q.10. Let s = {(x, y) : |x| + |y| = 2}. Then, the diameter of S is

(a) 2 (b) $4\sqrt{2}$

(c) 4 (d) $2\sqrt{2}$

Q.11. Through the centroid of an equilateral triangle, a line parallel to the base is drawn. On this line, an arbitrary point P is taken inside the triangle. Let h denote the distance of P from the base of the triangle. Let h_1 and h_2 be the distance of P from the other two sides of the triangle. Then

(a) $h = \frac{h_1 + h_2}{2}$ (b) $h = \sqrt{h_1 h_2}$

(c) $h = \frac{2h_1 h_2}{h_1 + h_2}$ (d) None of these

Q.12. Suppose ABCD is a cyclic quadrilateral within a circle of radius r. The bisector of the angle A cuts the circle at point P and the bisector of angle C cuts the circle at point Q. Then

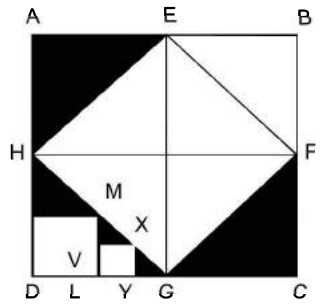
(a) AP = 2r (b) PQ = 2r
(c) BQ = DP (d) PQ = AP

Q.13. Let A (x_1, y_1), B (x_2, y_2) C (x_3, y_3), D (x_4, y_4) be four points such that x_1, x_2, x_3, x_4 and y_1, y_2, y_3, y_4 are both in an AP. If Δ denotes the area of the quadrilateral ABCD, then

(a) Δ = 0
(b) Δ = 1
(c) Δ < 1
(d) Δ depends on the coordinates A, B, C, and D.

Direction for questions 14 to 17: Read the passage below and solve the questions based on it.

In the figure given below, EFGH is a square formed by joining the mid-points of the sides of the square ABCD. KMID and VXYL are the squares formed inside the right angled triangles HDG and MLG respectively. The side of the square ABCD is equal to 'a' cm.



Q.14. What is the ratio of areas of ABCD and KMID?

(a) 4:1 (b) 8:1 (c) 16:1 (d) 12:1

Q.15. What is the ratio of areas of EFGH and VXYI?

(a) 32:1 (b) 64:1 (c) 16:1 (d) 8:1

Q.16. What is the ratio of areas of the inscribed circles of DXVM, DMKH and DGDH?

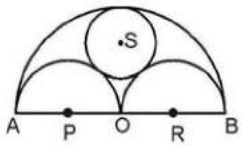
(a) 1:4:16 (b) 1:2:4 (c) 1:3:9 (d) 1:2:√2

Q.17. What is the area of shaded region (in cm^2)?

(a) $\frac{5}{16}a^2$ (b) $\frac{19}{64}a^2$

(c) $\frac{7}{16}a^2$ (d) $\frac{27}{64}a^2$

Q.18. Three horses are grazing within a semi-circular field. In the diagram given below, AB is the diameter of the semi-circular field with centre at O. Horses are tied up at P, R and S such that PO and RO are the radii of semi-circles with centres at P and R respectively and S is the centre of the circle touching the two semi-circles with diameters AO and OB. The horses tied at P and R can graze within the respective semi-circle and the horse tied at S can graze within the circle centered at S. The percentage of the area of the semi-circles with diameter AB that cannot be grazed by the horses is nearest to

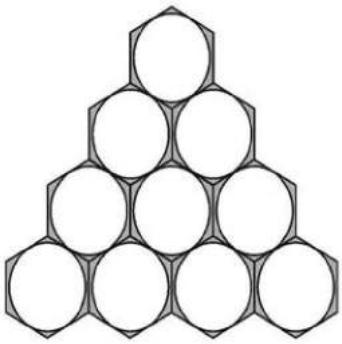


(a) 20 (b) 28
(c) 36 (d) 40

Q.19. Inradius of a right-angled ΔABC is 32 units and the sum of the length of its two sides other than the hypotenuse is 42 units. What is its circumradius?
(a) 26
(b) 52
(c) More than one value possible
(d) Triangle ABC is not possible

Direction for questions 20 and 21: Read the passage below and solve the questions based on it.

In the figure given above, each hexagon is regular and has side measuring 'a' cm. it is also given that the last row has 'n' such regular hexagons and all the hexagons are inscribed by circles.



Q.20. Find the ratio of area of the shaded region to the area of unshaded region.

(a) $\frac{3a^2}{2}[n(n+1)]\left(\sqrt{3}-\frac{\pi}{2}\right)$

(b) $\frac{a^2}{2}[n(n+1)]\left(\frac{3\sqrt{3}}{2}-\pi\right)$

(c) $\frac{3a^2}{2}[n(n+1)]\left(\sqrt{3}-\frac{\pi}{2}\right)$

(d) None of these

Q.21. Suppose the above figure is a 3-D figure and each hexagon is inscribed by a sphere of maximum volume. The height of the hexagonal figure is just enough to contain the sphere, then find the volume of the shaded region (there are 'n' hexagonal figures of side 'a' in the last row).

(a) $\frac{\sqrt{3}}{2}a^3\left(\frac{3\sqrt{3}}{2}-\pi\right)[n(n+1)]a^3$

(b) $\frac{\sqrt{3}}{4}a^3\left(\frac{3\sqrt{3}}{4}-\pi\right)[n(n+1)]$

(c) $\frac{1}{2}\frac{\sqrt{3}}{2}a^3\left(\frac{3\sqrt{3}}{4}-\pi\right)[n(n+1)]$

(d) $\frac{\sqrt{3}}{2}a^3(3\sqrt{3}-\pi)[n(n+1)]$

Q.22. If a sphere of the maximum volume is placed inside a hollow right circular cone with radius 'r' and slant height 'λ' such that the base of the cone touches the sphere, then the volume of the sphere is

(a) $\frac{4}{3}\pi\left(\frac{\ell+r}{\ell-r}\right)^3$ (b) $\frac{4}{3}\pi r^3\left(\frac{\ell-r}{\ell+r}\right)^{\frac{3}{2}}$

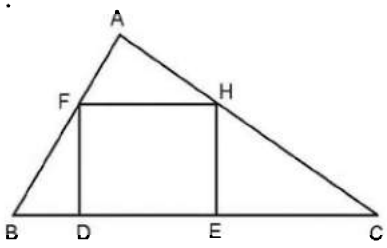
(c) $\frac{4}{3}\pi\left(\frac{\ell-r}{\ell+r}\right)^3$ (d) $\frac{4}{3}\pi r^3\left(\frac{\ell+r}{\ell-r}\right)^{\frac{3}{2}}$

Q.23. A hollow right circular cylinder of radius r and height 4r is standing vertically on a plane. If a solid right circular cone of radius 2r and height 6r is placed with its vertex down in the cylinder, then the volume of the portion of the cone outside the cylinder is

(a) $\frac{8}{3}\pi r^3$ (b) $2\pi r^3$

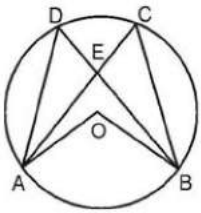
(c) $\frac{9}{8}\pi r^3$ (d) $7\pi r^3$

Q.24. Consider the following figure: AB = 10 cm, AC = 17 cm, BC = 21 cm and EHFD is a square. Find the length of the side of square (in cm).



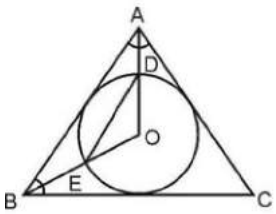
(a) 10.5 (b) 12
(c) 13.5 (d) None of these

Q.25. In the given figure, O is the centre of the circle and ∠AOB = 120°. What is the mesure of ∠AEB?



(a) 100° (b) 90°
(c) 110° (d) Cannot be determined

Q.26. For the equilateral triangle ABC, D & E are mid points of AO and OB, and DE = 3 units. The circumference of the circle is equal to



(a) 4π units (b) π√2 units
(c) 2π√3units (d) None of these

Q.27. The length of the sides CB and CA of a triangle ABC are given by a and b, and the angle C is 2π/3. The line CD bisects the angle C and meets AB at D. Then the length of CD is

(a) $\frac{a^2+b^2}{2(a+b)}$ (b) $\frac{ab}{2(a+b)}$

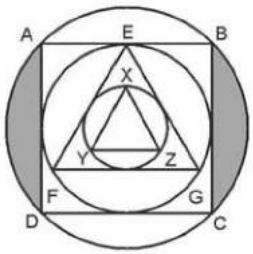
(c) $\frac{1}{a+b}$ (d) $\frac{ab}{a+b}$

Q.28. For a regular octagon inscribed in a circle of radius 1 cm, the product of the distance from a fixed vertex to the other seven vertices is

(a) 4 (b) 8
(c) 12 (d) 16

Direction for questions 29 to 31: Read the passage below and solve the questions based on it.

In the figure given below a square ABCD is inscribed in a circle of radius 5 cm. The square ABCD is inscribed by a circle which is inscribed by a triangle EFG. The triangle EFG is inscribed by a circle which is again inscribed by a triangle XYZ.



Q.29. Find the ratio of the radius of the outermost circle to the radius of the radius of the circle inscribed in triangle XYZ.

(a) 4:1 (b) $2\sqrt{2}:1$
(c) $4\sqrt{2}:1$ (d) 8:1

Q.30. Find the ratio of the area of square ABCD and the area of triangle XYZ.

(a) $16:3\sqrt{3}$ (b) $64:3\sqrt{3}$
(c) 16:3 (d) $64:\sqrt{3}$

Q.31. Find the area of the shaded region (in cm^2).

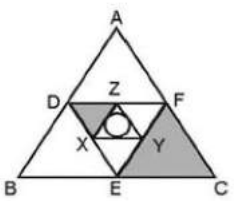
(a) $\frac{500\pi-800-75\sqrt{3}}{16}$

(b) $\frac{900\pi-1600-75\sqrt{3}}{}$

(c) $\frac{400\pi-800+75\sqrt{3}}{32}$

(d) None of these

Q.32. Find the area of the shaded region, if D, E and F are the mid-points of the sides AB, BC and CA respectively and X, Y and Z the are midpoints of DE, EF and FD respectively. It is given that the circumradius of triangle ABC is $8\sqrt{3}$ cm and triangle XYZ is inscribed by a circle. (in cm^2)



(a) $48\sqrt{3}-3\pi$ (b) $48\sqrt{3}-\pi$

(c) $54\sqrt{3}-3\pi$ (d) $192\sqrt{3}-4\pi$

Q.33. The diameter of a right conical tent is 6 m. If a pole of length 2 m can be fixed in the tent at half the distance of the radius from the centre of the base, then the area of the canvas required is (in m^2)

(a) 10π (b) 12π

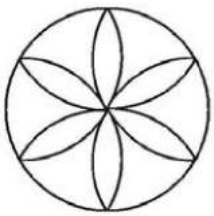
(c) 15π (d) 16π

Q.34. A pyramid has its base as an equilateral triangle, of each side being one m. Its slant edge is 3 m. The whole surface area of the pyramid is equal to.

(a) $\frac{\sqrt{3}+3\sqrt{13}}{4}m^2$ (b) $\frac{\sqrt{3}+2\sqrt{35}}{4}m^2$

(c) $\frac{\sqrt{3}+2\sqrt{13}}{4}$ (d) $\frac{\sqrt{3}+3\sqrt{35}}{4}m^2$

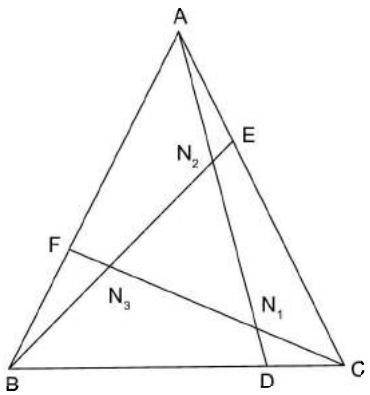
Q.35. In the figure given below a flower is inscribed in a circle of radius 1 cm, then find the area of the flower.



(a) $\left(\pi-\frac{3\sqrt{3}}{2}\right)cm^2$ (b) $(2\pi-3\sqrt{3})cm^2$

(c) $\left(\frac{5\pi}{3}-\frac{5\sqrt{3}}{2}\right)cm^2$ (d) $(4\pi-3\sqrt{3})cm^2$

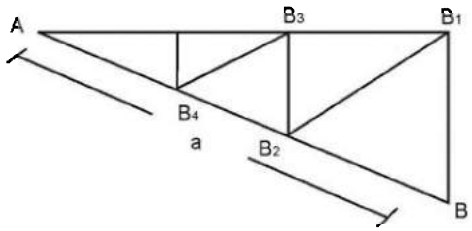
Q.36. In the figure given below CD, AE and BF are one third of their respective sides. It follows that $AN_2: N_2N_1:N_1D = 3:3:1$ and similarly for lines BE and CF. Then the area of triangle $N_1N_2N_3$ is



(a) $\frac{1}{7}\Delta ABC$ (b) $\frac{1}{6}\Delta ABC$

(c) $\frac{1}{9}\Delta ABC$ (d) $\frac{1}{10}\Delta ABC$

Q.37. In the figure given below, two rays are drawn through a point A at an angle of 30°. A point B is taken on one of them at a distance of 'a' from the point A. A perpendicular is drawn from the point B to the other ray, and another perpendicular is drawn from its foot to meet AB at another point from where the similar process is repeated indefinitely. The length of all such lines will be

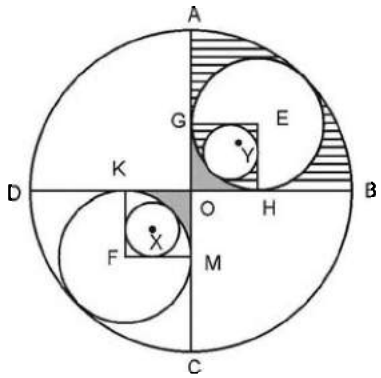


(a) $a(1-\sqrt{3})$ (b) $2a(2+\sqrt{3})$

(c) $\frac{3}{2a}$ (d) a

Direction for questions 38 to 40: Read the passage below and solve the questions based on it.

In the figure given below O is the centre of circles have radius (OB) + r. E and F are the centre of circles inscribed in the quarter circles AOB and COD respectively whereas X and Y are the centre of circles inscribed in the quarter circle KFM and GEH respectively.



Q.38. Find the area of the shaded region (not including the lined region).

(a) $\frac{3}{7}r^2(3-2\sqrt{2})$ (b) $\frac{3}{14}r^2(3-2\sqrt{2})$

(c) $3\frac{3}{14}r^2(3+2\sqrt{2})$ (d) None of these

Q.39. Find the ratio of the radii of the circle with centre Y, E and O.

(a) $1:(\sqrt{2}-1):(3+2\sqrt{2})$

(b) $1:(\sqrt{2}+1):(3-2\sqrt{2})$

(c) $1:(\sqrt{2}-1):(3-2\sqrt{2})$

(d) 1:2:4

Q.40. Find the area of the lined region.

(a) $\left(\frac{594\sqrt{2}-836}{14}\right)r^2$

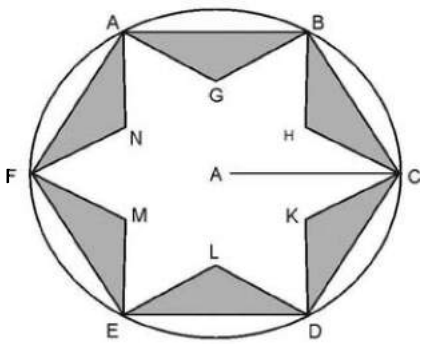
(b) $\left(\frac{600\sqrt{2}-836}{14}\right)r^2$

(c) $\left(\frac{594\sqrt{2}-845}{14}\right)r^2$

(d) $\left(\frac{600\sqrt{2}-845}{14}\right)r^2$

Direction for questions 41 to 43: Refer to the following information to answer the questions.

O is the centre of a circle having radiua (OC) = r. ABCDEF is a regular hexagon and AGBHCKDLEMFNA is a regular six pointed star.



Q.41. Find the ratio of the circumference of a circle to the perimeter of the hexagon ABCDEF to the perimeter of the star AGBHCKD LEMFNA.

(a) $\pi:\sqrt{3}$ (b) $\pi:3:2\sqrt{3}$

(c) $2\cdot\pi:6:4\sqrt{2}$ (d) $\pi:2\sqrt{3}$

Q.42. Find the ratio of the area of the shaded region to the area of the star.

(a) 1:1 (b) 1:2

(c) 1:3 (d) 2:3

Q.43. If the triangles ANG, BGH, CHK, DKL, ELM and FMN and the hexagon NGHKLM are inscribed by circles of the maximum area, then find the sum of the areas of all those circles.

(a) $\frac{\pi}{2}r^2$ (b) $\frac{\pi r^2}{3}$

(c) $\frac{5\pi r^2}{12}$ (d) $\frac{7\pi r^2}{12}$

Q.44. A triangle ABC with an obtuse angle B is inscribed in a circle. The altitude AD of the triangle is tangent to the circle. The side BC has length 12 cm and the segment BD has length 4 cm. Find the area of the triangle ABC (in cm^2).

(a) 48 (b) 60

(c) 72 (d) 36

Q.45. If the sum of the squares of the sides of a triangle is denoted by A and its perimeter by P, then which of the following is true about $\frac{A^2}{P}$?

(a) $\frac{A^2}{P}<2$ (b) $1\le\frac{A^2}{P}\le 4$

(c) $2\le\frac{A^2}{P}\le 3$ (d) $4\le\frac{A^2}{P}$

Q.46. A ring of diameter 10 cm is suspended from a point 12 cm vertically above the centre by six equal strings. The strings are attached to the circumference of the ring at equal intervals, thus keeping the ring in a horizontal plane. The value of cosine of the angle between the two adjacent strings lies in between

(a) 0 and 1/2 (b) 1/2 and 1

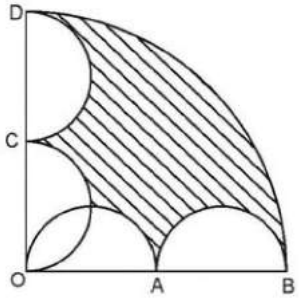
(c) −1/2 to 0 (d) −1 to −1/2

Q.47. A square of side 4 cm is drawn and a circle C_1 is inscribed in it. Now, four more circles C_2, C_3, C_4 and C_4, each of radius 2 cm, are drawn with centres as A, B, C and D respectively. In the resultant figure, what is the total area that is common to atleast two of the five circles (in sq. cms)?

(a) 6π−16 (b) 7π−16

(c) 8π−16 (d) 9π−16

Q.48. In the figure, find the area of the shaded portion, i.e., the portion of the quadrant DOB which is not included in the 4 semicircles. Given that OA = AB = OC = CD = 2 and ∠DOB = 90°.



(a) $5\pi - \frac{1}{2}$ (b) $5\pi - 1$

(c) $\frac{5\pi}{2} - \frac{1}{2}$ (d) $\frac{5\pi}{2} - 1$

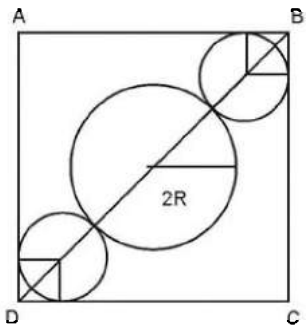
Q.49. Rajat cut out two identical triangular pieces of cardboard, each of area 300 sq. cm, and then placed them upon a table, one on top of the other such that the triangles completely coincide with each other. Now, if he rotated one of the two triangles by 180° about a vertical axis passing through its centroid, find the area that is common to both the triangles.

(a) 100 cm^2 (b) 150 cm^2

(c) 200 cm^2 (d) $133\frac{1}{3}$ cm^2

Q.50. See the figure given below:

Inside a square ABCD, three circles are drawn touching one another as shown in the figure. Radius of two smaller circles is R units and radius of the bigger circle is 2R units. Diagonal BD of the square passes through the centre of all the circles. What is the ratio of the radius of the smaller circles to the side of the square?



(a) 2:7 (b) 3:8

(c) 1:4 (d) None of these

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(a)	(a)	(c)	(b)	(d)	(c)	(d)	(c)	(b)	(c)	(d)	(a)	(b)	(a)	(b)	(b)	(a)	(a)
Q. No.	21	22	23	24	25															
Answer	(b)	(d)	(d)	(c)	(b)															

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(b)	(b)	(d)	(d)	(d)	(d)	(d)	(c)	(b)	(b)	(a)	(d)	(d)	(c)	(a)	(d)	(b)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(a)	(b)	(a)	(b)	(c)	(c)	(b)	(c)	(b)	(d)	(c)	(b)	(b)	(b)	(d)	(d)	(a)	(c)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(c)	(b)	(a)	(d)	(a)	(b)	(a)	(b)	(b)	(d)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(c)	(a)	(c)	(a)	(b)	(b)	(a)	(c)	(c)	(d)	(d)	(a)	(b)	(c)	(a)	(b)	(c)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(a)	(c)	(c)	(a)	(b)	(c)	(b)	(c)	(d)	(d)	(a)	(c)	(b)	(b)	(a)	(c)	(b)	(b)	(a)	(c)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(b)	(a)	(d)	(c)	(d)	(d)	(c)	(a)	(c)	(d)										

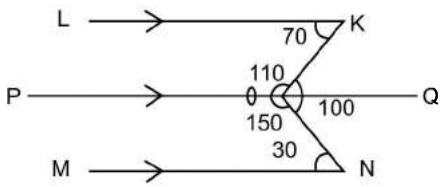
■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(b)	(b)	(c)	(a)	(c)	(a)	(b)	(c)	(a)	(c)	(a)	(c)	(a)	(a)	(b)	(b)	(d)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(d)	(b)	(d)	(d)	(d)	(c)	(d)	(b)	(c)	(b)	(d)	(b)	(c)	(d)	(b)	(a)	(c)	(a)	(c)	(d)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(b)	(b)	(c)	(a)	(c)	(b)	(c)	(d)	(c)	(d)										

HINTS AND SOLUTIONS

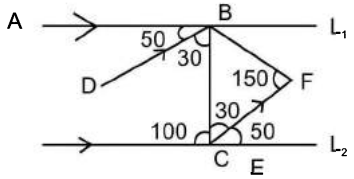
Warm up

1.



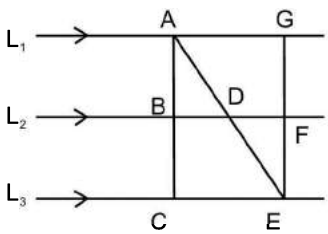
Hence, $\angle MNO = 30^\circ$

2.



$\angle FCE = 50^\circ$

3.



$$\frac{AB}{BC} = \frac{AD}{DE} \quad \frac{4}{6} = \frac{12}{DE}$$

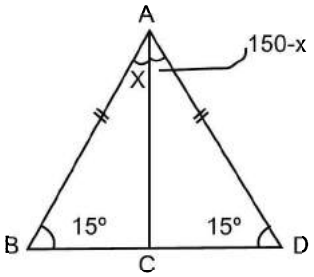
$$DE = \frac{12 \times 6}{4} = 18$$

$$\frac{ED}{EF} = \frac{DA}{FG}, or, \frac{18}{EF} = \frac{12}{18}$$

$$EF = 24$$

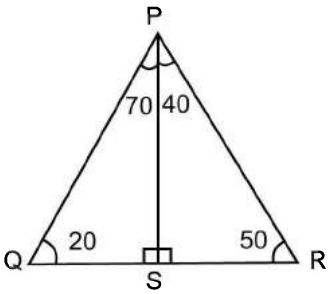
4. $x + 5x = 180^\circ$
$6x = 180^\circ$
$x = 30^\circ$

5.



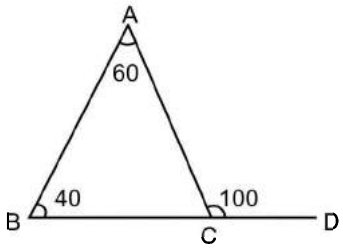
$150^\circ - x + 15^\circ = 95^\circ + x$
$2x = 70^\circ$
$x = 35^\circ$
Therefore, $\angle ACB = 95^\circ + 35^\circ = 130^\circ$

6.

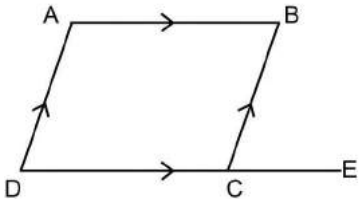


$\angle QPR = 110^\circ$

7.

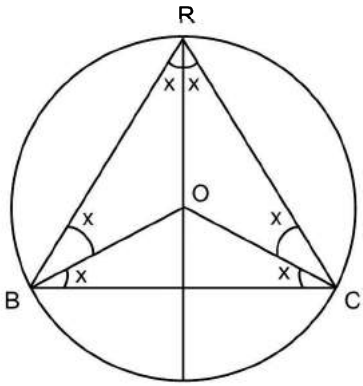


8.



Since A = 135°, So, B = D = 65°
We will find that (c) is the answer.

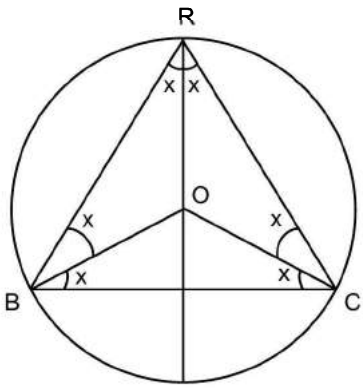
9.



Isosceles trapezium is always cyclic
The sum of opposite angles of a cyclic quadrilateral is 180°.
$\therefore \angle BCD + \angle BAD = 180^\circ$
$40^\circ + \angle BAD = 180^\circ$
$\angle BAD = 140^\circ$

10. $\frac{(2n-4)\times 90}{n} = 135$
$(2n - 4)\ 2 = 3\ n$
$4n - 8 = 3n$
$n = 8$
No. of diagonals $= \frac{1}{2}n(n-3) = 20$

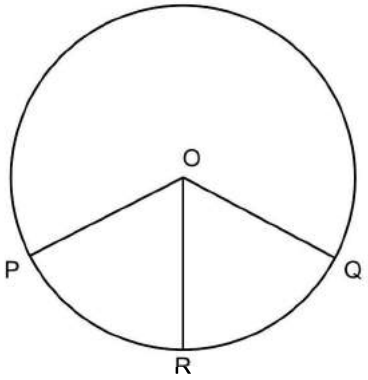
11.



$6x = 180^\circ$
$x = 30^\circ$
$\angle BAC = 2x = 2 \times 30 = 60^\circ$

12. AO × OB = OC × OD
or, 4 × 6 = 3 × OD
OD = 8 CM

13.



Let $\angle POR = x$

Therefore $\angle ROQ = \frac{2}{3}x$

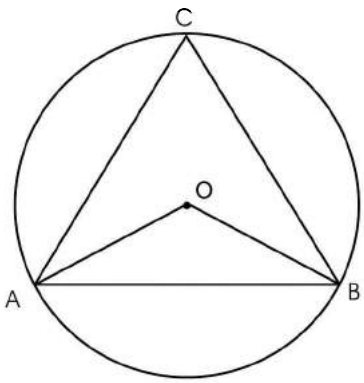
$x + \frac{2}{3}x = 100$

$\frac{5x}{3} = 100$

$x = 60^\circ$

$\therefore$ Therefore $\angle POR = 60^\circ$

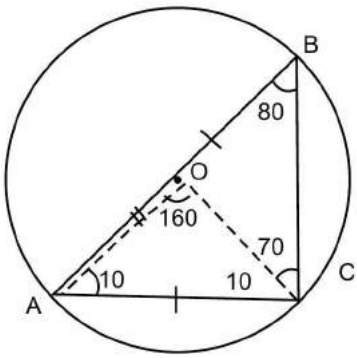
14.



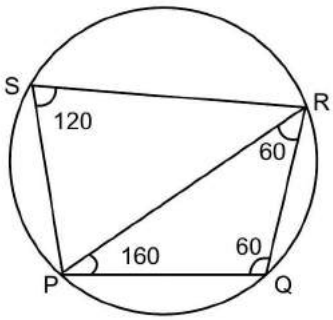
$\frac{1}{2} \angle AOB = \angle ACB$

$\angle ACB = 60^\circ$

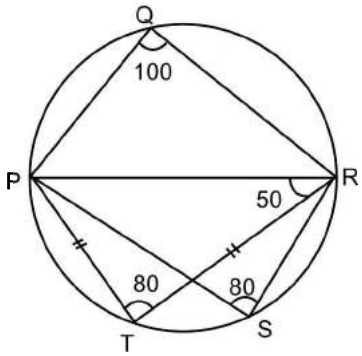
15.



16.

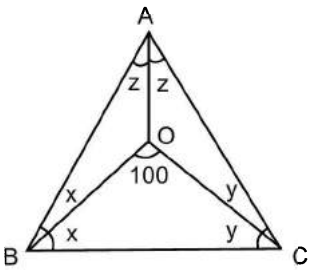


17.



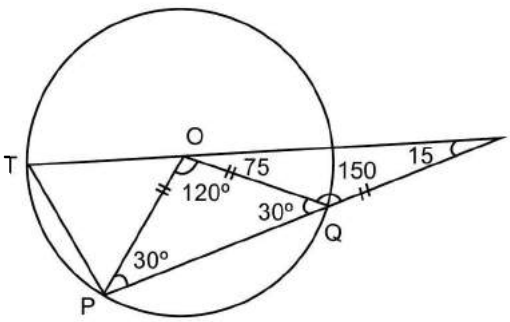
$\angle PRT = 50$

18.



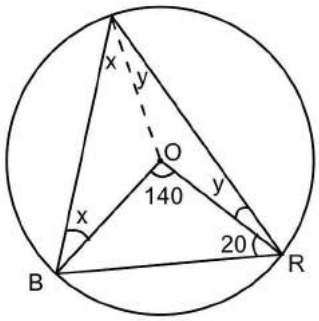
In $\angle BOC$, $x + y = 80$
$2x + 2y = 160^\circ$
Also, $2x + 2y + 2z = 180^\circ$
$2z = 20^\circ$
$\angle BAC = 2z = 20^\circ$

19.



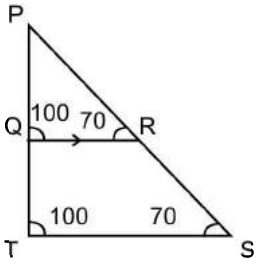
$\angle OQP = 30^\circ$
Therefore $\angle QOP = 120^\circ$

20.



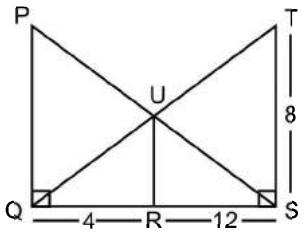
$\angle OQP + \angle ORP = 70^\circ$
$x + y = 70^\circ$
$\angle QPR = 70^\circ$
$\angle ORQ = 20^\circ$

21.



$PQ \times PS = PT \times PR$
$QR \parallel TS$
$\angle PQR = \angle PST + 30^\circ$
$\angle PST = 70^\circ$
$\angle PRQ = 70^\circ$

22.



In $\angle QST$ and $\angle QRU$ ($\angle QST$ and $\angle QRU$ are similar triangles)

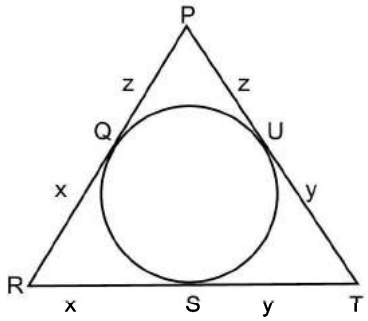
$$\frac{QR}{QS} = \frac{RU}{TS} \Rightarrow \frac{4}{16} = \frac{RU}{8} = RU = 2$$

Also, $\frac{QR}{PQ} = \frac{RU}{TS} \Rightarrow \frac{4}{16} = \frac{RU}{8} = RU = 2$

$$\frac{1}{PQ} = \frac{1}{2} - \frac{1}{8} = \frac{4-1}{8} = \frac{3}{8}$$

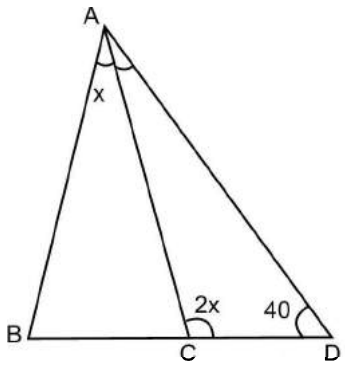
PQ = 8/3 cm

23.



$PR = RT + 3$
$x + z = x + y + 3$
$z - y = 3$ ……(1)
Similarly, $z + x = z + y + 1$
Or, $x - y = 1$ ……(2)
From 1 and 2
$z - 3 = x - 1$
$z - x = 2$ ……(3)
Also, $2x + 2y + 2z = 26$
$x + y + z = 13$ ……(4)
$3x + 1 = 13$
$3x = 12$
$x = 4$
z = 6
and $y = 3$
$QR + PT = x + y + z$
$= 4 + 6 + 3 = 13$

24.



$85^\circ - x + 2x + 40^\circ = 180^\circ$
$x = 55^\circ$
$\angle ACB = 70^\circ$

25. Only 8 such regular polygons are possible.

Foundation

1. The triangle is not possible. So, (d) is the correct options

2. Clearly, triangle is obtuse. So (d) is the correct options

3. Clearly, triangle is a right angled triangle

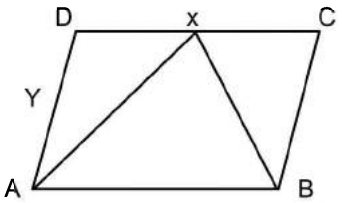
its area = $\frac{1}{2} \times 6 \times 8 = 24 \text{ cm}^2$

Now, $8 \times b = 24$

$b = 3$

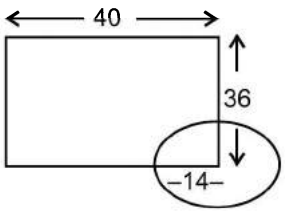
Required perimeter = 2 (8 + 3) = 22 cm

4.



We know that, between the same parallels and on the same base, the area of triangle is half the area of the parallelogram. Hence, Ans. (d)

5.



Required ratio = 3:1

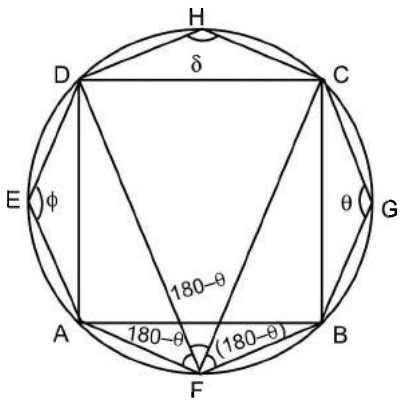
6. Triangle is not possible

7. Putting the value of n = 2, 3 and so on, we get (a) as answer

8. This is possible only when both length and breadth of rectangle are equal to the side of the square.

9. Clearly (d)

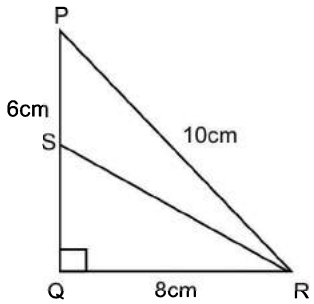
10.



Required sum

$= \theta + 180^\circ - \theta + 180^\circ - \theta + 180^\circ - \theta + \theta + 8 = 540^\circ$

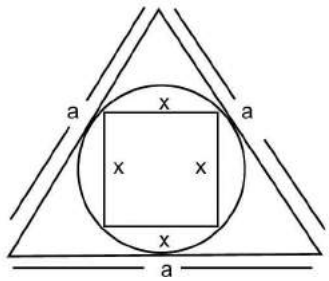
11.



Clearly, it is a right angle triangle at Q

$RS = \sqrt{8^2 + 3^2} = \sqrt{64+9} = \sqrt{73} \approx 8.5 \text{ cm}$

12.



Radius of circle $R = \frac{\frac{\sqrt{3}}{4}}{\frac{3a}{2}} = \frac{\sqrt{3}}{6}a$

Diagonal of square = diameter of circle

$= 2.\ \frac{\sqrt{3}}{6}a$

$\sqrt{2}x = \frac{\sqrt{3}}{3}a$

$x = \frac{\sqrt{3}}{3\sqrt{2}}a = \frac{\sqrt{6}}{6}a$

Area = $x^2 = \left[\frac{(\sqrt{6}a)}{6}\right]^2 = \frac{6}{36}a^2 = \frac{1}{6}a^2$

13. The two adjacent angles can be 30° and 60°, therefore others angle can be 150° and 120°

14. Going through options we find that only (d) satisfies the condition

15. 1 kg = 1000 cm³

$2700 = k.2^3$

$k = \frac{2700}{8}$

$6300 = k\ .\ r^3$

$r^3 = \frac{6300}{k} = \frac{6300}{2700} = \frac{8}{3} \quad r^3 = \frac{56}{3}, \quad r = 2.6 \text{ cm}$

16. $i(2,3) = 2(2+3) = 10$

$f(10) = 100$

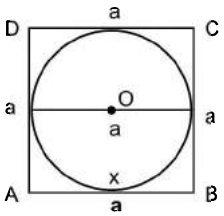
$h(100,2) = 200$

$g(200) = 80$

17. $$\frac{f(3)-g(2)}{h(4,2)-i(6,2)} \times \frac{g(6)}{f(1)} = \frac{9-8}{8-16} \times \frac{24}{1}$$

$$= \frac{1}{-8} \times 24 = -3$$

18.



$4a + \pi a = 10$

$$a = \frac{10}{4+\pi} = 1.4$$

Required Area $= (1.4)^2 - \pi (.7)^2$

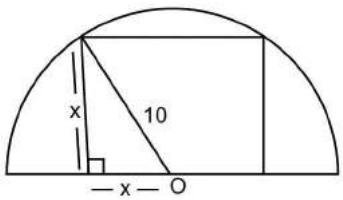
$$= 2.56 - \frac{22}{7} \times \frac{7}{10} \times \frac{7}{10} = 196 - 1.54$$

$= 0.42\ m^2 = 4200\ cm^2$

19. $$\frac{.7}{4 \times 1.42} = 1.8$$

20. Going through option, we can easily eliminate (a), (b) and (d)
Hence, (c) is the answer

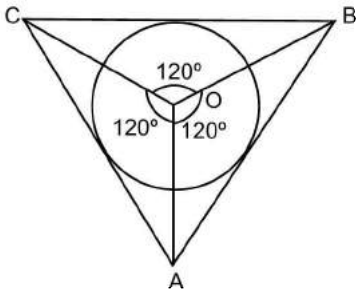
21.



$$x^2 + \frac{x^2}{4} = 100^\circ,\ x^2 = 80^\circ$$

22. No such point is possible

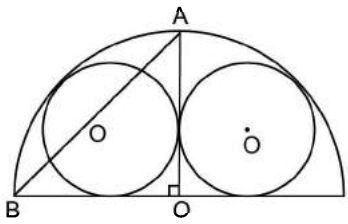
23.



Radius of disc $= \sqrt{3}$, OB = 20

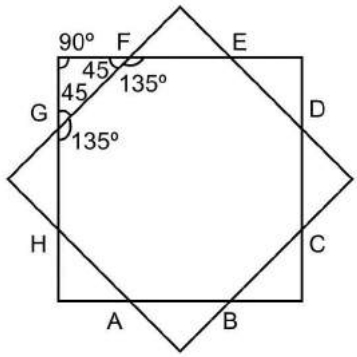
Use Sine law to find the length of AB = BC = AC

24.



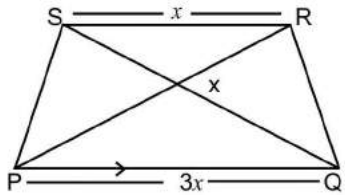
OA = OB = 2 cm, AB = 2 $\sqrt{2}$ cm

25.



Required angle = 90^0

26.



$$\frac{ar(\Delta P \times Q)}{ar(\Delta R \times S)} = \frac{PQ^2}{RS^2} = \frac{(3x)^2}{x^2} = 9:1$$

27. Let the angle be x, xr, xr^2 and xr^3

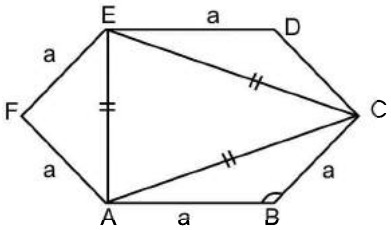
$x + xr + xr^2 + xr^3 = 360^\circ$

and $x(1 + r + r^2 + r^3) = 360^\circ$

therefore angles are 24°, 48°. 96°, 192° which satisfies the given condition.

Hence, answer to 27 is (c) and answer to 28 is (b)

30.



Clearly, Δ ACE is an equilateral triangle

$$\frac{AC}{Sin120^\circ}=\frac{a}{Sin30^\circ}, or, \frac{AC}{Sin60^\circ}=\frac{a}{Sin30^\circ}$$

$$\rightarrow AC=\frac{aSin60^\circ}{Sin30^\circ}=\frac{a\cdot\sqrt{3/2}}{1/2}=\sqrt{3a}$$

$$\text{Required ratio} = \frac{\frac{\sqrt{3}}{4}(\sqrt{3}a)^2}{6.\frac{\sqrt{3}}{4}a^2}=\frac{3a^2}{6a^2}=\frac{1}{2}$$

31. $\frac{1}{2}:\frac{1}{3}:\frac{1}{4}=6:4:3$

$6x + 4x + 3x = 52$, or, $13x = 52$ and $x = 4$

Required length = 9 cm

32. Required Area $=\pi\left(\frac{104}{2}\right)^2-\pi.50^2$

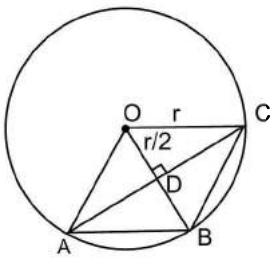
$=\pi.52^2-\pi.50^2=\pi(52-50)(52+50)$

$=\pi.2.102=204\pi$

33. Let h_1 = attitude of parallelogram and h_2 = altitude of triangle then $\frac{1}{2}h_2=h_1$

$h_2=2h_1=200\,m$

34.

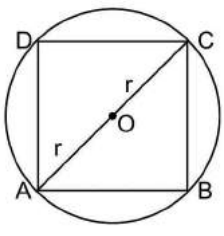


$$KD=\sqrt{r^2-\frac{r^2}{4}}=\frac{\sqrt{3}}{2}r$$

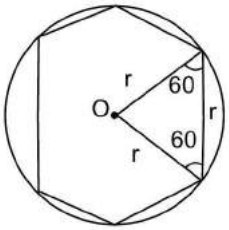
$\frac{1}{2}\cdot r\cdot\sqrt{3}r=32\times\sqrt{3}$, $r^2 = 32 \times 2$

$r^2 = 64$ and $r = 8$

35.



$A=\left(\sqrt{2}r\right)^2=2r^2$



$B=6\cdot\frac{\sqrt{3}}{4}r^2$

$\frac{B}{A}=\frac{\frac{3}{2}\sqrt{3}r^2}{2r^2}=\frac{3\sqrt{3}}{4}$

36. Let dimensions be x, $2x$ and $4x$

A/Q

$20.5\times2\left[2x^2+8x^2+4x^2\right]-20\times2\left[2x^2+8x^2+4x^2\right]$

= 120

or, $2 \times 14x^2 \times 0.5 = 126$

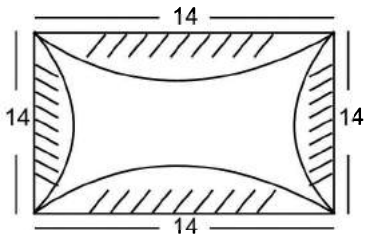
$x^2=\frac{63\times2}{14}\Rightarrow x=3$

therefore dimensions are 3, 6 and 12 meters.

37. Area $=\frac{1}{2}\cdot50\cdot130$ Sin 72°

$=\frac{1}{2}\cdot50\cdot130\cdot\times0.9510=3090.75\,m^2$

38.



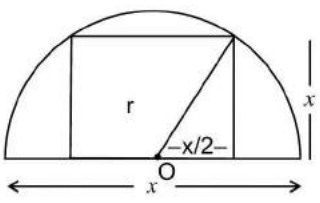
Area grazed by 4 horses $=2\cdot\pi\cdot7^2=308m^2$

Area ungrazed = (196 – 112) m^2 = 84m^2

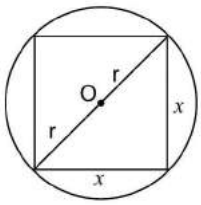
308 m^2 is grazed in – 11 days

84 m^2 is grazed in– 3 days

39.



Area of square $=\left(\frac{2}{\sqrt{5}}r\right)^2=\frac{4}{5}r^2$



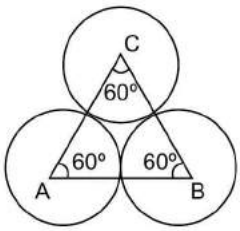
Area = $2r^2$

Required ratio = $\frac{\frac{4}{5}r^2}{2r^2} = 2:5$

40. $A = 8x^2 - 2x - 15$
$= 8x^2 - 12x + 10x - 15$
$= 4x(2x - 3) + 5(2x - 3)$
$A = (4x + 5)(2x - 3)$
Clearly, length = $(4x + 5)$
and breadth = $(2x - 3)$

41. Required ratio = $\frac{a^2}{(\sqrt{2}a)^2} = 1:2$

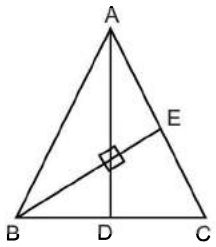
42.



Required area

$= \frac{\sqrt{3}}{4} \cdot 7^2 - 3 \cdot \frac{60}{360} \frac{22}{7} \times \left(\frac{7}{2}\right)^2 = 1.967 \text{ cm}^2$

43.



Area $\Delta AOB = \frac{1}{2} \cdot 8.6 = 24 \text{cm}^2$

Area of $\Delta BOD = \frac{1}{2} \cdot 8.3 = 12 \text{cm}^2$

Area of $\Delta ABD = 36 \text{ cm}^2$

Area of $\Delta ABC = 72 \text{ cm}^2$

44. $\frac{1}{2}(x+6)(y-4)+24 = \frac{1}{2}xy$

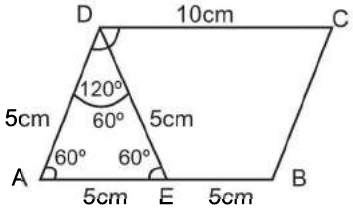
$\frac{1}{2}xy - \frac{1}{2}x \cdot 4 + \frac{1}{2}6 \cdot y - \frac{1}{2}6.4 + 24 = \frac{1}{2}xy$

or, $-2x + 3y - 12 + 24 = 0$

$2x - 3y = 12$

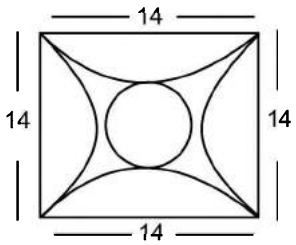
Now, putting the value of x or y from the options we get (d) as the answer.

45.



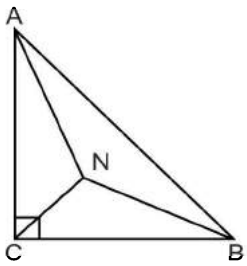
Perimeter = 30 cm

46.



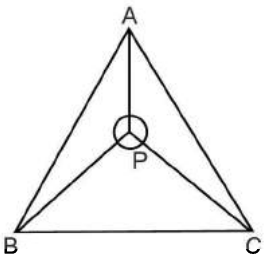
47. Going through option we get (a) as option

48.



49. Let AB = 4 and BC = 5 and AB is perpendicular to BC
then Area = $\frac{1}{2}$ AB . AC $= \frac{1}{2} \cdot 4.5 = 10$

50.



First three options do not have the properties as given in the question

Moderate

1. $\angle A : \angle B : \angle C : = 4:1:1 = x$ (let)
$4x + x + x = 180°$

$x = 30°$

Now, $\frac{\sin A}{\sin B} = \frac{A}{B}$ to solve get Ans. $\frac{A}{B} = \frac{\sqrt{3}}{1}$

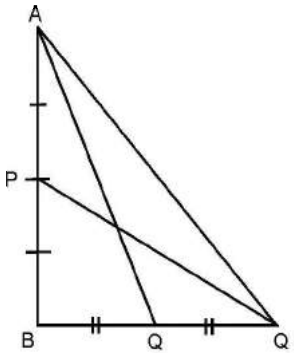
2. Use Pythagoras theorem, In right-angled ΔABQ

$$AQ^2 = AB^2 + \left(\frac{BC}{2}\right)^2 \quad (1)$$

and in right-angled ΔCBP

$$CP^2 = \left(\frac{AB^2}{2}\right) + (BC)^2 \quad (2)$$

add (1) and (2) to get the answer.

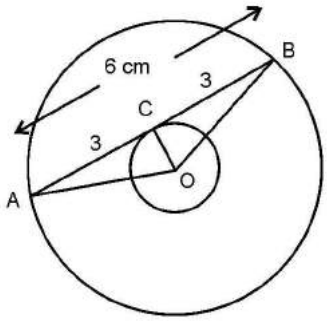


4. AC = 3 m

OC ⊥r AB

So, Δ AOI in right-angled triangle.

CO = 4 and AO = 5 cm



6. In ΔCOD

OC = OD = OB = OA = radius = r

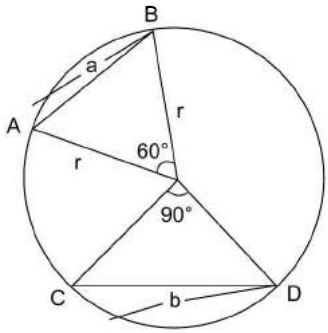
$b^2 = 2r^2$

$r = \frac{b}{\sqrt{2}}$

and ΔAOB= equilateral triangle

Hence, a = r

$r = \frac{b}{\sqrt{2}}$, hence $b = \sqrt{2a}$

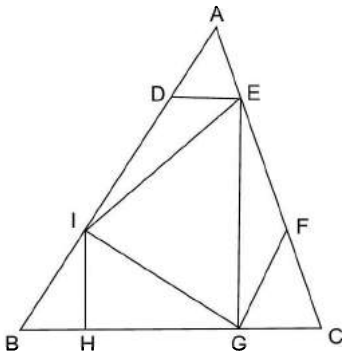


9. AQ = (16 + 5 = 21) cm and QB = 32 – 21 = 11 cm

12. If AB = 3a

then, DI = a = DE = EF = FG = GH = HI and IE = EG = IG = $\sqrt{3a}$

Now find the area.

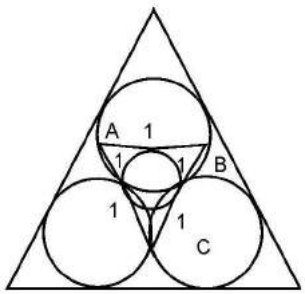


14. ΔABC is equilateral

$\tan 30 = r/x = x = \sqrt{3r}$ (where r = 1)

$AB = 2 + 2x = 2 + 2\sqrt{3} - 2\left(1+\sqrt{3}\right)$

$= \sqrt{3}\left(1+\sqrt{3}\right)^2$



15. Height of ΔABC $= \sqrt{\frac{3}{2}a}$

$\frac{\sqrt{3}}{2} \times 2h = h = \sqrt{3}$

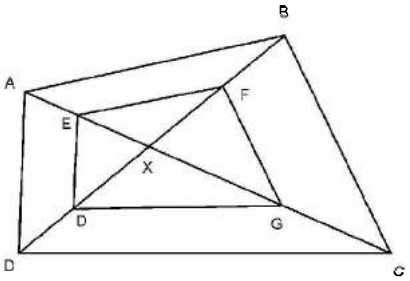
$\frac{2h}{3}$ is the radius of the circle inscribed in ABC

So radius of circumscribed circle

$$= \frac{2\sqrt{3}}{3} + 1 = \left(\frac{2}{\sqrt{3}} + 1\right)$$

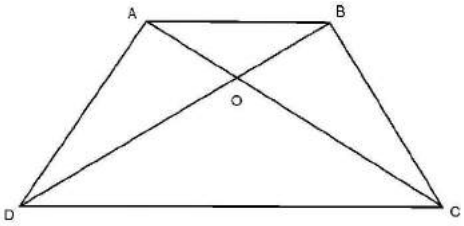
Hence, the required portion = $\pi\left(\frac{2}{\sqrt{3}} + 1\right)^2 - 3\pi$

17. This problem is based on mid-point theorem.

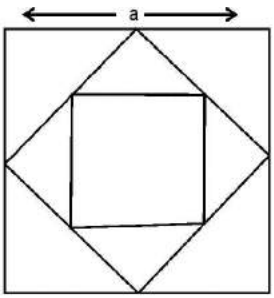


18. DAOB ~ DDOC

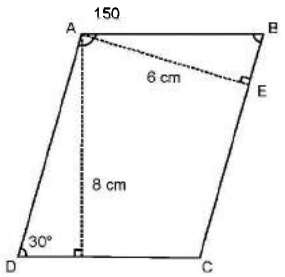
$$\therefore \frac{AO}{CO} = \frac{AB}{DC} = \frac{2}{3}$$



22. Find the sides of all the squares.
Then use geometric progression concept and get the answer.



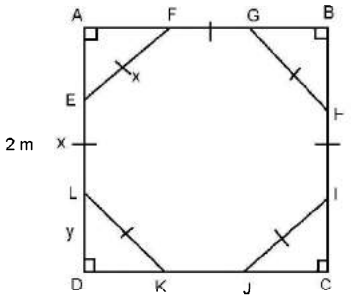
23. In ΔAEB, find AB and in ΔABF find AD.
then P = AB + BC + CD + AD.
Use this to get the answer.



24. $x + 2y = 2$ (1)

In ΔAEF, $x^2 = 2y^2$ (2)

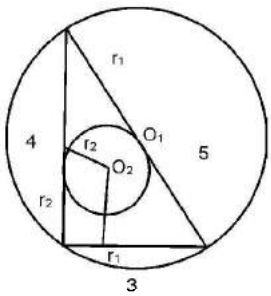
Use (1) and (2) to get the answer.



34. $r_1 = 2.5$ cm

r = 1 cm

Get the answer.



36. Use similarity theorem

40. Area of trapezium = $\frac{1}{2} \times$ height $(l_1 + l_2)$

SN⊥PQ and RM⊥PQ gives RS = MN = 3, So, PN + QM = 3

Assume PN = x.

Then, in ΔPNS and ΔQMR (SN = RM)

$(SN)^2 = (RM)^2 = 4^2 - x^2 = 5^2 - (3 - x)^2$

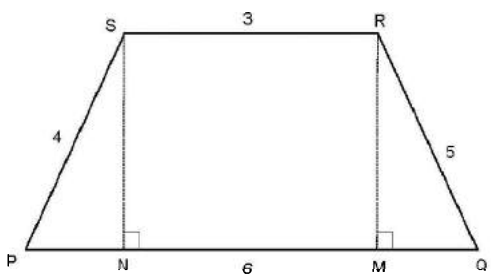
Finally, $x = 0$ = zero

That means,

$(SN)^2 = 4^2 - x^2$

SN = 4

So, Area = $\frac{1}{2} \times 4(3 + 6) = 18$ cm^2



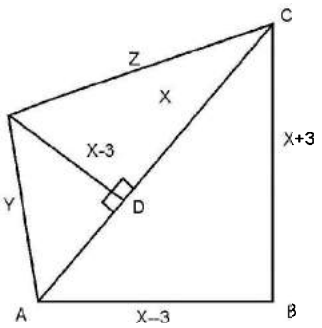
Alternatively,
In trapezium PQRS, PN + MQ = 3. If we join ΔPSN and ΔMRQ through their altitude, it forms a triplet (3, 4, 5) of right-angled Δ making right angle at P, i.e., ∠SPQ = 90°. Then altitude = 4 cm.

Area of trapezium = $\frac{1}{2}\times(3+6)=18\ \text{cm}^2$

41. Let side of square = a, then perimeter = $4a$
Area of square = a^2
According to the question, numerically $4a = a^2$
So, $a = 4$
Diagonal of square = $\sqrt{2}a = 4\sqrt{2}$

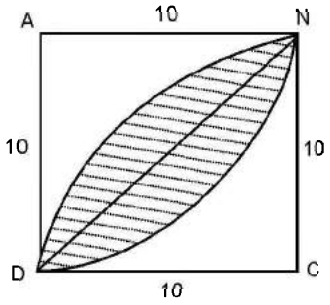
42. In square ADCE, area = x^2
So, side AC = x
In square AHIB, area = y^2
So, side AB = y
As $x^2 > y^2$ so, $x > y$
and in ΔABC (∠ABC = 90°)
$(AC)^2 - (AB)^2 = (BC)^2$
Hence area BCGF = $(AC)^2 - (AB)^2 = x^2 - y^2 = (x-y)(x+y)$

43. $AD = (y^2-(x-3)^2)^{1/2} = (100-(x-3)^2)^{1/2}$
$DC = x$
In ΔABC
$(x-3)^2 + (x+3)^2 = \left(x+\sqrt{100-(x-3)^2}\right)^2$



Now, putting the values of the given options, we can calculate the answer. It is none of these.
Alternatively, this question can be done very easily using the options.

44. In square ABCD
Area of shaded region = Area of sector BCD – Area of ΔBCD + Area of sector ABD – Area of ΔABD
Since, all sides of a square are same so,
Area of sectors and triangles will be the same having same radius (side of square) and angle 90°
So, area of shaded region = 2 (Area of Sector – Area of ΔABD)



$= 2\left(\frac{\pi}{4}(AB)^2 - \frac{1}{2}\times AD\times BA\right)$

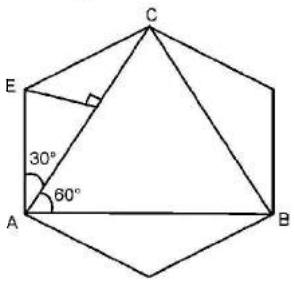
$= (AB)^2\left(\frac{\pi}{2}\right) = 100\left(\frac{\pi}{2}-1\right)$

$= 50\pi - 100$

45. As, hexagon is regular and AD = CD
So, ΔABC is equilateral triangle with

$AB = BC = AC = \frac{P}{3}$

$AD = \frac{P}{6}$



So,

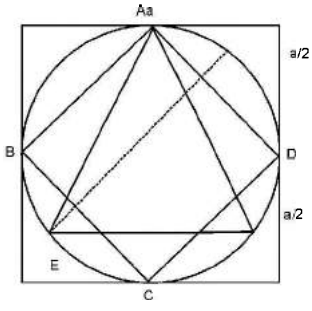
Cos $30^\circ \frac{\sqrt{3}}{2} = \frac{AD}{AE}$

$AE = AD\times\frac{2}{\sqrt{3}} = \frac{2p}{3\sqrt{3}}$

So, perimeter of hexagon = $6\times\frac{2p}{6\sqrt{3}} = \frac{2p}{\sqrt{3}}$

46. $AD = \left(\left(\frac{a}{2}\right)^2 + \left(\frac{a^2}{2}\right)\right)^{1/2}$

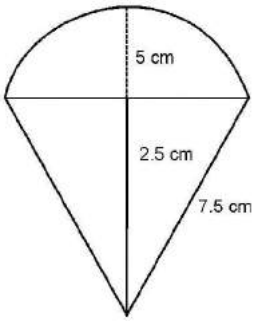
Then, radius of circle = $\frac{a}{2\sqrt{2}}$



Advanced

1. Radius of the circle cm $= \sqrt{(7.5)^2 = (2.5)^2} = 5\sqrt{2}$

   Hence, circumference $= 10\sqrt{2}\pi$ cm.

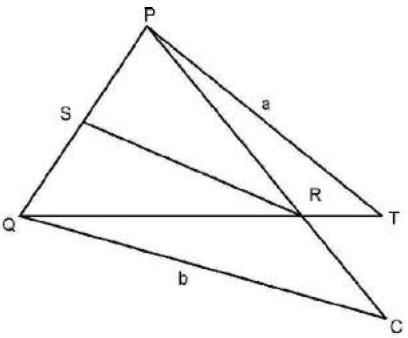


2. $\Delta PQU \sim PSR$

$$\Rightarrow \frac{PS}{PW} = \frac{SR}{QU} \quad (1)$$

$\Delta PQU \sim SQR$

$$\Rightarrow = \frac{SR}{PQ} = \frac{SR}{PT} \quad (2)$$



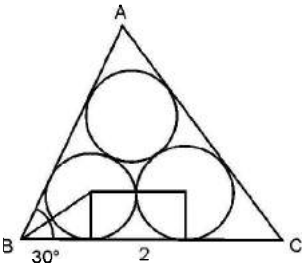
From (1) and (2)

$PQ \times SR = PS \times QU = SQ \times PT \Rightarrow = \frac{SQ}{PS} = \frac{b}{a}$

Now use componendo and equation (1) to obtain SR $= \frac{ab}{a+b}$

3. Use options.

4. It can be seen that the side of the triangle is



7. $\frac{\pi}{8} = S, (2+4+8+16+32+64+128)$

   Get S, from the above relation again $\frac{\pi\theta}{306^\circ} = S_1$

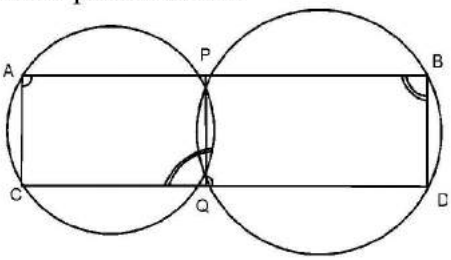
   Now θ can easily be calculated from here.

9. From alternate segment theorem

   $\angle PAC = \angle PQD$ and $\angle PBD = \angle PQC$

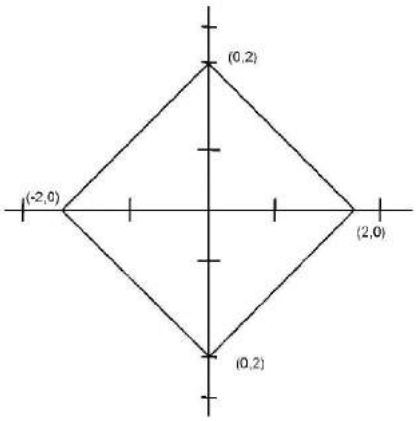
   But $\angle PQD$ and $\angle PQC$ are supplementary.

   Now since $\angle PAC$ and $\angle PBD$ are supplementary, hence AC is parallel to BD.



10. $x + y = 2$ when $x \geq 0$ and $y \geq 0$

    $x + y = -2$ when $x \leq 0$ and $y \leq 0$

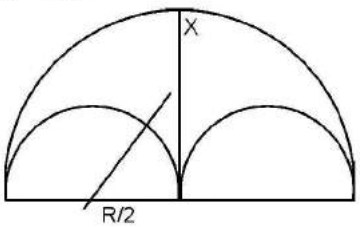


16. Ratio of radii $= \frac{a}{2\sqrt{2}} : \frac{a}{4\sqrt{2}} : \frac{a}{2\sqrt{2}}$

    Ratio of areas = 1:4:16

18. $\left(\frac{R}{2} + x\right)^2 = \left(\frac{R}{2}\right)^2 + (R - x)^2$

    x = R/3

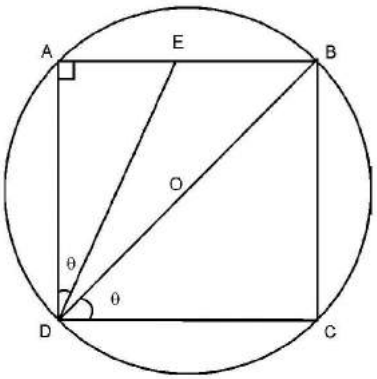


19. Let $\angle BDC = \theta$

    DB = 2r

BC = r

DE = r $\sqrt{3}$



Now area of ABCD $\frac{1}{2}\times\sqrt{3}\times r$

$=\frac{1}{2}\times\sqrt{3r}\times 2r\times\sin\theta$

$\Rightarrow \sin\theta=\frac{1}{2}$

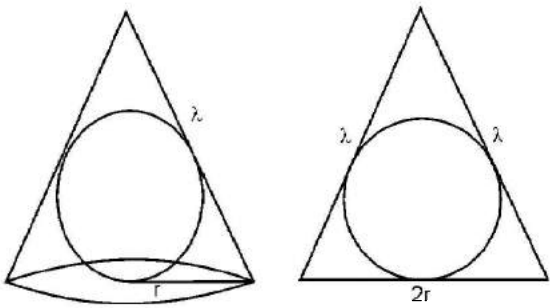
$\theta = 30°$

Now find AE.

21. Height of the octagonal prism will be the diameter of the sphere inscribed in it. Hence, the answer will be.

$$\frac{\sqrt{3}}{4}a^3\left(3\sqrt{3}-r\right)[n(n+1)]$$

22. Consider the biggest cross-section of the cone as a isosceles triangle therefore the circle inscribed in the triangle will be the biggest cross-section of the sphere.

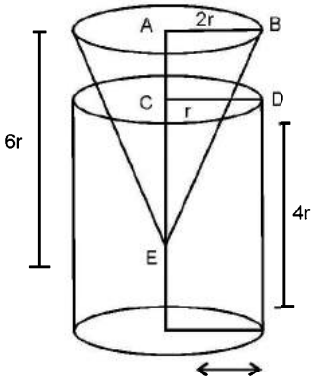


We know that in radius × semi perimeter = Area of the triangle.

A little calculation will lead to the answer, i.e.,

$$\frac{4}{3}\pi r^3\left(\frac{\lambda-r}{\lambda+r}\right)^{3/2}$$

23. ΔECD is similar to ΔEAB. Hence, EC = 3r



So the answer will be $7\pi r^3$.

24. Side of the square should always be less than the attitude of the triangle.

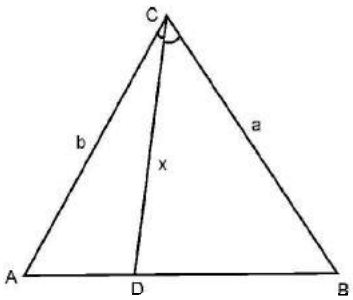
27. Let CD = x

Area ΔABC = ΔACD + ΔDCB

$\Rightarrow \frac{1}{2}ab\sin 120° bx\sin 60° + \frac{1}{2}ax\sin 60°$

Now it can be observed that x

$$x=\frac{ab}{a+b}$$



28. We have to calculate

AH × AG × AF × AE × AD × AC × AB

AE = 2

∠AOB = 45°

Use Cosine formula and get AB (= AH). ∠AHG = 135°

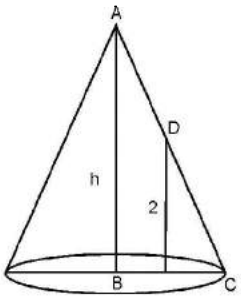
Again by Cosine formula get AG (= AC) and AF (= AD)

33. DE = 2

EC = 3/2

find DC and then find l

Thus, the answer will be 15 pm^2

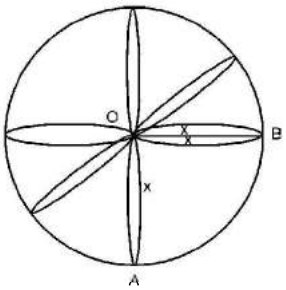


35. OA = OB = AB = 1 cm

$\angle OAB = 60^\circ$

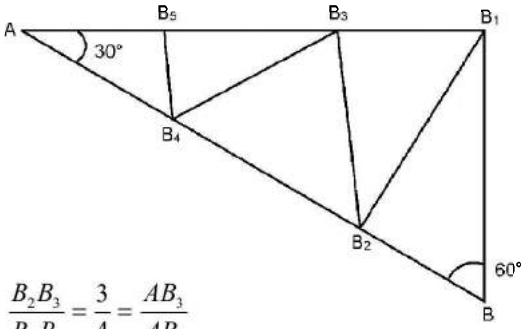
area x = area of sector OAB – DOAB

$12x$ = area of the flower



37. AB = a BB, = a/2

$B_2B = a/4$ $\Delta AB_2B_3 \sim \Delta ABB_,$

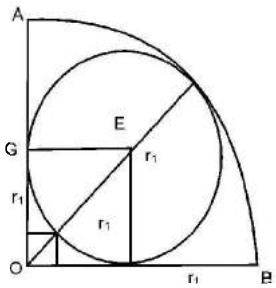


$$\frac{B_2B_3}{B_dB_1} = \frac{3}{4} = \frac{AB_3}{AB_1}$$

Now we have two infinite GP series having common ratio ¾. Calculate the sum of these series.

38. $r - r_1 = \sqrt{2}\, r_1$

Now find r, and solve to get the answer.



41. Let OC = r

ABCDEF is a regular hexagon

ED = r

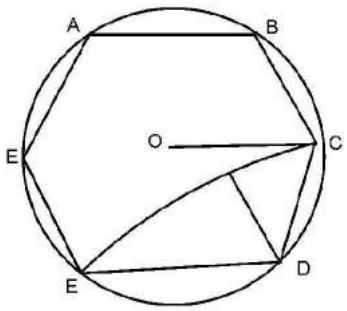
Let EC = x

In ΔEDC, using cosine formula we can get

$x = \sqrt{3}\, r$

Side of the star will be $r/\sqrt{3}$.

Now the desired ratio can be easily calculated.



46. Consider the problem as a pyramid to the base of a regular hexagon.

Now in Δ POR

OP = 12 cm

OR = 5 cm

PR = 13 cm

Now cosine formula will give the answer.

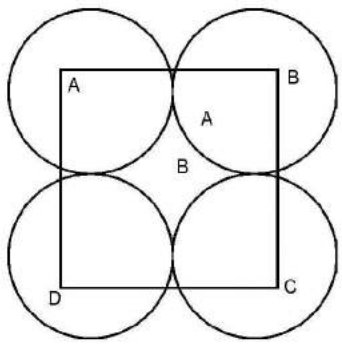
47. AB = 4 cm

$$a = \frac{16 - 4\pi}{4} = 4 - \pi \text{ cm}$$

area of common region = $b = \frac{4\pi}{4} - 4 + \pi$

$b = 2\pi - 4$

Total area = $4b = 8\pi - 16$



49. $\frac{\sqrt{3}}{4}a^2 = 300 \text{ cm}^2$

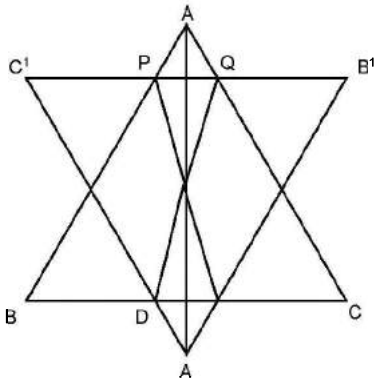
$a = \sqrt{\frac{1200}{\sqrt{3}}}$

So, AP will be equal to AB/3.

So the area common to both triangles

$= 300 - 3 \times \frac{\sqrt{3}}{4} \times \frac{1}{9} \times \frac{1200}{\sqrt{3}}$

$= 200 \text{ cm}^2$



18

CHAPTER

Mensuration

LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Relationship between 2-D and 3-D figures
- Definitions and properties
- Different structures
- Process of finding out volume, lateral surface area, curved surface area and total surface area
- Methods of solving questions

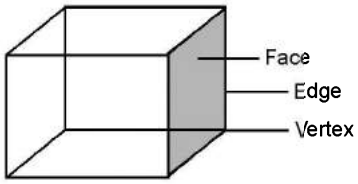
Introduction to the topic

There is a direct connection between geometry and mensuration. It can be seen through analogy that the relationship shared by 2-D and geometry is the same as the relationship shared by 3-D and mensuration in most cases. Mensuration has been an area from which questions have been asked in the CAT, albeit not as much as from geometry. A fair number of questions from this topic are being asked in the CAT. And therein lies the importance of this topic. Mensuration should be seen as a nice juxtaposition of 3-D figures which conforms to the properties of 2-D geometry as well as 3-D geometry.

SOLIDS

A solid has three dimensions, namely, length, breadth or width and height or thickness. The plane surfaces that binds it are called its faces and the solid so generated is known as a polyhedron.

The volume of any solid figure is the amount of space enclosed within its bounding faces. A solid has edges, vertices and faces which are shown in the figure given below:



A solid has two types of surface areas:

- *Lateral Surface Area (LSA)* LSA of a solid is the sum of the areas of all the surfaces it has except the top and the base.
- *Total Surface Area (TSA)* TSA of a solid is the sum of the lateral surface area and the areas of the base and the top.

Note In case of solids like the cube and cuboid, the lateral surface area consists of plane surface areas (i.e., area of all surfaces except the top and base) whereas in case of solids like cone and cylinder, it consists of curved surface areas (CSA). Thus, for such solids the LSA is also called CSA.

Euler's Rule

Euler's rule states that for any regular solid:
Number of faces (F) + Number of vertices (V) = Number of Edges (E) + 2

Cuboid

A cuboid is a rectangular solid having 6 rectangular faces. The opposite faces of a cuboid are equal rectangles. A cuboid has a length (l), breadth (b) and height (h).

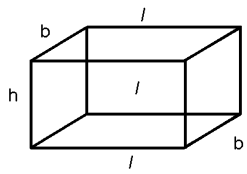


Fig. 1

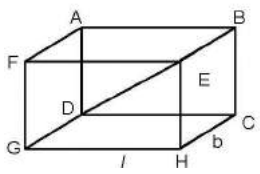


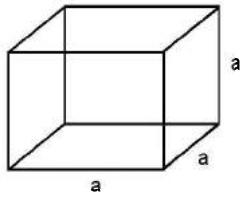
Fig. 2

In Fig. 2 ED is the diagonal of the cuboid. Moreover, the area of the surface GDCH is x, the area of the surface HEBC is y and the area of the surface GFEH is z.

- Volume = Area of base × Height = lbh
- Volume = $\sqrt{xyz}$
- Volume = $xh = yl = zb$
- Lateral surface area (LSA) or area of the 4 walls = 2 (l + b) h
- Total surface area (TSA) = $2(x+y+z)$ = 2 ($lb+bh$ + lh)
- Diagonal = $\sqrt{l^2+b^2+h^2}$

CUBE

A cube is a solid figure having six faces. All the faces of a cube are equal squares (let us say of the side 'a'). Thus the length, breadth and height of a cube are equal.



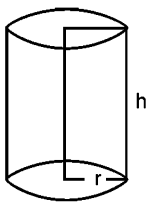
- Volume = a^3

(ii) Lateral surface area (LSA) or area of the 4 walls = $4a^2$

(iii) Total surface area (TSA) = $6a^2$

(iv) Diagonal = $a\sqrt{3}$

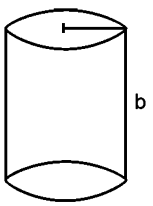
RIGHT CIRCULAR CYLINDER

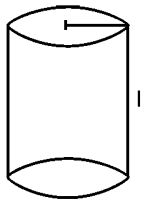


In the above figure, r is the radius of the base and h is the height of a right circular cylinder. A cylinder is generated by rotating a rectangle or a square by fixing one of its sides.

- Volume = area of base × height
  Volume = $\pi r^2 h$
- Curved surface area (CSA) = Perimeter of base × height
  LSA = $2\pi rh$
- Total surface area (TSA) = LSA + Area of the top + Area of the base
  TSA = $2\pi rh + \pi r^2 + \pi r^2$
  TSA = $2\pi r(r + h)$

Some important deductions



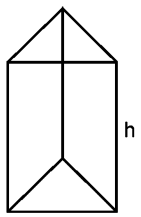
- If the above rectangular sheet of paper (ABCD) is rolled along its length to form a cylinder, then the radius (r) of the cylinder will be (L/2π) and its height will be b and Volume of this cylinder = $\frac{L^2 b}{4\pi}$, where L is the length of the rectangle.
- 

If the above rectangular sheet of paper (ABCD) is rolled along its breadth to form a cylinder, then the radius (r) of the cylinder will be $\frac{b}{2\pi}$ and its height will be L.

Volume of this cylinder = $\frac{b^2 L}{4\pi}$

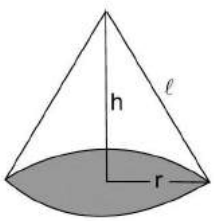
PRISM

A prism is a solid having identical and parallel top and bottom faces, i.e., they will be identical polygons of any number of sides. The side faces of a prism are rectangular and are known as lateral faces. The distance between two bases is known as the height or the length of the prism.



- Volume = Area of base × Height
- Lateral Surface Area (LSA) = Perimeter of the base × height
- Total surface Area (TSA) = LSA + (2 × Area of the base)

RIGHT CIRCULAR CONE



In the above figure 'r' is the radius of the base, h is the height and ℓ is the slant height of the right circular cone.

- Volume = $\frac{1}{3}$ × area of the base × height

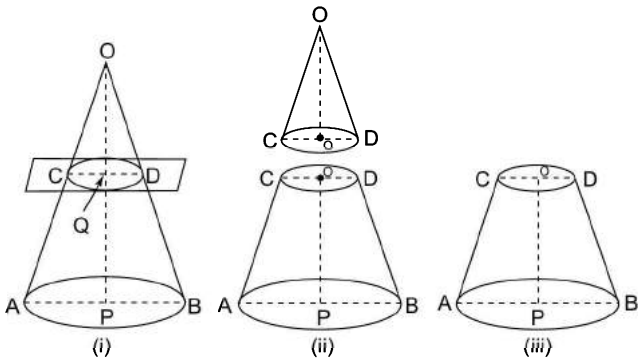
  Volume = $\frac{1}{3}\pi r^2 h$

(ii) Slant height = $\ell = \sqrt{r^2 + h^2}$
(iii) Curved surface area (CSA) = $\pi r \ell$
(iv) Total surface area (TSA) = (CSA + Area of the base)
TSA = $\pi r \ell + \pi r^2$

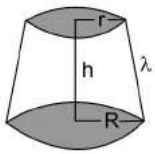
Frustum of Cone

A cone whose top portion is sliced off by a plane which is parallel to the base is called frustum of cone.

Formation of frustum:



However, for the sake of representing the formula, we will use another form of frustum right now as given below:



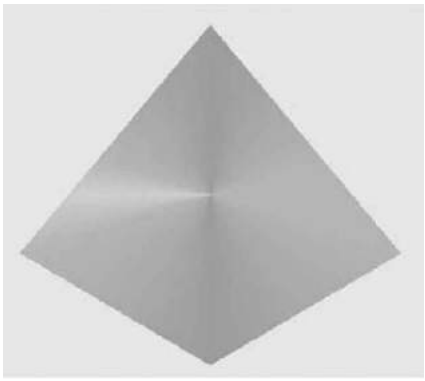
In the above figure, r is the radius of the base, h is the vertical height of the frustum and λ is the slant height of the frustum.

- Volume = $\frac{\pi h}{3}\left(R^2 + r^2 + Rr\right)$

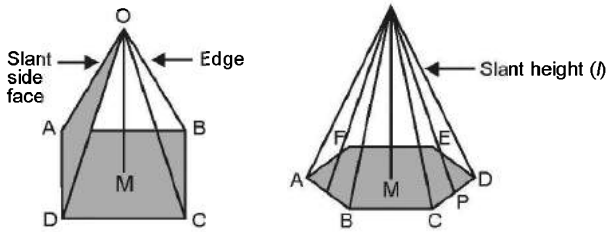
(ii) Slant height = $\lambda = \sqrt{(R-r)^2 + h^2}$
(iii) Curved Surface Area (CSA) = π(R + r) λ
(iv) Total Surface Area (TSA) = CSA + Area of the Top + Area of the base
TSA = π(R + r) λ + πr² + πR²
TSA = π(Rλ + rλ + r² + R²)
(v) To find the height (H) of original cone.

$$H = \frac{Rh}{R-r}$$

PYRAMID



A pyramid is a solid having an n-sided polygon at its base. The side faces of a pyramid are triangular with the top as a point.



In the above figures OM is the height of the pyramid.

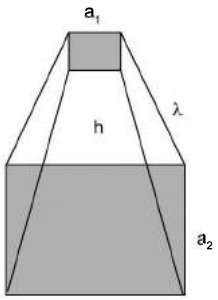
- Volume = $\frac{1}{3}$ × Area of the base × height

(ii) Lateral Surface Area (LSA) = $\frac{1}{2}$× (Perimeter of the base) × slant height

(iii) Total Surface Area (TSA) = LSA + Area of the base

Frustum of Pyramid

A pyramid whose top portion is sliced off by a plane which is parallel to the base is called the frustum of a pyramid.



In the above figure, A_1 is the area of the top face of the frustum, A_2 is the area of the bottom face of the frustum, h is the height of the frustum and l is the slant height of the frustum.

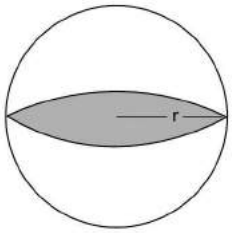
- Volume = $\frac{1}{3}h\left(A_1 + A_2 + \sqrt{A_1 A_2}\right)$

(ii) Lateral Surface Area (LSA) = $\frac{1}{2}(P_1 + P_2)\,\ell$

where P_1 and P_2 are perimeters of the top and the bottom faces.

(iii) Total surface Area (TSA) = $^{\ell}$S.A. + A_1 + A_2

SPHERE

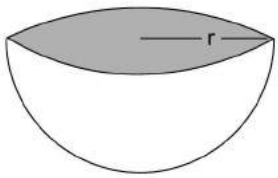


In the above figure r is the radius of the sphere.

- Volume = $\frac{4}{3}\pi r^3$

(ii) Surface Area = $4\pi r^2$

HEMISPHERE



- Volume = $\frac{2}{3}\pi r^3$

(ii) Curved surface area (CSA) = $2\pi r^2$

(iii) Total surface area (TSA) = LSA + Area of the top face (read circle)

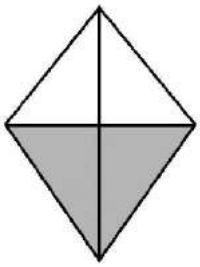
TSA = $2\pi r^2 + \pi r^2$

TSA = $3\pi r^2$

SOME MORE SOLIDS

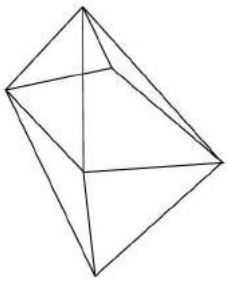
Tetrahedron

A tetrahedron is a solid which has 4 faces. All the faces of a tetrahedron are equilateral triangles. A tetrahedron has 4 vertices and 6 edges.



Octahedron

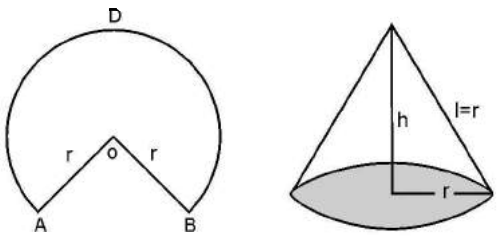
An octahedron is a solid which has 8 faces. All the faces of an octahedron are equilateral triangles. An octahedron has 6 vertices and 12 edges.



Inscribed and circumscribed solids

- If a sphere of the maximum volume is inscribed in a cube of edge 'a', then the radius of the sphere = $\frac{a}{2}$
- If a cube of the maximum volume is inscribed in a sphere of radius 'r', then the edge of the cube = $\frac{2r}{\sqrt{3}}$
- If a cube of the maximum volume is inscribed in a hemisphere of radius 'r', then the edge of the cube = $\sqrt{\frac{2}{3}} \times r$

Some important deductions



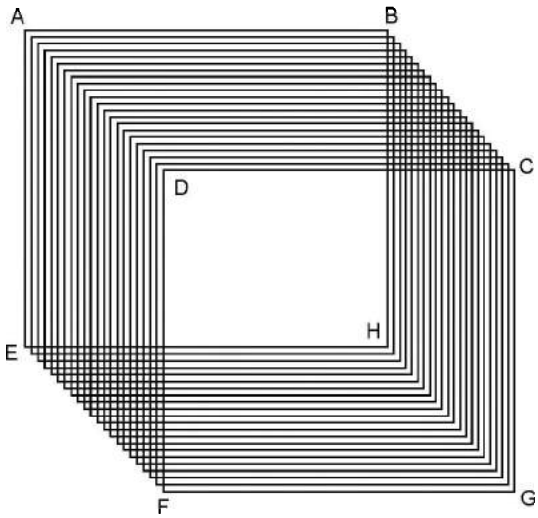
If a cone is made by a sector of a circle (AOBD), then the following two things must be remembered

- The area of the sector of a circle (AOBD) = The CSA of the cone
- Radius of the cirlce (r) = Slant height (l) of the cone.

VISUAL MENSURATION

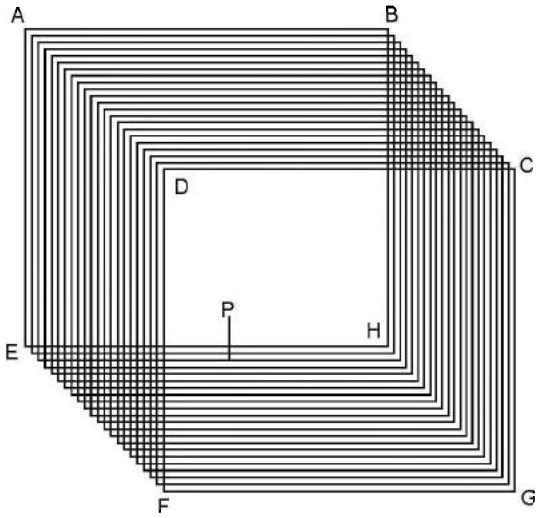
Under this section of mensuration, we will be required to visualize some of the unknown dimensions of any structure with the help of the given dimension. Some of the typical examples of Visual Mensuration are given below:

Given is a cube ABCDEFGH of side length 'a' units. Its top face is ABCD and its bottom face is EFGH. Since the side length of this cube is 'a' units, so, AB = BC = CD = AD = AE = EF = FD = FG = GH = EH = BH = CG = 'a'



Minimum length between vertex A and vertex G We can find the minimum distance between these two vertices in the following two ways:

- *Aerial distance* Aerial distance can be understood by assuming that there is a fly at vertex A and it has to reach vertex G through the minimum possible distance. This distance will be the diagonal distance between the vertices A and G = $\sqrt{a^2+a^2+a^2} = a\sqrt{3}$
- *Physical distance* Physical distance can be understood by assuming that there is an ant at vertex A and it has to reach vertex G through the minimum possible distance. Since it cannot fly, so ant will cover this distance along the two faces viz., face ABEH and face EFGH or face ADFE and face CDFG or face ABCD and face CDFG or face ABCD and face BCGH.



The shortest possible distance can be the diagonal only. Let us assume that the ant is going via face ABEH and face EFGH. So, the ant will cover first the diagonal distance between A and P, where P is the mid point of side EH, and then from P to vertex G.

Now $AP^2 = EP^2 + GP^2 = \frac{a\sqrt{5}}{2}$

And $GP^2 = HP^2 + GH^2 = \frac{a\sqrt{5}}{2}$

Hence, the minimum possible physical distance

$= 2\frac{a\sqrt{5}}{2} = a\sqrt{5}$

Let us find out why the distance AE + EG cannot be the shortest?

First calculate AE and EG.

AE = a

EG = $\sqrt{a^2 + a^2} = a\sqrt{2}$

So, AE + EG = $a + \sqrt{a^2 + a^2} = a\sqrt{2} + a = a\,(1+\sqrt{2})$

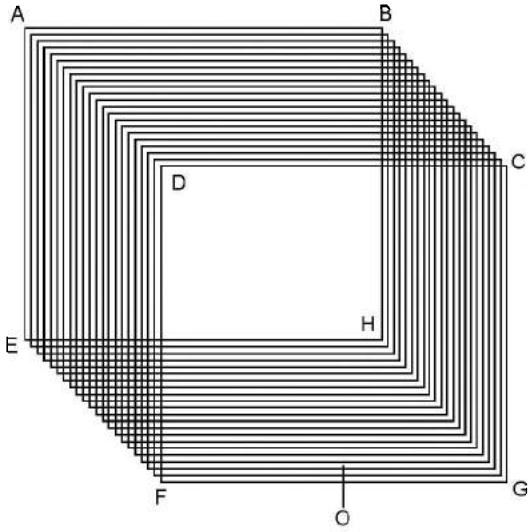
And obviously, $a\sqrt{5} < a(1 + \sqrt{2})$[For $a>0$]

We can understand this phenomenon by having a bit of *mental mapping*. As we have seen earlier that the minimum possible distance can be the diagonal distance only. Now let us cut open the face ABHE by making a cut mark at EH so that the faces ABHE and EFGH are in a plane, lying horizontal on the ground. Now the minimum possible distance between A and G will be the diagonal of the newly formed rectangle AFGB. This diagonal will pass through the mid-point of EH.

So, $AG^2 = AF^2 + FG^2 = (2a)^2 + (a)^2 = 5a^2$

AG = $a\sqrt{5}$

Minimum length between vertex A and vertex O, where O is the mid-point of FG

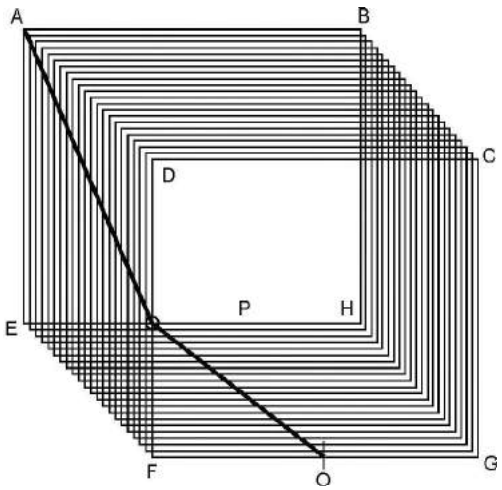


- *Aerial distance* We can see this situation vis-à-vis a cuboid of side lengths 'a' units, 'a' units and $\frac{a}{2}$ units.

  Minimum aerial distance between A and O

  $= \sqrt{\frac{a^2}{4} + a^2 + a^2} = \frac{3a}{2}$

- *Physical distance* Let us assume that the ant is moving through the faces ABHE and EFGH. Ant will first go the point Q, where Q is the mid-point of E and P. And then the ant will cover QO.

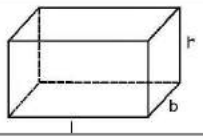
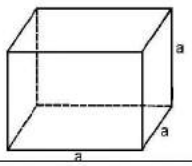
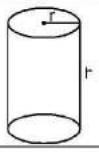
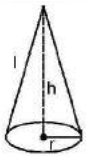
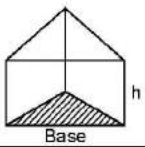
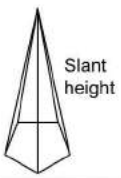
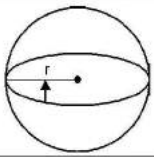
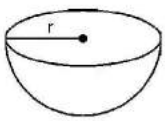
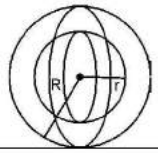


$AQ^2 = AE^2 + EQ^2 = a^2 + \frac{a^2}{16} = \frac{17a^2}{16}$

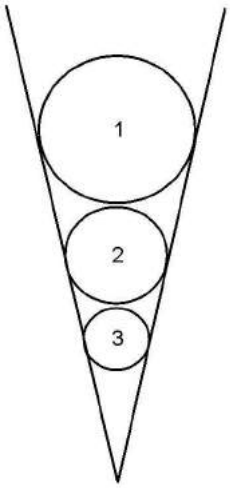
So, AQ = QO = $\frac{a\sqrt{17}}{4}$

So, the minimum distance between the vertex A and point O = AQ + QO = $\frac{a\sqrt{17}}{2}$.

FORMULAE

S.No.	Name	Figure	Nomenclature	Volume	Curved/ Lateral	Total Surface Area Surface Area
1.	Cuboid		l = length b = breadth h = height	lbh	2(l+b) h	2(lb+bh+hl)
2.	Cube		a = edge/side	a^3	$4a^2$	$6a^2$
3.	Right Circular Cylinder		r = radius of base h = height of the cylinder	πr^2h	$2\pi rh$	$2\pi r\ (r+h)$
4.	Right Circular Cone		r = radius h = height l = slant height $l=\sqrt{r^2+h^2}$	$\frac{\pi r^2h}{3}$	πrl	$\pi r\ (l+r)$
5.	Right Triangular Prism		–	Area of the base × height	Perimeter of the base × height	Lateral surface area + 2 (area of base)
6.	Right Pyramid		– height	1/3 × area of the base × of the base × slant height	½ × perimeter the base × slant height	Lateral surface area + area of base
7.	Sphere		r = radius	$\frac{4}{3}\pi r^3$	–	$4\pi r^2$
8.	Hemisphere		r = radius	$\frac{2}{3}\pi r^3$	$2\pi r^2$	$3\pi r^2$
9.	Spherical shell		r = inner radius R = outer radius	$\frac{4}{3}\ \pi(R^3-r^3)$	–	$4\pi(R^2+r^2)$
10.	Frustum of a Cone		–	–	–	Lateral surface area + Area of top + Area of base

Example 1 Three spheres are kept inside a cone, as given in the figure. Spheres are touching both the slant sides of the cone and the adjacent spheres. If the radius of the 1st sphere and the 3rd sphere are 5 units and 20 units respectively, find the radius of the 2nd sphere.



Solution We can see in Fig. 1, that the AOA′P, BOB′Q and COC′R will be similar.

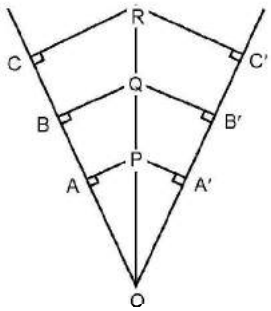
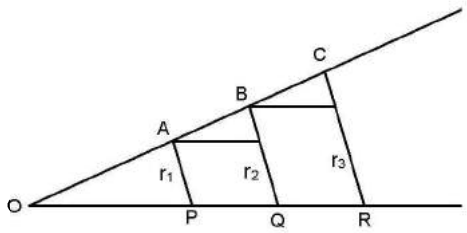


Fig. 1.



In Fig. 2, $(r_2 - r_1)/(r_2 + r_1) = (r_3 - r_2)/(r_3 + r_2) = K$

Using Componendo and Dividendo,

$r_2/r_1 = r_3/r_2$

Hence, the three radii are in a GP.

So, $r_2/20 = 5/r_2$, hence $r_2 = 10$ units.

Result of this question can be used as a formula also.

SOLVED EXAMPLES

Example 2 The diameter of a metallic sphere is 6 cm. The sphere is melted and drawn into a wire of uniform circular cross-section. If the length of the wire is 36 m, find its radius.

Solution The diameter of the metallic sphere is 6 cm. Hence, radius of the sphere is 3 cm. Now, let the radius of the cross–section of the wire be r cm. And as we know, metallic sphere is converted into a cylindrical shaped wire, then their volumes will be equal.

$$\text{So, } \frac{4}{3} \times \pi \times 3^3 = \pi \times r^2 \times 3600$$

$$\text{or, } 4 \times 9\,\pi \times 3^3 = 3600\,\pi r^2$$

$$\text{This gives } r^2 = 0.01$$

$$\text{i.e., } r = 0.1$$

Example 3 Sardar Sarovar Dam which is rectangular in shape, can produce electricity only if the height of the water level in it is atleast 7cm. Now the water is pumped in at the rate of 5 km per hour through a pipe of diameter 14 cm into the dam area of dimensions 50 m × 44 m. In what time the dam will be able to produce electricity?

Solution The volume of water flowing through the cylindrical pipe in one hour at the rate of 5 km (5000 m) per hour

$$= \frac{22}{7} \times \frac{7}{100} \times \frac{7}{100} \times 5000\,\text{m}^3$$

[Since radius = 7 cm = $\frac{7}{100}$ m

$= 77\ \text{m}^3$

Thus, 77 m^3 of water will fall into the tank in 1 h. Since the level of the water is required to be raised by 7 cm i.e., $\frac{7}{100}$ m,

Volume of the required quantity of water = $50 \times 44 \times \frac{7}{100}\ \text{m}^3 = 154\ \text{m}^3$

Since 77 m^3 of water falls into the tank in 1 h, therefore, 154 m^3 of water will fall into the dam in $\frac{154}{77}$ h, i.e., 2 h.

So, the level of water will rise by 7 cm in 2 h.

Example 4 A right angled triangle ABC, whose two sides other than the hypotonuse are 15 cm and 20 cm. The triangle is made to revolve about its hypotonuse. Find the volume of the double cone so formed.

Solution Let ΔABC be the right angled triangle right angled at A, whose sides AB and AC measures 15 cm and 20 cm, respectively.

The length of the side BC (hypotenuse)

$= \sqrt{15^2 + 20^2}$ cm = 25

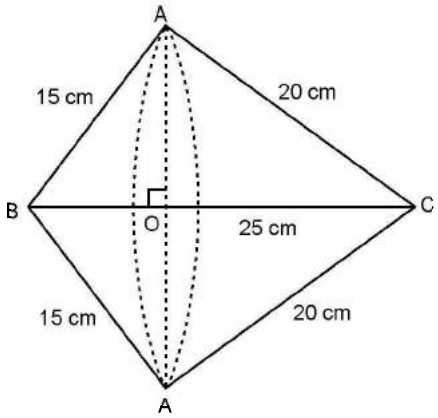
Here, AO (and A'O) is the radius of the common base of the double cone formed by revolving the ΔABC about BC.

Height of the cone BBA' is BO and the slant height is 15 cm.

Height of the cone CAA' is CO and the slant height is 20 cm

Using AA similarity,

Now, ΔAOB ~ ΔCAB



Therefore, $\frac{AO}{20} = \frac{15}{25}$

This gives $AO = \frac{20 \times 15}{25}\text{ cm} = 12\text{ cm}$

Also, $\frac{BO}{15} = \frac{15}{25}\text{ cm}$

This gives $BO = \frac{15 \times 15}{25}\text{cm} = 9\text{cm}$

Thus, CO = 25 cm – 9 cm = 16 cm

Now, volume of the double cone

$= \left(\frac{1}{3} \times 3.14 \times 12^2 \times 9 + \frac{1}{3} \times 3.14 \times 12^2 \times 16\right)\text{cm}^2$

$= \frac{3.14}{3} \times 12^2 \times (9+16)\text{cm}^2$

$= 3768\text{ cm}^3$

Example 5 When shopping in Big Bazar, I saw a peculiar solid toy in the form of a hemisphere surmounted by a right circular cone. Height of the cone was 2 cm and the diameter of the base was 4 cm. If a right circular cylinder circumscribed the solid, find out how much more space will it have, provided the height of the cone was 2 cm and diameter of the base was 4 cm respectively?

Solution See the figure below.

Assume BPC be the hemisphere and ABC is the cone standing on the base of the hemisphere.

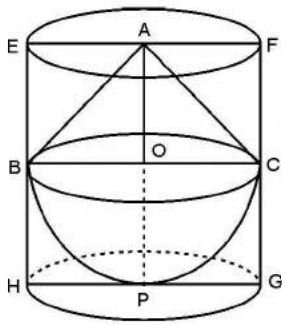
Radius BO of the hemisphere (as well as of the cone) = $\frac{1}{2} \times 4$ cm = 2 cm.

Now, let the right circular cylinder EFGH circumscribe the given solid.

Radius of the base of the right circular cylinder = HP = BO = 2 cm.

Height of the cylinder = AP = AO + OP = 2 cm + 2 cm = 4 cm

Now, volume of the right circular cylinder – volume of the solid



$= \left[\pi \times 2^2 \times 4 - \left(\frac{2}{3} \times \pi 2^3 + \frac{1}{3} \times \pi \times 2^3\right)\right]\text{cm}^3$

$= (16\pi - 8\pi)\text{ cm}^3$

$= 8\pi\text{ cm}^3$

Hence, the right circular cylinder is having 8p cm^3 more space than the solid.

Example 6 A toy consists of a base that is the section of a sphere and a conical top. The volume of the conical top is 30π sq. units and its height is 10 units. The total height of the toy is 19 units. The volume of the sphere (in cubic units), from which the base has been extracted, is

(a) $\frac{256}{3}\pi$ (b) $\frac{64}{3}\pi$

(c) $\frac{108}{3}\pi$ (d) $\frac{500}{3}\pi$

Solution Height of the cone = 10 units

Volume of the cone = 30π cubic units

= diameter of the cross section from where the sphere has been sectioned = 6 units

$= r + \sqrt{2^2 - 3^2} = 9$

= r = 5 units

= volume of the original sphere from which the base has been sectioned = $\frac{4}{3}\pi r^3 = \frac{500}{3}\pi$

PRACTICE EXERCISES

WARM UP

Q.1. If the side of a cube is increased by 100%, find by what percentage the surface area of the cube is increased?
(a) 150% (b) 200% (c) 300% (d) 350%

Q.2. A banquet hall has the dimensions 30 m×12 m×6 m. Each person should get 8 m^3 of space. Find the number of persons who can be accommodated in this hall.
(a) 240 (b) 250 (c) 270 (d) 300

Q.3. Find the height of the cylinder whose volume is 511 cm^3 and the area of the base is 36.5 cm^2.
(a) 3.5 cm (b) 10.5 cm
(c) 14 cm (d) None of these

Q.4. The ratio between the curved surface area and the total surface area is 2:3 and the total surface is 924 cm^2. What is the volume of the cylinder?
(a) 2156 cm3 (b) 2183 cm^3
(c) 2492 cm^3 (d) None of these

Q.5. If the curved surface area of a cylinder is 1320 cm^2 and its base radius is 21 cm, then what is its total surface area?
(a) 4092 cm2 (b) 2084 cm^2
(c) 5104 cm^2 (d) None of these

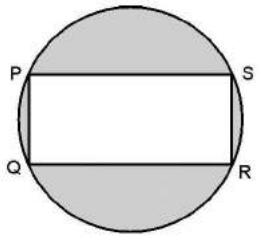
Q.6. The radius of an iron rod is decreased to one fourth of its actual radius. If its volume remains constant, then the length will become
(a) 2 times (b) 12 times
(c) 8 times (d) 16 times

Q.7. A reservoir is in the shape of a frustum of a right circular cone. It is 8 m wide at the top and 4 m wide at the bottom. If it is 6 m deep, then what is its volume?
(a) 224 m^3 (b) 176 m^3
(c) 204 m^3 (d) None of these

Q.8. A spherical metal ball of 6 cm radius is melted and recast into three spherical balls. The radii of two of these balls are 3 cm and 4 cm. What is the radius of the third ball?
(a) 4.5 cm (b) 5 cm
(c) 6 cm (d) 7 cm

Q.9. If the diagonals of a rhombus are 18 cm and 24 cm respectively, then find its perimeter.
(a) 15 cm (b) 42 cm
(c) 60 cm (d) 68 cm

Q.10. In the figure shown below, PQRS is a rectangle of the dimension 8 cm × 6 cm and is inscribed in a circle. Find the area of the shaded portion.



(a) 44 cm^2 (b) 34.25 cm^2
(c) 32.50 cm^2 (d) None of these

Q.11. The radius of the base and the volume of the right circular cone are doubled. What is the ratio of the length of the larger cone to that of the smaller cone?
(a) 1:4 (b) 1:2
(c) 1:3 (d) 4:1

Q.12. A cone and a hemisphere have equal base radius and equal volumes. The ratio of their heights is:
(a) 3:1 (b) 2:1
(c) 4:1 (d) None of these

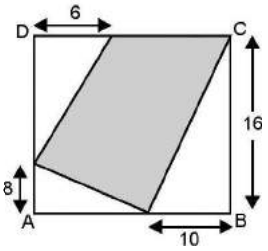
Q.13. If the right circular cone is cut into three solids of volumes V_1, V_2 and V_3 by two cuts which are parallel to the base and trisects the altitude, then $V_1:V_2:V_3$ is
(a) 1:2:3 (b) 1:4:6
(c) 1:6:9 (d) None of these

Q.14. Water flows at the rate of 10 m per min from a cylindrical pipe of radius 2.5 mm. A conical vessel whose diameter is 40 cm and depth 24 cm is filled with water flowing from this pipe. The time taken to fill the conical vessel is
(a) Less than 30 min
(b) Less than 50 min but more than 30 min
(c) More than 50 min but less than 75 min
(d) More than 75 min

Q.15. If a regular square pyramid has a base of side 8 cm and height of 30 cm, then what is its volume (in cc)?
(a) 360 (b) 480
(c) 640 (d) 800

Q.16. A cylinder circumscribes a sphere. The ratio of their volumes is
(a) 2 : 1 (b) 3 : 2 (c) 4 : 3 (d) 6 : 5

Q.17. Find the area of the shaded region in the given figure of square ABCD.



(a) 128 cm^2 (b) 184 cm^2
(c) 154 cm^2 (d) 168 cm^2

Q.18. An open box is made of wood 2 cm thick. Its internal dimension is 86 cm × 46 cm × 38 cm. What is the cost of painting the outer surface of this box @ Rs 10 per m^2?
(a) Rs 12.35 (b) Rs 8.85
(c) Rs 15.70 (d) Rs 16.50

Q.19. A copper wire when bent in the form of a square, encloses an area of 121 m^2. If the same wire is bent to form a circle, the area enclosed by it would be
(a) 124 m^2 (b) 136 m^2
(c) 154 m^2 (d) 184 m^2

Q.20. A spherical ball was painted black. After getting painted, it was cut into 4 similar pieces. What is ratio of the painted area to the non-painted area?
(a) 1 : 1 (b) 1 : 2
(c) 3 : 1 (d) 3 : 2

Q.21. Volume of a sphere, a cube, a tetrahedron and an octahedron is same. Find for which of the following structures, it will have the maximum surface area?
(a) Sphere (b) Cube
(c) Tetrahedron (d) Octahedron

Q.22. The ratio of diagonals of the two cubes is 3 : 2. What is the ratio of the surface areas of these two cubes respectively?
(a) 27 : 8 (b) 3 : 2
(c) 9 : 4 (d) 16 : 9

FOUNDATION

Q.1. A rectangular classroom has to be painted covering all its surfaces excluding floor. The cost of painting a similar room is Rs 1872. But its length, breadth and height are 10% less, 10% more and 10% less than the dimensions of the classroom respectively. How much will it cost to paint the classroom if its length, breadth and height are in the ratio of 1:2:3?
(a) Rs 1,956 (b) Rs 2,000
(c) Rs 2,100 (d) Cannot be determined

Q.2. The top of a conical container has a circumference of 308 m. Water flows in at a rate of 12 m^3 every 2 s. When will the container be half filled, if its depth is 12 m?
(a) 42 min (b) 68 min
(c) 54 min (d) 82 min

Q.3. What is the radius of a spherical ball in inches which is formed by melting a cylinder of base diameter 8 inches and height 160 inches, if the conversion wastage results in a 10% weight loss?
(a) 6 (b) 8
(c) 12 (d) 16

Q.4. A solid wooden toy is in the shape of a right circular cone mounted on the base of a hemisphere. If the radius of the hemisphere is 4.2 cm and the total height of the toy is 10.2 cm, find the volume of the wood needed to construct another such toy.
(a) 104 cm^3 (b) 162 cm^3
(c) 427 cm^3 (d) 266 cm^3

Q.5. A well is dug 20 ft deep and the mud which came out is used to build a wall of width 1 ft around the well on the ground. If the height of the wall around the well is 5 ft, then what is the radius of the well?
(a) $\frac{\sqrt{5}+1}{4}$ (b) $\sqrt{5}$ (c) 1 (d) $\frac{1}{4}$

Q.6. The foot of a ladder 15 m long, is 9 m from a house and its top reaches the upper part of a circular window. When the foot of the ladder is drawn to a distance of 3 m away from the house, the top reaches the lower edge of the window. The area of the window is
(a) $5\frac{2}{3}$ m^2 (b) 6 m^2
(c) $7\frac{1}{14}$ m^2 (d) $8\frac{2}{3}$ m^2

Q.7. A brick wall is to be built 20 m long, 3 m high and 40 cm thick. It has a door 3 m by 2 m. Suppose each brick is to be 15 cm long, $7\frac{1}{2}$ cm broad and 5 cm thick,

how much will be the cost of the bricks at the rate of Rs 800 per thousand bricks?
(a) Rs 55,270 (b) Rs 66,230
(c) Rs 30,720 (d) Rs 15,750

Q.8. A rectangular field is of the dimension 15.4 m × 12.1 m. A circular well of 0.7 m radius and 3 m depth is dug in the field. The mud, dug out from the well, is spread in the field. By how much would the level of the field rise?
(a) 1 cm (b) 2.5 cm
(c) 3.5 cm (d) 4 cm

Q.9. The dimension of a room is thrice as long as it is height, but only twice as long as it is wide. The total cost of painting its walls at the rate of Rs 2.50 m^2 is Rs 360. What is the cost of laying a carpet on its floor at the rate of Rs 3/m^2?
(a) Rs 81 (b) Rs 125
(c) Rs 216 (d) Rs 260

Q.10. A man, by walking diametrically across a circular grass plot, finds that it has taken 45 s less than if he had kept to the path around the outside if he walks at the rate of 80 m per minute, the diameter of the grass plot is
(a) 35 m (b) 65 m
(c) 105 m (d) 145 m

Q.11. Within a rectangular courtyard of length 60 feet, a graveled path, 3 feet wide, is laid down along all the four sides. The cost of gravelling the path is Rs 2 per sq ft^2. If the path had been twice as wide, the gravel would have cost Rs 984 more. The width of the courtyard is
(a) 24 feet (b) 40 feet
(c) 45 feet (d) 54 feet

Q.12. A vessel 2 m long, 1 m wide and 1.5 m deep contains 2 m^3 water. How many bricks of 20 cm by 10 cm by 7.5 cm can be put in it so that water does not overflow provided that a brick is supposed to absorb 1/7 of its own volume of water?
(a) 666 (b) 111
(c) 555 (d) None of these

Q.13. In Snehans Apartment there is a playground of the dimension $7\frac{1}{2}$ by 7 m. In the centre of the ground, there is a flower-bed cutout of the dimension 3 m by $2\frac{1}{2}$ m. What fraction of the whole ground is occupied by the flower-bed?
(a) $\frac{3}{7}$ (b) $\frac{3}{5}$ (c) $\frac{1}{7}$ (d) $\frac{1}{11}$

Q.14. The height of a room is $\frac{1}{5}$ th of the sum of its length and breadth. The cost of preparing its wall @ Rs 4 per m^2 is Rs 640. What is the height of the room?
(a) 4 m (b) 5 m
(c) 6 m (d) 7 m

Q.15. A cylindrical container of 32 cm height and 18 cm radius is filled with sand. Now all this sand is used to form a conical heap of sand. If the height of the conical heap is 24 cm, what is the radius of its base?
(a) 12 cm (b) 24 cm
(c) 36 cm (d) 48 cm

Q.16. The total area of the four walls of a room is 150 m^2. If the area of the floor is 50 m^2 and the width of the floor is 3 m, then find the height of the room.
(a) 2.6 m (b) 3.8 m
(c) 5.42 m (d) 7.32 m

Q.17. Anoop has a wooden box, each sides of which is an integer (in cm) and whose volume (neglecting the thickness of the box) is 216 cm^3. What is the minimum possible amount that Anoop will need to spend in order to get the outside of the box painted at the rate of Rs 2 per cm^2?
(a) Rs 320 (b) Rs 225
(c) Rs 236 (d) Rs 432

Direction for questions 18 and 19: Read the passage below and solve the questions based on it.

There is a rectangular table of the dimension 2.20 m × 1.4 m. A rectangular tablecloth of the dimension 2.4 m × 0.8 m is spread unfolded on the table so that the longer sides of the cloth and the table are parallel.

Q.18. If the larger side of the tablecloth and the smaller side of the table are parallel, then what fraction of the area of the top of the table remains uncovered by the cloth?
(a) 3/7 (b) 4/9
(c) 7/11 (d) None of these

Q.19. In the previous question, what fraction of the cloth hangs perpendicularly over the edge of the table?
(a) 5/9 (b) 2/3
(c) 7/13 (d) 5/12

Q.20. A solid consists of a right circular cylinder with the exact fitting right circular cone placed on its top. The height of the cone is N. If the ratio of the total volume of the solid and volume of the cone is 3 : 1, then the height of the circular cylinder is
(a) h/2 (b) 3h
(c) h/3 (d) None of these

Direction for questions 21 and 22: Read the passage below and solve the questions based on it.

There is an equilateral triangle of side 'a' units. Now we join any two sides of the triangle to form a cone.

Q.21. What is the slant height of the cone formed?

(a) $\frac{a}{2\pi}$ (b) a (c) $\frac{a}{2}$ (d) $2\pi a$

Q.22. What is the radius of the cone formed?

(a) a (b) $\frac{a}{4}$

(c) $\frac{a}{\pi}$ (d) None of these

Q.23. The area of three adjacent faces of a cuboidal box are p, q and r square cm. What is the volume of this box?

(a) $(p+q+r)$ cm^3 (b) $\sqrt{pqr}$ cm^3

(c) $\sqrt[3]{pqr}$ cm^2 (d) pqr cm^3

Q.24. Three equal cubes of unit side length are placed adjacent to each other in a row. Find the ratio of the total surface area of the new cuboid to that of the sum of the surface areas of all the three cubes.

(a) 3:5 (b) 4:5
(c) 6:7 (d) 7:9

Q.25. A classroom is to be built to accommodate 70 students. It should be done in such a way that for every student there is 2.2 m^2 of floor and 11 m^3 of space. If the length of the room is 14 m, then find the breadth and height of the room.

(a) 12 m 5.5 m (b) 11 m 5 m
(c) 13 m 6 m (d) 11 m 4 m

Q.26. Length, breadth and height of a rectangular parallelopiped are in the ratio of 6:5:4 and its total surface area is 33,300 m^2. Find length, breadth and height of the parallelopiped (in cm).

(a) 42, 35, 28 (b) 60, 50, 40
(c) 102, 85, 68 (d) 90, 75, 60

Q.27. A rectangular tank is of dimension 30 m × 20 m. Water is being flown into it through a square pipe of side length 5 cm. Find the speed of the water if the level of water in the tank rises by 1 m in 8 h?

(a) 30 km/h (b) 36 km/h
(c) km/h (d) None of these

Q.28. A cube of side length 3 cm weighs 12 kg. What is the weight of the similar cube of same material whose side length is 12 cm?

(a) 768 kg (b) 678 kg
(c) 964 kg (d) 864 kg

Q.29. A cube of side length 4 cm is cut into cubes of side 1 cm. Find the ratio of the sum of the surface area of all the small cubes to that of the large cube.

(a) 1:16 (b) 2:3
(c) 4:1 (d) 6:1

Q.30. A cylindrical well of depth 12 m with internal ratio of 1.75 m is dug up. The mud, so obtained, is spread evenly to form a platform of dimension 10.5 m × 8.8 m. What is the height of the platform?

(a) 2.25 m (b) 3.25 m
(c) 1.25 m (d) 4.25 m

Q.31. A hollow spherical ball of outer diameter 120 mm is cut into two equal hemispheres. One of the hemispheres is filled with honey and the total surface area of the other hemisphere is $389\frac{5}{7}$ cm^2. What is the volume of the honey put in the 1st hemi-sphere?

(a) 128 cm^3 (b) 134 cm^3
(c) 136 cm^3 (d) 138 cm^3

Q.32. The diameter of a cylindrical vessel is made twice as large. How should the height be changed so that it still contains the same quantity of water?

(a) $\frac{1}{2}$ × height of the two cylinders

(b) $\frac{1}{4}$ × height of the original cylinder

(c) Let it be the same

(d) None of these

Q.33. Two shopkeepers are offering Pepsi in two different types of cans. The 1st shopkeeper offers cylindrical cans with a circular base of radius 5 cm and 2nd shopkeeper offers the cans having a square base of length 10 cm. If height of both the cans is the same, then what is the difference in their capacities?

(a) 350 cm^3 (b) 450 cm^3
(c) 250 cm^3 (d) None of these

Q.34. A bucket is a standard example of frustum of a cone. The circumference of such a bucket of one end is 48 cm and of the other end is 34 cm. If the height of the bucket is 10 cm, its volume (in cm^3) is:

(a) 5400 (b) 1350 (c) 2700 (d) 4050

Q.35. The sum of the length, width and depth of a cuboid is s and its diagonal is d. What is its surface area?

(a) s^2 (b) d^2 (c) $s^2 - d^2$ (d) $s^2 + d^2$

Q.36. A solid right circular cylinder with height 10 cm and radius of the base 6 cm is given. A right circular cone of the same height and base is removed from this cylinder. Find the volume (in cm^3) of the remaining solid.

(a) 855.5 (b) 754.3
(c) 1294 (d) None of these

Q.37. At the Amul cooperative society, right cylindrical vessels were used to keep milk. But after the unceremonious exit of Mr Durien, it was decided that instead of right cylindrical vessels, right cones having the same diameter and height would be used. What would be the ratio of the number of the right cones to the number of right cylinders needed to hold the same quantity of milk?
(a) 2:1 (b) 3:1 (c) 4:1 (d) 5:1

Q.38. The radius of a right cylinder is doubled and its height is halved. What is the ratio between the new curved surface area and the previous curved surface area of the cylinder?
(a) 1:1 (b) 2:1 (c) 3:3 (d) 2:3

Q.39. A solid cone is converted into a solid cylinder of the same radius. If the height of the cylinder is 5 m, what is the height of the cone?
(a) 25 m (b) 15 m (c) 20 m (d) 10 m

Q.40. The radii of a cylinder and a cone are equal and the height of the cylinder is equal to the slant height of the cone. Find the ratio of the curved surfaces of the cylinder and the cone.
(a) 1:1 (b) 2:1 (c) 3:1 (d) 4:1

Q.41. A right circular cone is placed inside a cube such that the edges of the base of the cone are touching the edges of one of the faces of the cube and the vertex is touching the opposite face of the cube. If the volume of the cube is 343 cm^3, what is the volume of the cone (approx in cm^3)?
(a) 90 (b) 60 (c) 80 (d) 85

Q.42. When Natraj circus came to Patna for its show at Sonepur mela last year, I observed quite a few things. Its tent was cylindrical to a height of 3 m and conical above it and its diameter was 105 m and slant height of the conical portion was 53 m. But I was not able to calculate the length of the canvas 5 m wide to make the tent. Can you please bail me out by telling this?
(a) 1857 m (b) 1647 m
(c) 1947 m (d) 1847 m

Q.43. A cone and a cylinder have their height in the ratio of 3:2 and the radii of their bases in the ratio of 4:3. Find the ratio of their volumes.
(a) 9:1 (b) 9:2 (c) 8:9 (d) 3:1

Q.44. A cylindrical structure standing on its base with radius 1.5 m and height 5 m is cut with a saw in such a way that the cutting planes goes through all the points at a distance of 0.625 m from the base. Find the volume of the remaining piece.
(a) 5.62π (b) 9.24π
(c) 9.04π (d) None of these

Q.45. The height of a right circular cylinder is 6 m. Three times the sum of the areas of its two circular faces is twice the area of its curved surface. The radius of the base is
(a) 4 m (b) 2 m
(c) 6 m (d) 1.5 m

Q.46. A solid cone kept on its base is cut at 2/3rd of its height along a plane parallel to its circular base. The base radius and the slant height are 14 cm and 50 cm respectively. What is the ratio of the portion cut-out from the solid to the volume of the remaining solid?
(a) 1:20 (b) 1:25
(c) 1:36 (b) None of these

Q.47. A big cube of side 6 cm is formed by putting together 216 small identical cubes each of side 1 cm. Now, if the corner cubes in the topmost layer of the big cube are removed, what will be the impact on the total surface area of the big cube?
(a) Will decrease
(b) Will increase
(c) Will remain unchanged
(d) Cannot be determined

Q.48. Mr Sinha and Mr Verma are two brothers. They inherited their paternal land in a rectangular form having the same area but different dimensions. For Mr Sinha, rectangular plot, the length is increased by 10% and the breadth is decreased by 10%. For Mr Verma's rectangular plot, the length is decreased by 10% and the breadth is increased by 10%. Which of the following is true about the areas of the two rectangular plots now?
(a) The area of the first rectangle is 1% more than that of the second.
(b) The area of the second rectangle is 1% more than that of the first.
(c) The area of the two rectangles is the same.
(d) We cannot comment unless we know the dimensions of the individual rectangles.

Q.49. A big solid sphere of diameter 10 cm is melted and made into solid spheres of diameters 2 cm. What is the percentage increase/decrease in the surface area of smaller spheres over that of the big sphere?
(a) 200% increase (b) 400% increase
(c) 400% decrease (d) None of these

Q.50. A cone is made of a sector of a circle of radius 21 cm and an angle of 60°. What is the total surface are of the cone (in cm^2)?
(a) 269.5 (b) 308
(c) 312.5 (d) 231

MODERATE

Q.1. There is a right circular cone with base radius 3 units and height 4 units. The surface of this right circular cone is painted. It is then cut into two parts by a plane parallel to the base so that the volume of the top part (the small cone) divided by the volume of the frustum equals the painted area of the top part divided by the painted area of the bottom part. The height of the small cone is
(a) 7/3 (b) 5/4
(c) 5/2 (d) None of these

Direction for questions 2 and 3: Read the passage below and solve the questions based on it.

My grandfather owned plenty of land, which easily encompassed an area of 1000×200 m^2. He wanted to give some part of it to his servant Ramu. But he did not gift it directly. He supplied the material that could form a fence of length 100 m only. Then he allowed Ramu to take any part with four sides that could be encased with the help of the given fencing material.

Q.2. What is the maximum possible land Ramu can take away from my grandfather? (All the four sides are fenced).
(a) 100 m^2 (b) 2500 m^2
(c) 750 m^2 (d) 625 m^2

Q.3. Ramu had intimate knowledge of the land. Hence, he selected the site having a natural fencing of rocks on one side, because he could utilize the given material only on three sides of the plot. What is the maximum possible land that Ramu can claim now?
(a) 125 m^2 (b) 2500 m^2
(c) 625 m^2 (d) None of these

Q.4 The diameter of a road-roller is 42 cm and its length is 100 cm. It takes 400 complete revolutions moving once over to level the stretch of the road. If the cost of leveling is Rs 100 per m^2 then the total cost of leveling works out to
(a) Rs 52,800 (b) Rs 5,280
(c) Rs 5,28,000 (d) Rs 528

Q.5. Inside a triangular garden, there is a flower bed in the form of a similar triangle. Around the flower bed runs a uniform path of such a width that the sides of the garden are double of the corresponding sides of the flower bed. The areas of the path and the flower bed are in the ratio
(a) 1:1 (b) 4:1
(c) 1:3 (d) 3:1

Q.6. ABCD is a square of side 10 cm. What is the area of the least-sized square that may be inscribed in ABCD with its vertices on the sides of ABCD?
(a) 0 cm^2 (b) 25 cm^2
(c) 50 cm^2 (d) 66.66 cm^2

Q.7. A cylinder of the maximum possible size is made out of a solid wooden cube. How much material is lost in this process (approx.)?
(a) 20% (b) 22%
(c) 24% (d) Cannot be determined

Q.8. Two friends, ND and SD invited friends for a party. A hemispherical bowl full of ice cream of diameter 18 cm was served for dessert. Each of them had 3 scoops of the ice cream and it was exactly sufficient for all of them. If the ice cream was served in a hemispherical scoop of radius 1 cm, how many friends did they have for their party?
(a) 240 (b) 241 (c) 242 (d) 243

Q.9. The cost of white washing one m^2 is Rs 50. What will be the maximum amount saved in painting the room in the most economical way, if the sum of the length, breadth and height is 21 m and all the sides are integers (floor is not to be white washed)?
(a) Rs 1,08,050 (b) Rs 8,400
(c) Rs 9,300 (d) Rs 8,540

Q.10. A small bird is taking rest at the centre of the base of a hemispherical cage. Suddenly it stands up, flies to the topmost point in the cage, then in a straight line to the cage door at the intersection of the curved surface and the base. In the process it covers a total distance of 482 cm. What is the radius of the hemisphere?
(a) 100 cm (b) 150 cm
(c) 200 cm (d) 50 cm

Q.11. A cube is inscribed in a hemisphere of radius R, such that four of its vertices lie on the base of the hemisphere and the other four touch the hemispherical surface of the half-sphere. What is the volume of the cube?
(a) $0.25R^3$ (b) $0.67\sqrt{\frac{2}{3}R^3}$
(c) $0.5\sqrt{\frac{2}{3}R^3}$ (d) $0.67R^3$

Q.12. The biggest possible cube is taken out of a right solid cylinder of radius 15 cm and height 20 cm respectively. What will be the volume (in cm^3) of the cube?
(a) 375 (b) 800
(c) $375\sqrt{2}$ (d) None of these

Direction for questions 13 and 14: Read the passage below and solve the questions based on it.

A larger cube is made up of $N = n^3$ smaller cubes.
Let A = number of smaller cubes with no exposed surfaces.
B = number of smaller cubes with one exposed surfaces.
C = number of smaller cubes with two exposed surfaces.
D = number of smaller cubes with three exposed surfaces.

Q.13. What is the ratio of the number of unexposed smaller cubes to the total number of cubes?

(a) $\frac{1}{27}$ (b) $\frac{(n-2)}{n}$

(c) $\left(\frac{n-2}{n}\right)^3$ (d) $\frac{n}{(n-2)}$

Q.14. What is the total number of smaller cubes in a bigger cube for which A = B?

(a) 512 (b) 256 (c) 1000 (d) 729

Q.15. A ball is fitted into the mouth of an open cone whose base radius is equal to that of the ball. How far can the ball definitely go inside the cone?

(a) Half the ball will fit into the cone
(b) Less than half the ball will fit into the cone
(c) More than half the ball will fit into the cone
(d) One fourth of the ball will fit into the cone

Q.16. A well, 4 m in diameter and 35 m deep, is being dug out and the excavated soil is transported away in a truck trolley of size 5 m × 2 m × 0.5 m. Since the soil that is taken out is loose it occupies 20% more space. How many trips will the truck have to make to clear the excavated soil, if the truck can be filled only up to 80% of its height?

(a) 106 (b) 132
(c) 425 (d) 528

Q.17. The product of all the twelve edges of a rectangular box is equal to:

i. The fourth power of the volume of the box
ii. The product of the areas of the six faces of the box
iii. Sum of the squares of the areas of the six faces of the box

(a) None of the above three values
(b) The figure given by exactly one of the three statements
(c) The figure given by exactly two of the three statements
(d) The figure given by all the three statements

Q.18. A right circular cone is given with a height 6 cm and its slant side is making an angle of 60° with its base. Now a sphere of the maximum volume is placed inside this cone. If the radius of this sphere is 2.33 cm, then find the volume of the cone.

(a) 18π (b) 15π
(c) 24π (d) Cannot be determined

Q.19. A solid sphere of radius 7 cm is melted to form a number of small cones and cylinders. The requirement is such that the number of cones should be twice the number of cylinders. Also the radius of the cone must be equal to its height, which should be equal to the radius of the cylinder and also be half the height of the cylinder. If the height of one such cylinder is 4 cm, find the maximum number of cones which can be made out of the sphere.

(a) 18 (b) 21 (c) 36 (d) 42

Q.20. Let A and B be two solid spheres such that the surface area of B is 300% greater than the surface area of A. The volume of A is found to be K% lower than the volume of B. Then the value of K must be

(a) 85.5 (b) 92.5 (c) 90.5 (d) 87.5

Q.21. A large cube is formed from the material obtained by melting three smaller cubes of 3, 4 and 5 cm side. What is the ratio of the total surface areas of the smaller cubes and the large cube?

(a) 2:1 (b) 3:2 (c) 25:18 (d) 27:20

Q.22. The volume of a cube is V. Then the total length of its edges is

(a) $6V^3$ (b) $8\sqrt{V}$ (c) $12V^{2/3}$ (d) $12V^{1/3}$

Q.23. The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. The volume of this smaller cone is 1/27 of the given cone. What is the height of the smaller cone?

(a) 13.5 cm (b) 11 cm
(c) 10 cm (d) 12 cm

Q.24. The maximum distance between two points of the unit cube is

(a) $\sqrt{2}+1$ (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) $\sqrt{2}+\sqrt{3}$

Q.25. A sealed cylindrical drum of radius r is 9% filled with paint. If the drum is tilted to rest on its side, the fraction of its curved surface area (not counting the flat sides) that will be under the paint is

(a) Less than $\frac{1}{12}$

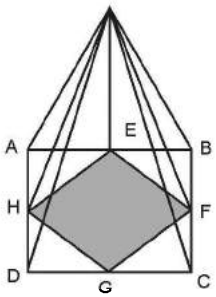
(b) Between $\frac{1}{12}$ and $\frac{1}{6}$

(c) Between $\frac{1}{6}$ and $\frac{1}{4}$

(d) Greater then $\frac{1}{4}$

Direction for questions 26 and 27: Read the passage below and solve the questions based on it.

In the figure given below, the base of the pyramid is a square ABCD of side 10 cm, another pyramid with base EFGH is formed where the points, E, F, G and H are obtained by joining the mid-points of the sides of the square ABCD. The height of the pyramid is 30 cm.



Q.26. Find the ratio of the volumes of pyramid with ABCD as base to the pyramid with EFGH as base.

(a) $\sqrt{2}:1$ (b) $2\sqrt{2}:1$
(c) $4:1$ (d) $2:1$

Q.27. Find the volume of the space left

(a) 1500 cm^3 (b) 500 cm^3
(c) 1000 cm^3 (d) 800 cm^3

Q.28. All five faces of a regular pyramid with a square base are found to be of the same area. The height of the pyramid is 3 cm. The total area of all its surfaces (in cm^2) is

(a) 8 (b) 10 (c) 12 (d) 16

Direction for questions 29 and 30: Read the passage below and solve the questions based on it.

A right-angled triangle of sides 5 m, 12 m and 13 m is made to spin about its hypotonuse.

Q.29. What kind of figure will be formed with this?

(a) Square (b) Double cone
(c) Sphere (d) None of these

Q.30. What is the volume of the figure thus formed?

(a) 270 m^3 (b) 290 m^3
(c) 320 m^3 (d) None of these

Q.31. A solid cube is cut into two halves by a plane passing through exactly two corners of the cube. What is the ratio of the total surface area of both the halves put together and the original total surface area of the cube?

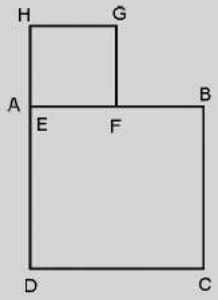
(a) $\sqrt{3}:\sqrt{3}+\sqrt{2}$ (b) $\sqrt{6}+1:\sqrt{6}$
(c) $6:6+\sqrt{6}$ (d) $3+\sqrt{2}:3$

Q.32. Sanjay wound a piece of string over a cylindrical block around its curved surface, and found that the string could be wound for exactly half a round more than an integral number of rounds. He also figured that if the string had been five times as long, the string would have had wound for exactly 10 cm more than an integral number of the rounds. Find the radius of the cylindrical block. (Assume that all windings are done in the same plane and neglect any increase in the radius due to the winding of the string).

(a) $2^{6}/_{22}$ cm (b) $3^{11}/_{22}$ cm
(c) $3^{4}/_{22}$ cm (d) $3^{9}/_{22}$ cm

Direction for questions 33 and 34: Go through the figure given below and solve the questions based on it.

Consider the figure given below:



A cube with surface EFGH is put on a cube with a surface ABCD in such a way that vertices E and A of the two blocks coincide and side EF coincides with side AB. EF = 1 unit and AB = 2 units. Now, keeping vertex F fixed, the smaller cube is rotated along the larger cube till G coincides with B once. Then keeping G fixed, the smaller block is again rotated till GH coincides with BC.

Q.33. How much is the aerial distance traversed by point E?

(a) $\sqrt{2}\pi$ units (b) π units
(c) $\sqrt{2}\pi + \frac{1}{2}\pi$ units (d) None of these

Q.34. In the above question, what is the total distance covered by point E?

(a) π units (b) $(1+\pi)$ units
(c) $(1+\sqrt{\pi})$ units (d) None of these

Q.35. How many cuboids of different dimensions can be made assembling 100 identical cubes?

(a) 9 (b) 8
(c) 12 (d) 10

Q.36. A cuboid is of the dimension $2\times2\times3$ units. Of the six surfaces of this cuboid, there are exactly x squares and

exactly y rectangles (rectangles which are not squares). Find the product of $x \times y$.
(a) 8 (b) 12
(c) 6 (d) None of these

Q.37. A box without a lid has a height of x cms, a square base of side y cm and a volume of 500 cm^3. For what value of x (in cms), the least quantity of the material will be needed to construct the box?
(a) $\sqrt{5}$ (b) $2\sqrt{5}$ (c) 5 (d) $5\sqrt{2}$

Q.38. Under the Indian Posts and Telegraph Act 1885, any package in the form of a right circular cylinder will not be accepted if the sum of its height and the diameter of its base exceeds 10 inches. The height (in inches) of a package of maximum volume that would be accepted is
(a) 10/3 (b) 20/3 (c) 10 (d)20

Q.39. A convex pentagon IJKLM is inscribed in a circle. If the angles subtended by the sides IJ, JK, KL, LM at the centre are 50°, 80°, 60° respectively, what is the measurement of the smallest interior angle of the pentagon?
(a) 120° (b) 95°
(c) 85° (d) None of these

Q.40. A right circular cone, with radius to height ratio as 12 : 5, is cut parallel to its base to get a smaller cone and a frustum. If the height of the smaller cone to the height of the frustum are in the ratio of 3 : 1, by what percentage is the combined total surface area of the smaller cone and frustum more with respect to the original cube?
(a) 22% (b) 32%
(c) 46% (d) None of these

ADVANCED

Q.1. A cuboid of length 20 m, breadth 15 m and height 12 m is lying on a table. The cuboid is cut into *two equal halves* by a plane which is perpendicular to the base and passes through a pair of diagonally opposite points of that surface. Then, a second cut is made by a plane which is parallel to the surface of the table again dividing the cuboid into *two equal halves*. Now this cuboid is divided into four pieces. Out of these four pieces, one piece is now removed from its place. What is the total surface area of the remaining portion of the cuboid?
(a) 1290 m^2 (b) 1380 m^2
(c) 1440 m^2 (d) Cannot be determined

Q.2. A metal is made up of a cylindrical base and a conical top with the base of radius 5 cm. The ratio of height of the cone and the cylinder is 2:3. A cylindrical hole is drilled through the metal solid with height two-third the height of the metal solid. What should be the radius of the hole, so that the volume of the hole is 1/3rd the volume of the metal solid after drilling?
(a) $\sqrt{45}$ cm (b) $\sqrt{\frac{55}{8}}$ cm
(c) $\sqrt{35}$ cm (d) $\sqrt{65}$ cm

Q.3. Sixteen cylindrical cans of Coke, each with a radius of 1 cm, are placed inside a wooden carton, four in a row. If the cans touch the adjacent cans and/or the walls of the box, then which of the following could be the internal area of the bottom of the carton (in cm^2)?
(a) 16 (b) 32
(c) 64 (d) None of these

Q.4. The areas of three adjacent faces of a cuboid are a, b and c. If the volume of the cuboid is N, then N^2 is equal to
(a) abc (b) $(ab + bc + ac)$
(c) $1/c$ (d) None of these

Q.5. A square tin sheet of side 12 inches is converted into a box with open top in the following steps: The sheet is placed horizontally. Then, equal-sized squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what value of x maximizes the volume of the box?
(a) 3 (b) 4 (c) 1 (d) 2

Q.6. A rectangular piece of cardboard 18 cm × 24 cm is made into an open box by cutting a square of 5 cm side from each corner and building up the side. What is the volume of the box (in cm^3)?
(a) 560 (b) 432
(c) 216 (d) None of these

Q.7. Consider the following function
$E = f(A) = \sqrt{\frac{A}{7}}$ where length of the edge E of a pyramid and the surface area A of the pyramid are the different terms used. How much longer is the edge

of the pyramid with a surface area 3087 square units than the edge of one with a surface area 2023 square units?

(a) 1064 (b) 152
(c) 4 (d) None of these

Q.8. The ratio of the volumes of the two cylinders are a:b, and their heights are in the ratio c:d. What is the ratio of their diameters?

(a) $\frac{ad}{bc}$ (b) $\frac{d^2}{c^2}$
(c) $\sqrt{\frac{ad}{bc}}$ (d) $\sqrt{\frac{a}{b}}\times\frac{c}{d}$

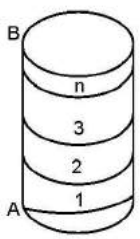
Q.9. From a solid right circular cone made of iron with base of radius 2 cm and height 5 cm, a hemisphere of diameter 2 cm and its centre, coinciding with the centre of the base of the cone, is cut out. The structure so obtained is then dropped in a right circular cylinder whose inner radius is 3 cm and inner height is 10 cm. Water is then poured into the cylinder to fill it up to its brim. What is the volume of the water required to fill it?

(a) 84π cm^3 (b) $\frac{250\pi}{3}$ cm^3
(c) $\frac{270\pi}{4}$ cm^3 (d) None of these

Direction for questions 10 to 12: Read the passage below and solve the questions based on it.

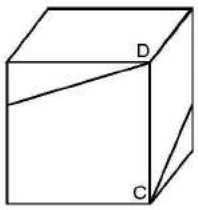
Consider a cylinder of height h cm and radius $r=\frac{2}{\pi}$ cm as shown in the figure (not drawn to scale). A string of a certain length, when wound on its cylindrical surface, starting at a point A and ending at point B, gives a maximum of n turns (in other words, the string length is the minimum length to wind n turns).

Q.10. What is the vertical spacing in cm between consecutive turns?



(a) $\frac{h}{n}$
(b) $\frac{h}{\sqrt{n}}$
(c) $\frac{h}{n^2}$
(d) Cannot be determined with the given information.

Q.11. The same string, when wound on the exterior four walls of a cube of side n cm, starting at point C and ending at point D, can give exactly one turn (see figure, not drawn to scale). The length of the string in cm, is



(a) $\sqrt{2}n$ (b) $\sqrt{17}n$ (c) n (d) $\sqrt{13}n$

Q.12. In the setup of the previous two questions, how is h related to n?

(a) $h=\sqrt{2}n$ (b) $h=\sqrt{17}n$
(c) $h=n$ (d) $h=\sqrt{13}n$

Q.13. An octagonal prism, of height 20 cm and a side of the base as $\sqrt{4\left(2-\sqrt{2}\right)}$ cm, is cut from its base to the top along the edges into two similar parts of equal volume. What is the total surface area of each part?

(a) $80+160\sqrt{2-\sqrt{2}}+4\sqrt{2}$ cm^2
(b) $80+40\sqrt{2-\sqrt{2}}$ cm^2
(c) $80+160\sqrt{2-\sqrt{2}}+8\sqrt{2}$ cm^2
(d) None of these

Direction for questions 14 to 16: Read the passage below and solve the questions based on it.

There are 300 coins, each coin having radius 2 cm and height 1 cm. The coins are so kept that each coin touches the other two. The base has 3 coins and the figure is built upon this base.

Q.14. Find the volume of the region enclosed by the coins.

(a) $600\left(2\sqrt{3}-\pi\right)$cm^3 (b) $400\left(2\sqrt{3}-\pi\right)$cm^3
(c) $100\left(2\sqrt{3}-\pi\right)$cm^3 (d) $200\left(2\sqrt{3}-\pi\right)$cm^3

Q.15. Find the number of spheres of the maximum volume that can be accommodated in the above region.

(a) 324 (b) 323
(c) 162 (d) 161

Q16. Find the volume of the prism circumscribing the whole structure of the coins.

(a) $100\sqrt{3}\left(7+4\sqrt{3}\right)\text{cm}^3$

(b) $2400\sqrt{3}\left(2+\sqrt{3}\right)\text{cm}^3$

(c) $800\sqrt{3}\left(2+\sqrt{3}\right)\text{cm}^3$

(b) None of these

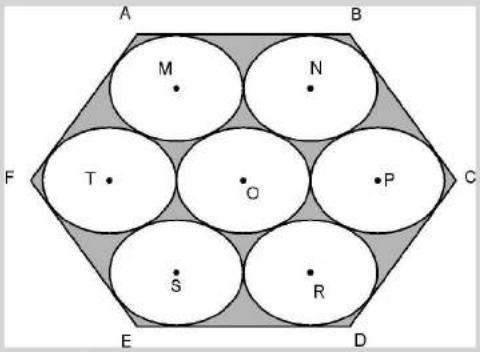
Q.17. A cylinder of height 'h' cm and radius 6/p cm is wound around with a string of width 'x' cm. The string covers the lateral surface area of the cylinder completely without keeping any space between two turns. What is the required length of the string.

(a) $\frac{12h}{x}\text{cm}$ (b) $\frac{12x}{h}\times\text{cm}$

(c) 12 h cm (d) $\frac{6h}{x}\text{cm}$

Direction for questions 18 to 20: Read the passage below and solve the questions based on it.

In the figure given below, a regular hexagon ABCDEF is inscribed by 7 circles. The radius of each circle is equal to 6 cm. The centres of the circles touching the sides are in, M, N, P, R, S, and T whereas the centre of the central circle is O.



Q.18. Find the ratio of the perimeter of the hexagon circumscribing the circle with centre O, the perimeter of hexagon formed by joining the points M, N, P, R, S and T and the perimeter of the hexagon ABCDEF.

(a) $6:12:(12+2\sqrt{3})$ (b) $2:2\sqrt{3}:(2\sqrt{3}+1)$

(c) $1:\sqrt{3}:(\sqrt{3}+1)$ (d) 1:2:3

Q.19. Find the area of the shaded region (in cm^2).

(a) 136.83 (b) 129.63

(c) 139.83 (d) 138.83

Q.20. If the above figure is a 3-D figure and seven spherical balls of radius 6 cm each are tightly arranged in a hexagonal box in a single layer, then what will be the volume of the box unoccupied by the balls in cm^3?

(a) $3456\sqrt{3}+5184-252\pi$

(b) $3456\sqrt{3}+5184-288\pi$

(c) $576\sqrt{3}+864-288\pi$

(d) None of these

Q.21. Four spheres each of radius 10 cm lie on a horizontal table so that the centres of the spheres form a square of side 20 cm. A fifth sphere also of radius 10 cm is placed on them so that it touches each of these spheres without disturbing them. How many cm above the table is the centre of the fifth sphere?

(a) $10\sqrt{6}$ (b) $10\left(1+\sqrt{2}\right)$

(c) $10\left(1+\sqrt{3}\right)$ (d) $10\left(4-\sqrt{2}\right)$

Q.22. A cube with an edge 'a' is cut from the angles by planes so that a regular octagon remains from each face, then the volume of the polygon so obtained will be

(a) $a^3\left(7\sqrt{2}-9\right)$ (b) $\frac{7}{3}a^3\left(\sqrt{2}-1\right)$

(c) $7a^3\left(\sqrt{2}-1\right)$ (d) $\frac{a^3}{3}\left(7\sqrt{2}-9\right)$

Q.23. A square of side x cm is cut from each corner of a rectangular sheet of metal of 10 cm by 14 cm. The resulting projections are folded up and the seams welded to construct an open box. What is the volume of the box thus obtained?

(a) $4x^3-48x^2+140x$ (b) $4x^3+48x^2+140x$

(c) x^3+24x^2+140x (d) None of these

Q.24. A sphere of radius 13 cm is cut by a plane whose distance from the centre of the sphere is 5 cm. What is the circumference of the plane circular section?

(a) 10π cm (b) 12π cm

(c) 24π cm (d) 26π cm

Q.25. The square of side 1 cm are cut from four corners of a sheet of tin (having length = l and breadth = b) in order to form an open box. If the whole sheet of tin was rolled along its length to form a cylinder, then the volume of the cylinder is equal to (343/4) cm^3. Find the volume of the box. (l and b are integers)

(a) 154 cm^3 (b) 100 cm^3

(c) 126 cm^3 (c) Insufficient data

Q.26. Two spheres of radii 6 cm and 1cm are inscribed in a right circular cone. The bigger sphere touches the smaller one and also the base of the cone. What is the height of the cone?

(a) 14 cm (b) 12 cm

(c) 85/6 cm (d) 72/5 cm

Q.27. If the total length of the diagonals of a cube is 12 cm, what is the total length of the edges of the cube?

(a) $6\sqrt{3}$ cm (b) 12 cm
(c) $12\sqrt{3}$ cm (d) 15 cm

Q.28. A cylindrical vessel open at the top contains water up to 1/3 of its height. A heavy sphere whose diameter is equal to the height of the cylinder is placed into the vessel touching its curved surface from all sides, then the water.
(a) Level will rise to half of its height
(b) Level will rise to 3/4th its height
(c) Level will rise to its height
(d) Water will overflow

Q.29. A sphere of maximum possible volume is to be completely immersed into a cylindrical container of radius 'a' containing water up to a height '2a'. What is the minimum height of the jar so that no water spills out of it?
(a) $\frac{10a}{3}$ (b) $\frac{11a}{3}$ (c) $\frac{12a}{3}$ (d) $\frac{13a}{3}$

Q.30. In a rectangular parallelopiped A, a, b and c are the lengths of the diagonals on three faces having a corner in common. Let d be the length of the largest diagonal that can be drawn in A. Then
(a) $2d^2 = a^2 + b^2 + c^2$
(b) $d^2 = a^2 + b^2 + c^2$
(c) $d^2 = 2(a^2 + b^2 + c^2)$
(d) None of these

Direction for questions 31 to 33: Read the passage below and solve the questions based on it.

A right-angled triangle with base and height measuring 15 cm and 20 cm is rotated along its hypotonuse. The structure so formed is placed by two spheres such that they are just touching the sides of the cone.

Q.31. Find the volume of the structure formed before the double cone is placed (in cm^3)?
(a) 1200π (b) 3600π
(c) 1000π (d) None of these

Q.32. Find the sum of the volumes of both the sphere (in cm^3).
(a) $\frac{364}{3}\pi$ (b) $\frac{140}{3}\pi$
(c) 12π (d) 47π

Q.33. Find the curved surface area of the figure (in sq. units).
(a) 840π (b) 400π
(c) 420π (d) None of these

Q.34. A square hole of cross-sectional area 4 cm^2 is drilled across a cube with its length parallel to a side of the cube. If an edge of the cube measures 5 cm, what is the total surface area of the body so formed?
(a) 158 cm^2 (b) 190 cm^2
(c) 166 cm^2 (d) 182 cm^2

Q.35. The radius and the height of a right solid circular cone are r and h respectively. A conical cavity of radius r/2 and height h/2 is cut out of the cone. What is the whole surface area of the rest of the portion?
(a) $\frac{\pi r}{4}\left(5\sqrt{\left(r^2+h^2\right)}\right)+3r$
(b) $\frac{5\pi r}{4}\sqrt{r^2+h^2}$
(c) $\frac{3\pi r}{4}\left(\sqrt{r^2+h^2}+r\right)$
(d) $\frac{3\pi r}{7}\left(\sqrt{r^3+h^2}+r\right)$

Q.36. John Nash, an avid mathematician, had his room constructed such that the floor of the room was an equilateral triangle in shape instead of the usual rectangular shape. One day he brought home a bird and tied it to one end of a string and then tied the other end of the string to one of the corners of his room. The next day, he untied the other end of the string from the corner of the room and tied it to a point exactly at the center of the floor of the room. Assuming that the dimensions of the room are relatively large compared to the length of the string, find the number of times, by which the maximum possible space in which the bird can fly, increase.
(a) 4 (b) 5
(c) 6 (d) 7

Direction for questions 37 to 40: Read the passage below and solve the questions based on it.

Homes of three of my friends–Bango, Mango and Tatto–are located at the three vertices of the municipal park near my house. This park is triangular in shape. Given below are the directions to reach each of their homes from my home:

- Bango's house–100 m to the east and 150 m to the north.
- Mango's house–200 m to the east and 50 m to the north.
- Tatto's house–50 m to the west and 25 m to the north.
- Neglect dimensions of all the house while calculating the distances.

Q.37. What is the area of the municipal park?
(a) 27,500 m^2 (b) 11,500 m^2
(c) 13,750 m^2 (d) Cannot be determined

Q.38. There is a highway (a straight road) passing through the houses of Bango and Mango. Tatto starts from his house and takes the shortest possible route to reach the highway. What is the distance travelled by Tatto?

(a) $\frac{125}{\sqrt{3}}$ (b) $\frac{275}{\sqrt{2}}$

(c) $\frac{150}{\sqrt{2}}$ (d) None of these

Q.39. Travelling along one of the following routes will take me from my house to the point (on the highway) where Tatto reached in the previous question. Which one is it?

(a) 100 m to the east, 200 m to the north
(b) 87.5 m to the east and 62.5 m to the north
(c) 67.5 m to the east and 167.5 m to the north
(d) None of these

Q.40. What is the approximate distance between Mango's house and the point Tatto reached in Q. 39?

(a) $\frac{225}{\sqrt{2}}$ m (b) $\frac{275}{\sqrt{2}}$ m

(c) $\frac{150}{\sqrt{2}}$ m (d) $\frac{325}{\sqrt{2}}$ m

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(c)	(a)	(a)	(d)	(b)	(b)	(c)	(d)	(b)	(b)	(d)	(c)	(c)	(b)	(a)	(c)	(c)	(a)
Q. No.	21	22																		
Answer	(c)	(c)																		

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(c)	(d)	(a)	(c)	(c)	(b)	(c)	(c)	(b)	(d)	(c)	(a)	(c)	(b)	(d)	(c)	(d)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(d)	(b)	(d)	(b)	(d)	(a)	(a)	(c)	(c)	(b)	(b)	(b)	(d)	(c)	(b)	(b)	(a)	(b)	(b)
Q. No.	41	42	43	44	45	46	47	48	49	50										
Answer	(a)	(c)	(c)	(d)	(a)	(d)	(c)	(c)	(b)	(a)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(d)	(a)	(d)	(c)	(b)	(b)	(c)	(c)	(b)	(d)	(c)	(a)	(b)	(b)	(c)	(c)	(d)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(c)	(d)	(c)	(c)	(a)	(d)	(a)	(c)	(b)	(d)	(d)	(c)	(c)	(b)	(b)	(a)	(c)	(a)	(b)	(d)

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(c)	(a)	(d)	(a)	(c)	(c)	(a)	(a)	(b)	(c)	(d)	(d)	(d)	(c)	(a)	(c)	(d)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Answer	(b)	(b)	(a)	(c)	(b)	(d)	(c)	(c)	(a)	(a)	(a)	(b)	(c)	(d)	(a)	(b)	(c)	(b)	(b)	(a)

HINTS AND SOLUTIONS

Moderate

1. Take the total surface area of the initial cone into consideration and then proceed.

6. The vertices of the smaller square will be on the mid-points of the sides of the larger square.

9. The amount saved would be maximum only when the difference of maximum area and minimum area to be painted is maximum.

11. Let ABCDEFGH be the cube of side a and O be the centre of the hemisphere.
AC = $\sqrt{2}$ a
OD = OC = R
Let P be the mid-point of AC
OP = a
Now in ΔAOC

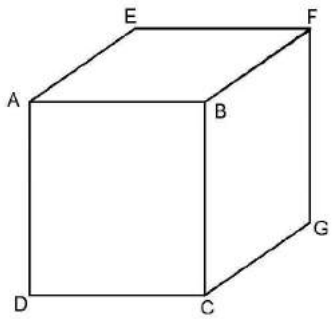
$$R^2 = a^2 + \frac{a^2}{2} \qquad \therefore a = \sqrt{\frac{2}{3}}R$$

13. Take a value of n, say 3 and then check options.

14. Go through options.

20. Get a relation between the radii of the two spheres.

24. The distance from any vertex at the base of the cube to the vertex that is perpendicular along height to the diametrically opposite vertex is required.
We have to calculate DF.



28. Equate the area of the square ABCD and triangle PDC and find a relation between the slant height and the length of the base of the pyramid

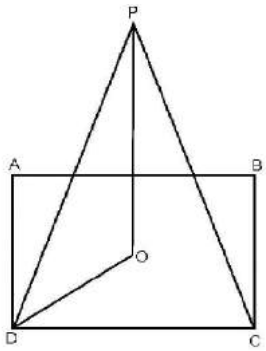
$$a^2 = \frac{a}{4}\sqrt{4x^2 - x^2}$$

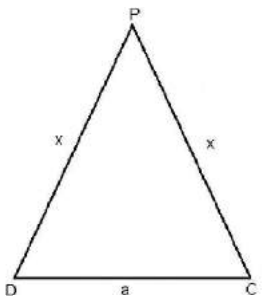
$16a^2 = 4x^2 - a^2$

$\Rightarrow 17a^2 = 4x^2$

$$\frac{a}{x} = \frac{2}{\sqrt{17}}$$

Let $a = 2\pi$; $x = \sqrt{17}$ P





Now in Δ POD

OD = $\sqrt{2}P$ PD = $\sqrt{17}P$

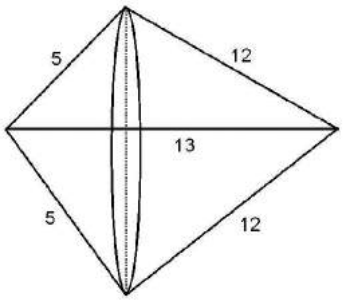
AO = $\sqrt{15}P$ But AO = 3 $\Rightarrow$ P = $\frac{3}{\sqrt{15}}$

$2P = a = \frac{6}{\sqrt{15}}$

Area of the base = $\frac{36}{15}$

Total surface area = $\frac{36}{15} \times 5 = 12$

29.



A structure will be formed like the figure given.

32. In the first operation, the string that is extra after winding an integral number of times will be equal to 10 cm.

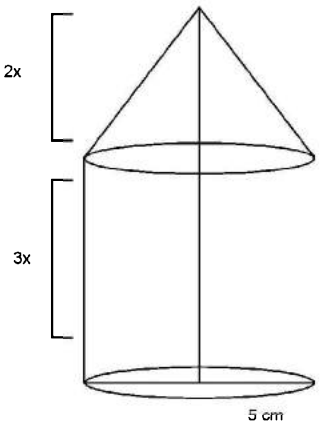
35. Use factors of 100.

38. Volume is maximum when radius is equal to height.

Advanced

1. Since we don't know that the cut is made parallel to which face, we cannot determine the surface area.

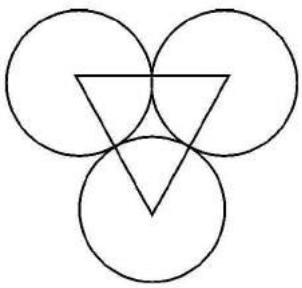
2. Suppose the height of the cone is 6 cm and that of cylinder is 9 cm. Thus, the height of the hole is10 cm now a straightforward calculation leads to the answer.



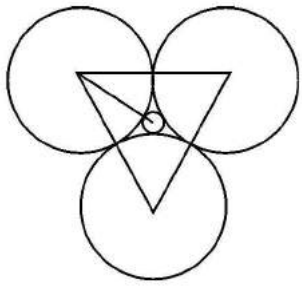
10. If the cylinder is cut along its length, it becomes a rectangle.

12. Equate the lengths of strings obtained from the above two questions.

14. Volume of hollow space = area of the triangle × height (100 cm) – ½ volume of the cylinder of base radius 2 cm and height 100 cm



15.

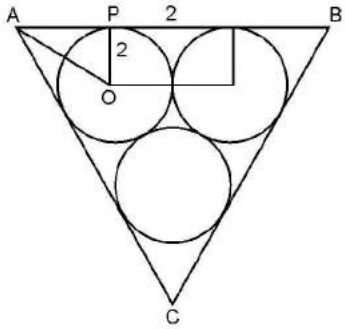


Volume of the hollow space = Area of the triangle – ½ Volume of the cylinder of base radius 2 cm and height 100 cm.

16. r = Radius of sphere

$$\frac{V_3}{2} \times 2\frac{2}{3} - 2\left(\frac{4}{\sqrt{3}-2}\right) \text{cm}$$

16.



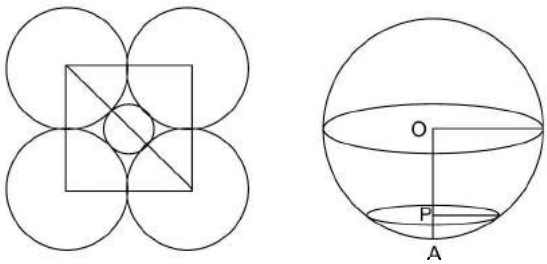
∠AOP = 30°

OP = 2 cm

Calculate AP

20. The 3D shape of the hexagon will be a hexagonal prism.

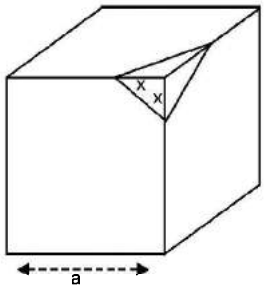
21. OA = 10



OA = 10

To find the value of PA, go through the options now.

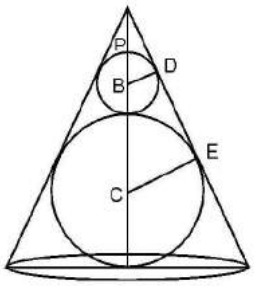
22. In this question we need to cut eight identical pyramids having a base of equilateral triangle of side equal to the side of the newly-formed octagon.



$A = 2x + \sqrt{2}x$

25. The length of the rectangle will be equal to the circumference of the base of the cylinder.

26. Let AP = x
$\Delta ABD \sim \Delta ACE$
Now x can be calculated.



36. Consider the length of the string less than or equal to the inradius of the floor. At the corner of the floor you will find a sixth part of a hemisphere and at the centre it will be a hemisphere.

Questions 37 to 40

Plot these points and direction on a graphical plane for a clear interpretation.

19

CHAPTER

Co-Ordinate Geometry

LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Types of sets and methods of their formation
- Definitions and properties
- Maxima and minima in set theory
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

The method of denoting points with the help of coordinates was proposed by French Mathematician Descartes. Further, he also proposed that lines and curves are nothing but the collection of points, and hence can be represented by equations derived out of Coordinate Geometry. To honour his work, the coordinates of a point are often referred to as its Cartesian coordinates.

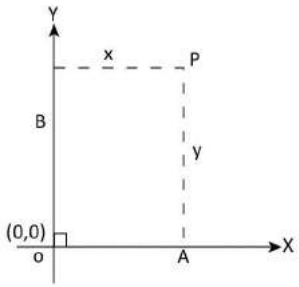
CONSIDER THE FOLLOWING CASE

Suppose there is an ant at point O in the figure and it wants to go to point P.

One way to reach P is that the ant travels along OX, reaches A and then travels along AP and reaches P.

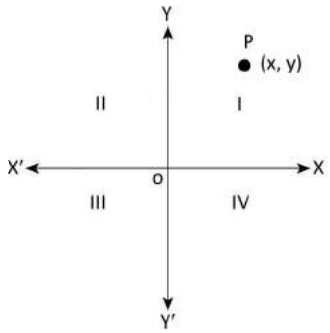
Hence, it first covers a distance 'X' horizontally and then covers a distance 'Y' vertically.

1st of all, we will assume a reference point, which is at a distance of 0 unit from both the X-axis and Y-axis and will call this Origin.



If we assume 'O' as the origin, the distances X and Y are the distances of this point P from Y–axis and X-axis respectively. These are known as coordinates of point P and it is written as $P(X, Y)$.

CO-ORIDNATE AXES AND REPRESENTATION OF A POINT



The figure given alongwith is called the X-Y Cartesian plane. The line XOX' is called the X-axis and YOY' the Y-axis.

If $P(x, y)$ is a point in this plane, then x is the X- coordinate of P or abscissa of P and y is called the Y-coordinate of P or the ordinate of P.

Remember that X-coordinate of the point is the distance of the point from Y-axis and Y-coordinate of the point is the distance of the point from X-axis.

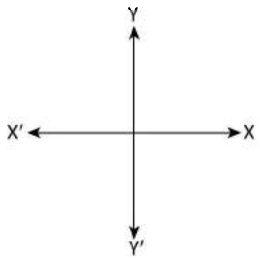
The X-Y Cartesian plane is divided into four equal parts called Quadrants (I, II, III, IV).

SIGN CONVENTION

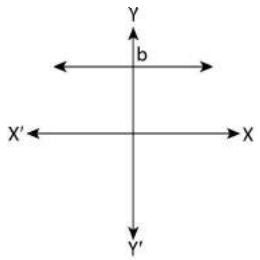
	1st Quadrant	2nd Quadrant	3rd Quadrant	4th Quadrant
X-axis	+ve	–ve	–ve	+ve
Y-axis	+ve	+ve	–ve	–ve

EQUATION AND GRAPH OF CO-ORDINATE AXES

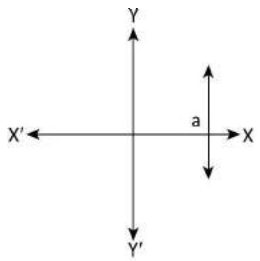
1. Equation of X and Y axes are $Y = 0$ and $x = 0$ respectively.



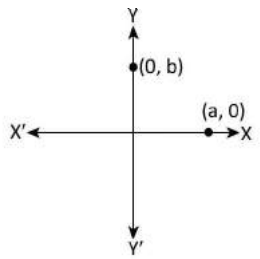
2. Equation of a line parallel to X axis is $Y = b$ (b is constant).



3. Equation of a line parallel to Y axis is X = a (a is constant).



4. Any point on the X axis can be taken as $(a, 0)$ and any point on the Y axis can be taken as $(0, b)$.



5. To find out X and Y intercepts of a line, we will put Y = 0 and X = 0 respectively in the equation of the line.

Some standard formulae

1. Distance between two points:
   If there are two points $A\ (x_1,\ y_1)$ and $B\ (x_2, y_2)$ on the XY plane, then the distance between them is given by

$$AB = d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

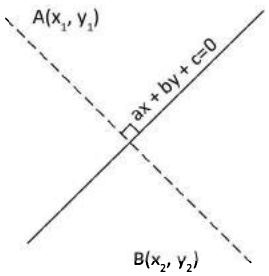
2. Division of a line segment:
   i. Internal - The Coordinates of a point P which divides the line joining $A\ (x_1\ y_1)$ and $B\ (x_2, y_2)$ internally in a ratio $l : m$ are given by

$$x = \frac{lx_2 + mx_1}{l+m}, y = \frac{ly_2 + my_1}{l+m}$$

   ii. External:
   The coordinates of a point P which divides the line joining the point $A(x_1, y_1)$ and $B(x_2, y_2)$ externally in the ratio $l : m$ are given by

$$x = \frac{lx_2 - mx_1}{l-m}, y = \frac{ly_2 - my_1}{l-m}$$

THE IMAGE OF A POINT ALONG THE MIRROR PLACE ON A STRAIGHT LINE



The image of $A\ (x_1, y_1)$ with respect to the line mirror $ax + by + c = 0$ be $B(x_2, y_2)$ is given by

$$\frac{x_2 - x_1}{a} = \frac{y_2 - y_1}{b}$$

$$= \frac{-2(ax_1 + by_1 + c)}{(a^2 + b^2)}$$

FOOT OF THE PERPENDICULAR

If the foot of the perpendicular from (x_1, y_1) to the line $lx + my + n = 0$ is (h, k) then

$$\frac{h - x_1}{l} = \frac{k - y_1}{m} = \frac{-(lx_1 + my_1 + n)}{l^2 + m^2}$$

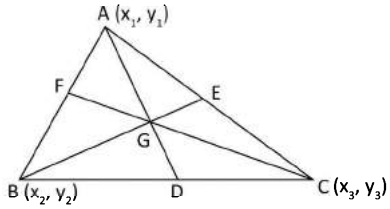
CENTROID OF A TRIANGLE

The point at which the medians of a triangle intersect is called the centroid of the triangle.

Let ABC be a given triangle with vertices $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$.

Since D is the mid-point of BC, its coordinates are $[(x_2 + x_3)/2, (x_2 + x_3)/2]$

Let G (x, y) be a point dividing AD in the ratio 2:1.



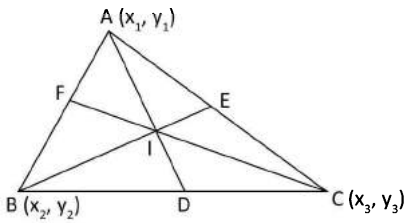
Then, $x = \dfrac{2\left(\dfrac{x_2 + x_3}{2}\right) + 1.x_1}{(2+1)} = \left(\dfrac{x_1 + x_2 + x_3}{3}\right)$

and $y = \dfrac{2\left(\dfrac{y_2 + y_3}{2}\right) + 1.y_1}{(2+1)} = \left(\dfrac{y_1 + y_2 + y_3}{3}\right)$

Similarly, the coordinates of a point which divides BE in the ratio 2:1 as well as those of the point which divides CF in the ratio 2:1 are

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

INCENTRE OF A TRIANGLE

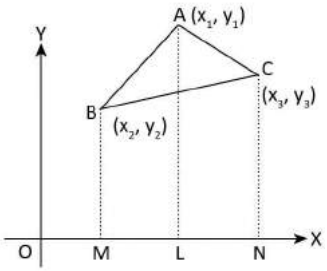


The point at which the bisectors of the angles of a triangle intersect, is called the incentre of the triangle. From geometry, we know that the bisector of an angle of a triangle divides the opposite side in the ratio of length of remaining sides. Hence, the bisectors of the angle of ΔABC are concurrent, and meet at a point, called incentre.

$$\left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$$

Area of triangle

Let ABC be a given triangle whose vertices are $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$.



Then the area of the triangle is given by

$$= \frac{1}{2}\left[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right].$$

If we interchange the order of any two vertices of the ΔABC, we obtain a negative value of the area. However, the area shall always be taken to be positive.

EQUATION OF A CURVE

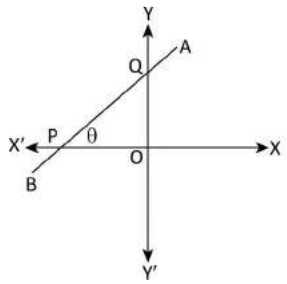
An equation in two variables X and Y with the degree of the equation being equal to or more than two is called the equation of a curve. If the graph of that equation plotted on the XY Cartesian plane, it will give a shape of a curve, and not a straight line

For example, $x^2 + y^2 = 16$, $y = x^2$.

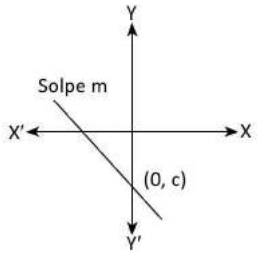
STRAIGHT LINE

Any equation with the degree of equation being one is known as an equation of straight line. General equation of straight line is given by $aX + bY + c = 0$, where X and Y are variables and a, b, c are constants.

Any point lying on this line will satisfy the equation of the line.



If AB is a straight line on the XY plane, then the angle θ which the line makes with the X axis in the anticlockwise direction is called the inclination of the line and tangent of this angle θ (tan θ) is called the slope of the line AB. It is denoted by 'm'. The lengths OP and OQ are respectively known as the intercepts on X-axis and Y-axis, made by the line.



DIFFERENT FORMS OF REPRESENTING A STRAIGHT LINE

1. Slope- intercept form

$$y = mx + c$$

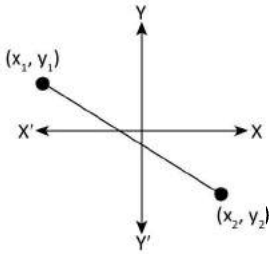
If 'm' is the slope of the line and 'c' the intercept made by the line on Y axis, the equation is

$$y = mx + c$$

2. Point-slope form

If 'm' is the slope of the line and it passes through the point, the equation is (x_1, y_1), then the equation of the line is given by:

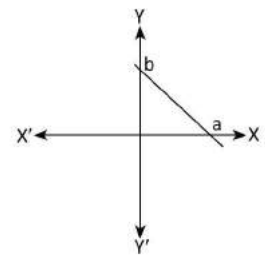
$$y - y_1 = m\,(x - x_1)$$



3. Two-point form

If the line passes through two points (X_1, Y_1) and (X_2, Y_2) the equation is

$$(Y - Y_1) = \frac{Y_2 - Y_1}{X_2 - X_1}(X - X_1)$$



Using Point-slope form and Two-point form, we can find out the formula for slope also. Comparing the two equations, we get m = $\frac{Y_2 - Y_1}{X_2 - X_1}$

4. Slope-intercept form
If the line makes an intercept of 'a' units on X-axis and b units on Y-axis, then the equation is:

$$\frac{X}{a}+\frac{Y}{b}=1$$

Angle between two intersecting lines

The angle between two lines whose slopes are m_1 and m_2 is given by a formula such that $\tan\theta = \frac{m_1 - m_2}{1+m_1m_2}$ (where θ is the angle between the lines)

Condition for two straight lines to be parallel

It can be visualized that two straight lines can be parallel only if they make an equal inclination with the X-axis. This will, in turn, ensure that their slopes are equal.

The lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are parallel if and only if $m_1 = m_2$.

Condition for two straight lines to be perpendicular

The lines $y = m_1x + c_1$ and $y = m_2x + c_2$ are perpendicular if and only if $m_1 m_2 = -1$.

The equation of a line parallel to a given line $ax + by + c = 0$ will be $ax + by + k = 0$, where k is any constant which can be found by additional information given in the question.

The equation of a line perpendicular to a given line $ax + by + c = 0$ will be $bx - ay + k = 0$, where k is any constant which can be found by additional information given in the question.

Point of intersection of two lines

The coordinates of the point of intersection of the two intersecting lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are

$$\left(\frac{b_1c_2 - b_2c_1}{a_1b_2 - a_2b_1}, \frac{a_2c_1 - a_1c_2}{a_1b_2 - a_2b_1}\right)$$

However, to obtain the point of intersection, we are required to just solve the equations of the straight lines given as we do in the case of Simultaneous Equations.

Condition of concurrency of three lines

Three lines are said to be concurrent, if they pass through a common point i.e. if they meet at a point. The condition for three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ is $a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$

Length of perpendicular

The length of perpendicular (p) from (X_1, Y_1) on the line $AX + BY + C = 0$ is:

$$P = \left|\frac{AX_1 + BY_1 + C}{\sqrt{A^2 + B^2}}\right|$$

Distance between two parallel lines

The distance between two parallel lines $AX + BY + C_1 = 0$ and $AX + BY + C_2 = 0$ is given by

$$\left|\frac{C_1 - C_2}{\sqrt{A^2 + B^2}}\right|$$

Conditions for points to be collinear

If three points A, B, C are co-linear, then any one of the following conditions should be true :

i. Area of triangle $ABC = 0$
ii. Slope of AB = slope of BC = slope of AC
iii. $AB + BC = AC$

Depending upon the points given, we can use any one of three to check if the points are collinear. It should also be mentioned that if one of these conditions is true, then other two will definitely be true.

WORKED OUT PROBLEMS

Example 1 Find the coordinates of the point which divides the line segment joining the point (5, –2) and (9,6) in the ratio 3:1.

Solution Let the required point be (x, y). Then,

$$x = \left(\frac{3\times 9 + 1\times 5}{3+1}\right) = 8$$

and

$$y = \left(\frac{3\times 6 + 1\times(-2)}{3+1}\right) = 4$$

The coordinates of the required point are (8, 4).

Example 2 Find the ratio in which the point $(2, y)$ divides the join of (–4, 3) and (6, 3) and hence find the value of y.

Solution Let the required ratio be k : 1.

Then, $2 = \frac{6k - 4\times 1}{k+1} \Rightarrow k = \frac{3}{2}$

$\therefore$ The required ratio is $\frac{3}{2}$:1, i.e., 3:2.

Also, $y = \frac{3\times 3 + 2\times 3}{3+2} = 3$

When asked for ratio m : n, for convenience we take ratio as m/n : 1 or k : 1.

Example 3 Two vertices of a triangle are (–1, 4) and (5, 2) and its centroid is (0, –3). Find the third vertex.

Solution Let the third vertex be (x, y). Then

$$\frac{x+(-1)+5}{3}=0 \text{ and } \frac{y+4+2}{3}=-3$$

$\therefore$ $x = -4$ and $y = -15$

Hence the third vertex of the triangle is (–4, –15).

Example 4 Find the co-ordinates of incentre of the triangle whose vertices are (4, –2), (–2, 4) and (5, 5).

Solution $a = BC = \sqrt{(5+2)^2+(5-4)^2} = \sqrt{50} = 5\sqrt{2}$

$$b = AC = \sqrt{(5-4)^2+(5+2)^2} = \sqrt{50} = 5\sqrt{2}$$

and $$c = AB = \sqrt{(-2-4)^2+(4+2)^2} = \sqrt{72} = \sqrt{2}6$$

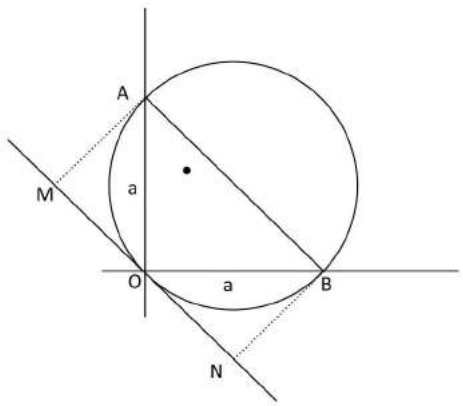
Let (x, y) be the coordinates of incentre of ΔABC, Then,

$$x = \frac{ax_1+bx_2+cx_3}{a+b+c} = \left(\frac{20\sqrt{2}-10\sqrt{2}+30\sqrt{2}}{5\sqrt{2}+5\sqrt{2}+6\sqrt{2}}\right) = \frac{40\sqrt{2}}{16\sqrt{2}} = \frac{5}{2}$$

$\therefore$ The coordinates of the incentre are $\left(\frac{5}{2}, \frac{5}{2}\right)$

Example 5 A line makes equal intercepts of length 'a' on the coordinate axes, intersecting the X axis and Y-axis at A and B respectively. A circle is circumscribed about the triangle OAB, where O is the origin of the coordinate system. A tangent is drawn to this circle at the point O, the sum of the perpendicular distances of the vertices A, B and O from this tangent is:

Solution



$$AM + BN + OO = \frac{a}{\sqrt{2}}+\frac{a}{\sqrt{2}}+0 = \sqrt{2}a$$

Example 6 What is the area of the triangle whose vertices are (4, 4), (3, –16) and (3, –2)?

Solution Let $x_1 = 4, x_2 = 3, x_3 = 3$ and $y_1 = 4, y_2 = -16, y_3 = -2$.

Then area of the EXAMPLE ven triangle

$$= \frac{1}{2}\left[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right]$$

$$= \frac{1}{2}\left[4.\{-16-(-2)\}+3.(-2-4)+3.\{4-(-16)\}\right]$$

$$= \frac{1}{2}\left[4.\{-14+3(-6)\}+3\times 20\right] = \frac{1}{2}[-56-18+60] = -7$$

Since area of a triangle cannot be negative, so area = 7 sq. units.

Example 7 What is the area of the triangle formed by the points (–5, 7), (–4, 5) and (1, –5)?

Solution Let $x_1 = -5, x_2 = -4, x_3 = 1$ and $y_1 = 7, y_2 = 5, y_3 = -5$.

Area of the triangle formed by given points

$$= \frac{1}{2}\left[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)\right]$$

$$= \frac{1}{2}[(-5)\{5-(-5)\}+(-4)(-5-7)+(7-5)]$$

$$= 0$$

Hence the EXAMPLE ven points are not forming any triangle, rather they are collinear.

Example 8 Find the equation of the straight line which passes through (3,4) and the sum of whose X and Y intercepts is 14.

Solution Let the intercepts made by the line x-axis and y-axis and y-axis be α and $(14-\alpha)$ respectively.

Then, its equation is $\frac{X}{\alpha}+\frac{Y}{14-\alpha}=1$ ... (i)

Since it passes through (3,4), we have:

$$\frac{3}{\alpha}+\frac{4}{14-\alpha}=1 \Rightarrow \alpha^2-13\alpha+42=0 \Rightarrow (\alpha-6)(\alpha-7)=0.$$

$\alpha = 6$ and $\alpha = 7$.

So, the required equation is: $\frac{x}{6}+\frac{y}{8}=1$ or $\frac{x}{7}+\frac{y}{7}=1$ i.e. $4x + 3y = 24$ or $x + y = 7$.

Example 9 What is the equation of a line which passes through the point (–1, 3) and is perpendicular to the straight-line $5x + 3y + 1 = 0$?

Solution The equation of any line perpendicular to the line $5x + 3y + 1 = 0$ is

$3x - 5y + K = 0$ ... (i)

Since the required line passes through the point (–1, 3), we have

$3 \times (-1) - 5 \times 3 + K = 0$, or, $K = 18$

Hence, the required equation is $3x - 4y + 18 = 0$.

Example 10 What is the point of intersection of the lines $2x + 3y = 5$ and $3x - 4y = 10$?

Solution To find out the point of intersection, we just need to solve the simultaneous equations.

$2x + 3y = 5$ ……(1)

$3x - 4y = 10$ ……(2)

Multiplying equation (1) by 3 and equation (2) by 2.

$2x + 3y = 5$ becomes $6x + 9y = 15$ (equation 3) And $3x - 4y = 10$ becomes $6x - 8y = 20$ (equation 4).

Doing equation (3) – equation (4) gives us:

$$\begin{array}{r} 6x+9y=15 \\ 6x-8y=20 \\ \hline -\quad + \\ \hline 17y=35 \end{array}$$

Hence $y = 35 / 17$. This gives us $x = -10/17$

So point of intersection = –10/17, 35/17.

PRACTICE EXERCISES

FOUNDATION

Q.1. The coordinates of the diagonals of a square are (2, 0) and (0, 5). What is the area of the square?

(a) 29sq.units (b) $\frac{\sqrt{29}}{\sqrt{2}}$sq.units

(c) $\frac{29}{4}$sq.units (d) 14.5sq.units

Q.2. Let P and Q be points on the line joining A (–2, 5) and B (3, 1) such that AP = PQ = QB. Then the mid-point of PQ is

(a) (1/2, 3) (b) (–1/2, 4)

(c) (2, 3) (d) (–1, 4)

Q.3. If each of the point $(x_1, 4)$,$(-2, y_1)$ lies on the line joining the points (2, –1),(5, –3), then the point $p(x_1, y_1)$ lies on the line

(a) $6(x + y) - 25 = 0$ (b) $2x + 6y + 1 = 0$

(c) $2x + 3y - 6 = 0$ (d) $6(x + y) + 25 = 0$

Q4. Find the area enclosed by the x-axis, y-axis and the graph $y = |x| - 4$ in the first quadrant (in sq. units).

(a) 8 (b) 16

(c) 14 (d) None of these

Q.5. The points (–a, –b),(0, 0),(a, b) and (a^2, ab) are

(a) collinear

(b) vertices of a parallelogram

(c) vertices of a rectangle

(d) none of these

Q.6. The points (0, 8/3), (1, 3) and (82, 30) are vertices of

(a) An obtuse angled triangle

(b) an acute angled triangle

(c) a right angled triangle

(d) none of these

Q.7. The points (0, –1), (–2, 3), (6, 7) and (8, 3) are

(a) collinear

(b) vertices of a parallelogram which is not a rectangle

(c) vertices of a rectangle, which is not a square

(d) none of these.

Q.8. Find the area enclosed by the graph $y = |X + 3|$ with the coordinate axes in square units.

(a) 9 (b) 4.5

(c) 0 (d) 12

Q.9. The coordinates of the middle points of the sides of a triangle are (4, 2) (3, 3) and (2, 2) then the coordinates of its centroid are

(a) (3, 7/3) (b) (3, 3)

(c) (4, 3) (d) none of these.

Q.10. The incentre of the triangle whose vertices are (–36, 7), (20, 7) and (0, –8) is

(a) (0, –1) (b) (–1, 0)

(c) (1/2, 1) (d) none of these.

Q.11. The triangle with vertices at (2, 4) (2, 6) and $(2+\sqrt{3}$, 5) is

(a) right angled (b) right angled and isosceles

(c) equilateral (d) obtuse angled

Q.12. The area of the triangle with vertices at the point $(a, b + c)$, $(b, c + a)$, $(c, a + b)$ is

(a) 0 (b) $a + b + c$

(c) $ab + bc + ca$ (d) none of these.

Q.13. The nearest point on the line $3x - 4y = 25$ from the origin is

(a) (–4, 5) (b) (3, –4)

(c) (3, 4) (d) (3, 5)

Q.14. If three vertices of a rhombus taken in order are (2, –1), (3, 4) and (–2, 3), then the fourth vertex is
(a) (–3, –2) (b) (3, 2)
(c) (2, 3) (d) (1, 2)

Q.15. If a,b,c be in AP, then $ax + by + c = 0$ represents
(a) a single line
(b) a family of concurrent lines
(c) a family of parallel lines
(d) none of these

Q.16. S_1 and S_2 are two light sources. Both the sources are originated from a single point (0, 1) and travel along a linear path. S_1 passes through (1, –2) and S_2 passes through (–1, 2). The path of S_1 & S_2 are:
(a) $x + 3y = 1, x + y = 1$
(b) $3x + y = 1, x - y = 1$
(c) $3x + y = 1, x + y = 1$
(d) $x - 3y = 1, x + 3y = 1$

Q17. A ray of light passing through the point (1, 2) is reflected on the x-axis at a point P, and then passes through the point (5, 3). Then the abscissa of the point P is
(a) $2\frac{1}{5}$; (b) $2\frac{2}{5}$;
(c) $2\frac{4}{5}$; (d) $2\frac{3}{5}$;

Q.18. Find the area of the region enclosed by the graphs of $y + \frac{1}{2}(|x-1|+(x-1)$ and $x = 4$ above the x-axis.
(a) 9 (b) 6
(c) 4.5 (d) 4.5

MODERATE

Q.1. There is a straight line S perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line S and the coordinate axes is 5 square units. Which of the following is the equation of straight line S?
(a) $x + 5y = 5\sqrt{2}$ (b) $x + 5y + 5\sqrt{2} = 0$
(c) Both 1 and 2 (d) None of these

Q.2. Find the distance of a point (3, 5) from the line $x + y = 2$, measured along a line making an angle 45° with the positive x-axis. (The anti clockwise direction is positive angle)
(a) $\sqrt{2}$ unit (b) $2\sqrt{2}$ unit
(c) $3\sqrt{2}$ units (d) $4\sqrt{2}$ units

Q.3. On an axis expressed in kilometers, the co-ordinates of 2 points are A (17, 11) and B (22, 23). In how much time does a man travelling at 91 kmph cover the shortest possible distance between the two points A and B?
(a) $6\frac{2}{7}$ min (b) $8\frac{3}{7}$ min
(c) $8\frac{4}{7}$ min (d) $8\frac{6}{7}$ min

Q.4. Find the distance between the parallel lines $3x + 4y - 7 = 0$ and $6x + 8y + 3 = 0$.
(a) 4 units (b) 11/10 units
(c) 17/10 units (d) 13/10 units

Q.5. The two vertices of the base of an isosceles triangle are (2p, 0) and (0, p). We also know that the equation of one of the sides of this triangle is $X = 2p$. What is the area of this isosceles triangle?
(a) 3.5 p^2 (b) 6.7 p^2
(c) 2.5 p^2 (d) 2 p^2

Q.6. Given four lines whose equations are $x + 2y - 3 = 0$, $2x + 3y - 4 = 0$, $3x + 4y - 7 = 0$ and $4x + 5y - 6 = 0$, then the lines are
(a) concurrent
(b) sides of a square
(c) sides of a rhombus
(d) none of these.

Q.7. Points on the line $x + y = 4$ that lie at a unit distance from the line $4x + 3y - 10 = 0$ are
(a) (3, 1) and (–7, 11) (b) (–3, 7) and (2, 2)
(c) (–3, 7) and (–7, 11) (d) none of these.

Q.8. The number of points on the lines $4x - 3y + 7 = 0$ or $x - y + 3 = 0$ which are at a distance 5 units from the point (2, 5) are
(a) 1 (b) 2
(c) 3 (d) 4

Q.9. If one vertex of an equilateral triangle is at (2, –1) and the base is $x + y - 2 = 0$, then the length of each side is
(a) $\sqrt{3/2}$ (b) $\sqrt{2/3}$
(c) 2/3 (d) 3/2

Q.10. If the points (1, 3) and (5, 1) are two opposite vertices of a rectangle and the other two vertices lie on the line $y = 2x + c$, then the value of c is
(a) 4 (b) –4
(c) 2 (d) None of these

Q.11. In the above question the coordinates of the other two vertices are

(a) (2, 0), (4, –4) (b) (2, 4), (4, 0)
(c) (–2, 0), (–5, 1) (d) None of these

Q.12. If the vertices of diagonal of a square are (–2, 4) and (–2, 2), then its other two vertices are at
(a) (1, –1), (5, 1) (b) (1, 1), (5, –1)
(c) (1, 1), (–5, 1) (d) None of these

Q.13. If the sum of the distances of a point from two perpendicular lines in a plane is 1, then it will form a structure-
(a) square (b) circle
(c) straight line (d) two intersecting lines.

Q.14. If a, b, c are in AP, then $ax + by + c = 0$ will always pass through a fixed point whose coordinates are
(a) (1, –2) (b) (–1, 2)
(c) (1, 2) (d) (–1, –2)

Q.15. The points (2, 1), (8, 5) and (x, 7) lie on a straight line. The value of x is
(a) 10; (b) 11;
(c) 12; (d) $11\frac{2}{3}$

Q16. The area of the bounded region enclosed between the curves $y^3 = x^2$ and $y = 2 - x^2$ is
(a) $2\frac{4}{25}$; (b) $1\frac{1}{15}$;
(c) $2\frac{2}{15}$; (d) $2\frac{14}{15}$

Q17. The area of the figure formed by the lines $ax \pm by \pm c = 0$ is
(a) $\frac{c^2}{ab}$ (b) $\frac{2c^2}{ab}$
(c) $\frac{c^2}{2ab}$ (d) None of these

Q.18. The algebraic sum of the perpendicular distances from $A(x_1, y_1), B(x_2, y_2)$ and $A(x_3, y_3)$ to a variable line is zero, then the line passes through
(a) the orthocentre of ∇ABC
(b) the centroid of ∇ABC
(c) the circumcentre of ∇ABC
(d) None of these

Q.19. Length of the median from B on AC where A (–1, 3), B (1, –1), C (5, 1) is
(a) $\sqrt{18}$ (b) $\sqrt{10}$
(c) $2\sqrt{3}$ (d) 4

Q.20. If A and B are two points having coordinates (3, 4) and (5, –2) respectively and P is a point such that PA = and area of triangle PAB = 10 square units, then the coordinates of P are
(a) (7, 4) or (13, 2) (b) (7, 2) or (13, 4)
(c) (2, 7) or (4, 13) (d) (2, 7) or (4, 13)

Q.21. If A and B are two fixed points, then the locus of a point which moves in such a way that the angle APB is a right angle is
(a) a circle (b) an ellipse
(c) a parabola (d) none of these.

Q.22. A point P with coordinates (x, y) is said to be good if both x and y are positive integers. The number of good points on the curve xy = 27027 is
(a) 8 (b) 16
(c) 32 (d) 64

Q.23. The point (4,1) undergoes the following three transformations successively
I. Reflection about the line $y = x$
II. Translation through a distance 2 units along the positive direction of x-axis.
III. Rotation through an angle $\pi/4$ about the origin in the clockwise direction.

The final position of the point is given by the coordinates
(a) $\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}$ (b) $(-2, 7\sqrt{2})$
(c) $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$ (d) $(\sqrt{2}, 7\sqrt{2})$

Q.24. Line L has intercepts a and b on the coordinate axes, When the axes are rotated through a given angle; keeping the origin fixed, the same line has intercepts p and q. Then
(a) $a^2 + b^2 = p^2 + q^2$ (b) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$
(c) $a^2 + p^2 = b^2 + q^2$ (d) $\frac{1}{a^2} + \frac{1}{p^2} = \frac{1}{b^2} + \frac{1}{q^2}$

ANSWERS

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Answer	(d)	(a)	(b)	(a)	(a)	(d)	(c)	(b)	(a)	(b)	(a)	(a)	(b)	(a)	(b)	(c)	(d)	(c)

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(c)	(c)	(c)	(d)	(a)	(d)	(a)	(b)	(a)	(c)	(a)	(a)	(b)	(c)	(b)	(b)	(b)	(b)
Q. No.	21	22	23	24																
Answer	(a)	(c)	(a)	(b)																

BENCHMARKING TEST 4

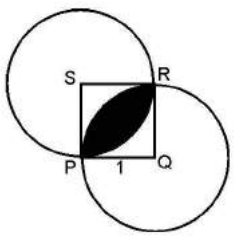
This test paper contains 30 questions of 4 marks each. One-third of the mark allotted to a particular question will be deducted in case of wrong answer.

Time Given: 70 Minutes **Total Marks: 120** **Qualifying Marks: 11**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
82+	99+
77	98.5+
71	98
51	92
37	85
27	80

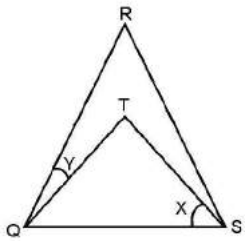
Q.1. Consider the figure given below:



PQRS is a square of side 1 unit and Q and S are the centers of the two circles. The area of the shaded portion is

(a) $\frac{\pi}{2}$ (b) $\frac{1}{2}$ (c) $\frac{\pi}{4}-\frac{1}{2}$ (d) $\frac{\pi}{2}-1$

Q.2. In the figure given below, if QRS is an equilateral triangle and TRS is an isosceles triangle and $x = 47°$, then the value of y (in degrees) will be

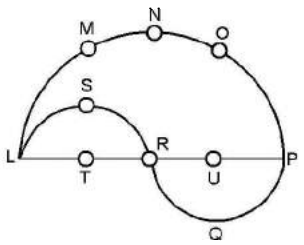


(a) 13° (b) 23° (c) 33° (d) 43°

Q.3. If the number representing the volume and the surface area of a cube are equal, then the length of the edge of the cube in terms of the unit of measurement will be

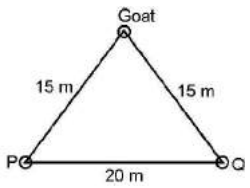
(a) 3 (b) 4 (c 5 (d) 6

Q.4. LMNOP is semicircle with the centre at R and diameter LP. LSR and RQP are also semicircles with centres at T and U respectively and diameters LR = RP = ½ LP. The ratio of the perimeter of LMNOP and LSRQP is

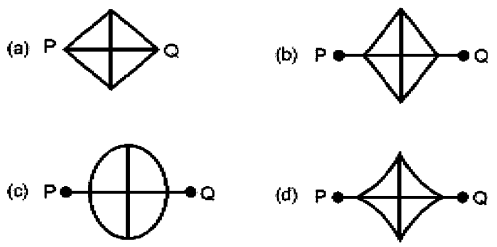


(a) 0.75:1 (b) 1:1 (c) 1:0.75 (d) 1.25:1

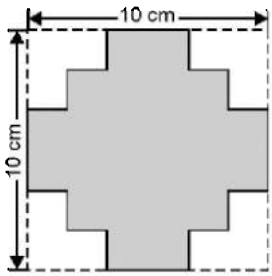
Q.5. A goat is tied to two poles P and Q with ropes that are 15 m long. P and Q are 20 m apart as shown in the given diagram:



Which one of the following shaded portions indicates the total area over which the goat can graze?



Q.6. In the given figure, all line segments of the shaded portion are of the same length and at right angles to each other. The same can be cut out of a board of side 10 cm. What is the area of the shaded portion?



(a) 36 cm^2 (b) 48 cm^2
(c) 52 cm^2 (d) 64 cm^2

Q.7. An equilateral triangle plate is to be cut into n number of identical small equilateral triangle plates. Which of the following can be a possible value of n?
(a) 196 (b) 216 (c) 256 (d) 296

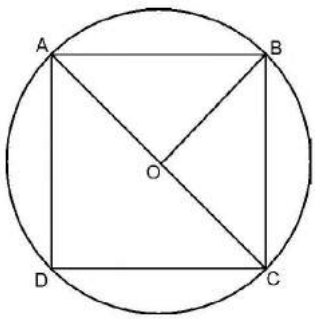
Q.8. In an isosceles trapezium ABCD, P and Q are the midpoints of diagonals BD and AC respectively. PQ = 4 cm, the perpendicular drawn from A on BD is 5 cm and that from C on BD is 6 cm and each of the diagonals measure 8 cm. Find the length of the sides AB and CD, given that the height of the trapezium is 4 cm?
(a) 12, 7 (b) 7, 15
(c) 5, 6 (d) None of these

Q.9. Consider a cylinder of radius 7 cm and height 14 cm. There is an intelligent ant that needs to travel on the surface of the cylinder. In order to travel to any point on the surface of the cylinder, the ant always chooses the shortest path. What is the distance, along the outer surface of the cylinder, of the point on the top of the cylinder for which the ant requires maximum time to reach?
(a) $\sqrt{340}$ (b) $\sqrt{680}$
(c) $\sqrt{240}$ (d) None of these

Q.10. There are 2 identical hollow cubical vessel A and B. A contains 64 identical solid spherical metallic balls. B contains 27 identical spherical balls of the same metal. In each cube the balls are packed in such a manner that no more balls can be packed (of course balls in A will not be the same as balls in B). Which cube is heavier?
(a) A (b) B
(c) A = B (d) Cannot be determined

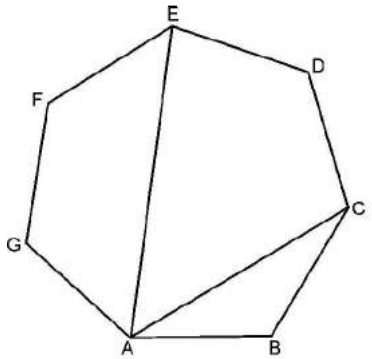
Q.11. There is a fan with 3 blades at 120° to each other. Its central circular disc has an area of 3À cm^2 and length of the blade is $\left(20-\sqrt{3}\right)$ cm. If the tips of the blades are joined so as to form an equilateral triangle, what will be its area?
(a) 900 cm^2 (b) $300\sqrt{3}$ m^2
(c) 900 + 9 cm^2 (d) None of these

Q.12. In the given figure, radius of the circle is 2.5 cm. O is the centre of the circle and OB bisects angle ABC. What can be the maximum area of ABCD?



(a) 2.5 (b) 12.5
(c) 12 (d) None of these

Q.13. In the given regular heptagon ABCDEFG, AB = a, AC = b, AE = c. Which of the following statements is true?



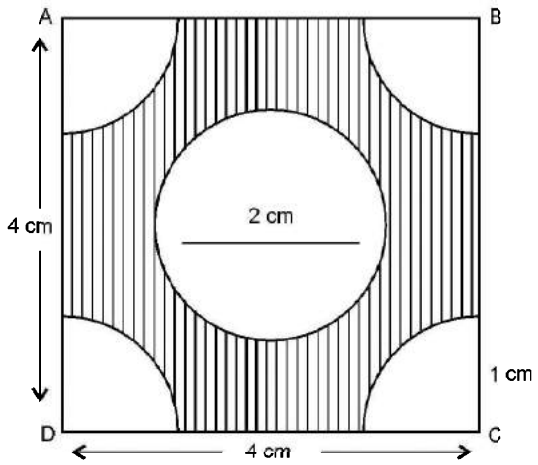
(a) $c = b - a$ (b) $\frac{1}{b} = \frac{1}{c} + \frac{1}{a}$

(c) $\frac{1}{a} = \frac{1}{b} - \frac{1}{c}$ (d) None of these

Q.14. An insect wants to travel from one bottom corner to the opposite top corner inside a room of dimension $6 \times 8 \times 10$ units. What is the minimum distance that the insect needs to travel?
[Common sense hint – Insect can fly]

(a) $10\sqrt{2}$ units (b) $20\sqrt{2}$ units
(c) $15\sqrt{2}$ units (d) $25\sqrt{2}$ units

Q.15. In the figure given below, ABCD is a square of side length 4 units which has four symmetric cuts at all its corners. Find the area of the shaded portion.



(a) 0.6 sq units (b) 0.3 sq units
(c) 0.9 sq units (d) None of these

Q.16. If $(9p^2 + 4)$, $(2p + 7)$ and $(8p^2 - 1)$ are the lengths of three sides of a triangle (p>1). Which of the following cannot be a value of p?

(a) 2 (b) $\sqrt{2}$
(c) $2\sqrt{2}$ (d) None of these

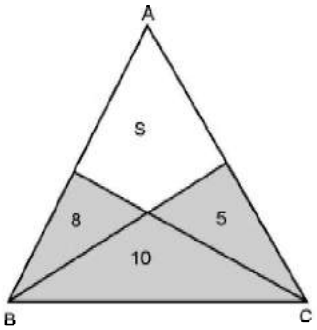
Q.17. AB is a line segment and P is its mid-point. Semicircles are drawn with AP, PB and AB as diameters on the same side of the line AB. A circle is drawn to touch all the three semicircles. Find the radius of this circle.

(a) 1/4 AB (b) 1/6 AB
(c) 1/7 AB (d) 1/8 AB

Q.18. In two circles C_1 and C_2, arcs of the same length subtend angles of 70° and 80° at their respective centers. What is the ratio of their radii (in the same order)?

(a) 7 : 8 (b 8 : 7
(c) 8 : 15 (d) Cannot be determined

Q.19. Consider the figure given below:



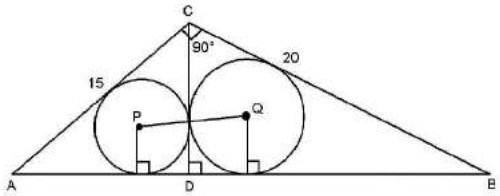
Δ ABC is divided into four parts by straight lines from two of its vertices. The areas of the three triangular parts are 8 sq units, 5 sq units and 10 sq units. What is the area of the remaining part S (in sq units)?

(a) 32 (b) 40
(c) 54 (d) None of these

Q.20. In a convex quadrilateral ABCD, the diagonal BD bisects neither the angle ABC nor the angle CDA. A point P lies inside ABCD and satisfies ∠PBC = ∠DBA and ∠PDC = ∠BDA. ABCD is cyclic quadrilateral if and only if

(a) AP = CP (b) AB = BC
(c) BP = AP (d) None of these

Q.21.



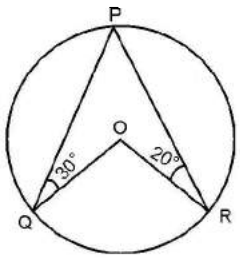
In the above figure, ACB is a right-angled triangle. CD is the altitude. Circles are inscribed within the ΔACD and ΔBCD. P and Q are the centres of the circles. The distance PQ is

(a) 5 (b) $\sqrt{50}$ (c) 7 (d) 8

Q.22. If a, b and c are the sides of a triangle, and $a^2 + b^2 + c^2 = bc + ca + ab$ then the triangle is

(a) Equilateral (b) Isosceles
(c) Right-angled (d) Obtuse-angled

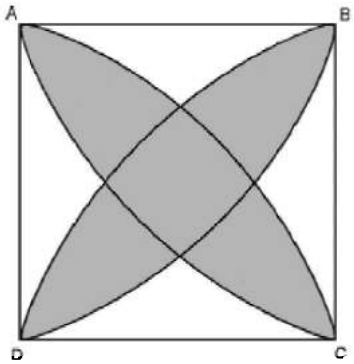
Q.23.



In the figure given about ∠OQP = 30° and ∠ORP = 20°, then ∠QOR is equal to

(a) 100° (b) 120° (c) 130° (d) 140°

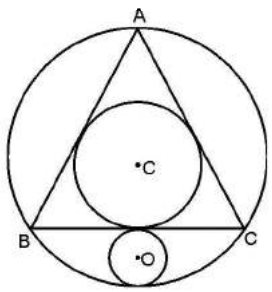
Q.24. In the figure given below, ABCD is a square of side length $4\sqrt{2}$ units. Shaded parts are the sectors of the different circles with A, B, C and D as the centers. What is the area (in sq. units) of the shaded region?



(a) $4\pi + 16$ (b) $8(\sqrt{2} + 1)\pi$

(c) $16\sqrt{2}\pi$ (d) None of these

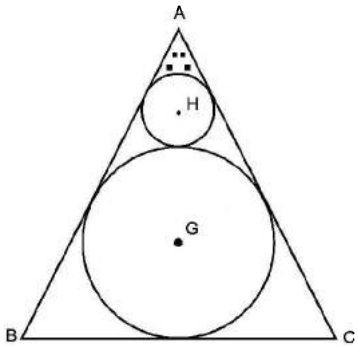
Q.25. In the figure given below, ABC is an equilateral Δ of side length 3 units and circle with centre C is the in-circle. Another circle with centre O is drawn in such a way that side BC of the triangle ABC is tangent to the circle and circumcircle is having a point of tangency with this circle. What will be the length of the radius of the circle with centre O?



(a) $\sqrt{3}$ (b) $\frac{\sqrt{3}+1}{2}$

(c) $\frac{\sqrt{3}-1}{2}$ (d) None of these

Q.26. In the figure given below, ΔABC is an equilateral triangle with side length 3 units. Triangles are arranged inside this triangle in such a way that the circle with centre G is the incircle of the triangle ABC and the circle with centre H is formed on the circle with centre H and so on. The next circle will be created on this circle with centre H, and so on the infinite circles are created one on the top of other.



What is the sum of the radius of all such circles formed?

(a) 4.5 units (b) 2.25 units

(c) 3 units (d) 1.5 units

Q.27. An octagon is inscribed in a circle. One set of alternate vertices forms a square of area 5 units. The other set forms a rectangle area of 4 units. What is the maximum possible area for the octagon (in sq. units)?

(a) $3\sqrt{5}$ (b) $\frac{3}{\sqrt{5}}$

(c) $5\sqrt{3}$ (d) None of these

Q.28. What is the largest number of the quadrilaterals formed by four adjacent vertices of an convex polygon of n sides that can have an inscribed circle?

(a) n (b) $n/2$

(c) $n/3$ (d) None of these

Q.29. There are two circles with centres at A and B respectively. The circle with center A has a radius of 8 units and the circle with center B has a radius of 6 units and the distance AB is 12 units. Both the circles meet at points P and S. A line through P meets the circles again at Q and R (with Q on the larger circle) in such a way that QP = PR. Find the length of QP.

(a) $\sqrt{160}$ units (b) $\sqrt{140}$ units

(c) $\sqrt{80}$ units (d) None of these

Q.30. A circle of 4 units is taken. Now n circles of the same radii are inserted in this circle ($1 \le n \le 10$, where n is a natural number) in such a way that they are encompassing the maximum possible areas of the circle and are inside the bigger circle along its circumference. [Obviously for $n = 1$, radius of the inside circle will be same as the radius of outside circle. Similarly for $n = 2$, radius of the inside circle will be half of the the outside bigger circle and so on]. For how many values of n, radius of the circle will be an integer

(a) 2 (b) 3 (c) 4 (d) 5

ANSWERS

■■■ Benchmarking Tests

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(d)	(b)	(b)	(c)	(c)	(b)	(b)	(a)	(b)	(b)	(d)	(a)	(b)	(c)	(b)	(b)	(d)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(b)	(a)	(a)	(d)	(d)	(b)	(a)	(b)	(d)	(b)										

Module 5

Miscellaneous

- Set Theory
- Miscellaneous
- Managerial Decision Making

20

CHAPTER

Set Theory

LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Types of sets and methods of their formation
- Definitions and properties
- Maxima and minima in set theory
- Kinds of questions asked in the CAT
- Methods of solving questions

Introduction to the topic

The concept of set theory has always been an important topic for the CAT, both in terms of direct and indirect application of this concept and as a principle of counting. Questions from this topic are asked very regularly in the CAT and other major B-school exams. However, this concept is still in an evolutionary phase, as can be seen from the last few years' QA paper for this examination. The questions from this topic are not conventional. Maxima and minima can emerge as important concept of set theory.

SET THEORY

A set is a collection of well-defined objects. The members of a set can be literally anything like letters of English alphabet, or different types of alphabets, or name of the countries, or numbers, or marks obtained by a student.

Given, capital letters are representing a standard set.

A = (a, b, c, d, e, f) – The first six letters of english alphabet

B = (US, China, Japan, India) – The names of the top four countries in terms of their GDP

C = (2, 4, 6, 8, 10) – The first five even natural numbers

Here, A, B and C are different sets which are representing different group of objects.

In this chapter, we will confine ourselves with

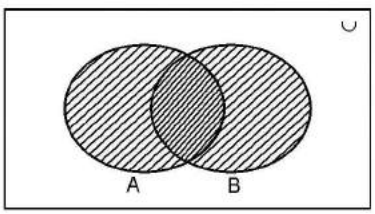
- Type of sets
- Solving techniques
- Maxima and minima

Different ways of representing a set:

1. **Roster Method** With the help of this method, a set is represented by all the elements of it written under the brackets separated by commas.
   For example, A = {1, 2, 3, 4, 5}
2. **Set Builder Method** With the help of this method, a set is represented by the common property of all its elements. It is written as
   A = {x| x, $p(x)$ holds}
   A = (x:x, $p(x)$ holds}; where $p(x)$ is the common property shared by all the elements of set A.
   For example, A = {x e N |$x \leq 5$), which can be written with the roster method as A = {1, 2, 3, 4, 5}

Types of Sets

1. *Empty or Null Set* A set having no element is known as a empty or a null set and it is denoted as ϕ or { }.
   For example, A = set of even prime numbers excluding 2.
2. *Singleton set* A set having only one element is known as a singleton set.
   For example, A = set of even prime number/s.
3. *Finite set* A set having a countable number of elements is known as the finite set.
   For example, A = set of odd numbers from 100 to 890.
4. *Infinite set* A set whose elements cannot be counted is known as infinite set.
   For example, A = set of all irrational numbers between 2 and 3.
5. *Equal sets* Two sets are said to be equal sets if all the elements of set A are included in set B and all the elements of set B are included in set A. If two sets A and B are equal then it is represented by A = B and if A and B are not equal then it is written as A^1 B, that is, all the elements of set A are not included in set B and all the elements of set B are not included in set A.
   For example, A = {a, b, c} and B = {c, b, a} are equal sets. Hence, in this case we can write Set A = Set B or simply A = B.
6. *Subsets Set* A is said to be the subset of another set B if all the elements of set A are included in set B. 'Set A is subset of Set B' is shown by A $\subseteq$ B. We can now say that every element of set A is a member of Set B.
   For example, If A = {a, b, c} and B = {a, b,c,d,e} then A $\subseteq$ B, or A is a subset of B.
   Some important results on subsets
   - Every set is a subset of itself.
   - Every set has an empty set as its subset.
   - Total number of subsets of a set having n elements in 2^n.
7. *Universal Set* A set which contains all the sets in a given context is a universal set.
   For example, when we are using sets containing natural numbers, N is the universal set.
   If A = {a, b, c,}, B = {b, c, d}, C = {c, d}
   Then we can take u = {a, b, c, d} as universal set.

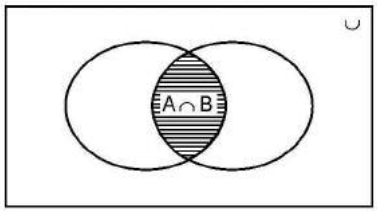


8. *Power Set* The collection of all the subsets of a set is known as the power set of that set. If A is the set, then a set containing all the subsets of A is known as the power set of A. It is denoted by P(A).
   Let A = {1, 2}, then the number of subsets of this set will be 2^2 and the subsets are {}, {1}, {2} and {1, 2} and the set containing all these four sets is known as the power set represented as P(A).
9. *Venn Diagram* Swiss mathematician Euler first gave the idea of representing sets by a diagram. Later on, British mathematician Venn brought this into practice. So, it is known as Euler–Venn diagram or simply Venn diagram. In this way of representing sets we use a closed curve, generally a circle, to denote sets and their operations.

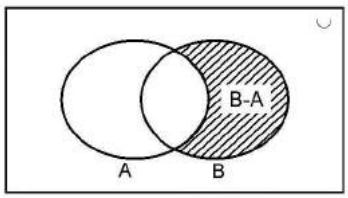
OPERATIONS ON SETS

1. *Union of Sets* If two sets are A and B, then union of A and B is defined as the set that have all the elements which belong to either A or B or both. It is represented by A $\cup$ B.

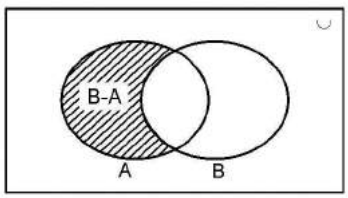
2. *Intersection of Sets* If two sets are A and B, then the intersection of A and B is defined as the set that have all the elements which belong to both A and B.
It is represented by $A \cap B$.
For example, find $A \cup B$ and $A \cap B$ if A = {1, 2, 3, 4, 5} and B = {2, 4, 6}
$A \cup B = \{1, 2, 3, 4, 5, 6\}$
$A \cap B = \{2, 4\}$



3. *Disjoint Sets* Two sets are said to be disjoint if $A \cap B = 0$, that is, not a single element is common to both of these two sets.
For example, If A = {Set of all odd numbers} and B = {Set of all even numbers} then set A and set B are disjoint sets.



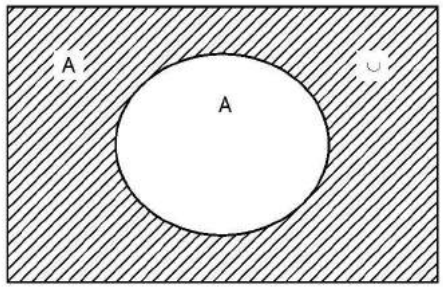
4. *Difference of Sets* For two sets A and B, A – B is the set of all those elements of A that do not belong to B. Similarly, B – A is the set of all those elements of B that do not belong to A.
For example, A = {1, 2, 3, 4, 5, 6} and B = {2, 4, 8}, then A – B = {1, 3, 5, 6,} and B – A = {8}



5. *Complement of a Set* If U is the universal set and a set A is such that $A \subseteq U$ then complement of the set A is defined as U – A and represented as A' or A^c
For example, U = {Set of all prime numbers} and A = {Set of all even prime numbers} then U – A = {Set of all odd prime numbers}

ALGEBRAIC LAWS IN SET THEORY

1. *Commutative Laws* For any two sets A and B
$A \cup B = B \cup A$ and $A \cap B = B \cap A$



2. *Associative Laws* For any three sets A, B and C,
$(A \cup B) \cup C = A \cup (B \cup C)$ and
$A \cap (B \cap C) = (A \cap B) \cap C$
3. *Distributive laws* For any three sets A, B and C,
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. *Identity laws*
$A \cup \phi = A$, where ϕ is the null set
$A \cap U = A$, where U is the universal set
5. *De-Morgan's law* If A and B are two sets then
$(A \cup B)' = A' \cap B'$
$(A \cap B)' = A' \cup B'$
6. $A \cup A = A$ and $A \cap A = A$

Some Important Results

If A and B are any two sets, then
1. $A - B = A \cap B'$
2. $B - A = B \cap A'$
3. $(A - B) \cup B = A \cup B$
4. $(A - B) \cap B = f$
5. $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$

Some Important Results on the Number of Elements in Sets

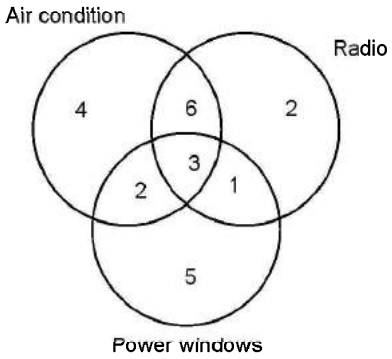
If A, B and C are three finite sets then
1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
2. $n(A \cup B) = n(A) + n(B)$ if set A and set B are two disjoint sets.
3. $n(A - B) = n(A) - n(A \cap B)$
4. $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$
5. $n(A' \cup B') = n((A \cap B)') = n(U) - n(A \cap B)$
6. $n(A' \cap B') = n((A \cup B)') = n(U) - n(A \cup B)$

Example 1 A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options—air conditioning, radio and power windows—were already installed. The survey found:
15 had air conditioning
2 had air conditioning and power windows but no radios
12 had radio
6 had air conditioning and radio but no power windows
11 had power windows
4 had radio and power windows
3 had all three options

What is the number of cars that had none of the options?

(a) 4 (b) 3 (c) 1 (d) 2

Solution From the given conditions, we have



When we add all the values, we get a total of 23 cars.

So, 2 cars don't have the air conditioning or the radio, or the power windows.

Example 2 If A = { 1, 3, 5, 7}, B = { 1, 2, 3, 4}

What is the value of (A U B) and (A ∩ B)?

Solution (A U B) = {1, 2, 3, 4, 5, 7}

(A ∩ B) = {1, 3}

Example 3 How many numbers from 1 to 100 are not divisible by either 2 or 4 or 5?

Solution Let us first understand the meaning of the statement given in the question—

It is given that the numbers are from 1 to 100, so while counting we will include both the limits, i.e., 1 and 100. Had this been "How many numbers in between 1 to 100 are … ", then we would not have included either 1 or 100.

Now to solve this question, we will first find out the number of numbers from 1 to 100 which are divisible by either 2 or 5 (since all the numbers which are not divisible by 2 will not be divisible by 4 also, so we do not need to find the numbers divisible by 4). And then we will subtract this from the total number of numbers i.e., 100. It can be seen below:

Total number of numbers = numbers which are divisible + numbers which are not divisible

So, $n(2 \cup 5) = n(2) + n(5) - n(2 \cap 5)$

Now, $n(2) = 50$

$n(5) = 20$

$n(2 \cap 5) = 10$

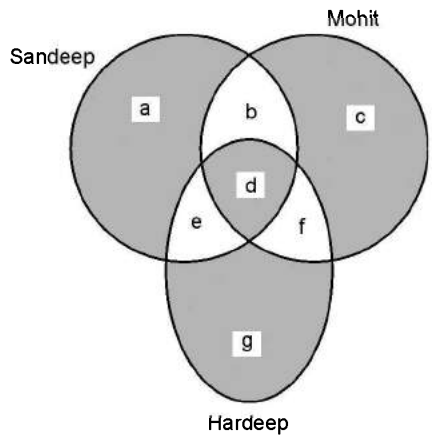
$n(2 \cup 5) = 50 + 20 - 10 = 60$

Numbers which are not divisible = total number of numbers-numbers which are divisible

$= 100 - 60 = 40$

Example 4 Sandeep, Mohit and Hardeep are three financial analysts working at Due North Consultants. The sum of the number of projects handled by Sandeep and Hardeep individually is equal to the number of projects in which Mohit is involved. All three consultants are involved together in 6 projects. Sandeep works with Mohit in 14 projects. Hardeep has 2 projects with Mohit but without Sandeep, and 3 projects with Sandeep but without Mohit. The total number of projects is one less than twice the number of projects in which more than one consultant is involved. In how many projects is Mohit involved alone?

Solution Let us see the situation given above through the Venn-diagram:



$a + g = b + d + f + c$ … (i)

And $d = 6$

$b + d = 14$, so, $b = 8$.

$f = 2$ and $e = 3$

And finally, $a + b + c + d + e + f + g$

$= 2(b + d + e + f) - 1$ … (ii)

Using (i) and (ii),

c = Number of projects handled by Mohit alone = 1

MAXIMA AND MINIMA IN SET THEORY

Possibility of maxima and minima occurs in a set theory due to the fact that intersections of two or more than two values, in most of the cases, have a finite range. In other words, they can lie in between x and y, where x and y can be any number depending upon the given conditions.

Let us understand this with the following example:

Example 5 In a class of 50 students, a test for 2 subjects maths and physics was conducted. 30 students passed in maths and 40 passed in physics.

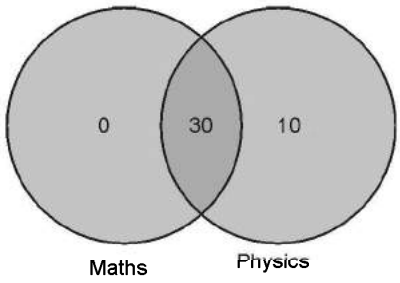
- What is the maximum number of students who passed in both the subjects?
- What is the minimum number of students who passed in both the subjects?

Solution

- Since 30 students passed in maths, so we can assume that these are the same 30 students out of 40 students who passed physics also. So, the maximum number of

students who passed in both the subjects = 30. (Remember that in this case there was no restriction regarding the number of students who passed any of the subject. Had it been given that all these 50 students are passing atleast one subject, then we would not have been in a position to do this.)

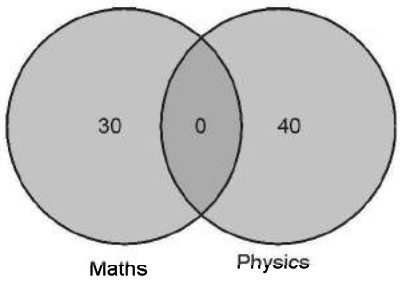
The Venn-diagram representation of the same is given below.



- To find out the minimum number of students who passed in both the subjects, intersection (math ∩ physics) should be minimum, i.e., individually math and physics each should be maximum.

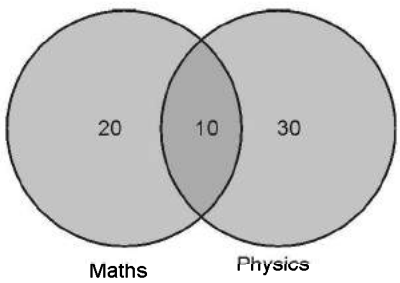
Let us do a bit of hit and trial before we finally move on to do it with its proper solution.

If we take (math ∩ physics) = 0, then the situation would be like



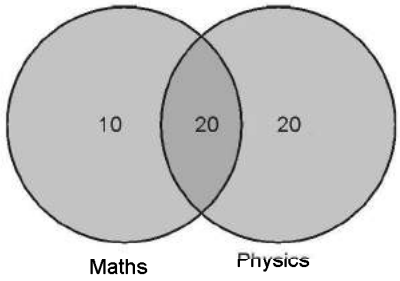
It can be seen that total number of students in this situation = 30 + 40 = 70, which is not true.

Similarly, if we take (math ∩ physics) = 10, then the situation would be like



It can be seen that total number of students in this situation = 20 + 10 + 30 = 60, which is again not true.

Hence, now we can conclude that since the number of students = 50, so (math ∩ physics) = 20. And this is the minimum value of (math ∩ physics).



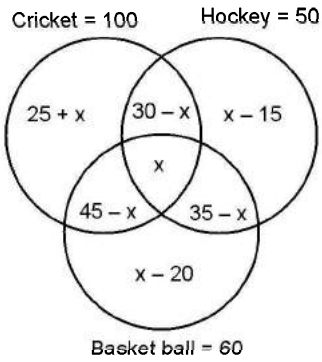
Alternatively, question (ii) can be done by the following set theory formula also: n (maths U physics) = n (maths) + n(physics) – n(maths ∩ physics)

Remember In case of two sets, by using this formula of set theory we get the minimum value, but if we use the same formula in case of three sets, then we obtain the maximum value.

Example 6 There are 200 students in a school. Out of these, 100 students play cricket, 50 students play hockey and 60 students play basketball. 30 students play both cricket and hockey, 35 students play both hockey and basketball, and 45 students play both basketball and cricket.

- What is the maximum number of students who play at least one game?
- What is the maximum number of students who play all the 3 games?
- What is the minimum number of students playing at least one game?
- What is the minimum number of students playing all the 3 games?

Solution



Consider the venn diagram given above:

At first we will convert all the values in terms of x, which can be seen above.

Since the number of students cannot be negative,

$x - 15 \geq 0$

$\therefore\ x - 20 \geq 0$

So, iv. For the minimum number of students playing all three games, i.e., $x = 20$

For the maximum value of x, again none of the categories should have –ve number of students.

$\therefore 30 - x \geq 0$

$x \leq 30$

If x is more than 30, $30 - x$ would be –ve which is not possible.

Total number of students playing at least one game,

$= 100 + x - 15 + 35 - x + x - 20$

$= 100 + x$

So, the minimum number of students playing at least one game = 100 + 20 = 120

Hence, the maximum number of students playing at least one game = 100 + 30 = 130

Example 7 In a class of 50 students, 70% students pass in QA and 60% pass in RC. What is the minimum percentage of students who pass in both the papers?

Solution We know that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Assuming that $n(Q)$ = number of students passing QA and $n(R)$ = number of students passing RC.

Hence, $n(Q \cup R) = n(Q) + n(R) - n(Q \cap R)$

Or, $n(Q \cap R) = n(Q) + n(R) - n(Q \cup R) = 70\% + 60\% - n(Q \cup R)$

Now the minimum value of $n(Q \cap R)$ will occur for the maximum value of $n(Q \cup R)$. The maximum possible value of $n(Q \cup R) = 100\%$

So, the minimum value of $n(Q \cap R) = 30\%$

Example 8 In a class of 50 students, 70% students pass in QA, 90% pass in EU and 60% pass in RC. What is the minimum percentage of students who pass in all the papers?

Solution We can do this question by using either the formula for three sets, or we can simply keep on applying the formula for two sets required a number of times.

We know that $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Assuming that $n(Q)$ = number of students passing QA, $n(E)$ = number of students passing EU and $n(R)$ = number of students passing RC.

Hence, $n(Q \cup R) = n(Q) + n(R) - n(Q \cap R)$

Or, $n(Q \cap R) = n(Q) + n(R) - n(Q \cup R) = 70\% + 60\% - n(Q \cup R)$

Now the minimum value of $n(Q \cap R)$ will occur for the maximum value of $n(Q \cup R)$. The maximum possible value of $n(Q \cup R) = 100\%$

So, the minimum value of $n(Q \cap R) = 30\%$

Now, the minimum value of EU, QA and RC will be obtained by finding the minimum of EU and $(QA \cap RC)$.

The minimum value of EU and $(QA \cap RC) = 90\% + 30\% - 100\% = 20\%$.

Example 9 In a class of 50 students, 70% students pass in QA, 90% pass in EU, 85% pass in DI and 60% pass in RC. What is the minimum percentage of students who pass in all the papers?

Solution This question is just the extension of the earlier question.

The minimum value of EU and $(QA \cap RC) = 90\% + 30\% - 100\% = 20\%$.

The minimum value of DI and EU and $(QA \cap RC) = 20\% + 85\% - 100\% = 5\%$

PRACTICE EXERCISES

WARM UP

Direction for questions 1 to 5: Read the passage below and solve the questions based on it.

A newspaper agent deliver the morning papers to 240 families. 90 buy only *Times of India*. 15 buy *Times of India* and *The Telegraph*. 100 buy The Telegraph. 30 buy *The Telegraph* and *The Hindu*. 30 buy only *The Hindu*. No family buys all the three papers.

Q.1. How many buy the *Times of India*?
(a) 120 (b) 110 (c) 125 (d) 150

Q.2. How many buy The *Telegraph*?
(a) 50 (b) 55 (c) 45 (d) 65

Q.3. How many people read only one paper?
(a) 165 (b) 175 (c) 145 (d) 195

Q.4. Which newspaper is the most popular?
(a) The Hindu (b) The Telegraph
(c) Times of India (d) None of these

Q.5. Which is the least popular newspaper?
(a) The Hindu (b) The Telegraph
(c) Times of India (d) None of these

Direction for questions 6 and 7: Read the passage below and solve the questions based on it.

Of the 400 members of a sport club, 144 play squash (S), 135 play badminton (B), 156 play cricket (C). Each member plays at least one game. The members who play badminton and cricket both must play squash also. For every three members who play at least two games, there are two members who play all the three games.

Q.6. Find the number of members who play all the three games.
(a) 14 (b) 23
(c) 11 (d) None of these

Q.7. Find the number of members who play only Squash.
(a) 125 (b) 122
(c) 101 (d) 123

Direction for questions 8 and 9: Read the passage below and solve the questions based on it.

The following are the statistics of 22 students of Career Zone who appeared for CAT. 15 students got calls from the IIMA, 12 students got calls from the IIMB and 8 students got calls from IIMC. 6 students got calls from the IIMA and IIMB, 7 students got calls from the IIMB and the IIMC and 4 students got calls from the IIMA and the IIMC if 4 students got calls from all three IIMs

Q.8. How many students of Career Zone got a call from IIMA only?
(a) 12 (b) 9
(c) 7 (d) None of these

Q.9. How many students of Career Zone got a call from IIMC only?
(a) 3 (b) 4 (c) 1 (d) 2

Direction for questions 10 to 12: Read the passage below and solve the questions based on it.

In order to study the breakfast habits of the people in the city of Longkong, a survey of 400 families was conducted. Of these 400 families, 156 had cereals, 288 had eggs and 300 had toast. It was found that 212 families had egg and toast and 104 had cereal and toast, and 128 had egg and cereal and 84 had all three.

Q.10. How many families had neither toast nor cereal nor egg?
(a) 9 (b) 16 (c) 32 (d) 64

Q.11. How many had egg or toast but not cereal?
(a) 196 (b) 252 (c) 228 (d) 238

Q.12. How many families had only two items?
(a) 192 (b) 182 (c) 212 (d) 312

Q.13. In a survey of 100 people, it was found that 60 people read the India Today, 55 read the Outlook and 10 read neither of the two magazines. How many people read both the magazines?
(a) 45 (b) 35
(c) 25 (d) None of these

Q.14. In a class of 50 students, 28 like pop music, and 20 like classical music. If 10 students like one of the two kinds of music, find the number of the students who like both kinds of music?
(a) 8 (b) 5 (c) 7 (d) 6

FOUNDATION

Direction for questions 1 to 5: Read the passage below and solve the questions based on it.

During the census 2001, 8,000 people were surveyed in an extremely backward village, where most of the people were suffering from a certain number of diseases. In that village, 1,264 people were suffering from japanese fever and dehydration, 2,976 people were suffering from dehydration, 3,472 people from japanese fever and cholera, 4,720 people from cholera and 1,008 people from cholera and dehydration. 300 people suffered from all the three given diseases.

Q.1. Find the number of persons who suffered from dehydration only.
(a) 586 (b) 368 (c) 492 (d) None of these

Q.2. Find the number of persons who suffered from dehydration or cholera but not japanese fever.
(a) 2552 (b) 2252 (c) 3516 (d) 3216

Q.3. Find the number of persons who suffered from japanese fever but neither cholera nor dehydration.
(a) 1258 (b) 1256
(c) 1254 (d) Cannot be determined

Q.4. Find the number of persons who suffered from dehydration but not cholera.
(a) 2268 (b) 2976 (c) 1968 (d) 2676

Q.5. Find the number of persons who suffered from cholera but not dehydration.
(a) 3712 (b) 4012
(c) 4920 (d) Cannot be determined

Q.6. In a class of 120 students, students are participating in the Sudoku or crossword classes. The ratio of students who participate in Sudoku to the number of students who participate in crossword is 2 : 1. 45 students do not participate in any of these two. What is the number of students who participate in Sudoku?
(a) 50 (b) 80 (c) 70 (d) 84

Q.7. In a group of 120 persons, there are 80 Biharis and 40 Punjabis. Further, 70 persons in the group are muslims and remaining are hindus. Then the number of Biharis who are also muslims in the group is
(a) Between 30 and 70
(b) Exactly 20
(c) Between 15 and 25
(d) Between 20 and 25

Q.8. 30 monkeys went to a picnic. 25 monkeys chose to irritate buffaloes, and 20 chose to irritate cows. How many monkeys chose to irritate both buffaloes and cows?
(a) 10 (b) 15 (c) 5 (d) 20

Q.9. A surveys shows that 89% of the Australians like butter whereas 79% like chocolate. It is known that they like at least one of the two. What is the percentage of Australians who like both butter and chocolate?
(a) 13% (b) 15%
(c) 11% (d) None of these

Q.10. The market research department of a certain breakfast cereal firm interviewing 100 persons found that on a certain morning for breakfast
i. 72 had cereals, 39 had fruits, 75 had toast.
ii. 32 had fruits and cereal, 53 had cereal and toast, 26 had toast and fruits
iii. 21 had all three.
How many of those interviewed had neither cereal nor toast nor fruits?
(a) 1 (b) 2
(c) 4 (d) None of these

Q.11. If S and T are two sets such that S has 21 elements, T has 32 elements and S∩T has 11 elements, how many elements does S∩T have?
(a) 36 (b) 38 (c) 42 (d) 40

Q.12. In a group of 52 persons, 16 drink tea but not coffee and 33 drink tea. Then what is the number of persons who take coffee but not tea?
(a) 11 (b) 19
(c) 23 (d) Cannot be determined

Direction for questions 13 to 16: Read the passage below and solve the questions based on it.

Out of 10,000 persons surveyed, 3,700 liked city A, 4,000 liked city B, and 5,000 liked city C. 700 persons liked city A and B, 1,200 liked A and C, and 1,000 liked city B and C. Each person liked at least one city.

Q.13. Find the number of persons who liked all the three cities.
(a) 100 (b) 200
(c) 300 (d) 400

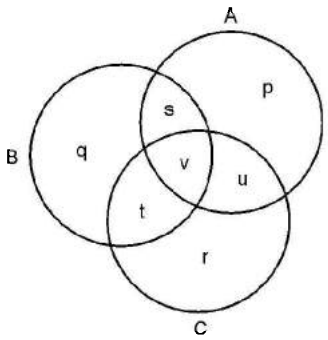
Q.14. Find the number of persons who liked at least two cities as a percentage of number of persons who liked exactly one city.
(a) 32.4% (b) 29.8%
(c) 20.83% (d) 33.33%

Q.15. Find the number of persons who liked exactly two cities as a percentage of the number of persons who liked at least one city.
(a) 21.5% (b) 19.16%
(c) 25.5% (d) None of these

Q.16. Find the number of persons who liked city A and city B but not city C.
(a) 400 (b) 5500
(c) 600 (d) None of these

Q.17. A group of 50 students appeared for the two examinations-physics and mathematics. 38 students passed in physics and 37 students passed in mathematics. If 30 students passed in both subject, what is the number of students who failed in both the subjects?
(a) 2 (b) 3 (c) 4 (d) 5

Q.18. In the given diagram, circle A represents teachers who can teach physics, circle B represents teachers who can teach chemistry and circle C represents those who can teach mathematics. Among the regions marked *p*, *q*, *r*, ... the one which represents teachers, who can teach physics and mathematics but not chemistry, is



(a) v (b) u (c) s (d) t

Q.19. In a group of 70 people, 37 like coffee, 52 like tea and each person likes at least one of the two drinks. How many people like both coffee and tea?
(a) 19 (b) 21 (c) 17 (d) 15

Direction for questions 20 and 21: Read the passage below and solve the questions based on it.

In a school with 727 students, 600 students take mathematics and 173 students take both mathematics and physics. Each student is taking at least one of the two subjects.

Q.20. How many students took physics?
(a) 250 (b) 300 (c) 280 (d) 290

Q.21. How many students took physics only?
(a) 300 (b) 280
(c) 127 (d) 173

Q.22. Of the members of three athletic teams in a certain school, 21 are on the basketball team, 26 on the hockey team, and 29 on the football team. 14 play hockey and football, 12 play football and basketball, 15 play hockey and football. 8 are on all the three teams. How many members are there altogether?
(a) 38 (b) 47
(c) 51 (d) 43

Direction for questions 23 and 24: Read the passage below and solve the questions based on it.

In a survey of 500 students, it was found that 255 play table tennis, 195 play lawn tennis, 115 play tennis, 45 play table tennis and tennis, 70 play table tennis and lawn tennis, 50 play lawn tennis and tennis. There are 80 students who do not play any of the three games.

Q.23. What is the number of students who play all the three games?
(a) 30 (b) 20
(c) 50 (d) None of these

Q.24. What is the number of students who play exactly one game?
(a) 295 (b) 300
(c) 325 (d) None of these

Q.25. In a party 70% of the people drank coffee, 65% drank soft drinks, 27% did not drink anything, whereas 248 people drank both. What is the number of people who attended the party?
(a) 400 (b) 600
(c) 800 (d) 900

Direction for questions 26 to 29: Read the passage below and solve the questions based on it.

In a class of 106 students, each student studies at least one of the three subjects maths, political science and chemistry. 48 of them study maths, 51 political science and 53 chemistry. 16 study maths and political science, 17 study maths and chemistry and 18 study political science and chemistry.

Q.26. What is the number of students who study, exactly two subjects?
(a) 31 (b) 32
(c) 33 (d) 36

Q.27. What is the number of students who study more than one subject?
(a) 39 (b) 41
(c) 40 (d) 42

Q.28. The number of students who study all the three subjects is
(a) 5 (b) 6
(c) 7 (d) 4

Q.29. The number of students who study exactly one subject is
(a) 45 (b) 55
(c) 65 (d) 70

Q.30. In a survey conducted in Patna, it was found that $\frac{3}{4}$th of the town owns colour TV, 85% of the people own refrigerator and every 4 in 5 in the town own music system. What is the minimum percentage of people who have all the three?
(a) 30% (b) 40%
(c) 55% (d) None of these

MODERATE

Q.1. In a referendum about three proposals, 78% of the people were against at least one of the proposals. 50% of the people were against proposal 1st, 30% against proposal 2nd and 20% against proposal 3rd. If 5% of the people were against all the three proposals, what percentage of people were against more than one of the three proposals?
(a) 10 (b) 12
(c) 17 (d) 22

Direction for questions 2 to 4: Read the passage below and solve the questions based on it.

In a recent survey conducted by Cable TV Association among the people who watch DD, Zee and Star TV, it is found that 80% of the people watched DD, 22% watched Star TV and 15% watched zee.

Q.2. What is the maximum percentage of people who can watch all the three channels?

(a) 12.5% (b) 8.5%
(c) 15% (d) Data insufficient

Q.3. If 5% of the people watched DD and Zee, 10% watched Star TV and DD, then what percentage of the people surveyed watched Star TV and Zee only?
(a) 2% (b) 5%
(c) 8.5% (d) Cannot be determined

Q.4. In the previous question, what percentage of people watched all the three channels?
(a) 3.5% (b) 0%
(c) 8.5% (d) Cannot be determined

Q.5. 80,000 students appeared in the last year's CAT consisting of three sections QA, DI and EU. 24,000 students passed in QA; 31,000 students passed in DI; 35,000 students passed in EU. Further 49,000 students passed in at least one of QA and DI; 59,000 students passed in at least one of DI and EU, 54,000 students passed in at least one of QA and EU. Calls for IIMs are sent to only those students who passed in all the three sections in CAT. If only 2,000 students got the IIM calls, then what is the numbers of students who failed in all three papers (either you can pass or fail)?
(a) 6,000 (b) 8,000
(c) 10,000 (d) 4,000

Direction for questions 6 to 9: Read the passage below and solve the questions based on it.

CAT 2010 was having four sections namely QA, LR, EU and RC. Following is the data pertaining to CAT 2010: Answer the following questions based on the followings information:

- The number of students who appeared 1867
- The number of students who passed in exactly two sections 232
- The number of students who passed in exactly one section 132
- The number of students who failed in all four sections 292
- The number of students who failed in QA only 200
- The number of students who failed in LR only 177
- The number of students who failed in EU only 108
- The number of students who failed in RC only 211

A student is assumed to be pass in CAT Examination if he is passed in all the four sections. Further, a student can either pass or fail in the exam.

Q.6. What is the number of students who passed CAT 2010?
(a) 508 (b) 1011
(c) 812 (d) None of these

Q.7. What is the number of students who failed because of failing in more than two sections?
(a) 108 (b) 324
(c) 672 (d) None of these

Q.8. What is the number of students who failed in QA?
(a) 200 (b) 492
(c) 711 (d) Cannot be determined

Q.9. What is the number of students who failed in EU?
(a) 400 (b) 108
(c) 732 (d) Cannot be determined

Direction for questions 10 to 13: Read the passage below and solve the questions based on it.

5% of the passengers who boarded Delhi-Lucknow Shatabdi express on 27th April 2005, do not like coffee, tea and lassi and 10% like all the three. 20% like coffee and tea, 25% like lassi and coffee and 25% like lassi and tea. 55% like coffee, 50% like tea, and 50% like lassi.

Q.10. The passengers who like only coffee is greater than the passengers who like only lassi by
(a) 25% (b) 100% (c) 75% (d) 0 %

Q.11. The percentage of passengers who like both tea and lassi but not coffee, is
(a) 15 (b) 25 (c) 40 (d) 75

Q.12. The percentage of passengers who like at least 2 of the coffee, tea and lassi, is
(a) 30 (b) 45 (c) 50 (d) 60

Q.13. If the number of passengers is 180, then the number of passengers who like lassi only, is
(a) 10 (b) 18 (c) 27 (d) 36

Q.14. Sandeep, Mohit and Hardeep are three financial analysts working at due North consultants. The sum of the number of projects handled by Sandeep and Hardeep individually is equal to the number of projects in which Mohit is involved. All three consultants are involved together in 6 projects. Sandeep works with Mohit in 14 projects. Hardeep has 2 projects with Mohit but without Sandeep, and 3 projects with Sandeep but without Mohit. The total number of projects is one less than twice the number of projects in which more than one consultant is involved. In how many projects Sandeep alone is involved?
(a) 1 (b) 0
(c) 3 (d) Cannot be determined

Q.15. There are a total of 78 persons in a big joint family. Each one of them watch atleast one out of Ramayana, Mahabharat, or Chanakya. Of these, 36 watch Ramayana, 48 watch Mahabharat and 32 watch Channakya. If 14 people watch both Ramayana and Mahabharat, 20 people watch both Mahabharat and

Chanakya and 12 people watch both Ramayana and Chanakya, find the ratio of the number of people who watch only Ramayana to the number of people who watch only Chanakya?
(a) 9:7 (b) 5:2 (c) 5:3 (d) 7:3

Q.16. In a class of 200 students, 70 played chess, 60 played hockey, and 80 played football. Thirty played chess and football, 30 played hockey and football, 40 played chess and hockey. If 130 people played at least one game, what is the number of people who play all the three games?
(a) 40 (b) 20
(c) 10 (d) None of these

Q.17. In a club, all the members are free to vote for one, two or three of the candidates. 20% of the members did not vote. 38% of the total members voted for at least two candidates. What percentage of the members voted for either one or three candidates if 10% of the total members who voted, voted for all the three candidates?
(a) 40% (b) 44%
(c) 36% (d) None of these

Q.18. In a class of 25 students, at least one of mathematics or statistics is taken by everybody. 12 have taken mathematics, 8 have taken mathematics but not statistics. Find the difference in the number of students who have taken mathematics and statistics and those who have taken statistics but not maths?
(a) 13 (b) 9
(c) 5 (d) Cannot be determined

Q.19. In a recent survey (conducted by HLL) of 1,000 houses, washing machine, vacuum cleaners and refrigerators were counted. Each house had at least one of these products. 400 had no refrigerators, 380 had no vacuum cleaners, 542 had no washing machines. 294 had both a vacuum cleaner and washing machine, 277 had both a vacuum cleaner and a refrigerator, and 120 had both a refrigerator and a washing machine. How many had only a vacuum cleaner?
(a) 132 (b) 234
(c) 342 (d) None of these

Q.20. A group of tourists (104 in all) speak Bengali or Spanish or Chinese. Out of these, 66 are Bengali speaking, 74 are Spanish speaking, 64 Chinese speaking, 44 speak Bengali and Spanish, 40 speak Bengali and Chinese, and 50 speak Chinese and Spanish. What is the number of tourists speaking all the three languages?
(a) 24 (b) 34
(c) 44 (d) 54

Q.21. In the above question the number of people speaking neither Spanish nor English, is
(a) 6 (b) 16
(c) 22 (d) Cannot be determined

ADVANCED

Q.1. We define, for two sets A and B, the following operations as:
A – B is the set of all elements of set A that are not the elements of set B,
$A \cup B$ is the set of all elements that appear in A or B or both A and B,
$A \cap B$ is the set of all elements that are common to both sets A and B.
If P, Q and R are three sets, then P-(Q-R) is equivalent to
(a) $(P-Q) \cup (P \cap R)$ (b) $P-(Q \cup R)$
(c) $P-(Q \cap R)$ (d) $(P-Q) \cup (P-R)$

Q.2. We define, for two sets A and B, the following operations as:
A – B is the set of all elements of set A that are not the elements of set B,
$A \cup B$ is the set of all elements that appear in A or B or both A and B,
$A \cap B$ is the set of all elements that are common to both sets A and B.
If P, Q and R are three sets, then (P-Q) – (Q-R) is equivalent to
(a) $P \cap (Q \cap R)$ (b) $P-(Q \cup R)$
(c) $(P \cap Q) \cup R$ (d) None of these

Q.3. A survey was conducted regarding the hobbies of a class having 200 students. The survey revealed that 85% of the students have played cricket, 80% have played basketball, 75% have played hockey, and 70% have played tennis. What is the minimum number of students who must have played all the four games?
(a) 15 (b) 25
(c) 40 (d) None of these

Q.4. There are three groups: A = {all men}, B = {all women}, and C = {men and women}, who have four members in common, and each pair of groups has 5 members in common. If each of the group has only eight members and if the common members are men only, what is the minimum number of men among the three groups taken together?
(a) 13 (b) 14
(c) 15 (d) None of these

Direction for questions 5 and 6: Read the passage below and solve these questions based on it:

Following is the result of an examination.
Out of 1,000 students who appeared

(a) 658 failed in physics
(b) 166 failed in physics and chemistry
(c) 372 failed in chemistry, 434 failed in physics and biology
(d) 590 failed in biology, 126 failed in biology and chemistry

Find the number of students who failed in (assuming that none of them passed in all the subjects)

Q.5. Chemistry but not physics.
(a) 312 (b) 226
(c) 266 (d) 206

Q.6. Physics or biology but not in chemistry.
(a) 560 (b) 710
(c) 620 (d) None of these

Direction for questions 7 and 8: Read the passage below and solve these questions based on it.

During break, 123 students go to the school canteen, which sells pastries, ice cream, and patties. 36 students buy patties and, 10 students buy only pastries. 15 students buy ice cream and patties, 10 students buy ice cream and pastries, 4 students buy pastries and patties but not ice cream and 11 students buy ice cream and patties but not pastries. 42 students buy ice cream.

Q.7. How many students buy nothing at all?
(a) 40 (b) 50 (c) 55 (d) 60

Q.8. How many students buy at least two items?
(a) 27 (b) 30 (c) 25 (d) 35

Direction for questions 9 to 12: Read the passage below and solve these questions based on it:

A and B are two sets (for e.g., A = mothers, B = women). The elements that could belong to both the sets (for e.g., women who are mothers) is given by the set $C = A \times B$. The elements which could belong to either set A or B or both is indicated by the set $D = A \cup B$. A set that does not contain any elements is known as a null set, represented by ϕ (for e.g., if none of the women in the set B is a mother, then $C = A \times B$ is a null set or $C = \phi$.

Let N signify the set of all the vertebrates; M the set of all the mammals; D the set of all the dogs; F the set of all the fish; A the set of all the alsatian and P is a dog named Party.

Q.9. Given that $X = M \times D$ such that $X = D$, then which of the following is true?
(a) All dogs are mammals
(b) Some dogs are mammals
(c) $X = \phi$
(d) All mammals are dog

Q.10. Given that $Y = F \times (D \times N) = \phi$. Which of the following is true?
(a) All fish are vertebrates
(b) All dogs are vertebrates
(c) Some fish are dogs
(d) None of these

Q.11. If $Z = (P \times D) \cup M$, then which of the following is true?
(a) The elements of Z consist of Party, the dog or any other mammal
(b) Z implies any dog or animal
(c) Z implies Party or any dog that is a mammal
(d) Z is a null set

Q.12. If $P \times A = \phi$, and $P \cup A = D$, then which of the following is true?
(a) Party and alsatians are dogs
(b) Party is an alsatian
(c) Party is not an alsatian
(d) D is a null set

Q.13. In a class of 200 students, 70 played cricket, 60 played hockey, and 80 played football. Thirty played cricket and football, 30 played hockey and football, 40 played cricket and hockey.
Find the maximum number of people playing all the three games and also the minimum number of people playing at least one game?
(a) 200,100 (b) 30, 110
(c) 30, 120 (d) None of these

Q.14. If class with n students is organized into four groups keeping the following conditions:
- Each student belongs to exactly two groups and
- Each pair of groups has exactly one student in common.

What is the value of n?
(a) $n = 11$ (b) $n = 7$
(c) $n = 9$ (d) None of these

Q.15. A survey show that 63% of the Americans like cheese whereas 76% like apples. If x% of the Americans like both cheese and apples then find the range of x.
(a) $0 \le x \le 23\%$ (b) $0 \le x \le 39\%$
(c) $4 \le x \le 35\%$ (d) $6 \le x \le 33\%$

Q.16. In Ranchi, only two newspapers *The Hindustan* and *The Indian Express* are published. It is known that 25% of the city population reads *The Hindustan* and 20% reads *The Indian Express*, while 8% read both newspapers. It is also known that 30% of those who read *The Hindustan* but not *The Indian Express* look into advertisements while 50% of those who read both

The Hindustan and *The Indian Express* look into advertisement. What percentage of the population reads an advertisement?
(a) 8.25% (b) 6.5%
(c) 11.75% (d) Cannot be determined

Q.17. In a certain examination of 3 subjects, 35 students failed in maths, 40 in physics and 40 in chemistry. 20 failed in maths and physics, 17 in physics and chemistry, 15 in maths and chemistry. If 132 students did not fail in any subject, what is the maximum number of students that may have appeared for the examination?
(a) 95 (b) 200 (c) 210 (d) 246

Direction for questions 18 to 20: Read the passage below and solve these questions based on it:

There are 120 players, having a chest number 1 through 120, seated in a circular pattern in the same order. Since the selector is involved in game fixing, he tries to ensure that none of the right candidate for right game should go. For this to happen, he starts selecting the players in a very arbitrary way for four games namely shot-put, javelin throw, discus throw and sprint: Player numbered 1 is selected for sprint and after that, every 11th player is selected for sprint. The counting continues around the circle repeatedly until it ends at player number 1. Similarly, starting with player number 1, every 9th player is selected for shot–put, every 7th player for javelin throw and every 8th player for discus throw.

Q.18. How many players are not selected for any of the four events?
(a) 45 (b) 65
(c) 33 (d) None of these

Q.19. How many players are selected for all the four events?
(a) 1 (b) 3
(c) 5 (d) None of these

Q.20. How many players were selected neither for discus throw nor shot-put?
(a) 12 (b) 14
(c) 16 (d) None of these

ANSWERS

■■■ Warm Up

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Answer	(c)	(b)	(b)	(c)	(b)	(a)	(d)	(b)	(c)	(b)	(c)	(a)	(c)	(a)	(d)

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(d)	(c)	(a)	(a)	(a)	(b)	(c)	(c)	(c)	(d)	(b)	(d)	(b)	(a)	(d)	(b)	(a)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(c)	(d)	(a)	(c)	(a)	(d)	(b)	(a)	(c)	(b)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(d)	(d)	(a)	(d)	(d)	(d)	(d)	(b)	(a)	(c)	(b)	(d)	(c)	(b)	(d)	(b)	(d)	(b)
Q. No.	21																			
Answer	(b)																			

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(d)	(d)	(d)	(d)	(d)	(b)	(c)	(a)	(c)	(a)	(c)	(b)	(d)	(b)	(d)	(c)	(a)	(b)	(d)

HINTS AND SOLUTIONS

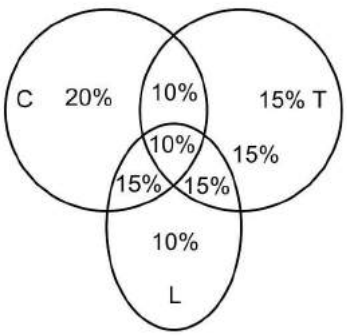
Moderate

1. $n(1 \cup 2 \cup 3) = n(1) + n(2) + n(3) - n(1 \cap 2) - n(2 \cap 3) - n(1 \cap 3) + n(1 \cap 2 \cap 3)$
   Or, 78% = 50% + 30% + 20% − $n(1 \cap 2)$ − $n(2 \cap 3)$ − $n(1 \cap 3)$ + 5%
   $n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)$ = 27%
   Only $[n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)]$ = 27% − 15% = 12%, since $n(1 \cap 2 \cap 3)$ is present in all the three.
   Percentage of people against more than one of the three proposals = Only $[n(1 \cap 2) + n(2 \cap 3) + n(1 \cap 3)] + n(1 \cap 2 \cap 3)$ = 12% + 5% = 17%

2. To find out the maximum percentage of people watching all the three channels, we can assume that the people who watch ZEE, which is minimum among all the three channels, are watching all the three channels. Any value of percentage of people more than this 15% will make the percentage of people watching ZEE negative. Hence, maximum percentage of people watching all the three channels = 15%

4. In this question, we can talk about a range of percentages who watch all the three channels. We cannot answer definitely what percentage of people watched all the three channels?

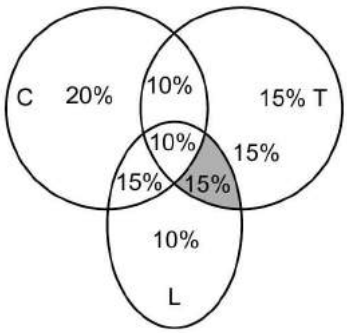
[redacted]



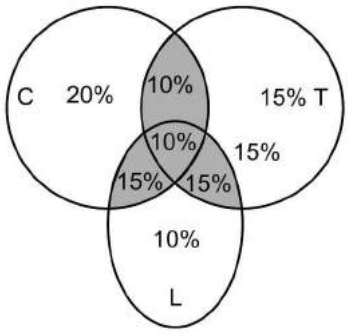
Where C = Coffee, T = Tea and L = Lassi.

10. The passengers who like only coffee = 20% and the passengers who like only lassi = 10%.

11. It can be seen that the percentage of passengers who like both tea and lassi but not coffee = 15%. This is the figure representing this area:



12. The percentage of passengers who like at least 2 of the coffee, tea and lassi can be seen in the below figure:



13. 10% of the people likes only lassi. So, the number of persons = 18

14. Go through the worked-out examples section.

15. Use the formula: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

17. The statement "12 have taken mathematics, 8 have taken mathematics but not statistics" means that 4 students have taken mathematics and statistics both.
    Total number of students = 25. Out of this, 8 have taken only mathematics, 4 have taken both. Hence 13 have taken only statistics.
    So, the difference = 13 − 4 = 9

18. Solution to this question requires only the application of: $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$.

Advanced

[redacted]

It is a simple application of the formula
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - c(A \cap B) - n(B \cap C) - (A \cap C) + (A \cap B \cap C)$

[redacted]

It is a simple application of the formula
$n(A \cup B \cup C) = n(A) + n(B) + n(C) - c(A \cap B) - n(B \cap C) - (A \cap C) + (A \cap B \cap C)$

9. $X = M \times D \rightarrow$ Mammals who are dogs. Now since X = D, so all the dogs are mammals.

15. $n(C) = 63\%$
$n(A) = 76\%$
$n(C \cup A) = n(C) + n(A) - n(C \cap A)$
$100\% = 63\% + 76\% - X\%$
$X\% = 39\%$

18. $n(\text{sprint}) = 19$
$n(\text{shot}) = 23$
$n(\text{javelin} = 28$
$n(\text{discus}) = 25$
$n(\text{sprint} \cap \text{short}) = 2$
$n(\text{shot} \cap \text{javelin}) = 3$
$n(\text{javelin} \cap \text{discus}) = 3$
$n(\text{discus} \cap \text{sprint}) = 0$
$n(\text{sprint} \cap \text{shot} \cap \text{javelin} \cap \text{discus}) = 0$

21

CHAPTER

Miscellaneous

LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- Concept of Calendars, Inequality and Height and Distance
- Definitions and properties
- Kind of questions which are asked at CAT
- Method of solving questions

Introduction to the topic

If we go into the history of CAT, we will conclude that CAT has never spared the above collection of miscellaneous concepts while asking questions. 1-3 questions have been asked from these topics every year.

Let us see each one of these concepts one-by-one.

CALENDAR

We know that

Any non-leap year contains 365 days = 52 weeks + 1 day

And leap-year contains 366 days = 52 weeks + 2 days

This 1-day and 2 days extra added to any year create changes in the calendar and this is the reason why calendar of Nth year will not be same as N + 1th year.

Before we proceed ahead, we should be very clear about two things:

Which years are leap years?

It takes the earth about 365.2422 days to go around the sun, but a normal calendar year is only 365 days. The extra fraction of a day added up four times makes four years (or, four revolution of earth around sun) 1460.9688 days, but four calendar years would only be 1460 days. That 0.9688 is almost a whole day, so every four years we add an extra day to our calendar, February 29. We call that year leap year. To make things easier, leap years are always divisible by four: 2004 and 2008 will both be leap years.

For hundreds of years, people used a calendar called the Julian calendar that followed this rule, adding a leap year every four years. However, because 0.9688 isn't *exactly* a whole day, the Julian calendar slowly began to disagree with the real seasons. In 1582, Pope Gregory fixed this problem by ordering everyone to use a new set of rules. These rules are named the Gregorian calendar, after him. They work like this:

The Gregorian Calendar

Rule	Examples
Every fourth year is a leap year.	2004, 2008, and 2012 are leap years.
However, every centenary year is *not* a leap year.	1900 and 2100 are *not* leap years.
In case of centuries, every four hundred years, there's a leap year after all.	2000 and 2400 are leap years.

In layman terms, all the century years divisible by 400 will be leap years and all the non-century years divisible by 4 will be leap years. So, leap year next to 2096 AD is 2104 AD and not 2100 AD.

Because 2000, 4000, 6000, etc. are leap years and 1000, 3000, 5000, etc. are not, the number of leap days in each millennium alternates between 242 and 243, with the first, third, etc. millennia (i.e., 1–1000, 2001–3000, etc.) having 242 leap days, and the second, fourth, etc. (i.e., 1001–2000, 3001–4000, etc.) having 243 leap days.

How the days of consecutive years change?

Due to any non-leap year, calendar of next year go ahead by 1 day and due to any leap year, calendar of next year goes ahead by 2 days, but this change in calendar will be there only before 29th February.

It can be seen through the example given below:

	1991	1992	1993
1st January	Sunday	Monday	Wednesday
28th February	Tuesday	Wednesday	Friday
1st March	Wednesday	Friday	Saturday

In the above example, we have assumed that 1st January of 1991 is Sunday. 1991 and 1993 are non-leap years and 1992 is a leap year.

If now we try to find out the symmetricity of calendars, we can see this in the following way:

i. For any leap-year
Let us see, for example, the case of 1972.

Year	1972	1973	1974	1975	1976
Excess days	2	1	1	1	2

Since no. of excess days are 7, so the days of the year 1972 and year 1977 will be same from 1st of January and 28th of February.

ii. For any (leap-year+1) year

Year	1973	1974	1975	1976	1977	1978
Excess days	1	1	1	2	1	1

Since no. of excess days are 7, so calendar of year 1973 and 1979 will be same for whole year.

iii. For any (leap-year + 2) year

Year	1974	1975	1976	1977	1978	1979
Excess days	1	1	2	1	1	1

Since excess days are 7, so calendar of year 1974 and 1980 will be same till 28th of February.

iv. For any (leap-year + 3) year

Year	1975	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985
Excess days	1	2	1	1	1	2	1	1	1	2	1

Since no. of excess days are 14, so calendar of year 1975 and 1986 will be same for whole year.

This whole mechanism can be summed up in following way:

Nature of year	No. of years after which 1st January will be same
Leap year	5
Leap year + 1	6
Leap year + 2	6
Leap year + 3	11

So, if 1st January of 1972 and 1st January of 1977 will be on same day.

If 1st January of 1973 and 1st January of 1979 will be on same day and so on.

Exception

No century year, which is not a leap year, should be included in this calculation.

Example 1 Sum of dates of last Monday of previous month and 1st Thursday of next month is 38. If both the dates are of the same year, then which month is the current month?

Solution Sum of dates of last Monday of previous month and 1st Thursday of next month is 38 is possible only if last Monday is 31st and 1st Thursday is 7th. (Since if we take 30 + 8 = 38, then 30 can be last Monday of any month but 8th can not be the 1st Thursday of any month)

So, 31st of last month is a Monday. Hence 7th of current month, 14th of current month, 21st of current month and 28th of current month will be a Monday.

Now, if current month is a month with 30 days, then 5th of next month will be a Monday, so 7th of next month cannot be a Thursday.

If current month is a month with 31 days, then 4th of next month will be a Monday, so 7th of next month will be a Thursday.

Finally we can conclude that previous month and current month, both are having 31 days. Since both the dates are of the same year, so current month is August.

Finding day of a date by using a reference date

Let us see this with the help of an example: If 9th Dec of 1972 is Sunday, then which day it will be on 14th December 1998?

Process: There are several processes to do this calculation: 1. Year method, 2. Days method, 3. Actual calculation method

1. **Year Method** – We use the above given table to find out about any of the years.

   9.12.1972 – Sunday
   1.1.1973 – Tuesday (It is a Leap year + 1 year)
   So, 1.1.1979 – Tuesday, (It is a Leap year + 3 year)
   So, 1.1.1990 – Tuesday, (It is a Leap year + 2 year)
   So, 1.1.1996 – Tuesday
   Now, we can find out all the next years one-by-one.
   1.1.1997 – Thursday
   1.1.1998 – Friday – 31.12.1998 – 24.12.1998 – 17.12.1998
   So, 14.12.1998 – Tuesday

2. **Days method** – We use the no. of excess days every year to find out the no. of days calendar will move ahead by.
   1.1.1973 – Tuesday
   Due to 1973, calendar will go ahead by 1 day, similarly due to 1974 – 1 day, due to 1975 – 1 day and due to 1976- 2 days.
   So, in four years, calendar will go ahead by 5 days.
   Using unitary method, in four years, calendar will move ahead by 5 days.
   So, in 24 years calendar will move ahead by 30 days.
   Hence calendar will move ahead by 2 days.
   So, 1.1.1997 will be two days ahead of Tuesday i.e. Thursday.
   Now, it is calculation as given in **Year Method.**

3. **Actual Calculation method** – With the help of this method, we can find the actual day of any date of 20th century. To use this method effectively, we need to remember the Month Codes of all the months.
   Let us learn this method by finding the date of 15th August 1947 –
   At 1st, add the Date + Month code of August + Last two digits of year + $\left[\frac{\text{Last two digits of year}}{4}\right]$
   (Where $[x]$ represents the greatest integer value of x.)
   So, corresponding to 15th August 1947 – 15 + 3 + 47 + 11 = 76
   Now, divide this value by 7 to find out the remainder.
   If the remainder is 0 → then it is a Saturday
   If the remainder is 1 → then it is a Sunday
   If the remainder is 2 → then it is a Monday
   If the remainder is 3 → then it is a Tuesday
   If the remainder is 4 → then it is a Wednesday
   If the remainder is 5 → then it is a Thursday
   If the remainder is 6 → then it is a Friday
   Here, remainder is 6, so 15th August 1947 was a Friday. (It should have been 'Free'day)

List of Month Code:

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Leap Year	0	3	4	0	2	5	0	3	6	1	4	6
Non-leap year	1	4	4	0	2	5	0	3	6	1	4	6

INEQUALITIES

While discussing Equations and Inequations in Polynomials, we have seen what Inequations are. However while discussing Inequalities here, our focus will be to discuss this concept in isolation with Equations, for real numbers only.

Tools of Inequality

'>' means 'greater than'
'<' means 'less than'
'≥' means 'greater than or equal to'
'≤' means 'less than or equal to'

If 'N' is any real number, then value of 'N' will be either positive or negative or zero.

When 'N' is positive, we write $N > 0$; which is read 'N is greater than zero'.

Similarly When 'N' is negative, we write $N < 0$; which is read as 'N is less than zero'. If 'N' is zero, we write $N = 0$ and in this case, 'N' is neither positive nor negative.

The two signs '>' and '<' are called the signs of inequalities.

Standard Definitions

For any set of real numbers M and N,

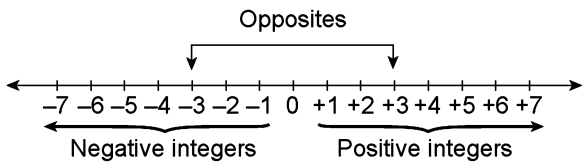
(i) M is said to be greater than N when $M - N$ is positive
⇒ $M > N$ when $M - N > 0$ and
As we can see, 10 is greater then 5 because $10 - 5 = 5$ and 5 is greater than zero.

(ii) M is said to be less than N when $M - N$ is negative.
⇒ $M < N$ when $M - N < 0$.
As we can see, -10 is less than -5 because $-10 - (-5) = -5$ and -5 is less than zero.

However, in case of numbers inequalities can be understood through Number Line also.

Number Line

The number line is used to represent the set of real numbers. Below is the brief representation of the number line:



Basics of Inequalities

It is quite pertinent here to understand some of the very basic properties related to inequalities. These properties should be seen as the building block of the concepts of Inequalities. Assume all the numbers used here are real numbers.

For any two real numbers M and N, either $M > N$ or $M < N$ or $M = N$.

If $M > N$, then $N < M$.

- If $M > N$ and $N > P$, then $M > P$.
- If $M < N$ and $N < P$, then $M < P$.
- If $M > N$, then $M \pm c > b \pm c$.
- If $M > N$ and $P > 0$ then $MP > NP$.
- If $M > N$ and $P < 0$, then $MP < NP$.
- If $M < N$ and $P > 0$, then $MP < NP$.
- If $M < N$ and $P < 0$, then $MP > NP$.
- If $M > N$ and $P > Q$, then $M + P > N + Q$.
- If $M < N$ and $P < Q$, then $M + P < N + Q$.
- However if $M > N$ and $P < Q$, Or, $M < N$ and $P > Q$ then we cannot comment about the inequality between $(M + P)$ and $(N + Q)$.
- If $M > N$ and $P > Q$, then we cannot infer the inequality sign between $(M - N)$ and $(P - Q)$. Depending on the values of M, N, P and Q, it is possible to have $(M - N) > (P - Q)$, $(M - N) = (P - Q)$ or $(M - N) < (P - Q)$
- The square of any real number is always greater than or equal to 0.
- The square of any NON-ZERO real number is always greater than 0.
- If $N > 0$, then $-N < 0$ and if $M > N$, then $-M < -N$.
- If M and N are positive numbers and $M > N$, then
  (i) $1/M < 1/N$
  (ii) $M/P > N/P$ if $P > 0$ and
  (iii) $M/P < N/P$ if $P < 0$
- For any two positive numbers M and N
  If $M > N$ then $M^2 > N^2$.
  If $M^2 > N^2$, then $M > N$.
  If $M > N$, then for any positive value of n, $M^n > N^n$.
- **For two positive numbers M and N**
  If $M/N < 1$ then $M < N$
  If $M/N = 1$ then $M = N$
  If $M/N > 1$ then $M > N$
- **Relationship between a number and its square root**
  Let N be a natural number.
  $\sqrt{N} = N$, for $N = 0$ or $N = 1$
  $\sqrt{N} > N$, for $0 < N < 1$
  $\sqrt{N} < N$, for $N > 1$
- Let A, G and H be the Arithmetic Mean, Geometric Mean and Harmonic Mean of n positive real numbers, then $A \geq G \geq H$. Equality occurs only when all the numbers are equal.
- If the sum of two positive quantities is given, their product is greatest when they are equal; and if the product of two positive quantities is given, their sum is least when they are equal.
- For any positive number, the sum of the number and its reciprocal is always greater than or equal to 2,
  i.e., $x + \frac{1}{x} \geq 2$ where $x > 0$.
  The equality in this relationship will occur only when $x = 1$.

Cauchy–Schwarz Inequality

If $a_1, a_2, \ldots\ldots a_n$ and $b_1\ b_2 \ldots\ b_n$ are $2n$ real numbers, then

$$(a_1b_1 + a_2b_2 + \ldots.. + a_nb_n)^2$$
$$\leq \left(a_1^2 + a_2^2 + \ldots\ldots a_n^2\right)\left(b_1^2 + b_2^2 \ldots\ldots + b_n^2\right)$$

With the equality holding if and only if

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \ldots\ldots = \frac{a_n}{b_n}$$

Despite all the points given above, however, we should not let the LOGIC die. Most of the questions asked in CAT can be solved by using options and we won't be in need of using any concept of Inequalities. But this should not be seen as a case in support of not-going through concepts.

Example 2 If $R = \dfrac{30^{65} - 29^{65}}{30^{64} + 29^{64}}$, then (CAT 05)

(a) $0 < R \leq 0.1$ (b) $0.1 < R \leq 0.5$
(c) $0.5 < R \leq 1.0$ (d) $R > 10$

Solution $\dfrac{30^{65} - 29^{65}}{30^{64} + 29^{64}} > 1$

As, $30^{65} - 29^{65} > 30^{64} + 29^{64}$

Or, $30^{64}(30 - 1) > 29^{64}(29 + 1)$

Or, $30^{64} \times 29 > 29^{64} \times 30$

Or, $30^{63} > 29^{63}$

Hence option (4)

Example 3 If $13x + 1 < 2z$, and $z + 3 = 5y^2$, then (CAT 03)

1. x is necessarily less than y.
2. x is necessarily greater then y.
3. x is necessarily equal to y.
4. None of the above is necessarily true.

Solution We have

$13x + 1 < 2z$ .... (1)

and $z + 3 = 5y^2$ .... (2)

From (1) and (2) we get

$13x + 1 < 2(5y^2 - 3)$

$\Rightarrow\ 13x + 1 < 10y^2 - 6$

$\Rightarrow\ 13x < 10y^2 - 7$

$\Rightarrow\ x < \dfrac{10y^2 - 7}{13}$

If $y = 1$ then, we get $x < 0.230$

i.e. $y > x$.

If $y = 2$. Then, we get $x < 2.538$.

i.e. $x > y$. This is not possible. Hence correct answer is (4).

Example 4 If $|b| \geq 1$ and $x = -|a|\,b$, then which one of the following is necessarily true? (CAT 03)

(a) $a - xb < 0$ (b) $a - xb \geq 0$
(c) $a - xb > 0$ (d) $a - xb \leq 0$

Solution Lets start assuming the values of a and b.

Assume $b = 2$ and $a = 1/2$. Then $x = -1$.

Thus options 1 and 4 have been eliminated.

Assuming $a = -1$ and $b = -1$, we get $x = 1$. Putting the values in options 1 and 2, we get **Ans.(2)**

HEIGHTS AND DISTANCES

General Theory on Heights and Distances

Angle of elevation: If the object is above the horizontal level of the eyes i.e. if it is above the eye level), we have to turn our head upwards to vie the object. In this process, our eyes move through an angle. Such an angle is called the angle of elevation of the object from our eyes.

Angle of depression: Suppose a boy, standing on the roof of a building, observes a ball lying on the ground at some distance form the building. In this case, he has to move his head downwards to view the ball. In this process, his eyes again move through an angle. Such an angle is called the angle of depression of the object from the location of his eyes (figure.)

SOME TRIGONOMETRIC RATIOS AND FORMULAE

	0°	30°	45°	60°	90°
Sin	0	1/2	1/√2	√3/2	1
Cos	1	√3/2	1/√2	1/2	0
Tan	0	1/√3	1	√3	∞

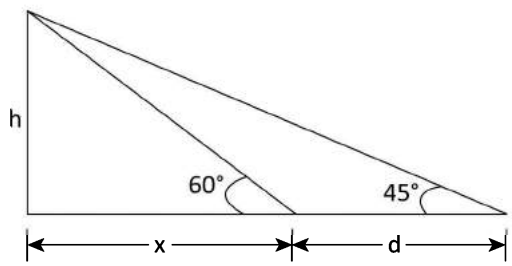
$\sin(-\theta) = -\sin\theta$; $\cos(-\theta) = \cos\theta$; $\tan(-\theta) = -\tan\theta$);

$\cot(-\theta) = -\cot\theta$; $\sec(-\theta) = \sec\theta$; $\text{cosec}(-\theta) = -\text{cosec}\theta$);

$\sin(90^0 - \theta) = \cos\theta$; $\cos(90^0 - \theta) = \sin\theta$

$\sin 2\theta = 2\sin\theta.\cos\theta$

$\cos 2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$



Example 5 A car is being driven, in a straight line and at a uniform speed, towards the base of a vertical tower. The top of the tower is observed from the car and, in the process, it takes 10 minutes for the angle of elevation to change from 45° to 60°. After how much more time will this car reach the base of the tower?

(a) $5(\sqrt{3}+1)$ (b) $6(\sqrt{3}+\sqrt{2})$

(c) $7(\sqrt{3}-1)$ (d) $8(\sqrt{3}-2)$

Solution

$$\tan 45^\circ = \frac{h}{x+d}$$

or, $h = x + d$

Similarly, $\tan 60^\circ = \frac{h}{x}$

or, $h = x\sqrt{3}$

Hence, $x + d = x\sqrt{3}$,

So, $x(\sqrt{3}-1) = d$

Car takes 10 mins in covering 'd' distance, so car will take $10/(\sqrt{3}-1)$ min in covering 'x'$[x = d/(\sqrt{3}-1)]$ distance.

Now, $t = 10/(\sqrt{3}-1)$

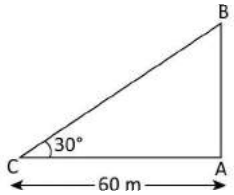
Multiplying both numerator and denominator by $(\sqrt{3}+1)$, we get $t = 5(\sqrt{3}+1)$

Example 6 The angle of elevation of the top of a tower from a point 60m from its foot is 30^0. What is the height of the tower?

Solution Let AB be the tower with its foot at A.

Let C be the point of observation.

Given $\angle ACB = 30^0$ and AC = 60m



From right triangle BAC:

$$\frac{AB}{AC} = \tan 30^0$$

$$\Rightarrow \quad AB = 60 \tan 30^0 = 60\left(\frac{1}{\sqrt{3}}\right) = 20\sqrt{3}$$

Example 7 Two pillars of equal height stand on either side of a road. At a point on the road between the pillars the elevation of the tops of the pillars are 60^0 and 30^0. Find the height of the pillars if it is given that the width of the road is 150m.

Solution $\tan 60^0 = \sqrt{3} = h/x$

$\tan 30^0 = 1/\sqrt{3} = h/(150 - X)$

Solving we get: $x = 37.5$m, $h = 64.95$m.

PRACTICE EXERCISES

FOUNDATION

Q.1. The total number of terms of in the expansion of $(x + a)^{100} + (x - a)^{100}$ after simplification is

(a) 202 (b) 51

(c) 50 (d) None of these

Q.2. The number of terms of terms in the expansion $(x + y + z)^{10}$ is

(a) 11 (b) 33

(c) 66 (d) None of these

Q.3. In the expansion of $(1 + x)^{50}$, the sum of the coefficients of odd powers of x is

(a) 0 (b) 2^{49}

(c) 2^{50} (d) 2^{51}.

Q.4. What is the no. of leap years in between any 100 consecutive years?

(a) 24 (b) 25

(c) 26 (d) cannot be determined

Q.5. Year 200X is a leap year. Which of the following year will definitely be a leap year again?

(a) 200X + 4 (b) 200X + 5

(c) 200X + 6 (d) cannot be determined

Q.6. June 10, 1979, was a SUNDAY. Then May 10, 1972, was a

(a) Wednesday (b) Thursday

(c) Tuesday (d) Friday

Q.7. In a certain year, the month of January had exactly four Wednesdays and four Sundays. Then January 1 of that year was a
(a) Monday (b) Thursday
(c) Friday (d) Saturday

Q.8. A. I was born in a century in which the last year of the previous century was leap year.
B. I was born in the first year after the first leap year of the century.
C. I was born after 1200 A.D. and before 2000 A.D.
What is the year of my birth?
(a) 1405 (b) 1505
(c) 1605 (d) 1705

Q.9. If $a \leq 22$ and $b \geq 12$, then which of the following is always true?
(a) $a > b$ (b) $a - b > 10$
(c) $a - b \leq 10$ (d) $a + b \geq 34$

Q.10. If x, y and z are such that $x < y$ and $z > 0$, then
(a) $\frac{x}{y} > 1$ (b) $\frac{z}{x} < \frac{z}{y}$
(c) $x > \frac{y}{z}$ (d) $xz < yz$

Q.11. Which of the following satisfies the condition: $\frac{1}{x} > -10$
(a) $x + 2 = 0$ (b) $x + 3 = 0$
(c) $x + 5 = 0$ (d) None of these

Q.12. Which of the following satisfies the condition: $\frac{1}{x} < -10$
(a) $x - 2 = 0$ (b) $x - 3 = 0$
(c) $x - 5 = 20$ (d) None of these

Q.13. $x + 2 < \sqrt{x+14}$
(a) $-2 \leq x < 14$ (b) $-14 \leq x < 2$
(c) $x \leq -2$ or $x > 14$ (d) $x < -14$ or $x > 2$

Q.14. $(x-1)(3-x)(x-2)^2 > 0$
(a) $1 < x < 3$ (b) $1 < x < 3, x \neq 2$
(c) $-1 < x < 3, x \neq 2$ (d) $-1 < x < 3$

Q.15. $\frac{x}{x-5} > \frac{1}{2}$
(a) $-5 < x < 5$ (b) $x < -5$ or $x > 5$
(c) $-5 < x \leq 0$ (d) $0 < x \leq 5$

Q.16. $\frac{3}{x-2} < 1$
(a) $2 < x < 5$ (b) $-2 \leq x < 5$
(c) $-5 \leq x \leq 2$ (d) $x < 2$ or $x > 5$

Q.17. For how many values of x is the following equation true $|2x + 3| + x = 10$?
(a) 1 (b) 2
(c) 0 (d) None of these

Q.18. From the top of a lighthouse 60 metres high with its base on the sea level, the angle of depression of a boat is 15°. Find the distance of the boat from the foot of the lighthouse.
(a) $60(2+\sqrt{3})$m (b) $60(1+\sqrt{3})$m
(c) $60\sqrt{3}$m (d) $100(2+\sqrt{3})$m

Q.19. From the top of a cliff 200 metres high, the angles of depression of the top and bottom of a tower are observed to be 30°and 60° respectively. Find the height of the tower.
(a) $400\sqrt{3}$ m (b) $200\sqrt{3}$ m
(c) $100\sqrt{3}$ m (d) None of these

Q.20. The angle of elevation of a moon when the length of the shadow is equal to its height is
(a) 30° (b) 60°
(c) 45° (d) none of these

Q.21. The ratio of the length of a rod and its shadow is $1 : \sqrt{3}$. The angle of elevation of the sun is
(a) 30° (b) 45°
(c) 60° (d) 90°

Q.22. A tower stands on a horizontal plane. A man on the ground 100 m from the base of the tower finds the angle of elevation of the top of the tower to be 30°. What is the height of the tower?
(a) 100 m (b) $100\sqrt{3}$
(c) $100/\sqrt{3}$ (d) None of these

Q.23. When the sun is 30° above the horizontal, the length of shadow cast by a building 50 m high is
(a) $\frac{50}{\sqrt{3}}$ m (b) $50\sqrt{3}$ m
(c) 25 m (d) $25\sqrt{3}$ m

Q.24. A pole beimg broken by the wind, the top a struck the ground at an angle of 30° and a distance of the 21 m from the foot of the pole. Fimd the total height of the pole.
(a) 21 m 2.21/√3
(c) $21\sqrt{3}$ m (d) None of these

Q.25. The shadow of a tower standing on a level plane is found to be 50 m longer when the altitude of the sun is 30° than when it is 60°. Find the height of the tower.
(a) $20\sqrt{3}$m (b) $25\sqrt{3}$ m
(c) $25/\sqrt{3}$ m (d) None of these

Q.26. From the top of a cliff 25 m high, the angle of elevation of a tower is found to be equal to the angle of depression of the foot of the tower. Find the height of the tower.

(a) 40 m (b) 48 m
(c) 50 m (d) 52 m

Q.27. When the length of the shadow of a pole is equal to the height of the pole, then the elevation of source of light is
(a) 30° (b) 45°
(c) 60° (d) 75°

Q.28. If I set out from my office at quarter past six and reach the station at half past seven, I will be ten minutes earlier than the person who is fifteen minutes late to the train. At what time does the train start?
(a) 7.25 p.m. (b) 7:35 p.m.
(c) 7:40 p.m. (d) 7:20 p.m.

Q.29. If June 8, 2006 is Thursday, then which day would be 15th August 1947?
(a) Sunday (b) Saturday
(c) Friday (d) Thursday

Q.30. 50 kids are standing in a circle and one of them is having a ball. The ball is passed counter clockwise, skipping 0, 1, 2, 3, 4 and so on kids respectively, i.e., the 1st kid gives it to the 2nd kid, then 2nd kid gives it to 4th kid, then 4th kid gives it to 7th kid and so on. After how many hands change will the ball be back to the 1st kid?
(a) 12 (b) 20
(c) 22 (d) None of these

MODERATE

Q.1. Jappi and Diksha are playing a card game involving green and red cards. They get 3 points every time they draw a green card and 5 points every time they draw a red card. The winner is the first one to reach 50 points. At one stage Diksha has 41 points. Which of the following could not be the total number of cards drawn by Diksha at that stage?
(a) 7 (b) 12
(c) 10 (d) None of these

Q.2. If the rate of change of the length of a rectangle is 5 m per second and the rate of change of breadth of the same rectangle is 3 m per second, then the rate of change of area of this rectangle, when length = 10m and breadth = 2 m, is:
(a) 30 m^2 per second (b) 20 m^2 per second
(c) 50 m^2 per second (d) 40 m^2 per second

Q.3. A cube of side of 4 cm is painted on all 6 faces with the same colour. It is then broken up into smaller cubes, each of side 1 cm. What is the proportion of the number cubes with no paint to the number of cubes with exactly one side painted to those with exactly two sides painted?
(a) 1:2:3 (b) 3:6:8
(c) 1:3:4 (d) 1:3:3

Q.4. The following data are available for the monsoon season of the Hyderabad racing club. The data is for a total of 'y' days.
- There were races on 11 days – morning or evening.
- Whenever there was a race in the morning, there was no race in the evening.
- There were 8 mornings without any race.
- There were 5 evenings without any race.

What is the value of y?
(a) 10 (b) 12
(c) 14 (d) Cannot be determined

Q.5. A gathering of a certain number of families consists of people to two generations only. It is known that the number of families is less than the number of girls, the number of girls is less than the number of boys and that the number of boys is less than the number of parents. If the minimum number of single parent families is two, then what is the minimum number of families in the gathering, given that no family has more than three children?
(a) 4 (b) 5
(c) 3 (d) None of these

Q.6. 1st January of Year 19XY and 1st January of year 19PQ are same day. It is also given that no other year before 19PQ has its 1st January on the same day as that of 1st January of 19XY.What is the minimum possible value of | PQ-XY|, where PQ and XY are the last two digits of the year?
(a) 5 (b) 6
(c) 11 (d) 13

Q.7. In the above question, what is maximum possible value of | PQ-XY|, where PQ and XY are the last two digits of the year?
(a) 5 (b) 6
(c) 11 (d) 13

Direction for questions 8 and 9: Read the passage below and solve the questions based on it.

January 1 of the year 1 of a century falls on a Sunday. Let Year X be the earliest year by which January 1 has fallen on all days of the week.

Q.8. What is year X?
(a) 7 (b) 10
(c) 8 (d) 9

Q.9. Which of the following occurs the most number of times by year X (including year X)?
(a) Mondays
(b) Sundays, Mondays and Wednesdays
(c) Mondays and Wednesdays
(d) None of these

Q.10. Given that $-3 \le a \le 2$, $0.5 \le b \le 3$ and $-2.5 \le 0.5 - 0.5$ If $x = \frac{a^2}{bc}$, then which of the following is necessarily true?
(a) $-16 \le x \le 0$ (b) $-6 \le x \le 16$
(c) $-18 \le x \le 35$ (d) $-36 \le x \le 0$

Q.11. Given that $-3 \le x \le 4$ and $1 \le y \le 6$. Which of the following expressions can have the least possible value?
(a) $x^2y + y^2x$ (b) $xy^3 - yx^2$
(c) $x^2y^2 - xy^3$ (d) $xy^2 + y^2x^3$

Q.12. n is a real number. It is given that $n!^2 > n^n$. For what values of n relationship holds good?
(a) All even nos. (b) All natural nos.
(c) All whole nos. (d) None of these

Q.13. If $|x^2 - 7x + 12| > x^2 - 7x + 12$, then
(a) $x \le 3$ or $x \ge 4$
(b) $3 \le x \le 4$;
(c) $3 < x < 4$
(d) x can take any value except $x = 3$ and $x = 4$.

Q.14. The inequality
$$\frac{|x|^2 - |x| - 2}{2|x| - |x|^2 - 2} > 2$$
holds if and only if
(a) $-1 < x < -2/3$ or $2/3 < x < 1$;
(b) $-1 < x < 1$;
(c) $\frac{2}{3} < x < 1$;
(d) $x > 1$ or $x < -1$ or $-\frac{2}{3} < x < \frac{2}{3}$

Q.15. For natural nos. N, the inequality $2^N > N^2$ holds when
(a) $N > 2$
(b) $N = 1$ and $N > 4$
(c) $N > 1$
(d) None of these

Q.16. What are the values of x satisfying
$|x - 2| \le 2$ and $|x + 3| \ge 4$.
(a) $1 < x < 4$
(b) $1 \le x \le 4$
(c) $1 \le x$ or $x \ge 4$
(d) $1 < x$ or $x > 4$

Q.17. $\frac{x^2 - 5x + 6}{|x| + 7} < 0$
(a) $2 \le x < 5$ (b) $-2 \le x \le 3$
(c) $2 < x < 3$ (d) $2 \le x \le 3$

Q.18. $|x^2 + 3x| + x^2 - 2 \ge 0$
(a) $-1/2 \le x < ½$ (b) $x \le -½$ or $x \ge ½$
(c) $x \ge ½$ and $x \le ½$ (d) None of these

Q.19. $\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$
(a) $2 < x < 3$ (b) $x < 2$ or $x > 3$
(c) $2 \le x < 3$ (d) $2 \le x \le 3$

Q.20. In a group of 120 persons there are 80 Bengalis and 40 Gujratis. Further, 70 persons in the group are Muslims and the remaining are Hindus. Then the number of Bengali Muslims in the group is
(a) between 10 and 14
(b) between15 and 19
(c) exactly 20
(d) 25 or more

Q.21. 300 persons are participating in a meeting in India, out of which 120 are foreigners, and the rest are Indians. Out of the Indians, there are 110 men who are not judges; 160 are men or judges, and 35 are women judges. There are no foreign judges. How many Indian women attended the meeting?
(a) 60 (b) 45
(c) 55 (d) 40

Q.22. There are six persons: A, B, C, D, E and F.
A has 3 items more than C.
D has 4 items less than B.
E has 6 items less than F.
C has 2 items more than E.
F has 3 items more than D.
Which one of the following figures can not be equal to the total number of items possessed by all the six persons?
(a) 41 (b) 53
(c) 47 (d) 58

Q.23. The angle of elevation, of the top of a tower from a point 20 m away from its base is 45°. The height of the tower is
(a) 10 m (b) 20 m
(c) 40 m (d) $200\sqrt{3}$ m

Q.24. The angle of elevation of the sun when the length of the shadow of a pole is $\sqrt{3}$ times the height of the pole will be

(a) 30° (b) 60°
(c) 90° (d) 45°

Q.25. A person walking along a straight road towards a hill observes at two points, distance $\sqrt{3}$ km, the angles of elevation of the hill to be 30° and 60°. The height of the hill is

(a) $\frac{3}{2}$ km (b) $\sqrt{\frac{2}{3}}$ km

(c) $\frac{\sqrt{3}+1}{2}$ km (d) $\sqrt{3}$ km

Q.26. The tops of two poles of height 20 m and 14 m of are connected by a wire. If the wire makes an angle 30° with the horizontal, then the length of the wire is

(a) 12m (b) 10m
(c) 8m (d) None of these

Q.27. The angles of elevation of the top of a tower from points at distance A and B from the base and in the same line with it are complementary. If $A > B$, then height of the tower is

(a) $\sqrt{B-A}$ (b) $\sqrt{B/A}$
(c) $\sqrt{B+A}$ (d) $\sqrt{BA}$

Q.28. A man walks 10m towards a lamp post and notices that the angle of elevation of the top of the post increases from 30^0 to 45^0. Find the height of the post.

(a) 10 m (b) $5(\sqrt{3}+1)$ m
(c) $5(\sqrt{3}-1)$ m (d) None of these

Q.29. A 6-ft-tall man finds that the angle of elevation of the top of a 24-ft-high tower and the angle of depression of its base are complementary angles. The distance of the man from the pillar is

(a) $2\sqrt{3}$ft. (b) $8\sqrt{3}$ft.
(c) $6\sqrt{3}$ft. (d) None of these

Q.30. A vertical pole PO is standing at the centre O of a square ABCD. If AC subtends an angle 90^0 at the top, P, of the pole then the angle subtended by a side of the square at P is

(a) 45^0 (b) 30^0
(c) 60^0 (d) None of these

Q.31. A vertical lamp-post of height 9 meters stands at the corner of rectangular field. The angel of evaluation of its top from the fare farthest corner is 30^0, while from another corner it is 45^0. The area of the field is

(a) $81\sqrt{2}$ metre (b) 9 metre
(c) 81 metre (d) 9 metre

Q.32. A vertical lamp-post, 6 m high, stands at a distance of 2m from a wall, 4 m high. A 1.5 m,-tall man starts to walk away form the wall on the other side of the wall, in line with the lamp – post. The maximum distance to which the man can walk remaining in the shadow is

(a) $\frac{5}{2}$m (b) $\frac{3}{2}$m
(c) 4 m (d) None of these

Direction for questions 33 and 34: Read the passage below and solve the questions based on it.

Mayank had left his home at the age of about 14 years. He remembers that the day was Monday. Since then he has been fasting on every Tuesday. Today he is celebrating his 60th birth anniversary in a five-star hotel with his friends. As today is Tuesday again, so he is not taking anything except wine. At the end of the party he discloses that it is his 2400th Tuesday of fasting.

Q.33. Today is 9th Oct, 2001. On which date had he left his home?

(a) 10.10.1956 (b) 8.8.1955
(c) 10.10.1955 (d) 8.8.1955

Q.34. He was born on

(a) Sunday (b) Monday
(c) Wednesday (d)Thursday

Q.35. All the possible two-character codes are formed using the digits 0 to 9 and the capital letters of the English alphabet A to Z. How many such codes are there which, upon writing the code on a piece of paper and then turning the paper around to read to code upside down will be mistaken for another code? There is no confusion between zero and the letter O or the number 1 and the letter I.

(a) 72 (b) 90
(c) 94 (d) 100

ADVANCED

Q.1. Let us express date in the DDMMYYYY, where DD represents the day of the month, MM represents the month, YYYY represents the year. If the last possible date in the 20th century with all the eight digits (in the date as expressed in the above format) being odd is a Sunday, then what day of the week will the first date of the 21st century with all the eight digits even will be?
(a) Monday (b) Saturday
(c) Friday (d) Tuesday

Q.2. A computer science student was using the following algorithm on his computer:

Step 1 : Read A
Step 2 : B = 1
Step 3 : If A > 1, then B = B × A
Step 4 : If A ≤ 1, then print B and stop
Step 5 : A = A – 1
Step 6 : go to step 3

If A is a non-negative integer, what is the student trying to find using the algorithm?
(a) Square root of A
(b) Factorial of A
(c) Sum of first A natural numbers
(d) None of the above

Direction for questions 3 to 5: Read the passage below and solve the questions based on it.

Pankaj has six friends, and during a certain vacation he met them at several dinners. He found that he dined with all six friends exactly one day; with every five of them of 2 days; with every four of them of 3 days; with every three of them of 4 days; with every two of them of 5 days. Further, every friend was present at 8 dinners and every friend was absent at 8 dinners.

Q.3. How many dinners did Pankaj have during the vacation?
(a) 220 (b) 173
(c) 16 (d) 12

Q.4. How many dinners did Pankaj have alone during the vacation?
(a) 6 (b) 1
(c) 7 (d) 2

Q.5. When Pankaj had dinner with every three of them of 4 days and every two of them of 5 days, what was the minimum number of friends who had dinner twice in these 9 days?
(a) 2 (b) 4
(c) 6 (d) 8

Q.6. A club with x members is organized into four committees according to the following rules:
1. Each member belongs to exactly two committees.
2. Each pair of committees has exactly one member in common.

Then which of the following is true regarding the value of x?
(a) $x = 4$ (b) $x = 6$;
(c) $x = 8$ (d) x cannot be determined.

Q.7. In a football league, a particular team played 60 games in a season. The team never lost three games consecutively and never won five games consecutively in that season. If N is the number of games the team won in that season, then N satisfies
(a) $24 \le N \le 50$ (b) $20 \le N \le 48$
(c) $12 \le N \le 40$ (d) $18 \le N \le 42$.

Q.8. From a group of seven persons, seven committee are formed. Any two committees have exactly one member in common. Each person is in exactly three committees. Then
(a) at least one committee must have more than three members;
(b) each committee must have exactly three members;
(c) each committee must have more than three members;
(d) nothing can be said about the sizes of the committees.

Q.9. How many integers satisfy the inequality

$$\frac{(x-1)(x+4)(x+2)}{(x-3)(x+8)(x+3)} < 0?$$

(a) 2 (b) 4
(c) 6 (d) Infinite

Q.10. If x satisfies $|x-1| + |x-2| + |x-3| \ge 6$, then
(a) $0 \le x \le 4$ (b) $x \le -2$ or $x \ge 4$
(c) $x \le 0$ or $x \ge 4$ (d) none of these

Q.11. For real values of x, the expression $\frac{(x-c)(x-b)}{(x-a)}$ will assume all real values provided
(a) $a \le c \le b$ (b) $b \le a \le c$
(c) $b \le c \le a$ (d) $a \le b \le c$

Q.12. Let a_n = product of the first n natural numbers. Then for all $n \ge N$
(a) $n^n \ge a_n$ (b) $\left(\frac{n+1}{2}\right)^n \ge n!$
(c) $n^n \ge a_{n+1}$ (d) None of these

Q.13. Solve for the values of y:
$\frac{z}{y} < 4, z \times y > 3, -3 < z < 3$
(a) $y < -\frac{\sqrt{3}}{2}$ (b) $y > \frac{\sqrt{3}}{2}$
(c) all the real values (d) None of these

Q.14. If $a^2 + b^2 + c^2 = 1$, then $ab + bc + ca$ lies in between(including)
(a) –2 and 1 (b) –1 and –2
(c) 0 and 1 (d) None of these

Q.15. If a, b, c and d are real numbers such that $b > 0$, $d > 0$ and $\frac{a}{b} < \frac{c}{d}$, then which of the following is true?
(a) $\frac{a}{b} < \frac{a-c}{b+d} < \frac{c}{d}$ (b) $\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}$
(c) $\frac{a}{b} < \frac{a-c}{b-d} < \frac{c}{d}$ (d) $\frac{a}{b} < \frac{a+c}{b-d} < \frac{c}{d}$

Q.16. Let x, y, z, w be positive real numbers, which satisfy the two conditions that
(i) If $x > y$, then $z > w$ and
(ii) If $x > z$, then $y < w$
One of the following statements given below is a valid conclusion. Which one is it?
(a) If $x < y$ then $z < w$
(b) If $x < z$ then $y > w$
(c) If $x > y + z$ then $z < y$
(d) If $x > y + z$ then $z > y$

Q.17. If $xy + x + y = 2$, then which of the following is true?
(a) $\left(\frac{x+1}{y+1}\right) > 0$ (b) $\left(\frac{x+2}{y+2}\right) > 0$
(c) $\frac{x-1}{y-1} > 0$ (d) $\frac{x-2}{y-2} > 0$

Q.18. Let $N = \frac{3 \times 5 \times 7 \times \ldots\ldots\ldots\ldots \times 99}{2 \times 4 \times 6 \times \ldots\ldots\ldots\ldots \times 100}$
Then which of the following is true regarding the value of N?
(a) $1/3 < N < ½$ (b) $1/5 < N < ¼$
(c) $1/15 < N < {}^1/_{10}$ (d) $1/10 < N < {}^1/_5$

Q.19. The number of points (x, y) satisfying (i) $3y - 4x = 20$ (ii) $x^2 + y^2 \leq 16$ is
(a) 0; (b) 1;
(c) 2; (d) infinite

Q.20. A sequence of terms, a_1, a_2, a_3 .... a_i, is generated as follows. The first term (a_1) is chosen as (a) To obtain every subsequent term a coin is tossed. If a heads (H) appears, the next term is obtained by adding 2 to the previous term. If a tails (T) appears, only 1 in added to obtain the next term. For example, the sequence corresponding to HTH is 1, 3, 4, 6.
If $a_{11} = 15$, what is the maximum number of consecutive tails that could have appeared up to that term?
(a) 6 (b) 9
(c) 7 (d) Cannot be determined.

Q.21. The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 15 seconds, the angle of elevation changes to 30°. If the jet plane is flying at a constant height of $1500\sqrt{3}$m find the speed of jet plane.
(a) 720 kmph (b) 600 kmph
(c) 540 kmph (d) None of these

Q.22. A club has 108 members. Two third of them are men and the rest are women. All members are married except for 9 women members. How many married women are there is the club?
(a) 20 (b) 24
(c) 27 (d) 30

Q.23. I see a bird flying away at a constant speed of 1.7568 kmph in the sky. The angle of elevation is 60°. After $\frac{1}{2}$ minute, I see the bird again and this time the angle of elevation is 45°. What is the perpendicular (horizontal) distance of the bird from me?
(a) 10 m (b) $10\sqrt{3}$ m
(c) 20 m (d) $20\sqrt{3}$ m

Q.24. Amitabh makes a road trip to the Hooghly beach. If he averaged his speed 5kmph more, he could have decreased his time by 10%. On the other hand, if he averaged his speed 5 kmph less, the journey to the beach would have taken an extra hour. What is the distance from Amitabh's home to the Hooghly beach?
(a) 380 kms (b) 420 kms
(c) 275 kms (d) 360 kms

Direction for questions 25 and 26: Read the passage below and solve the questions based on it.

Two Supersonic planes are flying in a straight line horizontally on different altitudes such that the distance between them is 200 km and the pilot of upper plane observes the lower plane at an angle of depression of 30°. After half an hour, the angle of depression becomes 60°.

Q.25. Find the distance between them after half an hour with respect to initial point of reference.
(a) $\frac{100}{\sqrt{3}}$km (b) $200\sqrt{3}$km
(c) $\frac{200}{\sqrt{3}}$km (d) Cannot be determined

Q.26. If they are travelling in opposite directions and the ratio of speed of upper plane to speed of lower plane is 2:1, find the speed of lower plane.

(c) $\frac{400}{\sqrt{3}}$ km/hr (b) $\frac{200}{\sqrt{3}}$ km/hr

(c) $\frac{800}{\sqrt{3}}$ km/hr (d) Cannot be determined.

Q.27. The length of a ladder exactly equals the height of a wall. If the ladder is placed on a 2 ft. tall stool placed 10ft. away from the wall, then its tip can just touch the top of the wall. The height of the wall is
(a) 15ft. (b) 26ft.
(c) 28ft. (d) 32ft.

Q.28. A tree of height 21m on a roadside broke at a certain height and fell in such a way that its top touched the other edge of the road. If the breadth of the road is 12m, then the height above the ground at which the tree broke was
(a) 12m (b) 6m
(c) 9m (d) None of these

Q.29. If a flagstaff of 6 metres high placed on the top of a tower throws a shadow of $2\sqrt{3}$ metres along the ground then the angle(in degrees) that the sun makes with the ground is
(a) 30° (b) 60°
(c) 45° (d) none of these

Q.30. In a chess tournament, each of the 5 players plays against every other player. No game results in a draw and the winner of each game gets one point and the loser gets zero. Then which one of the following sequences cannot represent the scores of the five players?
(a) 3,3,2,1,1 (b) 3,2,2,2,1
(c) 2,2,2,2,2 (d) 4, 4,1,1,0

Q.31. We consider the relation "a person x shakes hand with a person y". Obviously, if x shakes hand y, they y shakes hand with x. In a gathering of 99 persons, one of the following statements is always true, considering 0 to be an even number. Which one is it?
(a) There is at least one person who shakes hand exactly with an odd number of persons.
(b) There is at least one person who shakes hand exactly with an even number of persons.
(c) There is even number of persons who shake hand exactly with an even number of persons.
(d) None of the above.

Q.32. Let P, Q, R, S and T be statements such that if P is true then both Q and S are true, and if both R and S are true then T is false. We then have:
(a) If T is true then both P and R must be true.
(b) If T is true then both P and R must be false.
(c) If T is true then at least one of P and R must be true.
(d) If T is true then at least one of P and R must be false.

Q.33. If A, B, C and D are statements such that if at least one of A and B is true, then at least one of C and D must be true. Further, both A and C are false. Then
(a) If D is false then B is false
(b) Both B and D are false
(c) Both B and D are true
(d) If D is true then B is true

Q.34. If x, y and z are eliminated from the equations $\frac{x}{y}=\frac{a+1}{a-1}, \frac{y}{z}=\frac{b+1}{b-1}$ and, $\frac{z}{x}=\frac{c+1}{c-1}$ then which of the following is true regarding the relationship between a, b and c?
(a) $a+b+c=-1$ (b) $abc=-1$
(c) $ab+bc+ca=-1$ (d) $a^2+b^2+c^2=-1$

ANSWERS

■■■ Foundation

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(b)	(d)	(d)	(a)	(b)	(c)	(c)	(d)	(d)	(d)	(b)	(b)	(b)	(d)	(b)	(a)	(a)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(a)	(c)	(b)	(c)	(b)	(c)	(b)	(a)	(c)	(d)										

■■■ Moderate

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(d)	(b)	(b)	(a)	(c)	(b)	(b)	(a)	(a)	(d)	(c)	(a)	(b)	(b)	(c)	(d)	(a)	(d)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
Answer	(c)	(b)	(b)	(a)	(a)	(a)	(d)	(b)	(c)	(c)	(a)	(a)	(c)	(d)	(c)					

■■■ Advanced

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(b)	(a)	(d)	(d)	(b)	(b)	(b)	(c)	(b)	(a)	(d)	(d)	(b)	(d)	(a)	(c)	(b)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34						
Answer	(a)	(c)	(d)	(d)	(d)	(b)	(b)	(d)	(b)	(d)	(b)	(d)	(a)	(c)						

22

CHAPTER

Managerial Decision Making

LEARNING OBJECTIVES

After completion of this chapter, you should have a thorough understanding of the following:

- When decisions are required to be taken
- Some of the typical decisions making situations
- Some of the standard questions and methods to solve the same

Introduction to the topic

Decision making is the process by which managers respond to opportunities and threats by analysing options, and making decisions about goals and courses of action.

WHEN DECISIONS ARE MADE

Decisions in response to opportunities: managers respond to ways to improve organizational performance.

Decisions in response to threats: occurs when managers are impacted by adverse events to the organization.

Types of Decision Making

- **Programmed Decisions—are routine decisions and follow almost automatic process. Following are some of the attributes of Programmed decisions:**
- Managers have made decision many times before.
- There are rules or guidelines to follow.
- Example: Deciding to reorder office supplies.
- **Non-programmed Decisions—are required to be taken when some new situation has arisen, or a unusual situations that have not been often addressed has arisen.**
- No rules or established norms to follow since the situation is new to the organization. In some of the cases, managers take the cue of decisions made by other firms facing similar situations. However, these "other firms" may be the closest competitor and hence following their strategy may not produce the desired results.
- Hence, these decisions are made based on qualitative information, quantitative information, intuition, and judgment.
- **Example:** Should the firm invest in a new echnology?
- Rather, looking from an organizational perspective, if we look on a personal level, question asked in CAT 2007 was a similar situation of decision making.

Let us have a look at the questions from CAT 2007:

Directions for questions 1 and 2

Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of –3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of –2.5%, while a fall will entail of +2%.

Q.1. The maximum guaranteed return to Shabnam is

(a) 0.20% (b) 0.15%
(c) 0.30% (d) 0.25%
(e) 0.10%

Q.2. What strategy will maximize the guaranteed return to Shabnam?

(a) 64% in option B and 36% in option C
(b) 1/3 in each of the three options
(c) 30% in option A, 32% in option B and 38% in option C
(d) 100% in option A
(e) 36% in option B and 64% in option C

(Solution to these questions given at the end in CAT section).

In this chapter, we would be discussing the following types/tools of decision making:

1. Basics of Linear programming
2. Scheduling problems

BASICS OF LINEAR PROGRAMMING (LP)

At the onset, I would like to clarify that we are not getting into the depth of formulation of Linear programming. Our focus is to learn the methods of solving the questions using the concepts of LP.

Example 1 There is a carpenter who sells only tables (with two legs only) or chairs (with four legs only). The top part of table and top part of chair (the wooden plank) are same and one wooden plank is required to configure one chair or one table. Carpenter gets the legs and top wooden plank separately and then assembles it to make either chair or table.

Its also given that Profit per table = Rs 1000/unit and profit per chair = Rs 1500/unit.

Q.1. If there is limited supply of 1200 legs and top wooden plank, which is more profitable to produce—table or chair to maximize the overall profit?

Solution In this situation, for any limited supply of legs, it is more profitable to make table (if at all production to be made). We can simply see it through the concepts of average that in case of table, profit per leg = Rs 500, and for chair profit per leg = Rs 375/leg. We are not considering the top wooden plank because number of unit required for each of table and chair is same = 1 top wooden plank.

Now we are adding another aspect to this situation:

Carpenter takes 4 hours to assemble a table and 3 hours to assemble a chair.

Q.2. If there is limited supply of 1000 man-hour labor, which is more profitable to produce—table or chair to maximize the overall profit?

Solution Calculating again the per unit time profit of table and chair,

Profit per hour for table = Rs 250 and

Profit per hour for chair = Rs 500. Hence it is more profitable to produce chair.

Q.3. If we club the information given in Q.1 and Q.2, then which is more profitable to produce—table or chair to maximize the overall profit?

Solution It is difficult to tell by only looking at the average profit per leg or per unit of man-hour labor because we are getting contradictory results. It shows that there will be some midway solution, possibly some units of table and some units of chair.

Concept of Linear Programming comes handy in this situation. As I have stated earlier too, we will be concerned only with our primary purpose of getting the solution of our question, and not the concepts of LP in depth.

FORMULATION OF LP

It is simple linear equation representation of the constraints given in the question in terms of limited supply of legs and man-hour labour.

Assume that 'x' table and 'y' chairs are to be made.

Equation for legs $\Rightarrow$

$2x + 4y \leq 1200$ (we are taking '$\leq$' and not '=' sign because we may not be using all the units supplied)

Equation for man-hour labour $\Rightarrow$

$4x + 3y \leq 1000$ (we are taking '$\leq$' and not '=' sign because we may not be using all the units supplied)

Total profit = $1000x + 1500y$

Objective is to maximize total profit $\Rightarrow$ maximize (1000 x + 1500 y)

Process to solve:

Step 1: Solve all the pairs of equations (convert the inequations into equation by putting "=" sign instead of inequality sign) formed from the constraints (the way we solve simultaneous equations).

Step 2: Put all the point of intersection obtained into total profit one by one.

Step 3: The point which gives maximum total profit will give the units to be made.

In this question, we have only one pair of equation formed from the constraints viz., of legs and man-hour labour.

$2x + 4y = 1200$ ----------(1)

$4x + 3y = 1000$ ----------(2)

Solving these two equations, we get $y = 280$, $x = 40$

Hence carpenter should make 40 tables and 280 chairs to maximize his profit.

Total profit = $1000x + 1500y = 1000 \times 40 + 1500 \times 280 = 460{,}000$

Please note that if at all a solution can be found in case of LP, the solution has to be in the 1st quadrant ($x \geq 0, y \geq 0$).

SCHEDULING

Scheduling specifies the order in which the given tasks should be performed. In other words, it gives the precedence relationship among the tasks—which task will be done first, which task will be done second and so on. In a production environment, scheduling becomes very critical owing to the cost constraint and the offerings of the competitor. Even at B-Schools, scheduling constitute a substantial part of the course curriculum.

For the sake of CAT preparation, CAT has asked scheduling problems quite a few times in the last couple of years. We will now learn the methods of scheduling through some typical problems.

Example 2 Go through the following passage and solve the questions based on it.

Heartkare hospital is specialized in heart related surgeries. Hospital boasts of the following facilities and conditions of surgery:

(i) surgery is performed by one of the 12 senior doctors assisted by one of the seven junior doctors. Senior doctors generally take about one hour to prepare for and perform each surgery, and they operate on four patients per day.

(ii) Senior doctors day ends at 4 p.m., although they can be asked to come every 14th night and every 10th weekend.

(iii) Each patient goes through a series of tests and reporting before going for surgery. Tests are done between 9 a.m. to 3.30 p.m. (Monday to Friday), 10 a.m. to 2 a.m. (on Saturday).

(iv) Patients come to hospital between 1 p.m. to 3 p.m. the day before their surgery.

(v) 1st operation begins at 7.30 a.m.

(vi) The hospital is having five operating rooms, a patient recovery room, a laboratory, and five test examination rooms. Hospital performs 150 surgeries per week on an average with patients staying at the hospital for three days. Although surgeries are performed only five days a week, hospital is needed to be in operation for whole of the week to take care of recovering patients. Table above gives the current scenario of number of beds required in the hospital:

Current Scenario Beds required : Operations with 90 beds (30 patients per day)*
*** - Patients stay in the hospital for three days requiring beds.**

Check in day		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Monday		30	30	30				
Tuesday			30	30	30			
Wednesday				30	30	30		
Thursday					30	30	30	
Friday								
Saturday								
Sunday		30	30					30
Total		60	90	90	90	60	30	30

For example, 30 patients checked in the hospital on Tuesday and they stayed on Tuesday–Wednesday–Thursday. On Thursday, they were discharged from the hospital.

Hospital management is seriously considering now to expand its number of surgeries to generate more revenue. There are two options before the management right now: 1. Start surgeries on one more day (Saturday) ⇒ Leading to a flat 20% rise in surgeries done.

2. Add one more premises to increase the number of beds by 50%.

   Assume that there is sufficient demand of the facilities of the hospital.

Q.1. What is the 'bed' utilization of the hospital?

Q.2. How would the above given table look like if management decides to perform surgery on Saturday too (assume 30 surgeries per day still)? What will be the 'bed' utilization in that case?

Q.3. What problems can occur, if any, if management decides to add another floor to expand the operation (Assume that in one operating room, only one surgery can be done at one point of time)?

Solution

Q.1. Hospital is not able to use its beds on all the days at its maximum level. Only on three days of the week—Tuesday, Wednesday and Thursday—all the beds in the hospital are utilized. For two of the remaining days of the week, only 30 beds are utilized and for the left two days 60 beds are utilized.

Hence 'bed' utilization for the whole week = 450/(90×7) = 450 / 630 = 71.4%

Q.2. Following will be the occupancy table in the new case:

Proposed Scenario

Check in day		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Monday		30	30	30				
Tuesday			30	30	30			
Wednesday				30	30	30		
Thursday					30	30	30	
Friday						30		
Saturday							30	
Sunday		30	30					60
Total		60	90	90	90	90	60	60

New 'bed' utilization for the whole week = 540/(90×7) = 540 / 630 = 85.7%

Q.3. If the capacity of the number of beds in the hospital is increased by 50%, then on an average 45 operations should have been done using the existing resources.

The resources are:

Operation room, senior doctors and junior doctors.

Now, Operation starts at 7.30 a.m. and goes till 4 p.m.

In this period of 8.30 hours × 5 operation room, a maximum of 40 operations can be done. And this lack of available operating room is the bottleneck. Hence to fully utilize the beds, we need to increase the operating hours by 30 mins making it 9 hours per day (from 7.30 a.m. to 4.30 p.m. for example), and then we can do 45 operations daily on an average.

After increasing 30 mins of operating time per day, new bed utilization chart will look like:

Scenario								
Check in day		Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Monday		45	45	45				
Tuesday			45	45	45			
Wednesday				45	45	45		
Thursday					45	45	45	
Friday								
Saturday								
Sunday		45	45					45
Total		90	135	135	135	90	45	45

'Bed' utilization percentage for the whole week now = 675 /(135×7) = 71.4%

Example 3 Read the following passage and answer the questions based on it.

'Noodles' the fast food joint at North Campus, Delhi is famous for its value-for-money Chinese food. To understand and expand their business, management at 'Noodles' analyzed its processes and came out with following:

Customer enters→ Process 1→ Process 2 → Process 3 → Process 4 → Customer leaves

Any customer order has to go through all the four processes as given above.

Following is the time taken to do the processes:

Process 1 – 1 minute/order

Process 2 – 4 minutes/order

Process 3 – 3 minutes/order

Process 4 – 2 minutes/order

Noodles is open for 8 hours per day. A particular employee can work only in one process.

Q.4. What is the maximum number of customers that 'Noodles' can service daily?

Q.5. It has been found that only one new employee can be added to 'Noodles' looking upon cost-benefit analysis. Which process the employee should be added to? How many customers can be serviced after this addition of new employee?

Solution

Q.4. Total number of customers that can be serviced per day through different processes:

Process 1 – 480

Process 2 – 120

Process 3 – 160

Process 4 – 240

Since process 2 can service only 120 customers, hence total number of customers serviced = Min (480, 120, 160, 240) = 120

Q.5. Obviously process 2 is the bottleneck, so the new employee should be added to process 2. After adding new employee, total number of customers serviced/day = 160

PRACTICE EXERCISES

WARM UP

Direction for questions 1 to 3: Read the passage below and solve the questions based on it.

MUL is considering establishing a new factory for making its new C segment of cars—either at towns Ludhiana or Saharsa or Goa, or perhaps even at all the three towns. It is also considering building at most one new servicing station but the choice of location for servicing station is restricted to the city where a new factory is built i.e., servicing station can be built only at a town where factory is being built. Objective of the MUL is to get maximum benefit out of this whole exercise of establishing the factories and servicing station.

Following table gives the cost-benefit scenario for the two towns:

	Benefit INR (After deducting the cost)	Cost INR
Factory at Ludhiana	90 crores	60 crores
Factory at Saharsa	50 crores	30 crores
Factory at Goa	60 crores	40 crores
Service station at Saharsa	40 crores	20 crores
Service station at Goa	60 crores	30 crores
Service station at Ludhiana	40 crores	50 crores

MUL can invest a maximum of INR 100 crores for establishing the factories and service stations.

Q.1. At what location the factory should be built?

Q.2. At what location the service station should be built?

Q.3. What can be the maximum benefit (after deducting the cost) that MUL can derive?

Direction for questions 4 and 5: Read the passage below and solve the questions based on it.

An investment bank is considering 7 plans for investment. Following table gives the investment required and profit generated (after deducting the investment) in INR (Indian National Rupees) for each of the seven investment plans:

	Investment Required	Profit (after deducting the investment)
Investment plan 1	500	1600
Investment plan 2	250	800
Investment plan 3	350	1000
Investment plan 4	600	2000
Investment plan 5	700	2200
Investment plan 6	450	1200
Investment plan 7	300	800

Bank can investment a maximum of INR 1500 in all the investment plans added together.

Q.4. Which investment plans should be chosen to maximize the profit (after deducting the investment) by the bank?

Q.5. What maximum profit (after deducting the investment) can be generated under the given constraints?

Direction for questions 6 to 10: Read the passage below and solve the questions based on it.

ABC Limited is evaluating new copying machines to replace the current copier of the company. Three machines are available with the following estimates of operating costs. The labour costs on the copying operation is payable in proportion to the number of copies actually made:

Particulars	Copier A	Copier B	Copier C
Material cost per copy	0.15	0.12	0.06
Labour cost per copy	0.21	0.12	0.06
Annual lease rental	9000	17400	30000

(All figures are in Rs)

Q.6. Which of the copiers should be chosen by the company? Would your answer change for the different number of copies made per year?

Q.7. At what number of copies made per annum will be total cost of copier A and total cost of copier B per annum be same?

Q.8. If the total number of copies made is 45,000, then which copier should be chosen?

Q.9. What is the minimum number of copies made per annum so that copier C becomes the most profitable option?

Q.10. Company estimates that the copying volume will be 80,000 per annum. However the actual number of copies made is only 60,000. Company has already opted for copier B. What is the resultant loss (if any) with respect to the most efficient copier for this volume category?

Q.11. A company manufactures four variants of the same product and in the final part of the manufacturing process there are assembly, polishing and packing operations. For each variant the time required for these operations is shown below (in minutes) as is the profit per unit sold.

	Assembly	Polish	Pack	Profit (INR)
Variant 1	2	3	2	1.50
2	4	2	3	2.50
3	3	3	2	3.00
4	7	4	5	4.50

Given the current labour force, the company estimates that, each year, they have 100000 minutes of assembly time, 50000 minutes of polishing time and 60000 minutes of packing time available. How many of each variant should the company make per year and what would the associated profit?

HINTS AND SOLUTIONS

1. Make factory at Saharsa and Goa.
2. Make service station at Goa.
3. 170 crores.
4. Investment plans 1, 5 and 7.
5. INR 46,000.
6. Company should use different copiers for different volume of copies made:
   For 0 – 70, 000 copies – Copier A
   For 70, 000 – 1,05,000 copies – Copier B
   For 1,05,000+ copies – Copier C
7. At 70, 000 copies.
8. Copier A
9. 1,05,001 copies (For 1,05,000 copies, both copiers B and C are equally profitable.)
10. Rs 1200 (most efficient copier for 60,000 copies = copier A)
11. Total profit = Rs 58,000.
    And number of units made X_1= 0, X_2= 16000, X_3=6000 and X_4= 0

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PART 2

Data Interpretation

Module 1

- Data Interpretation

Module 2

- Test Your Mettle

Module 3

- Data Sufficiency

Module 1

Data Interpretation

1

CHAPTER

Introduction to Data Interpretation

LEARNING OBJECTIVES

- What do we understand with data
- What is data interpretation
- Different ways of representing the data
- Problem solving techniques

WHAT IS DATA INTERPRETATION?

Data Interpretation is the act of transforming data with the objective of extracting useful information and facilitating conclusions on the basis of the given data. Depending upon the type of data and the questions, we might be required to apply certain statistical tools with various methods to represent the given data.

Before we move ahead to see and learn the various methods of representing data, let us understand some basics.

WHAT IS DATA?

Data is a means to represent facts, concepts or instructions in a formalized manner suitable for communication, interpretation or processing by humans or other automatic modes.

In other words, propositions like measurements or observations of a variable, which may comprise numbers, words or images can be quantified with the help of data.

Data can be a numerical or non-numerical fact and figure related to a particular social, economic or political event etc.

Let us see some examples of data:

Yesterday the Sensex closed at 14,004 points.

25% of the total population of India is below The poverty line.

In the last general elections in India, 40% of the total seats were won by regional parties. Data can be further understood to be an agent to cause the sentiments and even sometimes it can be effect also.

If the data given is distinct and separate, i.e. It can be counted (1, 2, 3….), then it is known as Discrete Data. For example, population of a country, production of cars in a manufacturing plant.

However, if the data takes any value within a finite or infinite interval, then it is known as a Continuous Variable. For example, the percentage of sugar in mangoes or weight/height etc., are continuous variables.

Significance of Organized Data

It is said, "Trust only two—God and data." Since God cannnot be everywhere to prove a point, we have to resort to data to validate points or make inferences. The significance of data

can be further illustrated by the fact that data is the basis of an argument. In fact, data is the starting point for most arguments.

However, data if not properly managed is not of much use. People at the higher echelons of any organization will have no time to go through the details of all the reports. The need for organized data becomes more pertinent because data can be used to describe a current situation with respect to its future possibility. Data can establish a relationship between different phenomenon like expenditure government in the various sectors of its annual budget vis-a-vis the priority sectors; The per capita income of different countries in relation to the number of deaths due to a particular disease.

For maximum utilization of data, it should be formatted properly for easy interpretation and deduction. However, in CAT and other B-School entrance examinations, we should not expect easy-to-comprehend data. Rather, the students should be prepared to find data which is difficult to interrelate or comprehend.

Different Ways of Representing Data

Data representation in the DI section is primarily of two types:

Narration Based

Also known as caselets, these questions often involve stories that define a situation and give details of various parameters involved; including their inter-relationships.

E.g., Mittal has recently acquired four companies, viz., Bank of Bozoland (BOB), My Own Bank (MOB), Zany Obliterated Bank (ZOB) and Dogmatically Obscure Bank (DOB). He noticed that the sales of DOB are half than that of BOB, whereas, the profits of DOB are double than that of BOB. The expenses of ZOB are Rs 3 crores less than that of DOB, whereas, the profits of MOB is Rs 1 crore less than that of ZOB. The expenses of BOB are three times than that of DOB. It is also known that the sales of ZOB are Rs 15 crore or one-fourth of MOB's sales. All the figures are for 1992–93. An insider further informs Mittal that the sales of DOB are Rs 10 crores more than that of ZOB and the expenses of BOB are 90% of its own sales. Sales – Expenses = Profit

Q. The total sales of all the four companies is (Rs crores):
(a) 200 (b) 150
(c) 125 (d) 160

Pictorial

This is the most common form of data representation. In such problems, data is presented in various pictorial forms such as line graphs, bar diagrams, line charts etc.

The important point to remember pertaining to all these questions is the fact that each and every question asked in the CAT is based on some logic and reasoning meant to check your aptitude. Few questions that involve numbers may also require a basic level of calculation skills.

Let us see the various pictorial representation of data:

Table

Tabular method is the most fundamental way of representing data. In fact, most of the different kinds of data presentation formats like the bar charts, line charts etc. originate from the table. In other words, presenting the data in a tabular format is the first step in forming other types of data presentation formats.

E.g., the table given below shows the break-up of the percentage of people of different age groups frequenting bars in 4 different metro cities viz., Delhi, Hyderabad, Bangalore and Patna in the year 2002.

Cities	Percentage break-up for age groups (Years) in 2006						
	Up to 15	15-20	20-25	25-30	30-35	35-40	Above 40
Delhi	8	13	24	21	11	17	6
Hyderabad	3	8	35	23	10	16	5
Bangalore	4	21	27	11	8	14	15
Patna	1	7	43	32	9	5	3

The tabular format is considered to be the most versatile data presentation method. All data which can be expressed in any other format can also be expressed in the format of a table. On the other hand, it is quite possible that data that can be presented in a tabular format, cannot be presented in any other format like the pie chart etc.

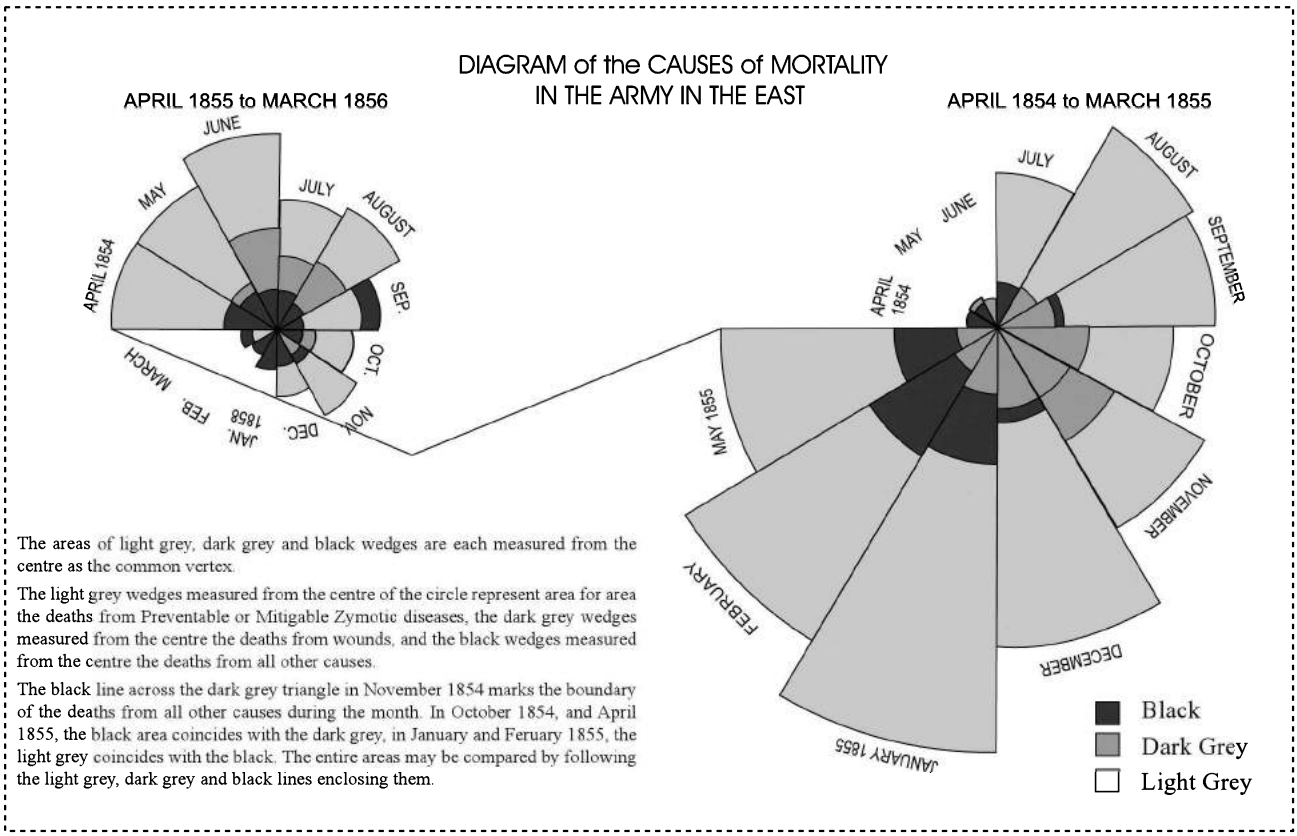
Pie Chart

Pie charts are a typical type of data representation where data is represented as a part of a circle. The circle represents the total value (or 100%), and the different parts represent certain proportions (or percentage) of the total. In a pie chart, the arc length of each sector (and in turn its central angle and area), is proportional to the part it represents.

The origin of the pie chart is traced back to Florence Nightingale in 1858. This was the year when she presented a paper on the causes of deaths in her army in the eastern part of the world.

Following is the pie chart originally developed by Florence Nightingale in the year 1858. (Actually called by her as the 'Polar Area Diagram'.)

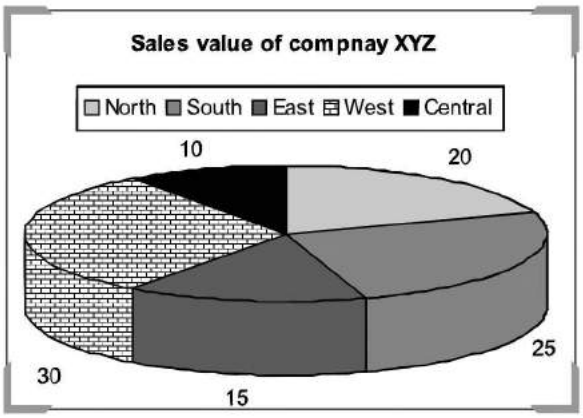
The below graphic gives the number of deaths that occured from diseases that could have been prevented (in light grey), those that were the results of wounds (in dark grey) and those due to other causes (in black).



There are two approaches constructing a pie chart from any given data:

(A) Degree Approach:
The central angle in a circle represents 360°, so any part or segment in a pie chart is calculated as a proportion of 360°.

(B) Percentage Approach:
In this case, any part or segment in a pie chart is calculated as a part of 100%.



If we convert the same pie chart into the degree format, we will be required to do the following conversions:

Total	=	100%	=	360°
Hence 1%			=	3.6°
Central	=	10%	=	36°
North	=	20%	=	72°
South	=	25%	=	90°
East	=	15%	=	54°
West	=	30%	=	108°

Limitations of Pie Charts

Despite the pie chart being one of the most important ways to represent data, it is marred by limitations of its own:

- Pie charts can be used only when the sum of all categories is given, for example if the categories represent proportions or percentage of a total.
- A single pie chart can represent only one continuous variable.

Significance of Pie Charts

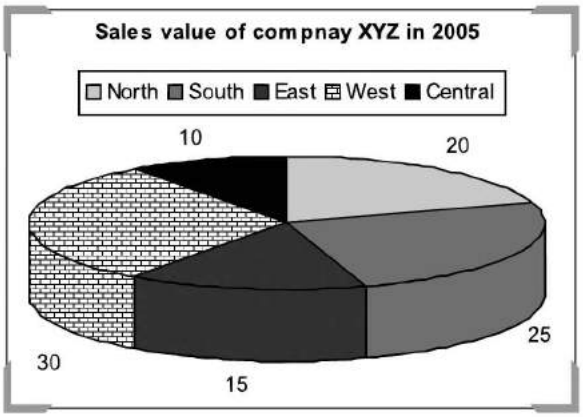
The pie chart has gained prominence due to the following reasons:

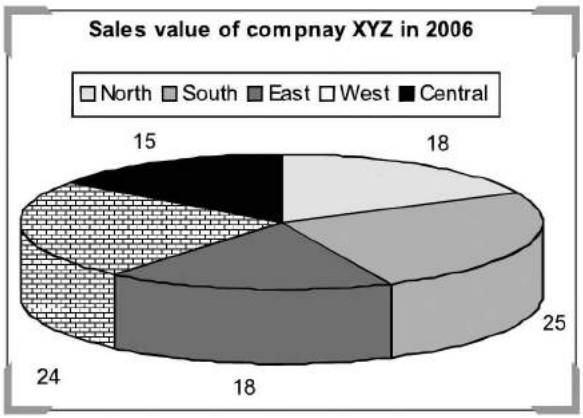
- In a pie chart, we get a clear picture of the contribution of different sectors to the build up of the total. E.g., presentation of budgets.

■ Comparing two pie charts is easier than comparing two bar charts or any other format of data representation.

Example 1

Let us see the following data:





Sales value in 2005 = Rs 180 crores

Sales value in 2006 = Rs 204 crores

What is the percentage increase in the sales value of the East zone?

Solution

There are two percentage increases (A) The total sales value of company XYZ is increasing. (B) The percentage contribution of the East zone is increasing.

Percentage increase in The total sales value of the company XYZ = 30%

Percentage increase in the percentage contribution of the East zone = 20%

Hence, the net percentage increase = 56%

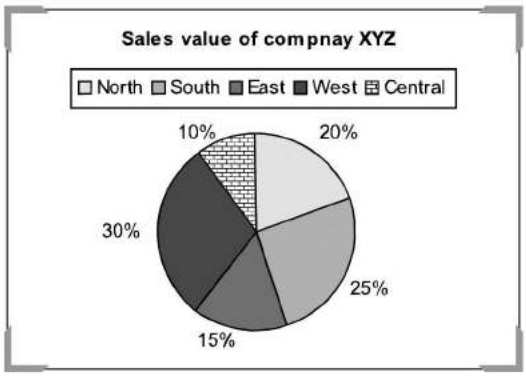
(Successive increase of 20% and 30%)

Types of Pie Charts

There are two types of pie charts:

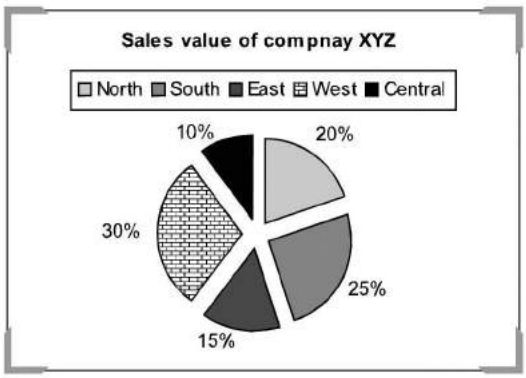
(A) Normal Pie Chart

This displays the contribution of each component of the pie.



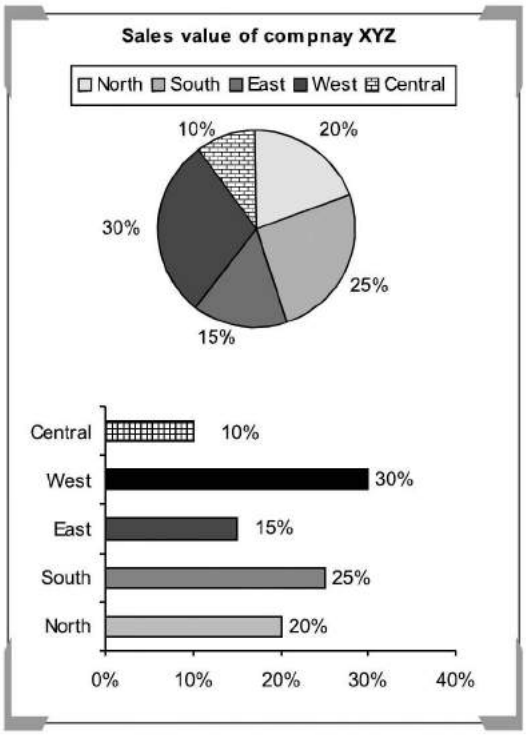
(B) Exploded Pie Chart

This pie chart has all the characteristics of a normal pie chart, the only addition is that the contribution of individual segments is highlighted.



Bar Chart/Bar Graph

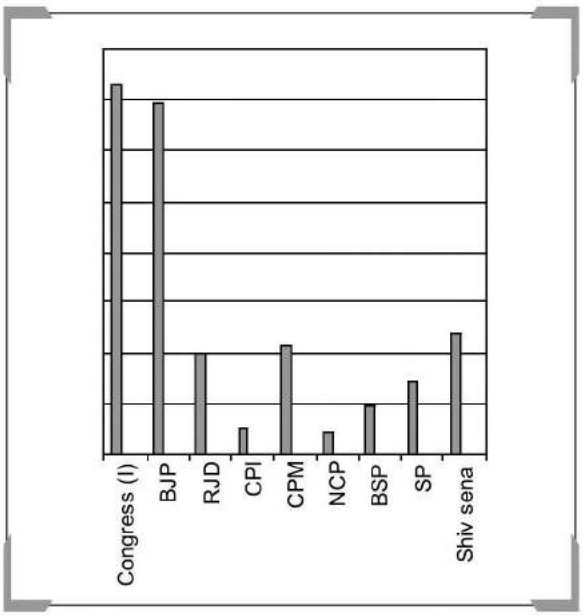
The bar chart, in comparison to the pie chart is more versatile in representing data. It has been proven that representation using lengths as in the case of bar charts is a better indicator of data vis-a-vis pie charts wherein data is categorized in terms of areas.



In the above diagram, the same data has been represented length-wise in the bar chart and area-wise in the pie chart. Obviously, it is easier to see the contribution of the various segments in the bar chart than in the pie chart.

A therefore, is a chart with rectangular bars of lengths proportional to the values which they represent. Usually, the terms 'bar chart' and 'bar graph' are used interchangeably. It should also be noted that in a bar chart, what matters is the length of the bar and not the width of the bar.

The following bar chart represents the number of seats won by different parties in the last general election.

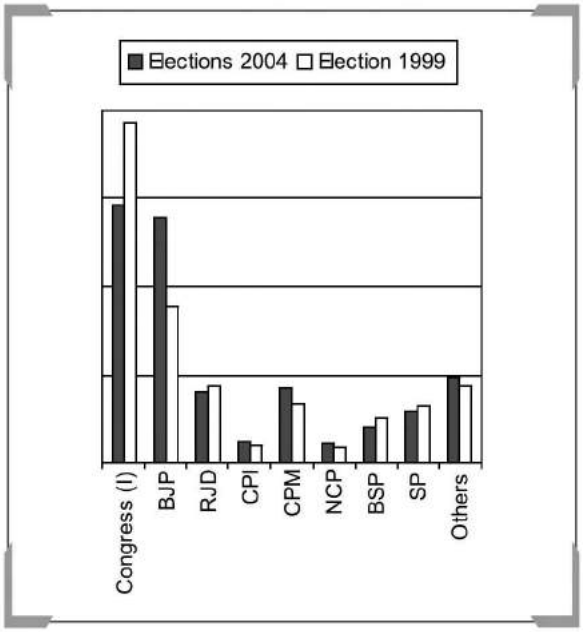


Significance of Bar Charts

Following are the major specialities of bar charts:

Unlike the pie chart, a single bar chart can be used to compare two, or more than two continuous variables.

The following bar chart represents the number of seats won by different parties in two general elections:



Since the length parameter is easier to study than the area parameter, a bar chart gives a quick understanding of the various ranks. Hence the time taken to understand the data becomes condiserably lower in a bar chart as compared to other formats of data presentation.

Types of Bar Charts

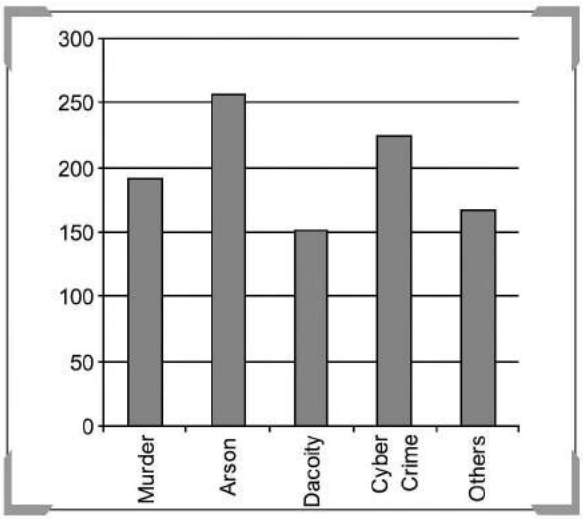
There are three types of bar charts:

(A) Normal Bar Chart

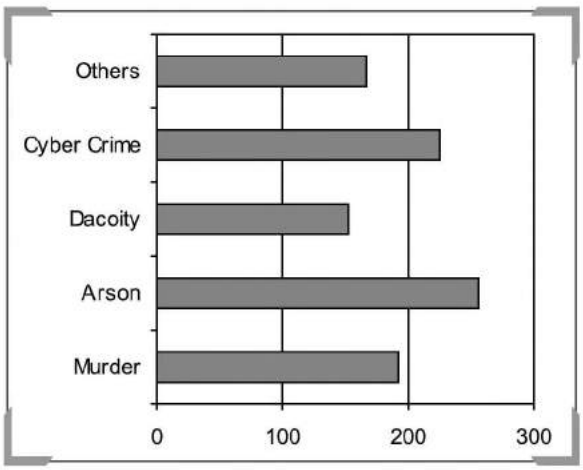
This is a simple bar chart with the values of different segments represented in the form of bars,which could be either horizontal, vertical or both.

The following bar chart represents the crime cases reported in Delhi in 2007:

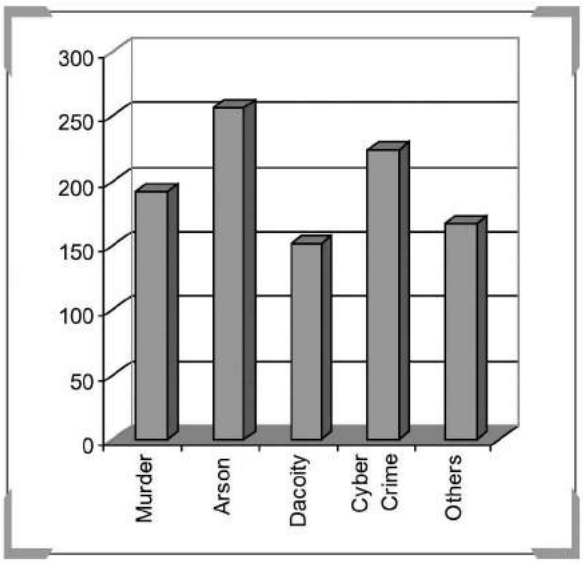
Bar Chart with Vertical Bars



Bar Chart with Horizontal Bars



Bar Charts with 3–D Effects



(B) Stacked Bar Chart Value-wise

When the same variable is to be represented on more than one parameters like; year etc., then we can have a stacked bar chart.

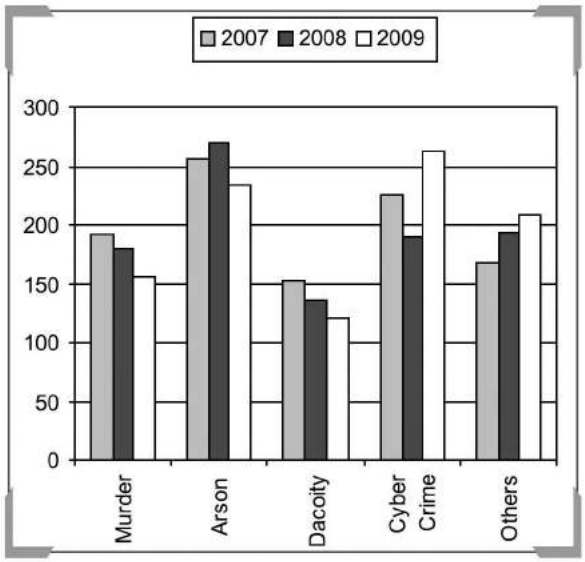
(C) Stacked Bar Chart Percentage-wise

The only difference between value-wise and percentage-wise bar charts is that in the former we apply values to stack the bars and in the latter we apply percentages.

Stacked Bar Charts are also known as Cumulative Bar Charts.

Here, we will see the same data in a normal bar chart, value-wise stacked bar chart and stacked bar chart percentage-wise

Normal Bar Chart



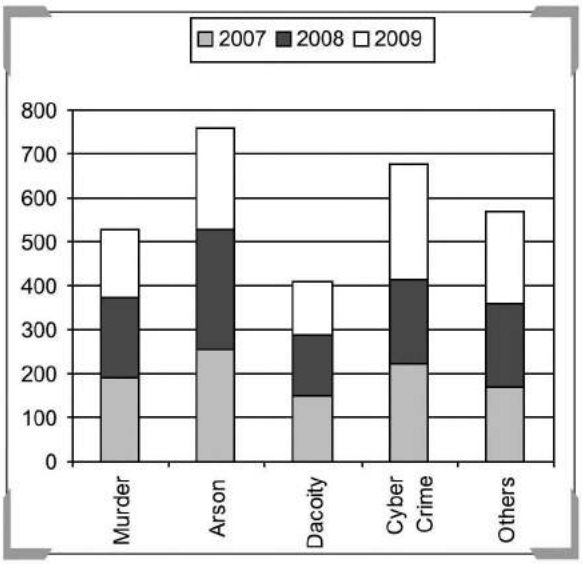
Following things can be observed in the above bar chart:

Bars representing different crimes in different years are proportional to the crimes reported.

The above bar chart takes the minimum value as 100, however the same bar chart could take the minimum value as 0 or 50 or anything else as well. The question here is, what will happen if we take the minimum value as 150?

Since some of the values are less than 150, what will happen to those? Find out yourself without using a computer.

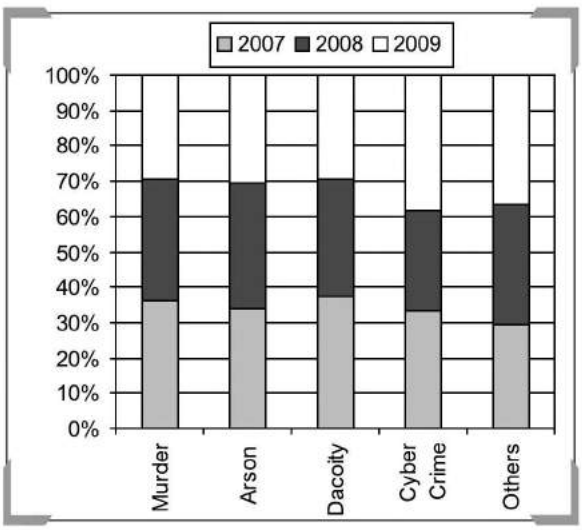
Stacked Bar Chart Value-wise



Following things can be observed in the above stacked bar chart:

- The total number of murders have been added up in one bar and the different years are shown as a part of that total.
- The same data could also have been represented by taking the sum of all the crimes reported in a particular year as a total, and the individual crimes as a part of that total.

Stacked Bar Chart Percentage-wise



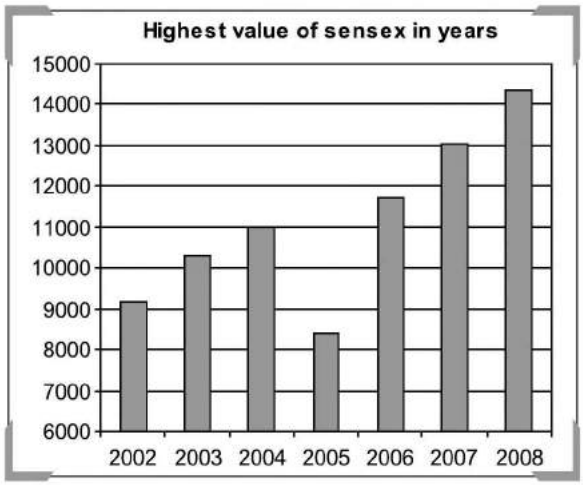
Following things can be observed in the above stacked bar chart:

- The total number of murders have been added up and they equal to 100%; and the murders reported in different years have been taken as a percentage of that total.
- The same data could have been represented by taking the sum of all the crimes reported in a particular year as a total, and the individual crimes as a part of that total.

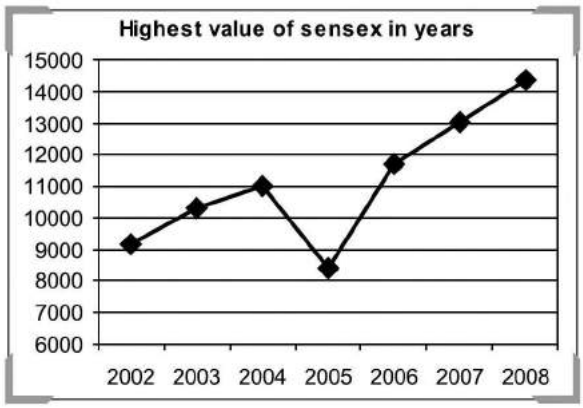
Line Chart/X–Y Chart

Line charts are seen as simplified forms of the normal bar chart.

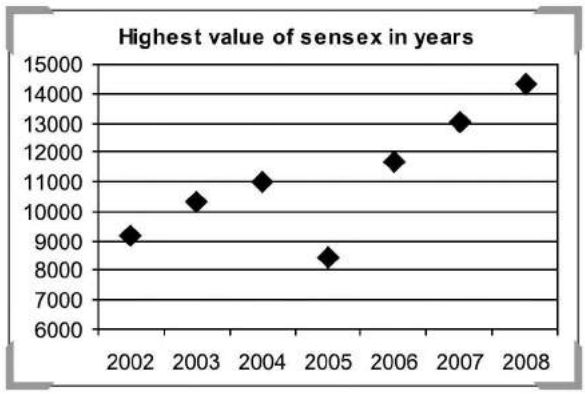
The bar chart given below represents the highest values of the sensex in the given years.



If we convert the same data into a line chart, it will look like this:



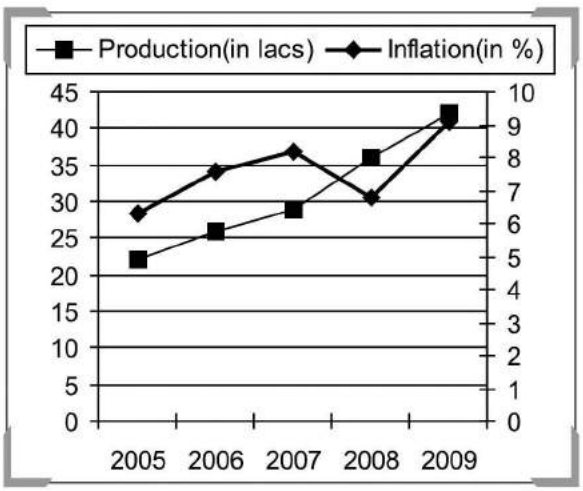
Sometimes in case of line charts, the lines are not given and only the dots are indicated in the graph. Let us see an example with the same data:



Significance of Line charts

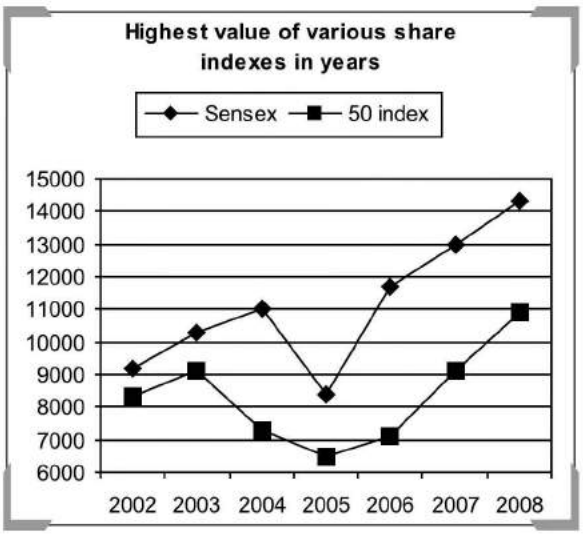
- Generally Line charts are used in case time is one of the variables in the data. The 'time' variable can be in the form of hour, day, months years or anything that represents chronological order of events.
- It is easier to calculate the percentage changes in a line chart and thereby understand the trends of the data in a better way.
- A line chart becomes very handy in case of data with two different scales.

The line chart given below represents the production of soaps by a company over a period of five years and the inflation for the same period.

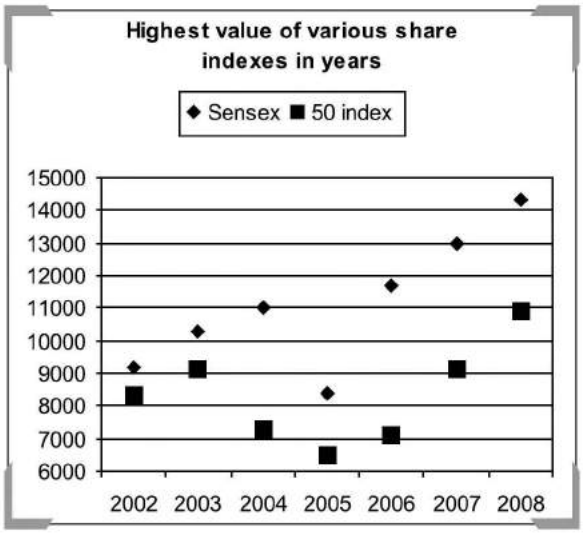


- Two or more than two variables can be represented on a line chart very easily. Besides, we can see the movement of data very easily in case of a line chart.

Below given line chart represents the movements of the highest value of two indices over a given period:



The same data can be represented using dots only also:



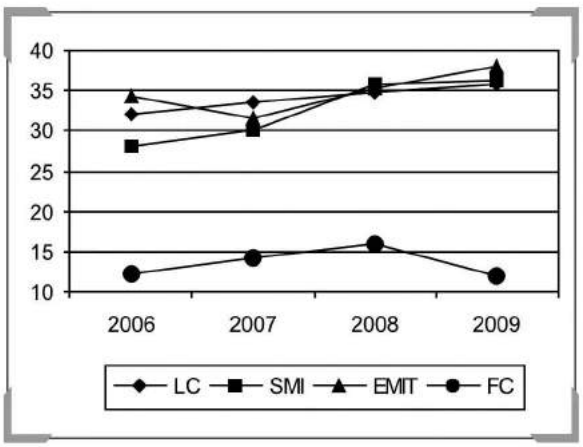
Types of Line Charts

Broadly, line charts are of three types:

Normal Line Chart

This is a simple line chart representing two or more than two variables.

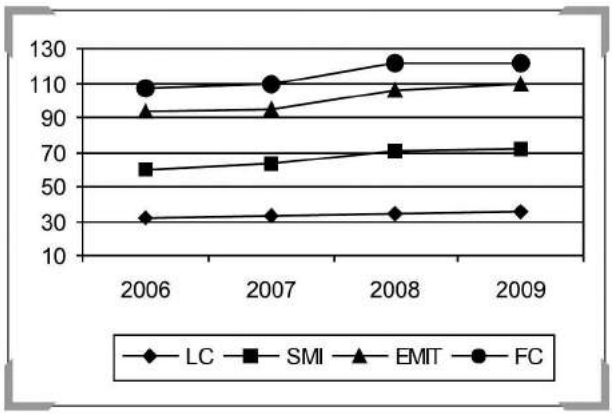
In the chart given below, The total number of enrolments for four different years of for four coaching institutes are represented:



Stacked Line Chart

In a stacked line chart, the values keep on getting added to obtain the next value.

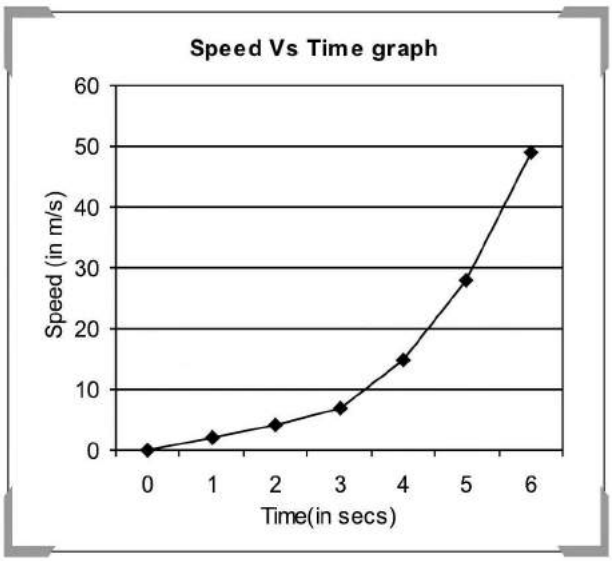
Here, the bottom line (of LC) gets added with the values of SMI to represent the value of SMI. Now this represented value of SMI is added to the actual value of EMIT to give the value of EMIT on the chart. And finally, the same is done with the value of FC.



Hence, in a stacked line chart, to obtain the values of different constituents/segments, either start with the top line or the bottom line and then keep on subtracting the values to obtain the next value.

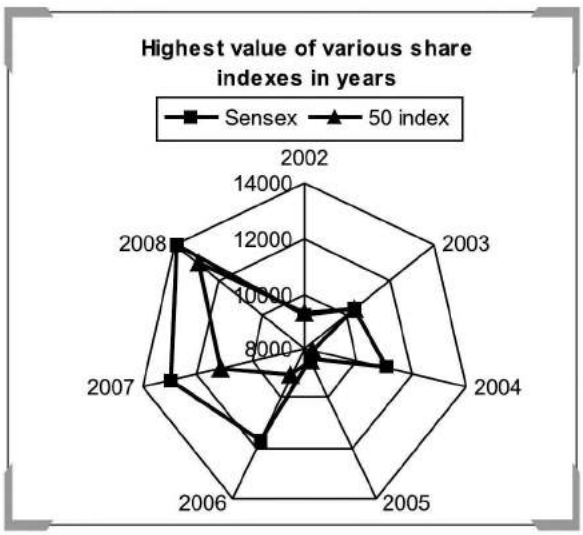
Time-Speed-Distance Line Chart

This line chart is used in tables of data collected from experiments on physical processes.



Radar Diagram

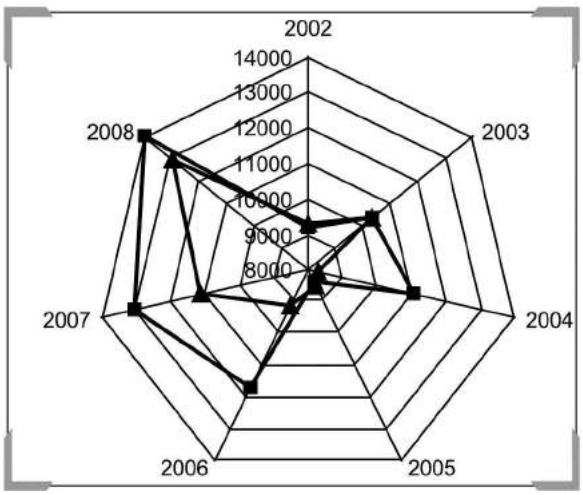
This diagram every value is represented with respect to a central point. All the changes in the values are expressed in the form of distance from this center point.



The radar diagram given below represents the highest value of two indices over a given period.

It can be seen that the centre value = 8000. With every passing circle the value increases by 2000. Since here are seven years, the diagram takes the shape of a heptagon.

Had there been only six years, it would have taken the shape of a hexagon, as shown below:



Similarly, had there been only four years, the diagram would have been in the shape of a square.

Area Diagram

In case of an area diagram, the values are represented in terms of areas.

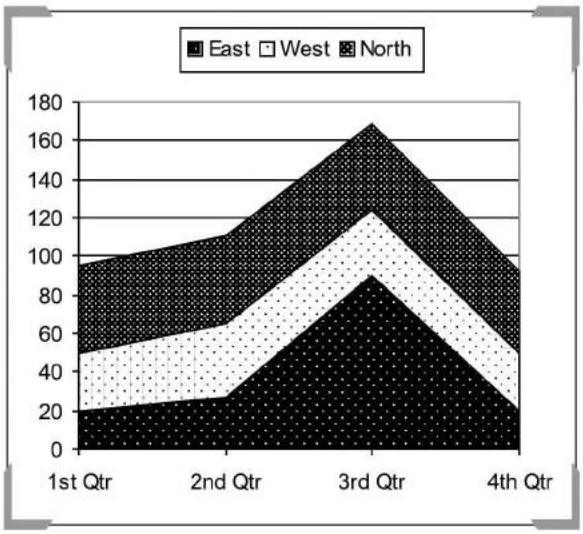
Example 2

Convert the data given below to a stacked area diagram:

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
East	20	27	90	20
West	30	38	34	30
North	45	46	45	42

Solution

The stacked area diagram corresponding to the table given above will be like this:



Floating Diagram

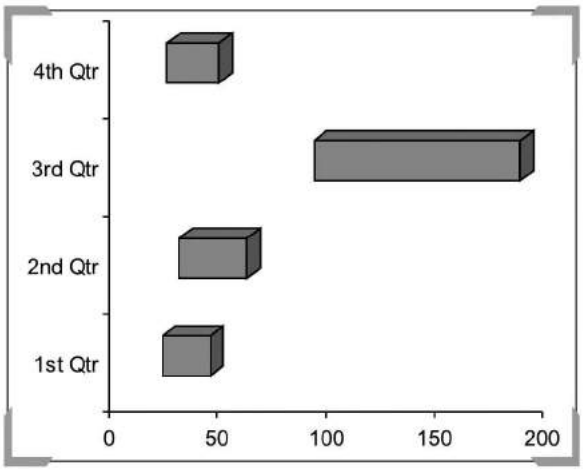
A floating diagram is used to represent the difference in any given variable between two different periods.

Example 3

Convert the following table into a floating diagram:

	1st Quarter	2nd Quarter	3rd Quarter	4th Quarter
2000	20	28	90	22
2001	22	31	95	24

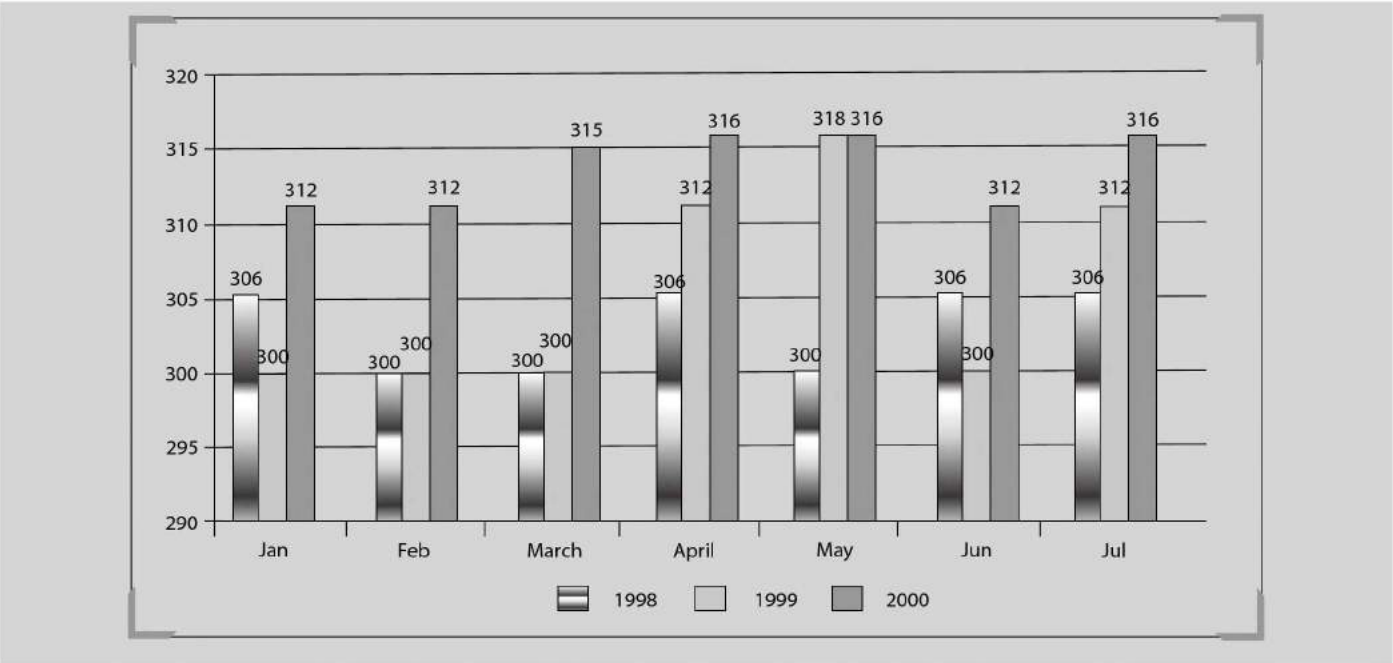
Solution



PRACTICE EXERCISE

BAR CHART

The following bar chart shows the monthly expenditure of a family comprising of five persons over a period of seven months during three different years.



Q 1. In any of the given years, which month sees the maximum percentage increase in expenses with respect to the previous month?
(a) Feburary (b) March
(c) April (d) June

Q 2. What is the average monthly expenditure of the whole family in the year 2000?
(a) Rs 314000 (b) Rs 315000
(c) Rs 316000 (d) Cannot be determined

Q 3. In April 2000, what was the percentage increase in expenditure over April 1999?
(a) 1.2% (b) 2%
(c) 6% (d) 9.4%

Q 4. Which period has shown the same change in trend across all the given three years?
(a) February-March
(b) May-June
(c) March-April
(d) January-February

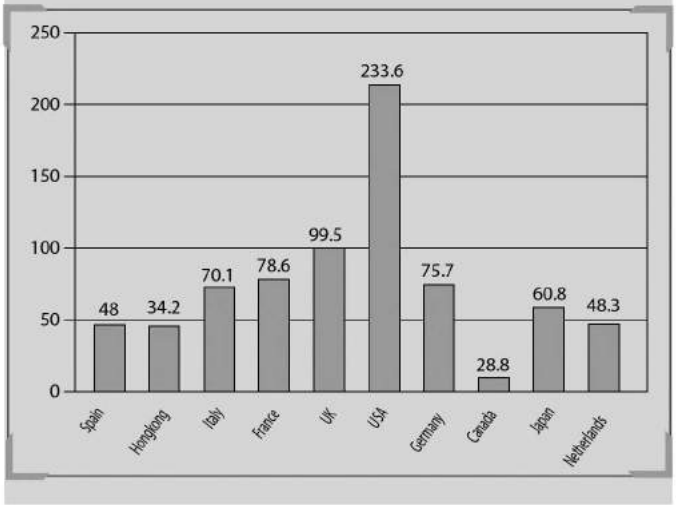
Q 5. Which of the following statements is correct?
(a) In 1998, May-June were the two consecutive months during which the expenditure was the maximum.
(b) During May-June 1999, the expenditure was the maximum for the year.
(c) Expenditure during January-Februrary was the same in 1998 as well as in 1999.
(d) None of these

Q 6. For how many months of the given years the expenditure has been consistently increasing or decreasing?
(a) 0 (b) 1
(c) 2 (d) 3

Q 7. Out of the following months in the options, Which month accounts for the maximum combined expenditure for three years?
(a) March (b) May
(c) April (d) Feburary

Direction for questions 8 to 14: Refer to the following bar graph and solve the based on it.

The bar chart given below represents top 10 exporters and their export value (in billion $) in 2008.



Q 8. If India also joins the top ten exporters list by becoming the 11th largest exporter, then what will be the effect on the average exports of these 11 countries; provided the export value of India is $ 10 billion?
(a) The average will increase by 1 billion dollars
(b) The average will decrease by 1 billion dollars
(c) The average will increase by 1.1 billion dollars
(d) None of these

Q 9. The ratio of exports between Spain and Japan is (approximately)
(a) 3 : 4 (b) 19 : 15
(c) 15 : 19 (d) 4 : 5

Q 10. What is the average export of all the countries in billion dollars for the year 2008?
(a) 77.66 (b) 76.76
(c) 76.67 (d) Cannot be determined

Q 11. If the exchange rate per dollar is Rs 43 then find the difference between the values of exports by Hong Kong and UK in Rupee.
(a) 28.179×10^9
(b) 28.179×10^{10}
(c) 28.079×10^{10}
(d) 28.079×10^{11}

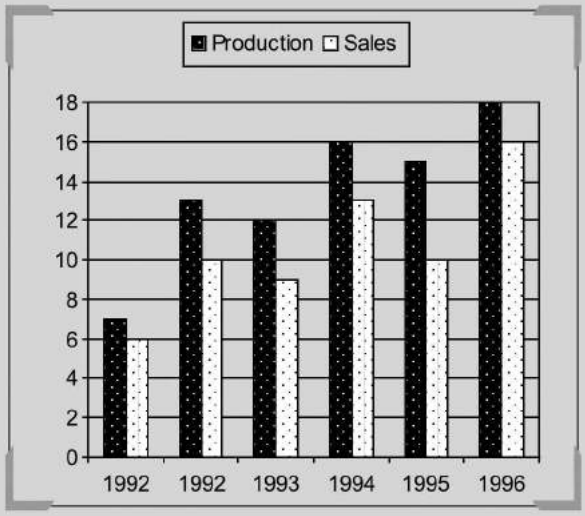
Q 12. If the average export of the top ten exporters is calculated how many of the top ten exporting countries have higher export than this average?
(a) 3 (b) 4
(c) 5 (d) Cannot be determined

Q 13. If the average export of the top ten exporters is calculated, how many countries have higher export than this average?
(a) 3 (b) 4
(c) 5 (d) Cannot be determined

Q 14. To fight the US hegemony, the countries have decided to merge their operations by adding their exports value. What is the minimum number of countries needed to merge their operation so that their combined exports is more than the exports value of the US? (None of these ten countries export to each other.)
(a) 3 (b) 4
(c) 5 (d) None of these

Direction for questions 15 to 20: Refer to the following bar graph and solve the questions based on it.

Amount of production and sales by a company over the years (in lacs tonnes)



Q 15. What is the difference between the sales of 1993 and 1995 (in thousand tonnes)?
(a) 1000 (b) 50
(c) 100 (d) 500

Q 16. Total sales of 1991 and 1992 together are approx-imately what percentage of the sales in 1994?
(a) 120% (b) 100%
(c) 50% (d) 200%

Q 17. What is the approximate percentage increase in production from 1993 to 1994?
(a) 20% (b) 33%
(c) 70% (d) 100%

Q 18. The percentage of sales to production is maximum in which of the following years?
(a) 1992 (b) 1994
(c) 1996 (d) 1991

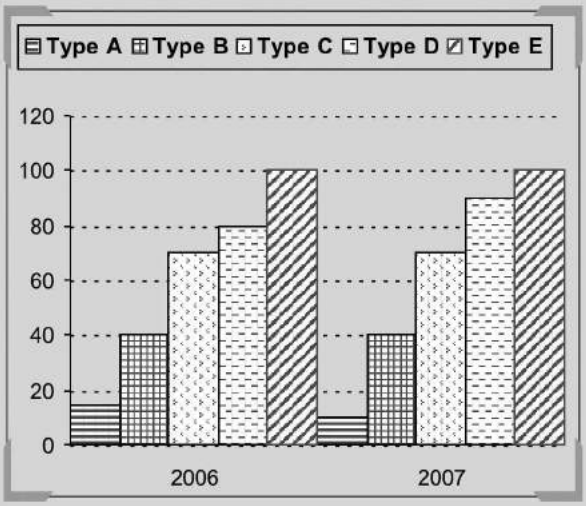
Q 19. What is the percentage drop in sales from 1992 to 1993?
(a) 5% (b) 40%
(c) 10% (d) 60%

Q 20. In which year is the difference between the production and sales the maximum?
(a) 1992 (b) 1993
(c) 1995 (d) 1996

Direction for questions 21 to 25: Refer to the following bar graph and solve the questions based on it.

The following bar chart gives the cumulative percentage of five different types of bikes produced by Kajaj during the given two years:

Total Bikes produced in 2006 is 4,50,000



Total bikes produced in 2007 is 520,000.

Q 21. What is the difference in the production of the C type bikes between 2006 and 2007?
(a) 5,000
(b) 7,500
(c) 10,000
(d) None of these

Q 22. If 85% of the E type bikes produced during 2006 and 2007 are being sold by the company, then how many E type bikes are left unsold?
(a) 1,42,800
(b) 21,825
(c) 29,100
(d) 21,300

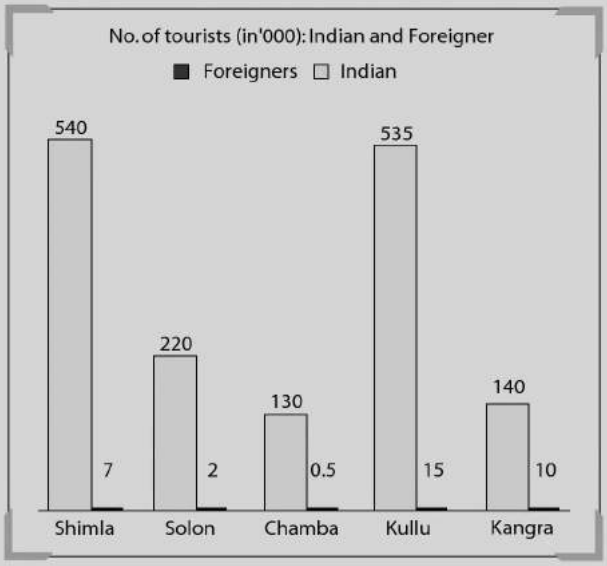
Q 23. What is the approximate percentage share of the A type bikes in the total production of (2007 + 2006)?
(a) 11 (b) 13
(c) 15 (d) 9

Q 24. In the case of which of the following types of bikes is the percentage increase from 2006 to 2007 the maximum?
(a) A (b) E
(c) D (d) B

Q 25. Production of B type bikes in 2007 is what percentage of bikes produced in 2006?
(a) 30% (b) 40%
(c) 35% (d) 25%

Direction for questions 26 to 30: Refer to the following bar graph and solve the based on it.

The bar graph given below shows the five districts of the state of Himachal Pradesh and the number of tourists visiting them. For the given bar chart, total number of Indian tourists is more that the total number of foreign tourists. None of the tourists can be Indian as well as foreigner.



Q 26. By what percentage are the Indian tourists visiting Chamba less than those visiting Shimla?
(a) 30% (b) 50%
(c) 60% (d) 75%

Q 27. Approximately what percentage of the total tourists visiting Himachal Pradesh are foreigners?
(a) 2% (b) 8%
(c) 4% (d) 5%

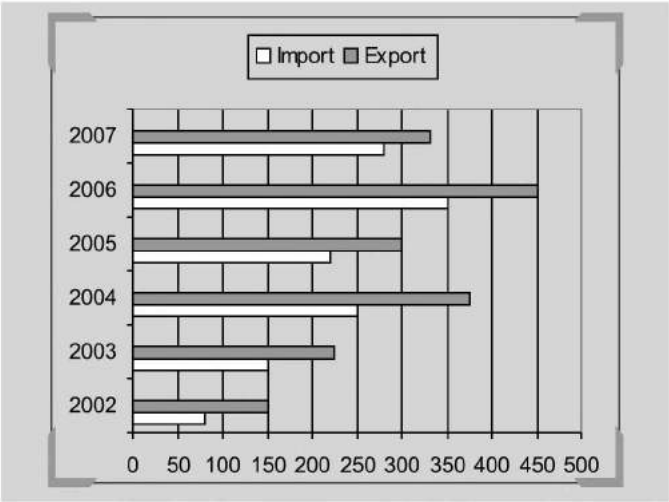
Q 28. What is the ratio between the Indian tourists and the foreign tourists visiting Kullu?
(a) 105 : 3 (b) 70 : 3
(c) 107 : 3 (d) 35 : 1

Q 29. Which of the following districts has the maximum ratio of Indian and foreign tourists?
(a) Shimla (b) Solon
(c) Chamba (d) Kangra

Q 30. Which of the following districts has the minimum ratio of Indian and foreign tourists?
(a) Shimla (b) Solon
(c) Chamba (d) Kangra

Direction for questions 31 to 35: Refer to the following bar graph and solve the questions based on it.

The following bar charts represents the value of exports and imports (in Rs crores) of a country for the given period:



(All the values in the above bar chart are the multiples of 10).

Q 31. The value of exports in 2006 was what percentage of the average value of imports in the years 2004, 2005 and 2007?
(a) 200% (b) 100%
(c) 150% (d) None of these

Q 32. The value of exports in 2004 was exactly what percentage of the value of imports in the same year?
(a) 200%
(b) 100%
(c) 150%
(d) None of these

Q 33. What is the approximate difference (in Rs crores) between the average exports and the average imports for the given years?
(a) 65 (b) 85
(c) 105 (d) 135

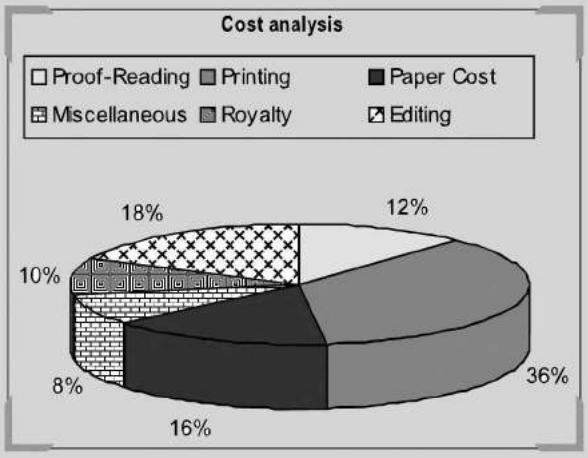
Q 34. In which year is the difference between the exports and the imports closest to Rs 100 crores?
(a) 2005
(b) 2006
(c) 2007
(d) 2008

Q 35. What is the percentage increase in the value of exports in 2006 over 2005?
(a) 30% (b) 50%
(c) 90% (d) 120%

PIE CHART

Direction for questions 1 to 5: Refer to the following pie chart and solve the questions based on it.

Following is the cost analysis of a book "Pearson's Guide to *Quantitative Aptitude* for CAT".



Q 1. What is the central angle showing the cost of paper?
(a) 42.8° (b) 32.6°
(c) 36.8° (d) 57.6°

Q 2. If the cost of printing is Rs 23,400 what would the cost of royalty be?
(a) Rs 6500 (b) Rs 2340
(c) Rs 4680 (d) Rs 7840

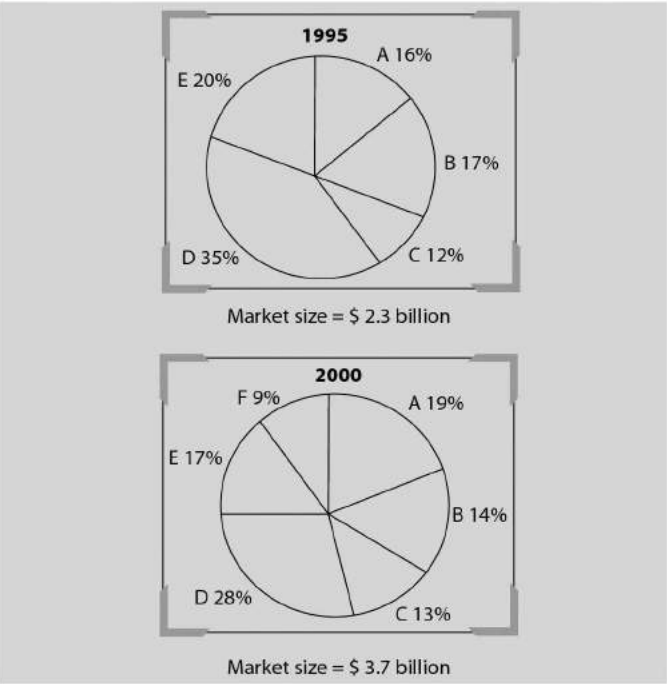
Q 3. If the miscellaneous expenditure amounts to Rs 18,000 then what is the expenditure on editing?
(a) Rs 8000 (b) Rs 14400
(c) Rs 46800 (d) None of these

Q 4. The royalty on the book is less than the editing expenditure by
(a) 8% (b) 80%
(c) 44.44% (d) None of these

Q 5. If 5500 copies of the book are published and the miscellaneous expenditure amounts to Rs 36,960 and the marked price is 40% above the cost price, then the marked price of each copy is
(a) Rs 122.50
(b) Rs 117.60
(c) Rs 126.40
(d) Rs 92.40

Direction for questions 6 to 10: Refer to the following pie charts and solve the questions based on it.

The pie charts given below represent the market share of different players in the electronics market in the year 1995 and 2000. A new player F has entered the market in the given period.



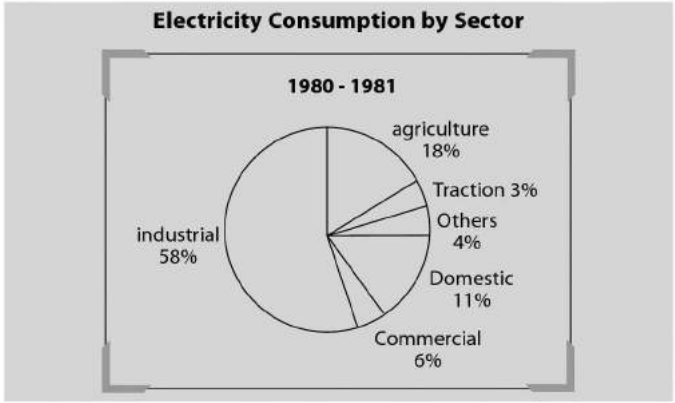
Q 6. What is the percentage increase in the sales of A in the year 2000 over 1995?
(a) 90% (b) 60%
(c) 80% (d) 50%

Q 7. Find the CAGR with which the market has grown for the period 1995-2000?
(a) 18%
(b) 10%
(c) 5%
(d) 12%

Q 8. If the new entrant F has grown at the expense of B, C, and E, what would be the change in the net value of sales of E in the year 2000, had F not entered the market?
(a) $ 111 million
(b) $ 340 million
(c) $ 247 million
(d) Cannot be determined

Q 9. What is the increase in sales (in $ million) of A, B and C put together, over the 5 years period?
(a) 103
(b) 1202
(c) 668
(d) 854

Q 10. If the market grows at 10% p.a. over the next 2 years and D captures 1/3rd of the new market, what would be its market share in 2002?
(a) 29% (b) 27%
(c) 25% (d) 32%



Q 11. By what percentage has the consumption of electricity by agriculture increased in 1993-1994 over 1980-1981?
(a) 66% (b) 33%
(c) 133% (d) Cannot be determined

Q 12. The electricity consumption of how many sectors has definitely increased over the given period?
(a) 2 (b) 3
(c) 4 (d) Cannot be determined

Q 13. If the total electricity consumption in 1993-94 is 1.2 times of the total electricity consumption in 1980- 1981, then how many sectors have definitely increased by more than 50% during the same period?
(a) 1 (b) 2
(c) 3 (d) 4

Q 14. If the total electricity consumption in 1993-1994 is 1.5 times of the total electricity consumption in 1980-1981, then what is the percentage increase in the electricity consumption of the agricultural sector in the given period?
(a) 50% (b) 100%
(c) 150% (d) None of these

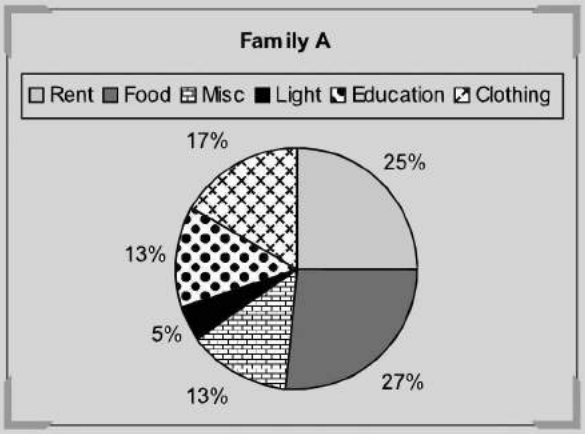
Q 15. What is the minimum number of sectors required to be added up in order to be more than 50% of the consumption for the period in 1993-1994?
(a) 1 (b) 2
(c) 3 (d) 4

Q 16. The agricultural consumption of electricity doubled from 1980-81 to 1993-94. By how much percentage has the total electricity consumption grown from 1980-81 to 1993-94?
(a) 20% (b) 25%
(c) 50% (d) Cannot be determined

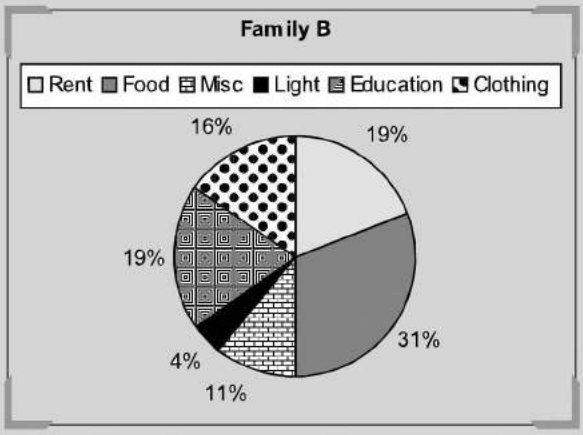
Q 17. If the electricity consumption of the 'others' category has remained constant over the period, then what is the percentage increase in the electricity consumption of the domestic category?
(a) 63% (b) 38%
(c) 58% (d) Cannot be determined

Direction for questions 18 to 23: Refer to the following pie charts and solve the questions based on it.

The two pie charts given below provide the expenses of two families:



Total expenses = Rs 48, 000



Total expenses = Rs 72,000.

Q 18. Both the families decide to double the total expenditure keeping the pattern of spending the same as given above. What will be the new ratio of expenditure on food between family A and family B?
(a) 18:31 (b) 31:27
(c) 2:3 (d) 3:2

Q 19. If the total expenses of family B increase three-fold, keeping the expenses on education the same as given above, what will be the expense on education?
(a) 6.33% (b) 57%
(c) 19% (d) None of these

Q 20. What will be the expenses on light by family A, as a percentage of expense on light by family B?
(a) 120% (b) 83.33%
(c) 62.5% (d) 66.66%

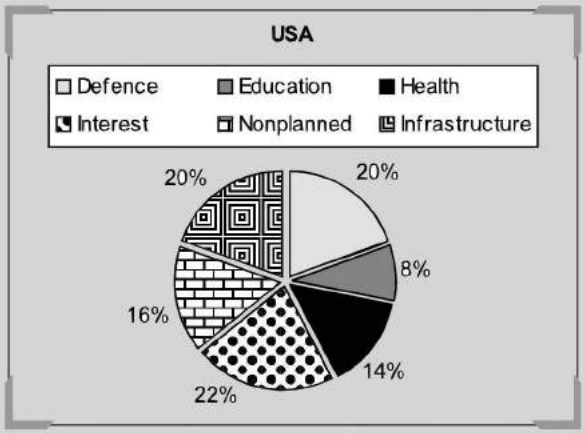
Q 21. If family A and family B decide to combine their expenses, then which one of the following heads will be responsible for the highest expenses?
(a) Rent (b) Misc
(c) Food (d) Education

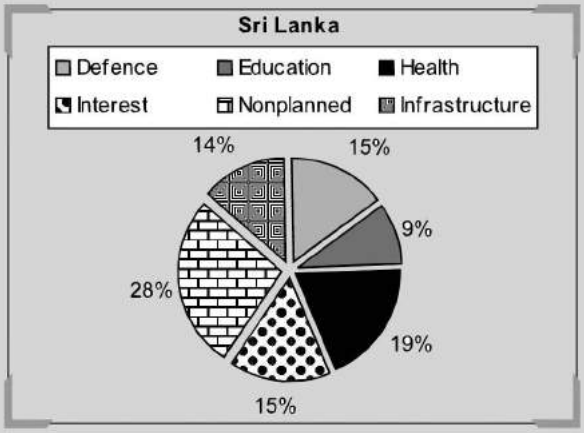
Q 22. In the above question, how many heads will have a lower percentage share in the combined total expenses of both the families than the percentage share of family B under the same head?
(a) 1 (b) 2
(c) 3 (d) 4

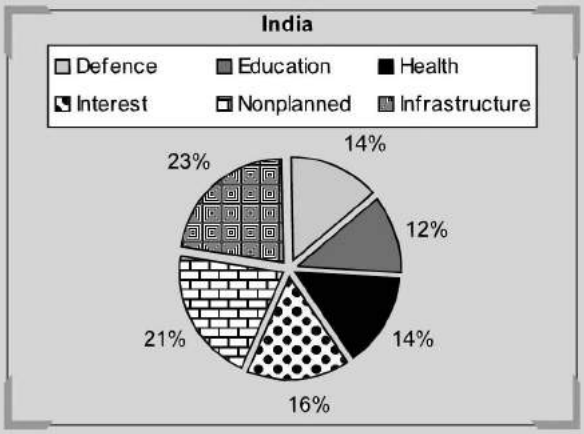
Q 23. Under how many heads are the expenses of family B more than the expenses of family A?
(a) Less than 3
(b) More than 3
(c) Equal to 3
(d) Cannot be determined

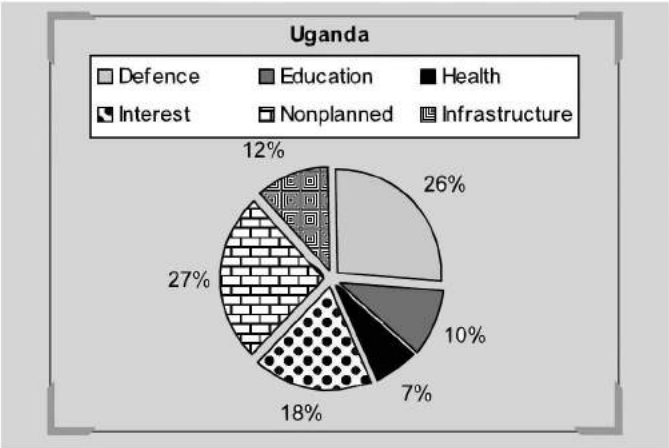
Direction for questions 24 to 30: Refer to the following pie charts and solve the questions based on it.

The following pie charts represent the budget expenditure of certain countries on various sectors for the year in 2007:









Q 24. A country is said to be progressive if its education, health and infrastructure expenditures are in the top four expenditure sectors. How many of the four countries are progressive?
(a) 0 (b) 1
(c) 2 (d) 3

Q 25. A country is said to be developing if its combined expenditure on education, health and infrastructure is atleast 50% of the total expenditure. How many of the four countries are developing nations?
(a) 0 (b) 1
(c) 2 (d) 3

Q 26. If a country is under military rule, it will spend maximum on defence and minimum on either health or education. How many of the above countries are under military rule?
(a) 0 (b) 1
(c) 2 (d) Cannot be determined

Q 27. Which of the following sectors will have same ranking in the countries budget expenditures?
(a) Nonplanned (b) Education
(c) Health (d) None

Q 28. If the budgetary expenditures of all the four countries are combined, which sector will account for the maximum expenditure?
(a) Non-planned (b) Interest
(c) Defence (d) Cannot be determined

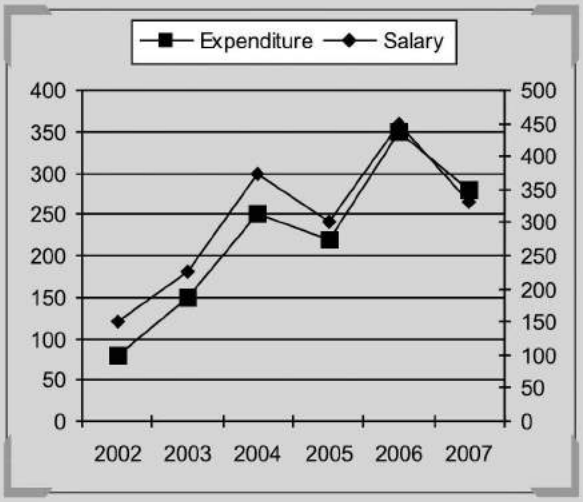
Q 29. The total budgetary expenditure of USA is four times the total budgetary expenditure of India; and the population of India is twice the population of USA. Then the per capita budgetary expenditure of USA is how many ….. times per capita expenditure of India.
(a) 8 (b) 4
(c) 1/8 (d) 1/4

Q 30. The total budgetary expenditure of Sri Lanka is twice the total budgetary expenditure of Uganda. Under how many sectors is the budgetary expenditure in Uganda more than the budgetary expenditure in Sri Lanka?
(a) 0 (b) 1
(c) 2 (d) 3

LINE CHART

Direction for questions 1 to 7: Refer to the following line chart and solve the questions based on it.

The line chart given below represents the salary and expenditure (in Rs) of Guru Gulab Khatri for the given period.



It is given that the salary is greater than the expenditure atleast for four years. All the values are multiples of 5.

Q 1. By what percentage has the salary of Guru Gulab Khatri increased in 2004 over 2002?
(a) 210% (b) 150%
(c) 100% (d) 250%

Q 2. By what percentage has the expenditure of Guru Gulab Khatri increased in 2004 over 2003?
(a) 66.66% (b) 133.33%
(c) 150% (d) None of these

Q 3. For how many years the salary is more than expenditure?
(a) 4 (b) 5
(c) 6 (d) Cannot be determined

Q 4. Which year is the percentage increase in the salary of Guru Gulab Khatri maximum?
(a) 2003 (b) 2004
(c) 2006 (d) 2003 or 2006

Q 5. A year is said to be comfortable, if Guru Gulab Khatri is able to save atleast 30% of his salary. How many years are comfortable during the given period?

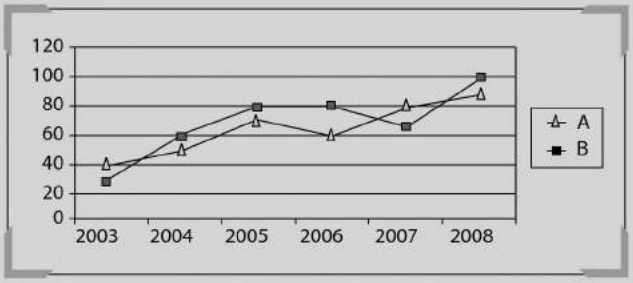
(a) 2 (b) 3
(c) 4 (d) 5

Q 6. What is the maximum difference (in Rs) between the salary and the expenditure for any year?
(a) 70 (b) 75
(c) 125 (d) 100

Q 7. For how many years is the difference between the salary and the expenditure more than Rs 80?
(a) 2 (b) 3
(c) 4 (d) 5

Direction for questions 8 to 12: Refer to the following line chart and solve the questions based on it.

Percentage profit earned by the two companies A and B over the years



Q 8. If the income for company A in the year 2004 was Rs 35 lacs what was the expenditure for company B in the same year?
(a) Rs 123.5 lacs
(b) Rs 128 lacs
(c) Rs 132 lacs
(d) Cannot be determined

Q 9. If the income of company A in 2006 and the income of company B in 2007 are equal, what will be the ratio of expenditure of company A in 2006 to the expenditure of company B in 2007?
(a) 26 : 7 (b) 17 : 16
(c) 15 : 170 (4) None of these

Q 10. During which of the following years is the ratio of percentage profit earned by company A to that of company B, the maximum?
(a) 2003 and 2006
(b) 2005 and 2007
(c) 2003 only
(d) 2008 only

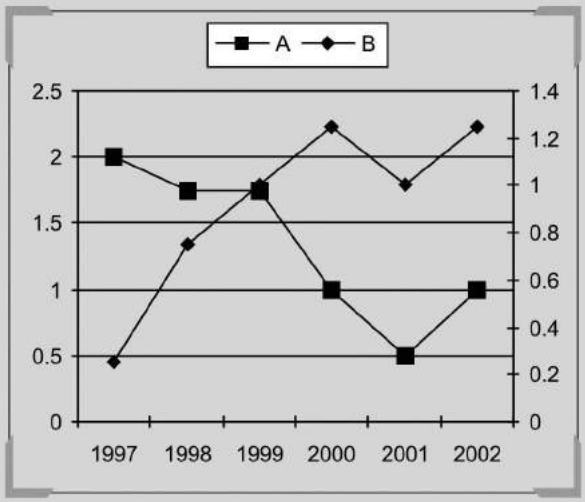
Q 11. If the expenditure of company B increases by 20% from 2005 to 2006, the income in 2006 will be how many times the income in 2005?
(a) 2.16 times
(b) 1.2 times
(c) 1.8 times
(d) None of these

Q 12. If the income of company A in 2006 was Rs 36 lacs, what was the expenditure of company A in 2006?
(a) Rs 12.5 lacs
(b) Rs 18.8 lacs
(c) Rs 20 lacs
(d) None of these

Direction for questions 13 to 17: Refer to the following line chart and solve the questions based on it.

The following line chart represents the ratio of exports (in Rs) to imports (in Rs) of two countries A and B over the given period:

Left Axis = Country A ; Right Axis = Country B



All the values are multiples of 0.25.

Q 13. For how many years do both the countries have their respective exports more than their imports?
(a) 1 (b) 2
(c) 4 (d) None of these

Q 14. Due to some compulsions, one year during the period given these two countries have to do business between themselves only and they are not allowed to export to or import from any other countries. Which out of the following years is possibly that year?
(a) 1997 (b) 1998
(c) 1999 (d) None of these

Q 15. What can be said regarding the imports/exports of the two countries?
(a) The value of exports of country A is constantly on the decline from 1997 to 2001.
(b) The value of imports of the country B is constantly on the rise from 1997 to 2000.
(c) The value of imports of country A is equal to the value of import in 2002 (ratio = 1 for both the years)
(d) none of these

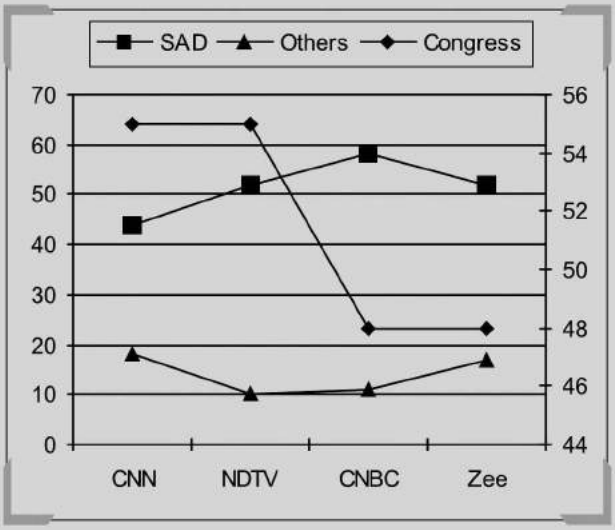
Q 16. For how many years is the value of exports of country B is constantly on the rise over the previous year for country B?

(a) 2 (b) 3
(c) 4 (d) Cannot be determined

Q 17. For how many years is the value of imports of country A is constantly on the rise over the previous year?
(a) 2 (b) 3
(c) 4 (d) Cannot be determined

Direction for questions 18 to 20: Refer to the following line chart and solve the questions based on it.

The line chart given below represents the projections of seats for the coming Punjab assembly elections according to surveys done by four different agencies.



The following observations have also been made regarding the projected seats:

(i) The others have won less than 20 seats according to the projections made by each of the agencies.
(ii) According to NDTV, the Congress' projected seats are more than the projected seats for the SAD.
(iii) The total number of seats have been won by only three parties—Congress, SAD and the others.
(iv) According to CNBC, the difference between projected seast of SAD and Congress is 20.

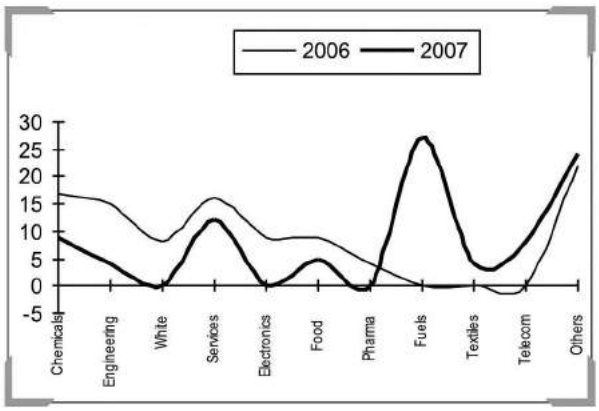
Q 18. If we add the projected seats of each of the parties individually, then which of the following will have the maximum number of projected seats?
(a) Congress
(b) SAD
(c) Congress or SAD
(d) Others

Q 19. According to how many agencies' projections is the Congress getting more seats than the SAD?
(a) 0
(b) 1
(c) 2
(d) 3

Q 20. For how many agencies' projection about seats has the sum of the two bottom ranked parties been more than the top ranking party?
(a) 0
(b) 1
(c) 2
(d) 4

Direction for questions 21 to 25: Refer to the following line chart and solve the questions based on it.

The following line charts give the FDI (in $ billion) in 2006 and 2007 for the different sectors in India:



Q 21. For how many sectors, is there an increase in the FDI in 2007 over 2006?
(a) 2 (b) 3
(c) 4 (d) 5

Q 22. Which of the following sectors has seen the maximum percentage increase in its value in 2007 over 2006 (only for those sectors which have their presence in both the years)?
(a) Fuel (b) Others
(c) Chemical (d) None of these

Q 23. Which of the following sectors has seen the maximum percentage decrease in its value in 2007 over 2006 (only for those sectors which have their presence in both the years)?
(a) Fuel
(b) Others
(c) Chemical
(d) None of these

Q 24. Which of the following sectors will have the maximum FDI added in both the years?
(a) Fuel (b) Others
(c) Chemical (d) None of these

Q 25. Which of the following sectors will have the minimum FDI added in both the years?
(a) Fuel (b) Others
(c) Chemical (d) Textile

Table

Direction for questions 1 to 4: Refer to the following table and solve the questions based on it.

The following table gives the exports of top six industries (in value terms) from India in 2007:

	Exports (Rs crore)	Growth (in %)
Glass industry	1093	11.7
Coal Industry	1080	7
Ruby Industry	913	–8.7
Nickel Industry	663	7.1
Software Industry	605	–12.3
Diamond Industry	505	–1.5

The overall exports from India increased by 20% to Rs 19,500 crore in 2007.

Q 1. What is the share of the Glass industry in the total exports for 2007?
(a) 2.5%
(b) 4%
(c) 5.6%
(d) 7%

Q 2. For how many of the given six industries has the share in exports increased in 2007?
(a) 0
(b) 1
(c) 2
(d) 3

Q 3. Which of the following has the maximum share among the given industries in 2006?
(a) Glass Industry
(b) Ruby Industry
(c) Software Industry
(d) Coal Industry

Q 4. What is the percentage increase in the exports of the given six industries?
(a) 0.1%
(b) 0.5%
(c) 1.05%
(d) 1.75%

Direction for questions 5 to 9: Refer to the following table and solve the questions based on it.

The following table gives the number of vehicles (in '000s) of different models and colours sold in two cities—Patna and Lucknow–in a given year.

Type	Patna					Lucknow				
	Colour					Colour				
	Black	Red	Blue	White	Silver	Black	Red	Blue	White	Silver
A	40	25	55	75	15	45	32	40	60	20
B	20	35	60	80	20	30	37	39	81	35
C	35	30	50	90	35	40	42	41	6	37
D	45	40	45	85	40	35	39	37	90	42
E	50	35	35	60	30	50	44	43	77	22
F	55	42	40	65	52	47	34	45	87	17

Q 5. For which of the following models, difference between white—coloured vehicles sold in the two cities is minimum?
(a) A
(b) C
(c) D
(d) None of these

Q 6. The total number of blue—coloured vehicles of model E and D sold in Lucknow is exactly equal to the number of white-coloured vehicles of which model in Patna?
(a) B (b) F
(c) C (d) A

Q 7. What is the difference between the number of blue-coloured vehicles of model C sold in Patna and the number of red coloured vehicles of model F sold in Lucknow?
(a) 8,000
(b) 10,000
(c) 12,000
(d) None of these

Q 8. The total number of silver-coloured vehicles sold in lucknow is approximately what percentage of that in Patna?
(a) 130% (b) 140%
(c) 90% (d) 100%

Q 9. In Patna the number of vehicles sold was the maximum for which of the colour–model combinations?
(a) white-C
(b) blue-B
(c) silver-B
(d) white-D

Direction for questions 10 to 14: Refer to the following table and solve the questions based on it.

The table given below gives the marks obtained by six students in six subjects:

Student	History (150)	Science (200)	Geography (150)	English (200)	Hindi (100)	Math (200)
A	75	110	90	140	75	170
B	105	130	75	130	80	140
C	95	105	80	150	90	160
D	85	115	95	125	65	135
E	115	135	110	145	70	125
F	120	160	96	110	55	145

Marks in bracket are Total Marks.

Q 10. What is the difference between the percentage marks obtained by student B in History and the percentage marks obtained by student C in Hindi?
(a) 25 (b) 20
(c) 35 (d) 30

Q 11. The marks obtained by student A in Math are how many times the percentage marks obtained by student F in Science?
(a) 2.5
(b) 4.125
(c) 1.125
(d) 2.125

Q 12. What is the difference between the percentage marks obtained by student C in English and the average percentage marks of all the six subjects?
(a) 82
(b) 38
(c) 7
(d) 14

Q 13. In how many of the given subjects has student D got more than seventy percent marks?
(a) none
(b) one
(c) two
(d) three

Q 14. Approximately, what is the average percentage marks obtained by the six students in English?
(a) 67%
(b) 72%
(c) 80%
(d) Data inadequate

Direction for questions 15 to 18: Refer the following table and solve the questions based on it.

The following table gives the number of students from different locations who appeared and passed in the Xth standard examination conducted by CBSE over the years:

Year	Rural		Semi-urban		State Capitals		Metro-politans	
	Appeared	Passed	Appeared	Passed	Appeared	Passed	Appeared	Passed
1990	1651	209	7897	2511	5057	1464	9537	3212
1991	1832	314	8561	2932	7163	3242	10152	4015
1992	2154	935	8133	2466	8251	3151	9697	3033
1993	5035	1792	9436	3525	8526	3627	11245	5157
1994	4911	1656	9782	4012	9013	4315	12512	6323
1995	5626	2391	9965	4266	1729	4523	13625	6411

Q 15. For the students from which of the following locations was there a continuous increase in both the appeared and passed member of candidates?
(a) semi-urban
(b) state capital
(c) state capital and rural
(d) None of these

Q 16. In which of the following years was the percentage passed to appeared candidates from the semi-urban area the least?
(a) 1991 (b) 1993
(c) 1990 (d) 1992

Q 17. By what approximate value did the percentage drop in the number of semi-urban candidates who appeared from 1991 to 1992?
(a) 5 (b) 10
(c) 15 (d) 8
(e) 12

Q 18. The total number of candidate who passed from the rural locations in 1993 and the semi-urban locations in 1990 was exactly equal to the total number of candidates who passed from the state capitals in which of the following years?
(a) 1990
(b) 1993
(c) 1994
(d) None of these

Direction for questions 19 to 21: Refer to the following table and solve the questions based on it.

The following table gives the average marks obtained by 20 boys and 20 girls in five subjects from five different schools P, Q, R, S and T:

Subject	Maximum Marks	P		Q		R		S		T	
		B	G	B	G	B	G	B	G	B	G
English	200	85	90	80	75	100	110	65	60	105	110
History	100	40	55	45	50	50	55	40	45	65	60
Geography	100	50	40	40	45	60	55	50	55	60	65
Math	200	120	110	95	85	135	130	75	80	130	135
Science	200	105	125	110	120	125	115	85	90	140	135

In the above table, B = Boys and G = Girls

Q 19. What were the total marks obtained by the boys in History from school Q?
(a) 900 (b) 1000
(c) 800 (d) 1300

Q 20. In which of the following subjects did the girls have the highest average percentage marks in all the schools?
(a) Science
(b) Geography
(c) English
(d) History

Q 21. What is the average mark obtained by all the students of school P taking all the subjects into account?
(a) 82
(b) 84
(c) 80
(d) Cannot be determined

Direction for questions 22 to 28: Refer to the following table and solve the questions based on it.

The following table shows the domestic sales of cars of five manufacturers from 1995 to 2000. (All the figures are in thousands)

Manufacturer	1995	1996	1997	1998	1999	2000
A	440	480	470	500	520	510
B	400	410	415	415	420	430
C	380	390	390	400	420	495
D	360	380	400	415	440	500
E	480	440	440	420	425	435

Q 22. Considering the period given, the domestic sales of which manufacturer are the highest?
(a) A (b) B
(c) C (d) D

Q 23. During 1996, what is the approximate share of domestic sales of cars of the manufacturer B?
(a) 10.5%
(b) 25.5%
(c) 15.5%
(d) 19.5%

Q 24. During 2000, the sales of which manufacturer has shown the maximum percentage increase over the previous year?
(a) A (b) B
(c) C (d) D

Q 25. With respect to which of the following combinations, is the sales of cars the highest over the given period?
(a) D, 2000
(b) A, 1998
(c) A, 2000
(d) A, 1999

Q 26. Which of the following manufacturers, has a consistent increase across the years given?
(a) A (b) B
(c) C (d) D

Q 27. How many of the manufacturers have never shown a decline in the number of cars sold in any particular year over the previous year?
(a) 1 (b) 2
(c) 3 (d) 4

Q 28. How many times have the number of cars sold been more than 450 for any manufacturer in any year more than once? (A manufacturer can achieve this feat)
(a) 6 (b) 7
(c) 8 (d) 9

Direction for questions 29 and 30: Refer to the following table.

The following table captures the comparative performance of LIC and HDFC in the housing sector in the last four years. (All figures are in Rs crores).

Attribute	1994-1995		1993-1994		1992-1993		1991-1992	
	LIC	HDFC	LIC	HDFC	LIC	HDFC	LIC	HDFC
Loan Sanctions	618.0	1495.0	564.0	1025	607	859	511	758
Dues	24.2	29.6	20.0	27.8	16.1	25.17	5.32	19.57
Total Loans	1672.0	3747.0	1283.0	3071	852	2561	401	2125
Default Percentage	1.45	0.79	1.56	0.91	1.89	.98	1.32	0.93
Total Revenue	251.0	780.0	175.0	608	109	473	44.4	371
Net Profit	41.4	146.1	21.53	105	13.5	55.55	6.71	45.6

Q 29. For how many years are the LIC loan sanctions 70% or more of HDFC's loan sanctions?
(a) 0 (b) 1
(c) 2 (d) 3

Q 30. The ratio of total revenue to the loan sanctions is the least for LIC in which of the following years?
(a) 1992-1993 (b) 1993-1994
(c) 1994-1995 (d) 1991-1992

ANSWERS

■■■ Bar Chart

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(a)	(c)	(d)	(b)	(b)	(d)	(d)	(d)	(d)	(a)	(a)	(a)	(c)	(a)	(b)	(c)	(c)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35					
Answer	(d)	(d)	(b)	(c)	(c)	(d)	(a)	(c)	(c)	(c)	(d)	(c)	(b)	(b)	(b)					

■■■ Pie Chart

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(d)	(c)	(b)	(a)	(b)	(d)	(c)	(a)	(d)	(d)	(b)	(c)	(b)	(a)	(a)	(a)	(a)	(b)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(c)	(b)	(b)	(a)	(a)	(d)	(d)	(d)	(a)	(a)										

■■■ Line Chart

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(a)	(c)	(b)	(b)	(c)	(a)	(d)	(b)	(c)	(b)	(d)	(d)	(d)	(d)	(d)	(d)	(c)	(c)	(d)
Q. No.	21	22	23	24	25															
Answer	(c)	(b)	(d)	(b)	(d)															

■■■ Table

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(a)	(d)	(c)	(d)	(a)	(d)	(c)	(a)	(b)	(d)	(c)	(a)	(a)	(d)	(d)	(a)	(d)	(a)	(a)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(d)	(a)	(d)	(c)	(d)	(d)	(c)	(c)	(b)	(d)										

HINTS AND SOLUTIONS

Bar Chart

1. This question can be solved by visual inspection.
2. To find the average monthly expenditure during the year 2000, the total income for the year has to be known. In the chart, the income of only seven months is known. Therefore it cannot be determined.
3. Value (April 2000) = 316

   Value (April 1999) = 312

   Approximate % age increase = 1.2%
4. The same increasing trend is seen every year during the period March-April.
5. In 1998, the period during which the expenditure was maximum was June-July.

   In 1999, the period of maximum expenditure was April-May.

   So, option (a) and (b) are incorrect. Option (c) is obviously incorrect.

   Hence option (d) has to be true.
6. Only April satisfies these conditions.
7. Visual inspection we can see that option (b) is correct. It should also be noted that we have to find out of the options given.
8. The average will decrease by 6.11 billion dollars. Hence, option (d) is the correct answer.
9. Required ratio = 4 : 5
10. Cannot be determined because the data is given only for the top ten exporters.
11. Required difference = (99.5 – 34.2) × 43 × 10^9 = Rs 28.079 × 10^{11} (1 billion = 10^9)
12. The average is 77.6. Now it can be seen that only three countries have more than this.
13. There is no difference between this and the last question. Understand that the average is calculated for the top ten countries only, and the only countries which can have exports higher than the average exports can be the countries which are in the list of the top ten exporter countries.
15. The difference between sales of 1993 and 1995 = 1 lacs or 100,000
16. Sum of sales of 1991 and 1992 = 6 + 10 = 16 lacs

    Required Answer $= \frac{16}{13} \times 100 = 123\%$
17. Percentage increase in production from 1993 to 1994

    $\frac{16-12}{12} \times 100 = 33.33\%$

18. Percentage of sales to production is maximum in the year when the difference between them is minimum.

19. Percentage drop in sales from 1992 to 1993

$$= \frac{1}{10} \times 100 = 10\%$$

20. By visual inspection we can see that the correct answer is option (c).

[redacted]

22. Production of E type bikes in 2006
= (100 – 80)% of 4,50,000
= 20% of 4,50,000 = 90,000
And in 2007 = 10% of 5,20,000 = 52,000
Total production = 90,000 + 52,000 = 1,42,000
So, the number of bikes
= 15 % of 1,42,000 = 21,300

23. Production of A type bikes in 2006 = 67,500
Production of A type bikes in 2007 = 52,000

$$\text{Required \%} = \frac{67,500 + 52,200}{9,70,000} \times 100 = 12.3\%$$

$\cong$ Closest to option (b)

24. By visual inspection we can say that option (C) is correct answer

25. Percentage production of B type bikes in 2007 = that in 2006 (given) = (40 – 15) = 25% of 5,20,000 = 1,30,000.

[redacted]

26. $$\text{Required \%} = \frac{540\text{-}130}{540} \times 100 = 75\%$$

27. Percentage share of foreign tourists visiting HP

$$\frac{34.5}{1600} \times 100 = 2\%$$

28. Ratio = 535 : 15 = 107 : 3

[redacted]

31. $$\text{Required \% age} = = \frac{450}{250} \times 100 = 180\%$$

Pie Chart

[redacted]

1. Required Angle = (16/100 × 360) = 57.6°

4. If the editing charges are Rs 18, royalty is Rs 10. On Rs 18, it is less by 8. On Rs 100, it is less by (8/18 × 100)% = 44.44%

5. Let the total expenditure be Rs x. Then, 8 : 100 = 36960 : x. So x = Rs 462000. Therefore Cost Price of 5500 copies of the book = Rs 4,62,000. Cost Price of each copy = Rs (4,62,000/500) = Rs 84. So, marked price = 140% of Rs 84 = Rs 117.60.

[redacted]

6. There are two successive increases—First is the total increase of 60% approx.($ 3.7 bn in 2000 from $ 2.3 billion in 1995) and second is the percentage share of A by 20% approx. Hence total percentage increase = 92%.

7. Market size in 2000 = $ 3.7 billion
Market size in 1995 = $ 2.3 billion
Therefore, growth rate
= $[(3.7/2.3)^{1/5} - 1] \times 100 = 10\ \%$
Hence, option (b) is the correct answer.
[3.7/2.3= 1.6, now to find $(1.6)^{1/5}$, $1.1^3 = 1.331$ and $1.1^2 = 1.21$ and $1.3 \times 1.2 = 1.6$
Thus, $1.6^{1/5} = 1.1$]

8. Since we do not know what part of market share of B. C and E has been eaten up by F, so cannot be determined.

9. Sales value for A + B + C in 2000 = (13 + 14 + 19)% of 3.7 billion = 46 % × 3.7 = 1.702 billion
Sales value for A + B + C in 1995 = (16 +17 + 12) % of 2.3 billion = 1.035 billion.
Hence increase in sales value = 1703 – 1035 = $ 668 million.
Hence, option (c) is the correct answer.

[redacted]

11. Since the total consumption for the given years is not known, we cannot determine it.

12. Since the total consumption for the given years is not known, we cannot determine it.

13. Total consumption has increased by 20%. So, any sector that shows an increase of even 25% will give us a net increase of 50%. There are only two sectors–agriculture and domestic. Therefore, option (b) is the correct answer.

14. There are two increases—one is the increase in the share of agriculture (66.66%) and the other is an increase of 50% in total. Net increase = 150% Therefore, option (c) is the correct answer.

15. By visual inspection. We can see that the correctansever is option (b)

16. There is an increase in the share of agriculture (66.66%), to make the agricultural consumption double, total consumption is to be increased by 20%.

17. The electricity consumption of the 'others' category has remained constant over the period. It means that the total consumption of 1980-1981 = Total consumption of 1993-1994. So, the only increase = [(18—11)/11] × 100 = 63.64%

18. If the percentage increase in the expenditure of both the families is the same then the ratio will be the same.

19. The total consumption has become 3 times more keeping the expenses on education the same. Hence the percentage consumption on education will become 1/3rd of the present.

21. By visual inspection we can see that option (c) is the correct answer.

22. It should be understood that the final percentage of expenditure will always be in between the percentage of family A and family B (It is true of any mixture that the percentage composition of the mixture will be always in between the percentage compositions of the components).

23. By visual inspection we can see that the correct answer is option (b).

24. By visual inspection we can see that option (a) is the correct answer.

25. By visual inspection we can see that option (a) is the correct answer.

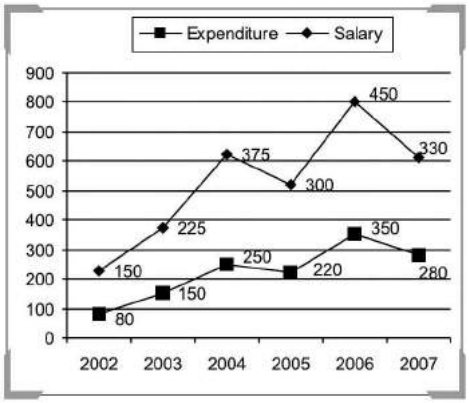
26. Understand the logic that "If a country is under military rule, it will spend maximum on defence and minimum on either health or education" does not mean that if a country is spending maximum on defence and minimum on either health or education, then it will under military rule.
[For more–See Logical Links in Part 1 of this book]

27. By visual inspection. we can see that option (d) is the correct anser.

28. Since individual expenditures of the countries are not given, we cannot determine it.

Line Chart

For the sake of convenience, use the following chart (with the same data) to find out the solution:



Now all the questions can be done with ease.

8. Income-expenditures of company A and B cannot be correlated. Hence cannot be determined.

9. Expenditure of company A in 2006 =

Income in 2006 of (A) × (100/160) = Income in 2006 of (A) × 5/8

Expenditure of Company B in 2007 =

Income in 2007 of (B) (100/170) = 10/17 of Income in 2007 of B

Now, [Expenditure in 2006 of (A)/Income in 2007 of (B)] = (5/8) ÷ (10/17) [Since Income in 2006 of (A) = Income in 2007 of (B)]

$$= \frac{5}{8} \times \frac{17}{10} = \frac{17}{16}$$

10. Ratio A : B is greater than 1 in only 2003 and 2007. It is 1.33 in 2003 and 1.1 in 2007.

11. Percentage profits are the same for two years, hence if expenditure increases by 20% the income should also increase by 20%. Hence, the required ratio = 120/100 = 1.2

13. This never happens. Understand that if export is more than import, then the ratio will be more than 1. If export is equal to import, then the ratio will be equal to 1 and if export is less than import, then the ratio will be less than 1.

14 . The ratio of one to the other should be reciprocal.

15. Despite the ratio being on a decline for country A, we cannot comment about the value of exports. it might be a possibility that imports have increased keeping exports constant. Similarly both the other options (b) and (c) can be refuted on the same grounds.

16. Since only the ratio is given, we cannot comment on the values.

17. Since only the ratio is given, we cannot command on the values.

Following is the table pertaining to the data set given:

	CNN	NDTV	CNBC	Zee
SAD	44	52	58	52
Congress	55	55	48	48
Others	18	10	11	17

Following is the table pertaining to the data given:

	Chemical	Engineering	White Goods	Services	Electronics	Food	Phamacentional	Fuel	Textiles	Telecommunication	Others
2006	17	15	8	16	9	9	4	0	0	0	22
2007	9	4	0	12	0	5	0	27	4	8	24

Table

	Export (Rs Crore)	
	2006	**2007**
Glass Industry	978	1093
Coal Industry	1009	1080
Ruby Industry	1000	913
Nickel Industry	619	663
Softwares Industry	690	605
Diamond Industry	513	505
Total	16250	19500

1. The share of Glass in the total exports for 2007 = 1093/19500 × 100 = 5.6%.

2. Overall exports have increased by 20% while none of the given six industries have grown by 20%. So, we can conclude that the market shares for all the six industries will decrease in 2003. Hence option (a) is the correct answer.

3. Coal has the maximum exports in 2006. Hence, its market share is the maximum.

4. Exports for the given six industries in 2006 = 4809 crore
Exports for the given six industries in 2007 = 4859 crore
Therefore, percentage increase in the exports = [(4859 – 4809)/4809] × 100 = 5000/4809 =1.05%.

5. The difference between the white-coloured cars sold is the minimum in the B type model.

6. Blue (E + D) = 43 + 37 = 80 = White (B)

7. Required difference = (50 – 34) × 1000 = 16,000

8. Required Percentage = 173/192 × 100= 90%

9. White-c = 90,000

10. The difference in percentage marks of B (History) and C (Hindi) = 90 – 70 = 20

11. The percentage of marks obtained by F in Science = 160/200 × 100 = 80%
Required Value = 170/80 = 2.125

12 Average percentage of marks of C = 680/1000 × 100 = 68%
Percentage of marks of C in English = 150/200 × 100= 75%
Required difference = 75 – 68 = 7

14. Average percentage of marks in English

$$=\left(\frac{140+130+150+125+145+110}{6}\right)\times\frac{100}{200}$$

= 67%

15. By visual inspection.

19. Average marks obtained by 20 boys in History from school Q = 45
Therefore, Total marks = 20 × 45 = 900

20. From visual inspection it is clear that Science is the desired subject.

21. The table gives the data pertaining to only 40 students of the school and it is not given if this is the total number of students in the school. Therefore it cannot be determined

[redacted]

22. It is a simple calculation. Alternatively, it can be observed that A has always been more than the others except but for one year.

23. Required share
= (410/480 + 410 + 390 + 380 + 440) × 100 = 19.5%

24. It can be observed that the difference is the highest in case of manufacturer C. We can further observe that the largest percentage growth would be for C, as the base is the smallest.

26 By visual inspection we can say that option (d) is the correct answer.

27. By visual inspection we can say that option (c) is the correct answer. Manufacturers are B, C, D.

28. By visual inspection we can say that option (d) is the correct answer.

[redacted]

29. It happens only in 1992–1993

2

CHAPTER

Developing the Skills

LEARNING OBJECTIVES

- Different constituents of Data Interpretation
- Various skill-sets required and the methods to acquire the same
- Calculations required
- How to excel in these calculations and the calculations specific to a particular data-set
- How to interpret the data
- Connecting diferent data-sets

CONSTITUENTS OF DATA INTERPRETATION (DI)

Data Interpretation is primarily aimed at assessing the candidate's ability to analyse the given data, arrive at meaningful conclusions and to take appropriate decisions. In many ways, this section truly reflects a business situation; where one is required to take a decision on the basis of available information and future trends.

In any DI question, students will be required to go through the data-set and make necessary relationships between "what is given?" and "what is asked?" These questions simulate the kinds of detailed analyses of relationships that management students must perform in solving problems under strict time restrictions.

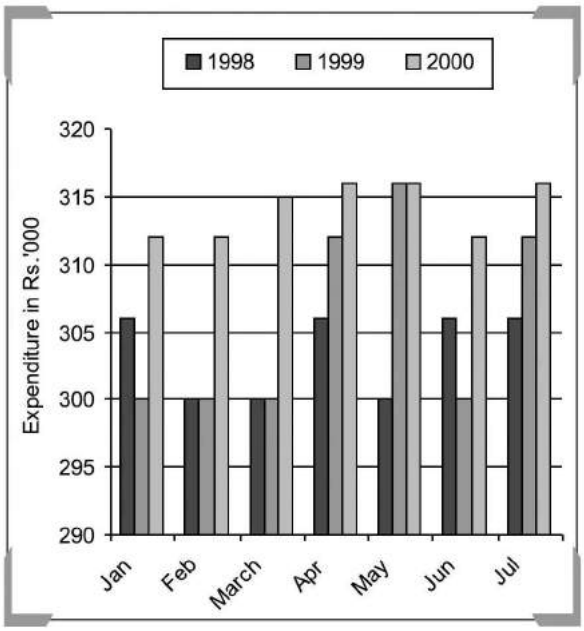
Each DI question contains three separate parts —

(A) Data-set,
(B) Explanations and restrictions given for the data and
(C) Question/s.

Data-Set

The data pertaining to the question can be represented in various formats—pie chart, bar chart, line chart etc. The data-set introduces a sample or a report of some activity, done in the past, present or the future.

For example,



In the above data-set, we see the expenses of a family for three years from January to July.

The same data-set can be represented by other formats like the line chart etc. keeping the essence of the data, the same.

We will learn more ways of presenting the data and how they can be inter-related, ahead in this chapter.

Explanations and Restrictions

These should be seen as the guiding principles or the helping factors associated with the data-set. Explanations help us in knowing and understanding the data better; on the other hand, restrictions narrow down our focus to what is being asked and what is to be extracted.

For example, the following are the final results of standard X. The students were tested in English, Hindi, Science, Math, Social Science, Drawing, Craft, PT and Scout-Guide. For a better presentation of the mark sheet, the scores in the first four subjects are consolidated into A1, the next three into A2 and last three into A3. The final score is the total of these three scores A1, A2 and A3. On the basis of the final score the students get their ranks. The table below gives the scores of the top eleven students of the class.

Name	Grade	Final Score	A1	A2	A3	Drawing	PT	Social service
Anana	A	725	282	217	226	80	81	69
Pankaj	B	864	310	87	267	85	86	92
Vamshi	C	812	300	290	222	70	91	74
Ram	C	805	270	262	273	65	79	97
Shann	D	900	391	292	217	96	86	81
Saket	D	876	301	297	278	65	87	93
Mini	C	825	296	296	233	50	89	91
Rashi Singh	E	905	320	290	295	72	99	99
Parul	B	895	310	290	295	72	99	98
Rakesh Rathi	B	870	276	296	298	63	99	99
Gauri	C	888	361	270	257	82	78	92

Without the given explanation, we would not have any idea about how to arrive at A1/A2/A3 and how the grades are being awarded.

Example 1 And now finally, the tools to check how much you have been able to interpret the data and relate it to the explanations/restrictions given.

Here, are a few questions based upon the data-set given above:

1. How many students have got grade A and more than 250 marks in all the group scores (A1, A2 and A3)?
   (a) 0
   (b) 1
   (c) 2
   (d) None of these

2. In the drawing subject, what is the difference between the total marks of grade A students and grade C students?
   (a) 50
   (b) 52
   (c) 48
   (d) None of these

We will see the solution of the questions given above in the next part. Let us now go through the skill-set required to excel in the Data Interpretation section.

SKILL-SET REQUIRED

Till now, we have understood that DI questions test the following:

- Ability to understand the data and retain that in mind.
- Exposure to various types of data presentation formats
- Understanding of how restrictions limit and order behaviour
- The 'logic' inside a data-set and the algorithm to extract it to arrive at a solution

It should be noted, that even if a student has not gone through a formal training or practice of DI for any aptitude exam including CAT, can have an understanding of data and exposure to the various formats of presenting the data, just by virtue of going through any standard newspaper regularly.

However, while discussing the skill-set required, we will assume that the student is at the basic level and has just started going through DI.

Skill 1—Knowing the Terminology

There can be various economic or non-economic terms like CAGR, inflation etc., involved in the data-set, and it is a prerequisite to have a good understanding of these terms.

Example 2 If the production of wheat in 2004 was 100 million tonnes and in 2007 it was 133 million tonnes, then find the (A) Percentage Growth, (b) SAGR, (c) CAGR.

Solution

(A) Percentage Growth = [Growth/Initial value] × 100 = 33%

(B) SAGR (Simple Annual Growth Rate)–It is the average growth rate over the given period. It is calculated by dividing the growth percentage by the number of years involved.
In the above case, SAGR = 33/3 = 11%

(C) CAGR (Compounded Annual Growth Rate)–It is the year to year growth rate over the given period. To calculate this, we take the help of the fundamentals of compound interest. We take the final value as the amount after the given period and the initial value as the principal.
In the above case,

$$133 = 100\left(1+\frac{r}{100}\right)^3$$

Hence, r = 10% (approx.)

Skill 2—Knowing the Different Formats of Presenting the Data

Having discussed earlier, the significance of organized data, we can automatically conclude that the same data when presented in an unorganized way makes difficult reading and is not of much use. The same data can be presented differently. In one format, it might make easy reading; in another format, it might make difficult reading.

Example 3

	Location 1	Location 2	Location 3	Location 4	Location 5
2004	112	128	98	132	102
2005	118	147	120	120	108
2006	120	144	126	124	113

A company is selling its products at five different locations and the sales value (Rs thousands) of different locations for three consecutive years is given.

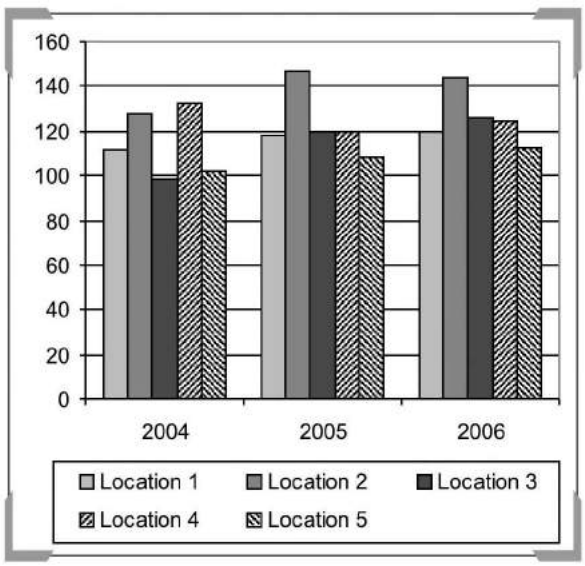
Question

1. Which of the following locations has the maximum sum of sales value for the three years?
   (a) Location 1 (b) Location 2
   (c) Location 3 (d) Location 4

Solution

This question simply requires the sum of the sales value of each of the locations for three years. Obviously, location 2 is the answer.

However, if we convert the same data into a bar chart, then it can be seen by simple observation that location 2 is the answer.



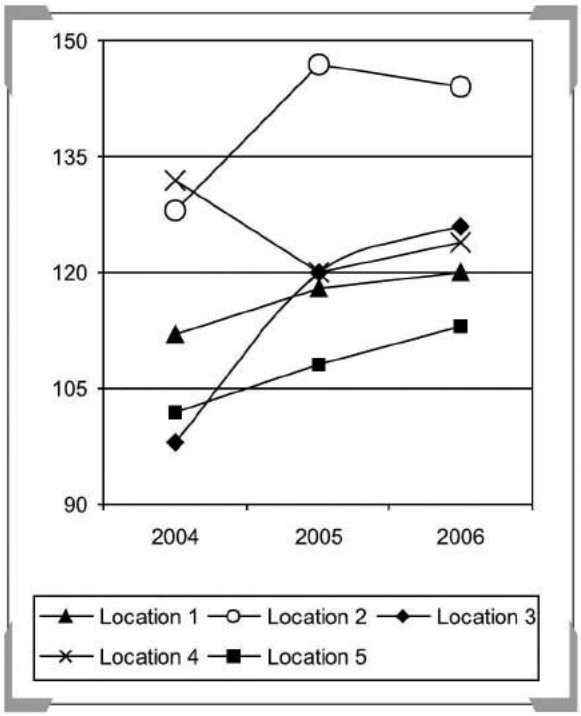
Since the bar of location 2 is above the other two for two years and in 2004, it is just a bit lower than one of the bars; we can simply say that location 2 produces the maximum combined sales value for the three years.

Example 4 Which of the following locations has the minimum sum of sales value for the three years?

(a) Location 1
(b) Location 2
(c) Location 3
(d) Location 5

Solution It can be seen that the bar of location 5 is lower than all the bars for two years and once it is equal to the bar of location 3; hence, location 5 produces the minimum combined sales value for the three years.

If we convert the same data into a line chart, it becomes more obvious than the bar chart that location 2 and location 5 give the maximum and minimum sales value for the given period.



Example 5 Which of the following locations has seen the maximum percentage increase in the sales value in 2005 over 2004?

Solution If we try to solve this question with the help of bar chart, it becomes a bit difficult without having the actual values. So, we are left with either the line chart or the given table.

In case of a table, this question can definitely be solved by calculation and observation. But if we see the same situation in case of a line chart, we can simply rule out location 1/location 4/location 5 as the answer. Now, from the remaining two locations 2 and 3, despite the slope being almost the same, the base in location 3 is very low with respect to the base in location 2. Hence, location 3 will have more percentage increase.

We will learn more about the different formats of presenting data, ahead in this chapter.

Hence, we can see that with different forms of data representation, one can gather different conclusions just by visual inspection. However, the speed with which this conclusion can be drawn depends heavily on the kind of chart given.

Skill 3—Knowing What is to be Found Out

In my experience of dealing with students from varied backgrounds, many a times I have seen that students are not able to solve a question because they have got very impressed with the data. And as a result, they lose track of what is to be fond out.

Remember, never ever get 'emotionally attached' to a question.

As an examinee, you are not supposed to get into the technicalities of the data. Rather, you should focus upon what is being asked in the question, and how to extract information from the data-set relevant to the given question.

However, during practice, one needs to focus not just on solving the problems on hand, but also on building an algorithm to solve similar problems whenever they occur.

Cracking the questions require a very systematic approach. The primary focus should be on understanding the question. It would be worthwhile to read the question patiently and carefully. At this juncture, efforts towards building your vocabulary and improving your reading skills might help you. One must pay attention to otherwise innocuous words and phrases such as "at most" or "at least", "cannot be less than" , "cannot be more than" etc.

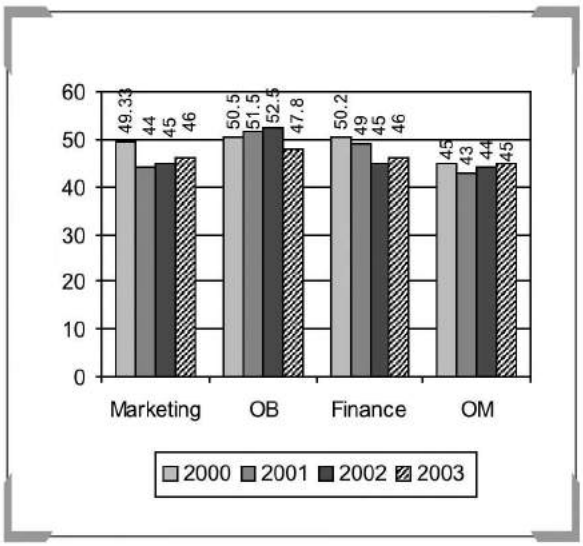
Till now we have discussed data, the significnace of organized data and the skill set required to solve the questions.

Let us see a few questions from an actual CAT DI paper:

CAT 2005

A management institute was established on 1 January 2000 with 3, 4, 5, and 6 faculty members in the Marketing, Organizational Behaviour (OB), Finance, and Operations Management (OM) areas respectively, to start with. No faculty member retired or joined the institute in the first three months of the year 2000. In the next four years, the institute recruited one faculty member in each of the four areas. All these new faculty members, who joined the institute subsequently over the years, were 25 years old at the time of

their joining the institute. All of them joined the institute on April 1. During these four years, one of the faculty members retired at the age of 60. The following diagram gives the area-wise average age (in terms of number of completed years) of faculty members as on April 1 of 2000, 2001, 2002, and 2003.



Example 6 From which area did the faculty member retire?
(a) Finance
(b) Marketing
(c) OB
(d) OM

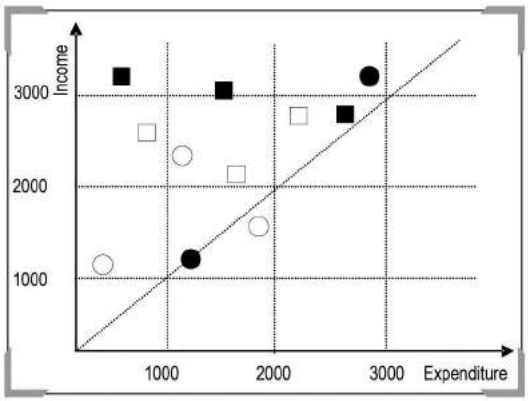
Solution To solve this question, following things were required:

- Understanding the passage given at the beginning of the DI set.
- Understanding the above bar chart.
- Understanding the question and the process of how to inter-relate the data.

Of course we can calculate the sum of ages of all the faculty members in all the areas and for all the years. There were 3 faculty members in Marketing in 2000 (inception of the institute), hence, the sum of completed years of these three faculty members in 2000 = 49.33 × 3 = 148 years. But the main problem lying ahead is, for how many years and areas can we do the same thing? As this will involve a good amount of calculation, it will require a good amount of time and patience.

CAT 2004

The data points in the figure below represent the monthly income and expenditure data of individual members of the Ahuja family (■), the Bose family (□), the Coomer family (○), and the Dubey family (●). For these questions, saving is defined as



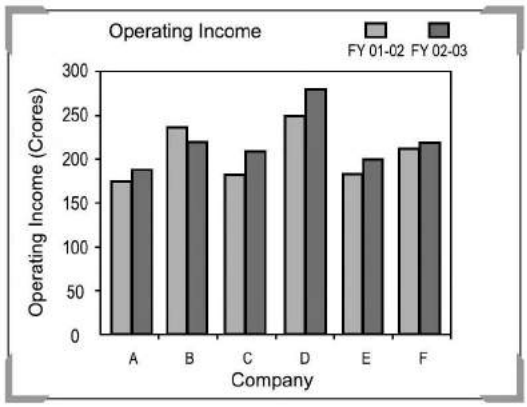
Example 7 Which family has the lowest average income?
(a) Ahuja (b) Bose
(c) Coomar (d) Dubey

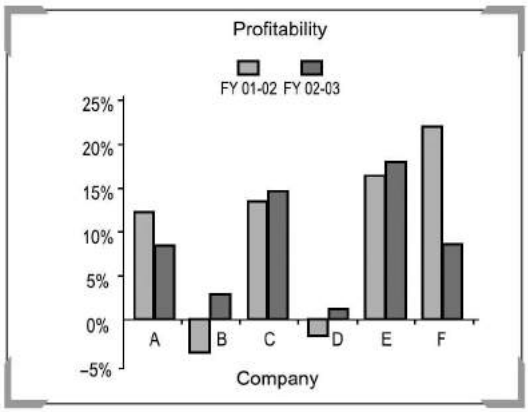
Solution To solve this question, following things are required—

- Understanding the data as given above.
- Understanding the question and the process of how to inter-relate the data.

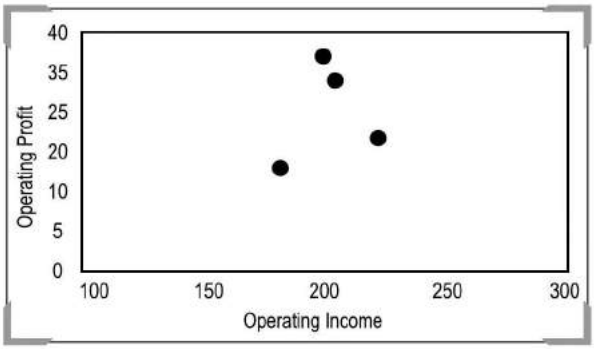
CAT 2003 (Re-Test)

The profitability of a company is defined as the ratio of its operating profit to its operating income, typically expressed in percentage. The following two charts show the operating income as well as the profitability of six companies in the Financial Years (FYs) 2001-2002 and 2002-2003.





The operating profits of four of these companies are plotted against their respective operating income figures for the FY 2002-2003, in the third chart given below.



Example 8 What is the approximate average operating profit, in FY 2001-2002, of the two companies excluded from the third chart?

(a) –7.5 crore (b) 3.5 crore
(c) 25 crore (d) cannot be determined

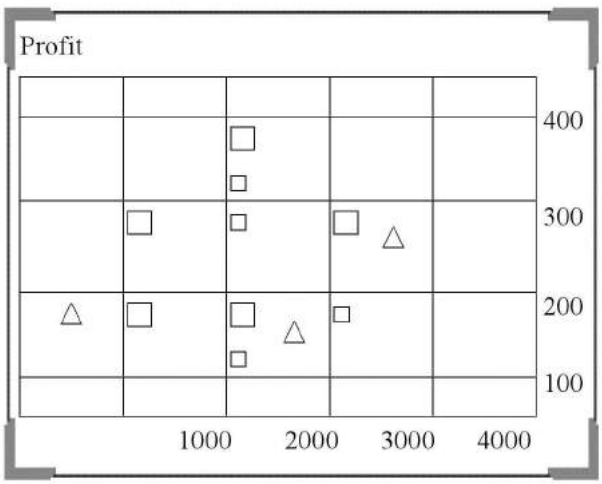
Solution To solve this question, following things are required—

- Understanding of the definitions given above
- Understanding of the above representations of the data-set

The biggest problem one would face while solving this question is how to inter-relate the data given.

CAT 2003 (Leaked)

Each point in the graph below shows the profit and turnover data for a company. Each company belongs to one of the three industries: textile, cement and steel.



Example 9 For how many companies does the profit exceed 10% of the turnover?

(a) 8 (b) 7
(c) 6 (d) 5

Solution To solve this question, following things are required:

- Understanding of the above representations of the data-set.
- Finding out which one is textile, cement and steel.

(For solution of all these questions refer to CAT papers given in this book.)

- The above examples illustrate a few simple facts:
- Data is an important tool for decision-making and drawing inferences.
- However, having only the data does not serve the purpose; it should be presented in a proper format.
- Despite the data being presented in a proper way, the most important thing is having the capability to interpret it.

CALCULATION TECHNIQUES

Despite the trend at CAT having shifted from easy-to-understand-data-with-difficult-calculations to more logical questions with lesser calculations, it is not advisable to sit for CAT without being reasonably good at quick calculations and approximations.

The importance of a good calculation speed can be understood with the following statistics:

We are taking a sample of a student who attempts 15 questions in DI at CAT. All the figures taken ahead are 'on an average' basis.

Number of calculations/questions

= 4–5 calculation

Hence, the total number of calculations done

= 60 – 75 calculations

If a calculation takes 6 seconds on an average, then time invested in doing calculations only

= 360 – 450 secs

= 6 – 7:30 minutes

Now if we double our calculation speed, we would have 3-4 minutes of extra time. What leverage can be taken from this 'generated time' depends totally upon the caliber of the student. Even if a student solves only one question than what he actually would have, and gets it right; the percentile in DI might differ by 1–2 percentile.

Following are the most frequent type of calculations done at CAT—

Addition

This is the most important calculation and takes the lion's share of the whole calculation section. Addition is also treated as the building block for all the other calculations like subtraction, multiplication etc.

Unfortunately, we don't have any scut for additions. The only thing that can be done is, practice.

I would suggest a 4 × 4 table for this practice:

	Column 1	Column 2	Column 3	Column 4
Row 1				
Row 2				
Row 3				
Row 4				

Fill in this table with two digit numbers and add all the columns and rows. To obtain a faster speed, it is advisable that both the digits are added simultaneously. For this, we need to train our mind to retain both the digits of a number simultaneously and then add it in one go with the two digits of the other number.

I would suggest, that rather than adding the unit digit first, add the tens place digit first; and while your brain is adding the tens place digit, your eyes should have captured the unit digit.

Sample table:

	Column 1	Column 2	Column 3	Column 4	
Row 1	36	42	67	98	243
Row 2	46	54	28	72	200
Row 3	58	45	82	89	274
Row 4	49	58	36	92	235
	189	199	213	351	

Rate your calculation speed:

Equal to or less than 15 seconds— Excellent

More than 15 seconds— Less than or equal to 30 seconds— Good

More than 30 seconds— Less than or equal to 60 seconds— Average

More than 60 seconds— A lot work is needed

The above given time includes time for writing the sum and it does not consider the time taken for creating the table and putting values inside it.

The idea is that if, 3 + 4 = 7 can be done without any effort (ask yourself after doing the addition if did you put any effort), then we could be in a position to do 79 + 68 = 147 also without any effort. And with this practice exercise, we want to achieve the same.

Subtraction

Once we attain a good speed in addition, subtraction too will become fast as a natural outcome.

Multiplication

Here, again addition is the key. However, it is advisable to know the basics of Vedic maths for multiplication.

Some of the methods are given below:

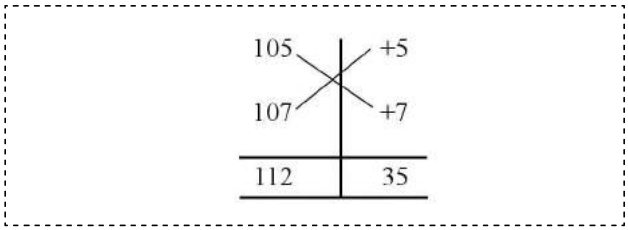
Base Method

In this method of multiplication, we use a number as a base, like 10, 50, 100 etc. We should try to assume the base that is closer to both the numbers.

Examples 10 105 × 107

Solution Both the numbers are close to 100, so let us assume 100 as the base. We will now find the deficit/surplus from the base.

Base = 100, Surplus = 5 and 7



Right part (after slash) is the product of the surplus. Since base = 100 and surplus are 5 and 7, so product would be 5 × 7 = 35.

Left part (before slash) could be either number plus surplus of the other multiplicand. Hence left part would be either (105 + 7) or (107 + 5) = 112 (both will always be the same) i.e., 112.

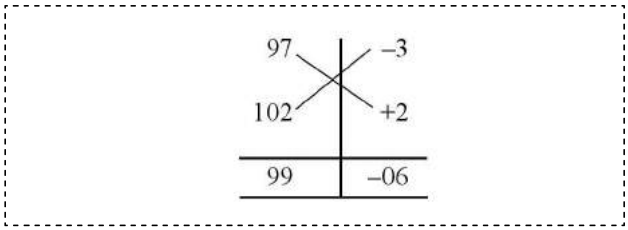
Left part would be the equivalent to Number × 100. In this case, 112 × 100 = 11200

Now we will add both the right part and the left part = 11200 + 35 = 11235

Hence, the result of the multiplication would be 11235.

Example 11 97 × 102

Solution



97 × 102

Base = 100, Deficit = 97 – 100 = – 3,

Surplus = 102 – 100 = 2

Right part will now be (–3) x 2 i.e., –06. To take care of the negative we will borrow 1 from the left part, which is equivalent to borrowing 100 (because we are borrowing from the hundred digits of the left part). Thus this part will be 100 – 06 = 94.

So, the answer = 9894

Place value Method

In this method of multiplication, every digit is assigned a place value and we do the multiplication by equating the place values of multiplicands with the place value of the product.

Examples 14

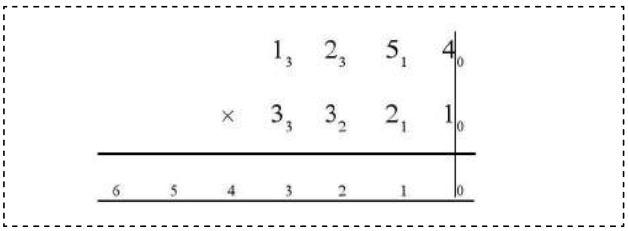
$$\begin{array}{r} 1_3\ 2_3\ 5_1\ 4_0 \\ \times\ 3_3\ 3_2\ 2_1\ 1_0 \\ \hline 6\ \ 5\ \ 4\ \ 3\ \ 2\ \ 1\ \ 0 \end{array}$$

Solution Conventionally, the unit digit is assigned a place value of 0, tens place digit is assigned a place value of 1, hundreds place digit is assigned a place value of 2, thousands place digits is assigned a place value of 4 and so on.

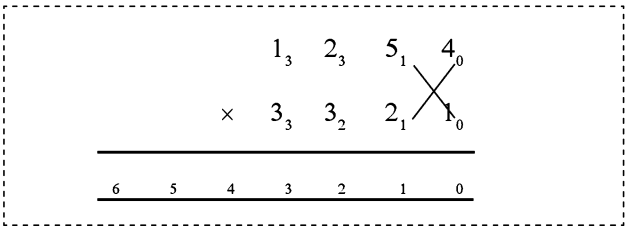
Now, this multiplication is a two-step process:

Step 1–Add the place values of digits of the numbers given (1254 × 3321) to obtain the place value of the digits of the product.

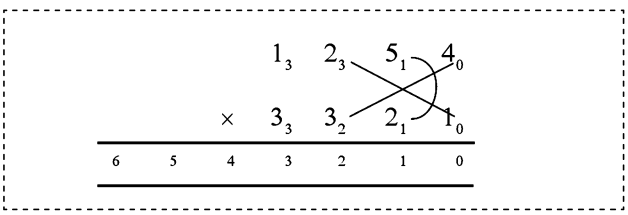
For example, using the place values of the multiplicands i.e., using 0, 1, 2 and 3 of the number 1254 and the same place values 0, 1, 2 and 3 of the other multiplicand 3321, we can get 0 place value in the product in just one way i.e., adding 0 and 0.



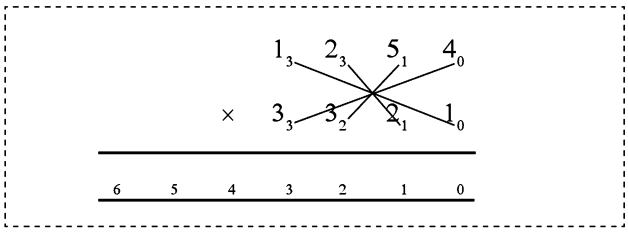
Place value 1 in the product can be obtained in two ways:



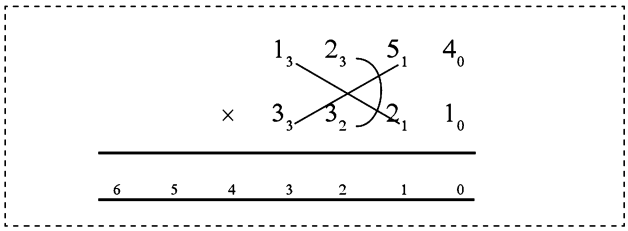
Place value 2 can be obtained in three ways:



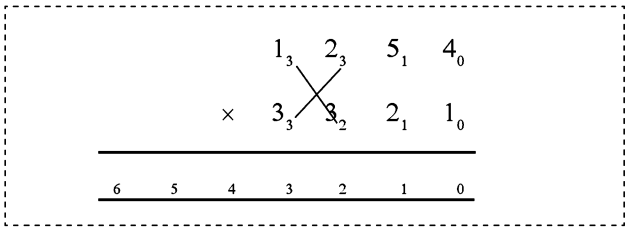
Place value 3 can be obtained in four ways:



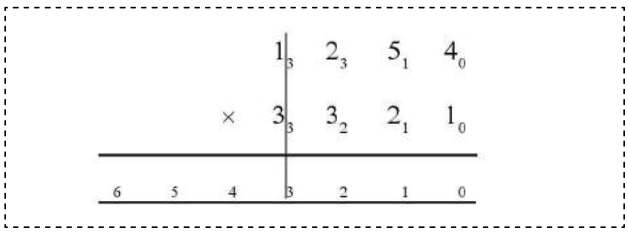
Place value 4 can be obtained in three ways:



Place value 5 can be obtained in two ways:

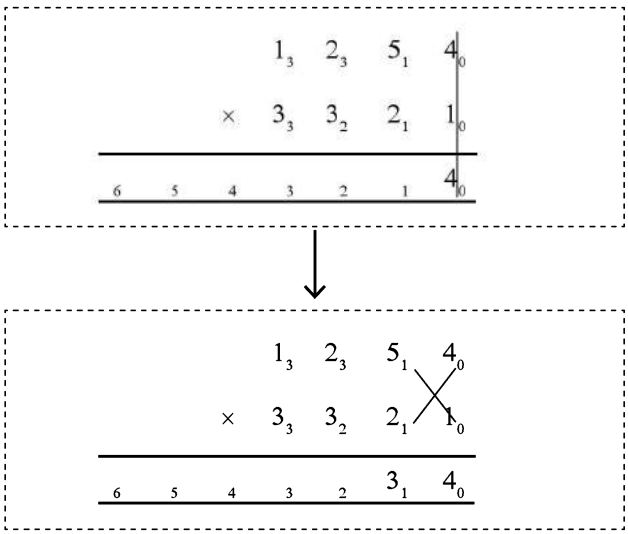


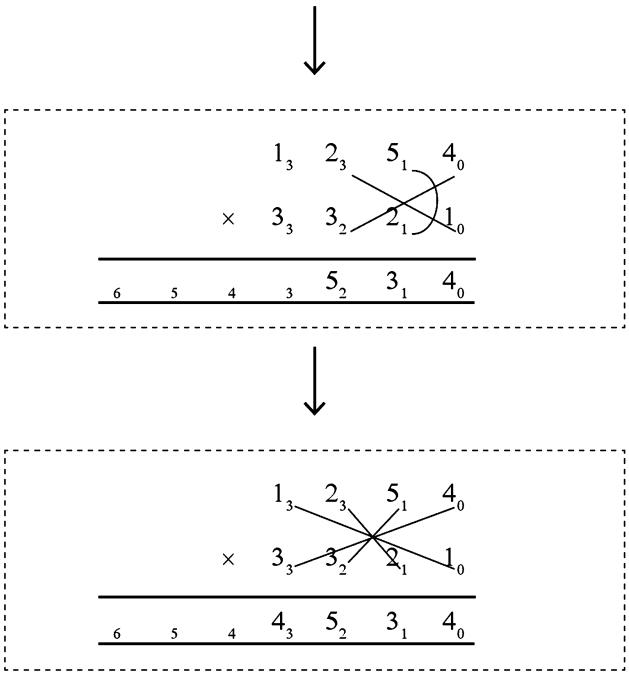
Place value 6 can be obtained in one way:



And this is the maximum place value that can be obtained.

Step 2–Now multiply the corresponding numbers one by one.



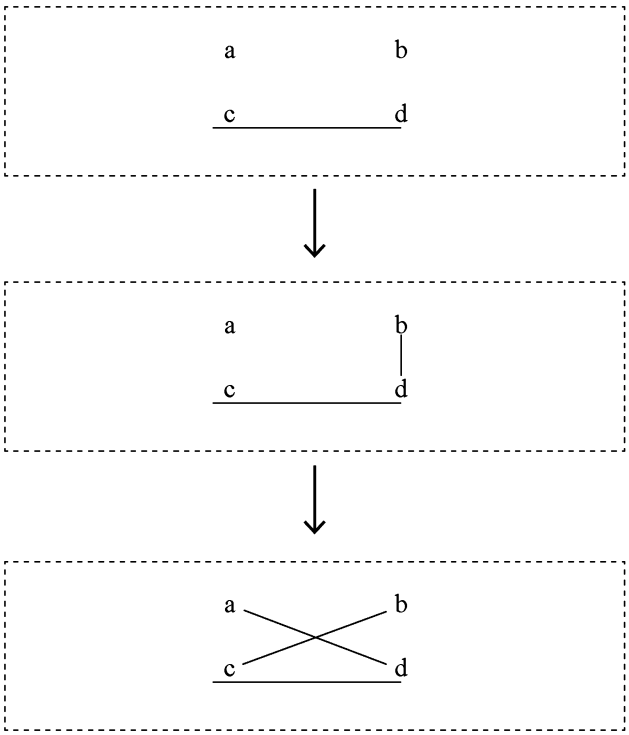


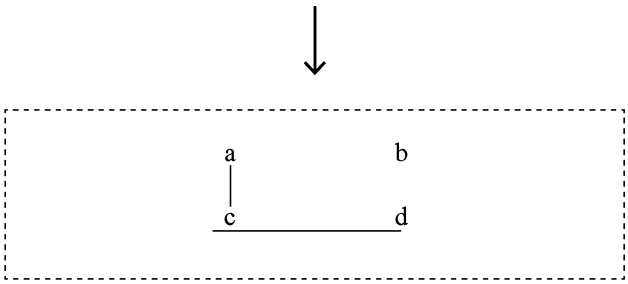
And so on, we can find the product = 4164534

This method is most useful in case of multiplications of 2 digits × 2 digits or 2 digits × 3 digits or 3 digits × 3 digits multiplication.

Example 13 ab × cd

Solution





Similarly, we can have a proper mechanism for multiplication of 2 digits × 3 digits or 3 digits × 3 digits using the above given place value method.

Division

Most of the division calculations can be done by approximation techniques without actually finding the original value.

Example 14

$$\frac{3548}{4182}$$

Solution Let us approximate this value.

There are three golden rules of approximation:

1. Range of approximation should be ± 2% with respect to the actual result.
2. Approximation should be done only if the options are at a considerable gap, say ± 5% atleast.
3. One of the key things to be kept in mind while doing approximation is the direction of the approximation i.e., one should know if the actual result will be more or less than the approximated result.

We should know that in any division, if the numerator and the denominator are both either increased or decreased by the same percentage, then the result will be exactly the same. While it is practically not possible to find such situations always, we will use this to know the direction while doing approximation for the above calculation:

Approximation 1

Eliminate the unit digit of both the numerator and the denominator

$$\frac{3548}{4182} \cong \frac{354}{418}$$

Now, the objective is to break the numerator into the parts of the denominator.

354 = 209(50% of 418) + 145

145 = 105(25% of 418 approx.) + 40

40 = 10% of 418

Hence, net value = 85% = 0.85

Is this value more or less than the actual value?

Let us compare it with the actual result = 0.8483, which is a slightly less than the approximated result of ours.

Approximation 2

Eliminate the unit and tens place digit of both the numerator and the denominator

$$\frac{3548}{4182} \cong \frac{35}{41}$$

35 = 20.5(50% 0f 41) + 14.5

14.5 = 12(30% of 40 approx.) +2.5 = 30% of 41 + 6% of 41(approx.)

[Since 0.41 = 1% of 41]

Hence, net value = 86% = 0.86. This is slightly more than the actual value.

Ratio Comparison

This is one of the most important calculations and is very extensively sought after in DI. On an average, while performing 100 odd calculations in DI, atleast 8-10 calculations will be on comparing the ratios. Normally there are two methods for comparing two or more than two ratios:

Cross Multiplication Method

This is one of the conventional methods of comparing two ratios.

Example 15 Let us compare 11/15 and 13/18.

$$\frac{11}{15} \qquad \frac{13}{18}$$

Solution Cross multiplying the numerator of the 1st fraction with the denominator of the 2nd fraction and denominator of the 1st fraction with the numerator of the 2nd fraction,

$$\frac{11\times18}{198} \qquad \frac{13\times15}{195}$$

Since, 198 is greater than 195; so the 1st fraction (11/15) is greater than the 2nd fraction (13/18).

Decimal Calculation Method

$$\frac{11}{15} = 0.7333 \qquad \frac{13}{18} = 0.722$$

Obviously, here the 1st fraction $\left(\frac{11}{15}\right)$ is greater than the 2nd fraction $\left(\frac{13}{18}\right)$.

However, if we have to compare 3156/5438 and 3423/5822, then using any of the above two methods becomes cumbersome and a good amount of time will be required.

Here, we will compare ratios with the help of percentage.

Percentage Comparison Method

Let us first understand this with the help of the following ratios:

1st Case

$$\frac{10}{15} \xrightarrow[100\%]{} \xrightarrow{100\%} \frac{20}{30}$$

2nd Case

$$\frac{10}{15} \xrightarrow[100\%]{} \xrightarrow{200\%} \frac{30}{30}$$

3rd Case

$$\frac{10}{15} \xrightarrow[300\%]{} \xrightarrow{200\%} \frac{30}{60}$$

In the 1st case, percentage change in numerator (100%↑) = percentage change in denominator (100%↑), So ratios are equal.

In the 2nd case, percentage change in the numerator (200%↑) > percentage change in the denominator(100%↑). So the 2nd ratio is greater than the 1st ratio.

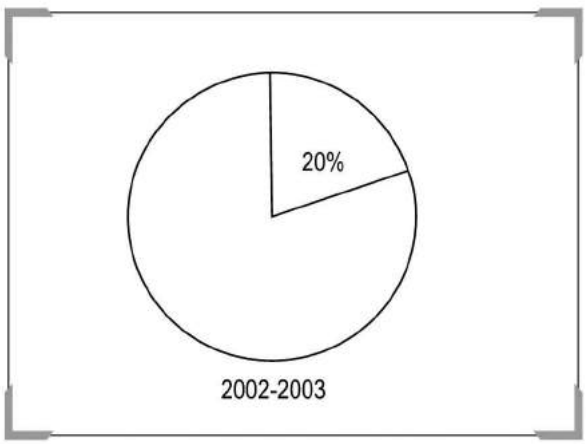
In the 3rd case, percentage change in the numerator (200%↑) < percentage change in the denominator (300%↑). So the 1st ratio is greater than the 2nd ratio.

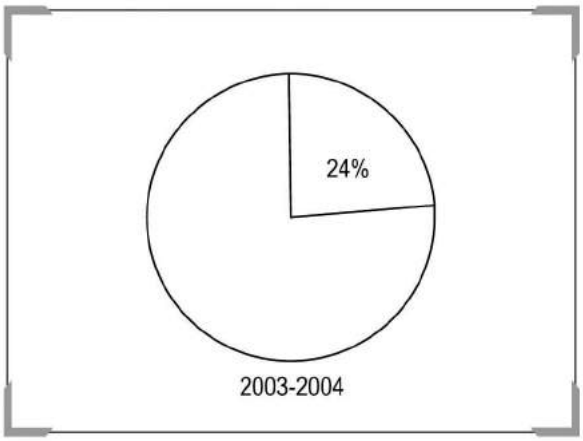
This particular example can also be seen as a general rule for determining the order of ratios.

Besides these general calculation techniques, there are certain techniques specific to particular types of data presentation formats:

Pie Chart Calculation Technique

The following figure shows the sales of *Due North Inc.* for the year 2002-2003 and 2003-2004. It is also given that the share of North India in the total sales figure is 20% and 24% for the two respective years.





The total sales for 2002-2003 is Rs 1202 crores and for 2003-2004, it is Rs 1381 crores.

Example 16 What is the percentage growth in the sales value of North India in 2003-2004 over the sales value of 2002-2003?

Solution The normal way of doing this question is –

Sales value of North India in 2002-2003
= Rs 240.4 crores

Sales value of North India in 2003-2004
= Rs 331.4 crores

So, growth = Rs 91 crore

So, Percentage growth =

$$\frac{Growth}{Initial\ Value} \times 100 = 37.93\%$$

Alternatively, there are percentage increases in two factors of the given pie-charts, i.e., share is increasing from 20% to 24% and then the total sales value is increasing from Rs 1202 crores to Rs 1381 crores.

Rather than calculating the sales value of the individual years, we can directly calculate the net percentage change by calculating the percentage changes independently in the above mentioned two factors.

Percentage growth in share = 20%

$$\left(\frac{Growth \times 100}{Initial\ Value}; \text{and remember that it is not 4\%}\right)$$

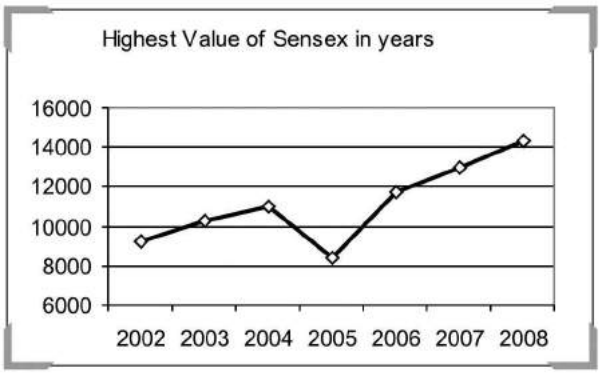
Percentage growth in total value ≅ 15% approx.

So, 100 → 20%↑ → 120 → 15%↑ → 138, OR
100 → 15%↑ → 115 → 20%↑ → 138

So, net percentage growth = 38%

Line Chart/Bar Chart

In case of line charts bar charts, understanding the process of slope calculation between two consecutive periods/ constituents/segments, can give us the highest or lowest percentage change in the whole chart.



Example 17 When does the highest percentage increase occur between any two years during the whole period?

Solution Let us start by going through the slopes of different lines joining the two values.

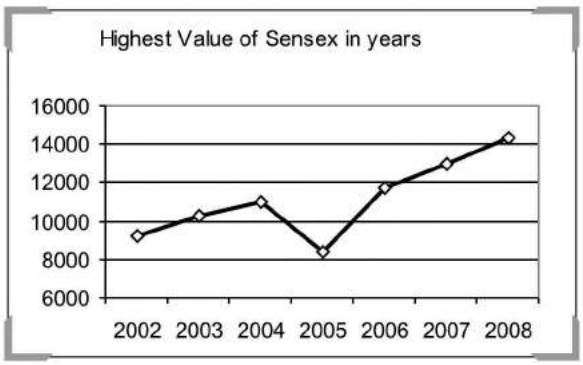
(Slope is defined as the tangent of the angle formed by the line in an anti-clockwise direction with the X-axis.)

If we find out the angle at the place of slope, it will serve the purpose.

It can be seen in the above line chart that the angle between 2005 and 2006 is the highest. Hence, the maximum percentage increase occurs during 2005-2006.

What is to be done if the angle formed between the two different periods is almost the same?

Let us see this through a final example:



In the line chart given above, the line between 2006-2007-2008 is almost straight; hence the slope is almost the same. But, as we can see, the base (or, denominator) is lower in 2005 than in 2006, hence, the percentage increase in 2005-2006 will be higher than 2006-2007.

Whatever we have discussed till now about line charts is true for bar charts as well.

CONNECTING DIFFERENT DATA-SETS

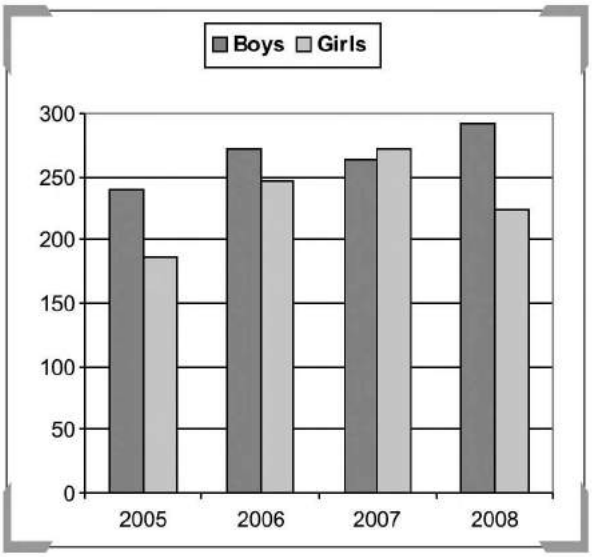
We have seen in Chapter 1, Introduction to Data Interpretation, that there exists a certain relationship among almost all the data presentation formats, it is quite possible to convert a data-set given in one format to a data-set in another format.

A number of boys and girls passed out from a school during 2005-2007:

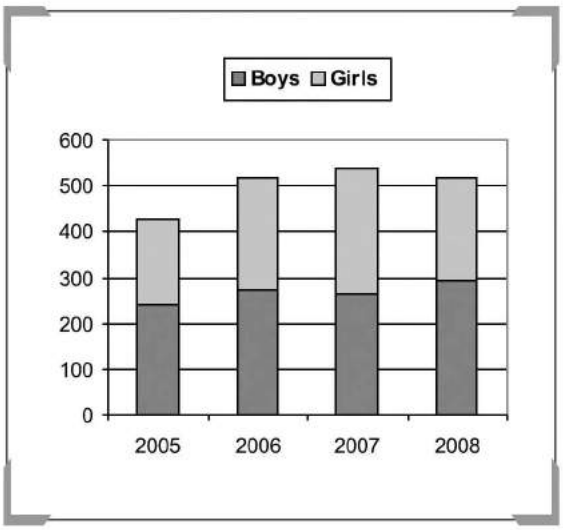
Tabular Presentation

	Boys	Girls
2005	240	186
2006	272	246
2007	264	272
2008	292	224

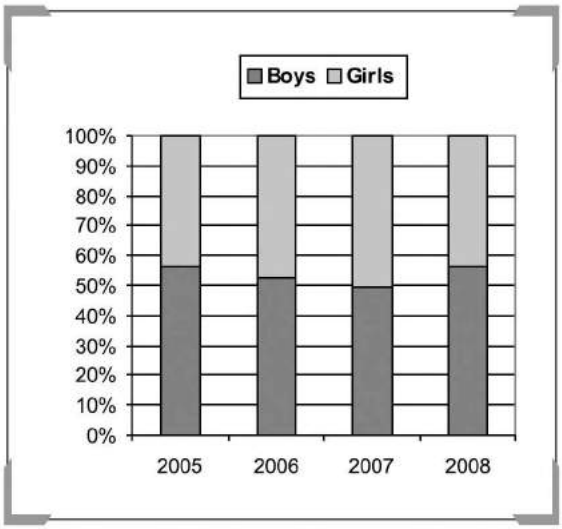
Bar Chart



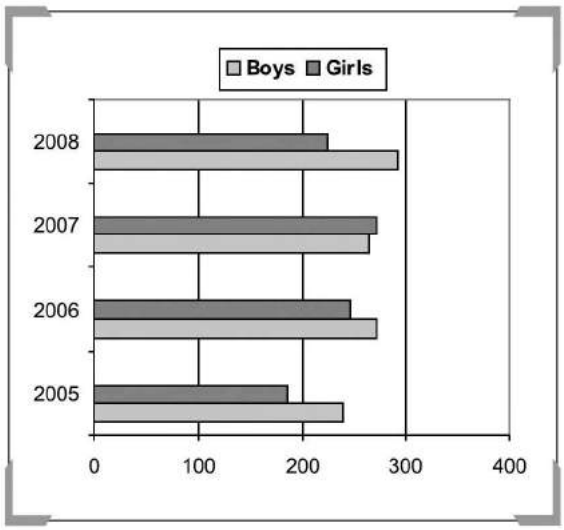
Value-Wise Stacked Bar Chart



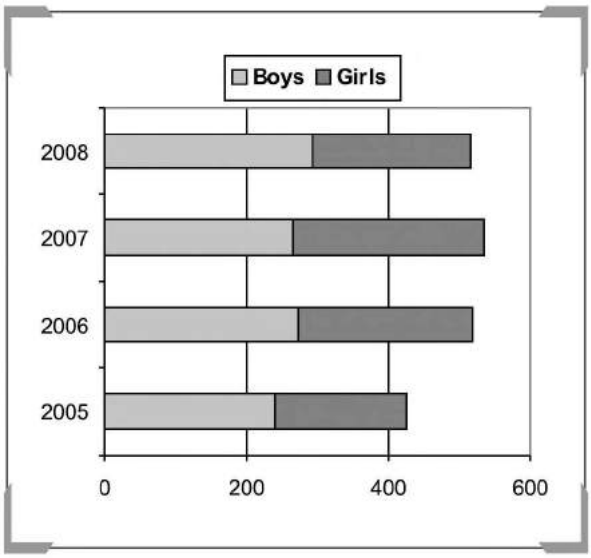
Percentage-wise Stacked Bar Chart



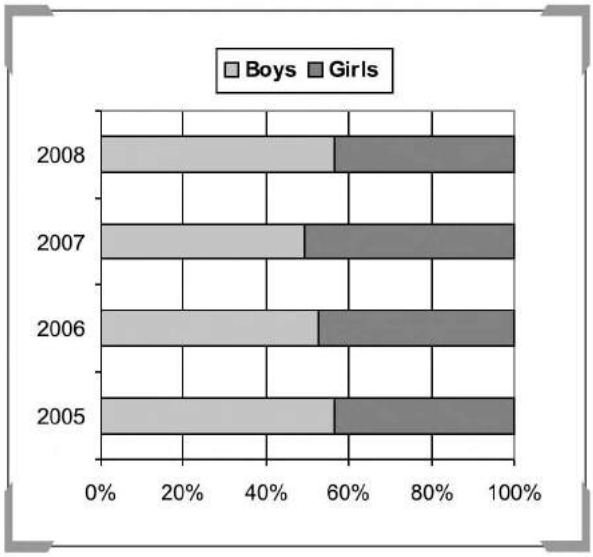
Horizontal Bar Chart



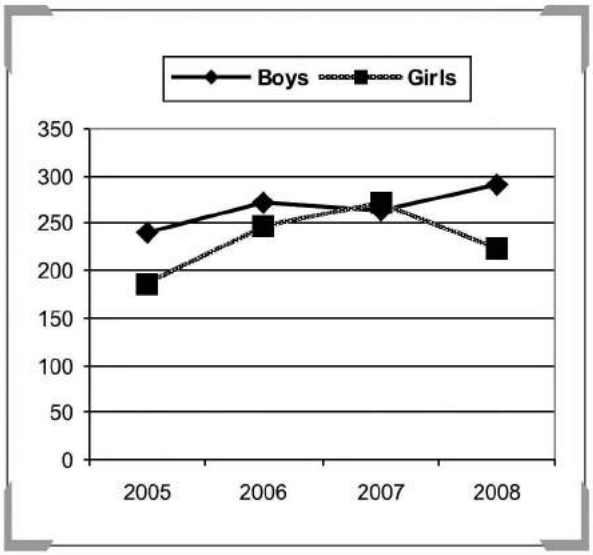
Horizontal Value-wise Stacked Bar Chart



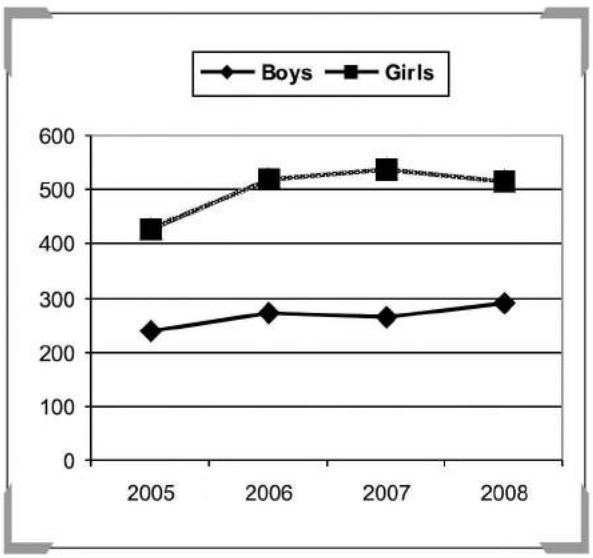
Horizontal Percentage-wise Stacked Bar Chart



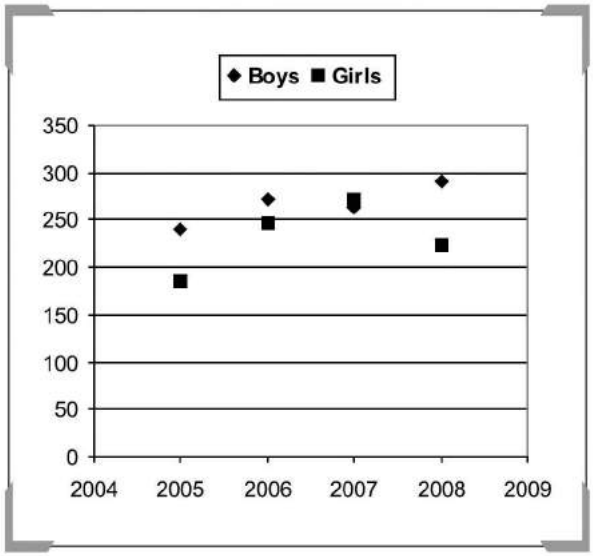
Line Chart



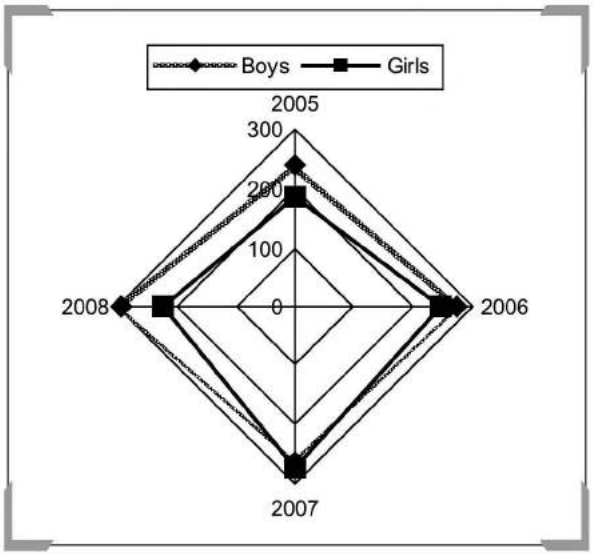
Stacked Line Chart



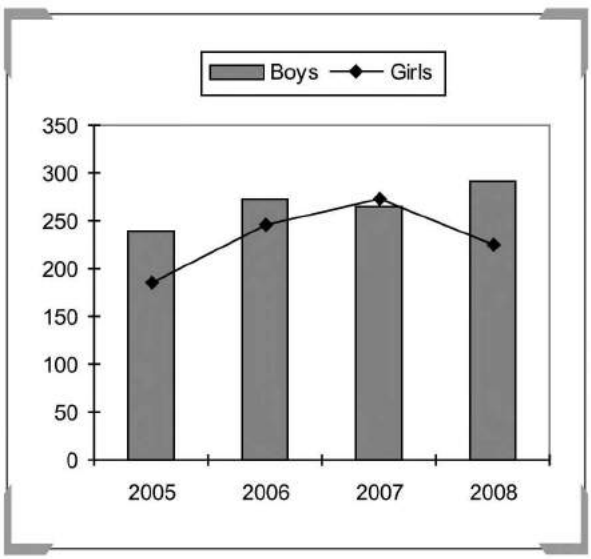
Line chart (with dots)



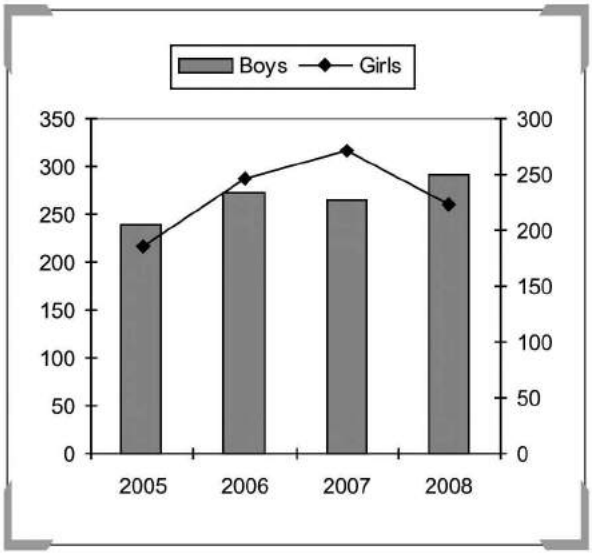
Radar Diagram



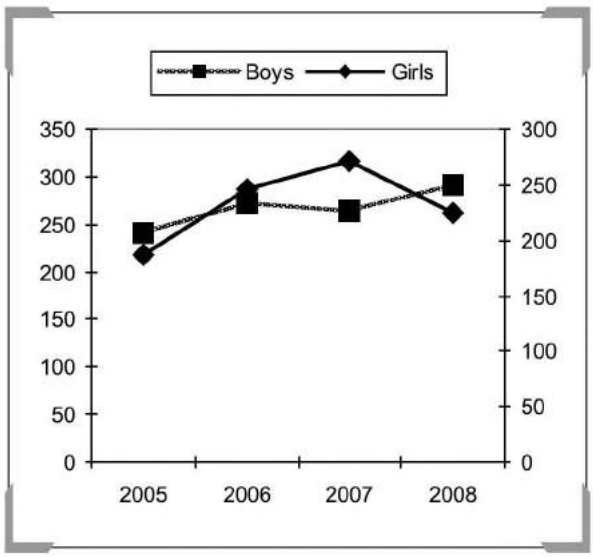
Combination Chart (Line Chart + Bar Chart)



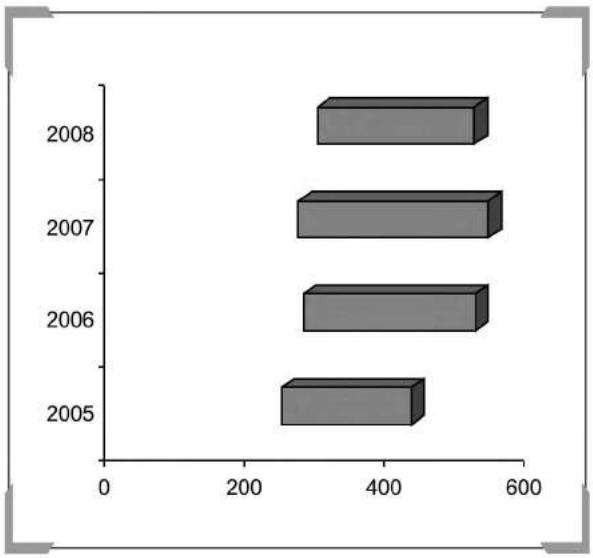
Combination Chart (Line Chart + Bar Chart) with Independent Axes



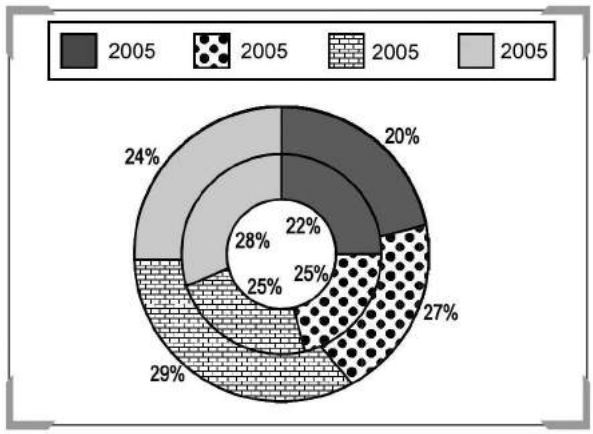
Line Chart with Independent Axes



Floating Diagram



Doughnut Graph



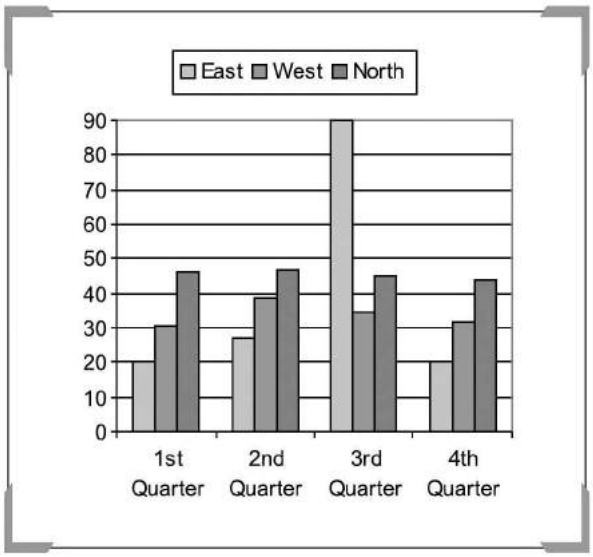
PRACTICE EXERCISE

Q 1. Convert the following table into (i) bar chart (ii) line chart (iii) set of pie charts.

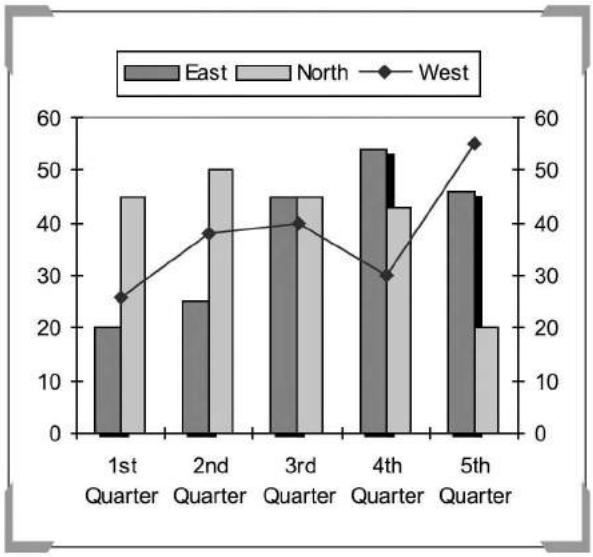
	BJP+	Congress	Communist	RJD+	Others
1996	154	214	38	32	72
1999	198	172	46	46	56
2004	234	146	52	42	34

Q 2. Convert the above given table into a line chart with two different axes—BJP on one axis and all the other parties on the second axis.

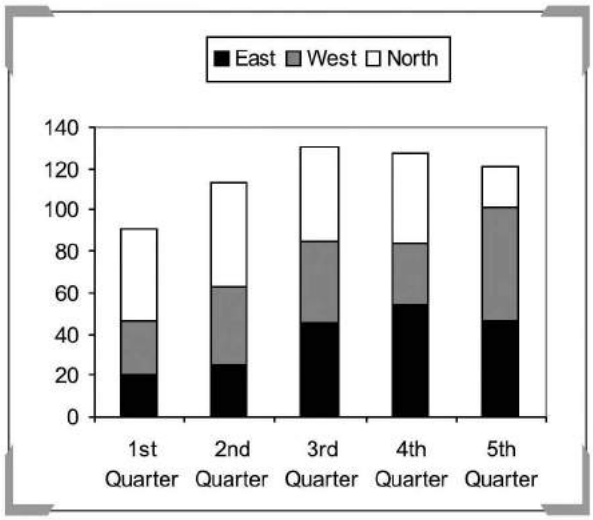
Q 3. Convert the below given bar chart into (i) line chart (ii) table



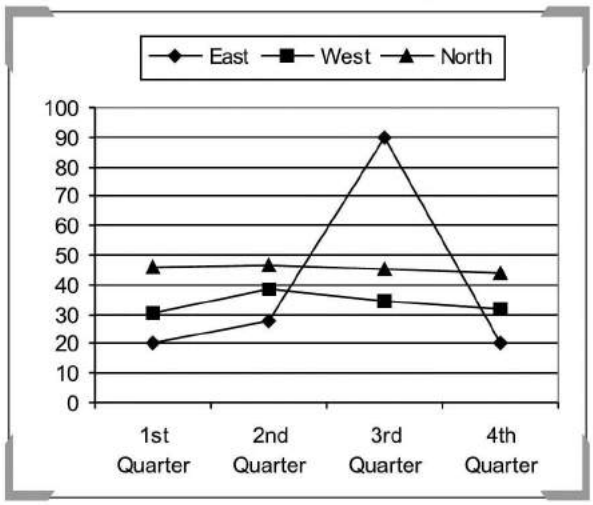
Q 4. Convert the below given line chart on two axes into (i) line chart on a single axis (ii) table (iii) bar chart.



Q 5. Convert the below given stacked bar chart into (i) normal bar chart (ii) line chart (iii) pie charts.



Q 6. Convert the below given line chart into (i) bar chart (ii) stacked bar chart (iii) pie charts.



Q 7. Rank the following ratios in descending order (without actual calculation):

3426/4536 3754/4765
3954/4865 4024/5128
3136/4129

Q 8. Rank the following ratios in ascending order(without actual calculation):

7324/6453 7212/6123
8345/7128 8523/7027
6873/5896

3

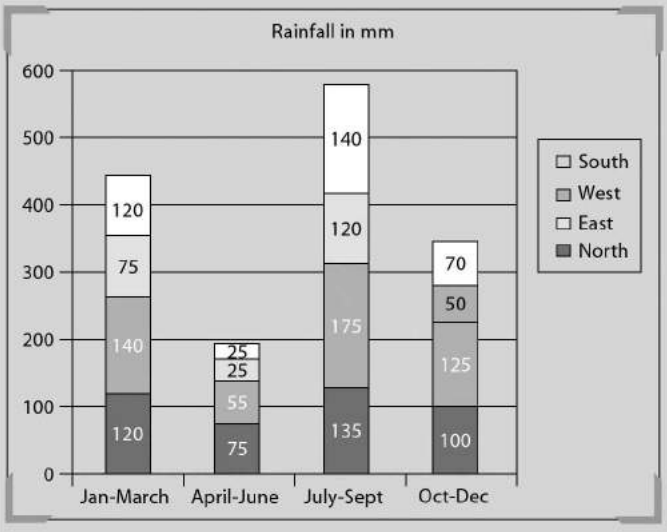
CHAPTER

Foundation Exercises

PRACTICE EXERCISE 1

Direction for questions 1 to 5: Refer to the bar chart below and answer the question based on it.

Below given bar chart shows the quantity of rainfall (in mm) in the four sectors of India in the four different periods of the year:



Q 1. Which sector received the maximum rainfall in the given year?
(a) North (b) East
(c) South (d) West

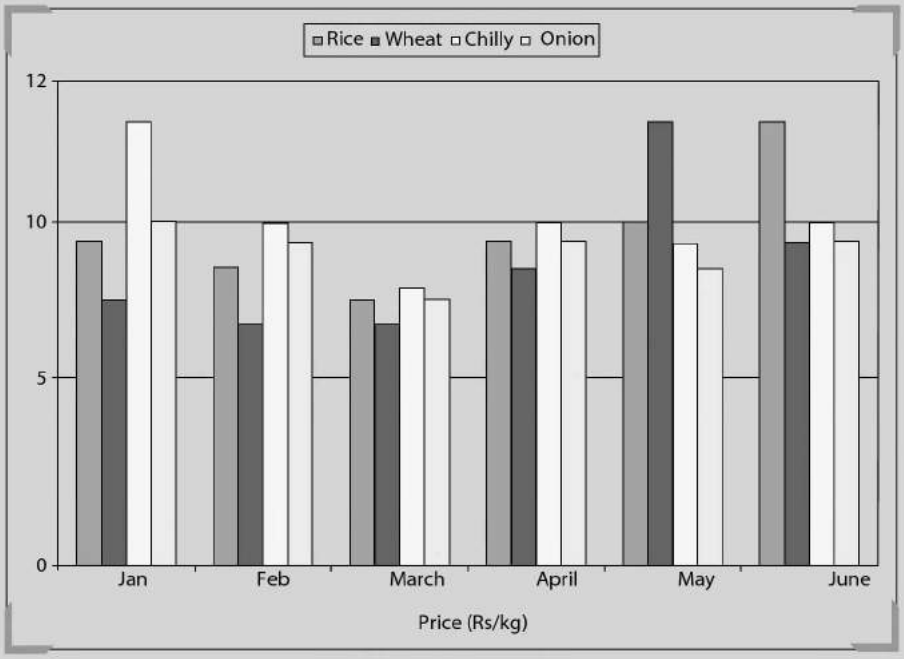
Q 2. During which period there was minimum rainfall?
(a) Jan-March (b) April-June
(c) July-Sep (d) Oct-Dec

Q 3. The average rainfall for the given year (rainfall/month) in the north sector was (approx)
(a) 33.8 (b) 34.6
(c) 35.3 (d) None of these

Q 4. In how many periods did the East receive more rainfall than the South?
(a) 0 (b) 1
(c) 2 (d) 4

Q 5. Which of the following statements is false?
(a) The rainfall followed a similar pattern for all the four sectors.
(b) The western sector received the least rainfall in Jan-March across all the regions.
(c) Rainfall was the least during the months of April-June.
(d) None of these

Direction for questions 6 to 10: Refer to the bar chart below and answer the questions based on it.



Q 6. Which of the following statements is correct?
(a) Price of rice is maximum in the month of May
(b) In all the months price of onion is less that that of chilly
(c) Price of wheat is minimum in January
(d) None of these

Q 7. Months of May and June differ from the rest of months for
(a) Price of onions is less than that of chilly
(b) Price of rice is more than Rs 8 per kg
(c) Price of wheat is more than Rs 8 per kg
(d) None of these

Q 8. Which month saw the maximum combined price of all the items?
(a) January (b) April
(c) May (d) None of these

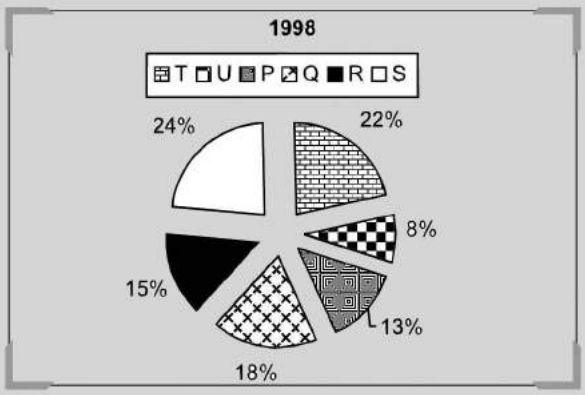
Q 9. Of the four given commodities how many have seen a drop in the price for two consecutive months?
(a) 0 (b) 1
(c) 2 (d) 3

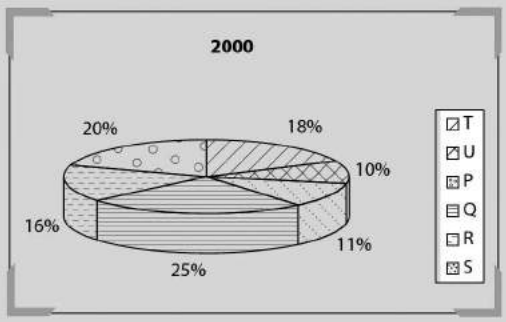
Q 10. A commodity is said to be 'Vary-Most' if its price keeps fluctuating between increase and decrease or vice-versa on alternate months. How many of the four commodities are 'Vary-Most'?
(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 11 to 15: Refer to the pie charts below and answer the questions based on it.

The following pie charts show the number of workers of different categories of a factory for the two different years.

The total number of workers in 1998 is 2000. Number of workers in 2000 grew by 20% over the number of workers in the year 1998.





Q 11. Find the percentage increase in the number of workers in category U in the year 2000.
(a) 25%
(b) 50%
(c) 33.33%
(d) 75%

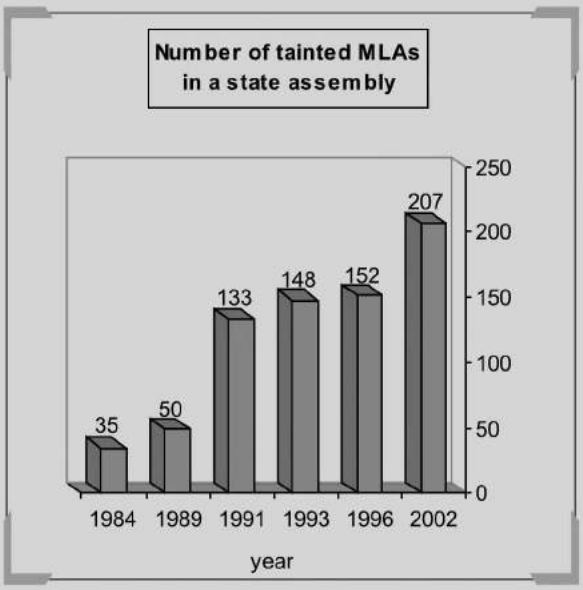
Q 12. What is the total number of increased workers for the categories in which the number of workers has increased?
(a) 468 (b) 404
(c) 382 (d) None of these

Q 13. Which of the following categories has shown a decrease in the number of workers from the year 1998 to 2000?
(a) P (b) R
(c) S (d) None of these

Q 14. Find the maximum difference between the number of workers of any two categories taken together for any one year and that of other two categories for the other year.
(a) 660 (b) 636
(c) 416 (d) 502

Q 15. For how many categories, number of workers have increased?
(a) 6 (b) 3
(c) 4 (d) 5

Direction for questions 16 to 19: Refer to the bar chart below and answer the questions based on it.



Tainted MLAs of different parties in the 2002 state assembly.

SPA	BP	SBJP	Congred	Lokmat	Others
84	54	41	7	7	14

Total number of MLAs in 2002 state assembly = 403

Q 16. Find the approximate percentage increase in the number of tainted MLAs from 1991 to 1993.
(a) 11.1 (b) 11.3
(c) 12.2 (d) 11.7

Q 17. During which year was the percentage increase in the no. of tainted MLAs from previous recorded year the lowest?
(a) 1989 (b) 1991
(c) 1993 (d) 1996

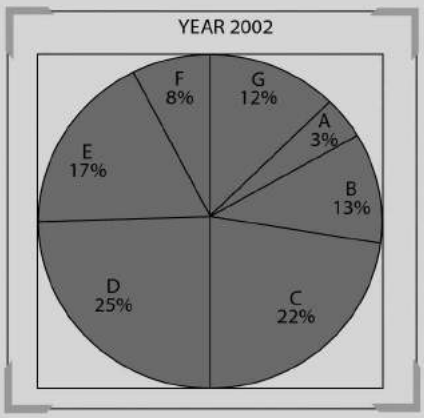
Q 18. Which of the following statements is not true?
(a) In the 1991 elections, there were 133 tainted MLAs–more than a double over 1989.
(b) Of the 403 legislators in 2002, there were 207 tainted MLAs of which a maximum of 84 were from SPA.
(c) The maximum percentage increase of tainted MLAs from the previous given year was for the year 1991.
(d) The maximum increase in the number of tainted MLAs from the previous given year was for the year 2002.

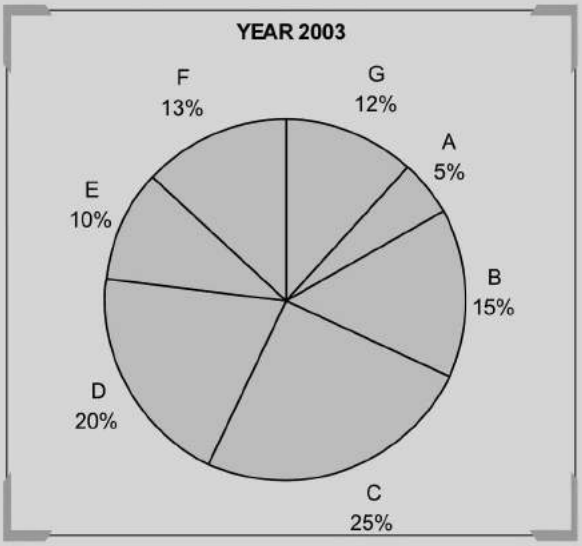
Q 19. After 2002 assembly elections, the ratio of tainted MLAs of BP to those of SPA is
(a) 14:9 (b) 9:21
(c) 21:9 (d) 9:14

Q 20. Which year exhibits the maximum percentage increase in the tainted MLAs over the previous recorded year?
(a) 1989 (b) 1991
(c) 1993 (d) 1996

Direction for questions 21 to 25: Refer to the pie charts below and answer the questions based on it.

There are seven types of employees in an organization doing seven different jobs. Following pie charts show the percentage of these employees in the organization across two years—2002 and 2003.





It is also given that the total number of B type employees in 2002 and 2003 were 1560 and 2250 respectively.

Q 21. C and D types of employees are known as grade IV employees. Find the total no. of grade IV employees in both the given years.
(a) 12390
(b) 11590
(c) 12790
(d) 11290

Q 22. A, E and F type of employees belong to higher income group (HIG). What is the number of HIG employees in 2003?
(a) 3600 (b) 4000
(c) 4200 (d) 3200

Q 23. Approximately what percentage of B type employees in 2002 is the number of B type employees in 2003?
(a) 72 (b) 73
(c) 144 (d) 142

Q 24. Which type of employees have recorded maximum increase in the year 2003?
(a) B (b) F
(c) C (d) None of these

Q 25. For how many types of employees, there is an absolute decrease in the number of employees in 2003 over 2002?
(a) 0 (b) 1
(c) 2 (d) 3

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(d)	(d)	(b)	(b)	(c)	(d)	(d)	(a)	(b)	(d)	(d)	(a)	(c)	(b)	(d)	(d)	(d)	(b)
Q. No.	21	22	23	24	25															
Answer	(a)	(c)	(c)	(c)	(b)															

HINTS AND SOLUTIONS

Let us have a tabular representation:

Sector/ Period	Rainfall in mm				
	North	East	West	South	Total
Jan-March	120	140	80	120	460
April-June	70	50	20	20	160
July-Sept	140	180	120	140	580
Oct-Dec	100	120	50	60	330
Total	430	490	270	340	1530

1. East

2. All four values for April–June season are less than those of other season so, the rainfall was minimum.

3. The answer is none of these

4. East received more rainfall than South in all the four periods.

5. Western sector received the least rainfall in April-June. So, statement b is false.

11. There are two increases–one in total number of workers (20%) and another in the percentage share of the workers(20%).
Using successive percentage change, the net change = 50%.

12. Visual inspection.

13. Only T has decreased.

14. Case I: Two categories having the highest number of workers in 1998 and simultaneously the two categories having the least number of workers in 2000.
Therefore, Difference = (480 + 440) – (240 + 264) = 416
Case II: Two categories having the highest number of workers in 2000 and simultaneously the two categories having the least number of workers in 2000.
Therefore, Difference = (600 + 480) – (160 – 260) = 660.

15. There are four categories with an increase in number of workers–U, P, Q and R. S remains constant and the T decreases.

21. Total no. of C and D type employees in 2002 = 5640
Total no. of C and D type employees in 2003 = 6750
Hence total number of employees = 12390

22. No. of A, E and F type employees in 2003
= (2250 /15) × (5 + 10 + 13) = 4200

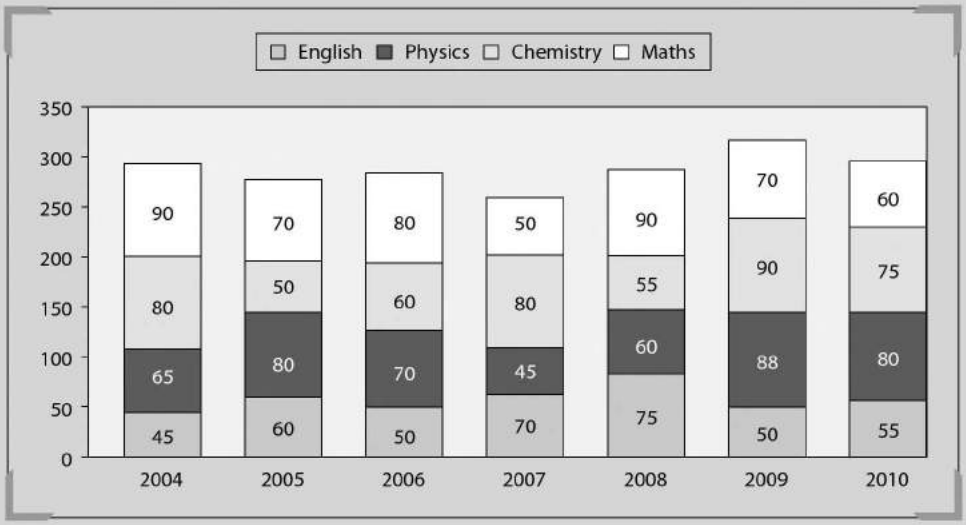
23. Required percentage = (2250/1560) × 100 ≈ 144%

24. Here we are talking about the number of employees only.

25. Only for E.

PRACTICE EXERCISE 2

Direction for questions 1 to 4: Go through the bar chart given below and solve the questions based on it.



The bar chart shows average marks obtained in four different subjects for different years at A R Academy. Marks obtained in any subject in any year can be the multiples of five only.

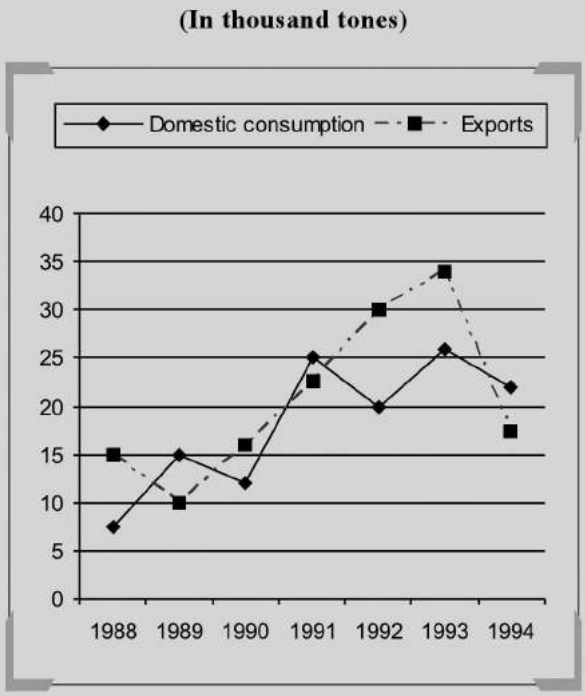
Q 1. Which of the following subjects shows maximum increase in average marks over any two consecutive years?
(a) Chemistry
(b) Maths
(c) Physics
(d) English

Q 2. The percentage increase in the average marks in Maths in 2006 with respect to the previous year = N% and the percentage increase in the average marks in Chemistry in 2006 over the previous year = M%. Find M – N.
(a) 5.7%
(b) 6.5%
(c) 4.3%
(d) 3.5%

Q 3. Find the year representing the maximum average marks, combining all the subjects.
(a) 2004
(b) 2008
(c) 2009
(d) 2010

Q 4. The total average marks in the year 2010 is decreased with respect to the total average marks in year 2004 by
(a) 2% (b) 3%
(c) 3.7% (d) 4.5%

Direction for questions 5 to 9: Go through the line chart given below and solve the questions based on it.



Q 5. In which of the following years was the total of the exports and domestic consumption highest among the given year?
(a) 1991 (b) 1993
(c) 1992 (d) None of these

Q 6. The difference between the domestic consumption between 1990 and 1991 was exactly equal to the difference in exports between which of the following pairs of year?
(a) 1991 and 1992
(b) 1989 and 1991
(c) 1988 and 1990
(d) 1991 and 1994

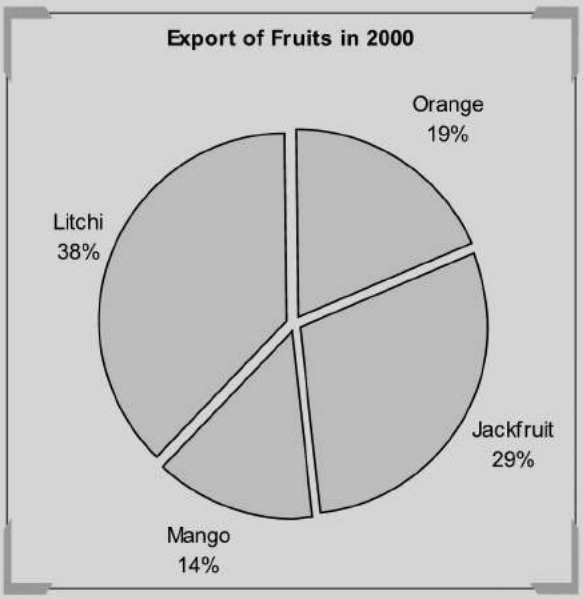
Q 7. If there was a 40 percent increase in the domestic consumption from 1994 to 1995, what would have been the difference between the domestic consumption and exports in 1995 (in tonnes)?
(a) 5,000 (b) 7,000
(c) 8,500 (d) Cannot be determined

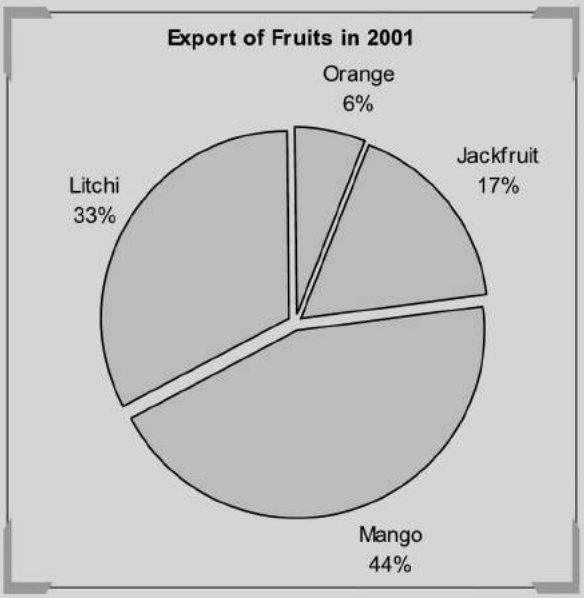
Q 8. In which of the following years was the percentage fluctuation (increase/decrease) in exports the maximum from the previous years?
(a) 1989 (b) 1991
(c) 1994 (d) 1990

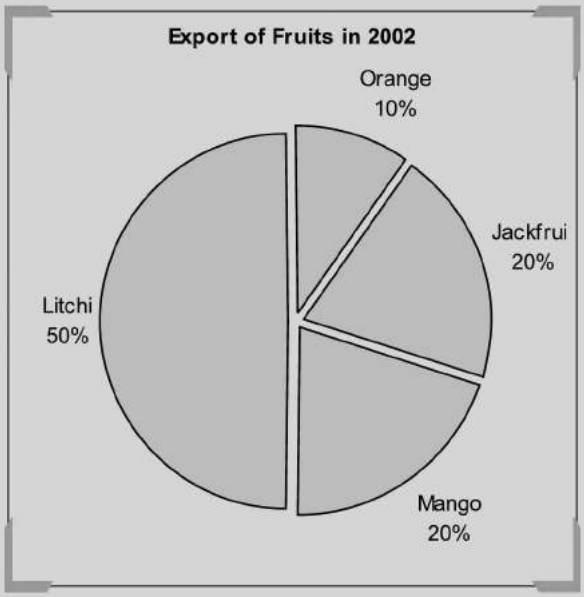
Q 9. What was the difference (in tonnes) between the average domestic consumption and the average exports during the given years?
(a) 2,000 (b) 3,000
(c) 2,500 (d) 4,000

Direction for questions 10 to 14: Refer to the charts below and answer the questions based on it.

The charts given below give the share of export of four fruits from India in three consecutive years. (in volume)







Q 10. In which year were the exports of Jackfruit maximum?
(a) 2000 (b) 2001
(c) 2002 (d) Cannot be determined

Q 11. In which year were the orange exports minimum?
(a) 2000 (b) 2001
(c) 2002 (d) Cannot be determined

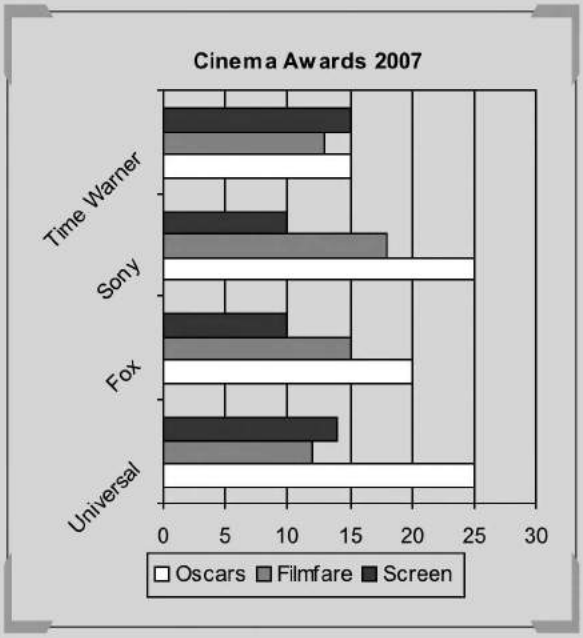
Q 12. In 2001, which fruit has shown maximum percentage increase in export over the last year?
(a) Orange (b) Jackfruit
(c) Mango (d) Litchi

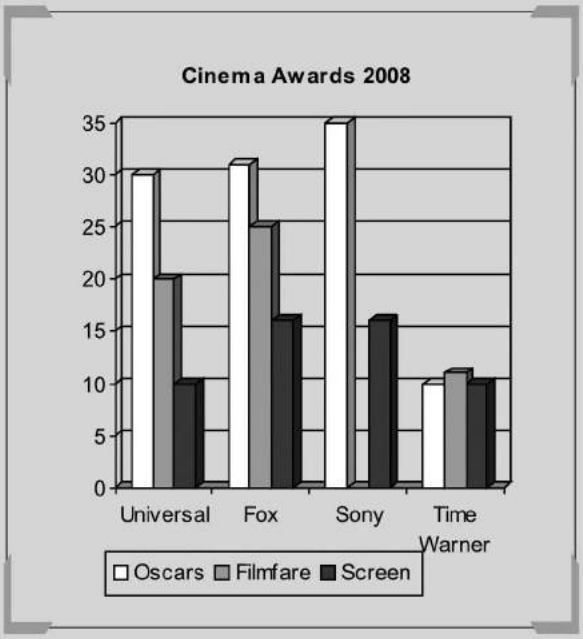
Q 13. Which fruit has the maximum percentage increase in export value in 2002 over the export value in 2000?
(a) Orange
(b) Mango
(c) Jackfruit
(d) Cannot be determined

Q 14. Export of which fruit has constantly been on a rise?
(a) Orange
(b) Mango
(c) Jackfruit
(d) Cannot be determined

Direction for question 15 to 19: Refer to the bar charts below and answer the questions based on the same.

The following bar charts give the details of three different awards won by the movies produced by the four studios:





Q 15. Which studio won the maximum total number of Screen awards in 2007 and 2008?
(a) Universal
(b) Fox
(c) Sony
(d) Time Warner

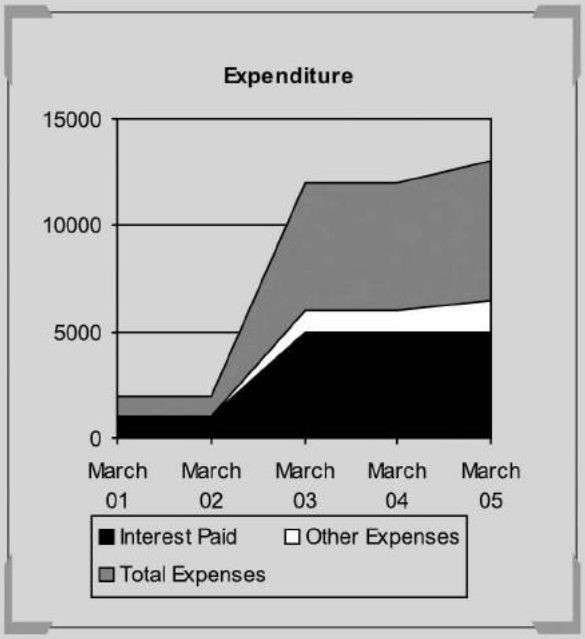
Q 16. Which studio saw the maximum percentage increase in the total number of awards?
(a) Universal
(b) Fox
(c) Sony
(d) Time Warner

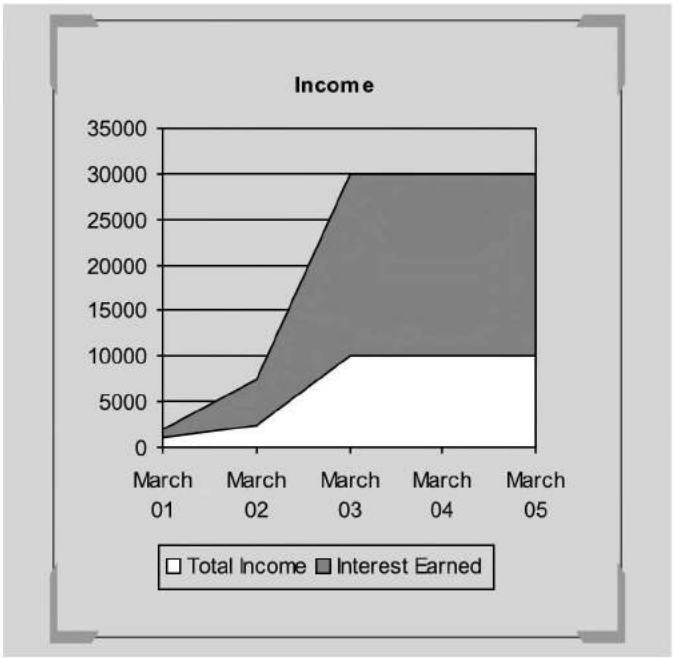
Q 17. Which award had the maximum number of categories in 2008?
(a) Oscars
(b) Filmfare
(c) Screen
(d) Filmfare and Screen

Q 18. Which studio had the maximum drop in the Filmfare awards from 2007 to 2008?
(a) Universal (b) Fox
(c) Sony (d) None of these

Q 19. Which award had the maximum percentage decrease/ increase in the number of awards won in 2008 from 2007?
(a) Oscars (b) Filmfare
(c) Screen (d) None of these

Direction for questions 20 to 24: Go through the charts given below and solve the questions based on it.





Q 20. The ratio of interest earned to the total income shows
(a) Increasing trend
(b) Decreasing trend
(c) Erratic trend
(d) No particular trend

Q 21. Total income and interest earned reported similar decline between the years
(a) 2001 and 2002
(b) 2002 and 2003
(c) 2003 and 2004
(d) 2004 and 2005

Q 22. The difference between the total income and the total expenditure is maximum in the year
(a) 2002
(b) 2003
(c) 2004
(d) 2005

Q 23. The ratio of total interest paid to expenditure is minimum in the year
(a) 2002 (b) 2003
(c) 2004 (d) 2005

Q 24. Which of the following statement is correct?
(a) Total income and total expenditure follow a similar trend
(b) Total income and interest paid follow a similar trend
(c) Total income and total expenditure do not follow a similar trend
(d) Other expenses and interest paid follow a similar trend

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(a)	(c)	(c)	(b)	(b)	(d)	(d)	(c)	(d)	(d)	(c)	(d)	(d)	(a)	(b)	(a)	(c)	(a)	(a)
Q. No.	21	22	23	24																
Answer	(c)	(b)	(d)	(c)																

HINTS AND SOLUTIONS

1. Maximum increase in average marks is observed for Maths from 2007 to 2008.

2. Percentage increase in average marks in Maths in 1996 over the previous year = 14.28%

Percentage increase in average marks in Chemistry in 1996 over the previous year = 20%

Difference = 20% – 14.28% = 5.72%.

3.

Year	Total marks
1994	280
1995	260
1996	260
1997	245
1998	280
1999	295
2000	270

4. Average marks in 2004 = 280
Average marks in 2010 = 270
Percentage decrease = 3.7%.

7. Growth or decline of exports is not given.

8. Percentage change in exports is maximum in 1990.

10. Since the value is not given, we cannot have the numbers. Hence cannot be determined.

11. Since the value is not given, we cannot have the numbers. Hence, cannot be determined.

12. Mango is the only fruit which has shown growth. Understand that total exports in 2001 might be increasing but it will impact the answer to this question because exports of all the fruits will be equally affected by it.

14. Since the value is not given, we cannot have the numbers. Hence, cannot be determined.

19.

Award	2007	2008	% Increase
Oscars	25 + 20 + 3+ 15 = 63	30 + 31 + 35 + 10 = 106	> 70%
Film-fare	12 + 15+18+13 = 58	20 + 25 + 0 + 11 = 56	= 4%
Screen	14 + 10 +10 + 15 = 49	10 + 16 + 16 + 10 = 52	= 6%

20. From the income graph, we see that the interest to the total income is showing an increasing trend.
Hence, option (a), is the answer

21. From the income graph, we can see that the interest earned and the total income reported a similar decline in 2003 and 2004, however, the decline as seen is very imperceptible.

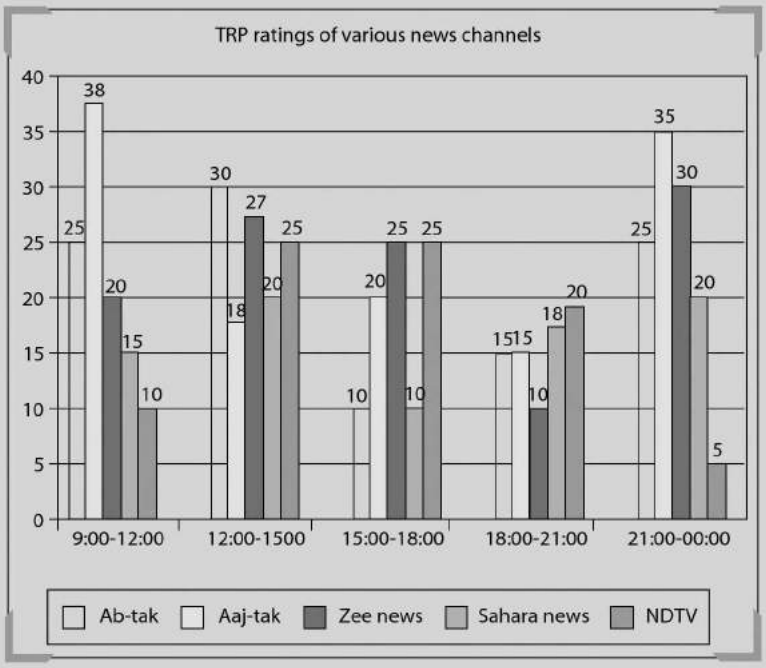
22. From both the graphs we see that the difference between the total income and the expenditure is maximum in 2003, So (b).

23. As seen from the expenditure graph, Ratio for
2002 =1/2
2003 = 8/22 = 0.33
2004 = 7/19 = 0.36
2005 = 5/19 = 0.26

PRACTICE EXERCISE 3

Direction for questions 1 to 5: Go through the bar chart given below and solve the questions based on it.

Following bar chart gives the TRP rating of five news channels in five time slots.



Q 1. Which time-slot is the most popular for watching news?
(a) 9:00-12:00 (b) 12:00-15:00
(c) 15:00-18:00 (d) 18:00-21:00

Q 2. Which TV channel has the maximum difference in the TRP ratings for any two time-slots?
(a) Ab-tak
(b) Aaj-Tak
(c) Sahara News
(d) None of these

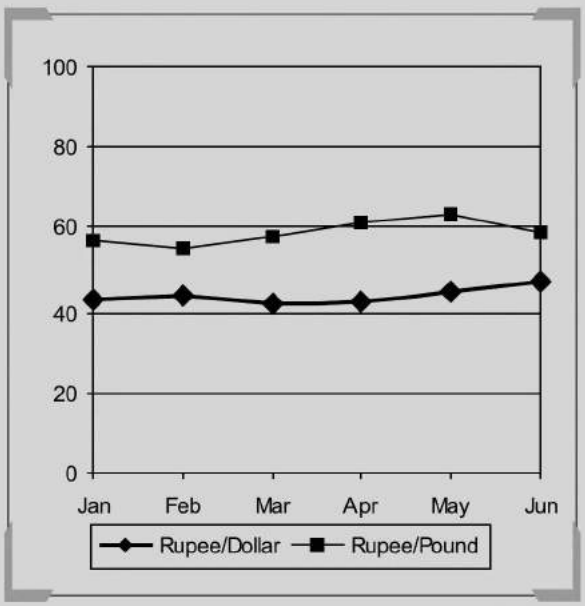
Q 3. For which time-slot is the difference between the maximum and minimum TRP rating the minimum?
(a) 9:00-12:00 (b) 12:00-15:00
(c) 15:00-18:00 (d) 18:00-21:00

Q 4. For how many time-slots does Zee news have a higher TRP rating than at least two more channels?
(a) 2 (b) 3
(c) 4 (d) None of these

Q 5. Under a new points scheme, news channels are given the points equal to their rank in the various time slots. The lower the sum of rank, the more popular the channel. Which channel is the most popular channel in accordance with this points scheme?
(a) Ab-tak (b) Aaj-Tak
(c) Sahara News (d) Zee news

Direction for questions 6 to 10: Refer to the graphs below.

Following line charts present exchange rates of three currencies for 6 months as in the month of January (Jan) 1 Dollar = Rs 43; 1 Pound = Rs 57.



Q 6. In which month can a person get least number of pounds with $ 100?
(a) Feb (b) Mar
(c) Apr (d) May

Q 7. If a person had 65 pounds, then in which month would he have exchanged them to get maximum dollars?
(a) Jan (b) Feb
(c) Jun (d) Apr

Q 8. $1 in Mar is worth 130 yens. How many yens can I purchase, if I sold 100 pounds to purchase rupees and that rupees to purchase the dollars to purchase yens?
(a) 18,000
(b) 16,000
(c) 5800
(d) 17,250

Q 9. If I purchased $ 75 in Jan paying in rupees, then in which month should I sell the dollars to get rupees so that the loss would be highest?
(a) March
(b) April
(c) May
(d) Any month

Q 10. For how many months, the exchange price of pound was more than the exchange price of dollar?
(a) 4
(b) 5
(c) 6
(d) None of these

Direction for questions 11 to 15: Go through the pie charts given below and solve the questions based on it.

There are six types of employees working in an organization. Following is the percentage breakup of different types of employees in a company in two successive years 2007 and 2008. It is also given that the total no. of employees in 2007 and 2008 is 42,980 and 48,640 respectively.
(All the percentage value are rounded off).

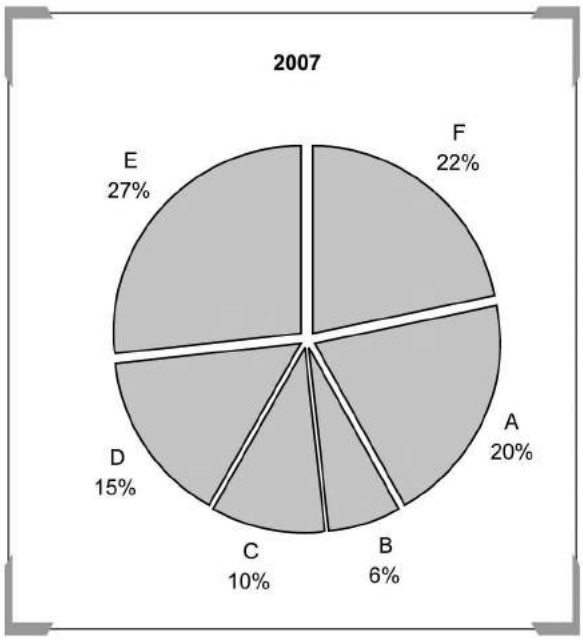
Q 11. The number of A type employees in 2008 was approximately what percentage of A type employees in 2007?
(a) 115 (b) 140
(c) 125 (d) 130

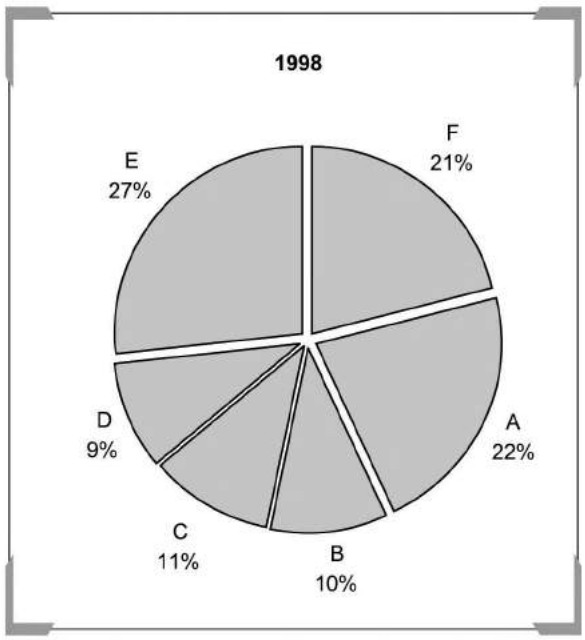
Q 12. Number of which of the following types of employees have increased maximum in 2008 over 2007?
(a) B (b) D
(c) C (d) None of these

Q 13. Approximately, what was the difference in the number of B type employee between 2007 and 2008?
(a) 2285 (b) 2325
(c) 2620 (d) 1825

Q 14. The total number of which of the following pairs of types of employees in 2007 was approximately equal to A type employees in 2008?
(a) B and C
(b) A and C
(c) D and E
(d) C and D

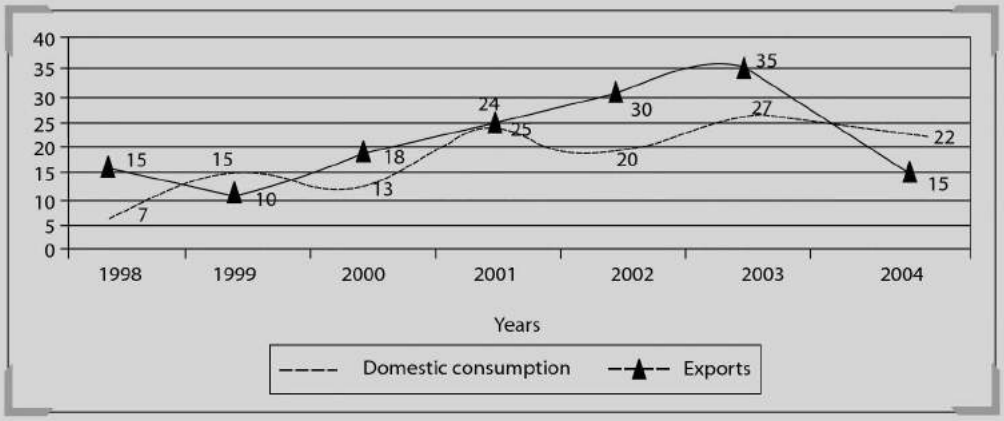
Q 15. Which type of employees have recorded maximum percentage growth in 2008 over 2007?
(a) B (b) D
(c) C (d) A





Direction for questions 16 to 20: Go through the line chart given below and solve the questions based on it.

Below given line chart represents the domestic consumption and export (In thousand tones) of rice from India in seven consecutive years.



Q 16. In which of the following years was the total of the exports and domestic consumption highest among the given years?
(a) 2001 (b) 2003
(c) 2004 (d) 2002

Q 17. The difference between the domestic consumption between 2000 and 2001 was approximately equal to the difference in exports between which of the following pairs of years?
(a) 2001 and 2002
(b) 1999 and 2001
(c) 1998 and 2000
(d) 2001 and 2003

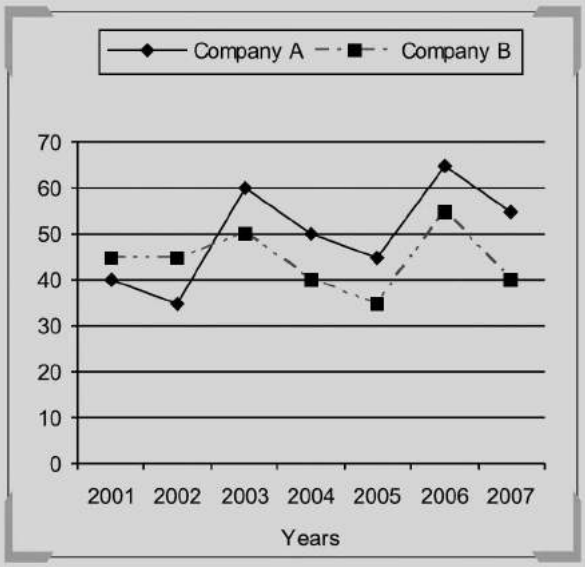
Q 18. If there was 40% increase in the domestic consum-ption from 2004 to 2005, what would have been the difference between the domestic consumption and exports in 2005 (in tonnes)?
(a) 5,000
(b) 7,000
(c) 8,500
(d) Cannot be determined

Q 19. In which of the following years was the percentage fluctuation (increase/decrease) in exports the maximum from the previous years?
(a) 1999 (b) 2000
(c) 2004 (d) 2002

Q 20. What was the difference between the average domestic consumption and the average exports during the given years?
(a) 2,000 tonnes
(b) 3,000 tonnes
(c) 4,000 tonnes
(d) None of these

Direction for questions 21 to 25: Go through the line chart given below and solve the questions based on it.

Following line chart represents percentage net profit of two companies over the years



$$\text{Percentage Profit} = \frac{\text{Income - Expenditure}}{\text{Expenditure}} \times 100$$

Q 21. If the total income in 2002 for company B was 140 crores, what was the total expenditure in that year?
(a) 10 cr
(b) 110 cr
(c) 98 cr
(d) None of these

Q 22. If the total expenditure of 2003 and 2004 together of company B was Rs 279 crores, what was the total income in these years?
(a) Rs 121.5 Cr
(b) Rs 135 Cr
(c) Rs 140 Cr
(d) Cannot be determined

Q 23. In how many of the given years the percentage of expenditure to the income of Company A was less than fifty?
(a) One
(b) Two
(c) Three
(d) None of these

Q 24. If the total expenditure of Company B in 2004 was Rs 200Cr, what was the total income?
(a) Rs 160 Cr
(b) Rs 200 Cr
(c) Rs 260 Cr
(d) None of these

Q 25. In which of the following years was the total income more than double the total expenditure in that year for Company B?
(a) 2005
(b) 2003
(c) 2007
(d) None of these

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(b)	(c)	(b)	(a)	(d)	(b)	(c)	(c)	(c)	(a)	(a)	(d)	(a)	(b)	(d)	(d)	(b)	(d)
Q. No.	21	22	23	24	25															
Answer	(d)	(d)	(d)	(d)	(d)															

HINTS AND SOLUTIONS

Let us tabulate the given info in the bar chart:

TRP	Slots						
Channels	**9:00 - 12:00**	**12:00 - 15:00**	**15:00 - 18:00**	**18:00 - 21:00**	**21:00 - 00:00**	**Total**	**Max. Difference in TRP any two time slots**
Ab-tak	25	30	10	15	25	105	20
Aaj-tak	38	18	20	15	35	126	23
Zee News	20	27	25	10	30	112	20
Sahara News	15	20	10	18	20	83	10
NDTV	10	25	25	20	5	85	20
Total	108	120	90	78	115	511	-
Max.Diff. in TRP for any 2 channels	28	12	15	10	30	-	-

Let us first establish the relation between dollar and pound:
Dollar-pound rate = Rupee-pound rate/Rupee-dollar rate

Jan	$\frac{57}{43} = 1.32$
Feb	$\frac{55}{44} = 1.25$
Mar	$\frac{58}{42} = 1.38$
Apr	$\frac{61}{42.5} = 1.44$
May	$\frac{63}{45} = 1.4$
Jun	$\frac{59}{47} = 1.26$

11. Required percentage

$$= 48{,}640 \times 22 \times 100 \approx 125\% \; 42{,}980 \times 20$$

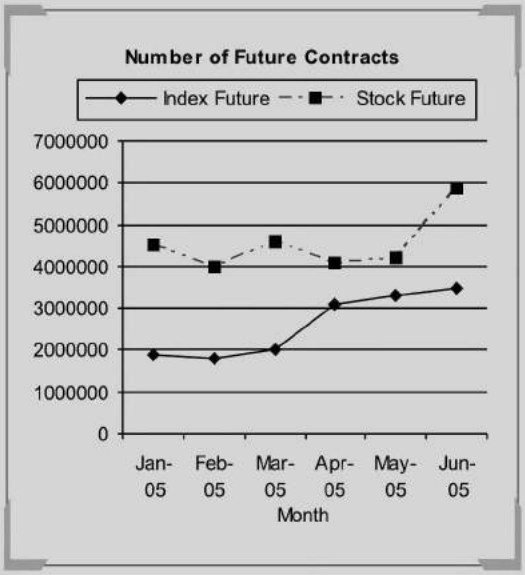
13. Required difference = 4864 – 2579 = 2285.

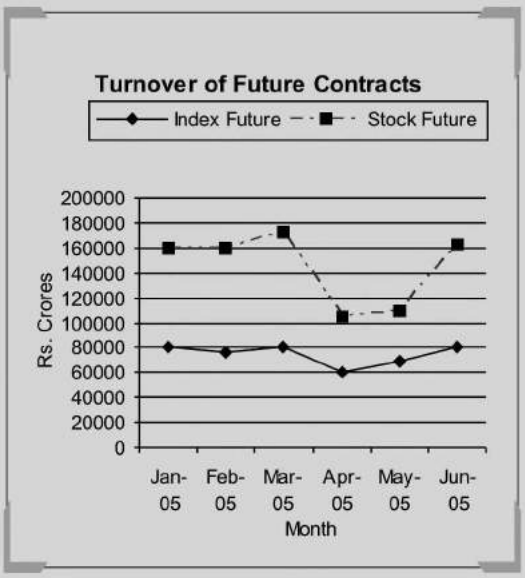
24. $I = E\left[\frac{100+\%Profit}{100}\right] = 200 \times 1.4 = 280$

25. I > 2E, so we can conclude that profit percentage is more than 100%. This is not correct for any of the given years.

PRACTICE EXERCISE 4

Direction for questions 1 to 7: Go through the charts given below and solve the questions based on it.





Q 1. The trend of stock and index futures contracts diverged in the month of
I. March to April II. April to May
III. May to June
(a) I and II (b) II and III
(c) I and III (d) Cannot be determined

Q 2. The difference in the growth or decline rates between number and turnover of futures contract in any given month is the highest for
I. Stock futures II. Index Futures
(a) I only (b) II only
(c) I and II (d) Insufficient Data

Q 3. The highest monthly percentage growth or decline occurred in the case of
(a) Number of stock futures contracts
(b) Number of Index futures contracts
(c) Turnover of stock futures contracts
(d) Turnover of index futures contracts

Q 4. The value of a contract is defined as the ratio of turnover to the number of contracts. The value of the contract was highest for both stock and index futures in
(a) January (b) February and March
(c) March (d) Insufficient data

Q 5. The highest growth rate between January and June occurred in the case of
(a) No. of stock futures contracts
(b) No. of index futures contracts
(c) Turnover of stock futures contracts
(d) Turnover of index futures contract

Q 6. Consistent growth pattern is observed for
(a) No. of stock futures contracts
(b) No. of index futures contracts
(c) Both stock and index futures contracts
(d) None of the above

Q 7. Which of the following is true?
(a) No. and turnover of stock futures follow a similar pattern
(b) No. of turnover of index futures follow a similar pattern
(c) No. of stock futures and turnover of index futures follow a similar pattern
(d) No. of index futures and turnover of stock futures follow a similar pattern

Direction for questions 8 to 11: Go through the table given below and solve the questions based on it.

All India Installed capacity (in MW) of power

Region	Sector	Total	Hydro	Thermal	Nuclear
Northern Region	State	17898	6401	11491	0
	Central	13642	3810	8652	1180
Western Region	State	19611	4180	15405	0
	Central	7387	875	5752	760
Southern Region	State	18744	10395	8318	0
	Central	7120	0	6340	780

All India Installed capacity (in MW) of power					
Region	**Sector**	**Total**	**Hydro**	**Thermal**	**Nuclear**
Eastern Region	State	7915	2256	5656	0
	Central	8552	204	8348	0
North Eastern Region	State	1098	274	824	0
	Central	1235	860	375	0
All India	State	65265	23505	41692	0
	Central	37936	5749	29467	2720

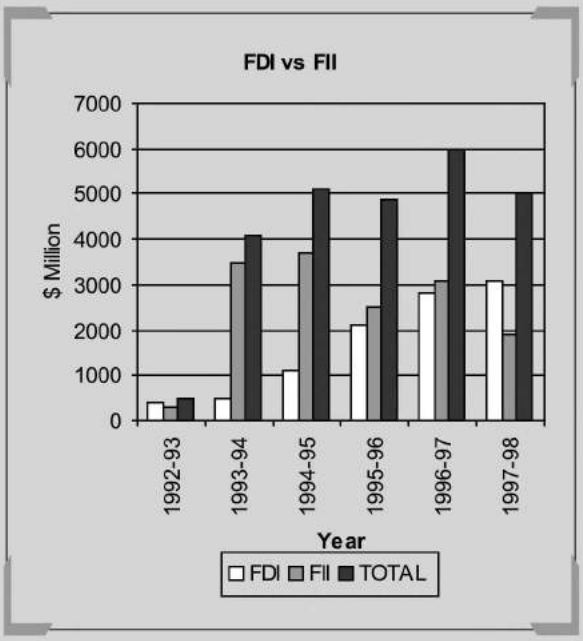
Q 8. Which region has the highest installed capacity by the central sector as compared to the state sector?
(a) Northern region (b) Southern region
(c) Eastern region (d) North Eastern region

Q 9. What is the percentage of nuclear power in total installed capacity in the country?
(a) 2.74 (b) 2.81
(c) 2.51 (d) 2.63

Q 10. In any sector, across the regions, the highest contribution from anyone of the three sources to the total installed capacity is
(a) 79% (b) 78%
(c) 89% (d) 90%

Q 11. Hydro power is the major source of energy in
(a) Northern and Southern regions
(b) Eastern and Western regions
(c) Southern and North Eastern regions
(d) None of these

Direction for questions 12 to 16: Go through the charts given below and solve the questions based on it.



Q 12. FII flows recorded negative growth rates in
(a) One Year
(b) Two Years
(c) Three Years
(d) Four Years

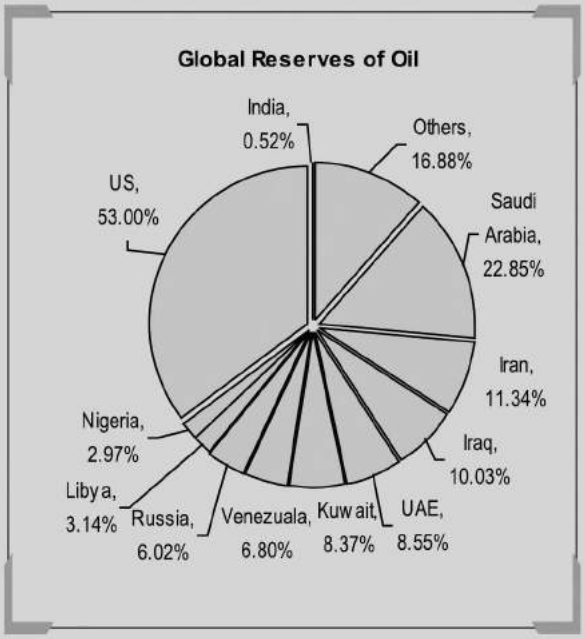
Q 13. FDI flows recorded highest growth between
(a) 1992–93 and 1993–94
(b) 1993–94 and 1994–95
(c) 1994–95 and 1995–96
(d) 1995–96 and 1996–97

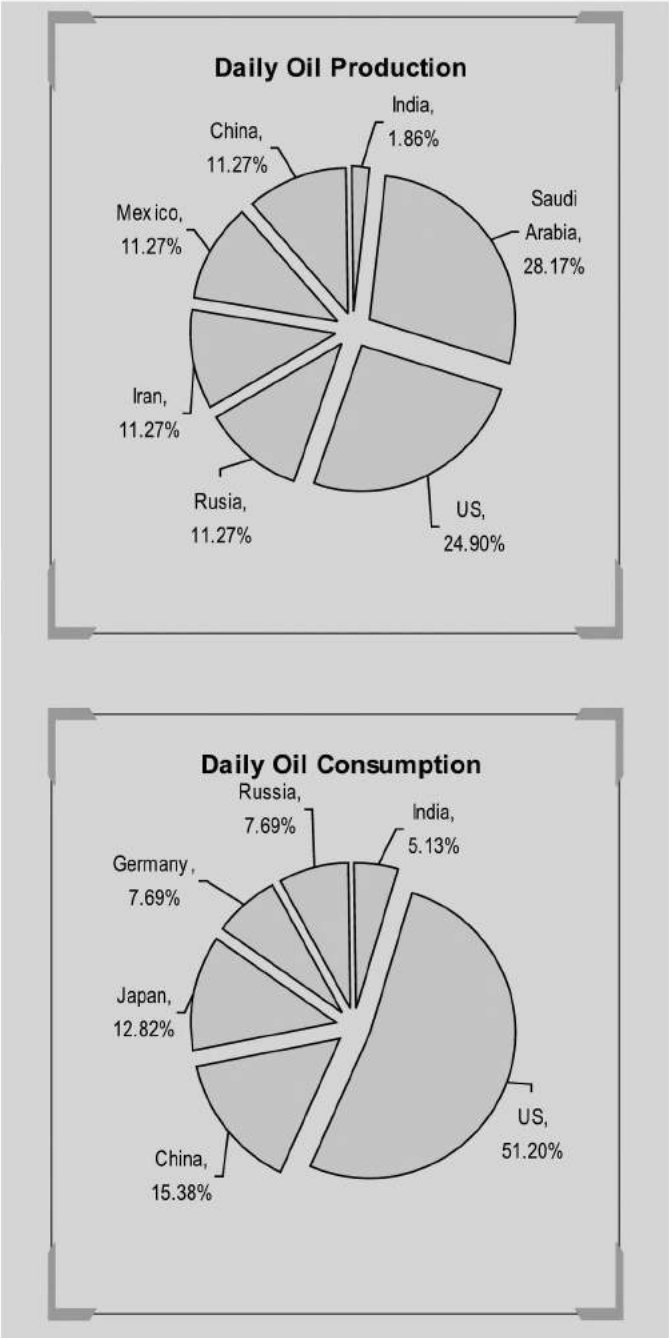
Q 14. FDI flows were greater than FII flows in
(a) 1992–93 and 1993–94
(b) 1996–97 and 1997–98
(c) 1993–94 and 1997–98
(d) 1992–93 and 1997–98

Q 15. Over the years, which of the following flows are highly volatile?
(a) FII flows
(b) FDI flows
(c) Total flows
(d) Both FII and total

Q 16. Difference between FII and FDI flows is maximum in the year
(a) 1992–93
(b) 1993–94
(c) 1994–95
(d) 1995–96

Direction for questions 17 to 25: Go through the charts given below and solve the questions based on it.





Q 17. If estimated global reserves of oil is 1146.49 billion barrels, the level of reserves located in the US (in billion barrels) would be approximately?
(a) 29 (b) 34
(c) 36 (d) 98

Q 18. The daily production and consumption of the US is 8.84 and 20 million barrels respectively. Assuming that this level remain constant, how long would the oil reserves in the US last? (Year = 360 days, Global reserves = 1146.49 billion barrels)
(a) 4 years (b) 6 years
(c) 8 years (d) 9 years

Q 19. The daily production and consumption figures of India are 0.66 and 2.2 million barrels. For which of the following countries the difference between consumption and production is the lowest?
(a) Iran (b) India
(c) Saudi Arabia (d) Cannot be determined

Q 20. How many days of India's oil consumption is equivalent to daily consumption of US?
(a) 8 (b) 10
(c) 11 (d) Insufficient data

Q 21. Which of the charts truly represent the global position?
(a) All the charts
(b) Two of the charts
(c) One of the charts
(d) Cannot be determined

Q 22. The reserves and annual production figures for India are 5.58 and 0.24 billion barrels respectively. Which country has the lowest ratio of annual production to reserves?
(a) USA
(b) Iran
(c) Saudi Arabia
(d) India

Q 23. The current reserves and the annual consumption of India is 5.58 and 0.72 billion barrels. If the expected annual growth rate of consumption for India is 2%, the oil reserves of India (in billion barrels) after 5 years would be?
(a) 1.74
(b) 1.76
(c) 1.78
(d) Cannot be determined

Q 24. The reserves located in Libya as percentage of reserves located in Russia is
(a) 50
(b) 52
(c) 54
(d) 56

Q 25. Which of the following is true?
(a) Saudi Arabia and US are the highest consumer and producer respectively
(b) Saudi Arabia and US are the highest producer and consumer respectively
(c) Russia is the lowest producer and consumer both
(d) None of these

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(b)	(c)	(b)	(d)	(a)	(d)	(d)	(d)	(d)	(b)	(c)	(d)	(d)	(b)	(a)	(c)	(d)	(b)
Q. No.	21	22	23	24	25															
Answer	(d)	(c)	(d)	(b)	(d)															

HINTS AND SOLUTIONS

1. As it is not given whether we need to scan the graph with respect to number or turnover of future contracts, so the answer cannot be determined.

2 We need to find out the individual growth rate or declined rate for number of future contracts (both stock and index) and turnover of future contracts (both stocks and index). After we have calculated this, we need to find out the difference in their growth/ declined rates.

Difference for index futures in between March–April is highest. So, (b) is the answer.

8. In this question, we have to simply find the value of (Central/State). And North-Eastern region is more than 1.5, which is highest.

9 Simple calculation involved. Percentage of nuclear power in total installed capacity = (Nuclear power/ total installed capacity) = (2720)/(103201) = 2.63%

10. It can be seen that highest contribution comes from thermal in eastern region, central sector.

12. 1995–1996 and 1997–98 are the two years.

15. Visual inspection. Bar of FII is sometimes getting up and down. Similar is the case with bar of total flows.

17. $\text{Estimated reserved} = 1146.49 \times \frac{2.53}{100} = 29$

18. As calculated, US estimated reserve
= 29 billion barrels.
Daily net consumption of USA
= 8.84 – 20 = 11.16 million barrels.
So no. of years that the oil will cost
$= \frac{29 \times 1000}{11.16 \times 360} = 8 \text{ years}$

19. Difference between production and consumption figures for India = 2.2 – 0.66
= 1.54 billion barrels.
Russia = –0.7
Consumption for Saudi Arabia and Iran is not given.
So answer cannot be determined (d).

20. No. of days equivalent to one day of consumption of US
$= \frac{51.28}{5.13} = 10$

21. No information is given about the total reserves, daily oil production, daily oil consumption. So, we cannot infer the given.

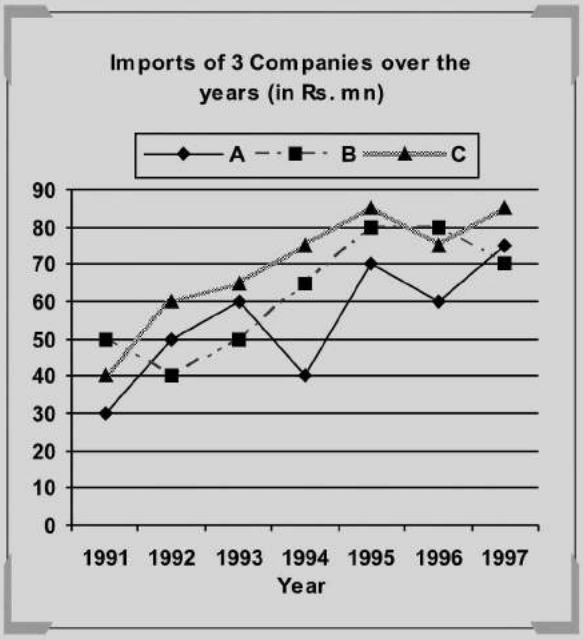
23. Since the production figures five years from now is not given, we cannot say any thing about the oil reserves.

24. $\text{Required Percentage} = \frac{3.14}{6.02} = 52.15$

25. Looking at the charts, we see that none the first three options is true.

PRACTICE EXERCISE 5

Direction for questions 1 to 5: Go through the line chart given below and solve the questions based on it.



In the above line chart, all the figures are in the multiples of five.

Q 1. In which of the following years, the imports made by company A was exactly equal to average imports made by it over the given years?
(a) 1992 (b) 1993
(c) 1994 (d) None of these

Q 2. In which of the following years was the difference between the import made by the company B and C the maximum?
(a) 1995 (b) 1994
(c) 1991 (d) 1992

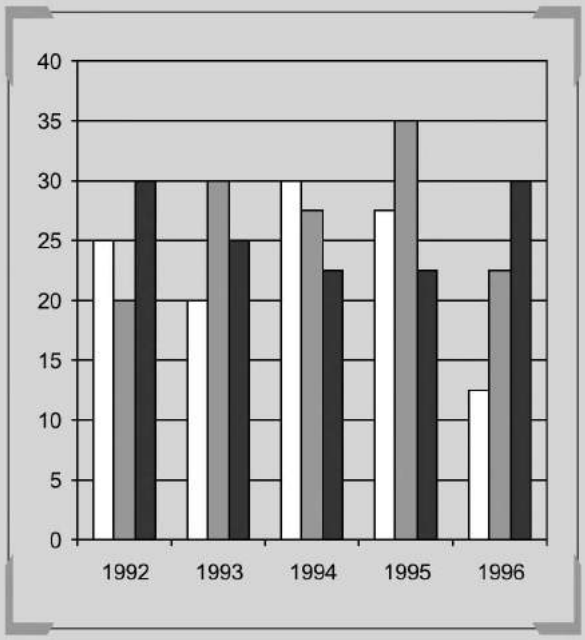
Q 3. In which of the following years was the import made by company A exactly half of the total imports made by company B and C together in that year?
(a) 1992 only (b) 1993 only
(c) 1992 and 1993 (d) 1995 only

Q 4. What was the percentage increase in imports by company B from 1992 to 1993?
(a) 10 (b) 25
(c) 40 (d) 50

Q 5. In which of the following years, the total exports made by all the three companies the maximum?
(a) 1994 (b) 1995
(c) 1996 (d) 1997

Direction for questions 6 to 10: Go through the bar chart given below and solve the questions based on it.

Production of three types of tea leaves over the five years (in mn kg)



All the values in the above chart are in the multiples of 2.5.

Q 6. What was the percentage drop in the C type tea leaves produced from 1993 to 1994?
(a) 25 (b) 10
(c) 20 (d) 25

Q 7. What was the difference between the B type tea leaves produced (in kg) in 1994 and 1995?
(a) 10,00,000 (b) 20,000,000
(c) 10,000,000 (d) None of these

Q 8. The total production of all the three types of tea leaves was the least in which of the following year?
(a) 1995 (b) 1996
(c) 1992 (d) 1994

Q 9. In which of the following years was the percentage production of B type tea leaves over the production of C type tea leaves the maximum?
(a) 1994 (b) 1992
(c) 1996 (d) 1995

Q 10. Which year the production of tea leaves of all the types the maximum?
(a) 1994 (b) 1992
(c) 1996 (d) 1995

Direction for questions 11 to 15: Go through the table given below and solve the questions based on it.

Following table gives the details of five leading opener's performance in 20 one-day cricket international matches:

Openers	Total runs	Highest Runs	No. of matches with runs		
			100 or more	50-99	0's
A	994	141	5	3	1
B	751	130	1	8	2
C	414	52	-	2	2
D	653	94	-	4	1
E	772	85	-	7	-

Q 11. What is the difference between the average runs of top two openers, if matches in which 0's are scored are not to be counted?
(a) 4.7
(b) 13.7
(c) 11.1
(d) None of these

Q 12. If matches having zero runs and the one with highest runs are ignored, what will be the average runs for opener C?
(a) 21.29
(b) 21.79
(c) 20.17
(d) 20.19

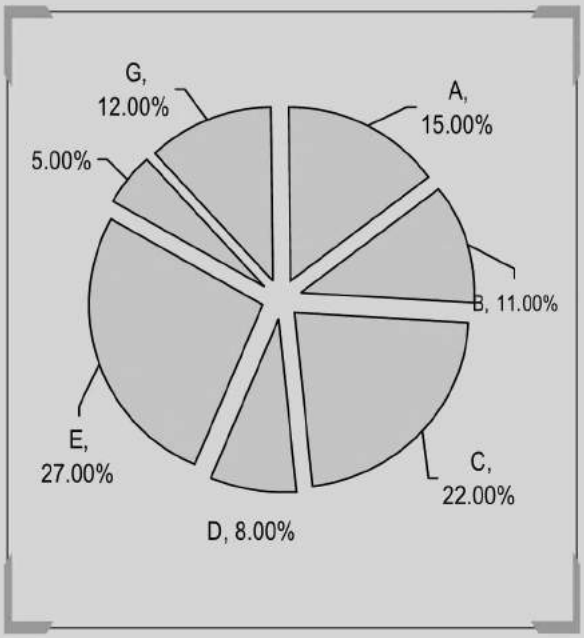
Q 13. What is the maximum number of 25 runs inning that could have been played by B?
(a) 6
(b) 8
(c) 9
(d) Cannot be determined

Q 14. Which of the given pairs of openers have ratio 3:2 in their highest runs?
(a) B and D (b) B and C
(c) A and D (d) D and C

Q 15. If each of the above given five openers have scored their highest run in a single match, then what is the average score per member of the team (there are eleven playing members in a cricket team)?
(a) 25 (b) 15
(c) 30 (d) Cannot be determined

Direction for questions 16 to 19: Go through the charts given below and solve the questions based on it.

There are seven companies—A, B, C, D, E, F and G, engaged in the production of naphtha of two types I and II. The comparatives data about production of these types by the seven companies is given in the following chart and table.
Percentage of the total production produced by the seven companies.



Cost of the total production (Both types together) by seven companies = Rs 25 crores and the production cost of each of the companies is same. Cost of individual company can be calculated using above pie-chart.
Ratio of production between types I and II and percentage profit earned for the two types:

Company	Ratio of Production		Percentage profit earned	
	Type I	Type II	Type I	Type II
A	2	3	25	20
B	3	2	32	35
C	4	1	20	22
D	3	5	15	25
E	5	3	28	30
F	1	4	35	25
G	1	2	30	24

Q 16. What is the total cost of the production of type I by companies A and C together in Rs Crore?
(a) 9.25 (b) 5.9
(c) 4.1625 (d) 4.9

Q 17. What is the amount of profit earned by company D on type II?
(a) Rs 3.125 cr
(b) Rs 31.25 cr
(c) Rs 3.125 lakh
(c) Rs 31.25 lakhs

Q 18. Cost of production of type I by company F is what percentage of the cost of production of type II by company D?
(a) 16%
(b) 33.33%
(c) 66.67%
(d) None of these

Q 19. What is the total profit (in Rs. lacs) earned by company G for types I and II together?
(a) 78 (b) 162
(c) 780 (d) 16.2

Direction for questions 19 to 22: Go through the table given below and solve the questions based on it.

The percentage change in the sales of six departmental stores from 1993 to 1995 is shown in the following table:

Store	Percentage change	
	From 1993 to 1994	From 1994 to 1995
UCB	+20	–20
VAN	–20	+9
Wrangler	+5	+12
X-tra	–7	–15
Yo	+17	–8
Zodiac	+21	+9

Q 19. In 1995 which of the stores had greater sales than any of the other stores?
(a) Van (b) Wrangler
(c) Zodiac (d) Cannot be determined

Q 20. Sales in store UCB amounted to Rs 80 lacs in 1993. What was the sales amount of the same store in 1995?
(a) Rs 96 lacs (b) Rs 74 lacs
(c) Rs 80 lacs (d) None of these

Q 21. In store Yo the sales for 1994 amounted to about what percentage of the sales for 1995?
(a) 117 (b) 109
(c) 101 (d) 113

Q 22. If in 1993 the sales in store Wrangler amounted to Rs X, the sales for that store in 1995 amounted to:
(a) 1.5X (b) 1.17X
(c) 1.32X (d) 1. 61X

Direction for question 23 to 26: Refer to the table below and answer the questions based on it.

Maruti has recently slashed the price of its car Alto. The following is the information on its previous and present price (ex. show room, Delhi)

	Old (Rs in Lakh)	New (Rs in Lakh)
Showroom price	6.30	4.90
(NAV)	3.98	3.12
Excise	1.59	1.25
Sales Tax	0.44	0.34
Dealer Commission	0.12	0.12
Company Margin	0.17	0.07

Q 23. What has been the fall in the showroom price of cars?
(a) 15% (b) 16.67%
(c) 22.2% (d) 25%

Q 24. Excise is levied as percentage of NAV. If NAV of a car is Rs 6.4 lakh, then what is its excise?
(a) Rs 1.4 lakh
(b) Rs 2.8 lakh
(c) Rs 2.2 lakh
(d) Rs 2.6 lakh

Q 25. As a percentage of the showroom price, the dealer commission has changed by how many percentage point?
(a) –0.35% (b) +0.55%
(c) –0.75% (d) +1.00%

Q 26. What is the drop in the sales tax collections per car?
(a) 16% (b) 18%
(c) 20% (d) 23%

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(d)	(a)	(b)	(d)	(b)	(d)	(b)	(d)	(d)	(d)	(a)	(b)	(c)	(d)	(b)	(c)	(d)	(d)	(d)
Q. No.	21	22	23	24	25	26														
Answer	(b)	(b)	(c)	(d)	(b)	(d)														

HINTS AND SOLUTIONS

1. Average imports made by company A = (30 + 50 + 60 + 40 + 70 + 60 + 75)/7 = 55
   In none of the given years the imports is equal to Rs 55 mn.

3. 1992 only is the desired year. What we are required to see here is the fact that A should lie exactly midway B and C.

4. Required percentage increase $= \frac{50-40}{40} \times 100 = 25\%$

6. Required percentage Drop $= \frac{25-22.5}{25} \times 100 = 100\%$

7. Required difference = (35–22.5) lakh
   = 12,50,000

9. Production of B type leaves is more than the production of C type leaves only in the years 1993, 1994 and 1995. We see the largest difference exists in the year 1995. So, the answer is 1995.

11. Average runs of A $= \frac{994}{19} = 52.31$

    Average runs of B $= \frac{751}{18} = 41.72$

    Difference = 52.31 – 41.72 = 10.59

12. Average runs of C $= \frac{414-52}{17}$

    $= \frac{362}{17} = 21.29$

14. Ratio of A and D = 141 : 94 = 3:2

16. Total cost for A $= \frac{2}{5} \times 15\% \times 25$ crores

    Total cost for C $= \frac{2}{5} \times 15\% \times 25$ crores

    So, total cost of A & C = 5.9 crores

17. Amount of profit earned by company D on type II

    $= \frac{5}{8} \times \frac{8}{100} \times 25 \times \frac{25}{100} = 31.25$ lakhs

18. Cost of production of type I by company F

    $= \frac{1}{5} \times \frac{5}{100} \times 25 =$ Rs. 0.25 cr

    Cost of production of type II by company D

    $= \frac{5}{8} \times \frac{8}{100} \times 25 =$ Rs.1.25 cr.

    Therefore, required % $= \frac{0.25}{1.25} \times 100 = 20\%$

20. In 1993 the sales in store UCB amounted to Rs 80 lacs. In 1994 the sales in store UCB amounted to 120% of Rs 80 lacs = 96 lacs. In 1995 the sales in store U amounted to 80% of Rs 96 lacs = Rs 76.8 lacs.

21. If x is the amount of sales for store Yo in 1994, then 0.92x is the amount of sales for 1995. So, the answer is: 1/0.92 = 1.086 = 1.09

23. Percentage decrease in showroom price = 22.22%

24. Excise as the % of NAV = 1.59/ 3.98 × 100 = 40%
    Therefore, Excise is 40% of NAV.
    When NAV = Rs 6.4 lakhs,
    Excise = 40/100 × 6.4 = 2.56 lakhs.

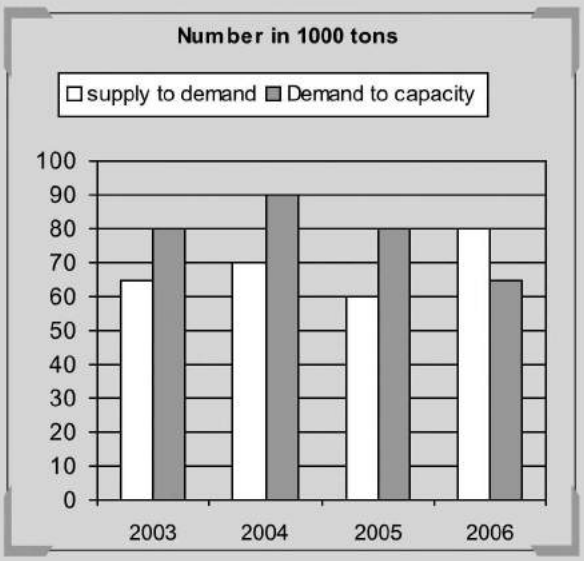
25. Initial dealer commission as percentage of show-room price = 0.12/6.30 × 100 = 1.90%,
    And similarly final dealer commission on showroom price = 2.45%
    Therefore, the dealer commission has risen by
    2.45 – 1.90 = 0.55%

26. Percentage decrease in sales tax
    = (0.44 – 0.34)/0.44 = 23% (Approx.)

PRACTICE EXERCISE 6

Direction for questions 1 to 6: Go through the chart given below and solve the questions based on it.

The following bar chart represents supply as a percentage of demand of a product as well as the demand as a percentage of capacity of the industry to produce the same product.



All the values are in the multiple of 5.

Q 1. If the product supply for the year 2005 is 6000 kgs, then the total capacity of the product is (in kg)
(a) 12500
(b) 16500
(c) 10000
(d) None of these

Q 2. If the capacity of the industry to produce the product remains constant at 10000 kgs, what is the approximate average supply (in kgs) for the given period?
(a) 5100 (b) 5800
(c) 4400 (d) 6200

Q 3. If the product supply remains unchanged for a given period then what does the trend of the industry capacity signify?
(a) The industry capacity steadily increases over the given period.
(b) The industry capacity steadily decreases over the given period
(c) The industry capacity steadily fluctuates over the given period.
(d) None of these

Q 4. Which year shows the highest increase in the difference between the industry capacity and supply over the previous year, if demand for the product is 10,000 kgs for each of the given years?
(a) 2003
(b) 2004
(c) 2005
(d) 2006

Q 5. If we interchange the bars of 'supply to demand' and 'demand to capacity' with each other, then which of the bars will represent inconsistency in data?
(a) 2003
(b) 2004
(c) 2005
(d) None of these

Q 6. If the supply is constant for each of the years given above, then for which year is the capacity highest?
(a) 2003
(b) 2004
(c) 2005
(d) 2006

Direction for questions 7 to 9: Given below is a table showing percentage out of a total of 700 employees ranking six attributes that help in promotion. Rank I is the highest. Study the table carefully and answer the questions that follow:

No employee can give the same rank to two or more than two attributes.
Rank 1 = 6 points. rank 2 = 5 points, and so on rank 6 = 1 point.
Higher the points, more is the chance of promotion.

Attribute	Percentage of Employees Giving Different Ranks					
	I	II	III	IV	V	VI
Seniority	32	17	22	19	5	5
Perservance	14	19	17	9	27	14
Efficiency	15	19	21	14	14	17
Intelligence	10	14	10	14	17	35
Honesty	24	17	7	9	27	16
Sociability	5	14	23	35	10	13

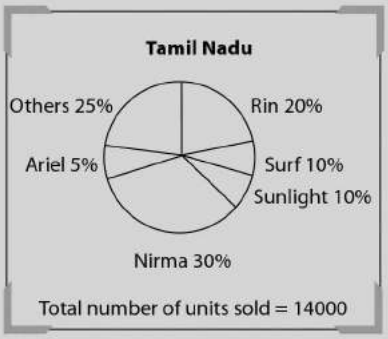
Q 7. Which attribute for promotion has received the highest rank?
(a) Perservance (b) Seniority
(c) Honesty (d) Sociability

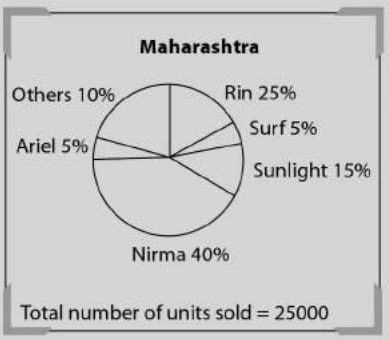
Q 8. How many employees gave rank III to intelligence?
(a) 119 (b) 98
(c) 77 (d) 70
(e) 10

Q 9. Which attribute is considered the least importance for promotion?
(a) Honesty (b) Intelligence
(c) Perservance (d) Efficiency
(e) Sociability

Direction for questions 10 to 12: Go through the pie charts given below and solve the questions based on it.

The market shares of various detergent brands in the year 2008 in the states of Tamil Nadu and Maharashtra are given below:





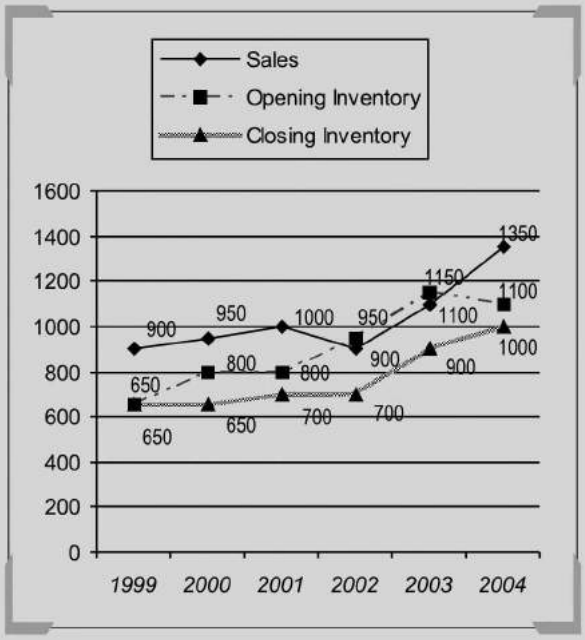
These are the top brands in these 2 states.

Q 10. Which brand has the largest difference in the units sold in the two given states?
(a) Rin
(b) Surf
(c) Sunlight
(d) None of these

Q 11. Total sale of Sunlight in the two states is:
(a) 4,850 units
(b) 5,000 units
(c) 5,150 units
(d) 5,300 units

Q 12. Which detergent is having the maximum sales in both the states taken together?
(a) Ariel
(b) Nirma
(c) Sunlight
(d) Cannot be determined

Direction for questions 13 to 15: Go through the chart given below and solve the questions based on it.



Sales = Opening Inventory + Production – Closing Inventory

Q 13. What is the percentage growth in average annual sales of the last 3 years versus average sales of first 4 years?
(a) 21% (b) 19%
(c) 24% (d) 23%
(e) None of the above

Q 14. What is the difference between the average closing inventory and the average opening inventory over the given period?
(a) 57
(b) 54
(c) 59
(d) 53
(e) None of the above

Q 15. If production figures are plotted on the graph, then the shape of the production line
(a) forms a linear
(b) resembles more like the sales curve
(c) resembles more like the opening inventory curve
(d) resembles more like the closing inventory curve
(e) None of the above

Direction for questions 16 to 20: Percentage of marks obtained by seven students in six subjects

	History	Geography	Maths	Science	English	Hindi
	(75)	(75)	(150)	(200)	(100)	(50)
A	84	72	81	67	62	68
B	76	58	87	73	65	72
C	59	64	75	61	71	56
D	69	78	64	58	61	66
E	88	68	93	83	74	78
F	82	66	89	91	77	68
G	78	72	66	71	75	80

Note: The figures in the brackets are the maximum marks for each subject.

Q 16. What is the total marks obtained by student 'A' in History, Geography and Science together?
(a) 261 (b) 245
(c) 342 (d) 251
(e) None of these

Q 17. What is the percentage of total marks obtained by student 'B' in all subjects together?
(a) 71.83 (b) 73.54
(c) 76.84 (d) 72.75
(e) None of these

Q 18. What is the average marks obtained by the seven students in Maths?
(a) 126
(b) 119
(c) 115
(d) 111
(e) Cannot be determined

Q 19. What is the percentage of marks obtained by student 'E' in History Geography, Maths and Hindi together?
(a) 93.2
(b) 89.6
(c) 81.4
(d) 84.4
(e) Cannot be determined

Q 20. What is the difference of the marks obtained by C & G is all subjects?
(a) 42.75
(b) 96.25
(c) 112.75
(d) No difference
(e) None of these

Direction for question 21 to 25: Study the following table carefully and answer the questions given below:

Number of students appeared and qualified in an examination from five states over the years 1993 to 1998

	A		B		C		D		E	
	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.	App.	Qual.
1993	950	230	880	200	1075	315	975	260	1120	325
1994	1090	320	960	250	1125	290	990	305	1208	308
1995	860	240	1100	330	950	275	1400	375	1375	380
1996	1200	350	1250	290	1240	325	1175	310	1060	265
1997	1150	300	1300	325	1220	340	1080	295	995	235
1998	980	280	1080	336	1140	305	1150	280	1075	290
Total	6230	1720	6570	1731	6750	1850	6770	1825	6833	1803

Q 21. Approximately, what is the percentage of students that qualified from the states B and C together in the year 1996?
(a) 32% (b) 20%
(c) 25% (d) 30%
(e) 35%

Q 22. The total number of students that appeared from state 'A' from 1996 to 1998, is approximately what percentage of the total number of students appeared from that state from 1993 to 1995 i.e.,
(a) 98% (b) 105%
(c) 85% (d) 87%
(e) 115%

Q 23. The total number of students the qualified from all the states in 1998 is approximately what percentage of the number of students appeared from all the states in that year i.e.,

(a) 25%
(b) 28%
(c) 32%
(d) 30%
(e) 35%

Q 24. The number of students that qualified from state 'D' from 1993 to 1995 together is approximately what percentage of the students appeared from the state 'D' in these three years i.e.,
(a) 27% (b) 32%
(c) 30% (d) 28%
(e) 25%

Q 25. The percentage of students qualified is the highest for which of the following states for all the six years together?
(a) A (b) B
(c) C (d) D
(e) E

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(a)	(c)	(c)	(d)	(c)	(b)	(d)	(e)	(d)	(c)	(b)	(b)	(e)	(e)	(d)	(b)	(b)	(d)	(e)
Q. No.	21	22	23	24	25															
Answer	(c)	(e)	(b)	(d)	(a)															

HINTS AND SOLUTIONS

1. Supply is 60% of the demand and demand is 80% of the production. Supply = 6000 kgs, Hence demand = 10,000 kgs, Hence capacity = 12,500 kgs

5. Data becomes inconsistent if the supply exceeds the production. This will not be observed for any of the years.

13. Average annual sales of the last three years
= 3350/3 = 1116.67
Average annual sales of the first four years
= 3750/4 = 937.5
Percentage growth = 19.11%

14. Average closing inventory = 766.66
Average opening inventory = 908.33
Difference = 141.66

15. The corresponding figures for the production is – 900, 800, 900, 650, 850, 1350 over the corresponding years. Obviously graph pertaining to these values will not resemble to any of the given graphs.

17. Marks obtained by B in history = 57
Marks obtained by B in Geography = 43.5
Marks obtained by B in Maths = 130.5
Marks obtained by B in Science = 146
Marks obtained by B in English = 65
Marks obtained by B in Hindi = 36
Total marks obtained by B = 478
Maximum marks = 650
Percentage marks obtained = 73.54%

19. Marks obtained by E in history = 66
Marks obtained by E in Geography = 51
Marks obtained by E in Maths = 139.5
Marks obtained by E in Hindi = 39
Total marks obtained by E in the given four subjects
= 295.5
Maximum marks in these four papers = 350
Percentage marks obtained = 84.4 %

[redacted]

21. Total number of students appeared from states B and C together in 1996 = 2490
Total number of students passed from states B and C together in 1996 = 615

22. Total no. of students appeared from state A from 1996 to 1998 = 3330. Total no. of students appeared from state A from 1996 to 1998 = 2900

23. Total number of students qualified from all the states in 1998 = 1491. Total number of students appeared from all the states in 1998 = 5425. Percentage required = 28% (approx.)

24. Total number of students qualified from state D from 1993 to 1995 = 940. Total number of students appeared from state D from 1993 to 1995 = 3395

PRACTICE EXERCISE 7

Direction for questions 1 to 5: Read the passage below and solve the questions based on it.

Following is the points scored by all the teams participated in Benson and Hedges World Cup cricket. Total number of matches played by any team is equal to the total number of wins, total number of matches lost, total tied matches and no result matches. No result matches are those matches which could not be completed due to rains.

Team	Points	Played	Win	Loss	No result	Tied
New Zealand	14	8	7	1	0	0
England	11	8	5	2	1	0
South Africa	10	8	5	3	0	0
Pakistan	9	8	4	3	1	0
Australia	8	8	4	4	0	0
West Indies	8	8	4	4	0	0
India	5	8	2	5	1	0
Sri Lanka	5	8	2	5	1	0
Zimbabwe	2	8	1	7	0	0

Q 1. In how many matches, there was no result?
(a) 2 (b) 4
(c) 0 (4) Cannot be determined

Q 2. Top six teams will automatically qualify for the next world cup cricket to be played in India in 1996. To find out the top teams, following is the criteria to select the top six teams based upon the priorities:
Priority 1 – More Points
Priority 2 – More Win
Priority 3 – Less number of losses
Which of the following teams will not be selected?
(a) Australia (b) Sri Lanka
(c) West Indies (d) None of these

Q 3. What is the number of points scored by a win?
(a) 1 (b) 2
(c) 3 (d) Cannot be determined

Q 4. New Zealand must have won its matches against all the teams except:
(a) England (b) South Africa
(c) Australia (d) Cannot be determined

Q 5. In how many matches, was it possible to find the winner of the match?
(a) 36 (b) 32
(c) 34 (d) None of these

Direction for questions 6 to 10: Read the passage below and solve the questions based on it.

Following is the medals obtained by different countries participating in the 2004 summer Olympics.

There is only one gold medal, one silver medal and one bronze medal associated with any event.

Additional information for questions 6–8: Use the information given below to answer the given questions.

Ranking of the countries is done on the basis of gold medals won by the country. If the tally of the gold medals is same, then number of silver medals won decides the rank. If both the number of gold medals and silver medals won are same, then the ranking is decided on the basis of bronze medal. If all these three medals won by any two or more than two countries are same, then they are awarded the same rank.

Country	Gold	Silver	Bronze	Total
United States	12	9	7	28
Australia	7	5	3	15
Japan	3	1	4	8
Holland	2	3	2	7
Ukraine	2	0	1	3
France	1	2	3	6
Iceland	1	2	0	3
South Africa	1	1	1	3
Zimbabwe	1	1	1	3
China	1	1	0	2
Romania	1	0	1	2
Austria	0	2	0	2
Germany	0	1	4	5
Hungary	0	1	1	2

Country	Gold	Silver	Bronze	Total
Italy	0	1	1	2
Croatia	0	1	0	1
Russia	0	1	0	1

Q 6. Which of the following ranks will be obtained by more than one country?
(a) 7 (b) 8
(3) 10 (4) 11

Q 7. How many countries will be ranked 17?
(a) 0 (b) 1
(c) 2 (d) Cannot be determined

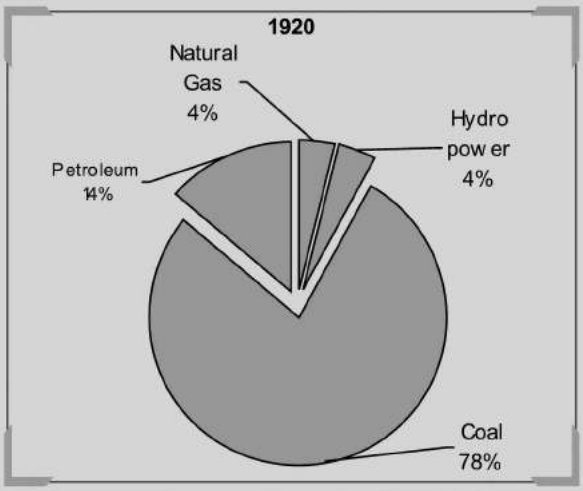
Q 8. At how many ranks, more than one country will stand at?
(a) 0 (b) 1
(c) 2 (d) None of these

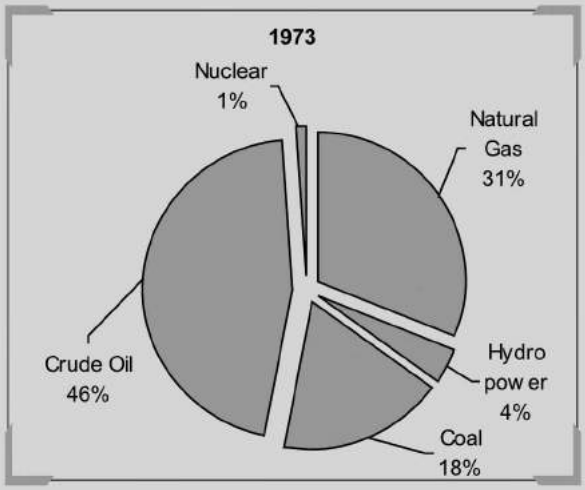
Q 9. Ranking of the countries is done on the basis of total medals won by the country.
Which country will be ranked 5 according to this scheme of ranking?
(a) Japan
(b) Ukraine
(c) France
(d) None of these

Q 10. In the above question, what will be the rank of Germany?
(a) 6
(b) 7
(c) 8
(d) None of these

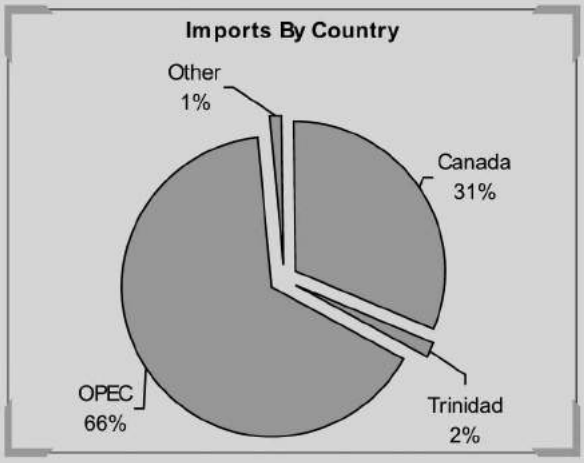
Direction for questions 11 to 14: Go through the charts given below and solve the questions based on it.

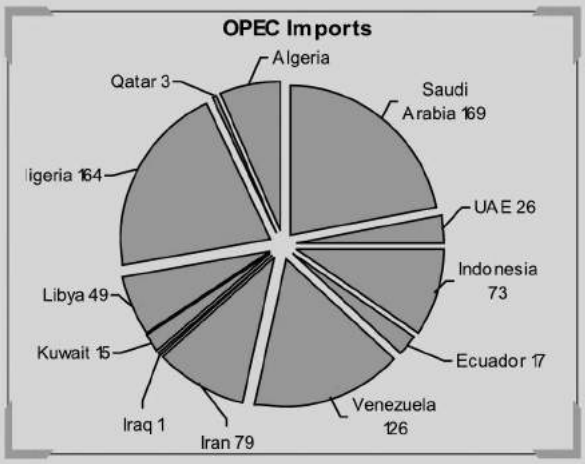
Sources of US Energy Supply: 1920 and 1973





1973 Crude Oil Imports to US (millions of barrels)





Q 11. What percentage of US energy supply in 1973 is fulfilled through the crude oil imports from OPEC?
(a) 46% (b) 65.5%
(c) 30.13% (d) cannot be determined

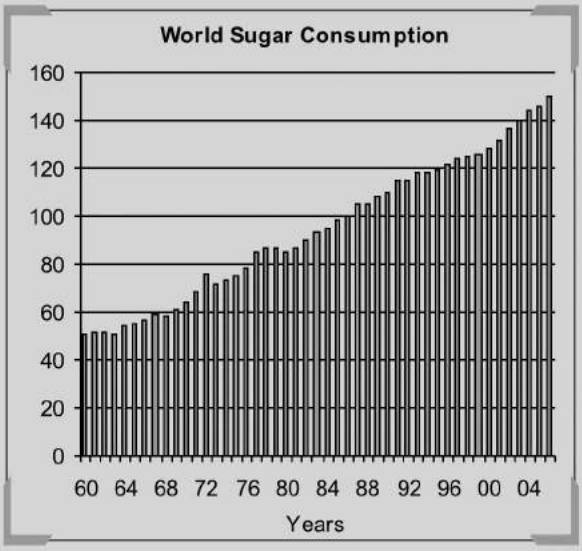
Q 12. Which of the following countries has a bigger share in crude oil imports to US in 1973 than the share of Canada same year?

(a) Saudi Arabia (b) Venezuela
(c) None (c) Cannot be determined

Q 13. Saudi Arabia supplies what percentage of total crude oil imports in 1973 to US?
(a) 28.7% (b) 34.4%
(c) 20.6% (d) Cannot be determined

Q 14. Which of the following can definitely be concluded after looking at the US energy supply in 1920 and 1973?
(a) No new source to generate hydropower has been identified.
(b) Total usage of coal has decreased over the period.
(c) At least one new source to generate nuclear power has been identified.
(d) None of these

Direction for questions 15 to 18: Go through the chart below and solve the questions based on it.



Q 15. For how many years during the given period consumption has increased over the previous year?
(a) 24
(b) 25
(c) 26
(d) None of these

Q 16. How many years the world sugar consumption has been more than 125 million tonnes per year?
(a) 7
(b) 8
(c) 9
(d) None of these

Q 17. Average sugar consumption per year is to be calculated for the given years. This value is most likely to be approximately equal the value of a year in between
(a) 75-80
(b) 80-90
(c) 90-95
(d) None of these

Q 18. For how many years the world sugar consumption has been more than 75 million tonnes and less than 100 tons per year?
(a) 9
(b) 10
(c) 11
(d) 12

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Answer	(a)	(b)	(b)	(d)	(c)	(b)	(a)	(c)	(c)	(a)	(d)	(c)	(d)	(c)	(d)	(d)	(b)	(c)

HINTS AND SOLUTIONS

Following is the medal table:

Rank	Nation	Gold	Silver	Bronze	Total
1.	United States	12	9	7	28
2.	Australia	7	5	3	15
3.	Japan	3	1	4	8
4.	Holland	2	3	2	7
5.	Ukraine	2	0	1	3
6.	France	1	2	3	6
7.	Iceland	1	2	0	3
8.	South Africa	1	1	1	3
9.	Zimbabwe	1	1	1	3
10.	China	1	1	0	2
11.	Romania	1	0	1	2
12.	Austria	0	2	0	2
13.	Germany	0	1	4	5
14.	Hungary	0	1	1	2
15.	Italy	0	1	1	2
16.	Croatia	0	1	0	1
17.	Russia	0	1	0	1

11. It is not given that what percentage of US energy supply is imported and what percentage is indigenous. Hence, cannot be determined.

12. Despite the value of Algeria is not given, it can be calculated. For the value to be more than that of Canada's value, it should be more than 31.3%, i.e., approximately a bit less than half of the OPEC countries share. Obviously it is not the case. Hence no such country is there.

13. Since the value for Algeria is not given, we do not have the data for total OPEC. Hence, cannot be determined.

14. Statement 1 and statement 2 need not be true always because it is percentage share of total.
Statement 3 has to be true because nuclear power was not present in 1920, and it is present in 1973.

15. After looking at the options, one is not expected to find the actual value in this question. And it can be seen very easily that the years have to be more than 26 years out of the given 60-06.

17. It is difficult to calculate the actual average without having the values of individual years. But looking at the options, it has to be between one of the years in the block of 80-90, as the values are constantly increasing and the mid-value will be obtained in between 80-90.

PRACTICE EXERCISE 8

Direction for questions 1 to 4: Go through the table given below and solve the questions based on it.

Table given below presents the different wars fought by US with total cost in dollars and cost per person per year (CPP). Cost per person per year is calculated by dividing total cost by average US population per annum during that period.

Today's Price of US wars

	Total Cost (Billions in $)	CPP	
Revolutionary War (1775-83)	2.4	481	Period 1
War of 1812 (1812-15)	1.2	129	Period 2
Mexican War (1846-48)	1.7	73	Period 3
Civil War (1861-65)	6.7	1,814	Period 4
Spanish-Americal war (1998)	10.3	118	Period 5
World War I (1917-18)	205.1	2,678	Period 6
World War II (1941-45)	3,116.4	21,937	Period 7
Korean War (1950-53)	361.4	2,438	Period 8
Vietnam War (1964-72)	531.9	2,372	Period 9
Gulf War (1990-91)	81.9	329	Period 10
Irag/Afganistan War (2001-)	549.0	1,829	

Q 1. Which period saw the maximum percentage increase in the population of US?
(a) Period 3 (b) Period 5
(c) Period 6 (d) Cannot be determined

Q 2. Price of dollar remains constant during this whole period. Assuming that any war started on 1st January of the given year and ended on 31st December of the given year, then which war resulted into maximum US cost per annum?
(a) World war 1
(b) World war 2
(c) Iraq/Afghanistan
(d) Cannot be determined

Q 3. The price of dollar remains constant during this whole period. Which period of war corresponds to the maximum US population per annum?
(a) Revolutionary war
(b) World war 1
(c) World war 2
(d) None of these

Q 4. Price of dollar remains constant during this whole period. Which period of war corresponds to the minimum US population per annum?
(a) Revolutionary war
(b) World war 1
(c) World war 2
(d) None of these

Direction for questions 5 to 9: Go through the table given below and solve the questions based on it.

Year		Export	% Growth over pervious year
2000-2001	Quantity (In Tonnes)		28.41
	Value (In Rs Crore)		25.94
	Value (In Millions US$)		19.11
2001-2002	Quantity (In Tonnes)	424470	(–) 3.63
	Value (In Rs Crore)	5957.05	(–) 7.56
	Value (In Millions US$)	1253.35	(–) 11.51

Year		Export	% Growth over pervious year
2002-2003	Quantity (In Tonnes)		10.09
	Value (In Rs Crore)		15.52
	Value (In Millions US$)		13.69
2003-2004	Quantity (In Tonnes)	267546	
	Value (In Rs Crore)	3911.00	
	Value (In Millions US$)	847.00	

Q 5. What was the approximate value of exports (in million US $) in 2000-2001?
(a) 1360
(b) 1415
(c) 1520
(d) 1484

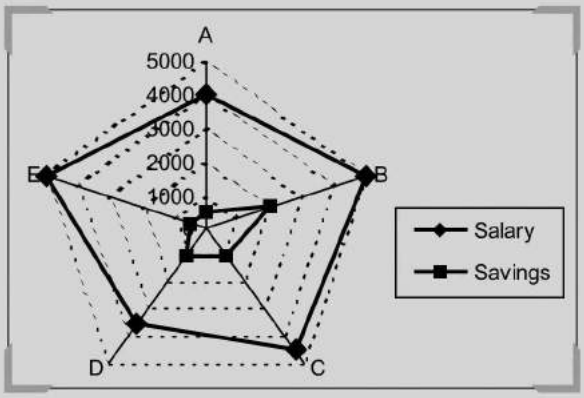
Q 6. What is difference between the quantity (in tonnes) exported in 2000-2001 and 2003-2004?
(a) 133000
(b) 153000
(c) 173000
(d) 193000

Q 7. For which of the following years, the value (in Rs) per unit weight is the lowest?
(a) 2000-2001
(b) 2001-2002
(c) 2002-2003
(d) 2003-2004

Q 8. For which of the following years, the price of dollar is maximum?
(a) 2000-2001
(b) 2001-2002
(c) 2002-2003
(d) 2003-2004

Q 9. What is the percentage decrease in the export value (in million US $) in 2003-04 over than that of 2000-2001?
(a) 20%
(b) 30%
(c) 40%
(d) 50%

Direction for questions 10 to 14: Go through the radar diagram given below and solve the questions based on it.



The above given radar diagram gives the salary and savings of five persons A, B, C, D and E for the month of July'06. All the values are in the multiples of Rs 500.
Salary – Savings = Expenditure

Q 10. Whose expenditure is the maximum for the month of July'06?
(a) B (b) C
(c) D (d) E

Q 11. Whose expenditure is minimum for the month of July'06?
(a) B (b) C
(c) D (d) E

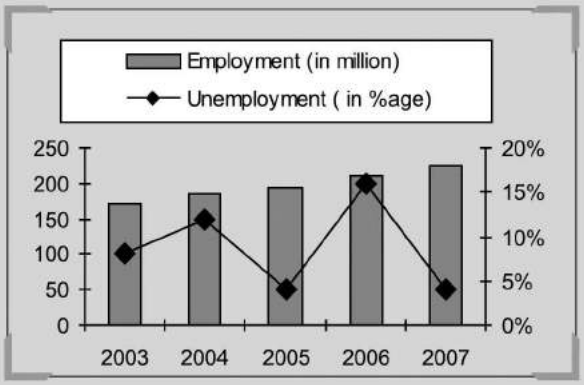
Q 12. Salary of B increases by 20% in the month for the month of August 06 than the month of July 06. However expenditure of B increases only by 10% in the month of August 06 than the month of July 06. By what percentage the savings of B has increased in the month of August 06 than the month of July 06?
(a) 33.33% (b) 23.66%
(c) 26.66% (d) None of these

Q 13. For who among the following, expenditure expressed as a percentage of salary for the given period the maximum?
(a) B (b) C
(c) D (d) E

Q 14. For how many of the given persons, the savings is more than 20% of the salary for the given period?
(a) 1.2 (b) 2.3
(c) 4 (d) 5

Direction for questions 15 to 20: Go through the chart given below and solve the questions based on it.

Following chart presents the employment scenario in India in different years:



Employment is in the multiples of 5 million and unemployment percentage is in the multiples of 4%. People can be either employed or unemployed.

Q 15. For which of the given years the number of persons unemployed is maximum?
(a) 2003 (b) 2005
(c) 2006 (d) 2007

Q 16. What is the total population of India in 2005 (in million)?
(a) 190 (b) 203
(c) 215 (d) 223

Q 17. For which year, the population of India is maximum?
(a) 2003 (b) 2005
(c) 2006 (d) 2007

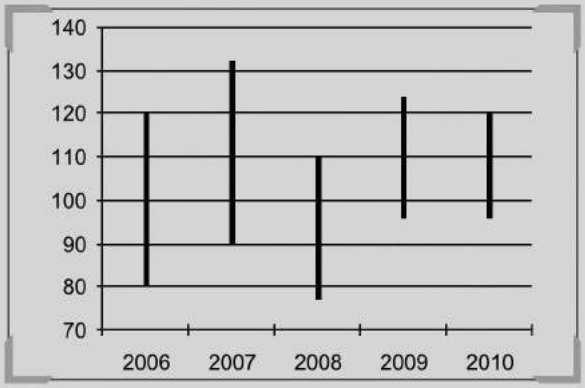
Q 18. If the percentage change in the population of India in 2008 is same as that of 2007 and the rate of unemployment in 2008 is 10%, then how many people in 2008 are employed in India (in million)?
(a) 185 (b) 191
(c) 197 (d) 207

Q 19. In the above question, total number of persons employed in the government sector to the non-government sector is in the ratio 2 : 3. What is the number of persons employed in government sector in 2007?
(a) 93 mn (b) 139 mn
(c) 162 mn (d) Cannot be determined

Q 20. How many of the following statement/s is/are correct?
I. The population of India is constantly on a rise during the given period every year.
II. The number of unemployed persons is constantly on a rise during the given period every year.
III. The total number of unemployed persons is decreasing every year by atleast 10% over the unemployed persons in the previous year.
(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 21 to 23: Go through the chart given below and solve the questions based on it.

In the chart given below, bottom of the bar represents the productivity of the company and the top of the bar presents the production (in million units) of the company for that particular year. Productivity is production as a percentage of production capacity.

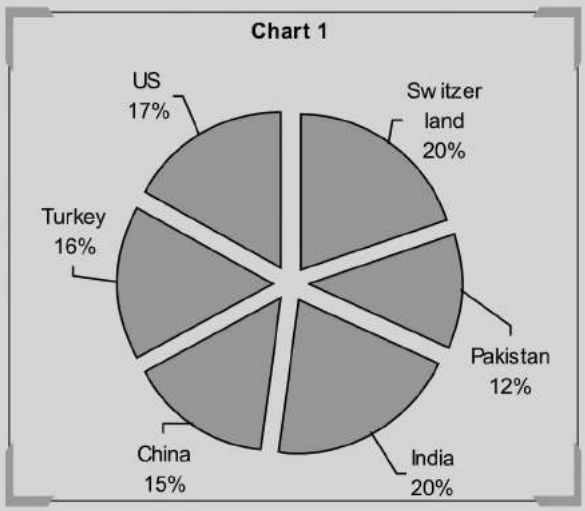


Q 21. For which year is the production capacity of the company the maximum?
(a) 2006
(b) 2007
(c) 2008
(d) None of these

Q 22. For which year is the production capacity of the company the minimum?
(a) 2007 (b) 2008
(c) 2009 (d) 2010

Q 23. Which of the following statement is true?
(a) The production capacity of the company is constantly decreasing year by year.
(b) The purnover of the company is constantly on a rise.
(c) Production capacity of the company is decreasing first and then increasing.
(d) None of these

Direction for questions 24 to 25: Go through the chart given below and solve the questions based on it.



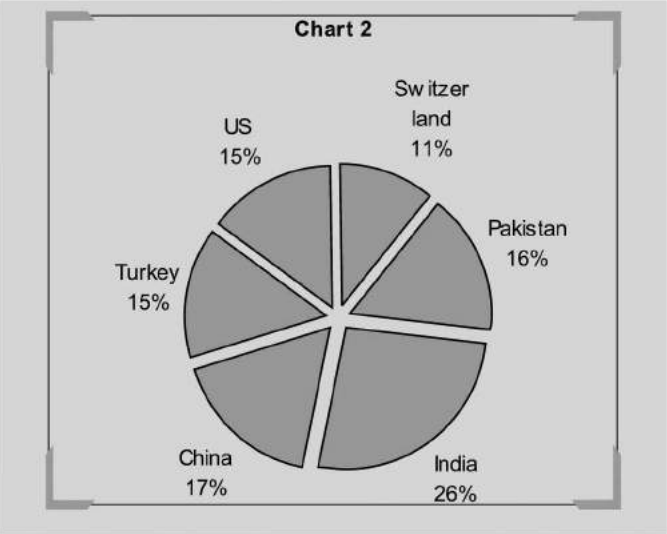


Chart 1 shows the distribution by value of the top 6 suppliers of MFA textiles in 1995. Chart 2 show the distribution by quantity of top 6 suppliers of MFA textiles in 1995. The total value is Rs 5760 million. The total is 1.055 million tonnes.

Q 24. The country which has the lowest average price is
(a) US
(b) India
(c) Pakistan
(d) None of these

Q 25. The average price in Rs per kilogram for Turkey is approximately
(a) 3.20
(b) 4.60
(c) 5.60
(d) 7.20

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(d)	(a)	(b)	(c)	(a)	(c)	(c)	(d)	(c)	(c)	(d)	(b)	(c)	(b)	(c)	(c)	(d)	(a)
Q. No.	21	22	23	24	25															
Answer	(a)	(d)	(a)	(c)	(c)															

HINTS AND SOLUTIONS

1. Since it is not given that price of dollar changes at what rate, so we cannot determine the population of US during different periods. Hence, cannot be determined.
2. It can be simply calculated by dividing total cost by number of years involved in war. Value for world war 2 is more than even Iraq/Afghanistan war(assuming it lasts only for one year).
3. To solve this question, we are supposed to divide total cost by CPP. It is maximum for Iraq/Afghanistan.
4. Using the process given in Q 3. Minimum is obtained for Revolutionary war.
   This is the table having all the data:

Year		Export	% Growth over pervious year
2000-2001	Quantity (In Tonnes)	440473	28.41
	Value (In Rs Crore)	443.89	25.94
	Value (In Millions US$)	1416.32	19.11

Year		Export	% Growth over pervious year
2001-2002	Quantity (In Tonnes)	424470	(–) 3.63
	Value (In Rs Crore)	5957.05	(–) 7.56
	Value (In Millions US$)	1253.35	(–) 11.51
2002-2003	Quantity (In Tonnes)	467297	10.09
	Value (In Rs Crore)	6881.31	15.52
	Value (In Millions US$)	1424.90	13.69
2003-2004	Quantity (In Tonnes)	267546	
	Value (In Rs Crore)	3911.00	
	Value (In Millions US$)	847.00	

8. We will divide the value (in Rs crores) by value (in million US $) of the corresponding year. It comes maximum for 2002—2003.

For

2001 – 2002 = Rs 47.5

For

2002 – 2003 = Rs 48.3

For

2003 – 2004 = Rs 46.1

9. Using the above table, it can be calculated that

$$\text{percentage decrease} = \frac{1416-847}{1416}\times 100 = 40\%$$

[redacted]

This is the table pertaining to the data given above:

	A	B	C	D	E
Salary	4000	5000	4500	3500	5000
Savings	500	2000	1000	1000	500
Expenditure	3500	3000	3500	2500	4500

Now all the questions can be answered.

[redacted]

This is the table pertaining to the data given.

	2003	2004	2005	2006	2007
Employment (in million)	170	185	195	210	225
Unemployment (in %age)	8%	12%	4%	16%	4%
Population (in million)	184	210	203	250	234

Now all the questions can be answered.

18. The percentage change in the population in 2007 over 2006 = 6.4%. Hence, the population in 2008 = 219 mn

10% unemployment means 90% people are employed.

Hence, number of persons employed = 0.9 × 219 = 197 mn

19. The data given in Q,18 is about 2008, whereas Q, 19 is about 2007. Hence, cannot be determined.

20. None of the statements is correct.

[redacted]

	2006	2007	2008	2009	2010
Production (In million units)	120	132	110	124	120
Productivity	80	90	77	96	96
Production Capacity (In million units)	150	146.66	142.85	129	125

PRACTICE EXERCISE 9

Direction for questions 1 to 5: Go through the data set given below and solve the questions based on it.

S. No.	Station Name	Arrival Time	Dep. Time	Distnace	Day
1.	New Delhi	Source	16:25	0	1
2.	Aligarh	19:05	19:10	132	1
3.	Tundla	20:15	20:25	210	1
4.	Etawah	21:28	21:30	302	1
5.	Kanpur Central	23:15	23:25	440	1
6.	Allahabad	02:00	02:10	634	2
7.	Mughal Sarai	04:25	04:40	787	2
8.	Buxar	05:52	05:54	881	2
9.	Ara	06:40	06:42	949	2
10.	Danapur	07:14	07:16	989	2
11.	Patna	07:45	07:55	999	2
12.	Bakhtiyarpur	08:33	08:35	1044	2
13.	Mokameh	09:18	09:20	1088	2
14.	Kiul	09:55	09:57	1122	2
15.	Jamui	10:18	10:20	1150	2
16.	Jhajha	10:57	11:02	1176	2
17.	Jaisidih	11:38		1220	2

Following is the time table of Poorva express which leaves New delhi station at 4.25 am. and reaches Jasidih station on next day at 11.38 pm. Below given table gives the arrival time and the departure time of Poorva express at different stations with the distance of that station from the source.

Q 1. According to the time-table given above, what is the maximum time for which Poorva express runs non-stop?
(a) 155 minutes (b) 160 minutes
(c) 135 minutes (d) None of these

Q 2. For how many of the stations from the list given above, the stoppage time is more than 5 minutes (stoppage time is the difference between the departure time and the arrival time)?
(a) 4 (b) 5
(c) 6 (d) None of these

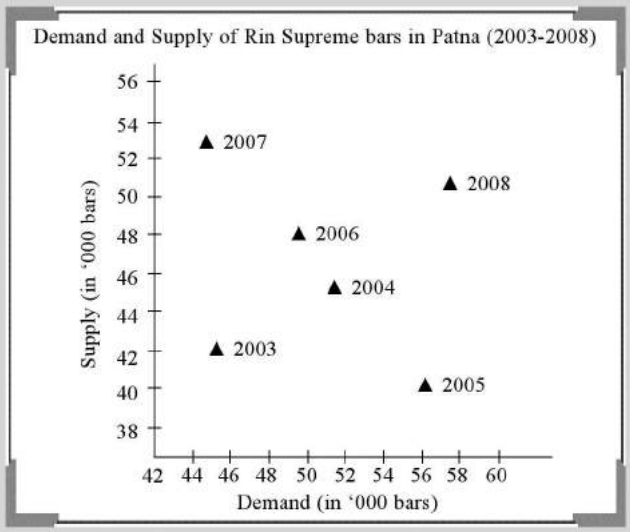
Q 3. In between which stretch the train obtains its maximum speed?
(a) Ara-Danapur (b) Buxar-Ara
(c) Kiul-Jamui (d) None of these

Q 4. If the speed of the train is increased by 20% in all the stretches, then at what time (approximate) the train will reach at Jasidih station keeping all the other things constant?
(a) 8.17 am (b) 8.39 am
(c) 9.12 am (d) 9.29 am

Q 5. What is the approximate average speed (kmph) of the train during the whole journey?
(a) 60 (b) 63
(c) 66 (d) 69

Direction for questions 6 to 10: Go through the data set given below and solve the questions based on it.

The following graph captures the demand and supply situation of Rin supreme bars in a town Patna over the years 2003-2008.



Q 6. For who many years, the supply was not less than the demand?
(a) 0 (b) 1
(c) 3 (d) 7

Q 7. If the price of one bar of Rin supreme is directly proportional to the ratio of demand to supply, in which year, in the time period 2003-2008, was the price of a bar the highest?
(a) 2007 (b) 2005
(c) 2006 (d) 2008

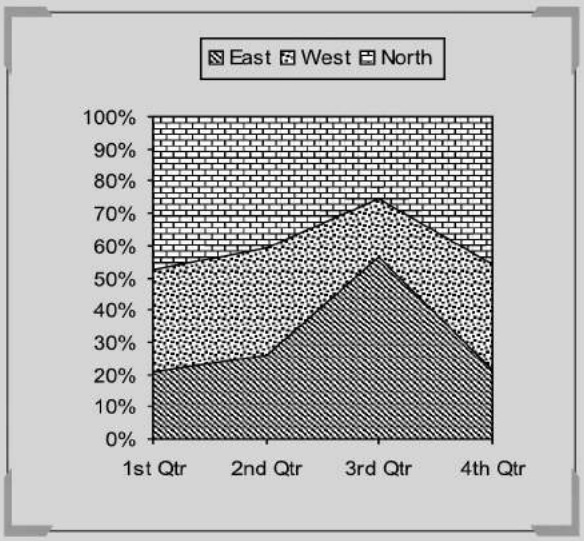
Q 8. Over the time period of 2003-2008, in how many years did both demand and supply show a similar trend (i.e., either both increased or both decreased) over the previous year?
(a) 1 (b) 2
(c) 3 (d) 4

Q 9. Patna had an inventory of 20,000 bars at the start of 2003. In which year was the deficit of bars the highest? [Deficit is defined as demand minus available quantity and available quantity is supply of year plus the inventory. In a year where availability is less than the demand, the demand goes unsatisfied in that year and is not carried forward. Excess availability over demand goes into inventory]
(a) 2008 (b) 2007
(c) 2006 (d) 2005

Q 10. The demand and supply given above is to be re-drawn using the line chart with the year along the X axis and the demand and supply along the Y axis (same scale being used for both demand and supply). At how many distinct points would the two lines cut?
(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 11 to 15: Go through the data set given below and solve the questions based on it.

Following is the area diagram given for the sales value of ABC Limited for the four quarters of the year 2007. ABC Limited has its operations in three zones—North, West and East. Below given data set represents the stacked contribution of these three zones in every quarter of the year 2007.



Q 11. If we rank the contribution of zones in each of the quarters, then how many zones will have the same rank across all the quarters?
(a) 0 (b) 1
(c) 2 (d) 3

Q 12. Which zone represents maximum sales value in the year 2007?
(a) North (b) West
(c) East (d) Cannot be determined

Q 13. If we convert the above given data set into a line chart (not stacked), then total how many times the lines will intersect?
(a) 0 (b) 2
(c) 4 (d) None of these

Q 14. There is an increase of 10% every quarter over the last quarter for the given period. Which of the following represents the maximum sales value?
(a) North zone Qtr 1
(b) East zone Qtr 3
(c) West zone Qtr 2
(d) North zone Qtr 4

Q 15. Data set given above is arranged in bars (in percentage) in the following way—First North zone, then West zone, then East zone of the 1st quarter and in the similar order for the 2nd, 3rd and 4th quarter. How many times there will be an increase in the height of bar with respect to the height of the previous bar?
(a) 1 (b) 2
(c) 3 (d) 4

Direction for questions 16 to 18: Go through the data set given below and solve the questions based on it.

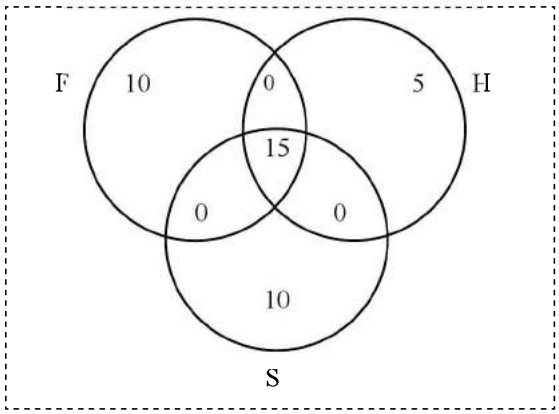
In a B-School, the students have to specialize in at least one of the areas—Finance, Marketing or Systems. The table given below represents the distribution of students in different specializations according to their subjects chosen. There were a total of 80 girls and 120 boys from arts background and 40 girls and 70 boys from science background comprising the entire batch.

	Arts		Science	
	Girls	Boys	Girls	Boys
Finance	25	45	25	65
Marketing	23	50	20	15
System	32	100	25	15

Q 16. What is the percentage of girls from arts specializing in all three areas?
(a) 16.5% (b) 16%
(c) 100% (d) 0%

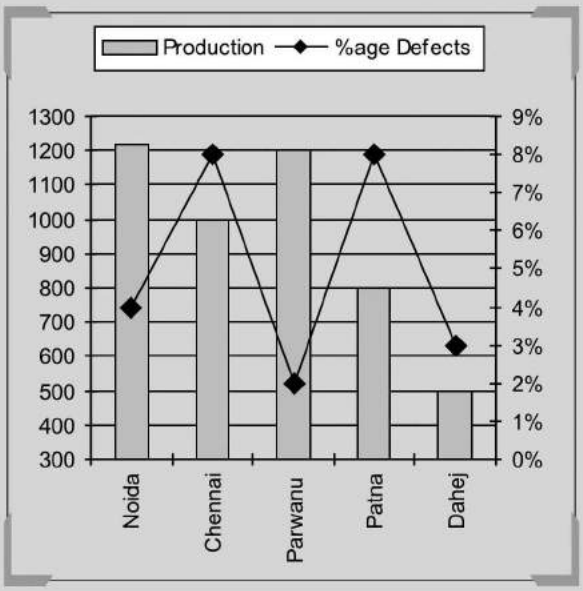
Q 17. If 60% of boys and 70% of girls are successful in at least one of the specializations taken by them, what is the percentage of students who have not passed in any subject?
(a) 33.9% (b) 36.1%
(c) 38.4% (d) Cannot be determined

Q 18. Find the maximum number of girls from science who could have specialized in all 3 subjects.
(a) 23 (b) 20
(c) 15 (d) Cannot be determined



Direction for questions 19 to 25: Go through the data set given below and solve the questions based on it.

Following bar chart represents the number of units produced (in millions) and percentage defective units produced out of the total units produced by the individual plants of an electronics company. These plants – Noida, Chennai, Parwanu, Patna and Dahej are the only plants of this company.
All the defective units are rejected and total units supplied from any of the plants
= Total units produced – total defective units.



All %age defects are in integers

Q 19. Which of the following plants can supply maximum units?
(a) Noida
(b) Chennai
(c) Parwanu
(d) Dahej

Q 20. Which of the following plants can supply minimum units?
(a) Noida
(b) Chennai
(c) Patna
(d) Dahej

Q 21. It has been observed that if the employees working at two plants are frequently given an opportunity to interact, then the combined percentage defect at these two plants becomes the harmonic mean of the individual percentage defect of these two plants. What is the total number of units (in lacs) that can be supplied from taking Chennai and Patna together?
(a) 1656
(b) 16560
(c) 1728
(d) 17280

Q 22. In the above question, which of the following two plants should be merged so that the number of units supplied from these two units combined is maximum?
(a) Noida + Chennai
(b) Parwanu + Patna
(c) Noida + Parwanu
(d) None of these

Q 23. The average of total number of units supplied by all the plants taken together will be closest to the number of units supplied by
(a) Noida
(b) Chennai
(c) Parwanu
(d) None of these

Q 24. What is the approximate total number of units (in millions) supplied by all the plants taken together?
(a) 4500
(b) 5000
(c) 4000
(d) Cannot be determined

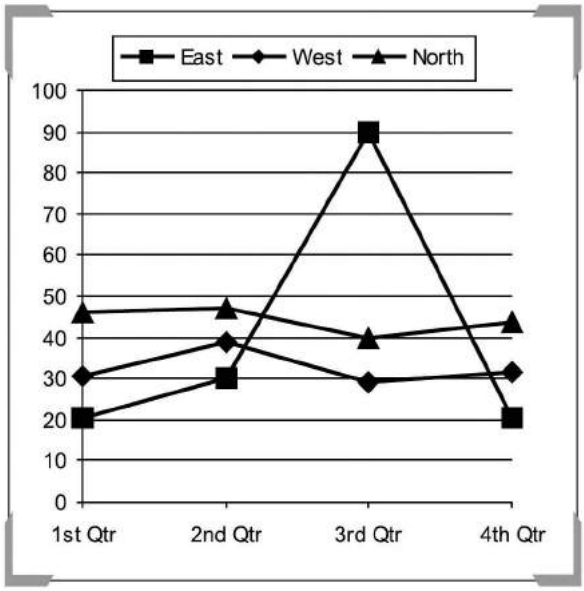
Q 25. The ratio of production of Parwanu plant and Patna plant is same as the ratio of production of Chennai plant and …… Plant.
(a) Noida
(b) Dahej
(c) Either Noida or Dahej
(d) None of these

ANSWERS

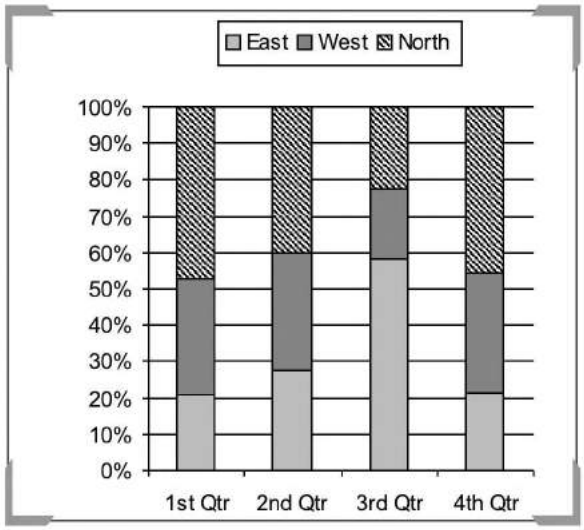
Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(d)	(b)	(b)	(b)	(b)	(a)	(d)	(c)	(a)	(d)	(c)	(b)	(b)	(d)	(b)	(c)	(c)	(d)
Q. No.	21	22	23	24	25															
Answer	(b)	(c)	(b)	(a)	(b)															

HINTS AND SOLUTIONS

1. This is obtained between New Delhi – Aligarh.
2. Visual inspection.
4. If speed of the train is increase by 20%, then the time taken (in which train is moving) will reduce by 16.66%. However there will not be any impact on the stoppage of the train (= 81 minutes).
   In new case, total time taken by train = 974 minutes.
   Hence 8:39 a.m.
5. Total time taken = 19: 13 mins and
   total distance = 1120 km
   Hence, average speed = 1120/19:13 = 63.3 kmph
6. Only 2007 is the year when the given condition in the question is fulfilled.
7. By visual inspection, it can be seen that the highest ratio of, it is for year 2005.
8. Only once in year 2004.
9. Highest deficit is in 2005 i.e., 6000 bars.
   In 2006, it is 2000 bars.
   In 2007, there is an inventory of 7000 bars.
   In 2008, the deficit is 1000 bars.
10. Between 2006 and 2007 and 2007 and 2008.
11. Visual inspection.
12. Since the value for each of the quarters is not given, hence we cannot determine it.
13. It will appear like this:
    We can see now that there are four points of intersection.



14. Start by assuming the value in 1st qtr = 100 and then for the subsequent quarters, the value will increase by 10% over the previous quarter.
15. Following bar chart gives the values of the quarters and zones. Now we will arrange the individual bars to see that the increase is appearing twice—2nd quarter North zone and 3rd quarter East zone.



16. Total number of girls in finance, Marketing and systems = 80.

Therefore, there are 0% girls who specialize in all three of areas.

17. 40% of boys and 30% of girls are not successful.

Hence, 40% of 190+30% of 120 = 76+36 = 112

Hence, percentage of students who are not successful = 36%

18. The maximum number is 15.

Following is the data-sheet pertaining to the data set given above:

	Noida	Chennai	Parwanu	Patna	Dahej
Production (in mn units)	1220	1000	1200	800	500
%age Defects	4%	8%	2%	8%	3%
Supplied Units (in mn units)	1171.2	920	176	736	485

21. Use the data sheet given above. Values are required in lacs.

22. Noida and Parwanu are the two highest producing plants with the percentage defects being in the lowest three. Hence, this combination can supply the maximum units.

23. Chennai is the median of all the plants and closely looking upon the values, value for Chennai is closest to the average of all the plants.

24. Use the data-sheet.

25. Visual inspection.

4

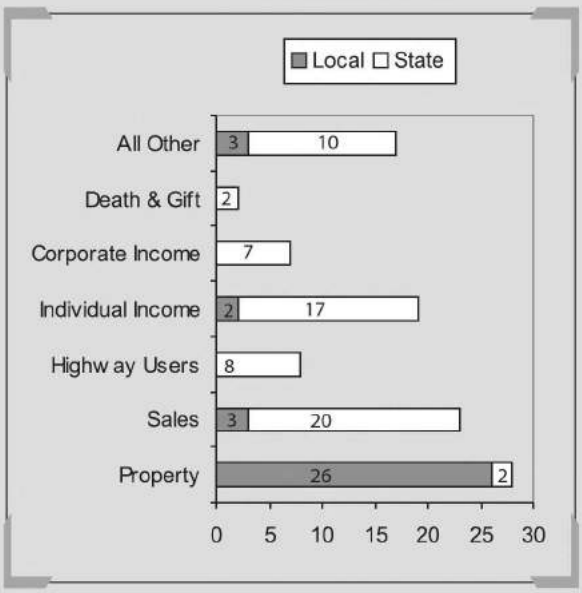
CHAPTER

Moderate Exercises

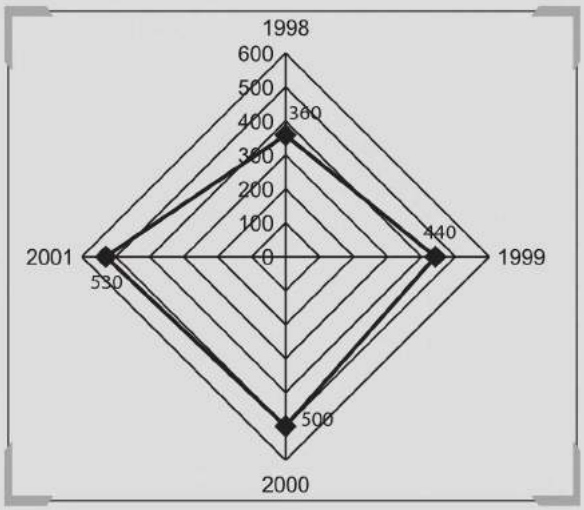
PRACTICE EXERCISE 1

Direction for questions 1 to 5: Refer to the charts below answer the questions that follow.

Chart given below presents the sources of local and state tax revenues for the year 2001 as a percentage of the total tax revenues.



The diagram shown below represents the combined state and local tax revenue (in $ mn).



Q 1. Find the difference between the tax revenue generated from individual income at local and state levels in 2001.

(a) $ 72 mn
(b) $ 78.75 mn
(c) $ 86.25 mn
(d) $ 97.25 mn

Q 2. What is the total tax revenue generated by the state in 1999?
(a) $ 149.6 mn (b) $ 210.3 mn
(c) $ 290.4 mn (d) Cannot be determined

Q 3. If revenue distribution in 2000 was same as in 2001 and tax from highway users was collected for the first time, then what was the percentage increase in tax revenue from source other than highway users in 2000 over the previous year?
(a) 4.5% (b) 7.3%
(c) 9.09% (d) 14.28%

Q 4. Find the compounded annual rate of growth during 1998-2001.
(a) 10.9% (b) 12.3%
(c) 13.4% (d) 14.7%

Q 5. What percentage of the total local tax revenue is generated from property tax in 2001?
(a) 52.4% (b) 76.4%
(c) 86.1% (d) 90.9%

Direction for questions 6 to 10: Refer to the table below and answer the questions that follow.

Length of service (years)	Knowing Shorthand	Knowing Typing	Knowing Both	Total staff
0 – 5	40/35	35/40	35/30	80/120
6 –10	60/15	60/20	40/10	150/75
More than 10	40/40	40/30	15/20	90/110

The chart below represents the statistics pertaining to the employees of Due North Inc.
Following is the representation:
40/35 – Graduate/Non-graduate

Q 6. What percentage of the employees with 10 years or less service know only one of shorthand or typing?
(a) 32 (b) 27
(c) 22 (d) 18

Q 7. What percentage of employees know shorthand only?
(a) 33 (b) 28
(c) 23 (d) 18

Q 8. How many graduate employees know neither shorthand nor typing?
(a) 115 (b) 120
(c) 135 (d) 130

Q 9. Out of those employees knowing shorthand only, how many are graduates?
(a) 63 (b) 58
(c) 53 (d) 48

Q 10. If 10% of those employees knowing shorthand only learn typing and 12% of those knowing typing only learn shorthand, by what % would the number of persons knowing both increases:
(a) 11 (b) 16
(c) 21 (d) 26.

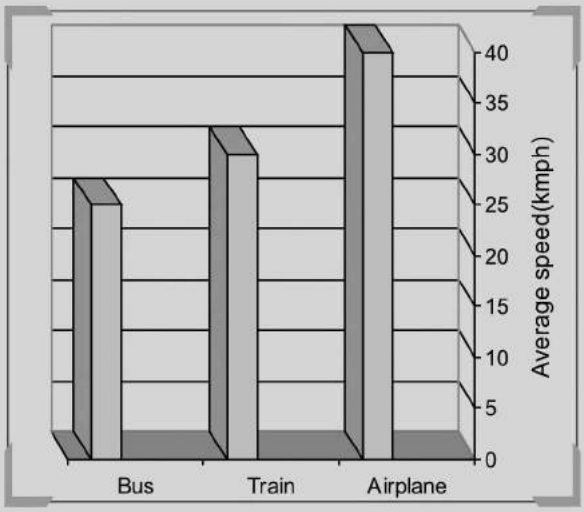
Direction for questions 11 to 14: Refer to the table given below and answer the questions based on it.

Table 1 – Mode of travel between two cities

	Patna	Mumbai	Calcutta	Chennai	Bangalore
Patna	-	Bus	Train	Air plane	Train
Mumbai	Bus	-	Train	Train	Air plane
Calcutta	Train	Train	-	Air plane	Bus
Chennai	Air plane	Train	Air plane	-	Bus
Banga-lore	Train	Air plane	Bus	Bus	-

Table 2 – Distance between two cities (in km)

	Patna	Mumbai	Calcutta	Chennai	Bangalore
Patna	-	100	150	150	300
Mumbai	100	-	200	300	500
Calcutta	150	200	-	500	300
Chennai	150	300	500	-	100
Banga-lore	300	500	300	100	-



All the values in above bar-chart are the multiples of 5.

Q 11. Which city can be reached from Patna in the least time?
(a) Mumbai (b) Calcutta
(c) Chennai (d) Bangalore

Q 12. For which of the following options, travel time is the least?
(a) Mumbai-Chennai
(b) Patna-Bangalore
(c) Bangalore-Chennai
(d) Patna-Bangalore

Q 13. If one can travel by bus or train only, then which two cities will have the least travel time?
(a) Patna-Mumbai
(b) Bangalore-Calcutta
(c) Bangalore-Mumbai
(d) None of these

Q 14. What is the maximum distance one can travel (approx. km) between Patna and Chennai in a day if the Stoppage time between any two consecutive trips is 2 h?
(a) 625 (b) 700
(c) 725 (d) 640

Direction for questions 15 to 19: Go through the table given below and solve the questions based on it.

Following is the final results of standard X. The subjects students tested in are—English, Hindi, Science, Maths, Social science, Drawing, Craft, PT, scout-guide and social service. To have a better presentation of the mark sheet, the scores in first four subjects are consolidated into A1, the next three into A2 and last three into A3. The final score is the total of these three scores A1, A2 and A3. On the basis of the final score students get their ranks. The table below gives the scores of top eighteen students of the class.

Name	Grade	Final Score	A1	A2	A3	Drawing	PT	Social service
Anana	A	725	282	217	226	80	81	69
Pankaj	B	864	310	87	267	85	86	92
Vamshi	C	812	300	290	222	70	91	74
Ram	C	805	270	262	273	65	79	97
Shann	D	900	391	292	217	96	86	81
Saket	D	876	301	297	278	65	87	93
Mini	C	825	296	296	233	50	89	91
Rashi Singh	E	905	320	290	295	72	99	99
Parul	B	895	310	290	295	72	99	98
Rakesh Rathi	B	870	276	296	298	63	99	99
Gauri	C	888	361	270	257	82	78	92
Kush Verma	A	-	315	-	296	62	99	99
Pragya	A	897	812	299	286	69	97	92
Shipra Gupta	D	907	319	289	299	71	100	100
Sachin	E	910	387	290	233	86	69	79
Sarita	A	901	389	270	242	87	79	86
Mahavir Sherawat	C	897	388	297	212	83	69	89
Ritu Sehgal	D	898	299	300	299	73	100	99

Q 15. How many students have got grade A and more than 250 marks in all the group scores A1, A2 and A3?
(a) 0 (b) 1
(c) 2 (d) None of these

Q 16. What is the difference between total marks in drawing of grade A students and grade C students?
(a) 50 (b) 52
(c) 48 (d) None of these

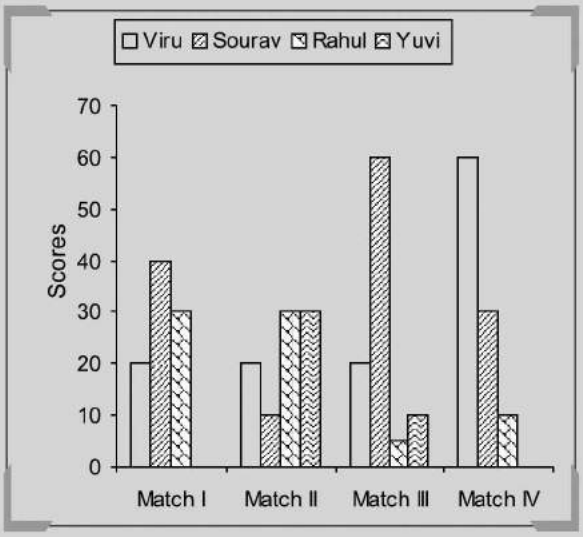
Q 17. What is the least that Kush Verma must get in A 2 that would ensure him 5th rank?
(a) 289 (b) 288
(c) 290 (d) Cannot be determined

Q 18. Who has scored maximum marks in Scout-guide provide the maximum marks in any subject = 100?
(a) Rakesh Rathi (b) Ritu Sehgal
(c) Parul (d) Rakesh Rathi and Ritu

Q 19. What is the difference between the sum of final scores of top five scorers and sum of final scores of bottom five scorers?
(a) 214 (b) 208
(c) 174 (d) None of these

Direction for questions 20 to 24: Go through the diagram below and answer the questions based on it.

Following diagram shows the percentage share of runs scored by four Indian Batsmen in total runs made by India in 4 matches:



Total runs scored by India in any match is in between 250 and 300.

Q 20. In which match is the share of runs scored by the four batsmen out of the total runs scored by India the maximum?
(a) Match I
(b) Match II
(c) Match III
(d) Match IV

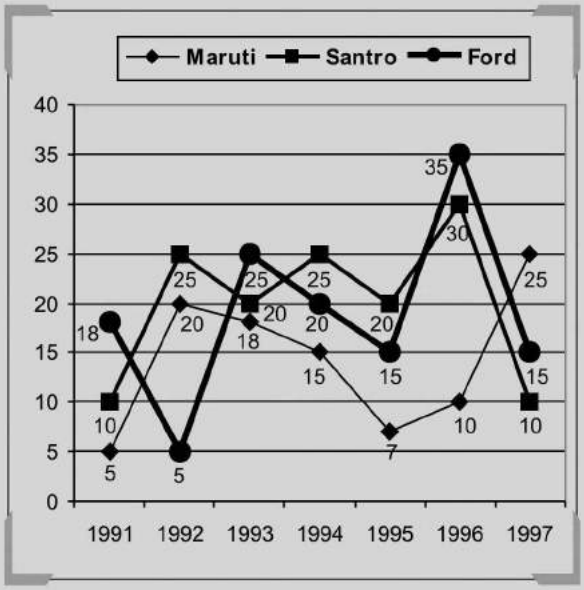
Q 21. What is the maximum possible difference between the runs scored by Sourav and Yuvi in any match?
(a) 25
(b) 30
(c) 50
(d) None of these

Q 22. Who is the lowest scorer in all the four matches taken together?
(a) Viru
(b) Sourav
(c) Rahul
(d) Yuvi

Q 23. Find the maximum possible difference in total runs by the four batsmen in any two matches?
(a) 25 (b) 50
(c) 75 (d) 100

Q 24. Among Viru, Sourav and Rahul, who has shown maximum percentage increase in the runs scored in a match over the previous match?
(a) Viru
(b) Sourav
(c) Rahul
(d) Cannot be determined

Direction for questions 25 and 26: Go through the graph below and answer the questions based on it.



The above graph gives the production of cars (in thousands) over the given years.

Q 25. In how many years is the number of cars manufactured by Maruti less than the average number of cars manufactured by Santro over the given years?
(a) 3 (b) 4
(c) 6 (d) None of these

Q 26. If the total average production of all cars increase at the rate of 10% every year from 1997 onwards, what will be the average approx. production of cars in the year 2000?
(a) 19000
(b) 21000
(c) 22500
(d) 24000

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(d)	(a)	(c)	(b)	(d)	(d)	(c)	(a)	(a)	(c)	(b)	(a)	(d)	(b)	(b)	(a)	(d)	(d)	(d)
Q. No.	21	22	23	24	25	26														
Answer	(d)	(d)	(c)	(b)	(c)	(c)														

HINTS AND SOLUTIONS

1. State tax revenue from individual income
   $= 17 \times 525/100 = 89.25$ mn
   Local tax revenue from individual income
   $= 2 \times 525/100 = 10.5$ mn
   Hence, Difference = 89.25 – 10.5 = $ 78.75 mn.

2. Percentage of state tax revenue in total tax revenue
   $= 2 + 20 + 8 + 17 + 7 + 2 + 10 = 66\%$
   Therefore, state tax revenue in 1999
   $= \frac{6 \times 440}{100} =$ $26.40 mn

3. Tax revenue from highway users in 2000
   $= \frac{8 \times 500}{100} =$ $40.0 mn
   Therefore, tax revenue from other sources in 2000
   $= 500 - 40 =$ $ 460 mn
   Therefore, % increase in tax revenue from other sources
   $= \frac{460 - 440}{440} \times 100 = 4.5\%$

5. Percentage of local tax revenue in total revenue
   $= 26 + 3 + 2 + 3 = 34$
   Percentage of local tax revenue from property source in total tax revenue = 26.
   Therefore, percentage of property tax in local tax revenue = $(26/34) \times 100 = 76.4\%$

6. Number of people knowing shorthand only with experience 10 year or less
   $= (40+35+60+15) - (35+30+40+10) = 35.$
   Numbers knowing typing only with experience 10 years or less
   $= (35+40+60+20) - (35+30+40+10) = 40.$
   No. of members knowing shorthand or typing and having experience of 10 years or less
   $= 35 + 40 = 75.$
   Percent required = 18.

7. No. of people knowing shorthand only
   $= (40 + 35 + 60 + 15 + 40 + 40) - (35 + 30 + 40 + 10 + 15 + 20) = 80$
   Percentage required= $80 \times 100/625 = 13$.

8. No. of graduates knowing neither shorthand nor typing
   $= 320 - (40 + 60 + 40 - 35 - 40 - 15) - (35 + 60 + 40 - 35 - 40 - 15) - (35+40+15) = 135.$

9. People knowing shorthand only
   $= (40 + 60 + 40 + 35 + 15 + 40) - (35 + 30 + 40 + 10 + 15 + 20) = 80.$
   Required percentage = (50/80) x 100 = 63.

10. No. of persons knowing shorthand only
    = 80 .....(A)
    No. of persons knowing typing only
    = 75 .....(B)
    If 10% of (A) learn typing and 12% of (B) learn shorthand, the total no. of those knowing both increases by 8 + 9 = 17; $17 \times 100/150 = 11\%$.

Let us make a table for the time taken:

	Patna	Mumbai	Calcutta	Chennai	Bangalore
Patna		4	5	3.75	10
Mumbai	4	-	6.666667	10	12.5
Calcutta	5	6.666667		12.5	12
Chennai	3.75	10	12.5		4
Banga-lore	10	12.5	12	4	-

14. Total time for Patna-Chennai trip
= 3.75 + 2 = 5.75 h
Hence, min. total stoppage = 2 × 4 = 8h
So, in 24 h, 4 trips get complete while 5th trip will be on.
Therefore, distance covered by airplane in 16 h
= 16 × 40 = 640 km.
Hence, option (d) is the answer.

[redacted]

15. Only one such person is there.

16. Total marks in drawing of division A students
= 298
Total marks in drawing of division C students
= 350.
Required difference = 350 – 298 = 52.

17. To stand 5th total score required = 900 marks. Hence marks required to be scored in A2 = 289 marks.

18. Marks in scout-guide can be calculated by subtracting marks of PT and social service from A3. Rakesh Rathi and Ritu Sehgal scored 100 marks in scout guide.

[redacted]

	Percentage runs scored in			
Batsmen	**Match I**	**Match II**	**Match III**	**Match IV**
Viru	20	20	20	60
Sourav	40	10	60	30

	Percentage runs scored in			
Batsmen	**Match I**	**Match II**	**Match III**	**Match IV**
Rahul	30	30	5	10
Yuvi	0	30	10	0
Total	90	90	95	100

20. It can be seen from the above chart that in match IV, the share of runs scored by these four batsmen = 100%, which is the maximum.

21. For maximum difference to occur, total runs scored by India and the contribution of these four batsmen, both should be maximum. and in the second case, runs scored by India and the score by individual batsmen should be minimum.

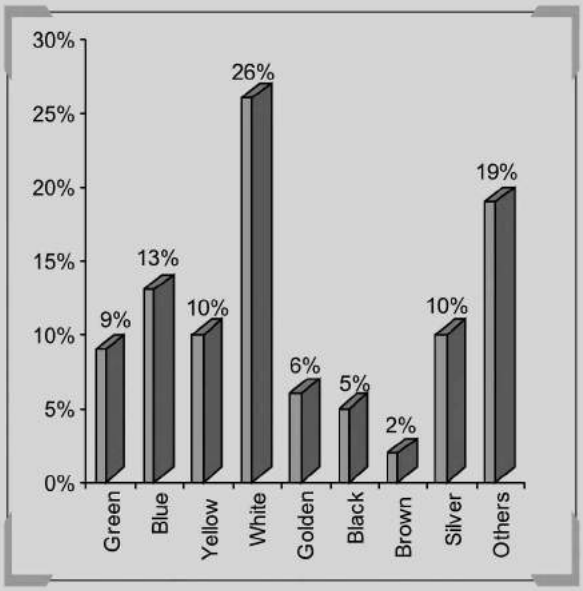
22. It is given that team score lies in between 250 and 300. We can thus see that the total score is more or less same in all the matches. So percentage share in total runs scored by the team is enough to answer who scored the minimum runs.
Yuvi has the lower percentage share.

24. Observe the data. Percentage runs scored by Sourav increased by 500% and the total runs scored by the team can vary by a maximum of 20%. And in no matches other batsmen showed such a high increase.

PRACTICE EXERCISE 2

Direction for questions 1 to 5: Study the following chart carefully and answer the questions based on it.

Following bar chart shows the sales of Cars in USA in 2008 in preference to different colors of the car:



Q 1. Which of the following colour of cars combined consist 50% of the total cars sold?
(a) Black, Golden, Blue, Silver
(b) Blue, Black, Green, Silver
(c) White, Golden, Blue, Black
(d) White, Blue, Green, Black

Q 2. Number of cars sold is directly proportional to the colour of car. Cars of how many colours are atleast 20% more popular than green coloured cars?
(a) 0 (b) 3
(c) 3 (d) Cannot be determined

Q 3. Brown coloured car is the least number of cars sold, and no other colour car's percentage share in total sale of cars is less than 4%. Now the sales figure of different coloured cars are arranged in ascending order. For how many colours' sales figure, we can definitely determine its rank?
(a) 0 (b) 1
(c) 6 (d) Cannot be determined

Q 4. A car can be painted multi-coloured too. However in that case, no colour used in making a car multi-coloured can be used individually. Total percentage share of a multi-coloured car = Sum of percentage share of all the individual colours used.

If a car is made multi-coloured by painting it by atleast two colours and at max by four colours, for how many different combinations it is possible to have 32% market share of total sales of the cars?
(a) 4 (b) 6
(c) 9 (d) None of these

Q 5. To increase the percentage share of all colour cars under 10% of total sales, a new exchange policy has been brought under practice. Following are the terms and conditions of this exchange policy:
i. Only those colour cars can be exchanged (to obtain a new colour car under 10% market share) which have a percentage share of more than 15%.
ii. No colour car whose market share is in between 10%–15% can be used in exchange.
How many different combinations are possible under this exchange scheme?
(A combination is defined as a set of a new car, which a person obtains after exchanging, and an old car defined as a car with the person before this exchange.)
(a) 6 (b) 8
(c) 12 (d) Cannot be determined

Direction for questions 6 to 10: Study the following chart carefully and answer the questions based on it.

Following table shows the percentage population of six states below poverty line and the proportion of male and female.

State	Percentage Population Below poverty Line	Proportion of male and female	
		Below Poverty line	Above poverty line
		M : F	M : F
A	12	3 : 2	4 : 3
B	15	5 : 7	3 : 4
C	25	4 : 5	2 : 3
D	26	1 : 2	5 : 6
E	10	6 : 5	3 : 2
F	32	2 : 3	4 : 5

Q 6. The total population of State A is 3000, then what is the approximate no. of females above poverty line in state A?
(a) 1200 (b) 2112
(c) 1800 (d) 1950

Q 7. If the total population of C and D together is 18000, then what is the total no. of females below poverty line in these two states?
(a) 5000 (b) 5500
(c) 4800 (d) Cannot be determined

Q 8. If the population of males below poverty line in state A is 3000 and that in state E is 6000, then what is the ratio of the total population of state A and E?
(a) 3 : 4 (b) 4 : 5
(c) 1 : 2 (d) None of these

Q 9. If the population of males below poverty line in state B is 500 then what is the total population of B?
(a) 14400 (b) 6000
(c) 8000 (d) 7600

Q 10. If in state E population of females above poverty line is 19800 then what is the population of males below poverty line in that state?
(a) 5500 (b) 3000
(c) 2970 (d) None of these

Direction for questions 11 to 15: Go through the table below and answer the questions based on it.

The table given below represents the Supply to demand ratio per 100 units of 15 items from the period Jan to Oct. For example, supply of item A is Jan was only 98 units where as the demand is of 100 units.

Month/ Items	Jan	Feb	March	April	May	June	July	August	Sep	Oct
A	98	99	98	98	96	99	97	97	96	95
B	91	91	89	87	88	87	86	90	91	92
C	106	105	102	99	102	100	105	96	92	90
D	109	110	112	108	108	112	85	101	86	99
E	95	94	94	96	101	102	96	97	96	92
F	86	83	84	96	94	87	93	99	101	105
G	88	88	89	94	86	97	93	101	92	89
H	98	89	89	96	94	91	86	102	94	84
I	100	98	96	101	104	98	104	99	111	100
J	97	100	101	99	96	94	100	97	101	92
K	98	96	95	97	98	94	93	98	96	100
L	103	97	96	96	95	92	91	99	101	104
M	83	105	100	103	100	99	98	104	107	102
N	104	78	80	82	86	92	91	100	98	87
O	94	91	95	96	99	92	93	92	97	100

Q 11. Which of the following items show the maximum increase in the supply to demand ratio from Feb to August?
(a) F (b) N
(c) G (d) H

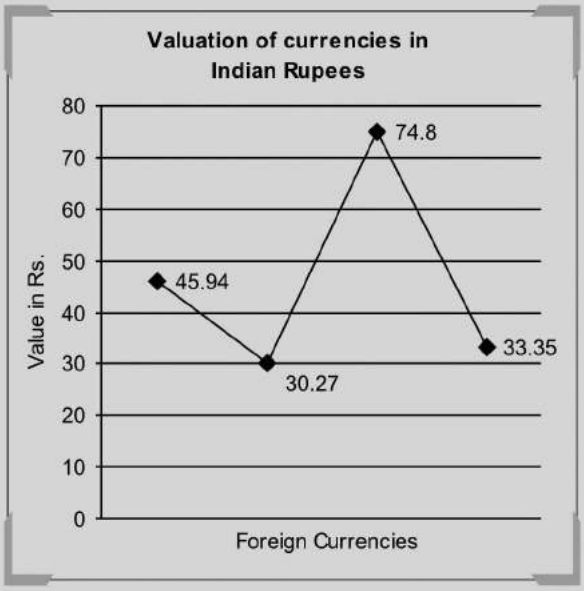
Q 12. For which item does supply exceed demand the maximum number of times?
(a) C (b) M
(c) D (d) J

Q 13. For the item with the least supply to demand ratio, what is the percentage increase in that ratio in the next month?
(a) 2.5% (b) 3%
(c) 4% (d) 2%

Q 14. How many items are constantly under supply through out the given period (under supply means demand is more than the supply)?
(a) 1 (b) 2
(c) 3 (d) 4

Q 15. How many items are constantly under demand through out the given period (An item is under demand if demand exceeds the supply in all the months during the given period)?
(a) 1 (b) 2
(c) 3 (d) 4

Direction for questions 16 to 20: Go through the chart below and answer the questions based on it.



Above given four currencies represents the following: US Dollar, Singapore Dollar, UK Pound and Euro. Following is the order of their valuations: UK pound > US Dollar > Euro > Singapore Dollar.

Q 16. A person wants to buy an article worth 65 Singapore Dollars. If Singapore Dollar loses 2% with respect to the above given exchange rates, then how many US dollars must be spend?

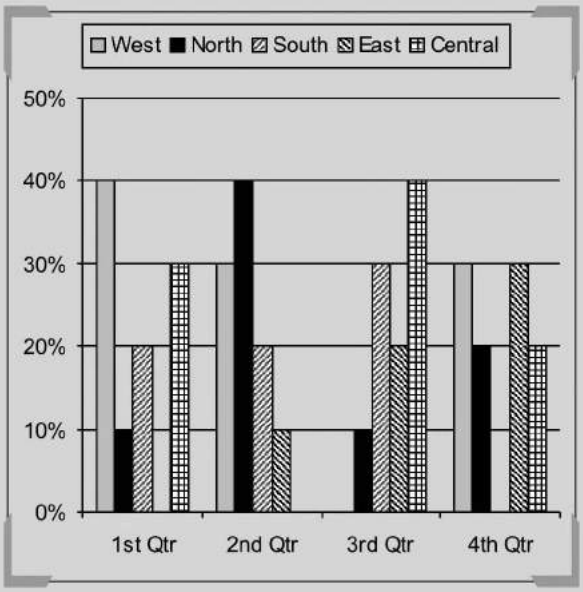
(a) 42.1 (b) 41.9
(c) 44.9 (d) 43.6

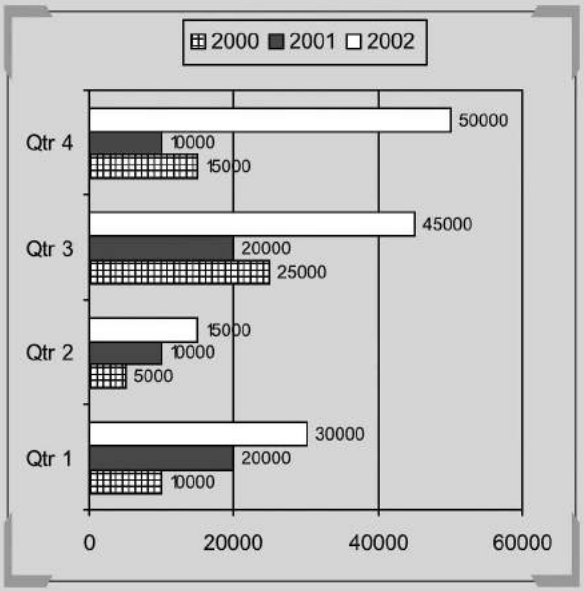
Q 17. A person wants to buy 800 barrels of oil. The payment can be made either in US dollars or in Euros. If the payment is made in Euros, then it would be 47 Euros per barrel while if one pays in US dollars, the rate would be 29 US dollars per barrel. How much money will he save if he goes for the cheaper deal?
(a) 29200 US dollars
(b) 34100 US dollars
(b) 37600 Euros
(d) None of these

Q 18. In the above question, if there is a discount of16% on paying in Euros and if the person goes for paying in Euros, how much does a person stand to gain or lose?
(a) Rs 12482 gain (b) Rs 12482 loss
(c) Rs 3987 loss (d) Rs 3987 gain

Q 19. US bonds give 9% return on investment. Which of the following when invested in US bonds gives maximum returns?
(a) 4000 US dollars
(b) 2500 UK Pounds
(c) 5300 Euros
(d) 7000 Singapore Dollars

Q 20. Value of rupee depreciates against US Dollar by 10%. What will be the new value of US Dollar in Rs?
(a) Rs 41.34
(b) Rs 51.04
(c) Rs 50.53
(d) Rs 40.83





The bar charts shown here represents the earnings of a Sales executive Mohit. Bar-chart (1) gives us the details of his earnings in the four quarters of the year across the five regions—West, North, South, East and Central. Bar chart (2) gives the details of his total earning in the four quarters in three different years. Bar-chart (1) is applicable for bar chart (2)and vice-versa.

Q 21. Across the given years, which quarter has given the maximum earning to Mohit?
(a) Qtr 1
(b) Qtr 2
(c) Qtr 3
(d) Qtr 4

Q 22. Which Region has brought maximum earning for Mohit in the year 2000?
(a) South (b) West
(c) East (d) Central

Q 23. During which year and quarter the earnings from the north Region have shown maximum change over the previous quarter in that year?
(a) Quarter III, year 2000
(b) Quarter II, year 2002
(c) Quarter II, year 2001
(d) Quarter IV, year 2002

Q 24. What is the maximum difference between any two regional earnings in any quarter?
(a) Rs 15500
(b) Rs 13000
(c) Rs 14500
(d) None of these

Q 25. Find the percentage change in Mohit's earning due to the south zone during 2001 over the previous year.
(a) 14% (b) 20%
(c) 25% (d) 33%

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(d)	(b)	(d)	(d)	(a)	(d)	(d)	(d)	(b)	(b)	(d)	(a)	(b)	(d)	(b)	(d)	(a)	(d)	(b)
Q. No.	21	22	23	24	25															
Answer	(c)	(d)	(d)	(d)	(a)															

HINTS AND SOLUTIONS

1. This question can be done by simple calculation using the options.
2. Since we do not know the colours of cars contained in 'Others' category, we cannot answer this question.
3. We can determine the ranks of only those colour cars which are having a share less than 4%. Reason being unavailability of the data pertaining to the colours contained in 'others' category.
5. We do not know the colours in 'Others' category. It might be a possibility that 'Others' category is containing 4-5 different colours and each one being around 4% -5%.
   Hence, cannot be determined.

7. From the given information, we cannot find the population of states C and D individually; so we can't find the required value.
8. Population of State A below poverty line
$$= \frac{3000\times5}{3} = 5000$$
Therefore, total population of state A
$$= \frac{5000\times100}{12}$$
And the population of state E below poverty line
$= 11000$
Therefore, total population of state E
$$= \frac{11000\times100}{10}$$
$$\text{Therefore, required ratio} = \frac{5}{12}\times\frac{10}{11} = \frac{25}{66}$$

11. By observation we can say that 'N' shows maximum increase in supply to demand ratio.
12. For D, supply exceeds demand the maximum number of times.
13. N has least supply to demand ratio in February
$$\%\text{ increase} = \frac{2\times100}{78} = 2.5\%$$
14. Only items A and B are in under supply.
15. Consider those items also which have a 100 supply viz., O.

16. Taking into consideration above rates.
1 US $ = Rs 45.94
1 Singapore $ = Rs 30.27
Hence, 1 US $ = 1.52 Singapore $
The Singapore $ has lost 2% with respect to the above given exchange rates
So, 1 US $ = 1.52 × 1.02 = 1.55 Singapore $
Hence, 65 Singapore $ = 41.9 US
Thus, the person should give 41.9 US $ to get 65 Singapore Dollars.
17. Price for 800 barrels = 37600 Euros = Rs 1253960.
However, if we are paying in US $,
we will be paying $ 23200 = Rs 1065808.
Thus, we are paying 188152 Rupees less
= 4096 US dollars.
18. If there is a discount of 16% on paying in Euros, then a person will have to pay 31584 Euros instead of earlier 37,600

dollars. To buy this much Euros, we will have to spend Rs 12482 less than the earlier case.

So, if we are paying in Euros we will be gaining Rs 12482.

19. Larger the amount larger the returns as rate is constant.

We are just supposed to find out that which amount given in the options is largest.

It can be seen that 7000 Singapore Dollars (4666.7 US dollars) is the largest investment and so it would give the maximum returns.

20. It should be understood that if rupee is depreciating by 10%, then dollar is appreciating by 11.11%.

[redacted]

21. Use the second bar chart. It is a visual inspection solution.

22

Region	Annual earnings for the region (in thousands)
West	$0.4 \times 10 + 0.3 \times 5 + 0.3 \times 15 = 10$
North	$0.1 \times 10 + 0.4 \times 5 + 0.1 \times 25 + 0.2 \times 15 = 8.5$
South	$0.2 \times 10 + 0.2 \times 5 + 0.3 \times 25 = 10.5$
East	$0.1 \times 5 + 0.2 \times 25 + 0.3 \times 15 = 10$
Central	$0.3 \times 10 + 0.4 \times 25 + 0.2 \times 15 = 16$

However, we could have done this question without using the actual calculation method too. Just see the major contributing quarters and major contributing Region in it. Major contributing quarters are quarter III and IV while central Region is main contributor in those quarters.

23. It can be seen that whenever percentage share of north Region in any quarter decreases, the total earning during that quarter increases and vice-versa with the only exception being quarter IV, year 2002.

So, the maximum change over the previous quarter must be during quarter IV, year 2002 when it's percentage share and total earnings both have increased.

24. Maximum quarterly zonal earnings are for central Region during quarter III, 2002 = Rs. 18000

Minimum quarterly zonal earnings are East Region during quarter II, 2000 = Rs 500

Hence, Maximum difference

= 18000 – 500 = Rs 17500

25. Let us tabulate the information given:

Region	Annual earning for the region
West	$0.4 \times Q1 + 0.3 \times Q11 + 0.3 \times QIV$
North	$0.1 \times Q1 + 0.4 \times Q11 + 0.1 \times Q111 + 0.2 \times QIV$
South	$0.2 \times Q1 + 0.2 \times Q11 + 0.3 \times Q111$
East	$0.1 \times Q11 + 0.2 \times Q111 + 0.3 \times QIV$
Central	$0.3 \times Q1 + 0.4 \times QIII + 0.2 \times QIV$

Earning in south Region in 2001 = $20000 \times 0.2 + 10000 \times 0.2 + 20000 \times 0.3 = 12000$

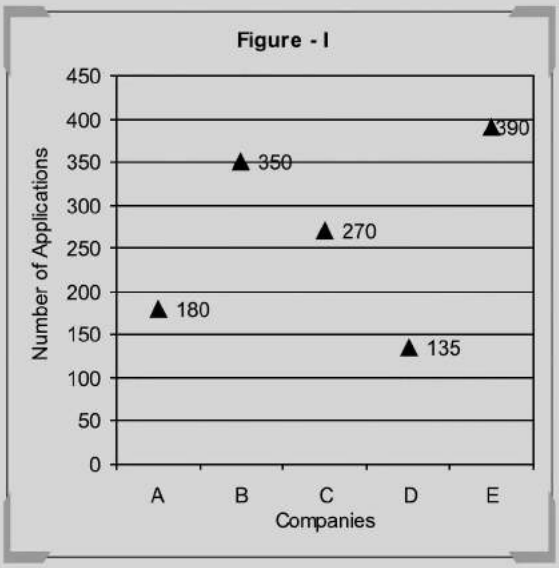
Earning in south Region in 2000 = $10000 \times 0.2 + 5000 \times 0.2 + 25000 \times 0.3$ = Rs 10500

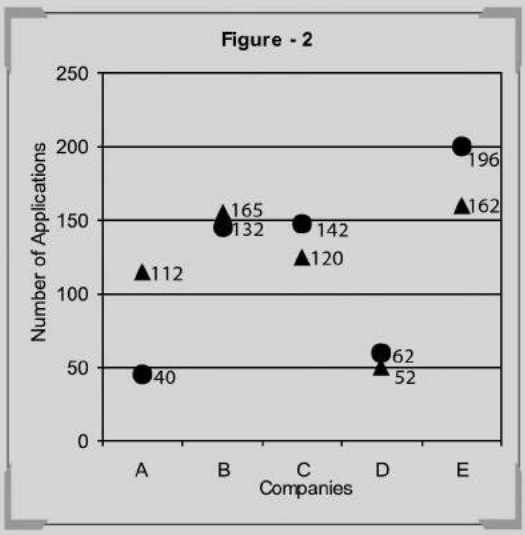
Therefore, percentage change = 14.28%

PRACTICE EXERCISE 3

Direction for questions 1 to 4: Go through the figures below and answer the questions based on it.

Fig. (1) represents the total number of applications received by five companies A, B, C, D and E in the year 2007. Fig (2) represents the total number of applications selected and total number of applications on waiting list. Total number of applications received = Number of applications selected + number of applications on rejected + number of applications on waiting list.





For companies A and B, no. of applicants selected is more than no. of applicants on waiting list.

Q 1. Which company has the minimum ratio of number of rejected applicants to the total number of applicants received?
(a) A (b) C
(c) B (d) E

Q 2. In company E, 29% of selected applicants are rejected due to health reasons and 36% of the waiting list applicants are selected. Then the percentage of selected applicants has increased/decreased by:
(a) 0.5% increase (b) 2% increase
(c) 0.5% decrease (d) 2% decrease

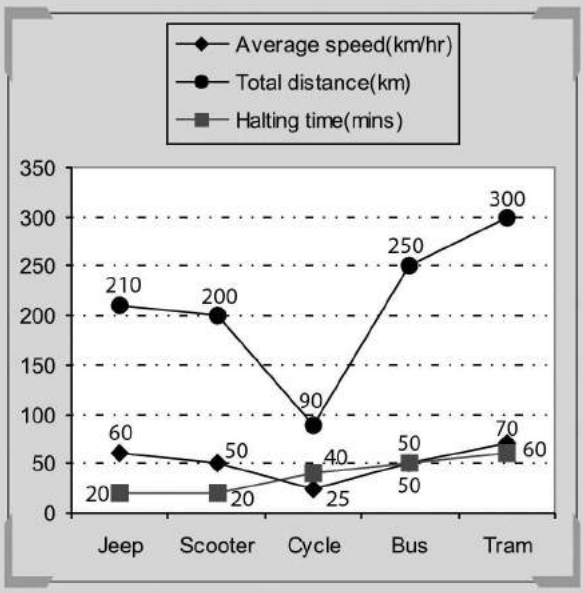
Q 3. Which company has maximum ratio of selected to rejected applicants, if the number of applicants selected and the number of applicants on waiting list are interchanged?
(a) E (b) A
(c) C (d) B

Q 4. For company A, 12.5% of the waiting list applicants are finally selected and 25% of the rejected candidates are added to the waiting list. Now, what percentage of applicants are on the waiting list?
(a) 60% (b) 45%
(c) 58% (d) 54%

Direction for questions 5 to 9: Refer to the line chart below and answer the questions based on it.

The below given line chart shows the total distance from Patna to Delhi, average speed and halting time for five different modes of travel.

It can be seen that distance from Patna to Delhi may change with the different mode of transport as different routes are taken. To calculate the average speed, we do not take halting time into consideration.



Q 5. Which mode of travel takes the least time after taking halting time into consideration?
(a) Jeep (b) Scooter
(c) Cycle (d) Tram

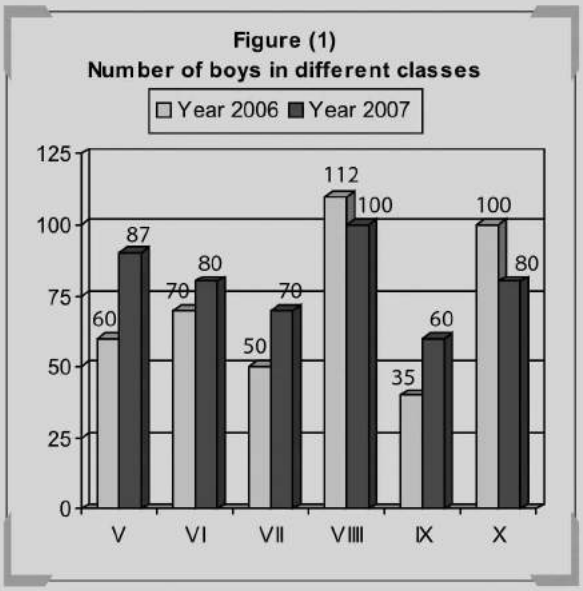
Q 6. What is the percentage change in the time taken to travel from Patna to Delhi by cycle due to halts taken during journey?
(a) 6.66%
(b) 11.1%
(c) 12.5%
(d) 16.66%

Q 7. What is the average speed of all the modes of transport during the journey without taking halting time into consideration?
(a) 46 kmph
(b) 54 kmph
(c) 50 kmph
(d) 58 kmph

Q 8. Find the maximum time taken to travel from Patna to Delhi taking halting time into consideration.
(a) 4.5 hours
(b) 5 hours
(c) 6.25 hours
(d) None of these

Q 9. If one can change the mode of transport at the halfway, then which of the following combinations takes the least time to reach B from A without taking halting time into consideration?
(a) Jeep and Scooter
(b) Cycle and Jeep
(c) Jeep and Tram
(d) Tram and Scooter

Direction for questions 10 to 13: Go through the bar charts given below and solve the questions based on it.



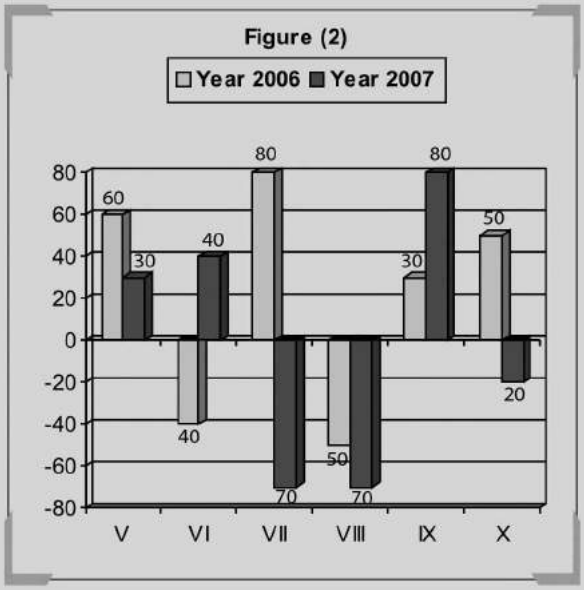


Fig. (1) gives the number of boys in different classes in year 2006 and year 2007 and fig. (2) gives the number of boys as a percentage of number of girls. If the value indicated in the chart is negative, then it is the number of girls as a percentage of number of boys.

Q 10. Find the difference in the total number of students in class VI in the year 2006 and 2007.
(a) 150 (b) 182
(c) 130 (d) 198

Q 11. Find the average number of girls studying in each standard in the year 2007.
(a) 92 (b) 103
(c) 118 (d) 136

Q 12. If 29% of the girls in class X passed in the year 2006 and 37.5% of girls passed in the year 2007, then find the approx. ratio of the no. of girls failing in the year 2006 to the no. of girls failing in the year 2007.
(a) 14 (b) 16
(c) 11 (d) 18

Q 13. Which of the following is false?
(1) Number of girls in class V is maximum in the year 2007.
(2) Number of girls in year 2006 is greater than in year 2007 in class VII.
(3) Number of girls in class VII is minimum in the year 2007.
(4) Number of girls in class X has decreased in 2007 as compared to that in 2006.

Direction for questions 14 to 17: Go through the table below and solve the questions based on it.

The chart gives details of fights operated by Indian airline in a week from Delhi to various destination and back.

From	To	Days	Departure	Arrival
DELHI	Coimbatore	Daily	6:20	7:35
DELHI	Coimbatore	2,4,6	19:15	21:15
DELHI	Cochin	Daily	9:45	11:20
DELHI	Madurai	1,3,5	20:00	21:40
DELHI	Mangalore	3,7	9:45	13:20
DELHI	Mangalore	1,3,5	7:00	9:25
DELHI	Bangalore	1,3,5	9:30	10:25
DELHI	Bangalore	2,4,6	9:30	10:25
DELHI	Bangalore	1,3,5	7:00	7:55
DELHI	Goa	2,4,6	9:30	14:00
DELHI	Hubli	1,3,5	9:30	12:05
DELHI	Huderabad	2,7	7:00	11:40
DELHI	Bombay	1,3,5	9:30	14:00
DELHI	Agatti	1,5	9:45	13:20
DELHI	Pune	2,4,6	9:30	15:20
DELHI	Trichy	2,4,6	19:15	20:10
DELHI	Trichy	1,3,5	20:00	20:50
DELHI	Vijaywada	2,6	7:00	8:20
DELHI	Vizag	2,6	7:00	9:40
DELHI	Bhuneshwar	4,6	7:00	11:20
DELHI	Trivandrrum	1,3,5	14:00	15:45
DELHI	Calicut	1,3,5	14:00	17:10

1. Monday
2. Tuesday
3. Wednesday
4. Thursday
5. Friday
6. Saturday
7. Sunday

Additional information:

In a week:

No. of departures delayed by less than 1 h = 12
No. of departures delayed by more then 1 h = 2
No of arrivals delayed by less than 1 h = 11
No. of arrivals delayed by more than 1 h = 3.

Q 14. An airlines is said to operate at 100% efficiency if 90% of its flights are on time. The efficiency of IA
(a) 80.9% (b) 88.23%
(c) 79.4% (d) 75.4%

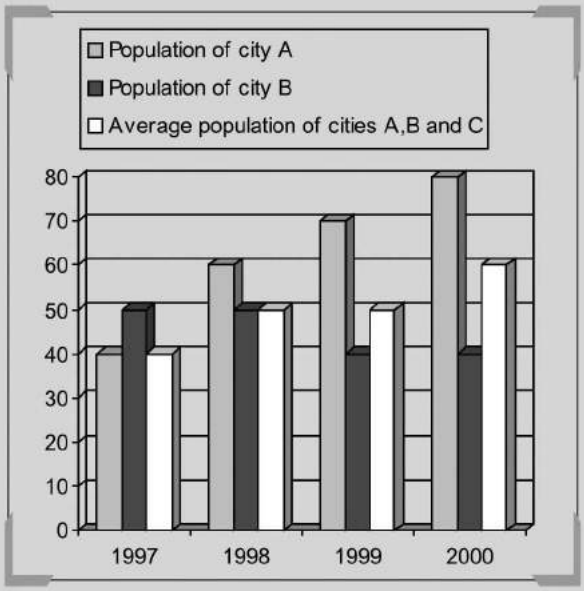
Q 15. If the airlines aims to achieve 90% efficiency on arrivals, then the no. of flights arriving on time must increase by (use data of the previous question)
(a) 3 (b) 2
(c) 4 (d) 5

Q 16. If the airlines suffers a loss of Rs 50000 on every flight delayed for more than one hour, the loss suffered by the airlines in a week is
(a) Rs 2 lakh (b) Rs 2.5 lakh
(c) Rs 3 lakh (d) Rs 3.5 lakh

Q 17. Average delay in departure is measured by delay occurring in each flight, adding up all delays and dividing by the total no. of delayed flights. The average delay in departure for flights operated by IA is
(a) 35 min (b) 25 min
(c) 40 min (d) Cannot be determined

Direction for questions 18 to 22: Go through the bar charts below and solve the questions that follow.

There are only three cities—A, B and C, in a state. Following bar chart gives us the population figures of these cities (Figure in '000):



Q 18. In which year did city C show the highest percentage growth in population?
(a) 1998 (b) 1999
(c) 2000 (d) cannot be determined

Q 19. In which year was the population of city A the highest in state's total population?
(a) 1997 (b) 1998
(c) 1999 (d) 2000

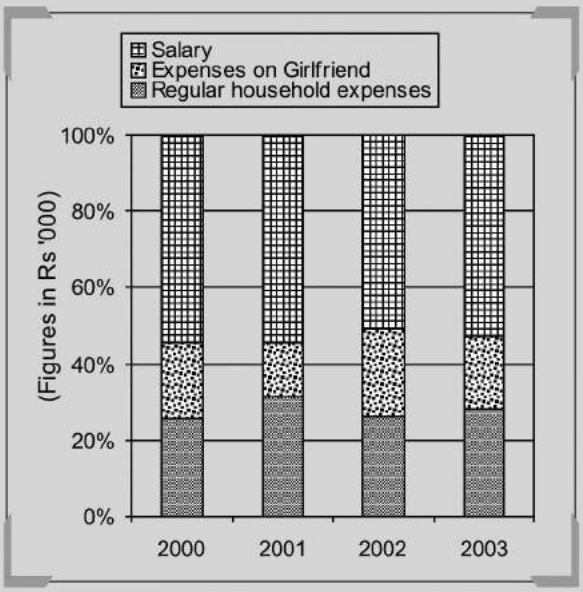
Q 20. In 1998, the ratio of men to women in the given state is 1:1 and in 1999, the same ratio is 2 : 3. What is the percentage increase in the women population from 1998 to 1999?

(a) 30% (b) 60%
(c) 20% (d) None of these

Q 21. If increase in the population of city can only be due to new births and decrease in population of city can only due to deaths, then number of new births in city A in year 2000 is (in '000):
(a) 20 (b) 40
(c) 38 (d) Cannot be determined

Q 22. For which year is the ratio of population of city A to the population of city B the highest?
(a) 1997 (b) 1998
(c) 1999 (d) 2000

Direction for questions 23 to 27: Go through the bar chart below and answer the questions based on it.



Mayank spends his yearly salary on two types of expenses only (given in the chart above) and saves the remaining amount. Given above is the graphical representation of his yearly salary and expenses over the four-year period.
Salary = Total Expenses + Savings

Q 23. If regular household expenses as a percentage of salary exceed 50%, then the living style in that year is said to be aristocratic. How many years Mayank has lived aristocratically in all the years given above?
(a) 0 (b) 1
(c) 2 (d) 3

Q 24. The ratio of expenses on girlfriend to salary has been the highest in which one of the following years?
(a) 2000
(b) 2001
(c) 2002
(d) 2003

Q 25. What is average monthly salary of Mayank over the four-year period (in Rs '000)?
(a) 10.8
(b) 12.7
(c) 14.8
(d) Cannot be determined

Q 26. In which year was the savings rate the highest [Saving rate = savings/salary × 100]?
(a) 2000
(b) 2001
(c) 2002
(d) 2003

Q 27. If his salary is same for the given four years, which year is the percentage increase in total expenses the highest?
(a) 2000 (b) 2001
(c) 2002 (d) 2003

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(a)	(c)	(c)	(a)	(b)	(a)	(d)	(c)	(b)	(c)	(a)	(c)	(b)	(b)	(b)	(d)	(a)	(c)	(c)
Q. No.	21	22	23	24	25	26	27													
Answer	(d)	(c)	(d)	(c)	(b)	(a)	(d)													

HINTS AND SOLUTIONS

1.

Company	Number of applicants Rejected/Number of applicants Received	Number of applicants Rejected
A	28/180 = 0.155	180 – (40 + 112) = 28
B	53/350 = 0.151	350 – (132 + 165) = 53
C	8/270 = 0.029	270 – (120 + 142) = 8
D	21/135 = 1.55	135 – (52 + 62) = 21
E	32/390 = 0.08	390 – (162 + 196) = 32

So, the ratio is minimum for company C.

2. For company E, Total applicants received = 390
Number of applicants selected = 196
Number of applicants on waiting list = 162
Number of applicants rejected = 32
Now, 29% of selected applicants are rejected. So, number of selected applicants retained
$= 0.71 \times 196 = 139$
And 36% of waiting list applicants are selected
Therefore, total selected applicants
$= 139 + 0.36 \times 162 = 197$
Hence, increase $= [1/196] \times 100 = 0.5\%$.

3.

Company	Selected Applicants/Rejected Applicants
A	112/28 = 4
B	165/53 = 3.11
C	120/8 = 15
D	52/21 = 2.4
E	162/32 = 5.1

Obviously the ratio is maximum for C.

4. For company A, no. of applicants received = 180
Number of selected applicants = 40
Number of waiting list applicants = 112
Number of rejected applicants = 28
Now, 12.5% of waiting list applicants are selected.
Therefore, waiting list applicants
$= 0.875 \times 112 = 98$
And 25% of rejected candidates list
$= 0.25 \times 28 = 7$
Therefore, number of applicants on waiting list
$= 7 + 98 = 105$
Required percentage = 58%

Let us tabulate the data given:

Mode of transport	Average speed (Km/h)	Total distance (km)	Time taken for actual Journey (h)	Halting Time (h)	Total Time taken (h)	Actual average speed (km/h)
Jeep	60	210	3.5	0.25	3.75	56
Scooter	45	195	4.33	0.25	4.58	42.6
Cycle	20	90	4.5	0.5	5	18
Bus	40	250	6.25	0.75	7	35.7
Tram	75	300	4	1	5	60
Total	-	1045	22.58	2.75	-	-

5. Jeep takes the least time after taking halting time into consideration.

6. Time taken cycle without taking halt = 4.5 h
Change in time due to halt = 0.5h
Hence, percentage change in time = 11%

7. Total distance covered = 1045 km.
Time taken for actual journey = 22.58h
Hence, average speed = 46 km/h

8. Maximum time taken to travel from Patna to Delhi is for Bus and it is equal to 7 h

9. Mode of transport — Total time taken (h)
Jeep + scooter = 1.75 + 2.165 = 3.915
Jeep + cycle = 1.75 + 2.25 = 4
Jeep + Tram = 1.75 + 2 = 3.75
Scooter + Tram = 2.165 + 2 = 4.165
Hence Jeep and Tram take minimum time for journey together.

10. In year 2000 number of boys in standard VI = 70
Number of girls = 0.4
Number of boys
Therefore, number of girls = 0.4 x 70 = 28
Total number of students = 98
In year 2001 number of boys = 80

$$= \frac{100}{0.5} = 200$$

Total number of students = 280
Difference = 280 – 98 = 182.

11.

Standard	Number of girls in year 2001
V	300
VI	200
VII	49
VIII	70
IX	75
X	16

12. In class X
Total number of girls in year 2000

$$\text{Ratio} = \frac{142}{10}$$

Total number of failed girls = 0.71 × 200 = 142
Total number of girls in year 2001 = 80 × 0.2 = 16
Total number of failed girls = 0.625 z 16 = 10

$$\text{Ratio} = \frac{142}{10} = 14.2$$

13. Only Statement [c] is wrong.

14. Total departure from Delhi = 68,
Total arrivals from Delhi = 68.
Total no. of delayed arrivals and departure = 28
10% of Total no. of flights that can be delayed = 136/10 = 13.6
No. of flights on time = 136 – 28 = 108 = 79.41% of flights operated. Efficiency
= 100 × 79.41/90 = 88.23%
Alternatively, for 100% efficiency, no. of flights to be on time = 90% of 136
Since only 108 flights are on time, so % age efficiency = 108/(90% of 136)

15. No. of arrivals = 68. For 100% efficiency no. of arrivals to be on time = 68 × 0.9 = 61.2.
For 90% efficiency no. of arrivals to be on time = 62.2 × 0.9 = 55.08 (nearly)
No. of present arrivals on time = 68–14 = 54.
Hence 2 more flights are to be on time.

16. In a week no. of departures and arrival involving a delay of more than one hour = 5.
Loss suffered = 5 × 50000 = 2.5 lakhs.

17. We don't have the exact time for which the flights are delayed. So, average delay cannot be determined.

18. The first step would be to approximate the lengths of various bars then one can prepare the table as follow:

Population	1997	1998	1999	2000
Average	40	50	50	60
Total (3 × average)	120	150	150	180
City A	40	60	70	80
City B	50	50	30	40
City C Total − (A+B)	30	40	50	60

The population of city C increases by 10000 every year.
Hence, highest percentage increase is in 1998 over 1997 as it has the lowest base.
So, [a].

19. We are supposed to find here the ratio of 1st bar and 3rd bar.

21. Population (2000) = Population (1999) +
No. of new births (2000) –
No. of deaths (2000).
The no. of births figure cannot be determined as the no. of deaths is not known.
Hence, [d]

22.

	2000	1997	1998	1999
City A	40	60	70	80
City B	50	50	30	40

The ratio is the highest for the year 1999.

23. Total income has never been the 50% of the total income. Hence, for none of the years this has happened.

24. Ratio of expenses on girlfriend to salary is tabulated below:

2000	55/150
2001	40/155
2002	60/135
2003	65/170

It is clear now that the ratio is the highest for year 2002.
Hence, [c].

25. Average yearly salary

$$= \frac{150+155+135+170}{4} = \frac{610}{4} = 152.5$$

Therefore, average monthly salary = 12.7

PRACTICE EXERCISE 4

Direction for questions 1 to 4: Go through the table below and solve the questions based on it.

Given below are the win loss histories of four tennis players in their career.

Result/Players	P	Q	R	S
Won	23	18	10	16
Won	13	16	13	14
Lost	8	13	16	13
Lost	6	8	20	17

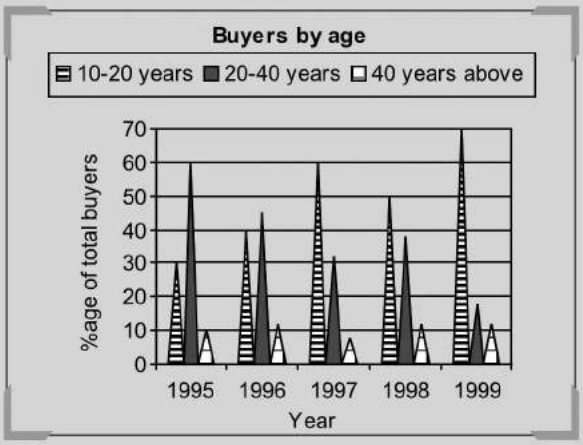
Q 1. If a win carries an award of Rs 1500 and a loss carries of Rs 1000, what is the largest difference in prize money earned among the four?
(a) 7000 (b) 14000
(c) 5000 (d) 9000

Q 2. As in the case above what is the lowest average prize money won per match by any player?
(a) 1135 (b) 1300
(c) 1195 (d) 1150

Q 3. As in the case above what is the highest average prize money won per match by any player?
(a) 1495 (b) 1360
(c) 1460 (d) 1410

Q 4. Who has lost the maximum percentage of matches played?
(a) P (b) Q
(c) R (d) S

Direction for questions 5 to 9: Go through the bar chart given below and solve the questions based on it.



Q 5. By how many percentage has the number of buyers in the age 20–40 years age group has declined from year 1995 to 1999?
(a) 26.6 (b) 73.4
(c) 44 (d) 20

Q 6. If the total number of people surveyed is 2000 in each of the five years given, then find the number of buyers changed from year 1995 to 1999 in the age group 10–20 years?
(a) 800 (b) 600
(c) 1200 (d) 1400

Q 7. Which of the following age group shows a declining trend in the purchases in the given five years?
(a) 10–20 years
(b) 20–40 years
(c) 40 years and above
(d) No particular trend in any group

Q 8. By how many has the percentage number of buyers in the age 10-20 years group has increased from year 1997?
(a) 13.6 (b) 15
(c) 16.7 (d) 10

Q 9. Which of the following age groups show no particular trend?
(a) 10-20 years
(b) 20-40 years
(c) 40 years and above
(d) no particular age group

Direction for questions 10 to 17: Go through the table given below and solve the questions based on it.

There are eight major locations in Sanjay Gandhi Botanical Garden in Patna. A survey was conducted to find out the movement of the people within the garden. Following results were obtained:

		To							
		A	B	C	D	E	F	G	H
From	A		48	46	66	24	4	68	64
	B	20		64	152	64	90	108	86
	C	28	108		88	86	46	130	130
From	D	54	92	66		42	82	20	4
	E	70	48	50	110		26	46	22
	F	24	74	92	68	112		120	60
	G	28	32	134	152	6	86		26
	H	50	70	90	108	68	110	86	

Q 10. The least number of people go to which location?
(a) G (b) D
(c) B (d) A

Q 11. What is the total number of people visiting Zoo?
(a) 3868 (b) 5000
(c) 4000 (d) Cannot be determined

Q 12. What percentage of total visiting E comes from D?
(a) 10 (b) 20
(c) 40 (d) 30

Q 13. What is the difference between the number of people going from B to F and G and D to F and G?
(a) 48 (b) 96
(c) 72 (d) 84

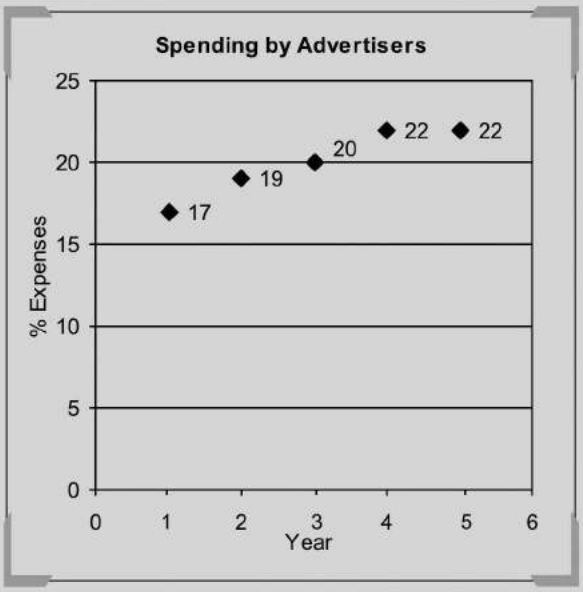
Q 14. The least number of people leave from which place?
(a) H (b) G
(c) A (d) E

Q 15. The number of people leaving F and C is …… the number of people leaving C and F.
(a) Greater than
(b) Less than
(c) Equal to
(d) Cannot say with certainty

Q 16. The number of people going to A from B and C is equal to the number of people going to ……from ……and ……
(a) E, A, C (b) E, D, C
(c) F, A, C (d) H, E, G

Q 17. The highest number of people leave from which place?
(a) C (b) D
(c) H (d) F

Direction for questions 18 to 24: Go through the chart given below and solve the questions based on it.



Total for all the five years = Rs 100 crores.

Q 18. In year 3, out of the total money spent by the advertisers, 12% had been spent on contingent expenses. What was the amount spent on contingent expenses (in Rs crore)?
(a) 2.4
(b) 2.1
(c) 2.5
(d) 2.6

Q 19. Expenses of year A added to one fourth the expenses of another year B equals the expenses of another year C. Identify A, B and C.
(a) 1-3-4
(b) 2-3-4
(c) 1-2-3
(d) 3-1-4

Q 20. In how many years, has the expenditure has got increased by the atleast 15% with respect to year 2?
(a) 1
(b) 3
(c) 2
(d) Insufficient data

Q 21. The difference in the expenditure between which two years equals the difference between the expenditure in year 2 to year 1?
(a) Year 5 and Year 3
(b) Year 4 and Year 3
(c) Both (1) and (2)
(d) Neither (1) and (2)

Q 22. If in year 6 the expenses increase at the same rate as they did from year 3 to year 5. what will be the expenditure in year 6 (Rs crore)?
(a) 28.2
(b) 24.2
(c) 25.2
(d) 23.2

Q 23. If in year 4, out of the money spent by the advertisers, 10% is spent on disaster recovery expenses, what was the amount spent on disaster recovery expenses?
(a) 2.2
(b) 2.6
(c) 2.3
(d) 2.4

Q 24. How many years, the spending was more than the average spending per annum for the given period?
(a) 0
(b) 1
(c) 2
(d) 3

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(d)	(b)	(c)	(b)	(a)	(b)	(d)	(c)	(d)	(d)	(d)	(a)	(b)	(c)	(a)	(a)	(a)	(a)	(c)
Q. No.	21	22	23	24																
Answer	(c)	(d)	(a)	(c)																

HINTS AND SOLUTIONS

1. Ee get the following table of earnings:

Persons	Win Total	Average
P	36 × 1500 + 14 × 1000 = 68000	1360
Q	34 × 1500 + 21 × 1000 = 72000	1310
R	23 × 1500 + 36 × 1000 = 68500	1160
S	30 × 1500 + 30 × 1000 = 60000	1000

Now we can see that the largest difference
= Max – Min = Rs 75000 – Rs 68000 = Rs 7000

2. Using the data from above question, lowest average is Rs 1160.

3. Highest average is Rs 1360

5. Required percentage = (60–16)/60 x 100 = 73%

6. Total no. of persons = 2000.
Change has been 40% (70–30).
So required answer = 2000 × 0.4 = 800

9. Required percentage = 16.7%

12. The information given is for 8 major locations in the Zoo. There can be many more minor locations in the Zoo for which information about movement of people is not given.

13 Total people visiting
E = 24 + 64 + 86 + 42 + 112 + 6 + 68 = 402.
So, required percentage = 42/402 × 100 = 10%.

14. Number of people going from
B to F and G = 90 + 108 = 198.
Number of people going from
D to F and G = 82 + 20 =102.
Since difference = 96.

17. Number of people going to A from B and C
= 20 + 28 = 48.
Go through the options now.

19. Required amount = 0.12 × 20 = 2.4 crore.

20. Use the options.

21. The difference between the expenditure in year 2 and year 1 is 2%. Use the options now.

22. The rate of increase in expenses from year 3 to year 5 is 10%. So expenditure in year 6
= 22 × 1.1 = 24.2 crore.

23. Required amount = 22 × 0.1 = 2.2 crore.

PRACTICE EXERCISE 5

Direction for questions 1 to 7: Go through the table given below and solve the questions based on it.

The XYZ manufacturing company produces a product which has a 6 month demand cycle as given below:

Month	Demand in Units	Working days/month
January	300	22
February	500	19
March	400	21
April	100	21
May	200	22
June	300	20

Company's policy is to retain safety stock of 20% of monthly demand and each month's safety stock becomes the opening stock for the next month, and opening stock for January is 50 units. Each unit requires 10 labour-hours and company works 8 hour/day.

Q 1. What is the opening stock for June?
(a) 20 (b) 40
(c) 60 (d) 100

Q 2. What is production requirement for March?
(a) 220 (b) 310
(c) 380 (d) 320

Q 3. In which month production requirement is highest?
(a) February (b) March
(c) January (d) June

Q 4. In which month Opening stock is lowest?
(a) February
(b) March
(c) January
(d) May

Q 5. If there are currently 20 workers employed, and how many workers are needed to be laid off in January to accommodate demand?
(a) 6 (b) 2
(c) 3 (d) 5

Q 6. If constant work force of 20 is retained, how many hours workers sit idle in January month?
(a) 300 h (b) 420 h
(c) 240 h (d) 200 h

Q 7. If constant work force of 20 is retained, how many total hours workers do overtime in February month?
(a) 2000 h (b) 3000 h
(c) 2360 h (d) 3360 h

Direction for questions 8 to 11: Go through the table given below and solve the questions based on it.

The following table gives the tariff (in paisa per kilowatt–hour) levied by the NTPC in 2004–05, in the four sectors and the regions within them. The table also gives the percentage change in the tariff as compared to 2001–02.

	Region 1		Region 2		Region 3		Region 4		Region 5	
Sector 1	425	+15	472	+5	420	–4	415	+8	440	+10
Sector 2	430	+12	468	+8	448	+7	423	–3	427	+11
Sector 3	428	+8	478	–4	432	+6	441	+10	439	+8
Sector 4	434	–5	470	+15	456	+10	451	+12	446	–12

Q 8. If the amount of power consumed by the various Regions in Sector 1 is the same, then, as compared to 2001–02, the net tariff in 2004-05...
(a) Increased by 9.5%
(b) Decrease by 1.5%
(c) Decrease by 13.3%
(d) None of these

Q 9. What approximately was the average tariff per sector in Region 3 in 2001–02?
(a) 402 (b) 424
(c) 433 (d) None of these

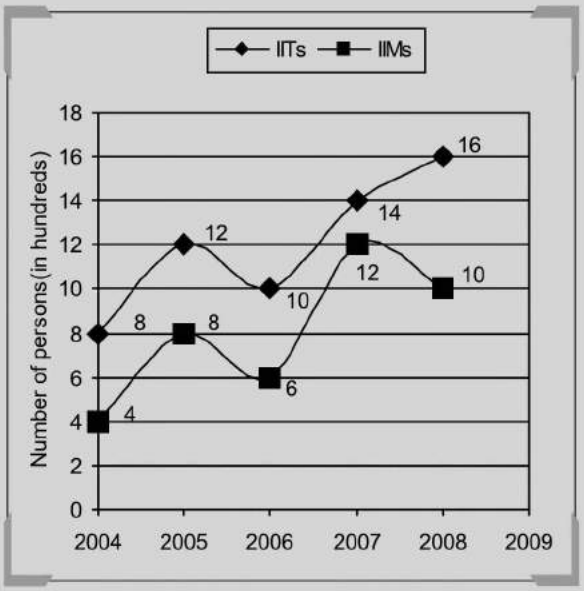
Q 10. Which of the following has got a maximum percentage increase for the given period?
(a) Region 1 – Sector 1
(b) Region 1 – Sector 2
(c) Region 5 – Sector 2
(d) None of these

Q 11. There are a total of 20 units of NTPC divided into regions and then subdivided into sectors, as given in the above table. For how many units of NTPC in 04–05, there is a net increase in the tariff?
(a) 15 (b) 5
(c) 8 (d) None of these

Direction for questions 12 to 17: Go through the chart given below and solve the questions based on it.

Chart given below represents total number of persons (in hundreds)who have passed out from IIMs and IITs

in each year from 2004 to 2008. The IITs and IIMs are among the best known professional institutes in India and offer courses in the areas of engineering and management respectively. Both the IIMs and IITs have six campuses located in various cities of India, none of the cities being common.



Q 12. The percentage increase in the number of IIMs pass outs in 2007 over 2004 is
(a) 300% (b) 100%
(c) 200% (d) 75%

Q 13. Compared to 2007, the number of management graduates in 2008 has
(a) Increased
(b) Decreased
(c) Remained same
(d) Cannot be determined

Q 14. How many times during the given period has the growth in pass outs from IITs exceeded 25% over previous year?
(a) Once (b) Twice
(c) Thrice (d) Never

Q 15. During the given period, the total pass outs from IIMs as a ratio of total pass outs from IIT is
(a) 41 : 60 (b) 10 : 13
(c) 6 : 9 (d) 4 : 5

Q 16. What is the ratio of IIM graduates in 1996 to that in 1998?
(a) 0.37 (b) 2.66
(c) 1.04 (d) 0.96

Q 17. On an average, every year 20% of the students who pass IITs go to IIMs the same year. What is the total number (in '000) of IITs pass outs who went to IIM in 2006?
(a) 2 (b) 2.5
(c) 1 (d) Cannot be determined

Direction for questions 18 to 21: Go through the table given below and solve the questions based on it.

The table given below gives the unit cost of transportation of material of three products A, B and C produced by ABC Ltd to five depots D1, D2, D3, D4 and D5. The production figures of the three products A, B and C are 240 units, 1,200 units and 840 units respectively.

Product	Transportation cost per unit (Rs)				
	D1	D2	D3	D4	D5
A	1.5	1.6	2.4	2	1
B	3	3.2	3	2.8	4
C	1.8	1.5	2	1.6	1.1

Q 18. What is the total cost of transportation of all units of all the products to D1?
(a) Rs 5,272
(b) Rs 5,472
(c) Rs 5,282
(d) Rs 5,742

Q 19. If 48 units of A are transported to each of the depots, what is the average cost of transportation?
(a) Rs 1.70
(b) Rs 1.60
(c) Rs 1.65
(d) Rs 1.75

Q 20. Which of the following would involve the least cost?
(a) Transporting 300 units of B to D4
(b) Transporting 200 units of B to D1
(c) Transporting 300 units of C to D2
(d) Transporting 400 units of C to D5

Q 21. By how much is the average transportation cost of B more than that of A?
(a) Rs 1.80 (b) Rs 1.50
(c) Rs 1.60 (d) Rs 1.40

Direction for questions 22 to 25: Read the passage below and solve the questions based on it.

ABC is a firm that deals with furniture. Manufacturing of table requires three levels of assembly. The finished table is at first level. The leg assembly and tabletop are second level. The pieces that go into the leg assembly are at the third level which consist of shortrails, longrails and legs. One unit of table requires one unit of tabletop and one unit of leg assembly. One unit of leg assembly requires 2 units of shortrails, 2 units of longrails and 4 units of legs. Orders are placed just in time to minimize storage.

The lead time for activities are (Lead time is waiting time required to complete one activity)

Parts	Weeks
Assemble table	1
Finished leg assembly	1
Purchase legs	1
Purchase shortrails	1
Purchase longrails	1
Purchase table top	2

The availability of parts at present time:

Parts	Units
Table	50
Leg assembly	100
Legs	150
Shortrails	50
Longrails	0
Table top	50

Demand of finished tables

Details	Week 4	Week 5	Week 6
Demand (units)	200	150	100

Q 22. For meeting the demand of 200 units of finished table of week 4, when would the first order of tabletops be placed?
(a) Week 1
(b) Week 3
(c) Week 5
(d) None of these

Q 23. What is the net requirement of legs for meeting the demand of week 4 finished table?
(a) 200
(b) 50
(c) 400
(d) 800

Q 24. When and how many units of shortrail would be placed for meeting the demand of finished table of week 6?
(a) 100 units in week 1
(b) 200 units in week 3
(c) 300 units in week 6
(d) Cannot be determined

Q 25. Due North Inc. needs a new kind of table in which 8 legs and 4 short rails would be needed for manufacturing a single table. What would be the net requirement of legs and shortrails for meeting the total demand of finished table (including the additional demand of 200 tables for Due North Inc.) in week 5?
(a) 1800, 900
(b) 2200, 1100
(c) 1600, 800
(d) None of these

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(a)	(d)	(c)	(b)	(c)	(d)	(d)	(a)	(a)	(c)	(d)	(b)	(c)	(d)	(d)	(b)	(a)	(d)
Q. No.	21	22	23	24	25															
Answer	(d)	(a)	(b)	(b)	(d)															

HINTS AND SOLUTIONS

We get the following table:

Month	O.S.	Demand	Reserve 20%	Prod. required	Working days
Jan	50	300	60	310	22
Feb	60	500	100	540	19
Mar	100	400	80	380	21
Apr	80	100	20	40	21
May	20	200	40	220	22
Jun	40	300	60	320	20

O.S. = Opening Stock

1. 20% of May demand (200) = 20% of 200 = 40
2. Production requirement for
   Mar = Demand + Reserve – opening stock
   = 400 + 80 – 1000 = 380
5. One worker in January (22 working day);
   Gives = 22 × 8 =1 76 man Hours;
   Total man hours required = 310 x 10 = 3100.
   No. of workers required to be laid off = 20 – 17 =3.
6. Required No. of man hours Available – No. of Man hours utilized = $20 \times 22 \times 8 - 310 = 3520 - 3100 = 420$.
7. No. of man hours required – No. of man hours Available = 540 × 10 – 20 × 19 × 8 = 2360 h

13. It is not given if IIMs are the only institutes providing management degrees, there can be more institutes providing management degree.

15. IITs = 60; IIMs = 40

Ratio of IIM to IIT pass out = $\frac{4}{6} = \frac{6}{9}$

16. $$\frac{\left(6/10\right)}{\left(10/16\right)} = 0.96$$

17. The answer is cannot be determined due to various reasons:
Reason 1 – Data is given as 'on an average' and not the exact data.
Reason 2 – There can be IITs pass outs from yesteryears making it to IIMs in a particular year.
Hence, cannot be determined.

18. Required cost
= (240 × 1.5) + (1200 × 3) + (840 × 1.8)
= Rs 5472.

19. Required average $= \frac{1.5+1.6+2.4+2+1}{5} =$ Rs 1.7

20. The cost of transportation for each options are:
(a) 300 × 2.8 = Rs 840
(b) 200 × 3 = Rs 600
(c) 300 × 1.5 = Rs 450
(d) 400 × 1.1 = Rs 440
Obviously, the cost of transportation of option (d) is the least.

21. We don't know how many units of which product are shipped to which depot.

5

CHAPTER

Advanced Exercises

PRACTICE EXERCISE 1

Direction for questions 1 and 2: Read the following passage and solve the questions based on it.

You are the head of a mutual fund, responsible for investing the available money in different funds/securities. From the market trends you find that blue-chip companies and government stocks will generally not do as well as corporate bonds, in the coming year. However, government regulations require that at least one-third of your mutual fund's capital must be in blue-chip stocks and another third in corporate bonds.

Q 1. Under current regulations, what seems to be the best way for you to invest your funds so as to maximize profit?

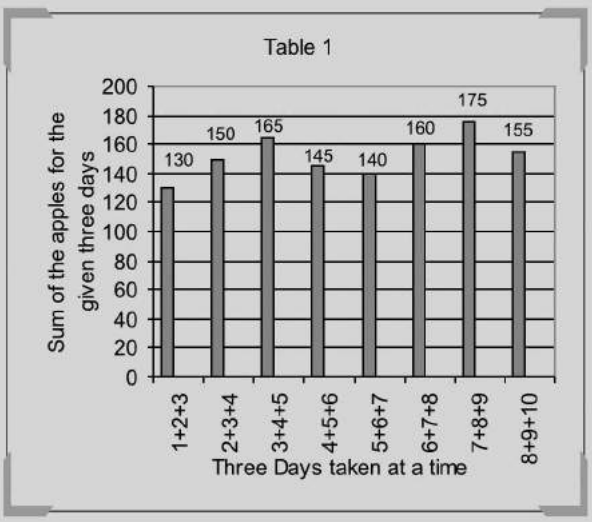
(a) two third government stocks, one-third blue-chip stocks
(b) two third government. stocks, one-third corporate bonds
(c) one third government. stocks, two-third corporate bonds
(d) one third each of government stocks, blue-chip stocks and corporate bonds

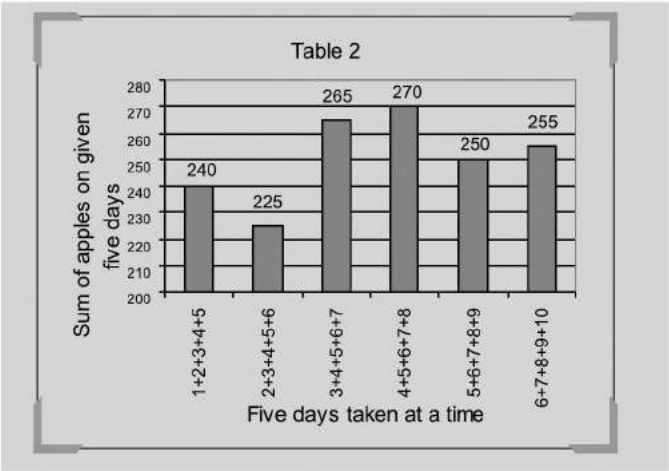
Q 2. If the return on government stocks suddenly goes up by 5%, you would probably:

(a) sell blue-chip stocks to buy government stocks.
(b) sell corporate bonds to buy government stocks.
(c) sell both corporate bonds and blue-chip stocks to buy government stocks
(d) no conclusive action can be taken

Direction for questions 3 to 8: Read the following passage and solve the questions based on it.

Munu, an intelligent baby girl, plucks apples from her father's orchard on ten consecutive days. When asked by her mother as to how many apples has she plucked from the orchard daily, she gives the following sheet to her mother:





Q 3. What is the number of apples plucked on day 6?
(a) 45 (b) 35
(c) 25 (d) Cannot be determined

Q 4. For how many days, is it possible to determine the exact number of apples plucked by Munu?
(a) 0 (b) 5
(c) 3 (d) None of these

Q 5. Disregarding table 2, how many of the following statements are definitely true?
I. Number of apples plucked on day 2 is more than the number of apples plucked on day 5.
II. Number of apples plucked on day 3 is more than the number of apples plucked on day 5.
III. Sum of the apples plucked on day 1 and day 2 is less than the sum of the apples plucked on day 4 and day 5.
IV. Sum of the apples plucked on day 2 and day 3 is less than the sum of the apples plucked on day 3 and day 4.
(a) 0 (b) 1
(c) 2 (d) 3

Q 6. In the above question, how many of the statements can be verified for being right or wrong (using table 1 only)?
(a) 0 (b) 1
(c) 2 (d) 3

Q 7. Disregarding table 2; using how many of the following statements individually can we find the number of apples plucked each day by Munu?
I. Total number of apples plucked from day 1 to day 10.
II. Sum of the number of apples plucked on all the possible pairs of days.
III. Difference in the number of apples plucked in between all the two consecutive days.
IV. Sum of the number of apples plucked in all the four consecutive day periods.
(a) 0 (b) 1
(c) 2 (d) 3

Q 8. Disregarding table 1; using how many of the following statements individually can we find the number of apples plucked each day by Munu?
I. Total number of apples plucked from day 1 to day 10.
II. Sum of the number of apples plucked on all the possible pairs of days.
III. Difference in the number of apples plucked in between all the two consecutive days.
IV. Sum of the number of apples plucked in all the four consecutive day periods.
(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 9 to 14: Read the following passage and solve the questions based on it.

A company 'Y' is engaged in the manufacturing and selling of instant breakfast food. It has a long standing in the consumer market and has captured a large share of the instant food market. As the company enjoys a long standing, and is quality oriented, it has over the years increased its overhead expenses substantially. However, as it has a large market share, it is surviving with good monetary results. The factory is situated in Jagroaonbad which is very far away from the main consumer market (which are the country's capital town and other bigger towns in the vicinity). This distance from the main consumer market has further attributed to an increased cost of the product, being marketed by company 'Y'.

Another company 'X' has come up a few years back, with its factory situated near Ambalpur, a few kilometers from the main capital town and other bigger towns in the area. The product being manufactured by this company is more or less the same in taste, favour, etc. as that of company 'Y'. In view of its proximity with the consumer market, the transportation cost being incurred by company 'X' is comparatively less. Furthermore, being a new enterprise, it has controlled overhead expenses. With this advantage to its credit, it has priced its product a little less than that of company 'Y' and has it slowly penetrated into the market share of company, 'Y'.

The company 'Y' is now faced with the problem of competing with this new company 'X' without, of course decreasing the quality of its product. Company 'Y' however, cannot reduce the price of its product that as will lower the image of the company in the market and it will also reduce its profits. In the present state the company cannot afford to do this. The chief executive of the company is now faced with the problem of competing with company 'X' without lowering the market reputation of his company and influencing its profits.

The chief executive has decided to undertake a detailed market survey and study all types of instant breakfast foods being marketed in the country. By examining the competitor's products it was revealed that the product being marketed by company 'Y' was a quick consumption item of the highest quality, having a much longer shelf life as compared to other products available in the market. This product was packed in tin containers which

after consumption of the contents became scrap without any resale value. The chief executive therefore, decided upon a pre-project study to go in for some different type of packing. The study revealed that they can either go in for glass jars or flexi-packs, or both. As the product is of a high quality and has a longer shelf life, it can withstand climatic conditions in flexi-packs. This, thereby drastically reduces the expenditure on packaging costs (in tin containers this is nearly 38 percent of the total cost of the product and by going in for flexi-packs, the same can be reduced to nearly 15 percent). The saving made in packaging costs can then be released to the consumers by way of reducing the price of the product. This reduction in price will not lower the image of the company as the price reduction will be given along with the background of regarding the change in packaging.

Q 9 Company 'Y', however, cannot outrighthy reduce the price of its product because:
I. of the company's policy
II. it lowers its image in the market
III. it will reduce its profit
(a) I only (b) III only
(c) I and II (d) II and III

Q 10. The selling price of the product manufactured by company 'X' was low because :
I. of less transportation costs
II. they had less overheads as compared to company 'Y'
III. the quantity per unit was less than that of their competitors
(a) I only (b) III only
(c) I and II (d) II and III

Q 11. Why did company 'Y' go for flexi-packs?
I. to be in tune with marketing trends
II. easy to handle on filling lines
III. to save on packaging costs
(a) I only (b) III only
(c) I and II (d) II and III

Q 12. Savings made in cost of packaging
I. helped reduce cost of production
II. helped reduce selling price
III. allow more production
(a) I only (b) III only
(c) I and II (d) II and III

Q 13. Why was company 'Y' selling the same type of product at a rate higher than that of company 'X'?
(I) Before company 'X' entered the market, company 'Y' enjoyed a monopoly in the market and took advantage of the same.
(II) As company 'Y' had a long standing, its overheads were comparatively higher.
(III) Their total cost of production was higher.
(a) I only (b) III only
(c) I and II (d) I, II and III

Q 14. The market survey conducted by company 'Y' revealed that their product was a quick consumption item of a higher quality with a longer shelf-life. What was the conclusion drawn from the survey?
I. Being a quick consumption item, it can have a cheaper packaging, as the problem of aging will not be there.
II. Longer shelf-life shows that the product has durability and therefore can stand for a longer period in the market without changes in the quality of the product.
III. Their product enjoyed better reputation than that of company 'X'.
(a) I only (b) III only
(c) I and II (d) II and III

Direction for questions 15 to 17: Read the following passage and solve the questions based on it.

The table below shows the total number of students who enrolled for five marketing courses over the last four years in a B-school. Answer the following questions based on the table.

Courses	1999	2000	2001	2002
Brand Management (BM)	24	29	30	33
Sales Management (SM)	25	30	26	29
Marketing Research (MR)	43	39	38	40
International Marketing (IM)	29	33	34	32
Consumer Behaviour (CB)	21	19	23	24

Q 15. If it is mandated that a student who wishes to enroll for BM or CB has to take MR, then which of the followings statements is necessarily true?
(a) In 1999, only one student must have enrolled for both BM and CB.
(b) In 2001, not more than 7 students could have enrolled both for BM and CB.
(c) In 2000, at least 9 students must have enrolled for both BM and CB.
(d) In 2002, not more than 5 students could have enrolled both for BM and CB.

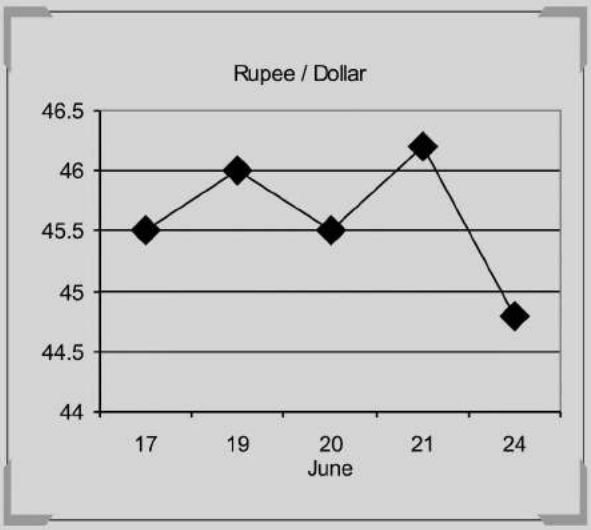
Q 16. It is mandated that those who enroll for IM should take SM or CB. Then, which of the statements is necessarily true?
(a) In 1999, the number of students who enrolled for SM, CB and IM is at least 4.
(b) In 2000, the number of students who enrolled for IM and CB and not SN is not more than 10.
(c) If at least four students who took IM and SM also took CB, then the maximum number of students who took CB but not IM is 17.

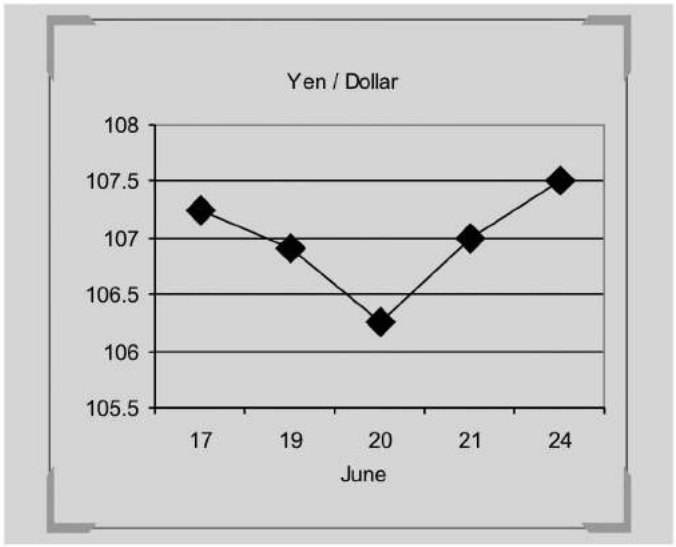
(d) In 2002, the number of students who enrolled for SM, CB and IM is at least 3.

Q 17. If the rules of this school prescribe that a student should take 3 and not more than 3 marketing area courses to specialize in the marketing stream, which of the following statements cannot be true?
(a) In 1999, all those who took CB also took BM and MR.
(b) In 2000, all those who took BM also took SM and IM.
(c) In 2001, all those who took BM also took MR and IM.
(d) all the above

Direction for questions 18 to 21: Refer to the following line charts and solve the questions based on it:

The following line charts show the currency ratio from 17th June to 24th June:





Q 18. On 19th June how much Yen would Rs 8000 amount to?
(a) 18600 (b) 19970
(c) 20200 (d) 17200

Q 19. If the Yen/Dollar ratio follows the same trend from 24th to 27th as from 21st to 24th, then what will be the value of $ 50 on 27th June?
(a) Rs 2000 (b) Rs 5400 Yen
(c) 6000 Yen (d) Rs 3000

Q 20. A jacket costs $ 400. Mohan has Rs 18,200, £ 225 and ¥ 35000 with him. If he can use only one currency to buy the jacket, he would be able to buy the stereo set on 20th June using:
(a) Rs 18,200 (b) £225
(c) ¥ 35000 (d) Cannot be determined

Q 21. On 15th June the Rupee/Dollar ratio was 46.5. On that day Rs 5000 would fetch_____ Pounds.
(a) 80.06 (b) 71.68
(c) 74.15 (d) Cannot be determined

Direction for questions 22 to 25: Read refer to the following table and solve the questions based on it.

The following table shows the highest and the average marks obtained by students in subjects over the years. The maximum mark in each subject is 100.

Q 22. What were the grand average marks of the five subjects in 1996?
(a) 63 (b) 64
(c) 65 (d) 68

Q 23. The difference in the average marks in History between 1994 and 1995 was exactly equal to the difference in the highest marks in Hindi between which of the following pair of years?
(a) 1992 and 1995 (b) 1993 and 1995
(c) 1992 and 1996 (d) none of these

Q 24. What was the approximate percentage increase in average marks in history from 1992 and 1993?
(a) 20%
(b) 25%
(c) 24%
(d) 16%

Q 25. The average highest marks in English in 1992, 1993 and 1996 were exactly equal to the highest marks in Hindi in which of the following years?
(a) 1996
(b) 1997
(c) 1994
(d) 1993

	SUBJECTS									
	English		Hindi		Maths		Sciences		History	
	High	Avg.	High	Avg.	High	Avg.	High	Avg.	High	Avg.
1992	85	62	75	52	98	65	88	72	72	46
1993	80	70	80	53	94	60	89	70	65	55
1994	82	65	77	54	85	62	95	64	66	58
1995	71	56	84	64	92	68	97	68	68	49
1996	75	52	82	66	91	64	92	75	70	58
1997	82	66	81	57	89	66	98	72	74	62

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(a)	(b)	(d)	(b)	(c)	(b)	(c)	(d)	(c)	(b)	(c)	(b)	(c)	(c)	(a)	(c)	(a)	(b)	(a)
Q. No.	21	22	23	24	25															
Answer	(d)	(c)	(a)	(a)	(d)															

HINTS AND SOLUTIONS

3. Do (1 + 2 + 3) + (4 + 5 + 6) in table 1 and then subtract (1 + 2 + 3 + 4 + 5) from it. Therefore, the correct answer is option (b) 35 apples.

4. We can find it out for all the days starting from day 1. See the solution of Q 3.

5. Only statement III is correct. To find the veracity of statement III, compare (1 + 2 + 3) with (3 + 4 +5)

6. Only statement I and statement III can be verified.

7. Except I, all give us the solution.

9. It is clearly given that company 'Y' cannot reduce the price as it lowers the image and reduces its profit also. Therefore, option (d) is the correct answer.

10. Transportation cost of company 'X' is comparatively less. Besides being a new entrant, it has kept its overhead expenses low. Therefore, option (c) is the correct answer.

11. Flexi-packs reduce the cost of packaging. Therefore, option (b) is the correct answer.

12. The savings made can be trickled down to the customers by way of reducing the price of the product. Therefore, option (c) is the correct answer.

13. Neither statement I nor II can be verified. However, statement III is mentioned very clearly. Therefore, option (b) is the correct answer.

14. Only statement I and II can be verified. It is given that quality of product 'Y' was better than all others, but it cannot be concluded from here that it enjoyed better reputation than company 'X'. Therefore, option (c) is the correct answer.

15. (29 + 19) – 39 = 9. Hence in 2000, at least 9 students must have enrolled for both BM and CM courses.

16. In 1999, difference of students in IM and SM = 29 – 25

In SM and CB = 25 – 21 = 4

Hence, atleast 4 students enrolled for SM, CB and IM.

17. Since, in 2001, number of students in BM > number of students in SM

22 $\text{Average} = \frac{52+66+64+75+58}{5} = \frac{315}{5} = 63$

23. The difference is 9. Therefore the correct answer is option (a)

24. $\text{Percentage Increase} = \frac{55-46}{46} \times 100 = 20\%$

25. Average highest marks

$$\frac{85+80+75}{3} = \frac{240}{3} = 80$$

Therefore the correct answer is option (d)

PRACTICE EXERCISE 2

Direction for questions 1 to 4: Refer to the following table and solve the questions based on it.

Table A
Manpower Costs for a Building Project in Narhi
(All costs in Rs)

Project Phase	Managers	Engineers	Skilled Workers	Unskilled Workers	Supervisors
Design	1	3	--	--	--
Foundation	1	2	5	23	2
Construction	1	2	12	43	4
Finishing (Internal)	1	3	15	31	6
Painting	1	0	5	23	3

Table B
Cost per month for Select Category of Workers

Manpower	Cost (in Rs)
Managers	10,000
Engineers	8,000
Supervisors	5,000
Skilled Workers	3,000
Unskilled Workers	1,800

Table C
Estimated Time and the Actual Time Required for Each Phase of the Project in Months

	Design	Foundation	Construction	Finishing	Painting
Actual	5	3	8	4	2
Estimated	3	2	6	3	2

Q 1. In which phase there was the highest absolute increase between the estimated and the actual cost?
(a) Foundation (b) Design
(c) Construction (d) Cannot be determined

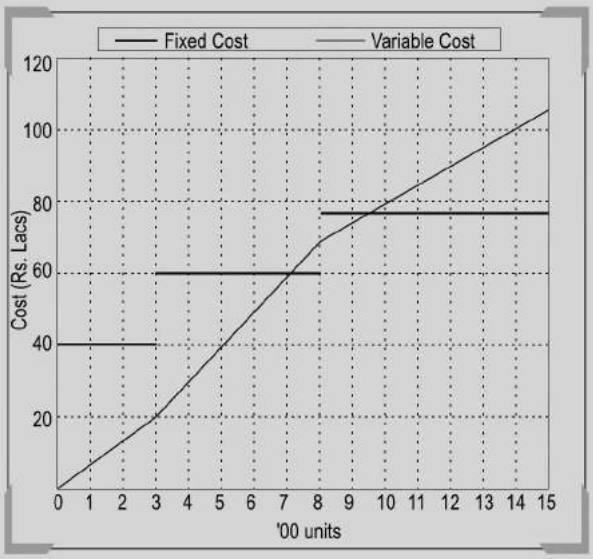
Q 2. For which phase of the project was the highest percentage increase in cost (as compared to the estimated cost)?
(a) Foundation (b) Design
(c) Construction (d) Cannot be determined

Q 3. What is the approximate value of the percentage overflow in costs during construction?
(a) 50% (b) 33%
(c) 66% (d) None of these

Q 4. If internal finishing and painting can be started simultaneously at the end of construction, then what is the overflow of the project as a percentage of the estimated total time for completion of the project?
(a) 1. 27% (b) 33%
(c) 42% (d) Cannot be determined

Direction for questions 5 to 9: Read the following passage and solve the questions based on it.

The following graph shows the variation in the variable and fixed cost of production of a TV. The fixed cost remains fixed in that particular slab and the variable cost varies according to the units produced. It can be seen that the fixed cost remains fixed for the first 300 units, then for the next 500 units and then, for any higher level of units produced. The total production cost is equivalent to the sum of the variable and the fixed cost. It is to be assumed that there is no other cost involved in producing the TVs.



Q 5. At a production level of 500 TVs, the selling price per unit is such that there is a net profit of 20%. What will be the impact on the net profit if the selling price per unit is reduced by 16.66% and the total number of units produced is increased by 40%?
(a) it will increase by less than 20% but more than 10%
(b) it will increase by less than 30% but more than 20%
(c) it will decrease by less than 15% but more than 5%
(d) none of these

Q 6. For production levels of less than 1500 units, the variation of the average cost of production per unit has the following relationship with the volume of production:
(a) decreases uniformly
(b) increases uniformly
(c) decreases, then increases and then increases again
(d) None of these

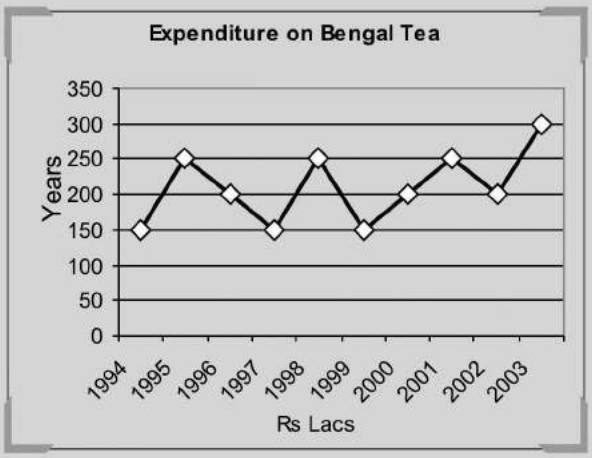
Q 7. The highest production cost per unit will be observed at the following level of production:
(a) 0 –100 (b) 300 – 400
(c) 800 – 900 (d) 1400 – 1500

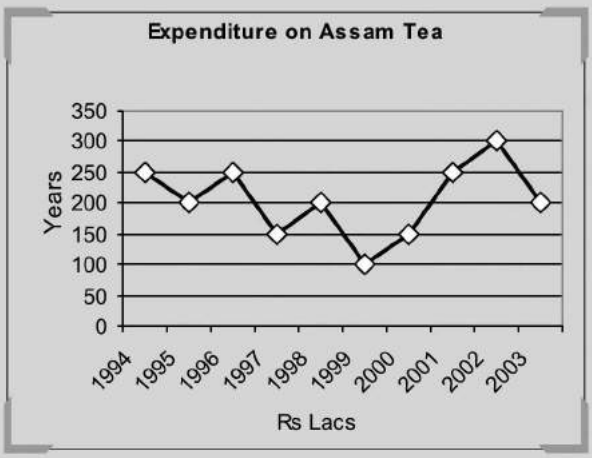
Q 8. What should be the selling price per unit (in Rs) To achieve a break-even if the total number of units produced is 1000?
(a) 12500 (b) 15500
(c) 17500 (d) cannot be determined

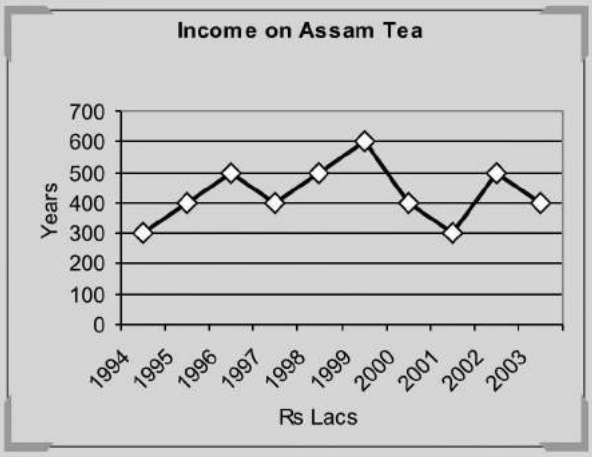
Q 9. What is the approximate additional cost for producing a TV after 920 units have been produced?
(a) 4000 (b) 5000
(c) 6000 (d) 7500

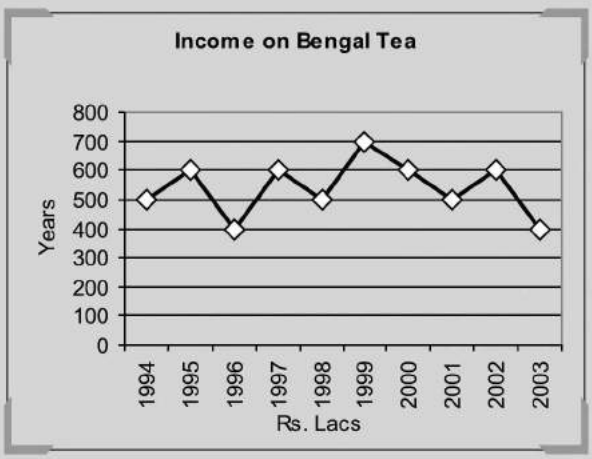
Direction for questions 10 to 14: Refer to the following line charts and solve the questions based on than.

The line charts given below give the expenditure and income of Tata Tea which produces and sells only two variants of its tea—Assam tea and Bengal tea.









Profit = Income – Expenditure

Q 10. In which of the following years, is the total profit of Tata Tea the highest?
(a) 1997 (b) 1999
(c) 2002 (d) 1994

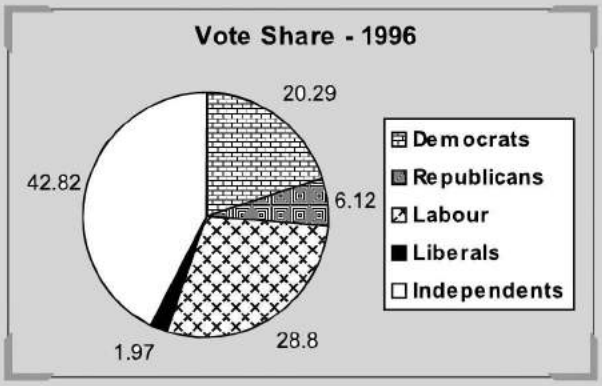
Q 11. For how many years during the period 1995 to 2003, for at least one of the two teas did the income and expenditure not follow the same trend, i.e., both of them did not decrease or did not increase together over the previous year ?
(a) 5 (b) 7
(c) 8 (d) 9

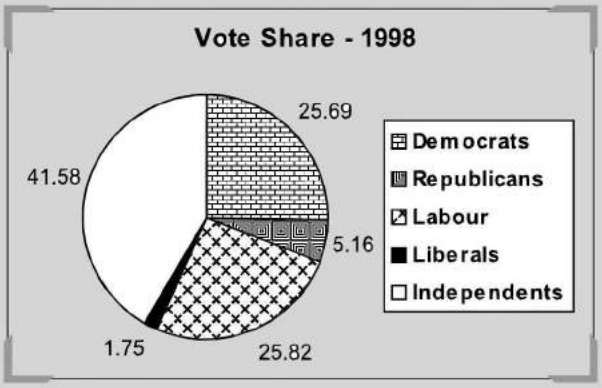
Q 12. In which of the following years was the percentage increase in the income of Bengal Tea over the previous year, the highest?
(a) 1995 (b) 1997
(c) 1999 (d) 2002

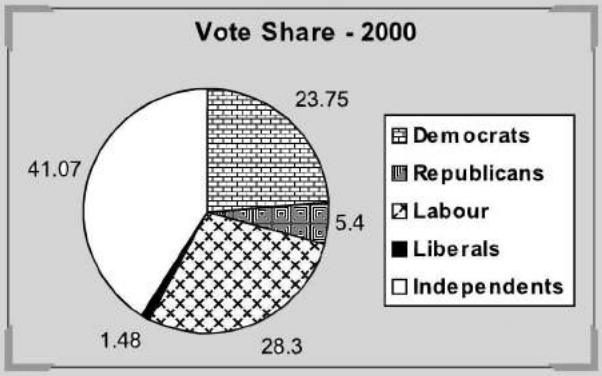
Q 13. For how many years during from 1995 to 2003, was there a growth in income compared to that of the previous year but there is a decrease in the income from the year to the next year for each of the two teas?
(a) 1 (b) 2
(c) 3 (d) 4

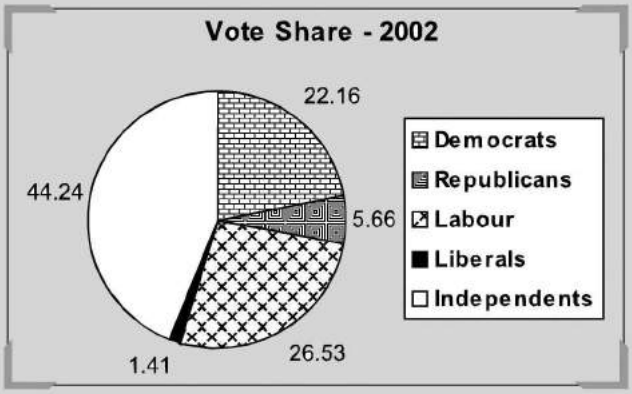
Q 14. In which of the following years during 1995 to 2003, did the profit of Assam Tea decrease by the maximum amount when compared to that of the previous year?
(a) 1999 (b) 2000
(c) 2001 (d) 2003

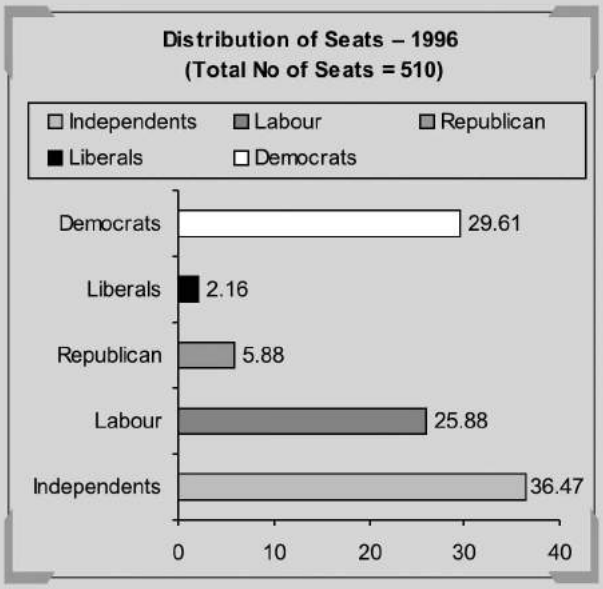
Direction for questions 15 to 18: Refer to the following graphs and solve the questions based on them.

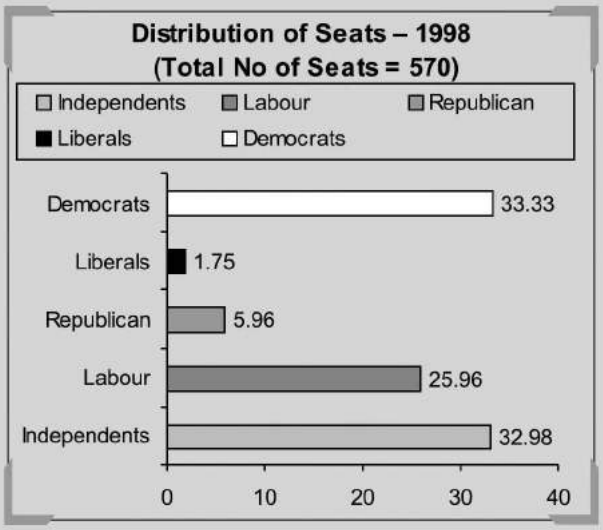






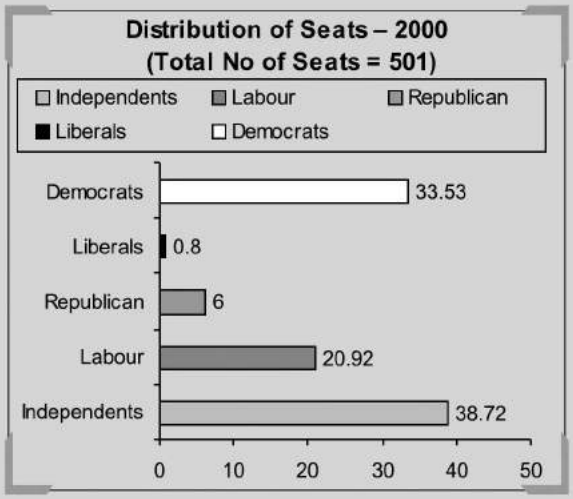


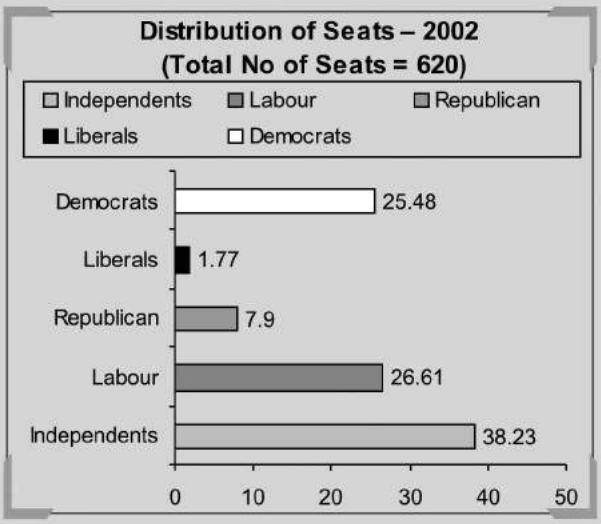




The given graphs show the vote share and distribution of seats of different parties in the election years during the period 1996-2002. Study the graphs and answer the questions, taking into account the following:

(i) Independents may be considered as a separate party





Q 15. How many of the following statements are correct?

I. In terms of vote share, Labour and Liberal parties taken together lost 3.2 percent in 1998 but gained a total of 16 seats in the same period.

II. Democrats, Republicans and 35% of the Independents could have formed the government only in two election years.

III. No party increased its vote share in every succeeding election.

IV. In the 2000 elections, as compared to the 1996 elections, Republicans and Democrats taken together, gained more in terms of vote share than in terms of percentage of seats.

(a) 0 (b) 1
(c) 2 (d) 3

Q 16. How many of the following statements are incorrect?

I. In the year 2002, in terms of percentage of seats obtained, Democrats and Labour together lost the maximum.

II. Liberals and Republicans taken together lost both in terms of vote share and the number of seats obtained in the year 2000.

III. In the 2000 elections, all the parties lost in terms of number of seats.

IV. The highest gain in the number of seats for Labour was in the year 2002.

(a) 0 (b) 1
(c) 2 (d) 3

Q 17. How many of the following statements are correct?

I. Democrats and Labour together improved their vote share by a larger margin in 1998 over 1996 as compared to the one in 2002 over 1996.

II. The number of seats lost by the Democrats in the 2002 elections was less than the number of seats gained by the Republicans in the same year.

III. In 2002, the Independents gained the most, both in terms of vote share and the number of seats obtained.

IV. If 70% of the Independents had joined Labour, they could have formed a government in 2002 but not in 1996.

(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 18 to 22: Refer to the following table and answer the questions based on it.

Unless otherwise stated, all changes (increases, decreases) indicated in the questions must be calculated over the immediately preceding year.

Q 18. Which of the following statements is not correct?

(a) Wipro's share in the total sales of the electronic sector was lower than the share of Tata Steel in the total sales of the iron and steel sector in each of the given years.

(b) From the group consisting of the textiles and the iron and steel sector, there were five companies whose salaries and wages as a percentage of the total salaries and wages of their respective sectors increased in 2007 as compared to 2005.

(c) Tata Steel's R & D expenditure In each of the given years as a percentage of the sales was lower than that of the iron and steel sector as a whole.

(d) From the group consisting of the pharmaceutical and the electronic sectors, there were five companies which experienced a decline in the growth rate of sales in 2007.

Q 19. In this question, a few statements given is/are right and a few statements is/are wrong. What is the ratio of the right to the wrong among the following statements?

(Rs crores)

Sector/Companies	Sales			Salaries and Wages			R & D			Profit / Loss		
	2005	2006	2007	2005	2006	2007	2005	2006	2007	2005	2006	2007
Textiles (Total)	53145	50184	52616	4296	4360	4315	26	36	39	–3770	–3437	–2441
Indo Rama Synthetics Ltd	2000	1953	2224	48	50	33	0	0	0	19	41	138
Arvind Mills Ltd	1970	740	1552	136	46	102	0	0	0	–499	20	129
Raymond Ltd	1477	992	1035	160	167	168	0	0	1	332	88	90
Century Enka Ltd	960	892	946	32	34	35	0	0	1	70	35	58
Pharmaceuticals (Total)	25245	30273	34731	2009	2291	2716	538	820	993	1758	2586	3069
Ranboxy Laboratory Ltd	2363	3461	4243	167	197	255	77	192	276	262	478	795
Dr. Reddy's Laboratories Ltd	991	1712	1705	83	121	137	42	102	163	144	460	392
Cipla Ltd	1064	1401	1573	50	63	73	41	47	0	179	208	248
Glaxosmithkline Ltd	1143	1197	1242	139	139	139	4	4	4	49	98	172
Electronic (Total)	26944	29323	30022	2164	2218	2323	200	263	247	830	692	–287
Infosys Technologies Ltd	1901	2604	3623	718	1119	1678	17	15	14	629	808	958
Videocon International Ltd	3244	4974	3602	43	63	53	0	0	0	155	156	–100
Bharat Electronic Ltd	1722	1947	2517	394	364	368	89	90	109	155	200	261
Iron and Steel (Total)	54396	55582	70341	5367	5643	6377	73	64	74	–2983	–4604	310
Steel Authority of India Ltd	17320	16624	20665	3135	3255	3728	52	50	55	–729	–1707	-304
Tata Steel Ltd	8491	8277	10517	922	1099	1218	10	8	16	553	205	1012
Rashtriyas Ispat Nigam Ltd	3585	4200	5187	408	376	407	0	3	0	–291	–75	521
Ispat Industries Ltd	3217	2812	4658	45	40	9	9	0	0	–312	–443	83

I. Total salaries wages for 2005-2007 as a percentage of the total sales of the same period had been the highest for the iron and steel sector.

II. In the year 2007, if all the companies were ranked in a descending order in terms of salaries and wages as a percentage of sales; Videocon International would have been ranked the lowest.

III. In the year 2006, if all the companies were ranked in a descending order in terms of R& D as a percentage of sales; Bharat Electronics Limited would have been ranked third.

IV. Of all the companies that made profits every year during 2005-2007, Wipro registered the highest growth in profits for the given period.

(a) 3:1
(b) 1:1
(c) 1:3
(d) All the statements are correct

Q 20 How many of the statements given below is/are correct?

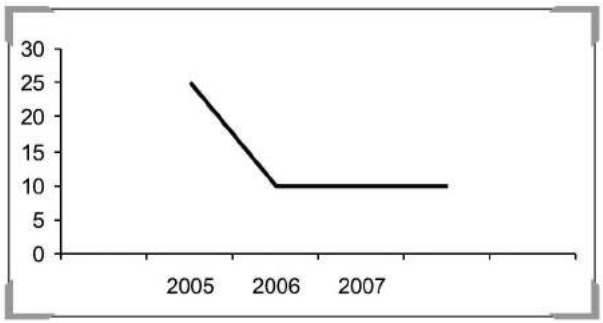
I. Salaries and wages of each company in the iron and steel sector were up to 13 times its losses reported in each of the loss-making years.

II. Total salaries and wages during 2005-2007 as a percentage of the total sales in the same period was minimum for the pharmaceutical sector.
III. Total salaries and wages during 2005-2007 as a percentage of the total sales in the same period for only two companies exceeded 20 percent.
IV. Total profits/Total sales for all the four sectors taken together was higher in 2006 than in 2005.

(a) 0 (b) 1
(c) 2 (d) 3

Q 21. Which of the following statements is incorrect?
(a) Cipla experienced the highest percentage decline in its R&D expenditure in any single year during the given period.
(b) Of all companies which incurred R&D expenditure every year during 2005-2007, the total R&D expenditure/total sales was the highest for Dr Reddy's Laboratories.
(c) During 2005-2007, in terms of sales growth, the best performer in the pharmaceutical sector fared better than the best performer in the iron and steel sector.
(d) Videocon International Ltd experienced the second largest percentage decline in salaries and wages in any single year during the given period.

Q 22. How many of the situations described in the options below, when plotted closely resemble the figure?



I. Raymond Limited 's share in the total sales of the textile sector
II. Tata Steel's share in the total sales of the iron and steel sector
III. Glaxosmithkline Ltd's salaries and wages as a percentage of sales
IV. Raymond Limited's profit as a percentage of sales

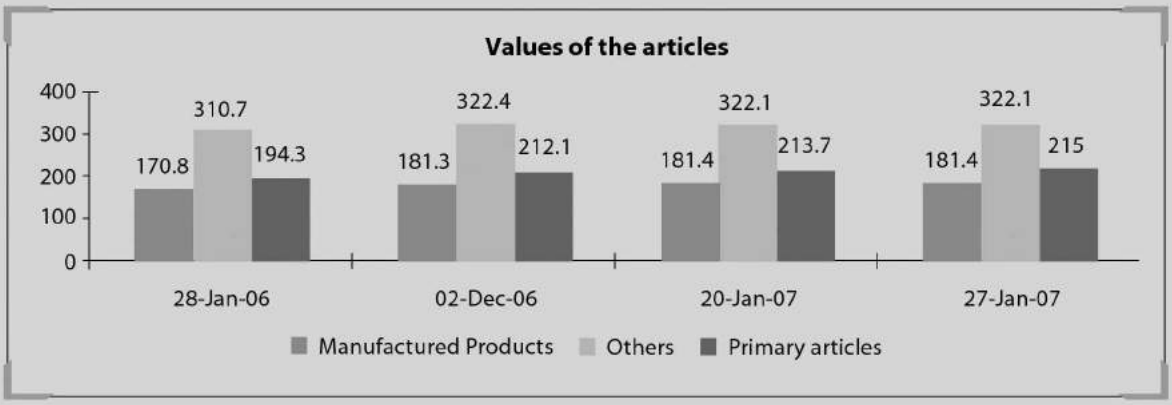
(a) 1 (b) 2
(c) 3 (d) 4

Direction for questions 23- 25: Go through the graphs below and solve the questions based on it.

Inflation is calculated on the basis of the wholesale price index. This index contains three groups of articles– Manufactured products, Primary articles and Others.

Following table represents the weightage of these three groups while calculating the inflation:

	Manufactured Products	Primary Articles	Others
Weightage	63.75%	22.02%	14.23%



Q 23 What is the rate of inflation on 20th Jan 07 over the previous given day?
(a) 5.21% (b) 6.11%
(c) 6.91% (d) 7.81%

Q 24 What is the rate of inflation on 27th Jan 07 over the previous given day?
(a) 5.12% (b) 2. 6%
(c) 6.58% (d) 7.91%

Q 25 What is the rate of inflation on 28th Jan 06 over the previous given day?
(a) 5.12%
(b) 6%
(c) 6.58%
(d) cannot be determined

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(c)	(d)	(d)	(d)	(d)	(d)	(b)	(b)	(c)	(b)	(b)	(b)	(b)	(b)	(c)	(c)	(b)	(a)
Q. No.	21	22	23	24	25															
Answer	(d)	(b)	(b)	(c)	(d)															

HINTS AND SOLUTIONS

5. For n = 500, CP/unit = Rs 20,000
SP/unit = Rs 24,000
Profit/unit = Rs 4000
Hence total profit = Rs 20 lacs
Now, if the production is increased by 40%, then the total number of units produced = 7000 units.
CP/unit = Rs 120000/7
SP/unit = Rs 24000
Total profit = Rs 20 lacs
So, the net profit will be the same.
Therefore, the correct answer is option (d)

6. It can be calculated that the average cost decreases from 0 to 300 units, then increases for the the 301st unit, then decreases till the 800th unit and then increases till the 801st unit and then decreases again.
Therefore, the correct answer is option (d)

7. From the solution of Q 6. We can see that option (a) is the correct answer.

8. Total cost of production for 1000 units = 155 × 105. Hence, to obtain the break even, the SP/unit = Rs 15,500. Therefore option (b) is the correct answer

PRACTICE EXERCISE 3

Direction for questions 1 to 4: Refer to the following tables and solve the questions based on them.

Table 1 gives the transportation cost of a product from six factories F1, F2, F3, F4, F5 and F6 to five warehouses W1, W2, W3, W4 and W5.

	W1	W2	W3	W4	W5
F1	60	25	45	40	80
F2	85	40	45	60	75
F3	80	20	45	50	90
F4	90	40	55	25	70
F5	95	20	40	45	80
F6	85	35	50	50	75

Table 2 gives the transportation cost of the same product from five warehouses W1, W2, W3, W4 and W5 to six retailers R1, R2, R3, R4, R5 and R6.

	R1	R2	R3	R4	R5	R6
W1	40	20	40	30	85	25
W2	50	50	45	60	70	30
W3	20	10	50	50	80	35
W4	40	30	45	40	90	40
W5	50	30	50	40	70	30

Q 1. What is the minimum transportation cost incurred in transferring the product from any factory to any retailer?
(a) Rs 45 (b) Rs 50
(c) Rs 55 (d) Rs 60

Q 2. What is the maximum transportation cost incurred in transferring the product from any factory to any retailer?
(a) Rs 175 (b) Rs 180
(c) Rs 170 (d) None of these

Q 3. What is the minimum transportation cost at which the product can be supplied to R4 from any factory?
(a) Rs 45 (b) Rs 50
(c) Rs 65 (d) Rs 70

Q 4. What is the maximum transportation cost at which the product can be supplied to R3 from any factory?
(a) Rs 145 (b) Rs 150
(c) Rs 140 (d) None of these

Direction for questions 5 to 10: Answer the questions based on the following information:

In addition to the above tables given, the following table gives the transportation cost incurred for transferring products from different retailers to the five different colonies C1, C2, C3, C4 and C5.

	C1	C2	C3	C4	C5
R1	10	25	10	10	10
R2	15	10	15	10	15
R3	10	20	10	15	10
R4	20	10	15	15	10
R5	15	10	20	20	20
R6	10	15	15	15	15

Q 5. What is the minimum transportation cost incurred in transferring the product from any factory to any colony?
(a) Rs 45 (b) Rs 65
(c) Rs 55 (d) none of these

Q 6. What is the minimum transportation cost incurred in transferring the product from any factory to the colony C2?
(a) Rs 45 (b) Rs 60
(c) Rs 55 (d) None of these

Q 7. What is the maximum transportation cost incurred in transferring the product from any factory to the colony C5?
(a) Rs 175 (b) Rs180
(c) Rs185 (d) None of these

Q 8. If one unit produced in each of the factories is to be sent to each of the colonies, then what will be the total transportation cost incurred?
(a) Rs 1705 (b) Rs 1980
(c) Rs 2185 (d) None of these

Q 9. $F_iW_jR_kC_l$ represents that the product has moved in this cycle – Factory i to Warehouse j to Retailer k to Colony l. If i = 2 and k = 2, then what is the sum of (j + l) for the lowest transportation cost incurred in one product transferred from i = 2 to any of the colonies?
(a) 4 (b) 5
(c) 7 (d) Cannot be determined

Q 10. In the above question, if j = 2, then how many different value/s of (i + k + l) is/are possible for the lowest transportation cost incurred in one product transferred from any of the factories to any of the colonies?
(a) 1 (b) 2
(c) 3 (d) Cannot be determined

Direction for questions 11 to 15: Refer to the following table and solve the questions based on it.

Given below are the forecasts of the world and Asian energy demand for the years 1991, 2001 and 2011. The demand is given in million barrels per day, crude oil equivalent.

	1991		2001		2011	
	World	Asia	World	Asia	world	Asia
Petrol	50.0	4.0	70.0	10.0	80.0	15.0
Naphtha	30.0	0.5	40.0	2.5	50.0	5.0
Solid Fuels	50.0	4.0	60.0	5.0	75.0	10.0
Nuclear Fuel	10.0	0.5	20.0	1.0	25.0	1.3
Hydro Energy	10.0	1.0	10.0	1.5	20.0	2.0
Total	150.0	10.0	200.0	20.0	250.0	33.3

Q 11. What can be inferred from the given table?
(a) Percentage increase in the number of petrol using vehicles/agents has increased in Asia rather than the world, in the decade 1991-2001.
(b) Exploration of Naphtha is low in Asia as compared to the exploration of Naphtha in the world throughout the period.
(c) Both (a) and (b)
(d) None of these

Q 12. During 1991-2011, which two fuels meet more than 60% of the total energy demand of both the world and Asia?
(a) Petrol and Naphtha
(b) Petrol and Solid fuels
(c) Naphtha and Solid fuels
(d) None of these

Q 13. Which fuel's proportion in the total energy demand increases over the decade 1991-2001 and decreases over the decade 2001-2011 for both the world and Asia?
(a) Petrol (b) Naphtha
(c) Solid fuels (d) Nuclear fuel

Q 14. In Asia for which fuel does the proportion in the total energy demand decrease continuously over the period 1991-2011?
(a) Naphtha (b) Solids fuels
(c) Nuclear fuel (d) Hydro energy

Q 15. For which fuel does the proportion in the total energy demand of the world remain constant over the period 1991-2011; but its proportion increases in the total energy demand in Asia?
(a) Solid fuels (b) Nuclear fuel
(c) Hydro energy (d) Naphtha

Direction for questions 16 to 20: Refer to the following table and solve the questions based on it.

The following table gives some incomplete information on the marks obtained by four persons Amit, Binit, Charu and Dilip in four different subjects.

	Physics	Chemistry	Maths	English	Total
Amit	80	70	-	80	-
Binit	60	-	60	-	-
Charu	-	-	60	-	220
Dilip	90	-	70	60	-

(i) Amit's average marks were 72.5, in all the four subjects.
(ii) The average marks in Physics for all four persons were 67.5.
(iii) Binit's average marks for all four subjects were 1.5 times of Charu's marks in Physics.
(iv) Binit and Dilip scored the same marks in English. The average English marks were the same as the average Physics marks.
(v) The total marks in Chemistry were 10 less than the total marks in Physics for all four persons.

Q 16. Who scored the lowest total?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Q 17. The approximate average marks for all four persons (for all the four subjects) are
(a) 65.625 (b) 81.225
(c) 99.5 (d) 105.6

Q 18. Who is the highest scorer in Chemistry?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Q 19. Who is the lowest scorer in Physics?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Q 20. Who is the highest scorer in English?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Direction for questions 21 to 23: *Refer to the following data-set and solve the questions based on it.*

The following table gives the price of petrol and diesel in India for the given period.

Date	Petrol Change (+/-)	Diesel Change (+/-)
May 2004	33.70	21.74
16 June 2004	35.71	22.74
1 August 2004	36.81	24.16
5 November 2004	39.00	26.28
16 November 2004	37.84	
1 April 2005 (VAT)	37.99	28.22
5 May 2005 (VAT on diesel)		26.45
21 June 2005	40.49	28.45
7 September 2005	43.49	30.45
1 April 2006	43.51	30.39
5 June 2006	47.51	32.47
June 16, 2006	46.85	32.25
November 30, 2006	44.85	31.25
February 15, 2007	42.85	30.25

Q 21. How many times did the price of petrol increase during the given period?
(a) 7 (b) 8
(c) 9 (d) 10

Q 22. How many times did the price of diesel increase during the given period?
(a) 7 (b) 8
(c) 9 (d) 10

Q 23. For how many times is the percentage increase in the price of petrol more than the percentage increase in the price of diesel over their respective previous date?
(a) 4 (b) 5
(c) 6 (d) 7

Direction for questions 24 to 25: Refer to the following data-set and solve the questions based on it.

REDUCTION IN FUEL PRICES (Rs Per Litre)

Feb 16, 2007	Petrol		Diesel	
	Old	New	Old	New
Delhi	44.85	**42.85**	31.25	**30.25**
Mumbai	50.85	**48.85**	36.11	**34.97**
Kolkata	48.99	**46.91**	33.92	**32.88**
Chennai	49.67	**47.51**	34.41	**33.22**

Q 24. In which city out of the given four cities has the price of petrol decreased by the maximum percentage?
(a) Delhi (b) Mumbai
(c) Kolkata (d) Chennai

Q 25. In which city out of the given four cities has the price of diesel decreased by the maximum percentage?
(a) Delhi (b) Mumbai
(c) Kolkata (d) Chennai

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(c)	(c)	(b)	(b)	(d)	(d)	(d)	(d)	(d)	(b)	(a)	(d)	(d)	(c)	(a)	(d)	(c)	(a)
Q. No.	21	22	23	24	25															
Answer	(b)	(b)	(b)	(a)	(c)															

HINTS AND SOLUTIONS

1. F3 to W2 to R6 is the lowest. Therefore, option (b) is the correct answer.
2. F5 to W1 to R5 is the maximum. Therefore, option (b)is the correct answer.
3. F4 to W4 to R4 is the lowest. Therefore, option (c) is the correct answer.
4. F3 to W5 to R3 is the maximum. Therefore, option (c) is the correct answer
5. F3 to W2 to R2 to C1 is the lowest. Therefore, option (b) is the correct answer
6. F3 to W2 to R6 to C2 is the minimum. Therefore, option (c) is the correct answer
9. F2 – W3 – R2 – either C2/ C4. Hence two different values are possible. Therefore, option (d) is the correct answer.
10. There can be two values of i = 3/5, one value of j = 2, one value of k = 6 and one value of l = 1. Hence, two different sums are possible. Therefore, option (d) is the correct answer.
11. Option (b) is obviously cannot be inferred. Option (a) seems to be probable, but number of vehicles/agents need not be increasing. There can be other reasons also like reduction in the efficiency of the vehicle etc. Therefore option (d) is the correct answer.

12.

		1991		200		2011	
		World	**Asia**	**World**	**Asia**	**World**	**Asia**
Total energy		150	10	200	20	250	33.3
Naphtha	Value	30	0.5	40	2.5	50	5
	Proportion	20%	5%	20%	12.50%	20%	15.01
Solid fuels	Value	50	4	60	5	75	10
	Proportion	**33.30%**	**40%**	**30%**	**25%**	**30%**	**30.03**
Petrol	Value	50	4	70	10	80	15
	Proportion	**33.30%**	**40%**	**35%**	**50%**	**32%**	**45.04**

Hence, solid fuels and petrol combined constitute more than 60 per cent of the total energy in both the world and Asia for the given period.

13. It can be seen from the table that Petrol is the fuel whose proportion in the total energy demand increases during 1991-2001 and decreases during 2001-2011 for both the world and Asia. Therefore option (a) is the correct answer

14. For the answer choices given and for Asia we can make the following table.

		1991	2001	2011
	Total energy	**10**	**20**	**33**
Naphtha	Value	0.5	2.5	5
	Proportion	**5%**	**12.50%**	**15.15%**

		1991	2001	2011
	Total energy	10	20	33
Solid Fuels	Value	4	5	10
	Proportion	**40%**	**25%**	**30.30%**
Nuclear Fuel	Value	0.5	1	1.3
	Proportion	**5%**	**5%**	**3.90%**
Hydro Energy	Value	1	1.5	2
	Proportion	**10%**	**7.50%**	**6.06%**

Hence, we can see that the proportion of Hydro Energy goes on decreasing over the given period.

15. For the answer choices given and for the world, we can make the following table.

		1991	2001	2011
	Total energy	**150**	**200**	**250**
Naphtha	**Value**	30	40	50
	Proportion	20%	20%	20%
Solid Fuels	**Value**	50	60	75
	Proportion	33%	30%	30%
Nuclear Fuel	**Value**	10	20	25
	Proportion	6.66%	10%	10%
Hydro Energy	**Value**	10	10	20
	Proportion	6.66%	5%	8%

Hence, we can see that the proportion of Naphtha remains constant over the given period.

From the given information the following table can be prepared.

	Physics	Chemistry	Maths	English	Total
Amit	80	70	60	80	290
Binit	60	60	60	60	240
Charu	40	50	60	70	220
Dilip	90	80	70	60	300

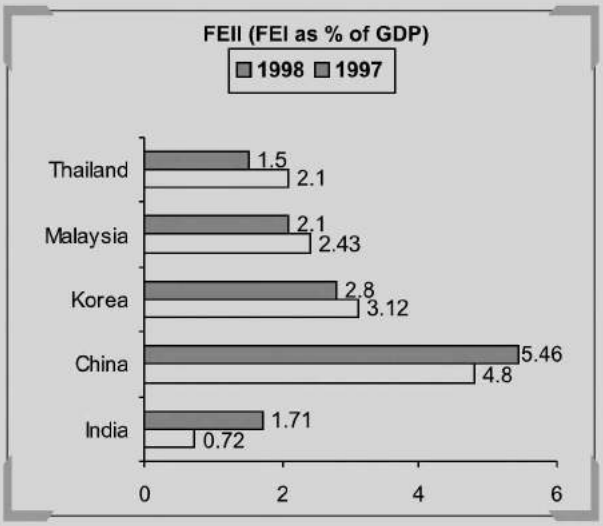
Following is the complete list:

Date	Petrol Change (+/–)	Diesel Change (+/–)
May 2004	33.70	21.74
16 June 2004	35.71 +2.01	22.74 (+1.00)
1 August 2004	36.81 +1.10	24.16 (+1.42)
5 November 2004	39.00 +2.19	26.28 +2.12
16 November 2004	37.84 –1.16	
1 April 2005 (VAT)	37.99 + 0.15	28.22 + 1.94
5 May 2005 (VAT on diesel)		26.45 (–1.77)
21 June 2005	40.49 + 2.50	28.45 (+2.00)
7 September 2005	43.49 +3.00	30.45 (+2.00)
1 April 2006	43.51 +0.02	30.47 (+0.02)
5 June 2006	47.51 +4.00	32.47 (+2.00)
16 June 2006	46.85 –0.66	32.25 (–0.22)
30 November 2006	44.85 –2.00	31.25 (–1.00)
15 February 2007	42.85 –2.00	30.25 (–1.00)

PRACTICE EXERCISE 4

Direction for questions 1 to 5: Refer to the following data set and solve the questions based on it.

The following graph gives the data on the foreign equity inflow index (FEII) for five countries during 1997 and 1998. FEII is taken as the ratio of the foreign equity inflow (FEI) to the country's GDP. FEII is expressed as percentage in the graph.



Q 1. Which country has shown the maximum percentage change in FEII in 1998 relative to 1997?
(a) India
(b) China
(c) Korea
(d) Malaysia

Q 2. If China's FEI is 10 times that of India in the year 1998, which of the following is true for 1998?
(a) China's GDP is 70% more than India's GDP
(b) China's GDP is 40% more than India's GDP
(c) China's GDP is 50% more than India's GDP
(d) Nothing can be inferred

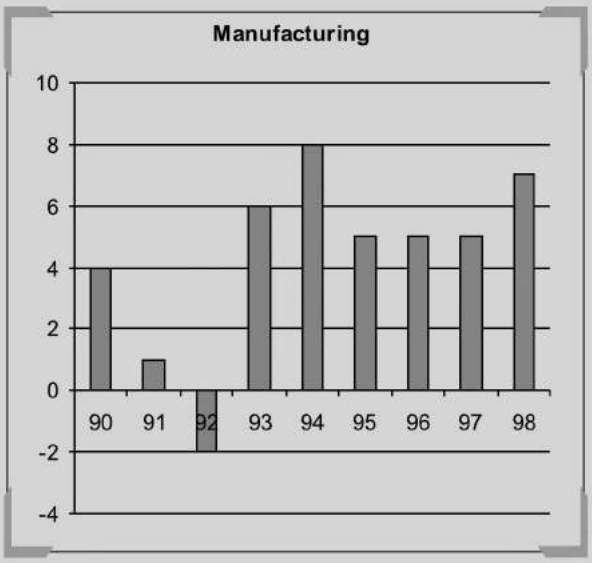
Q 3. If the GDP of India increased by 5%, GDP of China increased by 7% and that of Korea decreased by 2% 1997 to 1998, which of the following is/are true?
I. FEI to China was higher in 1998 than in 1997
II. FEI to China was lower in 1998 than in 1997
III. India's FEI has increased from 1997 to 1998
IV. FEI to Korea was lower in 1998 than in 1997
V. Korea's FEI has increased in 1998 from 1997
(a) I and III
(b) II and V
(c) II, III and V
(d) All of the above

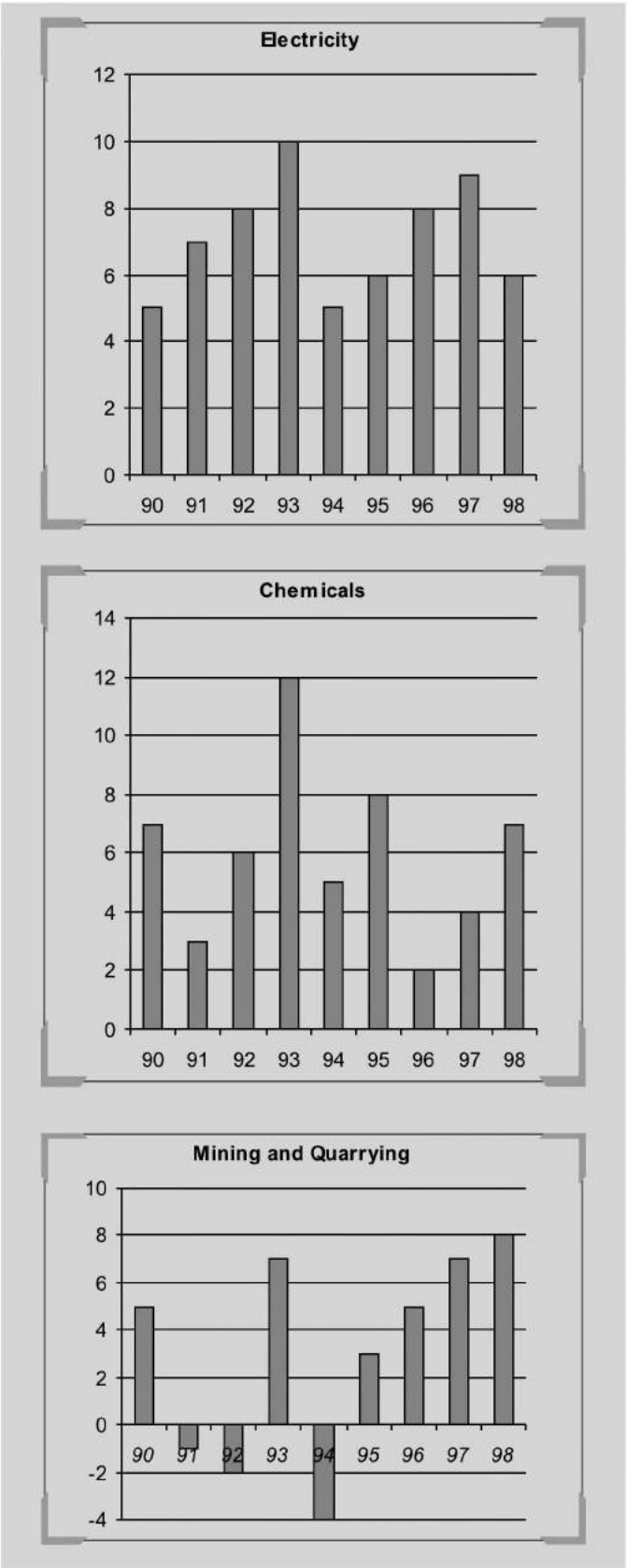
Q 4. Which of the folowing can be inferred from the given data?
(a) China's GDP is more than India's GDP
(b) China's GDP is less than India's GDP
(c) India's GDP is less than Malaysia's GDP
(d) Nothing can be deduced

Q 5. For how many countries has the FEI increased from 1997 to 1998?
(a) 1
(b) 2
(c) 3
(d) Indeterminate

Direction for questions 6 to 12: Refer to the following data-set and solve the questions based on it.

The following graphs provide the data on four industrial sectors in the Republic of Reposia. Manufacturing constitutes 20%, Mining and Quarrying 15%, Electricity 15% and Chemicals 10% of the Republic's total industrial production in 1989. For the period 1990-1998, the graphs give the percentage change in production over the previous year's production. 1989 production values have been assigned an index of 100 for each of the four sectors.





Q 6. What was the average annual growth rate in production for all four sectors combined between 1989 and 1998?
(a) 1.2% (b) 6%
(c) 56.6% (d) 104%

Q 7. During the given period, when was the maximum production level in the electricity sector achieved?
(a) 1989 (b) 1993
(c) 1995 (d) 1998

Q 8. During the given period, when was the minimum production level in the manufacturing sector achieved?
(a) 1989 (b) 1990
(c) 1993 (d) 1998

Q 9. For the given period, which sector showed the highest growth rate?
(a) Electricity (b) Manufacturing
(c) Chemical (d) Mining and Quarrying

Q 10. In 1992, the index for the Chemical sector as compared to the index for the Electricity sector was
(a) greater by 6 (b) less by 5
(c) greater by 20 (d) less by 20

Q 11. By what percentage was the output of the Manufacturing sector more than that of the Chemical sector in 1991?
(a) 84% (b) 91%
(c) 98% (d) Indeterminate

Q 12. When did the maximum percentage change in the production levels of the Mining and Quarrying sector take place?
(a) 1993-1994 (b) 1992-1993
(c) 1996-1997 (d) 1997-1998

Direction for questions 13 to 17: Refer to the following data-set and solve the questions based on it.

The following table gives the break-up of the revenue earned by Phistream Private Limited over a period of five years. All figures in the table are in Rscrores.

	1994-95	1995-96	1996-97	1997-98	1998-99
Hardware					
(a) Domestic	68	54	52	66	35
(b) Exports	540	600	730	1770	1827
Software					
(a) Domestic	142	127	150	320	422
(b) Exports	1100	1200	1339	2500	2215
Peripherals	25	10	25	20	37
Training	140	106	160	240	161
Maintenance	21	19	25	92	126
Others	12	10	19	40	24
Total	2048	2126	2500	5048	4847

Q 13. In which years did the hardware exports lie between 35%-40% of the total revenues for Phistream?
(a) 1997-1998, 1998-1999
(b) 1998-1999, 1995-1996
(c) 1997-1998, 1994-1995
(d) 1995-1996, 1996-1997

Q 14. Which of the following is true?
(a) The revenues from training consistently increased over the given period
(b) Hardware exports consistently increased over the given period
(c) Software exports consistently increased over the given period
(d) none of the above

Q 15. For how many years during the given period , was the total revenue from Training and maintenance less than 10% of the total revenue for the corresponding years?
(a) 2 (b) 3
(c) 4 (d) 5

Q 16. In which of the following years was the revenue from the hardware business more than 50% of that from the software business?
(a) 1994-1995
(b) 1995-1996, 1996-1997
(c) 1997-1998, 1994-1995
(d) None of these

Q 17. If in 1999-2000, the total revenue earned by Phistream decreases by 5% from the previous year but the revenue from software increases to Rs 2437 crores; then what is the approximate percentage decrease in the combined revenue from the rest of the heads(excepting software) in 1999-2000?
(a) 1.0% (b) 1.8%
(c) 2.1% (d) 3.3%

Direction for questions 18 to 23: Refer to the following data-set and solve the questions based on it.

Data Pertains to Marti Udyog Ltd.

Sector	Factories	Employment	Fixed Capital	Variable Cost	Value Added
Government	18	15	14	22	25
Central	8	6	5	7	10
State	4	6	4	8	3
Central/ State	6	3	5	7	12
Public	12	8	6	19	8
Private	55	65	72	54	62
Joint	15	12	8	5	5
Total	100	100	100	100	100

All data is percentage.

Q 18. If the total work force was 76 million, whereas the total values added were 225 million, then which of the following had the maximum value addition per worker?
(a) Central (b) State
(c) Central/State (d) Public

Q 19. Which of the following sectors has the maximum fixed capital invested per factory?
(a) Central (b) State
(c) Central/State (d) Public

Q 20. If the variable is proportional to the number of employees and the production per employee, then for which of the following is the production highest?
(a) Government (b) Private
(c) Joint (d) Public

Q 21. If the government has a fixed capital of $ 200 million in the Iron and Steel industry, which corresponds to 20.012% of its total investment as fixed capital, then how much did the government invest(in Rs million) in Maruti Udyog Ltd which forms 25% of the investment in the joint sector ? (1US $ = Rs 45)
(a) 6500 (b) 2500
(c) 143 (d) 145

Q 22. Maruti Udyog Ltd is a joint project of the Indian Government and Suzuki Motors, Japan; each having equal stake. One fine day, the Indian government decides to disinvest from the venture due to losses occurring from labour problems. How much money will be disinvested?
(a) Rs 246 million (b) Rs 6500 million
(c) $ 246 million (d) $ 6500 million

Q 23. Which of the following statements is true?
(a) The number of government employees is more than the number of factories in the joint sector
(b) The number of employees in the public sector is the same as the fixed capital of the joint sector
(c) Both (a) and (b)
(d) Cannot say

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(b)	(d)	(d)	(b)	(d)	(a)	(a)	(b)	(b)	(d)	(a)	(b)	(d)	(d)	(c)	(c)	(b)	(b)
Q. No.	21	22	23																	
Answer	(a)	(b)	(d)																	

HINTS AND SOLUTIONS

1. Percentage change for India

$$= \frac{0.72 - 1.71}{1.71} > 50\%$$

Percentage change for China

$$= \frac{5.46 - 4.8}{5.46} \approx 12\%$$

Percentage change for Korea

$$= \frac{3.12 - 2.8}{2.8} \approx 11\%$$

Percentage change for Malaysia

$$= \frac{0.33}{2.1} \approx 15\%$$

Hence, maximum % change is for India

2. For the year 1998

For China, $\frac{FEI_{China}}{GDP_{China}} = 0.048$ ……..(1)

For India, $\frac{FEI_{India}}{GDP_{India}} = 0.0072$ ……..(2)

$$\frac{FEI_{China}}{FEI_{India}} = 10$$

$\Rightarrow$ on dividing (1) by (2) :

$$\frac{GDP_{India}}{GDP_{China}} \times 10 = \frac{0.048}{0.0072} = \frac{20}{3}$$

$$\Rightarrow \frac{GDP_{China}}{GDP_{India}} = \frac{3}{2} = 1.5$$

So, percentage difference between China's GDP and India's GDP = 50%

3. Since the comparisons are for the same country in each case, assume the GDP as 100 in 1997 in all cases. So, GDP of China, India and Korea in 1998 is 105,107 and 98 respectively.

$$\frac{FEI_{1998}}{FEI_{1997}} = \frac{FEII_{1998} \times GDP_{1998}}{FEII_{1997} \times GDP_{1997}}$$

For China : $\frac{FEI_{1998}}{FEI_{1997}} = \frac{4.8 \times 107}{5.46 \times 100} < 1$

For India : $\frac{FEI_{1998}}{FEI_{1997}} = \frac{0.72 \times 105}{1.71 \times 100} < 1$

For Korea : $\frac{FEI_{1998}}{FEI_{1997}} = \frac{3.12 \times 98}{2.8 \times 100} > 1$

So, II and V are correct. Therefore option (b) is the correct answer

4. It can be seen that all the values (FEII) are expressed as a ratio of FEI to GDP, the absolute values of GDP cannot be determined.

5. Since only the ratios of GDP and FEI are known, the increase in one independent of the other cannot be determined.

6. You may note that apart from a few years for manufacturing and mining and quarrying, the annual growth rates in most years have been positive and near the 5%–6% mark for all the sectors. The other options (a), (c), and (d) are too far away to be considered. Alternatively, for ease of calculation take the 1989 values for Manufacturing as 400, mining and Quarrying and Electricity as 300 each and Chemicals as 200. then, the final value for manufacturing would be nearly 580; for Mining and Quarrying would be about 365; for Electricity would be around 550; and for chemicals would be around 335.

=> Initial value of production

= 400 + 300 + 300 + 200 = 1200

Final value of production

$= 580 + 365 + 550 + 335 = 1830$

$\Rightarrow$ Overall % increase $\approx 52.5\%$

$\Rightarrow$ Average annual growth $= 52.5/9 = 5.85\%$

7. Electricity production has increased throughout (as growth rates are positive throughout) and the maximum production would be achieved in the last year, i.e. 1998.

8. The production for Manufacturing can be minimum either in 1989 or in 1992 (when the growth rate is negative).

If Manufacturing started with a production of 100 in 1989, then production in 1992 will be $100 \times 1.04 \times 1.01 \times 0.97 > 1$

So, the production in 1989 was the least

9. Electricity has shown the highest growth rate in all the years and therefore, would have the highest growth rate for the entire period.

10. Electricity in 1992:

$100 \times 1.05 \times 1.07 \times 1.08 = 121.4$

Chemicals in 1992:

$100 \times 1.07 \times 1.03 \times 1.06 = 116.8$

=> In 1992, the index of Chemicals is less than that of Electricity by approximately 5.

11. Let Manufacturing in 1989 = 200 and

Chemicals in 1989 = 100

=> Manufacturing in 1991:

$200 \times 1.04 \times 1.01 = 210$

and Chemicals in 1991:

$100 \times 1.07 \times 1.03 = 110$

$$\text{Required \%} = \frac{210-110}{110} \times 100 = 90.91\%$$

12. As can be seen directly from the graph, the highest increase (that of 8%) for Mining and Quarrying was in the year 1998, relative to 1997.

13. Hardware exports as a percentage of total revenues for corresponding years are:

$$1994-1995: \frac{540}{2048} \times 100 < 33\%$$

$$1995-1996: \frac{600}{2126} \times 100 < 33\%$$

$$1996-1997: \frac{730}{2500} \times 100 < 33\%$$

$$1997-1998: \frac{1770}{5048} \times 100 = 35.06\%$$

$$1998-1999: \frac{1827}{4847} \times 100 = 37.7\%$$

So, hardware exports lie between 35%–40% of total revenue in the last 2 years.

14. Checking the values from the tables, you can verify that option (b) is true.

15. The combined revenue from training and maintenance is never greater than 10% of total revenue for the corresponding year throughout the given period.

16.

Year	50% of software Revenue	Hardware Revenue
1994-95	$1242 \times 0.5 = 621$	608
1995-96	$1327 \times 0.5 = 663$	654
1996-97	$1484 \times 0.5 = 744$	782
1997-98	$2820 \times 0.5 = 1410$	1836
1998-99	$2637 \times 0.5 = 1318$	1862

$\Rightarrow$ Hardware business provides revenues> 50% of the revenue from software business in 1996-1997, 1997-1998 and 1998-1999.

17. In 1999-2000:

Total revenue $= 0.95 \times 4847 = 4600$

Software revenue = 2437

Remaining revenue = 4600- 2437 = 2163

In 1998-1999, revenue from heads other than software = 4847 – 2637 = 2210

$$\text{\% decrease} = \frac{2210-2163}{2210} \times 100 = 2.12\%$$

18. Values added per worker for various sectors will be proportional to the following ratios:

Central	10/6
State	3/6
Central/state	12/3
Public	8/8

19. Fixed capital per factory will be proportional to the following ratios:

Central	5/8
State	4/4
Central/state	5/6
Public	6/12

20. If the variable is proportional to the number of employees and the production per employees, then we can say that the variable cost is directly proportional to the production. Since the variable cost is maximum for the private sector, it will also have the maximum production.

21. If 20% of the government fixed capital is equal to $ 200 million => Total governtment sector fixed capital = $ 1000 million which is equivalent to 14% of the total fixed capital. Now, 25% of government investment in joint sector i.e., 25% of the 8% of the total = 2% of total fixed capital will be given by:

(1000 × 2) /14 = $ 143 million

Value in Rs is given by:

143 × 45 = Rs 6450 million

22. From the above answer the money disinvested will be Rs 6450 million.

23. Neither of the statements (a) or (b) can be inferred, as we know only the percentages and not the actual values of the employees, factories or fixed capital.

6

CHAPTER

Mastering Data Interpretation

PRACTICE EXERCISE 1

Direction for questions 1 to 4: Refer to the following chart and solve the questions based on it.

The following table gives the career record in the One-Day Internationals of few of the Indian batsmen:

Name	Matches Played	Innings Played	Runs Scored	Average
Yuvraj	50	48	1800	40
Sachin	50	50	2034	45.2
Kapil	46	46	1679	36.5
Saurav	44	44	1539	40.5
Rahul	40	38	1806	51.6
Sehwag	42	40	1445	42.5

The average is calculated by dividing the total runs scored by the total number of matches in which the batsman got out.

Q 1. Who among the following did not get out for the maximum amount of time in the total innings played?

(a) Yuvraj (b) Sachin
(c) Kapil (d) Sehwag

Q 2. To reflect the performance of the players better, it was decided that the average would be calculated on the basis of the total innings played. Whose average will see the maximum change due to this?

(a) Yuvraj (b) Sachin
(c) Kapil (d) Sehwag

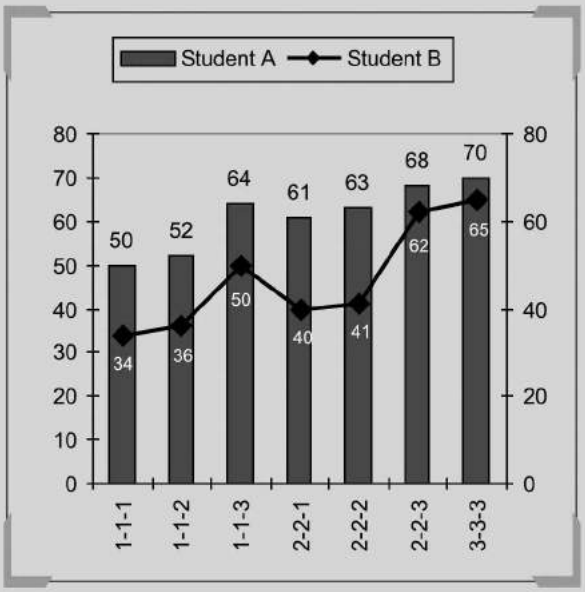
Q 3. How many of the batsmen got out in all the innings played?
(a) 0 (b) 1
(c) 2 (d) Cannot be determined

Q 4. To reflect the performance of the players better, it was decided that the average would be calculated on the basis of the total matches played. Whose average will see the minimum change due to this?
(a) Yuvraj (b) Sachin
(c) Kapil (d) Sehwag

Direction for questions 5 to 10: Refer to the following chart and solve the questions based on it.

The bar chart given below represents the performance of two students A and B while studying three different subjects with different groups of teachers. Ratings given to the teachers are 1, 2, 3 with 1 being the lowest and 3 being the highest.

The bar chart also represents the percentage marks obtained by A and B in three subjects. Each subject carries 100 marks. Students always get integral marks in each subject. For example, if A studies all the three subjects from a group of teachers so that all the three teachers have 1 rating (low rating); then he gets 50% marks of the total marks of the three subjects.



Q 5. What would be the change in the total marks obtained by student A when we compare the combination 1-1-1 and 1-1-3?
(a) 32 (b) 42
(c) 62 (d) Cannot be determined

Q 6. What is the percentage change in the total marks obtained by student B when we compare the combination 1-1-3 and 2-2-2?
(a) 15% (b) 16%
(c) 17% (d) 18%

Q 7. What can be the maximum marks obtained by student A in any of the three subjects with the combination of teachers being 1-1-3?
(a) 92
(b) 96
(c) 100
(d) None of these

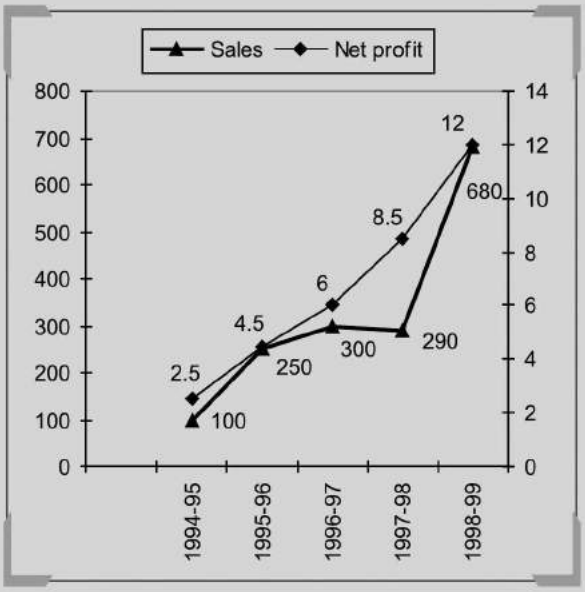
Q 8. For which of the following options is the difference in the marks scored by student B the maximum?
(a) 1-1-1 and 3-3-3 (b) 1-1-1 and 1-1-3
(c) 1-1-1 and 2-2-3 (d) 1-1-2 and 3-3-3

Q 9. For which of the following options are the marks scored by student A and student B the maximum?
(a) 1-1-1 (b) 1-1-3
(c) 2-2-3 (d) 2-2-2

Q 10. It is allowed to choose two different bars from the above given bars, and the percentage marks obtained become weighted average of the percentage marks obtained, in the ratio of the presence of 3-rating teacher in the bar. In the absence of 3-rating teacher, weighted average should be calculated in the ratio of the presence of 2-rating teachers. For example, if we mix the third bar and 7th bar, we will have to find out the weighted average of 64% and 70% in the ratio 1 : 3 for student A. What would be the percentage marks obtained by the student B if he opts for 4th bar and 5th bar?
(a) 40.2% (b) 40.6%
(c) 40.8% (d) None of these

Direction for questions 11 to 14: Refer to the following chart and solve the questions based on it.

The figure below represents sales and net profit (in Rs crore) of IVP Ltd, for five years from 1994-1995 to 1998-1999. During this period, the sales increased from Rs 100 crore to Rs 680 crore. Correspondingly, the net profit increased from Rs 2.5 crore to Rs 12 crore. (Net profit is defined as the excess of sales over total costs.)



Q 11. The highest percentage growth in sales, relative to the previous year occurred in
(a) 1995-1996 (b) 1996-1997
(c) 1997-1998 (d) 1998-1999

Q 12. The highest percentage growth in net profit, relative to the previous year, was achieved in
(a) 1998-1999 (b) 1997-1998
(c) 1996-1997 (d) 1995-1996

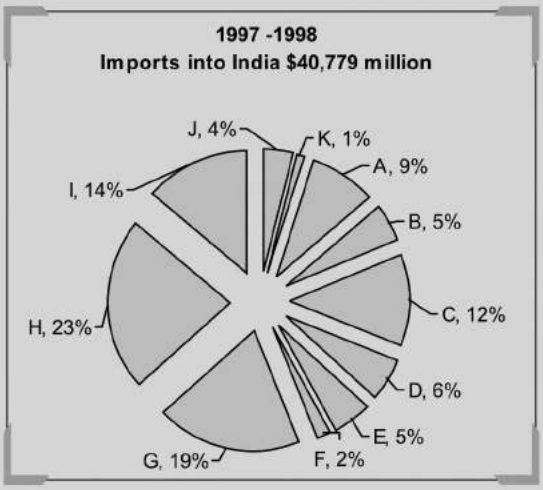
Q 13. Defining profitability as the ratio of net profit to sales, IVP Ltd recorded the highest profitability in
(a) 1998-1999 (b) 1997-1998
(c) 1994-1995 (d) 1996-1997

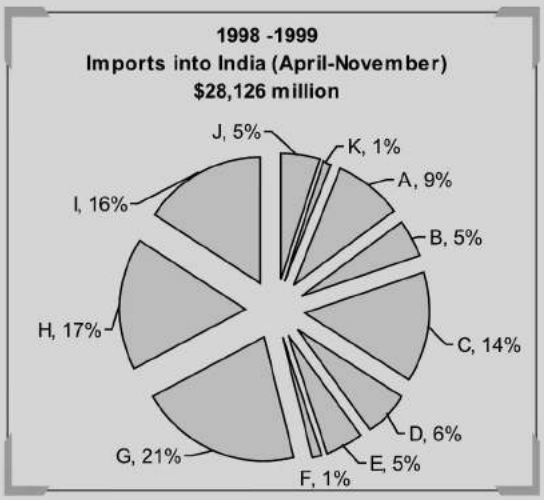
Q 14. With profitability as defined in question 3, it can be concluded that
(a) Profitability is non-decreasing during the five years from 1994-1995 to 1998-1999.
(b) Profitability is non-decreasing during the five years from 1994-1995 to 1998-1999.
(c) Profitability remains constant during the five years from 1994-1995 to 1998-1999
(d) None of above

Direction for questions 15 to 18: Refer to the following chart and solve the questions based on it.

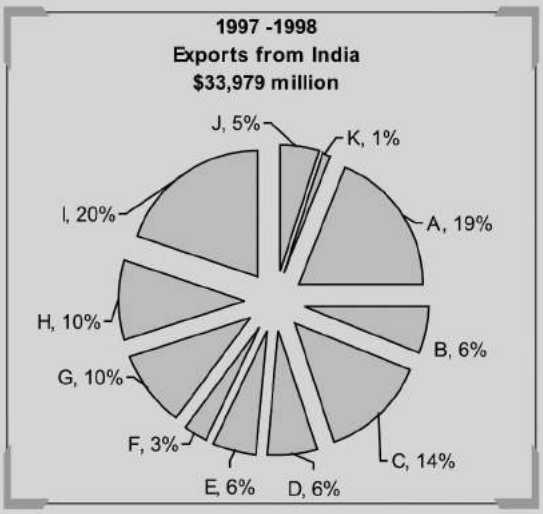
Consider the information provided in the figure below relating to India's foreign trade in 1997-1998 and the first eight months of 1998-1999. Total trade with a region is defined as the sum of exports and imports from that region. Trade deficit is defined as the excess of imports over exports; and it may be negative.

A. US B. Germany
C. Other EU D. UK
E. Japan F. Russia
G. Other East-European countries
H. OPEC I. Asia
J. Other LDCs
K. Other Source of Imports





Destination of exports



Q 15. With which region does India have the highest total trade in 1997-1998?
(a) US (b) Other EU countries
(c) OPEC (d) Others

Q 16. In 1997-1998, the amount of Indian exports (in million US $) to the region with which India has the lowest total trade, is approximately:
(a) 750 (b) 340
(c) 220 (d) 440

Q 17. In 1997-1998, the trade deficit (in billion US $) for the region with which India has the highest trade deficit is approximately equal to
(a) 6.0 (b) 3.0
(c) 4.5 (d) 7.5

Q 18. Which region has the lowest trade deficit with India in 1997-1998?
(a) US (b) Asia
(c) Others (d) Other EU countries

Direction for questions 19 and 20: Answer the questions based on the following information:

Assume that the average monthly exports and imports to India during the remaining four months of 1998-1999 are the same as that for the first eight months of the year.

Q 19. To which region did India's exports register the highest percentage growth between 1997-1998 and 1998-1999?
(a) Other East-European countries
(b) USA
(c) Asia
(d) Exports have declined, therefore no growth

Q 20. What is the percentage growth rate in India's total trade deficit between 1997-1998 and 1998-1999?
(a) 43% (b) 47%
(c) 50% (d) 40%

Direction for questions 21 to 25: *Refer to the following chart and solve the questions based on it.*

The table below presents the data on the percentage of population covered by drinking water and sanitation facilities in select Asian countries.

Percentage Population Covered by Drinking Water and Sanitation Facilities

	Drinking Water			Sanitation Facilities		
	Urban	Rural	Total	Urban	Rural	Total
India	85	79	81	70	14	29
Bangladesh	99	96	97	79	44	48
China	97	56	67	74	7	24
Pakistan	82	69	74	77	22	47
Philippines	92	80	86	88	66	77
Indonesia	79	54	62	73	40	51
Sri Lanka	88	52	57	68	62	63
Nepal	88	60	63	58	12	1

Country A is said to dominate B or A > B as A has a higher percentage in total coverage for both drinking water and sanitation facilities; and B is said to be dominated by A, or B < A.

A country is said to be on the coverage frontier if no other country dominates it. Similarly, a country is not on the coverage frontier if it is dominated by at least one other country.

Q 21. Which countries are on the coverage frontier?
(a) India and China
(b) Sri Lanka and Indonesia
(c) Philippines and Bangladesh
(d) Nepal and Pakistan

Q 22. Which of the following statements are true?
I. India > Pakistan and India > Indonesia
II. India > China and India > Nepal
III. Sri Lanka > China
IV. China > Nepal
(a) I and III (b) II and IV
(c) I, II and III (d) II, III and IV

Q 23. Using only the data presented under the 'Sanitation facilities' columns, it can be concluded that the rural population in India, as a percentage of its total population is approximately:
(a) 76 (b) 70
(c) 73 (d) Cannot be determined

Q 24. Again, using only the data presented under the 'Sanitation facilities' columns, China, Indonesia and Philippines are sequenced in an ascending order of the rural population as a percentage of their respective total population. The correct order is:
(a) Philippines, Indonesia, China
(b) Indonesia, China, Philippines
(c) Indonesia, Philippines, China
(d) China, Indonesia, Philippines

Q 25. India is not on the coverage frontier because
I. It is lower than Bangladesh in terms of the coverage of drinking water facilities.
II. It is lower than Sri Lanka in terms of the coverage of sanitation facilities.
III. It is lower than Pakistan in terms of the coverage of sanitation facilities.
IV. It is dominated by Indonesia.
(a) I and II (b) I and III
(c) IV (d) None of these

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12
Answer	(d)	(d)	(b)	(c)	(b)	(d)	(b)	(c)	(d)	(b)	(a)	(d)
Q. No.	13	14	15	16	17	18	19	20	21	22	23	24
Answer	(b)	(d)	(c)	(b)	(a)	(a)	(b)	(b)	(c)	(b)	(c)	(a)
Q. No.	25											
Answer	(d)											

HINTS AND SOLUTIONS

1. Yuvraj – 1800/40 = 45.
   Hence, not-out in 3 innings.
   Sachin – 2034/45.2 = 45.
   Hence, not-out in 5 innings.
   Kapil – 1679/36.5 = 46.
   Hence, not-out in 0 inning.
   Sehwag – 1445/42.5 = 34.
   Hence, not-out in 6 innings.
   Therefore, option (d) is the correct answer.
2. The difference is maximum in case of Sehwag.
3. Only Kapil got out in all the innings played. Take the help of the solution to Q 1.
4. Again, the difference is maximum in case of Sehwag.
5. The difference in the total score = 64 × 3 – 50 × 3 = 42 (Difference is caused by the teacher of the high rating only).
8. Go through the options.
9 Look at the bars for the maximum difference. It is the maximum in case of 2-2-2.
10. We would be required to find the weighted average of 40 and 41 in the ratio 2 : 3. Hence (40 × 2 + 41 × 3)/5 = 40.6%
11. Using the graph, the percentage growth in sales are:

	Previous Sales	Current Sales	Difference	Percentage
1995-96	100	250	150	150%
1996-97	250	300	50	20%
1997-98	300	290	–10	–3.33%
1998-99	290	680	390	134.5%

It is but obvious from the above table that the maximum percentage increase relative to previous year occurred in 1995-1996.

12. We can again calculate the growth in profits.

	Previous Profit	Current Profit	Difference	Percentage
1995-96	2.5	4.5	2	80%
1996-97	4.5	6	1.5	33.33%
1997-98	6.0	8.5	2.5	41.66%
1998-99	8.5	12	3.5	41.2%

The highest percentage growth in net profit relative to the previous year was achieved in 1995-1996.

13.

	Net Profit	Net Sales	Ratio
1994-95	2.5	100	0.025
1995-96	4.5	250	0.02
1996-97	6	300	0.02
1997-98	8.5	290	0.03
1998-99	12	680	0.018

Therefore, profitability is maximum for 1997-1998.

14. It can be seen that profitability does not follow a fixed pattern and the first three statements try to generalize the profitability.
Therefore, they are not applicable and option (d) is the correct answer.

15. Total trade with a region = Sum of exports and imports from that region. From the pie charts for 1997-1998, we have the following sectors occupying the maximum area.

	Country	Exports	Imports	Trade
H	OPEC	3397.9	9379.2	12795.1
I	Asia	6796	5709	12505
G	East Europe	3397	7748	11145.9
C	EU	4757	4893.5	9650.5
A	USA	6456	3670	10126

Therefore, H-OPEC has the maximum trade with India in 1997-1998

16. Using the pie chart, the region having the lowest trade is K =
1% of imports + 1% of exports
So, Indian exports are 1% of 3397.9 which is roughly 340 millon US $.

17.

	Imports	Exports	Trade Deficit
A	3670.11	6456.01	–2785.9
B	2038.95	2038.74	0.21
C	4893.48	4757.06	136.42
D	2446.74	2038.74	408
E	2038.95	2038.74	0.21
F	815.58	1019.37	–203.79
G	7748.01	3397.9	4350.11
H	9379.17	3397.9	5981.27
I	5709.06	6795.8	–1086.74
J	1631.16	1698.95	–67.79

So, it can be seen that region H has the highest trade deficit of approximately $ 6,000 million or $ 6 billion.

18. Using the pie chart for 1997-1998, we can see that US which is region A, has the lowest trade deficit.
(9% of imports – 19% of total exports)
(9/100 × 40779 – 19/100 × 33979)

19. Using the pie chart, we know that the exports have increased from three regions A, G and H as follows.

	Country	1998-99	1997-98
A	USA	7395.4	6456
G	East European Countries	3858.5	3397.9
H	OPEC	3215.4	3397.9

Now, the exports for 8 months have been given. According to new directions, we need to calculate exports for 12 months.

$\frac{21436 \times 12}{8}$ = $ 32,154 million

So, maximum percentage increase is from region A (USA).

20. Following is India's total trade deficit:

	Imports	Exports	Deficit
1997-98	40779	33979	6800
1998-99	42189	32154	10035

Total imports for 1998-1999
= 28126 × 12/8
= $ 42,189 million
Similarly, exports for 1998-1999
= 28126 x 12/8
= $ 32,154 million
Percentage growth rate
= 10035 – 6800/6800 x 100
= 47.6%
Therefore, option (b) is the correct answer.

21. Also, Bangladesh > Philippines (97 > 86) for drinking water.
And Philippines > any other country for sanitation facilities. Thus, countries are on the coverage frontier for two facilities.

22. Statement I > Statement II only if statement I has higher percentage in total coverage for both drinking water and sanitation facilities taken independently and not as a total of the two facilities.
Thus, only statement II and statement IV are true
India > China
(81 > 67 and 29 > 24)
India > Nepal
(81 > 63 and 29 > 18)
Also China > Nepal (67 > 63 and 24 >18)

23. Let the urban population be x and rural population be y.
Using the sanitation column, we have
0.7x + 0.14y = 0.29(x + y)
0.41x = 0.15y
So x = 15/41y
So, percentage of rural population
= y/x + y × 100
= y/15/41y + y × 100
= 41/56 × 100
= 73.2%

24. Use the method given in the last question to find the percentage of rural population for Philippines, Indonesia and China.

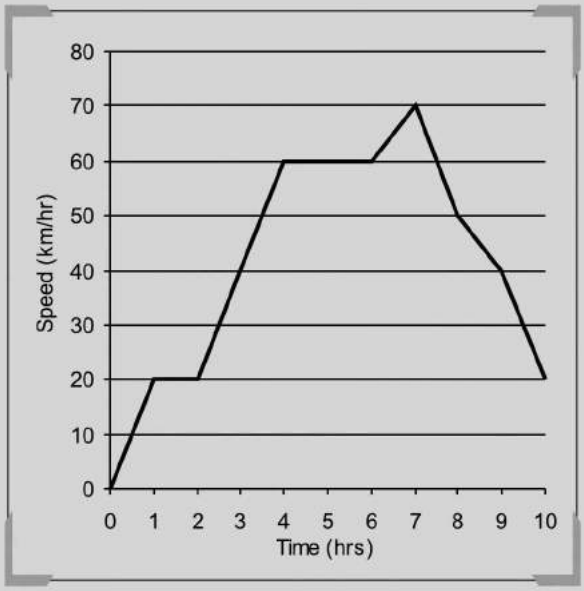
P	50%
I	66.66%
C	79.8%

Thus, P < I < C

25. India is not on the coverage frontier because
(i) It is below Bangladesh and Philippines for drinking water.
(ii) For sanitation facilities it is below Philippines, Sri Lanka, Indonesia and Pakistan.

PRACTICE EXERCISE 2

Direction for questions 1 to 5: Go through the data set given below and solve the questions based on it.



Q 1. What is the average speed represented in the above graph?
(a) 32 km/h
(b) 37 km/h
(c) 43 km/h
(d) None of these

Q 2. Considering two-hour slots from the beginning, the second highest distance covered in a two-hour slot was between
(a) 0 – 2 h (b) 2 – 4 h
(c) 4 – 6 h (d) 6 – 8 h

Q 3. What is the highest rate of acceleration achieved during the whole period?
(a) 25 km/h^2
(b) 20 km/h^2
(c) 15 km/h^2
(d) 10 km/h^2

Q 4. If the rate of deceleration of the 8th hour had continued further, the speed would have become zero in
(a) 9 hrs 10 mins
(b) 10 hrs 30 mins
(c) 11 hrs 10 mins
(d) None of these

Q 5. For which of the given hours in the graph was the rate of change of speed minimum?
(a) 1st–2nd
(b) 4th–5th
(c) 5th–6th
(d) All of these

Direction for questions 6 to 9: Go through the data set given below and solve the questions based on it.

In a management course, students of 2nd year are asked to select four optional subjects from the given list of seven optional subjects. There are ten students whose roll numbers are from 1 to 10. They have selected four optional subjects each. Their subjects selected are indicated by "X" marks.

Courses	Roll Number									
	1	2	3	4	5	6	7	8	9	10
TA	X		X	X		X		X	X	X
MBC		X	X		X	X		X		
YCA			X			X	X		X	X
DMBC	X	X		X	X		X	X		
XA		X					X		X	X
PG	X		X	X	X		X	X		
PM	X	X		X	X	X			X	X

Q 6. Which of the following triple courses have maximum number of students in common?
(a) TA, MBC, YCA (b) DMBC, TA, PG
(c) YCA, PM, XA (d) DMBC, PG, XA

Q 7. Which roll numbers have the maximum common number of optional subjects?
(a) 1 and 8 (b) 3 and 6
(c) 9 and 10 (d) None of these

Q 8. Which optional subject was chosen by maximum number of students?
(a) XA (b) TA
(c) MBC (d) DMBC

Q 9. Which roll numbers have the same optional subjects?
(a) 3 and 6 (b) 1 and 3
(c) 6 and 8 (d) 9 and 10

Direction for questions 10 to 14: Go through the data set given below and solve the questions based on it.

A rating company rates the performance of three companies producing shoes. The points are allotted according to their sales. The point index (PI) of each of the companies = The number of lakh units sold during the month × points allotted.

Table 1

The number of lakh units sold = y	Points allotted
$y < 5$	3
$5 < y < 8$	4
$8 < y < 11$	5
$y > 11$	6

The following table shows the number of lakh units sold of each of the brands—Moon, Warle and Enivi—from Jan 07 to June 07.

Table 2

Name of month	Moon	Warle	Enivi	Rank of months according to PI
Jan 07	7	4	-	4
Feb 07	-	13	-	1
Mar 07	-	-	-	5
Apr 07	-	-	-	6
May 07	-	-	-	2
Jun 07	-	-	10	3

Note:

1. The number of lakh units sold by all the three companies in each of the months is identical.
2. The sum of the number of lakh units sold by each of the companies in all the six months together is identical.
3. The number of lakh units sold by any of the companies in any one of the months is at least 1.
4. The number of lakh units sold by exactly two companies in each of the months Feb 07, Mar 07 and Apr 07 is identical.
5. The Point Index of Jan 07 and May 07 is 26 and 17 less than Feb 07 respectively. Also, the PI in Jan 07 is one more than that of March 07.
6. The number of lakh units sold by Moon in Mar 07 and Apr 07 together is equal to that in May 07.
7. The sum of PI in any month is not greater than 100.
8. The least possible Point Index (PI) is in Apr 07 for the sum of lakh units sold by all three companies.

Q 10. What is the number of lakh units sold by Enivi in Mar 07?
(a) 5 (b) 6
(c) 4 (d) 7

Q 11. Find the PI of all the three companies in Apr 07.
(a) 59 (b) 60
(c) 61 (d) 58

Q 12. What is the number of lakh units sold by Moon in all the six months together?
(a) 34 (b) 32
(c) 36 (d) 38

Q 13. What is the number of lakh units sold by Enivi in May 07?
(a) 7 (b) 4
(c) 5 (d) 8

Q 14. What is the PI of Warle in all the six months together?
(a) 139 (b) 153
(c) 138 (d) 148

Direction for questions 15 to 20: Go through the passage given below and solve the questions based on it.

Five cricket coaches B, D, J, L and W were being judged for the award of best coach of the year. The selection committee ranked them on five different parameters viz., Empathy, Vision, Focus, Creativity, and Intuitions. The five coaches were given points on these five parameters. The best coach gets 5 points and the worst coach gets 1 point. There are no ties. The coach with highest total gets the award of best coach. B gets the award by scoring 24 points. W gets 5 points in Creativity and 3 points in Intuition. J got same score in four of the given five parameters. Their final standings were in alphabetical order of their names.

Q 15. What is the overall score of J?
(a) 12 (b) 13
(b) 14 (c) Indeterminable

Q 16. What is the rank of D in Creativity?
(a) 1st (b) 2nd
(c) 5th (d) Indeterminable

Q 17. What are the points scored by J in Intuition?
(a) 1 (b) 2
(c) 4 (d) Indeterminable

Q 18. If L scored his highest points in Empathy then what is the score of D in Vision?
(a) 1 (b) 2
(c) 3 (d) 4

Q 19. For how many of the coaches given above, is it possible to determine the exact points obtained in different parameters?
(a) 2 (b) 3
(c) 4 (d) 5

Q 20. If L gets a score of 2 in each—Empathy, Vision and Focus, then what is the rank of D in intuition?
(a) 2nd (b) 4th
(c) 3rd (d) 5th

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12
Answer	(c)	(d)	(b)	(b)	(d)	(b)	(c)	(b)	(d)	(b)	(d)	(a)
Q. No.	13	14	15	16	17	18	19	20	21	22	23	24
Answer	(c)	(d)	(b)	(c)	(a)	(b)	(c)	(b)				

HINTS AND SOLUTIONS

1. The average speed in first hour =10 km/hr
Average speed represented by the graph
$= \frac{10 + 20 + 30 + 50 + 60 + 60 + 65 + 60 + 45 + 30}{10}$
$= \frac{430}{10} = 43\,\text{km/h}.$

2. Distance covered in 0–2 hrs slot = 10 + 20 = 30 km
Distance covered in 2–4 hrs slot = 80 km
Distance covered in 4–6 hrs slot = 120 km
Distance covered in 6–8 hrs slot = 125 km

3. Acceleration = Change in velocity/h
= 20 km/h² in 1st, 3rd and 4th h

4. Deceleration in 8th hour = 20 km/h²
Speed at the end of 8th hour = 50 km/h
Speed will be zero after 't' h where 0 = 50 – 20 × t
$= t = \frac{50}{20} = \frac{5}{2}\,\text{hours} = 2\frac{1}{2}\,\text{hrs}.$
= Speed will be zero at 10 h 30 mts.

6. (1) Roll nos 3 and 6
(2) Roll nos 1, 4 and 8
(3) Roll nos 9, 10
(4) Roll no 7

7. (1) Roll nos 1 and 8 have TA, DMBC and PG as common.
(2) Roll nos 3 and 6 have TA, MBC, YCA as common.
(3) Roll nos 9 and 10 have all the four common options.
(4) Roll nos 2 and 5 have MBC, DMBC and PM as common.

8. TA → 7 students
XA → 4 students
MBCA → 5 students
DMBC → 6 students

[redacted]

Following table can be formed with the data given in the question:

	B	D	J	L	W
Empathy	5				
Vision	5				
Focus	5				
Creativity	4				5
Intuition	5				3
Total	24		Sum = 75 – 24 = 51		

We can see now that the minimum possible sum for W = 11.
Sum of points of the coaches are in the following order – B > D > J > L > W
And sum of the points of (D + J + L + W) = 5(1 + 2 + 3 + 4 + 5) – 24 (points of B) = 51
Hence, the point obtained by the remaining four coaches D, J, L and W will be 11, 12, 13, 15 respectively.
(Points obtained by any of J, L and W can not be other than 11, 12 and 13 because in that case, the sum of the points of the four coaches will be more than 51.)
Hence, the final points tally will be:

	B	D	J	L	W
Empathy	5	2/4	3	2/4	1
Vision	5	2/4	3	2/4	1
Focus	5	2/4	3	2/4	1
Creativity	4	1/2	3	1/2	5
Intuition	5	2/4	1	2/4	3
Total	**24**	**15**	**13**	**12**	**11**

Now all the questions can be answered.

PRACTICE EXERCISE 3

Direction for questions 1 to 7: Refer to the following table and solve the questions based on it.

The following tables and charts show the details of the advertising expenses and corresponding results for XYZ Ltd.

During 2000–2004, the company issues ads in three different newspapers— TOI, HT and others.

Table 1 shows the year wise cost incurred by XYZ Ltd during the given period.

Table 2 shows the number of advertisements released in each publication during the given period.

Chart 1 shows the total number of enquiries generated at XYZ Ltd in response to the advertisements in each publication during the given period.

Chart 2 gives the number of clients (out of the total who enquire) who finally provide business to XYZ Ltd, as a result of its advertising in each publication during the given period.

Table 3 shows the average amount of business per new client for XYZ Ltd as a result of its advertising in each publication in during the given period. (Assume that all new clients in a year are a result of the advertising in that specific year only and the clients responding to advertising in different publications are mutually exclusive)

Table 1

Total Money Spent (Rs Lacs)

Year	TOI	HT	Others
2000	3.5	3.2	2
2001	5	3.4	2.8
2002	6.2	4	5.2
2003	8	4.8	4.4
2004	14	3.5	3.1

Table 2

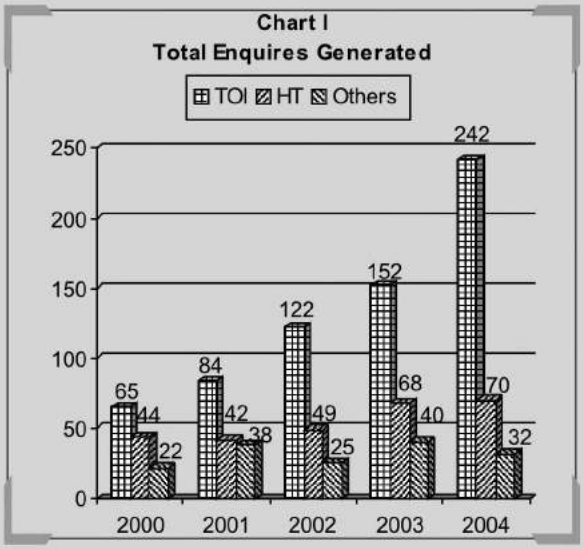
Number of Ads

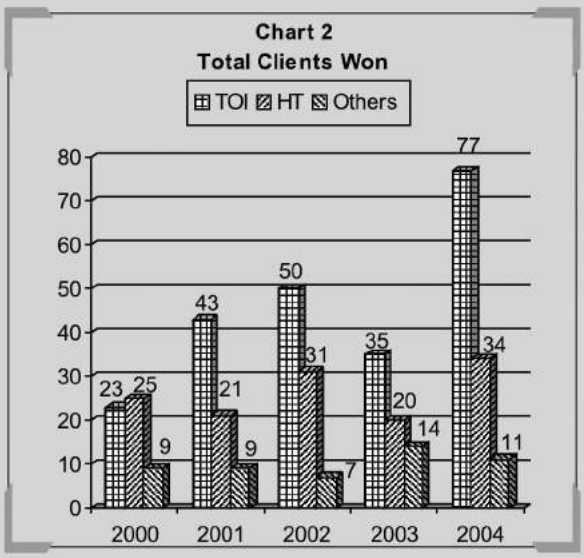
Year	TOI	HT	Others
2000	10	12	15
2001	12	15	17
2002	15	20	36
2003	19	24	22
2004	38	18	30

Table 3

Average Business/Client (Rs lacs)

Year	TOI	HT	Others
2000	0.25	0.19	0.12
2001	0.19	0.28	0.09
2002	0.28	0.15	0.18
2003	0.31	0.55	0.22
2004	0.44	0.38	0.15





For advertising in any publication in any year (or period):

(i) Response ratio is the average number of enquiries generated per advertisement.

(ii) Conversion ratio is the percentage of clients who provide business out of those who enquire.

(iii) Success ratio is the average number of clients won per advertisement.

(iv) Return ratio is the ratio of business generated to the money spent on advertising.

Q 1. During the given period, the average cost per advertisement for XYZ Ltd was highest in
(a) TOI (b) HT
(c) Others (d) Cannot be determined

Q 2. The response ratio in a single year during the given period was highest for advertising in
(a) TOI in 2003 (b) HT in 2004
(c) Others in 2001 (d) TOI in 2002

Q 3. The response ratio during the given period was the lowest in which of the publications?
(a) TOI (b) HT
(c) Others (d) Cannot be determined

Q 4. The lowest conversion ratio for advertising by XYZ Ltd in HT was observed in
(a) 2001 (b) 2003
(c) 2002 (d) 2000

Q 5. Which of the following shows the correct order of the conversion ratios for advertising by XYZ Ltd in the three publications during the given period?
(a) TOI > HT > Others (b) HT > Others > TOI
(c) HT < TOI < Others (d) Others < TOI < HT

Q 6. For XYZ Ltd, the success ratio for advertising in the TOI during the given period was approximately what percentage of the success ratio for advertising in others in 2003?
(a) 525% (b) 425%
(c) 350% (d) 285%

Q 7. In how many years during the given period was the success ratio for advertising in the TOI greater than the success ratio for advertising in the HT?
(a) 5 (b) 4
(c) 3 (d) 2

Direction for questions 8 to 13: Refer to the following table and solve the questions based on it.

The tables given below show the currency exchange rates in April 2005 and April 2004 respectively. For any currency row, the value corresponding to each column shows that currency's exchange rate with the currency column. For example, in April 2005, one dollar was equivalent to Rs 43.750

Chart 3 and Chart 4 show the percentage break-up of rupee equivalent of currency reserves of a company in April 2005 and April 2004 respectively. The rupee equivalent reserves of a currency are the currency reserves calculated in rupee terms based on the prevalent exchange rate between the rupee and that currency. For example, reserves of 1000 euros in April 2004 would constitute rupee equivalent reserves of Rs 53,000. The rupee equivalent of currency reserves for the company was Rs 5 crores in April 2004, and grew by 40% in the next one year.

Table 1

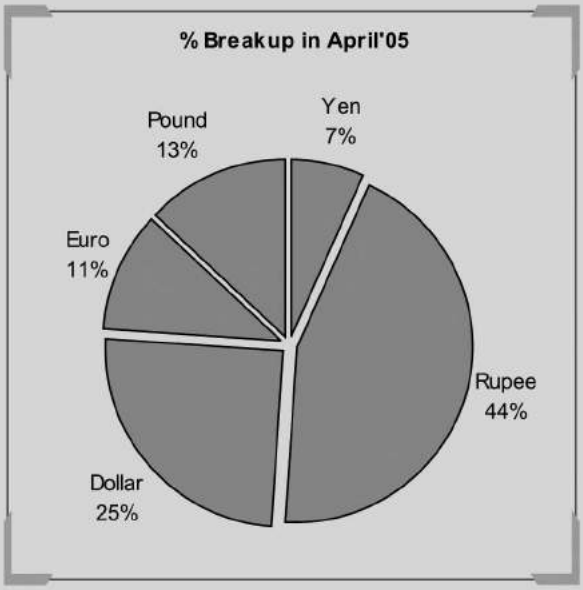
Currency Rates (April 2005)

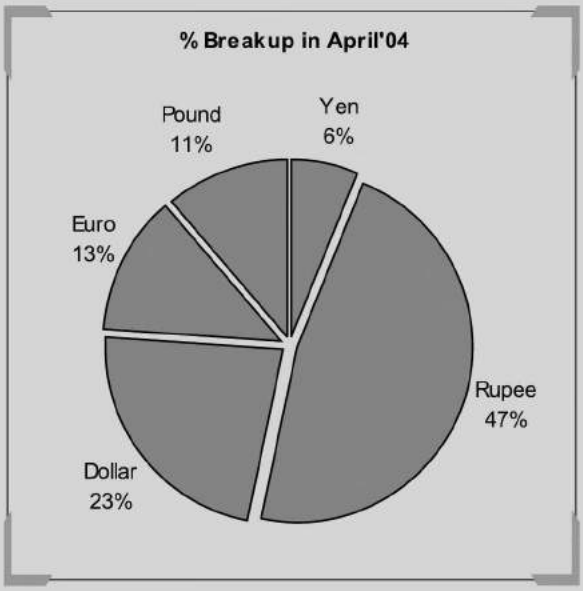
Currency	Rupee	Dollar	Euro	Pound	Yen
Rupee	----	0.023	0.018	0.012	0.024
Dollar	43.750	----	0.770	0.535	1.065
Euro	56.790	1.298	----	0.695	1.382
Pound	81.750	1.869	1.440	----	1.990
Yen	41.090	0.939	0.724	0.503	----

Table 2

Currency Rates (April 2004)

Currency	Rupee	Dollar	Euro	Pound	Yen
Rupee	----	0.020	0.019	0.013	0.029
Dollar	50.000	----	0.880	0.612	1.217
Euro	53.000	1.211	----	0.648	1.290
Pound	75.000	1.714	1.321	----	1.825
Yen	35.000	0.800	0.616	0.428	----





Q 8. The dollar currency reserves for the company (in $ million) in April 2004 was closest to

(a) 11.5 (b) 4.0

(c) 2.3 (d) 0.25

Q 9. The rupee equivalent (in rupees million) of the Yen reserves for the company in April 2005 was closest to
(a) 49 (b) 1.3
(c) 5 (d) 0.12

Q 10. The ratio of money in dollar reserves to the money in euro reserves for the company in April 2004 was
(a) 7 : 4 (b) 3 : 5
(c) 9 : 4 (d) Cannot be determined

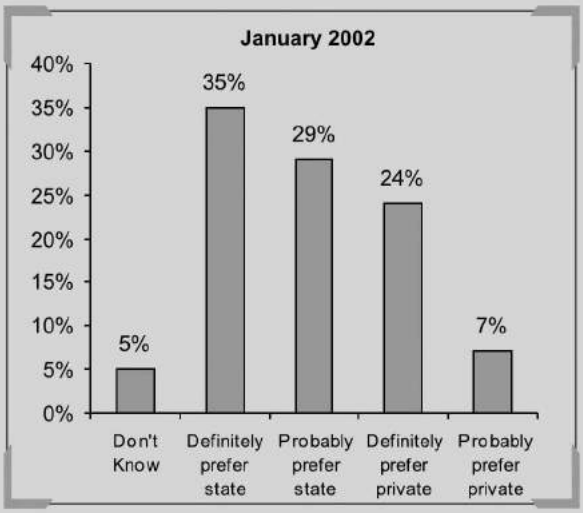
Q 11. The money in pound reserves in April 2005 was how many times the money in Yen reserves in April 2004?
(a) 1.65 (b) 1.82
(c) 3 (d) Cannot be determined

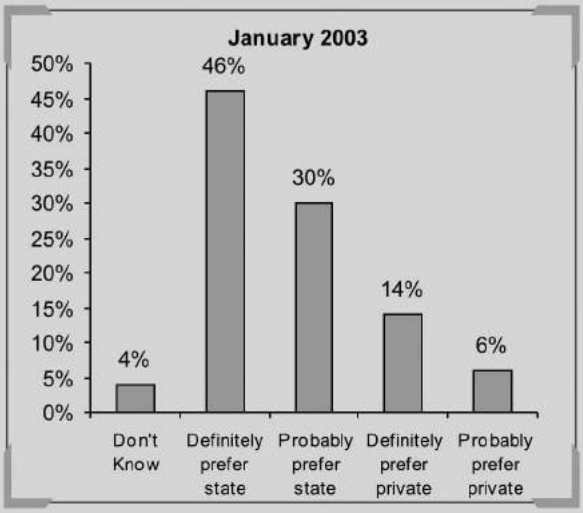
Q 12. Between April 2004 and April 2005, by what percentage did the rupee reserves grow?
(a) 22% (b) 31%
(c) 49% (d) Cannot be determined

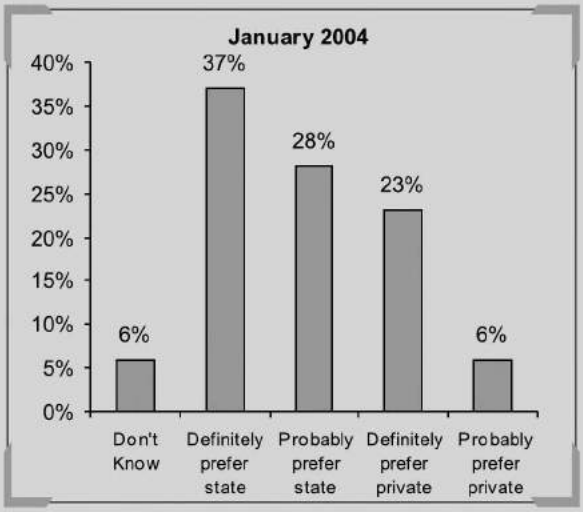
Q 13. Between April 2004 and April 2005, the company's reserves of which currency saw the maximum absolute increase (in rupee equivalent terms)?
(a) Dollar (b) Euro
(c) Pound (d) Rupee

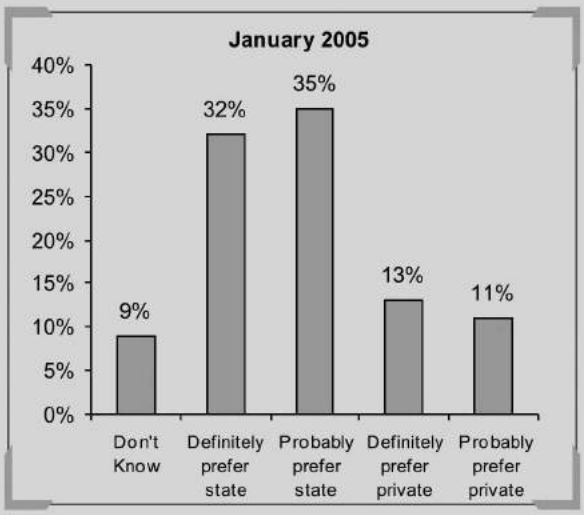
Direction for questions 14 to 20: Refer to the following table and solve the questions based on it.

The following charts show the responses of the population of the state of ABC over a four-year period, to the question—"Is State ownership the best way to run an enterprise OR an enterprise is best run by private entrepreneurs?" The eligible voting population of ABC was 2.4 million in January 2003 and can be assumed to have grown by 20% each year during the period January 2002–January 2005. The entire eligible population voted in each of the four years and every person casted a vote in favour of only one of the five options shown each year.









Q 14. For how many options during the given period did the number of people voting in favour of the options, increase in each year, relative to the the previous year?

(a) 2 (b) 3

(c) 1 (d) None

Q 15. For how many options during the given period did the number of people voting in favour of the options, decrease in each year, relative to the previous year.
(a) 2 (b) 3
(c) 1 (d) None

Q 16. The greatest absolute increase, between January 2002 and January 2005, in the number of people voting for an option was observed for:
(a) Definitely prefer private
(b) Probably prefer state
(c) Probably prefer private
(d) Don't know

Q 17. For which option did the number of people casting their vote decrease between January 2002 and January 2005?
(a) Definitely prefer state
(b) Probably prefer state
(c) Definitely prefer private
(d) None

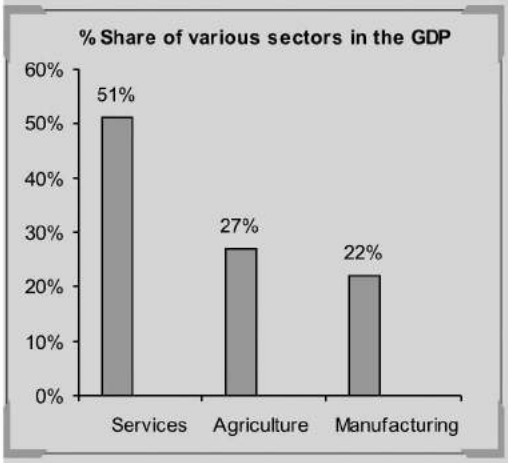
Q 18. The greatest percentage increase, between January 2002 and January 2005, in the number of people voting for an option took place for
(a) Definitely prefer private
(b) Probably prefer state
(c) Probably prefer private
(d) Don't know

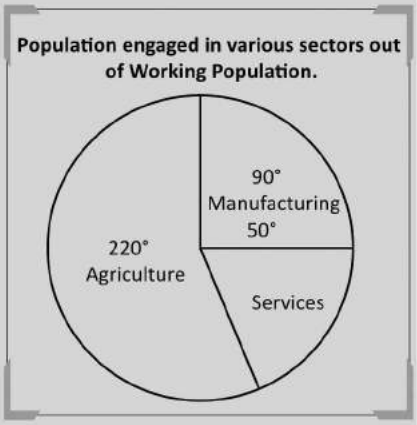
Q 19. The greatest number of votes cast in favour of an option in a single year were for
(a) Definitely prefer state
(b) Probably prefer private
(c) Probably prefer state
(d) Don't know

Q 20. What is (most nearly) the cumulative difference, over the entire period, between the number of votes in favour of 'state ownership'(definitely/probably) and the number of votes in favour of 'private ownership'(definitely/probably)?
(a) 2.2 million (b) 4.5 million
(c) 3.3 million (d) 6.3 million

Direction for questions 21 to 25: Refer to the following table and solve the questions based on it.

The following charts show the sectorwise percentage break-up of $ 650 billion, which is the GDP of the country with a population of 12 billion people. 70% of the population of the country constitutes its working population, engaged in the various sectors of the economy and contributing to the GDP.





Productivity is defined as the average amount contributed to the GDP per person in that group or sector.

Q 21. Which of the following groups/sectors has the lowest productivity?
(a) Agriculture sector workers
(b) Manufacturing sector workers
(c) Services sector workers
(d) Non-working population

Q 22. What is the approximate difference in the number of people involved in the agriculture sector and the services sector?
(a) 1.8 billion (b) 2.5 billion
(c) 4 billion (d) 5.7 billion

Q 23. What percentage of the country's total population is engaged in the manufacturing sector?
(a) 17.5% (b) 35.7%
(c) 25% (d) Indeterminate

Q 24. The productivity for how many sectors is more than the productivity of the entire population of the country?
(a) 0 (b) 1
(c) 2 (d) 3

Q 25. In the next year, the work force engaged in each of the sectors maintains its productivity at the same level. If there is an increase of 5% in the work-force for each of the sectors, then the productivity of the country has:
(a) Increased (b) Decreased
(c) Remained same (d) Cannot be determined

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12
Answer	(a)	(d)	(c)	(b)	(d)	(d)	(d)	(d)	(c)	(a)	(c)	(b)
Q. No.	13	14	15	16	17	18	19	20	21	22	23	24
Answer	(d)	(a)	(d)	(b)	(c)	(d)	(c)	(b)	(d)	(c)	(a)	(c)
Q. No.	25											
Answer	(d)											

HINTS AND SOLUTIONS

Table 1

Total Money Spent (Rs Lacs)

Year	TOI	HT	Others
2000	3.5	3.2	2
2001	5	3.4	2.8
2002	6.2	4	5.2
2003	8	4.8	4.4
2004	14	3.5	3.1
Total	36.7	18.9	17.5

Table 2

Average of Ads

Year	TOI	HT	Others
2000	10	12	15
2001	12	15	17
2002	15	20	36
2003	19	24	22
2004	38	18	30
Total	94	89	120

Table 3

Average Business/Client (Rs Lacs)

Year	TOI	HT	Others
2000	0.25	0.19	0.12
2001	0.19	0.28	0.09
2002	0.28	0.15	0.18
2003	0.31	0.55	0.22
2004	0.44	0.38	0.15
Total	1.47	1.55	0.76

Table 4

Total Client Won

Year	TOI	HT	Others
2000	23	25	9
2001	43	21	9
2002	50	31	7
2003	35	20	14
2004	77	34	11
Total	228	131	50

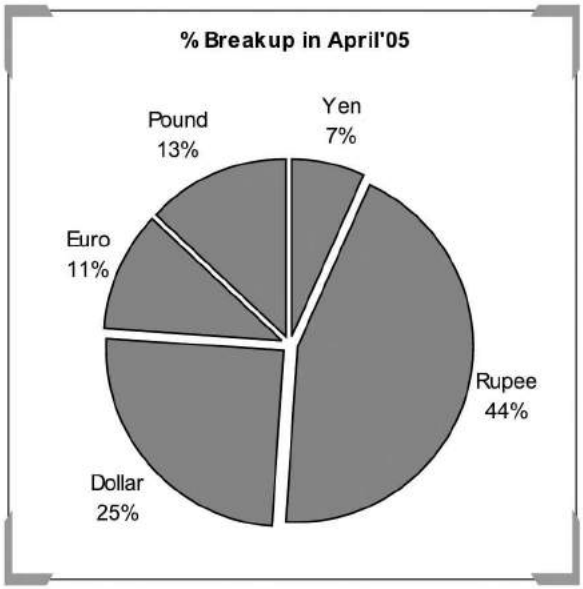
Table 5

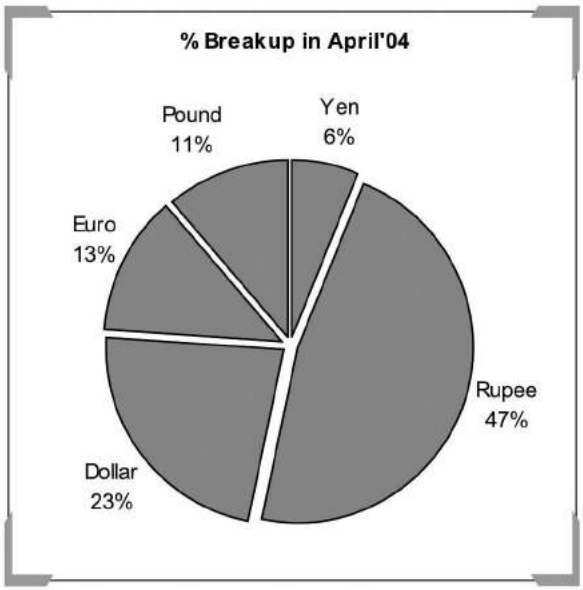
Total Enquiries Generated

Year	TOI	HT	Others
2000	65	44	22
2001	84	42	38
2002	122	49	25
2003	152	68	40
2004	242	70	32
Total	665	273	157

1. While the amount of money spent on ads in TOI is the maximum (nearly twice as that on ads in HT or others), the number of ads is nearly the same as that in HT and less than that in others.

2. Response ratio: Average number of enquiries generated (Table 4) per advertising (Table 2)
   (a) TOI in 2003:152/19 = 8
   (b) HT in 2004:70/18 < 4
   (c) Others in 2001:38/17 < 2.5
   (c) TOI in 2002:122/15 > 8
   So, amongst the given options, TOI in 2002 had the highest response ratio.
3. Amongst the three publications, the total number of clients (Table 5) to the number of enquiries generated was the least while the number of ads was the higher for others. So , the response ratio for others was the lowest during the given period.
4. Conversion ratio: Ratio of number of clients (Table 5) to the number of enquires generated (Table 4)
   For HT:
   2001:21/42 = 50% 2003: 20/68 < 33% 2002:31/49 > 60% 2000:25/44 > 55%
   So, amongst the given options, HT had the lowest conversion ratio in 2003.
5. Conversion ratio: Ratio of number of clients (Table 5) to the number of enquiries generated (Table 4) over the entire period
   TOI: 228/665 > 33% HT:131/273 = 48% Others: 50/157 < 33%
   So, the conversion ratios are in the order: Others < TOI < HT
6. Success ratio of number of clients (Table 5) to number of ads (Table 2)
   Success ratio for TOI over the entire period = 228/94 = 2.43; Success Ratios for others in 2003 = 14/22 = 0.63
   Required percentage = 2.43 – 0.63/0.63 x 100 = 285% (You can see that it is < 300%)
7. You need not calculate the success Ratios in 2001,2002 and 2003 to compare since the number of clients won by TOI is higher than HT and the number of ads in TOI is lower than HT in each of these years. So, the success ratio for TOI is higher than for HT in these 3 years. Comparing the success ratios for 2000 (when number of clients is less for TOI) and 2004 (when number of ads in TOI is more):
   2000: Success ratio for TOI = 23/10 = 2.3;
   Success ratio for HT = 25/12 = 2.083 < 2.3
   2004: Success ratio for TOI = 77/38 > 2;
   Success ratio for HT = 34/18 < 2
   So, the success ratio for TOI > the success ratio for HT in each of the 5 years.





8. Dollar reserves in April 2004 = 0.23 × 5 =Rs 1.15 crores = Rs 11.5 million
= $ 11.5/50 million = $ 0.23 million

9. Yen reserves in April 2005 = 7% of Rs 7 crores = Rs 0.49 crores
= Rs 4.9 million

10. The ratio of the amount of reserves in dollar to that in euro is the same as the ratio of their respective percentages in rupee equivalent terms i.e., 23 : 13 which is very close to 7 : 4.

11. Rupee equivalent in pound reserves in April 2005 = 13% of 7 crores = 0.91 crores
Rupee equivalent Yen reserves in April 2005 = 6% of 5 crores = 0.3 crores
⇨ the money in pound reserves in April 2005 is nearly three times the money in Yen reserves in April 2004.

12. Rupee reserves in April 2004 = 47% of 5 crores = 2.35 crores
Rupee reserves in April 2005 = 44% of 7 crores = 3.08 crores
Percentage increase in Rupee reserves = 308 – 235/235 × 100
= 73/235 × 100 = 31%
The following table summarizes the status of reserves of various currencies (in Lacs)

	2004	2005	Absolute Increase	% Increase
Rupee	235.00	308.00	73.00	31.06
Dollar	2.30	4.00	1.70	73.91
Euro	1.23	1.36	0.13	10.56
Pound	0.73	1.11	0.38	51.79
Yen	0.86	1.19	0.34	39.13

13. Absolute change for Dollar reserves
= 0.25 × 700 – 0.23 × 500 = Rs 60 lacs
Absolute change for Euro reserves
= 0.11 × 700 – 0.13 × 500 = Rs 12 lacs
Absolute change for Pound reserves
= 0.13 × 700 – 0.11 × 500 = Rs 36 lacs
Absolute change for Rupee reserves
= 0.44 × 700 – 0.47 × 500 = Rs 73 lacs
Absolute change for Yen reserves
= 0.07 × 700 – 0.06 × 500 = Rs 19 lacs
⇨ The maximum absolute increase is for the Rupee (change of 73 lacs)

The population of ABC in January 2002 was 2.4/1.2 = 2 million; in January 2003 was 2.4 million;

In January 2004 was 2.4 × 1.2 = 2.88 million; in January 2005 was 2.4 x $(1.2)^2$=3.46 million

14. Since the voting population increases by 20% each year, an increase in the number of people choosing an option would take place between any two years if:
(i) the percentage of people choosing an option would remain the same or increase between the two years
(ii) 1.2 × percentage in the second year ≥ percentage in the first year
Increase in each year, from 2003 to 2005, happens only for 'probably prefer state' and 'probably prefer private'

15. Using the reverse of the above conditions, you can see that the actual percentage voting for each one of the options increased in at least one of the years. Hence, none of the options saw a continuous decrease over the period.

16. Assume that P_1 and P_2 are the percentage votings for an option in January 2002 and January 2005 respectively. Then, the greatest absolute increase in the number of people would be for the option for which $(3.46P_2 – 2P_1)$ is the highest. This is the highest for the option 'probably prefer state'.

17. Assume that P_1 and P_2 are the percentage voting for an option in January 2002 and January 2005 respectively. A decrease in the number of people would take place for the option for which $(3.46\ P_2 – 2\ P_1) < 0$. This is so for the option 'definitely prefer private'.

18. Assume that P_1 and P_2 are the percentage votings for an option in January 2002 and January 2005 respectively. Then, the greatest percentage increase in the number of people would be for the option for which $3.46P_2 - 2\ P_1/2P_1$ is the highest or $1.73\ P_2/P_1$ or P_2/P_1 is the highest. This is clearly the case for the option 'don't know'.

Population = 12 billion = > working population = 0.7 X 12 = 8.4 billion.

	Contribution to GDP ($ billion)	Number of People (billion)	Productivity ($/person)
Agriculture	= 0.27 of 650 = 175.5	= (22/36) of 8.4 = 5.13	34.2
Manufacturing	= 0.22 of 650 = 143	= (9/36) of 8.4 = 2.1	68.1
Services	= 0.51 of 650 = 331.5	= (5/36) of 8.4 = 1.17	284.1
Working population	650	= 0.7 of 12 = 8.4	77.4
Total population	650	12	54.2

21. Non-working population contributes nothing to the GDP; hence, its productivity is zero.

22. Required percentage = $2.1/12 \times 100 = 17.5\%$

25. Since we do not know anything about the non-working population, we cannot determine the answer.

Module 2

Test Your Mettle

BENCHMARKING TEST 1

- There are 20 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd negative marking for every wrong answer.
- Time allotted = 45 minutes

Time Given: 45 Minutes **Total Marks: 60** **Qualifying Marks: 38**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
48+	99+
45	98.5+
43	98
38	92
30	85

Direction for questions 1 to 3: Read the passage below and solve the questions based on it.

Following table gives the result of Hero Cup football competition. Matches were played round robin (each team playing against another once) and the teams participated were Bihar, Punjab and Haryana. The table has some data missing:

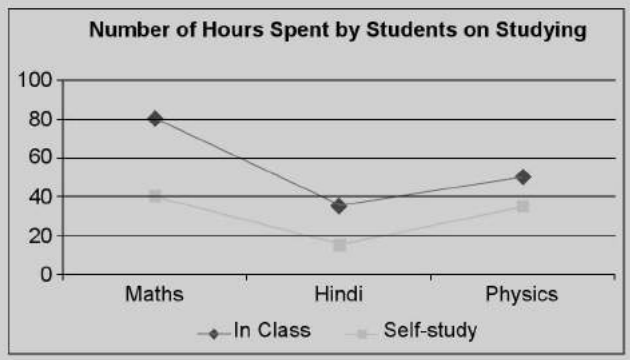
	Match played	*Won*	*Drawn*	*Lost*	*Goals for*	*Goals against*
Bihar	2		1		2	4
Punjab	2				3	7
Haryana	2					1

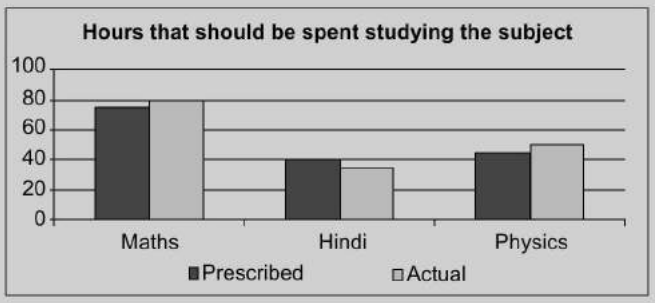
Q.1. What did the score card read in favour of Punjab in Punjab-Haryana match?
(a) 2 – 5 (b) 1 – 5
(c) 1 – 4 (d) 1 – 6

Q.2. How many matches did Punjab lose?
(a) 0 (b) 1
(c) 2 (d) 0 or 1

Q.3. How many goals did Haryana score?
(a) 5 (b) 1
(c) 3 (d) 7

Direction for questions 4 to 8: Go through the charts given below and solve the questions based on it.





Self-study is done always outside the class. All the values are the multiples of 5.

Q.4. Which of the following gives as many hours as those spent on Physics?
(a) Self-study - Math
(b) In Class - Physics and Hindi
(c) Self-study - Math and Hindi
(d) In Class - Math and Hindi

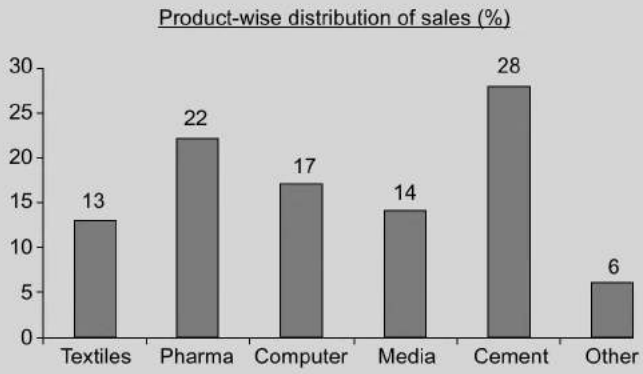
Q.5. What percentage of the hours is spent on learning in Class?
(a) 32% (b) 45%
(c) 65% (d) 80%

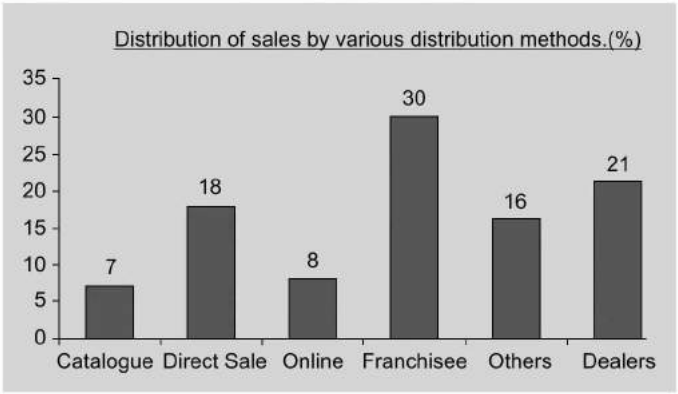
Q.6. The total number of hours spent on learning the subjects self-study is nearest to which of the following:
(a) The prescribed hours for Math
(b) The In Class hours for Math
(c) The sum of the Prescribed and Actual hours spent on Physics
(d) The total hours spent on Hindi

Q.7. If the total hours of study were to be 60 hours, then which of the following would account for approximately 30 hours?
(a) Physics (b) Math
(c) Hindi (d) In Class - Math and Hindi

Direction for questions 8 to 10: Read the passage below and solve the questions based on it.

The sales of a company in 2007–08 were Rs 1200 crore. Following charts represent the distribution of sales for various regions, product and distribution methods wise:





Q.8. If the computers are sold only through the dealers, how much of the dealers sales are made by products other than computers?
(a) Rs 48 crore (b) Rs 43crore
(c) Rs 252 crore (d) Rs 204 crore

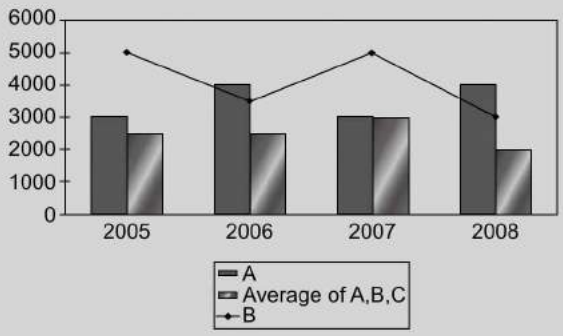
Q.9. In 2008–09 computers sales grow by 50 % while the sales of dealers grows by 30 %. If computers are sold only through dealers, what has been the percentage change in the sales of dealers of products other than computers?
(a) –20 % (b) –45 %
(c) –55% (d) +15%

Q.10. Cement sector has total sales of Rs 3215crore in the year. What is the market share of the company?
(a) 9.55% (b) 10.45%
(c) 8.71% (d) 11.55%

Direction for questions 11 to 15: Go through the bar chart below and solve the questions based on it.

There are three constituents of product D – A, B and C. When A, B and C are mixed in the ratio 1:2:3, D forms.

Following bar chart gives prices (in Rs) of 500 Kg of constituent A, 1000 Kg of constituent B and the average cost of 500 kgs each of constituent A, constituent B and constituent C. (All the values in the chart are in the multiples of 500).



Q.11. In which year was the price of C the highest?
(a) 2005 (b) 2006
(c) 2007 (d) 2008

Q.12. What was the ratio of price of C in 2005 to that in 2006?
(a) 13:14 (b) 8:7
(c) 27:29 (d) 4:3

Q.13. What was the percentage increase in price of C from 2006 to that in 2007?
(a) 100%
(b) 66.66%
(c) 124%
(d) 142%

Q.14. Vijay Shankar Pd. decides to buy 10 kg each of A, B and C. In which year will he pay the least amount of money?
(a) 2005 (b) 2006
(c) 2007 (d) 2008

Q.15. What was the cost of D in 2006 if 6000 kgs of D is to be made?
(a) Rs 17250 (b) Rs 36450
(c) Rs 25500 (d) Rs 20750

Direction for questions 16 to 20: Go through the table below and solve the questions based on it.

Following table gives the details pertaining to the company's employee appointment in three different wings at different levels and the salaries offered:

	Finance Wing			*Marketing Wing*			*H R D Wing*		
Details/Level of entry	***Entry Level***	***Middle Level***	***Top Level***	***Entry Level***	***Middle Level***	***Top Level***	***Entry Level***	***Middle Level***	***Top Level***
Average gross salary drawn per month (in Rs)	5000	8000	18000	4000	7000	17000	5500	7500	15000
Minimum Work Ex required for a graduate (in years)	0	2	NA	0	3	5	0	2	NA
Minimum Work Ex required for a PG (in years)	0	1	2	0	0	2	0	0	2
Number of Grads. Employed	1000	100	NA	1000	70	1	40	1	NA
Number of PG employed	200	100	5	Nil	80	2	80	4	1

Grads. Stands for graduate. A graduate has completed exactly 15 years of education. PG stands for post graduation and a post graduate has completed exactly 17 years of education.

A person is not allowed to shift from one wing to another wing.

Net salary = Gross Salary – taxes and other benefits

NA is given when it is not eligible.

Q.16. Ravi is a graduate and he joins in a wing at a particular level. What is the minimum time that Ravi will take to save Rs 200000, if 15% of his salary is deducted for tax and other benefits and he spends 50% of his net salary?
(a) 34 months (b) 44 months
(c) 56 months (d) None of these

Q.17. Ravi's friend Aftab, a post graduate joins at a middle management cadre. He joins that particular wing which will help him in saving maximum money in two years, given that 15% of his salary is deducted for taxation and other benefits and he spends 40% of his net income. Which wing he must have joined?
(a) Finance (b) Marketing
(c) HRD (d) None of these

Q.18. Ravi's another friend Indrajeet is a fresh graduate. He is confused about his next career move that should he go for post graduation or to start working. He finally decides to calculate her total earnings in 5 years' time in both cases for all wings and make his decision on that basis. Which of the following would be the best course of action assuming he keeps getting promoted immediately after meeting the minimum criteria? (cost of Post graduate program is Rs 250000 keeping all the other things constant.)
(a) Do post graduation and join marketing wing
(b) Do post graduation and join HRD wing
(c) Do not do post graduation and join HRD wing
(d) Do not do post graduation and join finance wing

Q.19. To facilitate the promotion of employees in a particular wing from lower level (L_1) to next higher level (L_2), a mechanism has been devised. The whole of this mechanism has been developed by Mr Thomas and hence, it is named as Thomas profile co-efficient. Thomas profile co-efficient (TPC) for any wing at any level is calculated as follows:

TPC = No. of the Employees in L_2 / No. of employees in L_1

Lower the TPC, higher is the chances of getting promoted.

What is the ratio of highest TPC of post graduate to the highest TPC of a graduate?

(a) 1:5 (b) 5:1
(c) 7:3 (d) 25:7

Q.20. Under a new policy to improve the profile of the company, it has been decided that no graduates will be recruited now and all graduates at entry level and middle level will be replaced with post graduates. Which wing has the best TPC at any level now (Use the definition of TPC from Q 19)?

(a) Finance
(b) HRD
(c) Marketing
(d) Finance or Marketing

My performance in this Test

Attempts	Right	Wrong	Net Marks	% Accuracy

ANSWERS

■■■ Benchmarking Test 1

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(d)	(b)	(c)	(c)	(b)	(a)	(c)	(b)	(c)	(b)	(a)	(d)	(c)	(d)	(a)	(d)	(b)	(b)

HINTS AND SOLUTIONS

	Match Played	*Won*	*Drawn*	*Lost*	*Goals For*	*Goal Against*
Bihar	2	0	1	1	2	4
Punjab	2	0	1	1	3	7
Haryana	2	2	0	0	7	1

Score Card:

Bihar – (2 – 2), (0 – 2)

Punjab – (2 – 2), (1 – 5)

Haryana – (5 – 1), (2 – 0)

4. It can be observed that the value is same as [2] as in Class Hindi has the same value as Self-study Physics. Hence [2]

5. Hours In Class = 165. Hours self-study = 90.
= 65% Hence [3]

6. Total Self-study = 90 hours. Option [3], 95 hours is the closest. Hence [3]

7. Total hours currently = 255. If we have 60 hours and are looking at 30 hours, we are looking at 50% of the time. This is around 127 hours.
We find that Math with 120 hours comes closest. Hence [2]

8. (1) Computer sales = 0.17 × 1200 = Rs 204 crore.
Dealer sales = 0.21 × 1200 = Rs 252 crore
Hence, dealers sales by other products = Rs 48 crore

9. (3) In 2008–09, computer's sales = 1.5 × 204 = Rs 306 crore, Dealer sales = 1.3 × 252 = Rs 327.6 cross

2008–09 sales by dealers of other products = 327.6 – 306 = Rs 21.6 crore

From 48 to 21.6, change is negative and exceeds 50%.

10. (2) $\frac{0.28 \times 1200}{3215} = \frac{336}{3215} \approx 10.45\%$

[redacted]

For 2005 : 2500x3 – (3000+2500) = Rs 2000; For 2006: 7500 – 5750 = Rs 1750

For 2007: 9000 – 5500 = Rs 3500 ; For 2008: 6000 – 5500 = Rs 500

12. 2000:1750 = 8:7. So [2]

13. (3500–1750)/1750 = 100%. So [1]

15. Cost of 1000 kg of A in 2006 = 8000
Cost of 2000 kg of B in 2006 = 7000 and
Cost of 3000 kg of C in 2006 = 10500.
So, Cost of 6000 kgs of D in 2006 is Rs 25500.
So [3]

[redacted]

16. Minimum time will be if he earns maximum salary. This is possible if he works at the top level of Marketing Wing. He saves Rs 7225. Hence, time taken to save Rs 200000 is 28 months.
Hence, option [5] is the answer.

17. At the middle level, finance wing gives highest salaries. Hence he is bound to save most in this wing. Hence option [1] is the answer.

19. Highest TPC for a post-graduate is from entry level to middle level in finance wing (50%). Highest TPC for graduate is also from entry level to middle level in finance wing (10%). Hence, ratio = 5:1. Hence, option [2]is the answer.

20. Option 2, from middle level to top level in HRD.

BENCHMARKING TEST 2

- There are 20 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd negative marking for every wrong answer.
- Time allotted = 45 minutes

Time Given: 45 Minutes **Total Marks: 60** **Qualifying Marks: 38**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
48+	99+
45	98.5+
43	98
38	92
30	85

Direction for questions 1 to 5: Go through the data set given below and solve the questions based on it.

Name of Stock	*Date of Auction*	*Date of Issue*	*Notified amount*	*Devolvement on PDs*	*Cut off price (in Rs)*	*Cut off yield (%)*	*Date of Maturity*
7.56% GS 2014	02-Apr-09	06-Apr-09	8000	0	103.47	6.80	03-Nov-14
7.94% GS 2021	02-Apr-09	06-Apr-09	4000	567	102.5	7.62	24-May-21
6.05% GS 2019	09-Apr-09	13-Apr-09	8000	0	95.05	6.75	02-Feb-19
7.50% GS 2034	09-Apr-09	13-Apr-09	4000	0	97.35	7.74	10-Aug-34
7.56% GS 2014	17-Apr-09	20-Apr-09	8000	0	106.75	6.10	03-Nov-14
8.24% GS 2027	17-Apr-09	20-Apr-09	4000	0	107.82	7.44	15-Feb-27
6.05% GS 2019	24-Apr-09	27-Apr-09	8000	0	99.38	6.13	02-Feb-19

Q.1. For how many stocks, date of maturity is minimum 5 years ahead of date of auction?
(a) All but 1 stock (b) All but 2 stocks
(c) All but 3 stocks (d) None of these

Q.2. Nomenclature of the stocks is done in the following way:
7.56% GS 2014 - It means that desired yield from the Government stock (GS) in 2014 is 7.56%.
For how many of the stocks Desired yield is more than cut-off yield?
(a) 3 (b) 4
(c) 5 (d) 2

Q.3. How many stocks with a cut-off price of more than Rs 100 have their cut-off yield more than its desired yield (use definition from above question if required)?
(a) 0 (b) 1
(c) 2 (d) 3

Q.4. Mr Robert is quite an ambitious person. He wants to obtain a minimum desired yield of 8%. For that to happen, he buys exactly two stocks in different ratio. How many combinations can he have, out of 7 stocks given, such that his minimum desired yield is atleast 8% (use information from previous qs)?

(a) Not possible (b) 6
(c) 7 (d) None of these

Q.5. If we arrange the stocks in the ascending order of their cut-off price, which stock will be ranked third?
(a) 7.50% GS 2034 (b) 6.05% GS 2019
(c) 7.94% GS 2021 (d) 7.56% GS 2014

Direction for questions 6 to 8: Go through the data set given below and solve the questions based on it.

Five women decided to go shopping to MG Road, Bangalore. They arrived at the designated meeting place in the following order: 1. Archana, 2. Chellamma, 3. Dhenuka, 4. Helen, and 5. Shahnaz. Each women spent at least Rs 1000. below are some additional facts about how much they spent during their shopping spree.

i. The woman who spent Rs 2234 arrived before the lady who spent Rs 1193.
ii. One woman spent Rs 1340 and she was not Dhenuka.
iii. One woman spent Rs 1378 more than Chellamma.
iv. One woman spent Rs 2517 and she was not Archana.
v. Helen spent more than Dhenuka.
vi. Shahnaz spent the largest amount and Chellamma the smallest.

Q.6. What was the amount spent by Helen?
(a) Rs 1193 (b) Rs 1340
(c) Rs 2234 (d) Rs 2517

Q.7. Which of the following amount was spent by one of them?
(a) Rs 1139 (b) Rs 1378
(c) Rs 2571 (d) Rs 2718

Q.8. The women who spent Rs 1193 is
(a) Archana (b) Chellamma
(c) Dhenuka (d) Helen

Direction for questions 9 to 11: Go through the data set given below and solve the questions based on it.

Five friends meet every morning at Sree sagar restaurant for an idli-vada breakfast. Each consumes a different number of idli and vada. The number of idli consumed are 1,4, 5,6 and 8, while the number of vadas consumed are 0,1,2,4 and 6. Below are some more facts about who eats what and how much:

i. The number of vadas eaten by Ignesh is three times the number of vadas consumed by the person who eats four idlis.
ii. Three persons, including the one who eats four vadas eat without chutney.
iii. Sandeep does not take any chutney.
iv. The one who eats one idki a day does not eat any vadas or chutney. Further, he is not Mukesh.
v. Daljits eats idli with chutney and also eats vada.
vi. Mukesh, who does not take chutney, eats half as many vadas as the person who eats twice as many idlis as he does.
vii. Bimal eats two more idlis than Ignesh, but Ignesh eats two more vadas than Bimal.

Q.9. Which of the following statement is true?
(a) Daljit eats 5 idlis (b) Ignesh eats 8 idlis
(c) Bimal eats 1 idli (d) Bimal eats 6 idlis.

Q.10. Which of the following statement is true?
(a) Sandeep eats 2 vadas.
(b) Mukesh eats 4 vadas.
(c) Ignesh eats 6 vadas.
(d) Bimal eats 2 vadas.

Q.11. Which of the following statements is true?
(a) Mukesh eats 8 idlis and 4 vada but not chutney
(b) The person who eats 5 idlis and 1 vada does not take chutney.
(c) The person who eats equal number of vadas and idlisalso takes chutney.
(d) The person who eats 4 idlis and 2 vada also takes chutney.

Direction for questions 12 to 16: Go through the data set given below and solve the questions based on it.

The table below presents data on percentage data on percentage covered by drinking water and sanitation facilities in selected Asian countries.

Population covered by drinking water and sanitation facilities (Percentage Coverage)

	Drinking water			Sanitation Facilities		
	Urban	Rural	Total	Urban	Rural	Total
India	85	79	81	70	14	29
Bangladesh	99	96	97	79	44	48
China	97	56	67	74	7	24
Pakistan	82	69	74	77	22	47
Philippines	92	80	86	88	66	77
Indonesia	79	54	62	73	40	51
Sri-Lanka	88	52	57	68	62	63
Nepal	88	60	63	58	12	1

Country A is said to dominate B or A>B if A has higher percentage in total coverage for both drinking water and sanitation facilities, and, B is said to be dominated by A, or B <A.

A country is said to be on coverage frontier if no other country dominates it. Similarly, a country is not on the coverage frontier if it is dominated by at least one other country.

Q12. Which countries are the countries on the coverage frontier?
(a) India and China
(b) Sri Lanka and Indonesia

(c) Philippines and Bangladesh
(d) Nepal and Pakistan

Q.13. Which of the following statements are true?
A India>Pakistan and India>Indonesia
B India>China and India> Nepal
C Sri Lanka>China
D China>Nepal
(a) A and C (b) B and D
(c) A, B and C (d) B, C and D

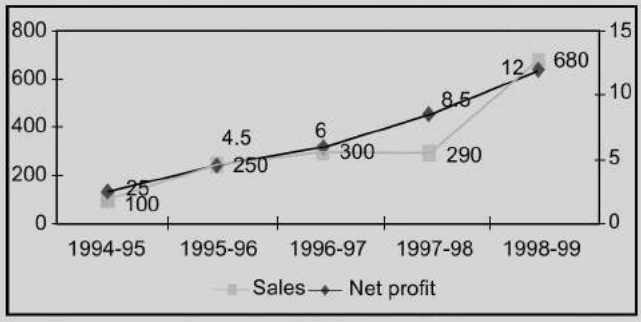
Q.14. Using only the data presented under 'sanitation facilities' columns, it can be concluded that rural population in India, as a percentage of its total population is approximately
(a) 76 (b) 70
(c) 73 (d) Cannot be determined

Q.15. Again, using only the data presented under 'Sanitation facilities' columns, sequence China, Indonesia and Philippines in ascending order of rural population as a percentage of their respective total population. The correct order is
1. Philippines, Indonesia, China
2. Indonesia, China, Philippines
3. Indonesia, Philippines, China
4. China, Indonesia, Philippines

Q.16. India is not on the coverage frontier because:
A. It is lower than Bangladesh in terms of coverage of drinking water facilities.
B. It is lower than Sri-Lanka in terms of coverage of sanitation facilities.
C. It is lower than Pakistan in terms of coverage of sanitation facilities.
D. It is dominated by Indonesia.
(a) A and B (b) A and C
(c) D (d) None of these.

Direction for questions 17 to 20: Go through the data set given below and solve the questions based on it.

The figure below represents states sales and net profit in Rs crore of IVP ltd For five years from 1994–95 to 1998–99. During this period the sales increased from Rs 100 crore to Rs 680 crore. Correspondingly, the net profit increased from Rs 2.5 crore to Rs 12 crore. Net profit is defined as the excess of sales over total costs.



Q17. The highest percentage of growth in sales, relative to the previous year occurred in
(a) 1995–96 (b) 1996–97
(c) 1997–98 (d) 1998–99

Q18. The highest percentage growth in net profit, relative to the previous year, was achieved in
(a) 1998–99 (b) 1997–98
(c) 1996–97 (d) 1995–96

Q.19. Defining the profitability as the ratio of net profit to sales, IVP Ltd, recorded the highest profitability in
(a) 1998–99 (b) 1997–98
(c) 1994–95 (d) 1996–97

Q.20. With profitability as defined in question 19, it can be concluded that:
(a) Profitability is non-decreasing during the five years from 1994–95 to 1998–99.
(b) Profitability is non-decreasing during the five years from 1994–95 to 1998–99.
(c) Profitability remained constant during the five years from 1994–95 to 1998–99
(d) None of above.

My performance in this Test

Attempts	Right	Wrong	Net Marks	% Accuracy

ANSWERS

■■■ Benchmarking Test 2

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(a)	(b)	(b)	(b)	(a)	(c)	(a)	(c)	(c)	(c)	(b)	(c)	(a)	(d)	(a)	(d)	(b)	(d)

HINTS AND SOLUTIONS

1. The least occurs in case of 7.56% GS 2014 which is more than 5 years. Hence, answer is – all the years. Hence answer is option (d) None of these.

2. For 4 stocks, the desired yield is more than cut off yield. Hence, option (b) is the answer.

3. Following stocks are having their cut-off price more than Rs 100:

Name of Stock	*Cut off price*
7.56% GS 2014	103.47
7.94% GS 2021	102.5
7.56% GS 2014	106.75
8.24% GS 2027	107.82

None of these stocks have their cut-off yield more than its desired yield. Hence, option (a) is the answer.

4. To obtain a minimum desired yield of 8%, Mr Robert has to take 8.24% GS, as for remaining all the stocks, minimum desired yield is 8%. (One cannot have above 8% yield by mixing two stocks with less than 7% stock. It means that one stock has to be higher than 8 % and other has to be less than 8%).
He can mix 8.24% GS with any one of the remaining 6 stocks to obtain minimum 8% desired yield.
Hence, option (b) is the answer.

5.

Rank 1	**6.05% GS 2019**	**95.05**
Rank 2	7.50% GS 2034	97.35
Rank 3	6.05% GS 2019	99.38
Rank 4	7.94% GS 2021	102.5
Rank 5	7.56% GS 2014	103.47
Rank 6	7.56% GS 2014	106.75
Rank 7	8.24% GS 2027	107.82

Hence option (b) is the answer.

One of the women spent 2517 – 1378 = 1139, she is Chellamma. This is the only possibility as we add Rs 1378 even to the least amount Rs 1193, we will not be able to satisfy all the conditions given simultaneously.
So, the table obtained is:

A	*C*	*D*	*H*	*S*
2234	1139	1193	1340	2517

Let us make a table of the information given:

	Idli	*Vada*
Ignesh	6	6
Sandeep	1	0
Mukesh	4	2
Daljit	5	1
Bimal	8	4

12. Bangladesh > Philippines (97>86) for drinking water. And Philippines > any other countries for sanitation facilities. Thus, these two countries are on the coverage frontier for two facilities.

13. Statement A > Statement B only if statement A has higher percentage in total coverage for both drinking water and sanitation facilities taken independently and not as a total of the two facilities.
Thus, only statement B and statement D are true
India > China (81>67 and 29>24)
India >Nepal (81>63 and 29>18)
Also china > Nepal (67>63 and 24>18)

14. Let the urban population be x and rural population be y.
From the sanitation column, we have $0.7x + 0.14y = 0.29(x+y)$, OR, $0.41x = 0.15y$

Therefore $x = \frac{15y}{41}$

Therefore percentage of rural population

$$\frac{y}{x+y} \times 100 = \frac{y}{\frac{15y}{41}+y} \times 100 = \frac{41}{56} \times 100 = 73.2\%$$

15. In the same way as the previous questions, we can find percentage of rural population for Philippines, Indonesia and China.

P	50%
I	66.66%
C	79.8%

Thus, Philippines < Indonesia < China.

16. India is not on coverage frontier because
(a) It is below Bangladesh and Philippines for drinking water.
(b) For sanitation facilities it is below Philippines *inter alia.*

BENCHMARKING TEST 3

- There are 20 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd negative marking for every wrong answer.
- Time allotted = 45 minutes

Time Given: 45 Minutes **Total Marks: 60** **Qualifying Marks: 38**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
48+	99+
45	98.5+
43	98
38	92
30	85

Direction for questions 1 to 5: Refer to the tables given below and answer the questions that follow.

Table 1 give the transportation cost of a product from six factories F1, F2, F3, F4, F5 and F6 to five warehouses W1, W2, W3, W4 and W5.

	W1	*W2*	*W3*	*W4*	*W5*
F1	60	25	45	40	80
F2	85	40	45	60	75
F3	80	20	45	50	90
F4	90	40	55	25	70
F5	95	20	40	45	80
F6	85	35	50	50	75

Table 2 give the transportation cost of the same product from five warehouses W1, W2, W3, W4 and W5 to six retailers R1, R2, R3, R4, R5 and R6.

	R1	*R2*	*R3*	*R4*	*R5*	*R6*
W1	40	20	40	30	85	25
W2	50	50	45	60	70	30
W3	20	10	50	50	80	35
W4	40	30	45	40	90	40
W5	50	30	50	40	70	30

In addition to the above tables given, following table gives the transportation cost incurred from different retailers to the five different colonies C1, C2, C3, C4 and C5.

	C1	*C2*	*C3*	*C4*	*C5*
R1	10	25	10	10	10
R2	15	10	15	10	15
R3	10	20	10	15	10
R4	20	10	15	15	10
R5	15	10	20	20	20
R6	10	15	15	15	15

Q.1. What is the minimum transportation cost incurred in transferring the product from any factory to any colony?
(a) Rs 45 (b) Rs 65
(c) Rs 55 (d) None of these

Q.2. What is the minimum transportation cost incurred in transferring the product from any factory to the colony C2?
(a) Rs 45 (b) Rs 60
(c) Rs 55 (d) None of these

Q.3. If one unit produced in each of the factories is to be sent to each of the colonies, then what will be the total transportation cost incurred?
(a) Rs 1705 (b) Rs 1980
(c) Rs 2185 (d) None of these

Q.4. $F_iW_jR_kC_l$ represents that the product has moved in this cycle – Factory i to warehouse j to retailer k to colony l. If i = 2 and k = 2, then what is the sum of (j + l) for the lowest transportation cost incurred in one product transferred from i = 2 to any of the colonies?
(a) 4 (b) 5
(c) 7 (d) cannot be determined

Q.5. In the above question, if j = 2, then how many different value/s of (i + k + l) is/are possible for the lowest transportation cost incurred in one product transferred from any of the factories to any of the colonies?
(a) 1 (b) 2
(c) 3 (d) cannot be determined

Direction for questions 6 to 10: Refer to the table given below and answer the questions that follow.

Given below are the forecasts of world and Asian energy demand for the years 1991, 2001 and 2011. The demand is given in million barrels per day, crude oil equivalent.

	1991		*2001*		*2011*	
	World	*Asia*	*World*	*Asia*	*World*	*Asia*
Petrol	50.0	4.0	70.0	10.0	80.0	15.0
Naphtha	30.0	0.5	40.0	2.5	50.0	5.0
Solid Fuels	50.0	4.0	60.0	5.0	75.0	10.0
Nuclear Fuel	10.0	0.5	20.0	1.0	25.0	1.3
Hydro Energy	10.0	1.0	10.0	1.5	20.0	2.0
Total	150.0	10.0	200.0	20.0	250.0	33.3

Q.6. What can be inferred from the given table?
(a) Percentage increase in the number of Petrol using vehicles/agents have increased in Asia than the world in the decade 1991–2001.
(b) Exploration of Naphtha is low in Asia is less than the exploration of Naphtha in the world throughout the period.
(c) Both (a) and (b)
(d) None of these

Q.7. Over 1991-2011, which two fuels meet more than 60% of the total energy demand of both the world and Asia?
(a) Petrol and Naphtha
(b) Petrol and Solid fuels
(c) Naphtha and Solid fuels
(d) None of these

Q.8. Which fuel's proportion in the total energy demand increases over the decade 1991–2001 and decreases over the decade 2001–2011 for both the world and Asia?
(a) Petrol (b) Naphtha
(c) Solid fuels (d) Nuclear Fuel

Q.9. Which is the fuel whose proportion in the total energy demand will decrease continuously over the period 1991–2011, in Asia?
(a) Naphtha (b) Solids fuels
(c) Nuclear Fuel (d) Hydro Energy

Q.10. Which is the fuel whose proportions to the total energy demand of the world will remain constant over the period 1991–2011 but whose proportion will increase in the total energy demand in Asia?
(a) Naphtha (b) Solids fuels
(c) Nuclear Fuel (d) Hydro Energy

Direction for questions 11 to 15: Refer to the table given below and answer the questions that follow.

Following table gives some incomplete information on the marks obtained by four persons Amit, Binit, Charu & Dilip in four different subjects.

	Physics	*Chemistry*	*Maths*	*English*	*Total*
Amit	80	70	–	80	–
Binit	60	–	60	–	–
Charu	–	–	60	–	220
Dilip	90	–	70	60	–

Additional Information:

(i) Amit's average marks was 72.5, in all the four subjects.
(ii) The average marks in Physics for all four persons was 67.5.
(iii) Binit's Average marks for all four subjects was 1.5 times Charu's marks in physics.
(iv) Binit & Dilip scored the same marks in English. The average English marks were the same as the average Physics marks.
(v) The total marks in Chemistry was 10 less than the total marks in physics for all four persons.

Q.11. Who scored the lowest total?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Q.12. The average marks for all four persons (for all the four subjects) is (approx.)
(a) 65.625 (b) 81.225
(c) 99.5 (d) 105.6

Q.13. Who is the highest scorer in chemistry?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Q.14. Who is the lowest scorer in physics?
(a) Amit (b) Binit
(c) Charu (d) Dilip

Directions for questions 15 to 17: In each question, there are two statements: A and B, either of which can be true or false on the basis of the information given below.

A research agency collected the following data regarding the admission process of a reputed management school in India.

Year	Gender	Number bought application forms	Number appeared for written test	Number called for interviews	Number selected for the course
2002	Male	61205	59981	684	171
	Female	19236	15389	138	48
2003	Male	63298	60133	637	115
	Female	45292	40763	399	84

Choose (1) if only A is true
Choose (2) if only b is true
Choose (3) if both A and B are true
Choose (4) if neither A nor b is true

Q.15. **Statement A:** The success rate of moving from written test to interview stage for males was worse than for females in 2003.

Statement B: The success rate of moving from written test to interview test to interview stage for females was better in 2002 than in 2003.

Q.16. **Statement A:** In 2002, the number of females selected for the course as a proportion of the number of females who bought application forms, was higher than the corresponding proportion for males.

Statement B: In 2002, among those called for interview, males had a greater success rate than females.

Q.17. **Statement A:** The percentage of absentees in the written test among females decreased from 2002 to 2003.

Statement B: The percentage of absentees in the written test among males was larger than among females in 2003.

Directions for questions 18 to 20: Answer the questions on the basis of the information given below.

The table below provides certain demographic details of 30 respondents who were part of a survey investigating the irregularities in NREGS. The demographic characteristics are: gender, number of children, and age of respondents. The first number in each cell is the number of respondents in that group. The minimum and maximum age of respondents in each group is given in brackets. For example, there are five female respondents with no children and among these five; the youngest is 34 years old, while the oldest is 49.

No. of children	*Male*	*Female*	*Total*
0	1(38,38)	5(34,49)	6
1	1(32,32)	8(35,57)	9
2	8(21,65)	3(37,63)	11
3	2(32,33)	2(27,40)	4
Total	12	18	30

Q.18. The percentage of respondents aged less than 40 years is at least
(a) 10% (b) 16.67%
(c) 20.0% (d) 30%

Q.19. Given the information above, the percentage of respondents older than 35 can be at most
(a) 30% (b) 73.33%
(c) 76.67% (d) 90%

Q.20. The percentage of respondents that fall into the 35 to 40 years age group (both inclusive) is at least
(a) 6.67% (b) 10%
(c) 13.33% (d) 26.67%

My performance in this Test

Attempts	Right	Wrong	Net Marks	% Accuracy

ANSWERS

■■■ Benchmarking Test 3

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(b)	(d)	(d)	(d)	(d)	(b)	(a)	(d)	(d)	(c)	(a)	(d)	(c)	(d)	(b)	(a)	(b)	(c)	(c)

HINTS AND SOLUTIONS

1. F3 to W2 to R2 to C1 is the lowest.
2. F3 to W2 to R6 to C2 is the minimum.
4. F2 – W3 – R2 – either C2/C4. Hence, two different values are possible.
5. There can be two values of i = 3/5, one value of j = 2, one value of k = 6 and one value of l = 1. Hence, two different sums are possible.

6. Obviously cannot be inferred. (1) seems probable, but number of vehicles/agents need not be increasing. There can be other reasons also like reduction in the efficiency of the vehicle etc.

7.

		1991		*2001*		*2011*	
		World	***Asia***	***World***	***Asia***	***World***	***Asia***
Total energy		150	10	200	20	250	33.3
Naphtha	Value	30	0.5	40	2.5	50	5
	Proportion	20%	5%	20%	12.50%	20%	15.01
Solid fuels	Value	50	4	60	5	75	10
	Proportion	33.30%	40%	30%	25%	30%	30.03
Petrol	Value	50	4	70	10	80	15
	Proportion	33.30%	40%	35%	50%	32%	45.04

Hence, solid fuels and Petrol combined constitute more than 60 percent of total energy in both the world and Asia for the given period.

8. It can be seen from the table that Petrol is the fuel whose proportion in the total energy demand increases during 1991-2001 and decreases during 2001-2011 for both the world and Asia.

9. For the answer choices given and for Asia we can take make the following table.

		1991	*2001*	*2011*
	Total energy	***10***	***20***	***33***
Naphtha	Value	0.5	2.5	5
	Proportion	5%	12.50%	15.15%
Solid fuels	Value	4	5	10
	Proportion	40%	25%	30.30%
Nuclear Fuel	Value	0.5	1	1.3
	Proportion	5%	5%	3.90%
Hydro Energy	Value	1	1.5	2
	Proportion	10%	7.50%	6.06%

Hence, we can see that the proportion of Hydro Energy goes on decreasing over the period.

10. For answer choices given and for the worlds, we can make the following table.

		1991	*2001*	*2011*
	Total energy	***150***	***200***	***250***
Naphtha	Value	30	40	50
	Proportion	20%	20%	20%
Solid Fuels	Value	50	60	75
	Proportion	33%	30%	30%
Nuclear Fuel	Value	10	20	25
	Proportion	6.66%	10%	10%
Hydro Energy	Value	10	10	20
	Proportion	6.66%	5%	8%

Hence, we can see that the proportion of Naphtha remains constant over the given period.

From the given information the following table can be prepared:

	Physics	*Chemistry*	*Maths*	*English*	*Total*
Amit	80	70	60	80	290
Binit	60	60	60	60	240
Charu	40	50	60	70	220
Dilip	90	80	70	60	300

19. (c) (23/30) × 100 = 76.67 %
20. (c) (4/30) × 100 = 13.33 %

BENCHMARKING TEST 4

- There are 20 questions in this test.
- Each question carries 3 marks.
- There is 1/3rd negative marking for every wrong answer.
- Time allotted = 45 minutes

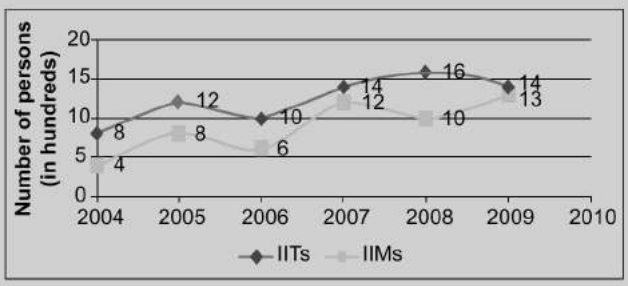
Time Given: 45 Minutes **Total Marks: 60** **Qualifying Marks: 38**

Break-up of the Expected Percentile
(assuming this to be a CAT paper)

Marks	*Expected Percentile*
48+	99+
45	98.5+
43	98
38	92
30	85

Direction for questions 1 to 4: Go through the chart given below and solve the questions based on it.

Chart given below represents total number of persons (in hundreds) who have passed out from IIMs and IITs in each year from 2004 to 2009. The IITs and IIMs are among the best known professional institutes in India (under top 10 rank) and offer courses in the areas of engineering and management respectively. Both the IIMs and IITs have six campuses located in various cities of India, none of the cities being common.



Q.1. How many times during the given period has the growth in pass outs from IITs exceeded 25% over previous year?
(a) Once (b) Twice
(c) Thrice (d) Never

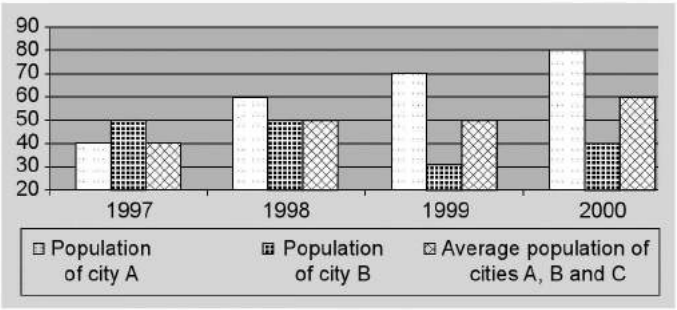
Q.2. During the 2004-2008, the total pass outs from IIMs as a ratio of total pass outs from IIT is
(a) 41/60 (b) 10/13
(c) 6/9 (d) 4/5

Q.3. What is the ratio of [ratio of IIM graduates/IIT Graduates] in 2006 to that in 2008?
(a) 0.37 (b) 2.66
(c) 1.04 (d) 0.96

Q.4. On an average, every year 20% of the students who pass IITs join IIMs the same year (An IITian can join IIM but not the vice versa). What is the total number (in '000) of IITs pass outs who went to IIM in 2006?
(a) 2 (b) 2.5
(c) 1 (d) cannot be determined

Direction for questions 5 and 6: Go through the chart given below and solve the questions based on it.

There are only three cities—A, B and C—in a state. Following bar chart gives us the population figures of these cities (Figure in '000):

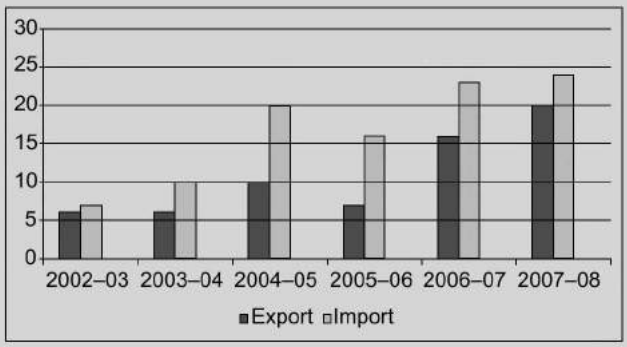


Q.5. If increase in the population of city can only be due to new births and decrease in population of city can only due to deaths, then number of new births in city A in year 2000 is (in '000):
(a) 20 (b) 40
(c) 38 (d) Cannot be determined

Q.6. For which year is the ratio of population of city A to the population of city B the highest?
(a) 1997 (b) 1998
(c) 1999 (d) 2000

Direction for questions 7 – 9: Go through the chart given below and solve the questions based on it.

Following Bar graph for the data related to the Export & Import in million Rupees of dueNorth Limited.



Q.7. Import in financial year 2007 – 08 has approximately
(a) Increased by 24% since base year of 2002 – 03
(b) Increased by 240% since base year of 2002 – 03
(c) Increased by 140% since base year of 2002 – 03
(d) Data is insufficient to find the answer.

Q.8. In which financial year difference between import and export is the highest?
(a) 2007 – 08 (b) 2006 – 07
(c) 2005 – 06 (d) 2004 – 05

Q.9. From year 2003 – 04 to 2007 – 08, the ratio of total Import during these five years to the total export in the same period is in the range of
(a) 0.75 – 1 (b) 1 – 1.25
(c) 1.25 – 1.5 (d) 1.5 – 1.75

Direction for questions 10 to 14: Go through the chart given below and solve the questions based on it.

A test was performed on various food items to check for protein content, fat content and Mineral content. Five food items viz., Butter, Apple, Dal, Chicken and Bournvita. The chart shows the quantity of foods checked:

Food Item	*Quantity Checked*	*Protein content*	*Fat Content*	*Mineral content*
Butter	2 gm	10 mg	1.0 gm	75 mg
Apple	1 gm	15 mg	35 mg	60 mg
Dal	2 gm	100 mg	15 mg	90 mg
Chicken	3 gm	75 mg	1 gm	150 mg
Bournvita	1 gm	25 mg	20 mg	80 mg
Veg Burger	2gm	5 mg	1.1 gm	15 mg

NOTE: 1gm = 1000 mg

Based on these results, the foods are classified into the following categories:

Healthy food	If percentage of proteins the highest and the percentage of fats is the least as compared to the other foods.
Junk food	If the percentage of fat is the highest and the percentage of protein is the least.
Not recommended	If the percentage of fats is more than the percentage of proteins and the mineral content is least as compared to the other foods

Answer the questions on the basis of the information given above.

Q.10. What is the highest percentage of protein any of the given foods?
(a) 0.5% (b) 5%
(c) 1% (d) 10%

Q.11. Which food can be referred to as Healthy?
(a) Butter (b) Apple
(c) Dal (d) Chicken

Q.12. Which food can be classified as junk from amongst the above mentioned foods?
(a) Veg burger (b) Apple
(c) Dal (d) Chicken

Q.13. Which food is not recommended as per the criteria defined above?
(a) Butter (b) Apple
(c) Veg Burger (d) Chicken

Q.14. In how many foods is the percentage of fat more than the percentage of protein?
(a) 1 (b) 2
(c) 3 (d) 4

Direction for questions 15 to 18: Go through the chart given below and solve the questions based on it.

Following grid presents the percentage composition of five verticals under different sectors.

Sector	*Factories*	*Employment*	*Fixed Capital*	*Variable Cost*	*Value Added*
Government	18	15	14	22	25
Central	8	6	5	7	10
State	4	6	4	8	3
Central/State	6	3	5	7	12
Public	12	8	6	19	8
Private	55	65	72	54	62
Joint	15	12	8	5	5
Total	100	100	100	100	100

Q15. If the total work force was 76 million whereas the total value added was 225 million, then which of the following had the maximum value addition per worker?
(a) Central (b) State
(c) Central/State (d) Public

Q16. Which of the following sectors has the maximum fixed capital invested per factory?
(a) Central (b) State
(c) Central/State (d) Public

Q17. If the variable is proportional to the number of employees and the production per employee, then for which of the following is the production highest?
(a) Government (b) Private
(c) Joint (d) Public

Q18. If the government has a fixed capital of $200 million in the Iron& Steel industry, which corresponds to 20.012% of its total investment as fixed capital, then how much did the government invest (in Rs million) in Maruti Udyog Ltd which forms 25% of the investment in the joint sector? (1US $ = Rs 45)
(a) 6500 (b) 2500
(c) 143 (d) 145

Direction for questions 19 to 20: Go through the chart given below and solve the questions based on it.

The following table shows the report card of a student of a management entrance exam. The exam had three sections Data Interpretation, Quantitative aptitude and Verbal Ability.

Test Area	*Number of questions*	*Questions Attempted*	*Incorrect attempts*
Data Interpretation	50	28	6
Quantitative Aptitude	50	30	11
Verbal Ability	50	38	13

For each correct answer the student gets 4 marks and for each wrong answer 1 mark is reduced from the total. Also, Accuracy is defined as (no. of questions attempted correctly)/(total no. of questions attempted).

Answer the following questions on the basis of the information given above.

Q.19. Arrange the three sections in the increasing order of the score obtained by the student?
(a) DI, VA, QA
(b) VA, DI, QA
(c) QA, DI, VA
(d) VA, QA, DI

Q.20. How much percentage of the total maximum score did the student score?
(a) 39% (b) 41.6%
(c) 47.2% (d) 50%

My performance in this Test

Attempts	Right	Wrong	Net Marks	% Accuracy

ANSWERS

■■■ Benchmarking Test 4

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(b)	(c)	(a)	(b)	(b)	(b)	(a)	(c)	(a)	(c)	(c)	(c)	(b)	(c)	(a)	(d)	(a)	(d)	(b)	(d)

HINTS AND SOLUTIONS

2. (c) IITs = 60; IIMs = 40 ⇒ Ratio of IIM to IIT pass out = 4/6 = 6/9
3. (d) (6/10) / (10/16) = 0.96
4. Answer is 'cannot be determined' due to various reasons: Reason 1 – Data is given as 'on an average' and not the exact data. Reason 2 – There can be IITs pass outs from yesteryears making it to IIMs in a particular year. Hence, cannot be determined.

Solution to Questions 5 – 6:

5. Population (2000) = Population (1999) + Number of new births (2000) – Number of deaths (2000). The no. of births figure cannot be determined as the no. of deaths is not known. Hence [d].
6. Look at the following table:

	1997	*1998*	*1999*	*2000*
City A	40	60	70	80
City B	50	50	30	40
Ratio	0.8	1.2	2.333333	2

Hence option (c) is the answer.

7. Import increased from 7 to 24, around 240% increase. Hence, option (b) is correct.
8. This question can be solved by looking the graph given. Hence, option (d) is correct.
9. $\frac{10+20+16+23+24}{6+10+7+16+20}$ x 100% = 157.6%. Hence, answer option is (4).
10. % of protein is highest in protein = (100/1000) × (1/2) × 100 = 5%. **Ans (b)**
11. % of protein is maximum in dal and % of fat is the least. Hence, is classified as 'Healthy'. **Ans (c)**
12. Clearly, Veg Burger has the highest percentage of fat and the least percentage of protein. Hence, butter is the junk food out of the above mentioned foods. **Ans (a)**
14. Butter, Apple, Veg Burger and Chicken. Hence, **Ans (d)**
15. [c] Values added per worker for various sectors will be proportional to following ratios:

Central	10/6
State	3/6
Central/state	12/3
Public	8/8

16. [b] Fixed capital per factory will be proportional to the following ratios:

Central	5/8
State	4/4
Central/state	5/6
Public	6/12

17. [b] If the variables are proportional to the number of employees and the production per employee, then we can say that the variable cost is directly proportional to the production. Since the variable cost is maximum for private sector, it will also have the maximum production.
18. [a] If 20% of govt. fixed capital is equal to $200 million => Total govt. sector fixed capital = $ 1000 m which is equivalent to 14% of the total fixed capital. Now, 25% of Govt. investment in joint sector i.e., 25% of the 8% of the total = 2% 0f total fixed capital will be given by: (1000 × 2) /14 = $ 143 million
Value in Rs is given by: 143 × 45 = Rs 6450 million
19. Marks obtained in the three sections are 65, 82 and 87 in QA, DI and VA respectively.
Hence, option (3) is correct.
20. No. of correct attempts => 22 + 19 + 25 = 66. No. of wrong attempts = 30.
Required percentage = $= \frac{66\times 4-30}{150\times 4} = \frac{234}{150\times 4} = 30\%$
Hence, option (1) is correct.

Module 3

Data Sufficiency

7

CHAPTER

Data Sufficiency

LEARNING OBJECTIVES

- How does a typical Data Sufficiency question look like
- Similarity/Dissimilarity between Data Sufficiency questions and Quantitative Aptitude questions
- Various do's and don'ts of solving Data Sufficiency questions
- Process of solving the questions

DATA SUFFICIENCY QUESTIONS

Data Sufficiency questions are designed to test the student's ability to reason with a limited set of data and to reach a valid conclusion. To solve data sufficiency questions, we should have a clear understanding of the different situations that might come with the questions. Besides, we should be clear with the format of the possible questions and various do's and don'ts related to the above situation and solution.

In most of the cases, a data sufficiency question will have the following structure:

Q. A question will be asked.

Statement 1 – One set of information will be given.

Statement 2 – Another set of information will be given.

On the basis of the statements, four or five different options will be provided. One of those options will be the most probable choice and hence the answer.

Significance of Data Sufficiency

In post MBA life, a manager is virtually burdened with the weight of the data—sometimes organized and sometimes unorganized. DS questions test the ability of the future managers to prune the data to make decisions. Apart from testing the decision making ability of the individual, DS questions also try to evaluate the ability of the student to see into the future that is visualize and predict what can happen in the next 2-3 minutes. Try to understand this—We might find the answer to the question asked in 2-3 minutes time with the help of the given statements, but can we find out the answer right away without actually investing those 2-3 minutes. So, the question now is—Can we predict correctly what is going to happen in the next 2-3 minutes without actually living those 2-3 minutes?

A data sufficiency question is designed—by virtue of its design and content—in such a way that it essentially checks students' ability to reason out. In most of the cases of

DS questions, basic concepts of Quantitative Aptitude will be required.

Various do's and don'ts of Data Sufficiency

The process to solve the DS questions is a bit different from the process involved in solving the QA or LR/DI questions. There are certain key things that we should keep in mind before solving the DS questions. Let us look at them one by one:

1. Understanding the options

This is the first step in solving the DS questions. One must go through the options and understand them clearly. Sometimes, the options given in data set themselves can become a perennial source of problem.

Let us understand the statements given in the options:

If the option statement reads like — "If the question can be solved by using only one of the statements alone, but not by the other statement alone".

It means—Out of the given two (or, more) statements, only one statement can answer the question given, and the other statement cannot give the answer to the question given. In this case either the first statement will give the answer or the second one, but not both simultaneously i.e., if the first statement can give the answer, then second statement cannot give the answer and vice-versa.

If the option statement reads like— "If the question can be solved by using either of the statements alone".

It means—Out of the given two (or more) statements, the question can be solved by using any one of the statements alone. In this case either first statement will give the answer or the second one used individually and independently.

If the option statement reads like — "If the question can be solved by using both (or all) the statements together, but not by either of the statements alone".

It means—Out of the given two (or more) statements, the question can be solved only if the information given in both (or all) the statements is used together. And, using only one of the statements alone cannot solve the given question.

If the statement reads like— "If the question cannot be solved by even using both the statements together".

It means—Out of the given two (or more) statements, the question cannot be solved even by using the information given in both (or all) the statements together.

Once you have understood the statements given in the options, you must not forget to see which statement corresponds to which option. CAT is known to introduce the surprise element quite often, and this also can be one of those surprises which might catch you unaware if you are not careful about this basic fact.

2. General awareness/assumptions are not allowed

As an unproclaimed rule, the element of general awareness cannot be used to solve the DS questions. However, this is also true for QA and LR /OI, but its importance in DS increases manifold.

Let us understand this with an example:

Example 1 Is New Delhi the most polluted city in India?

Statement 1 – The Capital of India is the most polluted city in India.

Statement 2 – New Delhi is the capital.

Solution Well, we can understand that statement 1 alone cannot answer the question, as it is not given that New Delhi is the capital of India.

If your answer is "Both the statements together can answer the question", think again. It is not given in statement 2 that New Delhi is the capital of India.

So, answer to this question is "Cannot be determined."

However, we should have a clear idea regarding the differences between a universal fact like "Pythagoras theorem" and general awareness like "New Delhi is the capital of India".

Universal facts like mathematical principles are not the general awareness stuff, and hence can be used.

Example 2 In how many days A and B working together can finish a work?

Statement I— A can finish the same work alone in 10 days.

Statement 2— B can finish the same work alone in 15 days.

Solution Understandably, we cannot have a unique answer to this question by using the statements alone. However, in this case we cannot solve this question even by using both the statement together.

FLAW DETECTOR—Had this question been a question in Quantitative Aptitude, we must have got the answer by using the methods of time and work. However, in this case of data sufficiency, since it is not given that A and B start working together and kept on working together till the work is finished, we cannot solve this question even by using both the statements together.

And we cannot make assumptions that they started working together and kept on working till the work is finished.

Answer should be unique

This is one aspect that makes the process of solving the DS questions different from the process of solving QA or DI/LR questions. While in case of QA, we can have multiple values of a variable; in DS there should be only one value of any variable or constant given.

Example 3 What is the value of x?

Statement 1— $x^2 - 5x + 6 = 0$

Statement 2— $x^2 + 2x - 15 = 0$

Solution Using statement 1, value of x = 2, 3

Since we are not getting unique value of x using statement 1 alone, we will say that statement 1 alone is not sufficient to give the answer.

Using statement 2, value of x = –5, 3

Since we are not getting unique value of x using statement 2 alone, we will say that statement 2 alone is not sufficient to give the answer.

Till now we have seen that using the statements alone is not sufficient to find the answer. Hence, we will move on to using both the statements together.

Using statement 1 and 2 simultaneously, we get a common unique value of x = 3.

Hence, both the statements together are needed to answer the question.

Example 4 What is the value of x, where x is the length of a rectangle?

Statement 1— $x^2 - 5x + 6 = 0$

Statement 2— $x^2 + 2x - 15 = 0$

Solution Using statement 1, value of x = 2, 3

Since we are not getting unique value of x using statement 1 alone, we will say that statement 1 alone is not sufficient to give the answer.

Using statement 2, value of x = –5, 3

x being the length of the rectangle, it cannot be negative. Hence, x = 3 is the unique answer that we are getting using statement 2 alone.

Concern for the answer, and not the technicalities

Quite a few times it has been observed that students, rather than solving the questions, tend to get into the technicalities of the data. This furthers students' 'emotional attachment' with the question ending up in the wastage of time and efforts.

Example 5 Can we fill up a big drum of 2000 litres capacity with exactly 88 litres of water (there is unlimited supply of water)?

Statement 1—There is a bucket having a of capacity 20 litres.

Statement 2—There is a bucket having a of capacity 50 litres and a mug of 3 litres capacity.

Solution Using statement 1, we can fill anything in the multiples of 20 litres only. And it can never be ascertained that the drum has exactly 88 litres of water. So, answer to the question is—We may or may not be able to fill up a big drum with exactly 88 litres of water. And whatever the quantity of the water in the drum is, we can not be sure if this is exactly 88 litres of water.

Using statement 2, we can fill the drum exactly with 88 litres of water (50 + 50 – 3 – 3 –3 –3). Hence, answer to the question is—Yes, we can fill up a big drum with exactly 88 litres of water.

So, we can find the answer using statement 2 alone.

Example 6 Is y > x ?

Statement 1— $x^2 > y^2$.

Statement 2— $x^3 > y^3$.

Solution Using statement 1, assuming some of the values of x and y can tell us that in some cases x > y and in some of the cases y > x.

x	y	Remarks
5	4	x > y
–5	–4	y > x

While assuming the values, we can take only those values of x and y that satisfy the given statement.

Using statement 2,

$x^3 > y^3$

$\Rightarrow$ $x > y$

Now look at the question. Obviously, we are getting an answer that y is not greater than x.

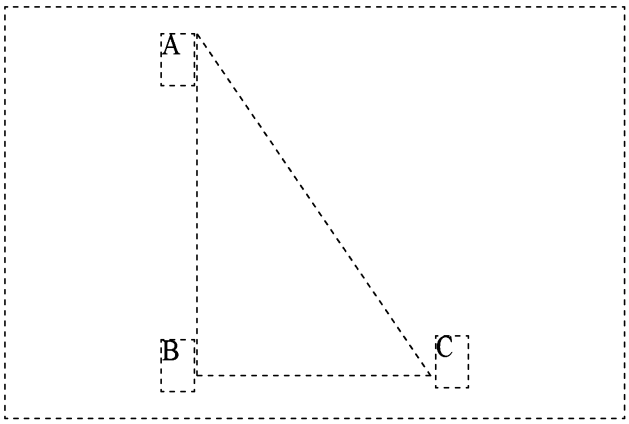
Hence, statement 2 alone is sufficient to answer the question.

Look for the obvious, but don't miss the hidden

Looking at any of the statements gives us an idea of the facts given, and sometimes we get so enamored with these obvious facts that we fail to see something hidden. Evidently, there is no point saying that "You should not make silly mistakes" (because nobody does it intentionally anyway. However, we can reduce the incidences of these silly mistakes or missing some hidden information.

Some of the precautions that must be taken while tackling any DS question:

1. While assuming the values, take different set of values (like positive, negative, fractions between 0 and 1) to make an informed decision.
2. Any geometrical figure should not be taken as accurate unless it is given. For example, the below given triangle ABC given below cannot be taken as a right-angled triangle even though it appears to be so.



Hence, no conclusion should be drawn by finding proportionate value of length or the angle in any geometrical figure unless such are clearly stated.

Example 7 Is Arvind the father of Puja?

Statement 1 — Puja is the daughter of Arvind.

Statement 2 — Arvind is the aunt of Abhishek.

Solution Using statement 1, despite the fact that Puja is the daughter of Arvind, we cannot conclude that Arvind is the father of Puja. Arvind can be the mother too.

Using statement 2, in first impression statement 2 seems to be irrelevant, but looking closely at this statement gives us an idea that Arvind is a woman (only women can be aunts). And hence, we can conclude that Arvind is not the father of Puja.

Therefore, statement 2 alone is sufficient to answer the question.

PRACTICE EXERCISE 1

Direction for questions 1 to 25: Each item is followed by two statements, A and B. Answer each question using the following instructions.

Choose (a): If the question can be answered by one of the statements alone and not by the other.

Choose (b): If the question can be answered by using either statement alone.

Choose (c): If the question can be answered by using both the statements together, but cannot be using either statement alone.

Choose (d): If the question cannot be answered even by using both the statements together.

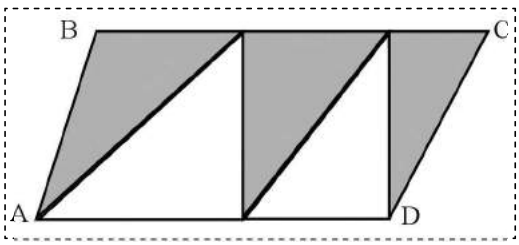
Q 1. Is the point P on the circle with center O?
- A. Q is a point on the circle and the distance from P to Q is equal to the distance from O to Q.
- B. Q is on the circle and PQO is an equilateral triangle.

Q 2. Is Arvind the father of Puja?
- A. Puja is the daughter of Arvind.
- B. Arvind is the brother of Abhishek.

Q 3. If the ratio of boys to girls attending school in 2000 was 1/3, what was the ratio of boys to girls attending school in 2001?
- A. 100 more boys were attending school in 2001 than in 2000.
- B. 150 more girls were attending school in 2001 than in 2000.

Q 4. If a and b are positive numbers, is $b > a$?
- A. $a^2 > b$
- B. $a^2 > b^2$

Q 5. What is the area of the unshaded region in the given figure (Points E, F are on the line BC and point G is on the line AD)?



- A. ABCD is a parallelogram.
- B. Area of ABCD is 50 square units.

Q 6. If the integer P divisible by 15?
- A. Sum of the digits of P equals 15.
- B. Unit digit of P is 6.

Q 7. Is Q a positive number?
- A. $4Q + 24 > 0$
- B. $4Q - 24 < 0$

Q 8. How many chocolate bars 2 inches wide and 4 inches long can be packed into carton C?
- A. The inside dimension of carton C are 8 cms by 12 cms.
- B. The width of carton C is equal to the height and 3/4 of the length.

Q 9. What are the values of x and y?
- A. $3x + 2y = 45$
- B. $y = 22.5 - 1.5x$

Q 10. Is x an even number?
- A. $4x + 3y$ is an even number.
- B. $3x + 4y$ is an even number.

Q 11. What is the ratio of the number of boys and girls in a school (Number of boys and girls are natural numbers)?
- A. Number of boys is 40 more than girls.
- B. Number of girls is 80% of the number of boys.

Q 12. What is the difference between two numbers?
- A. First number is 60% of the other number.
- B. 50% of the sum of first and second number is 24.

Q 13. How many marks did Arpit Sinha obtain in mathematics?
- A. Arpit Sinha secured on an average 55% marks in mathematics, physics, and chemistry together.
- B. Arpit Sinha secured 10% marks more in mathematics than the average of mathematics, physics and chemistry.

Q 14. Is $x > y$?
- A. $\frac{(x^4 - y^4)}{x^3 + y^3} > 0$
- B. $\frac{x^3 - y^3}{x^4 + y^4} > 0$

Q 15. What is a two digits number?
- A. The number obtained by interchanging the digits is smaller than the original number by 63.
- B. Sum of the digits is 11.

Q 16. x, y, and z are integers, is x an odd number?
- A. an odd number is obtained when x is divided by 5.
- B. (x+y) is an odd number.

Q 17. What is the number x?
- A. The LCM of x and 18 is 36.
- B. The HCF of x and 18 is 2.

Q 18. Is y greater than x?
A. x+y = 2
B. x/y = 2

Q 19. Which of the four numbers w, x, y, and z is the largest?
A. The average of w, x, y and z is 25.
B. The numbers w, x, and y are each less than 24.

Q 20. P, Q, R, and S are four consecutive even integers. What is the value of the largest integer among these?
A. The average of the four numbers is the first prime number greater than 10.
B. The ratio between the largest and the smallest of the number is less than 10.

Q 21. Is A greater than B?
A. (A + 3) is greater than (B + 2).
B. Fourth power of A is greater than third power of B.

Q 22. Which of the following is greater: $\frac{a+x}{b+x}$ or $\frac{a}{b}$
A. $\frac{a}{b} > 1$
B. x is positive.

Q 23. How many boys and girls are there in the family of 7 children?
A. Each boy in the family has as many sisters as brothers.
B. Each girl in the family has as many brothers as sisters.

Q 24. In how many days does Binod finish a certain work?
A. Binod is twice as efficient as Amar.
B. Akshay is twice as efficient as Binod.

Q 25. What is the equation of the straight line AB?
A. Straight line AB is perpendicular to another straight line 3x + 5y = 10
B. Straight line AB passes through the point (1, 2).

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(b)	(c)	(a)	(b)	(a)	(d)	(c)	(d)	(d)	(a)	(c)	(d)	(a)	(c)	(a)	(d)	(c)	(c)	(a)
Q. No.	21	22	23	24	25															
Answer	(d)	(c)	(b)	(d)	(c)															

HINTS AND SOLUTIONS

1. Using statement A alone, since it is given that the centre of the circle is O, hence OQ is the radius of the circle. But point Q can be inside the circle or on the circle or outside the circle and in all these cases OQ = PQ. Hence, statement A alone does not give the answer.
Using statement B alone, PQ = QO = PO = Radius of the circle.
Hence, point P is on the circle.

2. Using statement A alone, Arvind can be father or mother of Puja.
Using statement B alone, obviously it cannot be solved.
Using both the statements together, since Arvind is the brother of Abhishek, hence Arvind is a male. Hence Arvind is the father of Puja.

3. Using statement A alone, nothing can be concluded.
Using statement B alone, nothing can be concluded.
Using both the statements together, if the ratio of boys to girls in 2000 = x/3x, then the new ratio of boys to girls in 2001 = $\frac{x+100}{3x+150}$
Now everything depends upon the value of x. Hence cannot be determined uniquely.

4. Using statement A alone, depending upon the values of A and B, either can be greater (A = 10, B = 1 or A= 5, B = 10) according to the condition given in the statement.

   Using statement B alone, since both the numbers a and b are positive, and a2 > b2, hence a > b. So, the answer to the question is – No, b is not greater than a.

5. Using both the statements together, since the base and height of ABCD and unshaded region are same, hence the area of unshaded region is half of area (ABCD).

6. Using statement A alone, nothing can be concluded. Numbers can be 195 or 159 etc. Otherwise also, we know that if any number is divisible by 5 and 3 both, then the number will be divisible by 15. If the sum of the digits is 15, then the number is divisible by 3 but nothing can be said about the divisibility of this number by 5.
   Using statement B alone, since the unit digit of P is 6, hence this number is not divisible by 5, and hence not divisible by 15.
   Answer (a)

7. A. $4Q + 24 > 0$ Or, $4Q > -24$, or, $Q > -6$
   Hence, Q can be negative or positive or zero as well.
   B. $4Q - 24 < 0$
   Or, $4Q < 24$, or, $Q < 6$
   Hence, Q can be negative or positive or zero as well.

9. Using statement A alone, since the equation is in two variables and there is one equation nothing can be concluded.
   Using statement B alone, since the equation is in two variables and there is one equation nothing can be concluded.
   Using both the equations together, both the equations are same. Hence, cannot be determined.

10. Is x an even number?
    A. 4x + 3y is an even number.
    B. 3x + 4y is an even number.
    Using statement A alone, if 4x + 3y is an even number then both 4x and 3y should be either odd or both should be even. Both 4x and 3y cannot be odd (as 4x is even) and hence both of 4x and 3y are even.
    Now even if 4x is even, then we cannot conclude that x is even or odd.
    Using statement B alone, if 3x + 4y is an even number then both 3x and 4y should be either odd or both should be even.
    If each of 3x and 4y is even, then x can take values like x = 4/3, which is neither odd nor even.
    If each of 3x and 4y is odd, then 3x is odd. But using this we cannot conclude that x is also odd. X may be a fraction also like x = 1/3 or even it can be an odd number.

FLAW DETECTOR—Whenever we are required to assume the values, we should assume the different values viz. Positive numbers, negative numbers, fractional values and definitely the fractional values between O and 1 before validating our result.

11. Using statement A alone, we cannot determine the ratio of boys and girls.
    Using statement B alone, Number of girls = 4/5 Number of boys, now we can find out the ratio of boys and girls in the school.

12. Using statement A, we get the ratio between the two numbers but not the difference between the numbers. Statement B gives the sum of the numbers but not the difference. Using both the statements together gives the difference between the numbers.

13. In none of the statements given marks have been given in terms of numbers, hence we cannot find the marks obtained in mathematics.

14. Using statement A, we cannot determine if $(x^4 - y^4)$ is greater than zero or less than zero since we do not have the sign of $(x^3 + y^3)$. Hence we cannot determine anything.
    Using statement B, we cannot determine if
    $(x^3 - y^3) > 0$ since $(x^4 + y^4) > 0$.
    $(x^3 - y^3) > 0$, so, $x > y$.

15. Using statement A alone gives that the difference between the digits of the number is 7. This gives the possibility of many numbers like 81, 92 etc.
    Using statement B alone also gives the possibility of many numbers.
    Using both the statements together, we get a unique number.

16. Using statement A alone, we get that the quotient obtained when x is divided by an odd number is odd. Hence x has to be odd.
    Using statement B alone, one of x and y will be odd and other one will be even. So we cannot determine that which of x or y is even?

17. Using statement A, value of x can be $22 \times 30-1$. Hence, two values of x are possible.
    Using statement B, value of x can be
    $21 \times 30 \times 5N \times 7N \times \ldots$,
    where value of N can be any real number.
    Using both the statements together, we can see that no unique value of X is possible.

18. Using statement A alone cannot give us any result.
    Using statement B alone cannot give us any result since the values of x and y can be positive as well as negative too.

Using both the statements together will give us the result.

19. Using statement A alone cannot give us any result.
Using statement B alone gives that value of three of the given numbers is less than the average. Hence, fourth number will be greater than the average. So, this fourth number is largest.

20. Using statement A alone, average of the given four numbers = 11, hence, their sum = 44, hence, the numbers will be 8, 10, 12, 14. Hence, the value of largest integer, whichever is that, is 14.
Using statement B alone is not going to give us the result.

PRACTICE EXERCISE 2

Direction for questions 1 to 20: Each item is followed by two statements, A and B. Answer each question using the following instructions.

Choose (a): If the question can be answered by one of the statements alone and not by the other.

Choose (b): If the question can be answered by using either statement alone.

Choose (c): If the question can be answered by using both the statements together, but cannot be using either statement alone.

Choose (d): If the question cannot be answered even by using both the statements together.

Q 1. A, B, C, D are playing cricket and scored 96 runs together. Did A score the maximum runs?
A. A scored 23 runs.
B. B and C together scored 42 runs.

Q 2. Is ab an even number?
A. a is divisible by 3.
B. (b + 1) is divisible by 4.

Q 3. Three teams of woodcutters took part in a competition (to cut the maximum quantity of wood). Which team could win the competition?
A. The first and the third teams cut twice as much as the second team cut.
B. The second and the third teams cut three times as much as the first team cut.

Q 4. If m and n are integers, is m divisible by 11?
A. mn is divisible by 110.
B. n is divisible by 2.

Q 5. On a fishing trip, Sanju and Ajay each caught some fishes. Who caught more fish?
A. Sanju and Ajay caught fishes in the ratio 2 : 3.
B. After Sanju stopped fishing, Ajay continued fishing until he caught 12 fishes.

Q 6. What are the ages of three brothers (ages are natural numbers)?
A. The product of their ages is 21.
B. The sum of their ages is not divisible by 3.

Q 7. Is 5x + 25 is divisible by 50?
A. x is divisible by 5.
B. x is divisible by 10.

Q 8. How many of a, b, c, d are odd, given that all of them are positive integers?
A. ad + bc is odd.
B. ac + bd is odd.

Q 9. A is not shorter than B, who in turn, is not taller than C who is shorter than D. who among A, B, C and D is the tallest?
A. C and A are of equal height.
B. B and A are of equal height.

Q 10. If Bhushan or Charan passed the examination, then neither Sujan nor Tarun passed the examination. Did Bhushan pass the examination?
A. Tarun did not pass the examination.
B. Sujan passed the examination.

Q 11. Six people—A through F—sit around a circular table, not necessarily in the same order. B and E sit opposite each other. Does C sit opposite D?
A. If C and E interchange their positions, then E will be to the immediate left of B.
B. If A and B interchange their positions, then B will be to the immediate left of E.

Q 12. Average age of a, b, c, and d is 46 years. Who is the oldest among them?
A. c is 91 years old.
B. Ages (in years) of all of them are distinct natural numbers.

Q 13. Product of a, b, c, and d is 1003 (a, b, c and d are all natural numbers). Which one is the largest among them?
A. a > b > c
B. a > b > d

Q 14. What is the least common multiple of a x b, b x c and c x a (a, b, c are all natural numbers)?
A. least common multiple of a, b and c is 30.
B. The highest common factor of a, b and c is 2.

Q 15. What are the values of m and n?
A. n is an even integer, m is an odd integer, and m is greater than n.
B. Product of m and n is 30.

Q 16. On a given day a boat ferried 1500 passengers across the river in 12 hours. How many round trips did it make?
A. The boat can carry 200 passengers at any time.
B. It takes 40 min each way and 20 min of waiting time at each terminal.

Q 17. What is the value of x?
A. x and y are unequal even integers, less than 10, and x/y is an odd integers.
B. x and y are even integers, each less than 10, and product of x and y is 12.

Q 18. What are the ages of two individuals, x and y?
A. The age difference between them is 6 years.
B. The product of their ages is divisible by 6.

Q 19. If m and n are consecutive positive integers, is m > n?
A. m – 1 and n + 1 are consecutive positive integers.
B. m is an even integer.

Q 20. Which of a, b, c and d is/are odd?
A. ad + bc is odd.
B. ac + bd is odd.

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(c)	(d)	(a)	(d)	(a)	(c)	(b)	(a)	(c)	(c)	(a)	(d)	(d)	(a)	(a)	(d)	(a)	(a)

HINTS AND SOLUTIONS

1. Using statement A alone, since the average runs scored by all of them = 24, and runs scored by A = 23, hence we can conclude that atleast one of remaining three persons must have scored more than the score of A. So, A didn't score maximum runs.
Using statement B alone cannot give us any result.

2. Using statement A alone cannot give us any result.
Using statement B alone gives that b is an odd number.
Using both the statements together we get ab is an odd number.

4. Using statement A, atleast one of m or n will be divisible by 11, but we are not sure that which of m or n will be divisible by 11.
Using statement B, n is divisible by 2 but it does not give that which of m or n is divisible by 11.
Using both the statements together, we cannot find if m is divisible by 11 or not.

5. Using statement A alone, we can find out that Sanju caught more fishes than Ajay whereas using statement B alone, we cannot find that who caught more fishes.

6. Using statement A alone, ages of the brothers can be–21, 1, 1 or 1, 7, 3. Hence, using statement A alone will not give us the unique answer.
Using statement B alone also will not give us the result.
Even after using both the statements together, we cannot find the ages of the brothers as both the possibilities (21 + 1 + 1) and (1 + 7 + 3) are not divisible by 3.

7. Using statement A alone, assuming some values of x (= 5 or = 10) gives that 5x + 25 is sometimes divisible and sometimes not divisible.
Using statement B alone, whatever values of x we take, we see that 5x + 25 = 5 (x + 5) is not divisible by 10.

8. Using both the statements together, we can find the answer.

9. This question can be answered by using either statement alone.

10. Question can be answered by using the statement B alone but not by statement A alone.

11. Question can be answered by using both the statements together but not by any of the statements alone.

12. Sum of their ages = 184 years.
Using statement A alone, since the age of C = 91 years, hence sum of ages of rest of them = 93 years. Using this we cannot find that who is oldest of them.
Using statement B alone we cannot get the answer.
Using both the statements together, since the age has to be natural number, so minimum possible age of any one of them = 1 year. So, the youngest of them can be 1 year old, second youngest of them can be 2 years old, hence the maximum age of third person can be 90 years.
So, C is the oldest of them.

13. $1003 = 1 \times 1 \times 17 \times 59$
Using both the statements together, 'a' is the largest one.

14. Using statement A alone, values of a, b and c can be- 2, 3, 5 or 2, 3, 30. Hence, answer cannot be obtained using statement A only.

Using statement B alone, values of a, b and c can be 2, 4, 6 or 4, 6, 8 etc. Hence, answer cannot be obtained using statement B only.

Using both the statements together, values of a, b and c can be – 2, 6, 10 or 6, 10, 30. And we can see that we are not getting a unique value of $a \times b$, $b \times c$ and $a \times c$.

15. (Don't forget to take the negative values of m and n)

16. This question can be answered by using statement B alone but not by using statement A alone.

17. (Don't forget to take the negative values of x and y)

18. Using statement A alone will not give us the answer.
Using statement B alone will not give us the answer.
Using both the statements together also will not give us the answer.

19. This question can be solved by using statement A alone but not by using statement B alone.

PRACTICE EXERCISE 3

Direction for questions 1 to 17: Each item is followed by two statements, A and B. Answer each question using the following instructions.

Choose (a): If the question can be answered by first statement (statement A) alone but not by the second statement (statement B) alone.

Choose (b): If the question can be answered by second statement (statement B) alone but not by the first statement (statement A) alone.

Choose (c): If the question can be answered by using either statement alone.

Choose (d): If the question can be answered by using both the statements together, but cannot be using either statement alone.

Choose (e): If the question cannot be answered even by using both the statements together.

Q 1. Is the integer y greater than 4?
A. 2y is a factor of 8
B. 24 is a factor of y
(a) 1 (b) 3
(c) 4 (d) 2
(e) 5

Q 2. A sum of Rs 215 is distributed among three persons A, B and C. Who gets the least?
A. A gets 2/3rd of what (B + C) are getting.
B. B gets 1/5th of what (A + C) are getting.
(a) 1 (b) 3
(c) 4 (d) 2
(e) 5

Q 3. What is the HCF of natural numbers A and B?
A. LCM of the natural numbers A and B is 101.
B. A and B are distinct natural numbers.
(a) 1 (b) 4
(c) 2 (d) 3
(e) 5

Q 4. Ram is the son of Shyam and father of Dharam. All of these people—Ram, Shyam and Dharam are politicians. Is Shyam an honest politician?
A. Sons of all the honest politicians are honest politicians. Ram is an honest politician.
B. Son of only a dishonest politician is a dishonest politician. Dharam is a dishonest politician.
(a) 1 (b) 4
(c) 2 (d) 3
(e) 5

Q 5. y = f(x) is said to be an even function if f(x) = f(–x). Is y = f(x) an even function?
A. Graph of y = f(x) lies in only two quadrants and both these quadrants are adjacent to each other.
B. Graph of y = f(x) lies in only two quadrants and both these quadrants are opposite to each other.
(a) 1 (b) 2
(c) 3 (d) 4
(e) 5

Q 6. In an examination of five papers, marks obtained by Pranav are in the ratio 4 : 5 : 6 : 7 : 8 across the five papers. Each paper carries the same maximum marks. In how many papers did Pranav obtain more than 50% of the maximum marks?
A. Total marks obtained by Pranav in all the papers together is 300 marks.
B. Lowest percentage of marks obtained by Pranav in any of the papers of this examination is 30%.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 7. N is a natural number. Is N a perfect square?
A. When N is divided by 100, remainder obtained is 2.
B. Total number of factors of N is odd.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 8. What is the respective speed of two trains of lengths 70 m and 75 m?
A. They take 35 seconds to cross each other when they are running in the same direction.
B. They take 10 seconds to cross each other when they are running in the opposite direction.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 9. Vijay and Rupesh are two friends working in the same company. Sum of their monthly salaries is more than Rs 1 lac per annum. Who saves more per month?
A. Ratio of monthly income of Vijay and Rupesh is 3 : 5.
B. Ratio of monthly expenses of Vijay and Rupesh is 1 : 2.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 10. Is triangle ABC a right-angled triangle?
A. Ratio of length of side AB and side BC = 3 : 2
B. Ratio of length of side BC and side CA = 4 : 5

(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 11. A bus is running with a uniform speed from Patna to Gaya. It met with an accident somewhere enroute. What is the speed of this bus?
A. Due to this accident, speed of bus got reduced by 22%. And Due to this one bus reached Gaya 40 minutes late.
B. Had the accident occurred 5 kms ahead of the point where the accident occurred, it would have reached Gaya only 20 minutes late.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 12. There are five members—P, Q, R, S and T in a family.How many members of this family are male?
A. T has two sisters—Q and S.
B. P is the father of T and R is the only son of T.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 13. Which month is the current month?
A. Sum of dates of last Monday of previous month and 1st Thursday of next month is 38 and both the dates are of the same year.
B. Current year is a leap year and the month in the question is from current year.
(a) 5 (b) 4
(c) 3 (d) 2
(e) 1

Q 14. 1st January 19PQ is Monday, where P and Q are the last two digits of the year in such a way that maximum value of P and Q each is 6. Which is the next year when 1st January will be a Monday?
A. 19PQ is a leap year.
B. 19PQ is a non-leap year.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 15. A triangle is circumscribed by a circle. Is this triangle a right-angled triangle?
A. Triangle is isosceles triangle.
B. One side of the triangle is the largest possible chord of the circle.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 16. Triangle ABC is a right angled triangle. What is the distance between circum centre and the centroid of this triangle ABC?
A. Inradius of Δ ABC = 5 units
B. Two smaller sides of the Δ ABC are 3 units and 4 units.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Q 17. What is the HCF of (5a – 1) and (5b – 1)?
A. HCF of a and b = 5
B. a and b are natural numbers.
(a) 1 (b) 3
(c) 2 (d) 4
(e) 5

Direction for questions 18 to 20: Each item is followed by three statements, A, B and C. Answer each question using the following instructions.

Mark (a): If the question can be answered by using any two statements together but not by using any statement alone
Mark (b): If the question can be answered by using either of the statements alone.
Mark (c): If the question can be answered only by using all the three statements together and neither by any pair of statements nor by any statement independently.
Mark (d): If the question cannot be answered using the statements in any case.
Mark (e): If none of the above four options are correct.

Q 18. A football is kept inside a circular field. What is the probability of this ball being closer to the center of the circle than to the periphery of the ground?
(a) Radius of the circular field is 15 m.
(b) Radius of the football is 5 cm.
(c) Ratio of circumference and diameter of the circle is constant.

Q 19. A and B are running continuously around a circular track with respective uniform speed. They start running from the same point at the same time. After how much time will they meet at the diametrically opposite point of their starting point?
(a) Speed of A is 20 m/s and speed of B is 35 m/s.
(b) Length of the track is 1000 m.
(c) Both A and B are running in the same direction.

Q 20. At how many different points, graph of $f(x) = ax^2 + bx + c = 0$ will cut X- axis?
(a) $a > 0$, $b > 0$ and $c > 0$.
(b) Discriminant of $f(x) < 0$
(c) $a < 0$.

ANSWERS

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(d)	(b)	(d)	(c)	(b)	(c)	(b)	(e)	(d)	(d)	(e)	(e)	(e)	(a)	(c)	(c)	(a)	(e)	(e)	(e)

HINTS AND SOLUTIONS

1. Using statement A, y can have any integer value from 0-7, hence cannot be determined using statement A alone.

   Using statement B, minimum value of y can be 6. And hence y is greater than 4.

2. Obviously, using the statements alone will not give the result. However, if we use both the statements together, then we will get the result that who is earning the least.

3. Using statement A only gives us two different sets of values of A and B – (101, 101) and (101, 1). So, unique value of HCF cannot be obtained.

   Using the statement B only also does not give us a unique number.

   However, using both the statements together gives us the values of A and B as – (101, 1). And HCF of these two numbers = 1.

4. Using statement A, we cannot find if the fathers of all the honest politicians are honest or not? It is only said that if the father is honest, then the son will be honest; however, it does not mean that son of dishonest politicians cannot be honest. [For more, see the chapter—Logical Link in the 1st part of this book]. So, nothing can be determined on the basis of this statement alone.

   Using statement B, since son of only a dishonest politician can be a dishonest politician, hence we can conclude that if a politician is dishonest, then his father will also be dishonest. On the basis of this, if Dharam is dishonest politician then his father Ram is dishonest politician and if Ram is dishonest politician then his father Shyam is dishonest politician. And this answers our question.

5. Using statement A, nothing can be inferred about the even or odd nature of the function y = f(x).

   Using statement B, since the graph lies in only two quadrants opposite to each other, it cannot be the graph of any even function. Hence, y = f(x) is not an even function.

6. Using statement A, since it is not given that what are the maximum marks for each of the papers, hence just by knowing that 300 marks are obtained by Pranav does not provide us with any other information.

   Using statement B, Let us assume that the marks obtained by Pranav are 4x, 5x, 6x, 7x and 8x. Now, 4x = 30%. Hence, now we can find out that in how many papers Pranav has scored more than 50% marks.

7. Using statement A, if the remainder obtained when N is divided by 100 is 2, then the unit digit of N is 2. Hence, N cannot be a perfect square.

   Using statement B, since the total number of factors of N is odd, hence N is a perfect square. [Total number of factors of any perfect square is odd and vice-versa is also true.]

8. Using statement A and B independently, we cannot find the speed of the trains. Even using both the two statements together, we get two speeds but we cannot find that which speed corresponds to which train.

   Hence, the speed of the trains cannot be determined.

9. Obviously, it can be seen that using the statements alone will not give us the solution.

   Using both the statements together, since the ratio of expenses is more than the ratio of income and both the fractions are proper fractions, hence it can be concluded on the basis of the data given that Rupesh is saving more than Vijay in all the cases.

10. Obviously, it can be seen that using the statements alone will not give us the solution.

    Using both the statements together will give us the ratio of sides of the triangle. Using that we can find out if the triangle ABC is right-angled or not?

13. Using statement A alone,

    Sum of dates of last Monday of previous month and 1st Thursday of next month is 38 is possible only if last Monday is 31st and 1st Thursday is 7th. (Since if

we take 30 + 8 = 38, then 30 can be last Monday of any month but 8th can not be the 1st Thursday of any month)

So, 31st of last month is a Monday. Hence 7th of current month—14th of current month—21st of current month—28th of current month will be a Monday.

Now, if current month is a month with 30 days, then 5th of next month will be a Monday, so 7th of next month cannot be a Thursday.

If current month is a month with 31 days, then 4th of next month will be a Monday, so 7th of next month will be a Thursday.

Finally, we can conclude that previous month and current month, both are having 31 days. Since both the dates are of the same year, so current month is August.

We cannot find the answer using statement B alone.

14. We know that any leap year and leap year +5 will have same 1st January. Hence, 19(PQ + 5) will have its 1st January as Monday. Hence, question can be solved using statement A alone. However, statement B alone will not give us a unique result.

Nature of year	No. of years after which 1st January will be same
Leap year	5
Leap year + 1	6
Leap year + 2	6
Leap year + 3	11

15. Using statement A alone will not answer this question. Using statement B alone tells that one side of the triangle is the diameter of the circle and hence will be the hypotenuse of the circle. Hence, the triangle is right-angled triangle.

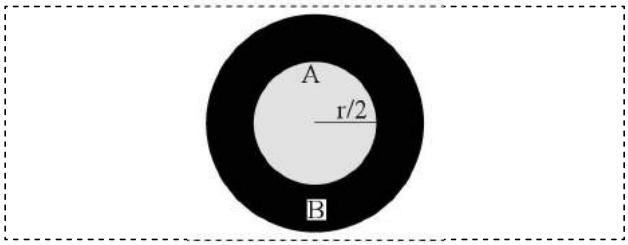
16. Using statement A alone, we cannot find the circum radius of the triangle ABC.

Using statement B alone, since the circum radius of the right-angled triangle is half the hypotenuse of the triangle, so we can find out the circum radius of the triangle. And 1/3rd of this circum radius will be the distance between circum centre and the centroid. In short, 1/6th of the hypotenuse will be the distance between circum centre and the centroid.

18. This question can be solved without using any of the statements given.

Football will be closer to the center than to the circumference if point is lying in the segment A.

Area of segment A = $1/4\pi r2$



So, probability of point being closer to the circumference = $1/4\ \pi r2/\pi r2 = 1/4$

[The probability of ball being closer to the center = Area of the smaller circle inside/Area of the larger circle (A/ A+ B) = 1/4]

19. This question can be solved by using statement A individually. Since ratio of speed of A and speed of B = 4 : 7, hence total number of meeting points = 3 (if they are running in the same direction) and total number of meeting points = 11 (if they are running in the opposite direction). In none of these cases, A and B will meet at diametrically opposite point of their starting point.

Using statement B or statement C alone will not give us the result.

20. This question can be solved by using statement B alone. Since discriminant < 0, hence this equation f(x) = 0 will not have any real root, and hence graph of f(x) = 0 will not cut X-axis.

So, graph of f(x) = 0 cuts X-axis at 0 point.

Using statement B or statement C alone will not give us the unique result.

PART 3

CAT Papers

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CAT 2002

SECTION III

There are 50 questions in this paper. All the questions carry equal mark.

There is negative marking.

Q.51. If there are 10 positive real numbers $n1 < n2 < n3 ... < n10$, How many triplets of these numbers ($n1$, $n2$, $n3$), ($n2$, $n3$, $n4$),... can be generated such that in each triplet the first number is always less than the second number, and the second number is always less than the third number?

(a) 45 (b) 90 (c) 120 (d) 180

Q.52. In ΔABC, the internal bisector of ΔA meets BC at D. If AB = 4, AC = 3 and DA = 60°, then the length of AD is

(a) $2\sqrt{3}$ (b) $\frac{12\sqrt{3}}{7}$

(c) $\frac{15\sqrt{3}}{8}$ (d) $\frac{6\sqrt{3}}{7}$

Q.53. The length of the common chord of two circles of radii 15 cm and 20 cm, whose centres are 25 cm apart, is

(a) 24 cm (b) 25 cm

(c) 15 cm (d) 20 cm

Q.54. If $f(x) = \log\left\{\frac{(1+x)}{(1-x)}\right\}$ then $f(x) + f(y)$ is

(a) $f(x+y)$ (b) $f\left\{\frac{(x+y)}{(1+xy)}\right\}$

(c) $(x+y)f\left\{\frac{1}{(1+xy)}\right\}$ (d) $\frac{f(x)+f(y)}{(1+xy)}$

Q.55. Four horses are tethered at four corners of a square plot of side 14 m so that the adjacent horses can just reach one another. There is a small circular pond of area 20 m^2 at the centre. Find the ungrazed area.

(a) 22 m^2 (b) 42 m^2

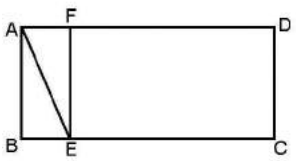
(c) 84 m^2 (d) 168 m^2

Q.56. On a straight road XY, 100 m long, five heavy stones are placed 2 m apart beginning at the end X. A worker, starting at X, has to transport all the stones to Y, by carrying only one stone at a time. The minimum distance he has to travel is

(a) 472 m (b) 422 m

(c) 744 m (d) 860 m

Q.57. In the figure given below, ABCD is a rectangle. The area of the isosceles right triangle ABE = 7 cm^2, EC = 3(BE). The area of ABCD (in cm^2) is



(a) 21 cm^2 (b) 28 cm^2

(c) 42 cm^2 (d) 56 cm^2

Q.58. The area of the triangle whose vertices are (a, a), (a + 1, a + 1) and (a + 2, a) is

(a) a^3 (b) 1 (c) $2a$ (d) $2^{1/2}$

Q.59. Instead of walking along two adjacent sides of a rectangular field, a boy took a short cut along the diagonal and saved a distance equal to half the longer side. Then the ratio of the shorter side to the longer side is

(a) $\frac{1}{2}$ (b) $\frac{2}{3}$ (c) $\frac{1}{4}$ (d) $\frac{3}{4}$

Q.60. Only a single rail track exists between stations A and B on a railway line. One hour after the northbound super fast train N leaves station A for station B, a

south-bound passenger train S reaches station A from station B. The speed of the super fast train is twice that of a normal express train E, while the speed of a passenger train S is half that of E. On a particular day, N leaves for B from A, 20 min behind the normal schedule. In order to maintain the schedule, both N and S increased their speeds. If the super fast train doubles its speed, what should be the ratio (approximately) of the speeds of passenger train to that of the super fast train so that the passenger train S reaches exactly at the scheduled time at A on that day?
(a) 1 : 3 (b) 1 : 4 (c) 1 : 5 (d 1 : 6

Q.61. On a 20 km tunnel, connecting two cities A and B, there are three gutters (1, 2 and 3). The distance between gutters 1 and 2 is half the distance between gutters 2 and 3. The distance from city A to its nearest gutter, gutter 1, is equal to the distance of city B from gutter 3. On a particular day, the hospital in city A receives information that an accident has happened at gutter 3. The victim can be saved only if an operation is started within 40 min. An ambulance started from city A at 30 km/h and crossed gutter 1 after 5 min. If the driver had doubled the speed after that, what is the maximum amount of time would the doctor get to attend the patient at the hospital. Assume 1 min is elapsed for taking the patient into and out of the ambulance?
(a) 4 min
(b) 2.5 min
(c) 1.5 min
(d) The patient died before reaching the hospital

Q.62. Number S is obtained by squaring the sum of the digits of a two-digit number D. If difference between S and D is 27, then the two-digit number D is
(a) 24 (b) 54 (c) 34 (d) 45

Q.63. The nth element of a series is represented as $X_n = (-1)^n X_{n-1}$
If $X_0 = x$ and $x > 0$, then which of the following is always true?
(a) X_n, is positive if n is even
(b) X_n, is positive if n is odd
(c) X_n is negative if n is even
(d) None of these

Q.64. If x, y and z are real numbers such that $x + y + z = 5$ and $xy + yz + zx = 3$, what is the largest value that x can have?
(a) $\frac{5}{3}$ (b) $\sqrt{19}$
(c) $\frac{13}{3}$ (d) None of these

Q.65. Neeraj has agreed to mow a lawn, which is a 20 m × 40 m rectangle. He mows it with 1 m wide strip. If Neeraj starts at one corner and mows around the lawn toward the centre, about how many times would he go round before he has mowed half the lawn?
(a) 2.5 (b) 3.5
(c) 3.5 (d) 4

Q.66. The owner of a local jewellery store hired three watchmen to guard his diamonds, but a thief still got in and stole some diamonds. On the way out, the thief met each watchman, one at a time. To each he gave $\frac{1}{2}$ of the diamonds he had then, besides 2 more. He escaped with one diamond. How many did he steal originally?
(a 40 (b) 36
(c) 25 (d) None of these

Q.67. Mayank, Mirza, Little and Jaspal bought a motorbike for $ 60. Mayank paid one-half of the sum of the amounts paid by the other boys. Mirza paid one-third of the sum of the amounts paid by the other boys. Little paid one-fourth of the sum of the amounts paid by the other boys. How much did Jaspal have to pay?
(a) $ 15 (b) $ 13
(c) $ 17 (d) None of these

Q.68. A rich merchant had collected many gold coins. He did not want anybody to know about his collection. One day, his wife asked, "How many gold coins do we have?" After a brief pause, he replied, "Well! if I divide the coins into two unequal numbers, then 48 times the difference between the two numbers equals the difference between the squares of the two numbers." The wife looked puzzled. Can you help the merchant's wife by finding out how many gold coins the merchant has?
(a) 96 (b) 53
(c) 43 (d) None of these

Q.69. Shyam visited Ram during his brief vacation. In the mornings, they both would go for yoga. In the evenings they would play tennis. To have more fun, they indulge in only one activity per day, i.e., either they went for yoga or played tennis each day. There were days when they were lazy and stayed home all day long. There were 24 mornings when they did nothing, 14 evenings when they stayed at home, and a total of 22 days when they did yoga or played tennis. For how many days Shyam stayed with Ram?
(a) 32 (b) 24
(c) 30 (d None of these

Q.70. Let S denotes the infinite sum $2 + 5x + 9x^2 + 14x^3 + 20x^4 + \ldots$, where $|x| < 1$ and the coefficient of x^{n-1} is $\frac{1}{2} n(n + 3)$, $(n = 1, 2, \ldots)$. Then S equals:

(a) $\frac{2-x}{(1-x)^3}$ (b) $\frac{2-x}{(1+x)^3}$

(c) $\frac{2+x}{(1-x)^3}$ (d) $\frac{2+x}{(1+x)^3}$

Q.71. If $x^2 + 5y^2 + z^2 = 2y\,(2x + z)$, then which of the following statements is/are necessarily true?

(a) $x = 2y$ (b) $x = 2z$ (c) $2x = z$

(a) Only A (b) B and C

(c) A and B (d) None of these

Q.72. Amol was asked to calculate the arithmetic mean of 10 positive integers, each of which had 2 digits. By mistake, he interchanged the 2 digits, say a and b, in one of these 10 integers. As a result, his answer for the arithmetic mean was 1.8 more than what it should have been. Then b–a equals:

(a) 1 (b) 2

(c) 3 (d) None of these

Q.73. A car rental agency has the following terms. If a car is rented for 5 h or less, then, the charge is Rs 60 per h or Rs 12 per km whichever is more. On the other hand, if the car is rented for more than 5 h, the charge is Rs 50 per h or Rs 7.50 per km whichever is more. Saurabh Gandhi rented a car from this agency, drove it for 30 km and ended up paying Rs 300. For how many hours did he rent the car?

(a) 4 h (b) 5 h

(c) 6 h (d) None of these

Q.74. A child was asked to add first few natural numbers (i.e., $1 + 2 + 3 + \ldots$) so long as his patience permitted. As he stopped, he gave the sum as 575. When the teacher declared the result as wrong, the child discovered he had missed one number in the sequence during addition. The number he missed was

(a) Less than 10 (b) 10

(c) 15 (d) More than 15

Q.75. Suppose for any real number x, $[x]$ denotes the greatest integer less than or equal to x. Let $L(x, y) = [x] + [y] + [x + y]$ and $R(x, y) = [2x] + [2y]$. Then it is impossible to find any two positive real numbers x and y for which

(a) $L(x, y) = R(x, y)$

(b) $L(x, y) \neq R(x, y)$

(c) $L(x, y) < R(x, y)$

(d) $L(x, y) > R(x, y)$

Q.76. Ten straight lines, no two of which are parallel and no three of which pass through any common point, are drawn on a plane. The total number of regions (including finite and infinite regions) into which the plane would be divided by the lines is

(a) 56 (b) 255

(c) 1024 (d) Not unique

Q.77. When 2^{256} is divided by 17, then the remainder would be

(a) 1 (b) 16

(c) 14 (d) None of these

Q.78. The number of real roots of the equation $\frac{A^2}{x} + \frac{B^2}{x-1} = 1$, where A and B are real numbers and simultaneously not equal to zero, is

(a) None (b) 1

(c) 2 (d) 1 or 2

Q.79. At a bookstore, 'Modern Book Store' is a flashed using neon lights. The words as individually flashed at the intervals of $2\frac{1}{2}$s, $4\frac{1}{4}$s and $5\frac{1}{8}$s respectively, and each word is put off after a second. The least time after which the full name of the bookstore can be read again is

(a) 49.5 s (b) 73.5 s

(c) 1744.5 s (d) 855 s

Q.80. Three pieces of cakes of weight $4\frac{1}{2}$lb, $6\frac{3}{4}$lb and $7\frac{1}{5}$lb respectively are to be divided into parts of equal weight. Each part must be as heavy as possible. If one such part is served to each guest, then what is the maximum number of guests that could be entertained?

(a) 54 (b) 72

(c) 20 (d) None of these

Q.81. After the division of a number successively by 3, 4 and 7, the remainders obtained are 2, 1 and 4 respectively. What will be the remainder if 84 divides the same number?

(a) 80 (b) 75 (c) 41 (d) 53

Q.82. Six persons are playing a card game. Suresh is facing Raghubir who is to the left of Ajay and to the right of Pramod. Ajay is to the left of Dhiraj. Yogendra is to the left of Pramod. If Dhiraj exchanges his seat with Yogendra and Pramod exchanges with Raghubir, who will be sitting to the left of Dhiraj?

(a) Yogendra (b) Raghubir

(c) Suresh (d) Ajay

Direction for questions 83 and 84: Answer the questions based on the following information.

A boy is asked to put one mango in a basket when ordered 'one', one orange when ordered 'two', one apple when ordered three, and is asked to take out from the basket one

mango and an orange when ordered four. A sequence of orders is given as: 1 2 3 3 2 1 4 2 3 1 4 2 2 3 3 1 4 1 1 3 2 3 4

Q.83. How many total oranges were in the basket at the end of the above sequence?
(a) 1 (b) 4
(c) 3 (d) 2

Q.84. How many total fruits will be in the basket at the end of the above order sequence?
(a) 9 (b) 8 (c) 11 (d) 10

Direction for questions 85 and 86: Answer the questions based on the following information.

Each of the 11 letters A, H, I, M, 0, T, U, V, W, X and Z appears same when looked at in a mirror. They are called symmetric letters. Other letters in the alphabet are asymmetric letters.

Q.85. How many four-letter computer passwords can be formed using only the symmetric letters (no repetition allowed)?
(a) 7,920 (b) 330
(c) 14,640 (d) 4,19,430

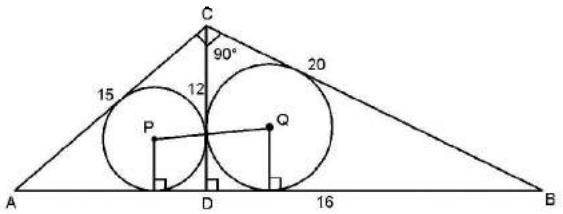
Q.86. How many three-letter computer passwords can be formed (no repetition allowed) with at least one symmetric letter?
(a) 990 (b) 2,730
(c) 12,870 (d) 15.600

Q.87. A train approaches a tunnel AB. Inside the tunnel is a cat located at a point that is $\frac{3}{8}$ of the distance AB measured from the entrance A. When the train whistles the cat runs. If the cat moves to the entrance of the tunnel A, the train catches the cat exactly at the entrance. If the cat moves to the exit B, the train catches the cat at exactly the exit. The speed of the train is greater than the speed of the cat by what order?
(a) 3:1 (b) 4:1
(c) 5:1 (d) None of these

Q.88. A piece of string is 40 cm long. It is cut into three pieces. The longest piece is three times as long as the middle-sized and the shortest piece is 23 cm shorter than the longest piece. Find the length of the shortest piece.
(a) 27 cm (b) 5 cm (c) 4 cm (d) 9 cm

Q.89. Three travellers are sitting around a fire, and are about to eat a meal. One of them has 5 small loaves of bread, the second has 3 small loaves of bread. The third has no food, but has 8 coins. He offers to pay for some bread. They agree to share the 8 loaves equally among the three travellers, and the third traveller will pay 8 coins for his share of the 8 loaves. All loaves were of the same size. The second traveller (who had 3 loaves) suggests that he will be paid 3 coins, and that the first traveller be paid 5 coins. The first traveller says that he should get more than 5 coins. How much should the first traveller get?
(a) 5 (b) 7
(c) 1 (d) None of these

Q.90.



In the above figure, ACB is a right-angled triangle. CD is the altitude. Circles are inscribed within the ΔACD and ΔBCD. P and Q are the centres of the circles. The distance PQ is
(a) 5 (b) 50
(c) 7 (d) 8

Q.91. If u, v is the weighted average and m is the natural number such that $u^m + v^m = w^m$, then which one of the following is true?
(a) $m \geq \min(u, v, w)$
(b) $m \geq \max(u, v, w)$
(c) $m < \min(u, v, w)$
(d) None of these

Q.92. In how many ways is it possible to choose a white square and a black square on a chessboard so that the squares must not lie in the same row or column?
(a) 56 (b) 896
(c) 60 (d) 768

Q.93. $7^{6n} - 6^{6n}$, where n is an integer > 0, is divisible by
(a) 13 (b) 127
(c) 559 (d) All of these

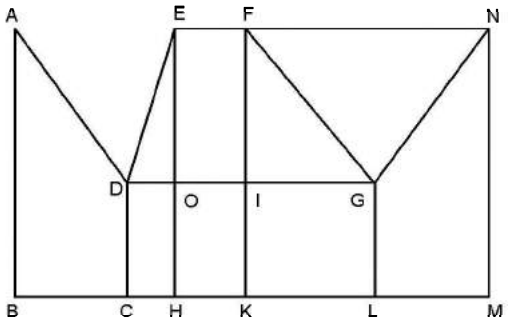
Q.94. If $pqr = 1$, the value of the expression $\frac{1}{1+p+q^{-1}} + \frac{1}{1+q+r^{-1}} + \frac{1}{1+r+p^{-1}}$ is equal to
(a) $p + q + r$ (b) $\frac{1}{p+q+r}$
(c) 1 (d) $p^{-1} + q^{-1} + r^{-1}$

Q.95. It takes six technicians a total of 10 h to build a new server from a direct computer, with each working at the same rate. If six technicians start to build the server at 11 am, and one technician per h is added beginning at 5 pm, at what time will the server be completed?

(a) 6.40 pm (b) 7 pm
(c) 7.20 pm (d) 8 pm

Q.96. Davji Shop sells samosas in boxes of different sizes. The samosas are priced at Rs 2 per samosa up to 200 samosas. For every additional 20 samosas, the price of the whole lot goes down by 10 paise per samosa. What should be the maximum size of the box that would maximize the revenue?
(a) 240
(b) 300
(c) 400
(d) None of these

Q.97. Three small pumps and a large pump are filling a tank. Each of the three small pump works at $\frac{2}{3}$rd the rate of the large pump. If all four pumps work at the same time, they should fill the tank in what fraction of the time that it would have taken the large pump alone?
(a) $\frac{4}{7}$ (b) $\frac{1}{3}$
(c) $\frac{2}{3}$ (d) $\frac{3}{4}$



In the above diagram, ∠ABC = 90° = ∠DCH = ∠DOE = ∠EHK = ∠FKL = ∠GLM = ∠LMN
AB = BC = 2CH = 2CD = EH = FK = 2HK = 4KL = 2LM = MN

Q.98. The magnitude of ∠FGO =
(a) 30° (b) 45°
(c) 60° (d) None of these

Q.99. What is the ratio of the areas of the two quadrilaterals ABCD to DEFG?
(a) 1:2 (b) 2:1
(c) 12:7 (d) None of these

Q.100. How many numbers greater than 0 and less than a million can be formed with the digits 0, 7 and 8?
(a) 486 (b) 1084
(c) 728 (d) None of these

ANSWERS

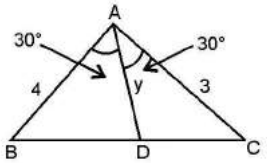
■■■ CAT 2002

Q. No.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
Answer	(c)	(b)	(a)	(b)	(a)	(d)	(d)	(b)	(d)	(d)	(c)	(b)	(d)	(d)	(c)	(b)	(b)	(d)	(c)	(a)
Q. No.	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
Answer	(c)	(b)	(c)	(d)	(d)	(a)	(a)	(d)	(b)	(d)	(d)	(c)	(d)	(c)	(a)	(c)	(a)	(c)	(b)	(b)
Q. No.	91	92	93	94	95	96	97	98	99	100										
Answer	(d)	(d)	(d)	(c)	(d)	(b)	(b)	(d)	(c)	(c)										

HINTS AND SOLUTIONS

51. (c) Total possible arrangements = $10 \times 9 \times 8$
Now 3 number can be arranged among themselves in 3! way = 6 ways
Given condition is satisfied by only 1 out of 6 ways hence, the required number of arrangements= $\frac{10\times9\times8}{6}=120$

52. (b)



Let BC = x and AD = y

As per bisector theorem, $\frac{BD}{DC}=\frac{AB}{AC}=\frac{4}{3}$

Hence, BD = $\frac{4x}{7}$; DC = $\frac{3x}{7}$

In ΔABD, $\cos 30° = \frac{(4)^2+y^2-\frac{16x^2 49}{49}}{2\times4\times y}$

$2\times4\times y\times\frac{\sqrt{3}}{2}=16+y^2-\frac{16x^2}{49}$

$4\sqrt{3}y=16+y^2-\frac{16x^2}{49}$ ...(i)

Similarly, from DADC, Cos 30°

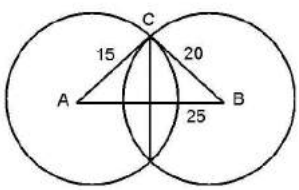
$=\frac{9+y^2-\frac{9x^2}{49}}{2\times3\times y}$

$\Rightarrow 3\sqrt{3}y=9+y^2-\frac{9x^2}{49}$ ...(ii)

Now (i) x 9 −16 x (ii), we get

$36\sqrt{3}y-48\sqrt{3}y=9y^2-16y^2\Rightarrow y=\frac{12\sqrt{3}}{7}$

53. (a)



Let the chord be = x cm

$\therefore \frac{1}{2}(15\times20)=\frac{1}{2}\times25\times\frac{x}{2}\Rightarrow x=24\,\text{cm}$

54. (b) $f(x)+f(y)=\log\left(\frac{1+x}{1-x}\right)+\log\left(\frac{1+y}{1-y}\right)$

$=\log\left(\frac{(1+x)(1+y)}{(1-x)(1-y)}\right)$

$=\log\left(\frac{1+x+y+xy}{1+xy-(x+y)}\right)$

$=\log\left(\frac{1+\left(\frac{x+y}{1-xy}\right)}{1-\left(\frac{x+y}{1-xy}\right)}\right)$

$=f\left(\frac{x+y}{1+xy}\right)$

55. (a) Total area=14 × 14 = 196 m²

Grazed area = $\left(\frac{\pi\times r^2}{4}\right)\times4=\pi r^2=22\times7$

= 154 m²

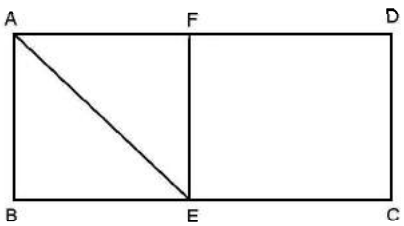
The ungrazed area is less than (196 – 154) = 42 m² for which there is only one option.

56. (d) Every trip will need more than 180 m and there are $4\frac{1}{2}$ trips. hence, the distance covered will be greater than 750 m for which there is only one option = 860.

Alternative Method

For the first stone, he will cover 100 m.
For second, 200 – 4 = 196
For third, 200 – 8 = 192
For fourth, 200 – 12 = 188
For fifth, 200 – 16 = 184
Hence, the total distance = 860 m

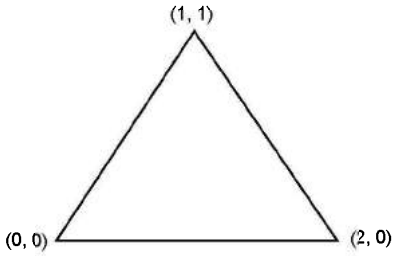
57. (d)



Area of ΔABE = 7cm²

Area of ABEF = 14 cm^2
Area of ABCD = 14 × 4
= 56 cm^2

58. (b)



Hence, area = $\frac{1}{2}(2)(1) = 1$

Note: Answer should be independent of a and area of the triangle does not have a square root.

59. (d) Check choices, e.g., $\frac{1}{2}$ Diagonal = $\sqrt{5}$

Distance saved = $3 - \sqrt{5} = 0.75 \neq$ Half the larger side hence, incorrect.

$\frac{3}{4} \Rightarrow$ Diagonal = 5

Distance saved = (4 + 3) – 5 = 2 = Half the larger side.

60. (d) If speed of N = 4, speed of g = 1,

Average speed = $\frac{2\times4\times1}{4+1} = 1.6$

Because time available is $\frac{2}{3}$, speed = $\frac{3}{2}$

Now average speed = 2, 4

Now speed of N = 8

Now speed of S = y

$\frac{2\times8\times y}{8+y} = 2.4 \Rightarrow y = 1.3$

Required ratio = 1.3 : 8 = 1:6

61. (c) AG_1=5 min at 30 km/h = 2.5 km
G_1G_3 = 15 km
Time for AG_1 = 5 min
Time for
$G_1G_3 + G_3A$ = 32.5 min = total of 37.5 min 1 min is taken for transferring the patient into and out of the ambulance.
Hence, (40 – 37.5 – 1) = 1.5 min is remaining.

62. (b) Check choices
Choice (b) 54 S = (5 + 4)2 = 81
⇒ D – S = 81 – 54 = 27. Hence, the number = 54

63. (d) $x_0 = x$
$x_1 = -x$
$x_3 = x$
$x_4 = x$
$x_5 = -x$
$x_6 = -x$
.....
⇒ Choices (a), (b), (c) are incorrect.

64. (d) $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + zy + zx)$
$\Rightarrow x^2 + y^2 + z^2 = 19$
$\Rightarrow y + z$ cannot be simultaneously = 0
else $xy + zy + zx = 0 \Rightarrow x^2 < 19 \Rightarrow x < \sqrt{19} \cong 4.4$

65. (c) Area = 40 × 20 = 800
If 3 rounds are done, area = 34 × 14 = 476
Area > 3 rounds
If 4 rounds ⇒ Area left = 32 × 12 = 347
Hence, the area should be slightly less than 4 rounds.

66. (b) Since thief escaped with 1 diamond,
Before 3rd watchman he had (1 + 2) × 2 = 6
Before 2nd watchman he had (6 + 2) × 2 = 16
Before 1st watchman he had (16 + 2) × 2 = 36
Alternative Method
Check with choices.

67. (b) Mayank paid $\frac{1}{2}$ of what others paid.

⇒ Mayank paid $\frac{1}{3}$ rd of the total amount
= $ 20.
Similarly, Mirza paid $ 15 and Little paid $ 12.
Remaining amount of $ 60 – $ 20 – $ 15 – $ 12 = $ 13 is paid by Jaspal.

68. (d) Let the number of gold coins = $x + y$
$48(x - y) = x^2 - y$?
$48(x - y) = (x - y)(x + y) = -x + y = 48$
Hence, the correct choice would be none of these.

69. (c) Let us assume that
p days : they played tennis
y days: they went for yoga
T days: total duration for which Ram and Shyam stayed together.
⇒ p + y = 22
(T – y) = 24
And (T – p) = 14
Adding all of them,
2T = 22 + 24 + 14 – T = 30 days.

70. (a) Coefficient of $x = -(n+1)(n+4)$

$S = 2 + 5x + 9x^2 + 14x^3 +$

$xS = 2x + 5x^2 + \ldots$

$S(1-x) = 23 + 3x + 4x^2 + 5x^3 + \ldots$

Let $S_1 = S(1-x) \Rightarrow S_1 = 2 + 3x + 4x^2 + \ldots$

$x S_1 = 2x + 3x^2 + \ldots$

$S_1(1-x) = 2 + x + x^2 + \ldots$

$$S_1(1-x^2) = 2 + \frac{x}{1-x}$$

$$S(1-x^2) = 2 + \frac{x}{1-x} \Rightarrow S = \frac{2-x}{(1-x)^3}$$

71. (c) $x^2 + 5y^2 + z^2 - 4yx + 2yz$

$(x^2 + 4y^2 - 4yx) + z^2 + y^2 - 2yz = 0$

$(x-2y)^2 + (z-y)^2 = 0$

It can be true only if $x = 2y$ and $z = y$

72. (b) Arithmetic mean is more by 18 means sum is more by 18. So ba-$ab = 18$

$b > a$ because sum has gone up, e.g., 31–13 = 18

Hence, $b - a = 2$

73. (c) By trial and error:

$30 \times 12 = 360 > 300$

$30 \times 7.5 = 225 < 300$

$50 \times 6 = 300$. Hence, he rented the car for 6 h.

74. (d) $575 = -x$

$1150 = n^2 + n - 2x$

$n(n+1) \geq 1150$

$n + n \geq 1150$

The smallest value for it is $n = 34$.

For $n = 34$

$40 = 2x \Rightarrow x = 20$

75. (d) $x - 1 \leq [x] \leq x$

$2x + 2y - 3 \leq L(x, y) \leq 2x + 2y \Rightarrow a - 3 \leq L \leq a$

$2x + 2y - 2 \leq R(x, y) \leq 2x + 2y \Rightarrow a - 2 \leq R \leq a$

Therefore, $L \leq R$

Note Choice (b) is wrong, otherwise choice (a) and choice (c) are also not correct. Choose the numbers to check.

76. (a) Number of regions $\frac{n(n+1)}{2} + 1$ where n

= Number of lines, i.e., for 0 line we have region = 1.

For 1 line we have region = 2.

If can be shown as:

Number of lines	0	1	2	3	4	5	...	10
Number of regions	1	2	4	7	11	16	...	56

Therefore, for $n = 10$, it is $2 + 1 = 56$

77. (a) $(2^4)^{64} = (17-1)^{64} = 17n + (-1)^{54} = 17n + 1$ Hence, remainder = 1

78. (d) $\frac{A^2}{x} + \frac{B^2 - 1}{x-1} \Rightarrow A^2(x-1)\, B^2 x = x^2 - x$

This is a quadratic equation. Hence, the number of roots = 2 or 1 (1 in the case when both roots are equal).

79. (b) Because each word is lit for a second,

$$\text{LCM}\left(\frac{5}{2}+1, \frac{17}{4}+1, \frac{41}{8}+1\right) = \text{LCM}\left(\frac{7}{2}, \frac{21}{4}, \frac{49}{8}\right)$$

$$\frac{\text{LCM}(7,21,49)}{\text{HCM}(2,4,8)} = \frac{49 \times 3}{2} = 73.5\text{S}$$

80. (d) $$HCF\left(\frac{9}{2}, \frac{27}{4}, \frac{36}{5}\right) = \frac{HCF(9,27,36)}{LCM(2,4,5)} = \frac{9}{20} lb$$

= weight of each piece total = weight = 18.45 lb

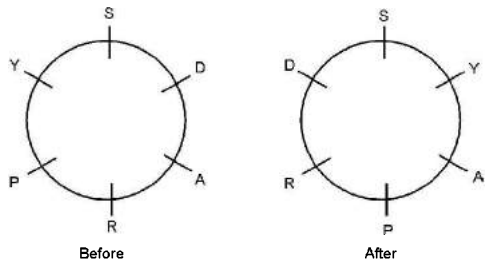
Maximum number of guests

$$= \frac{18.45 \times 20}{9} = 41$$

81. (d) $3(4(7x+4)+1)+2 = 84x + 53$

Therefore, remainder is 53.

82. (c)



Suresh is sitting to the left of Dhiraj.

83. (d) Number of oranges at the end of the sequence = Number of 2s – Number of 4s = 6 – 4 = 2

84. (c) Number of (1s + 2s + 3s) – 2 (Number of 4s) = 19 – 8

85. (a) $11 \times 10 \times 9 \times 8 = 7920$

86. (c) Total number of passwords using all letters – Total number of passwords using no symmetric letters = $(26 \times 25 \times 24) - (15 \times 14 \times 13) = 12870$

87. (a) Let tunnel be = 8 km and speed of cat = 1 km/h

Time taken to reach the entrance of the tunnel by the cat = 3 h

Time taken to reach the exit of the tunnel by the cat = 5 h

Train will cover the sum (length of tunnel) = 2 h

Therefore, the ratio of the speeds of the train and the cat = 4 : 1

Speed of the train is greater by 3:1 than that of the cat.

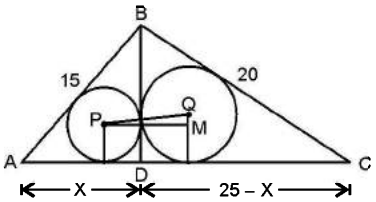
88. (c) Let the largest piece = $3x$ Middle = x
Shortest = $3x - 23$
or $3x + x + (3x - 23) = 40$
or $x = 9$
or the shortest piece = $3(9) - 23 = 4$
Check choices:
The shortest piece has to be < 20 cm.
27 is wrong choice.
The largest piece is a multiple of 3.
Or (23 + shortest) should be a multiple of 3.
Answer = 4 cm (Among other choices)

89. (b) Each traveller had $\frac{8}{3}$ loaves.

First traveller has given $5 - \frac{8}{3}$ loaves to the third.

Second traveller sacrificed only $3 - \frac{8}{3} = \frac{1}{3}$rd of a loaf. So, first should get 7 coins.

90. (b)



$(15)^2 - x^2 = (20)^2 - (25 - x)^2$
$\Rightarrow x = 9$
$\Rightarrow$ BD = 12

Area of $\Delta ABD = \frac{1}{2} \times 12 \times 9 = 54$

$s = \frac{1}{2}(15 + 12 + 9) = 18$

$r_1 = \frac{Area}{s} \Rightarrow r_1 = 3$

Area of $\Delta BCD = \frac{1}{2} = 96$

$s = \frac{1}{2}(16 + 20 + 12) = 24$

$r_2 = \frac{Area}{s} \Rightarrow r_2 = 4$

in ΔPQM, $PM = r_1 + r_2 = 7$ cm
$QM = r_2 - r_1 = 1$ cm
Hence, $PQ = \sqrt{50}$ cm

91. (d) $u^m + v^m = w^m$
$u^2 + v^2 = w^2$
Taking Pythagorean triplet 3, 4 and 5, we see in < mm (u, v, w)
Also $1^1 + 2' = 3'$ and hence in ≤ mm (u, v, w)

92. (d) A black square can be chosen in 32 ways. Once a black square is there, you cannot choose the 8 white squares in its row or column. So the number of white squares available = 24
Number of ways = $32 \times 24 = 768$

93. (d) $7^{6n} - 6^{6n}$
put $n = 1$
$7^6 - 6^6 = (7^3 - 6^3)(7^3 + 6^3)$
This is a multiple of $7^3 + 6^3 = 559$ and $7 + 6 = 13$

94. (c) If $p = q = r = 1$, then expression = 1
Check the choice only, one choice gives the value of expression = 1.

95. (d) Total amount of work = 60 man hours
From 11 am to 5 pm, 6 technicians = 36 man hours
From 5 pm to 6 pm, 7 technicians = 7 man hours
From 6 pm to 7 pm, 8 technicians = 8 man hours
From 7 am to 8 pm, 9 technicians = 9 man hours
Total = 60 man hours.

96. (b) Number of samosas = 200 + 20n,
n is a natural number.
Price per samosa = Rs (2 – 0.1n)
Revenue = (200 + 20n) (2 – 0.1n) = $400 + 20n - 2n^2$
For maxima $20 - 4n = 0$; by differentiation n = 5
$\Rightarrow$ Maximum revenue will be at
$(200 + 20 \times 5) = 300$ samosas

97. (b) Three small pumps = Two large pumps
Three small + One large pumps
= Three large pump $\Rightarrow \frac{1}{3}$

98. (d) It KL = 1, then lG = 1 and FI = 2
Hence, $\tan q = \frac{2}{1} = 2$
Thus, q none of 30. 45 and 600.

99. (c) Area of quadrilateral ABCD = $\frac{1}{2}(2x + 4x) \times 4x = 12x^2$

Area of quadrilateral DEFG = $\frac{1}{2}(5x + 2x) \times 2x = 7x^2$

100. (c) Number of ways for single digit = 2
2 digits = $2 \times 3 = 6$
3 digits = $2 \times 3 \times 3 = 18$
4 digits = $2 \times 3 \times 3 \times 3 = 54$
5 digits = $2 \times 3 \times 3 \times 3 \times 3 = 162$
6 digits = $2 \times 3 \times 3 \times 3 \times 3 \times 3 = 486$
Total = 728

CAT 2003

LEAKED PAPER

Direction for questions 101 and 102: Answer the questions on the basis of the information given below.

A certain perfume is available at a duty-free shop at the Bangkok International Airport. It is priced in the Thai currency Baht but other currencies are also acceptable. In particular, the shop accepts Euro and US Dollar at the following rates of exchange:

US Dollar 1 = 41 Bahts and Euro 1 = 46 Bahts

The perfume is priced at 520 Bahts per bottle. After one bottle is purchased, subsequent bottles are available at a discount of 30%. Three friends S, R and M together purchase three bottles of the perfume, agreeing to share the cost equally. R pays 2 Euros M pays 4 Euros and 27 Thai Bahts and S pays the remaining amount in US Dollars.

Q.101. How much does R owe to S in Thai Baht?
(a) 428 (b) 416 (c) 334 (d) 324

Q.102. How much does M owe to S in US Dollars?
(a) 3 (b) 4 (c) 5 (d) 6

Direction for questions 103 and 104: Answer the questions on the basis of the information given below.

New Age Consultants have three consultants Gyani, Medha and Buddhi. The sum of the number of projects handled by Gyani and Buddhi individually is equal to the number of projects in which Medha is involved. All three consultants are involved together in 6 projects. Gyani works with Medha in 14 projects. Buddhi has 2 projects with Medha but without Gyani, and 3 projects with Gyani but without Medha. The total number of projects for the New Age Consultants is one less than twice the number of projects in which more than one consultant is involved.

Q.103. What is the number of projects in which Gyani alone is involved?
(a) Uniquely equal to zero
(b) Uniquely equal to 1
(c) Uniquely equal to 4
(d) Cannot be determined uniquely

Q.104. What is the number of projects in which Medha alone is involved?
(a) Uniquely equal to zero
(b) Uniquely equal to 1
(c) Uniquely equal to 4
(d) Cannot be determined uniquely

[redacted]

Q.105. The number of non-negative real roots of $2^x - x - 1 = 0$ equals
(a) 0 (b) 1 (c) 2 (d) 3

Q.106. When the curves $y = \log_{10} x$ and $y = x^{-1}$ are drawn in the x-y plane, how many times do they intersect for values $x \geq 1$?
(a) Never (b) Once
(c) Twice (d) More than twice

Q.107. Let A and B be the two solid spheres such that the surface area of B is 300% higher than the surface area of A. The volume of A is found to be k% lower than the volume of B. The value of k must be
(a) 85.5 (b) 92.5 (c) 90.5 (d) 87.5

Q.108. Which one of the following conditions must p, q and r satisfy so that the following system of linear simultaneous equations has at least one solution, such that $p + q + r \neq 0$?
$x + 2y - 3z = p$

$2x + 6y - 11z = q$
$x - 2y + 7z = r$
(a) $5p - 2q - r = 0$ (b) $5p + 2q + r = 0$
(c) $5p + 2q - r = 0$ (d) $5p - 2q + r = 0$

Q.109. A leather factory produces two kinds of bags, standard and deluxe. The profit margin is Rs 20 on a standard bag and Rs 30 on a deluxe bag. Every bag must be processed on machine A and on Machine B. The processing times per bag on the two machines are as follows:

	Time required (h/bag)	
	Machine A	*Machine B*
Standard Bag	4	6
Deluxe Bag	5	10

The total time available on machine A is 700 h and on machine B is 1250 h. Among the following production plans, which one meets the machine availability constraints and maximizes the profit?
(a) Standard 75 bags, Deluxe 80 bags
(b) Standard 100 bags, Deluxe 60 bags
(c) Standard 50 bags, Deluxe 100 bags
(d) Standard 60 bags, Deluxe 90 bags

Q.110. The sum of the 3rd and the 15th elements of an arithmetic progression is equal to the sum of 6th, 11th and 13th elements of the same progression. Then which element of the series should necessarily be equal to zero?
(a) 1st (b) 9th
(c) 12th (d) None of the above

[redacted]

A city has two perfectly circular and concentric ring roads, the outer ring road (OR) being twice as long as the inner ring road (IR). There are also four (straight line) chord roads from E1, the east end point of OR to N2, the north end point of IR; from N1, the north end point of OR to W2, the west end point of IR; from W1, the west end point of OR, to S2, the south end point of IR; and from S1 the south end point of OR to E2, the east end point of IR. Traffic moves at a constant speed of 30 p km/h on the OR road, 20 p km/h on the IR road, and $15\sqrt{5}$ km/h on all the chord roads.

Q.111. The ratio of the sum of the lengths of all the chord roads to the length of the outer ring road is
(a) $\sqrt{5}:2$ (b) $\sqrt{5}:2p$
(c) $\sqrt{5}:p$ (d) None of these

Q.112. Amit wants to reach N2 from S1. It would take him 90 min if he goes on minor arc S1 – E1 on or, and then on the chord road E1 – N2. What is the radius of the outer ring road in kms?
(a) 60 (b) 40 (c) 30 (d) 20

Q.113. Amit wants to reach E2 from N1 using first the chord N1 – W2 and then the inner ring road. What will be his travel time in min on the basis of information given in the above question?
(a) 60 (b) 45 (c) 90 (d) 105

[redacted]

Q.114. A test has 50 questions. A student scores 1 mark for a correct answer, −1/3 for a wrong answer, and −1/6 for not attempting a question. If the net score of a student is 32, the number of questions answered wrongly by that student cannot be less than
(a) 6 (b) 12 (c) 3 (d) 9

Q.115. Twenty-seven persons attend a party. Which one of the following statements can never be true?
(a) There is a person in the party who is acquainted with all the twenty-six others.
(b) Each person in the party has a different number of acquaintances.
(c) There is a person in the party who has an odd number of acquaintances.
(d) In the party there is no set of three mutual acquaintances.

Q.116. Let $g(x) = \max(5 - x, x + 2)$. The smallest possible value of $g(x)$ is
(a) 4.0 (b) 4.5
(c) 1.5 (d) None of the above

Q.117. The function $f(x) = |x - 2| + |2.5 - x| + |3.6 - x|$, where x is a real number, attains a minimum at:
(a) $x = 2.3$ (b) $x = 2.5$
(c) $x = 2.7$ (d) None of these

Q.118. How many even integers n, where $100 \le n \le 200$, are divisible neither by seven nor by nine?
(a) 40 (b) 37 (c) 39 (d) 38

Q.119. A positive whole number M less than 100 is represented in base 2 notation, base 3 notation, and base 5 notation. It is found that in all three cases the last digit is 1, while in exactly two out of the three cases the leading digit is 1. Then M equals
(a) 31 (b) 63 (c) 75 (d) 91

Q.120. In a 4000 m race around a circular stadium having a circumference of 1000 m, the fastest runner and the slowest runner reach the same point at the end of the 5th min for the first time after the start of the race. All the runners have the same starting point and each runner maintains a uniform speed throughout the race. If the fastest runner runs at twice the speed of the slowest runner, what is the time taken by the fastest runner to finish the race?
(a) 20 min (b) 15 min
(c) 10 min (d) 5 min

Direction for questions 121 to 125: Each question is followed by two statements, A and B. Answer each question using the following instructions.

Choose (a): If the question can be answered by one of the statements alone but not by the other.
Choose (b): If the question can be answered by using either statement alone.
Choose (c): If the question can be answered by using both the statements together, but cannot be answered by using either statement alone.
Choose (d): If the question cannot be answered even by using both the statements together.

Q.121. Is $a^{44} < b^{11}$, given that $a = 2$ and b is an integer?
(A) b is even
(B) b is greater than 16

Q.122. What are the unique values of b and c in the equation $4x^2 + bx + c = 0$ if one of the roots of the equation is $(-1/2)$?
(A) The second root is 1/2.
(B) The ratio of c and b is 1.

Q.123. AB is a chord of a circle. AB = 5 cm. A tangent parallel to AB touches the minor arc AB at E.
What is the radius of the circle?
(A) AB is not a diameter of the circle
(B) The distance between AB and the tangent at E is 5 cm

Q.124. Is $\left(\frac{1}{a^2}+\frac{1}{a^4}+\frac{1}{a^6}+\ldots\right) > \left(\frac{1}{a}+\frac{1}{a^3}+\frac{1}{a^5}+\ldots\right)$?
(A) $-3 \le a \le 3$
(B) One of the roots of the equation $4x^2 - 4x + 1 = 0$ is a.

Q.125. D, E, F are the mid points of the sides AB, BC and CA of triangle ABC respectively. What is the area of DEF in square centimeters?
(A) AD = 1 cm, DF = 1 cm and perimeter of DEF = 3 cm
(B) Perimeter of ABC = 6 cm, AB = 2 cm, and AC = 2 cm.

Q.126. At the end of year 1998, Shepard bought nine dozen goats. Henceforth, every year he added p% of the goats at the beginning of the year and sold q% of the goats at the end of the year where $p > 0$ and $q > 0$. If Shepard had nine dozen goats at the end of year 2002 after making the sales for that year, which of the following is true?
(a) $p = q$ (b) $p < q$ (c) $p > q$ (d) $p = q/2$

Q.127. Each side of a given polygon is parallel to either the X or the Y axis. A corner of such a polygon is said to be convex if the internal angle is 90° or concave if the internal angle is 270°. If the number of convex corners in such a polygon is 25, the number of concave corners must be
(a) 20 (b) 0 (c) 21 (d) 22

Q.128. The 288th term of the series a, b, b, c, c, c, d, d, d, d, e, e, e, e, e, f, f, f, f, f, f...is
(a) u (b) v (c) w (d) x

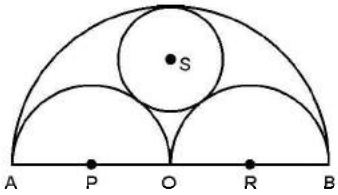
Q.129. Let p and q be the roots of the quadratic equation $x^2 - (a-2)x - a - 1 = 0$. What is the minimum possible value of $p^2 + q^2$?
(a) 0 (b) 3 (c) 4 (d) 5

Q.130. There are two concentric circles such that the area of the outer circle is four times the area of the inner circle. Let A, B and C be three distinct points on the perimeter of the outer circle such that AB and AC are tangents to the inner circle. If the area of the outer circle is 12 cm^2 then the area (in sq cm) of the triangle ABC would be
(a) $\pi\sqrt{12}$ (b) $\frac{9}{\pi}$ (c) $\frac{9\sqrt{3}}{\pi}$ (d) $\frac{6\sqrt{3}}{\pi}$

Q.131. Let a, b, c and d be four integers such that $a + b + c + d = 4m + 1$ where m is a positive integer. Given m, which one of the following is necessarily true?
(a) The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
(b) The minimum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$
(c) The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 - 2m + 1$
(d) The maximum possible value of $a^2 + b^2 + c^2 + d^2$ is $4m^2 + 2m + 1$

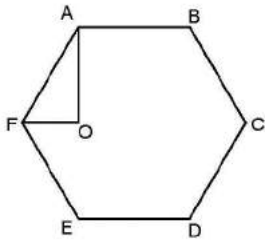
Q.132. Three horses are grazing within a semi-circular field. In the diagram given below, AB is the diameter of the semicircular field with center at O. Horses are tied up at P, R and S such that PO and RO are the radii of semicircles with centres at P and R respectively, and S is the center of the circle touching the two semicircles

with diameters AO and OB. The horses tied at P and R can graze within the respective semicircles and the horse tied at S can graze within the circle centred at S. The percentage of the area of the semicircle with diameter AB that cannot be grazed by the horses is nearest to



(a) 20 (b) 28 (c) 36 (d) 40

Q.133. In the figure given below, ABCDEF is a regular hexagon and ∠AOF = 90°. FO is parallel to ED. What is the ratio of the area of the triangle AOF to that of the hexagon ABCDEF?



(a) $\frac{1}{12}$ (b) $\frac{1}{6}$

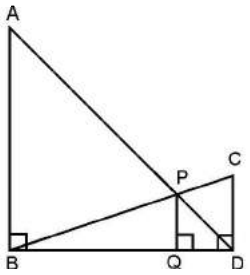
(c) $\frac{1}{24}$ (d) $\frac{1}{18}$

Q.134. How many three digit positive integers, with digits x, y and z in the hundreds, tens and unit's place respectively, exist such that $x < y$, $z < y$ and $x \neq 0$?
(a) 245 (b) 285 (c) 240 (d) 320

Q.135. A vertical tower OP stands at the centre O of a square ABCD. Let h and b denote the length OP and AB respectively. Suppose ∠APB = 60°, then the relationship between h and b can be expressed as
(a) $2b^2 = h^2$ (b) $2h^2 = b^2$
(c) $3b^2 = 2h^2$ (d) $3h^2 = 2b^2$

Q.136. In the triangle ABC, AB = 6, BC = 8 and AC = 10. A perpendicular dropped from B, meets the side AC at D. A circle of radius BD (with center B) is drawn. If the circle cuts AB and BC at P and Q respectively, the AP : QC is equal to
(a) 1:1 (b) 3:2 (c) 4:1 (d) 3:8

Q.137. In the diagram given below, ∠ABD = ∠CDB = ∠PQD = 90°. If AB:CD = 3:1, the ratio of CD: PQ is



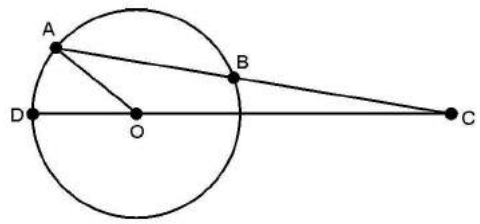
(a) 1:0.69 (b) 1:0.75
(c) 1:0.72 (d) None of the above

Q.138. There are 8436 steel balls, each with a radius of 1 cm, stacked in a pile, with 1 ball on top, 3 balls in the second layer, 6 in the third layer, 10 in the fourth, and so on. The number of horizontal layers in the pile is
(a) 34 (b) 38 (c) 36 (d) 32

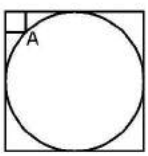
Q.139. If the product of n positive real numbers is unity, then their sum is necessarily
(a) A multiple of n (b) Equal to $n + \frac{1}{n}$
(c) Never less than n (d) A positive integer

Q.140. If $\log_3 2$, $\log_3(2^x - 5)$, $\log_3(2^x - 7/2)$ are in an arithmetic progression, then the value of x is equal to
(a 5 (b) 4 (c) 2 (d) 3

Q.141. In the figure given below, AB is the chord of a circle with center O. AB is extended to C such that BC = OB. The straight line CO is produced to meet the circle at D. If ∠ACD = y degrees and ∠AOD = x degrees such that x = ky, then the value of k is
(a) 3 (b) 2
(c) 1 (d) None of the above



Q.142. In the figure given below, the rectangle at the corner measures 10 cm × 20 cm. The corner A of the rectangle is also a point on the circumference of the circle. What is the radius of the circle in cm?



(a) 10 cm (b) 40 cm
(c) 50 cm (d) None of these

Q.143. Given that $-1 \le v \le 1$, $-2 \le u \le -0.5$ and $-2 \le z \le -0.5$ and $w = vz/u$, then which of the following is necessarily true?
(a) $-0.5 \le w \le 2$ (b) $-4 \le w \le 4$
(c) $-4 \le w \le 2$ (d) $-2 \le w \le -0.5$

Q.144. There are 6 boxes numbered 1, 2, ... 6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is
(a) 5 (b) 21 (c) 33 (d) 60

Q.145. Consider the following two curves in the x-y plane:
$y = x^3 + x^2 + 5$
$y = x^2 + x + 5$
Which of following statements is true for $-2 \le \times \le 2$?
(a) The two curves intersect once
(b) The two curves intersect twice
(c) The two curves do not intersect
(d) The two curves intersect thrice

Q.146. In a certain examination paper, there are n questions. For $j = 1, 2 \ldots n$, there are 2^{n-j} students who answered j or more questions wrongly. If the total number of wrong answers is 4095, then the value of n is
(a) 12 (b) 11 (c) 10 (d) 9

Q.147. If x, y, z are distinct positive real numbers the $\dfrac{x^2(y+z)+y^2(x+z)+z^2(x+y)}{xyz}$ would be:
(a) Greater than 4 (b) Greater than 5
(c) Greater than 6 (d) None of the above

Q.148. A graph may be defined as a set of points connected by lines called edges. Every edge connects a pair of points. Thus, a triangle is a graph with 3 edges and 3 points. The degree of a point is the number of edges connected to it. For example, a triangle is a graph with three points of degree 2 each. Consider a graph with 12 points. It is possible to reach any point from any point through a sequence of edges. The number of edges, e, in the graph must satisfy the condition
(a) $11 \le e \le 66$ (b) $10 \le e \le 66$
(c) $11 \le e \le 65$ (d) $0 \le e \le 11$

Q.149. The number of positive integers n in the range $12 \le n \le 40$ such that the product $(n-1)(n-2) \ldots 3.2.1$ is not divisible by n is
(a) 5 (b) 7 (c) 13 (d) 14

Q.150. Let T be the set of integers {3, 11 19, 27, 451, 459, 467} and S be a subset of T such that the sum of no two elements of S is 470. The maximum possible number of elements in S is
(a) 32 (b) 28 (c) 29 (d) 30

ANSWERS

■■■ CAT 2003

Q. No.	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
Answer	(d)	(c)	(d)	(b)	(c)	(b)	(d)	(a)	(a)	(c)	(c)	(c)	(d)	(c)	(b)	(d)	(b)	(c)	(d)	(c)
Q. No.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140
Answer	(a)	(b)	(a)	(a)	(b)	(c)	(c)	(d)	(d)	(c)	(b)	(b)	(a)	(c)	(b)	(d)	(b)	(c)	(c)	(c)
Q. No.	141	142	143	144	145	146	147	148	149	150										
Answer	(a)	(c)	(b)	(b)	(d)	(a)	(c)	(a)	(b)	(d)										

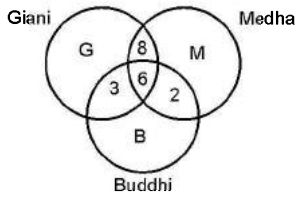
HINTS AND SOLUTIONS

101. (d)

102. (c)

103. (d) Putting the value of M in either equation, we get G + B = 17.
Hence, neither of two can be uniquely determined.

104. (b) As per the given data we get the following:



G + B = M + 16

Also, M + B + G + 19 = (2 × 19) − 1 i.e., (G + B) = 18 − M. Thus, M + 16 = 18 − M i.e., M = 1

105. (c) $2^x - x - 1 = 0$

$\Rightarrow 2^x - 1 = x$

If we put $x = 0$, then this is satisfied and if we put $x = 1$, then also this is satisfied.

Now we put $x = 2$, then this is not valid.

106. (b) For the curves to intersect, $\log_{10} x = x^{-1}$

Thus, $\log_{10} x = \frac{1}{x}$ or $x^x = 10$

This is possible for only one value of x ($2 < x < 3$).

107. (d) The surface area of a sphere is proportional to the square of the radius.

Thus, $\frac{S_B}{S_A} = \frac{4}{1}$ (SA of B is 300% higher than A)

5A 1

$\therefore \frac{r_B}{r_A} = \frac{2}{1}$

The volume of a sphere is proportional to the cube of the radius.

Thus, $\frac{V_B}{V_A} = \frac{8}{1}$

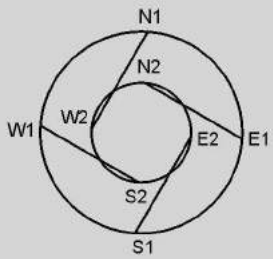
Or, V_A is $\frac{7}{8}$th less than B i.e., $\left(\frac{7}{8} \times 100\right) 87.5\%$

108. (a) It is given that $p + q + r \neq 0$, if we consider the first option, and multiply the first equation by 5, second by −2 and third by −1, we see that the coefficients of x, y and z all add up-to zero.
Thus, $5p - 2q - r = 0$
No other option satisfies this.

109. (a) Let 'x' be the number of standard bags and y be the number of deluxe bags.
Thus, $4x + 5y \leq 700$ and $6x + by \leq 1250$
Among the choices, (c) and (d) do not satisfy the second equation.
Choice (b) is eliminated as, in order to maximize profits the number of deluxe bags should be higher than the number of standard bags because the profit margin is higher in a deluxe bag.

110. (c) Let the 1st term be 'a' and common difference be 'd' then we have 3rd term = $a + 2d$
15th term = $a + 14d$
6th term = $a + 5d$
11th term = $a + 10d$
13th term = $a + 12d$
Since sum of 3rd and 15th term − sum of 6th, 11th and 13th term, therefore we have $2a + 16d = 3a + 27d$
$\Rightarrow a + 11d = 0$
Which is the 12th term

Questions 111 to 113



If the radius of the inner ring road is r, then the radius of the outer ring road will be 2r (since the circumference is double).

The length of IR = 2πr, that of OR = 4πr and that of the chord roads are r $\sqrt{5}$ (Pythagoras theorem)

The corresponding speeds are
20π, 30π and 15 $\sqrt{5}$ km/h.

Thus time taken to travel one circumference of IR = $\frac{r}{10}$ h, one circumference or or = $\frac{r}{7.5}$ h

111. (c) Sum of the length of the chord roads = $4r\sqrt{5}$ and the length of OR = $4\pi r$.
Thus the required ratio = $\sqrt{5}:\pi$

112. (c) The total time taken by the route given
$= \frac{r}{30}+\frac{r}{15}=\frac{3}{2}$ (i.e., 90 min.)
Thus, r = 15 km.
The radius of OR = 2r = 30 km

113. (d) The total time taken = $\frac{r}{20}+\frac{r}{15}=\frac{7r}{60}$
Since r = 15, the total time taken = $\frac{7}{4}$h
= 105 min.

114. (c) Let the number of correct answers be x', number of wrong answers be y and number of questions not attempted be 'z'.
Thus, $x + y + z = 50$ ... (i)
And $x-\frac{y}{3}-\frac{z}{6}=32$
The second equation can be written as,
$6x - 2y - z = 192$ ... (ii)
Adding the two equations we get,
$7x - y = 242$ or $x = \frac{242}{7}+y$
Since, x and y are both integers, y cannot be 1 or 2. The minimum value that y can have is 3.

115. (b) The number 27 has no significance here.
Statement b, will never be true for any number of people.
Let us take the case of 2 people.
If A knows B and B only knows A, both of them have 1 acquaintance each. Thus, B should be knowing atleast one other person.
Let us say he knows 'C' as well. So now 'B has two acquaintances (A and C), but C has only acquaintance (B), which is equal to that of A.
To close this loop, C will have to know A as well.
In which case he will have two acquaintances, which is the same as that of C.
Thus the loop will never be completed unless atleast two of them have the same number of acquaintances. Besides, statements 1, 3 and 4 can be true.
Note If we consider the situation other wise, to satisfy condition 2, the first person must have 26 acquaintances, the second 25, third 24 and so on. If we continue, the last one should have 0 acquaintance, which is not possible.

116. (d) We can see that $x + 2$ is an increasing function and $5 - x$ is a decreasing function. This system of equation will have smallest value at the point of intersection of the two i.e., $5 - x = x + 2$ or $x = 1.5$. Thus, the smallest value of $g(x) = 3.5$

117. (b) **Case 1:** If $x < 2$, then $y = 2 - x + 2.5 - x + 3.6 - x = 8.1 - 3x$.
This will be the least if x is the highest i.e., just less than 2.
In this case y will be just more than 2.1
Case 2: If $2 \le x < 2.5$, then $y = x - 2 + 2.5 - x + 3.6 - x = 4.1 - x$
Again, this will be the least if x is the highest in this case y will be just more than 1.6.
Case 3: If $2.5 \le x < 3.6$, then $y = x - 2 + x - 2.5 + 3.6 - x = x - 0.9$
This will be the least if x is least i.e., the $X = 2.5$.
Case 4: If $x \ge 3.6$, then
$y = x - 2 + x - 2.5 + x - 3.6 = 3x - 8.1$
The minimum value of this will be at $x = 3.6$ and $y = 2.7$
Hence, the minimum value of y is attained at $x = 2.5$

118. (c) There are 101 integers in all, of which 51 are even.
From 100 to 200, there are 14 multiples of 7, of which 7 are even.
There are 11 multiples of 9, of which 6 are even.
But, there is one integer (i.e., 126) that is a multiple of both 7 and 9 and also even.
Hence, the answer is (51 – 7 – 6 + 1) = 39

119. (d) Since the last digit in base 2, 3 and 5 is 1, the number should be such that on dividing by either 2, 3 or 5 we should get a remainder 1. The smallest such number is 31. The next set of numbers are 61, 91. Among these only 31 and 91 are a part of the answer choices.
Among these, $(31)_{10} = (11111)_2 = (1011)_3 = (111)_5$
Thus, all three forms have the leading digit 1.
Hence, the answer is 91.

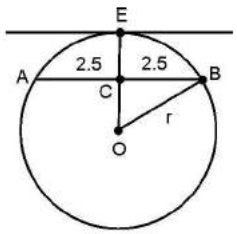
120. (c) The ratio of the speeds of the fastest and the slowest runners is 21. Hence they should meet at only one point on the circumference i.e., the starting point (as the difference in the ratio in the reduced form is 1). For the two of them to meet for the first time, the faster should have completed one complete round over the slower one. Since the two of them meet for the first time after 5 min, the faster one should have completed 2 rounds

(i.e., 2000 m) and the slower one should have completed 1 round. (i.e., 1000 m) in this time. Thus, the faster one would complete the race (i.e., 4000 m) in 10 min.

121. (a) Solution cannot be found by using only Statement A since b can take any even number 2, 4, 6…
But we can arrive at solution by using statement B alone.
If $b > 16$, say $b = 17$
Hence, $2^{44} < (16 + 1)^{11}$
$2^{44} < (2^4 + 1)^{11}$

122. (b) Solution can be found using Statement A as we know both the roots for the equation $\left(\text{viz.}, \frac{1}{2} \text{ and } -\frac{1}{2}\right)$
Also statement B is sufficient.
Since ratio of c and $b = 1$, c = b.
Thus the equation = $4x^2 + bx + b = 0$. Since $x = -\frac{1}{2}$ is one of the roots, substituting x we get $1 - \frac{b}{2} + b = 0$ or $b = -2$. Thus, $c = -2$.

123. (a)



We can get the answer by using the second statement only. Let the radius be r. AC = CB = 2.5 and using statement B, CE = 5, thus OC = (r – 5).
Using Pythagoras theorem, $(r - 5)^2 + (2.5)^2 = r^2$
We get r = 3.125
Note: You will realize that such a circle is not possible (if r = 3.125 how can CE be 5). However, we need to check for data sufficiency and not for data consistency. Since we are able to find the value of r uniquely using second statement the answer is (a).

124. (a) Both the series are infinitely diminishing series.
For the first series: First term $\frac{1}{a^2}$ and $r = \frac{1}{a^2}$
For the second series: First term = $\frac{1}{a}$ and $r = \frac{1}{a^2}$
The sum of the first series = $\frac{\frac{1}{a^2}}{1 - \frac{1}{a^2}} = \frac{1}{a^2 - 1}$
The sum of the second series = $\frac{\frac{1}{a^2}}{1 - \frac{1}{a^2}} = \frac{a}{a^2 - 1}$
Now, from the first statement, the relation can be anything (depending on whether 'a' is positive or negative).
But the second statement tells us, $4a^2 - 4a + 1 = 0$ or $a = \frac{1}{2}$. For this value of a, the sum of second series will always be greater than that of the first.

125. (b) The question tells us that the area of a triangle DEF will be the area of a triangle ABC. Thus by knowing either of the statements, we can get the area of the triangle DEF.

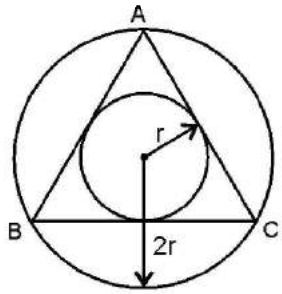
126. (c) The number of goats remain the same.
If the percentage that is added every time is equal to the percentage that is sold, then there should be a net decrease. The same will be the case if the percentage added is less than the percentage sold.
The only way, the number of goats will remain the same is if p > q.

127. (c) In this kind of polygon, the number of convex angles will always be exactly 4 more than the number of concave angles.
Note The number of vertices have to be even. Hence, the number of concave and convex corners should add up to an even number. This is true only for the answer choice (c).

128. (d) The number of terms of the series forms the sum of first n natural numbers i.e., $n(n + 1)$
Thus the first 23 letters will account for the first 23×24
= 276 terms of the series.
The 288 m term will be the 24 m letter which is x.

129. (d) $p + q = a - 2$ and $pq = -a - 1$
$(p + q)^2 = p^2 + q^2 + 2pq$,
Thus, $(a - 2)^2 = p^2 + q^2 + 2(-\alpha - 1)$
$p^2 + q^2 = \alpha^2 - 4\alpha + 4 + 2\alpha + 2$
$p^2 + q^2 = \alpha^2 - 2\alpha + 6$
$p^2 + q^2 = -2a + 1 + 5$
$p^2 + q^2 - (\alpha - 1)^2 + 5$ Thus, the minimum value of $p2 + q2$ is 5.

130. (c)



Since the area of the outer circle is 4 times the area of the inner circle, the radius of the outer circle should be 2 times that of the inner circle.

Since AB and AC are the tangents to the inner circle, they should be equal. Also, BC should be a tangent to inner circle. In other words, triangle ABC should be equilateral.

The area of the outer circle is 12. Hence, the area of the inner circle is 3 or the radius is $\sqrt{\frac{3}{\pi}}$. The area of the equilateral triangle $= 3\sqrt{3}\,r^2$, where r is the in radius. Hence the answer is c.

131. (b) $(a + b + c + d)^2 - (4m + 1)^2$

Thus, $a^2 + b^2 + c^2 + d^2 + 2(ab + ac + ad + bc + bd + cd)$

$- 16m^2 + 8m + 1$

$a^2 + b^2 + c^2 + d^2$ will have the minimum value if $(ab + ac + ad + bc + bd + cd)$ is the maximum.

This is possible if $a = b = c = d = (m + 0.25)$ ... since $a + b + c + d = 4m + 1$

In that case $2(ab + ac + ad + bc + bd + cd)$

$= 12(m + 0.25)^2 - 12m^2 + 6m + 0.75$

Thus, the minimum value of $a^2 + b^2 + c^2 + d^2$

$= (16m^2 + 8m + 1) - 2(ab + ac + ad + bc + bd + cd)$

$= (16m\,2 + 8m + 1) - (12m^2 + 6m + 0.75)$

$= 4m^2 + 2m + 0.25$

Since it is an integer, the actual minimum value $= 4m^2 + 2m + 1$

132. (b) If the radius of the field is r, then the total area of the field $= \frac{\pi r^2}{2}$

The radius of the semi-circles with centre's P and R $= \frac{r}{2}$

Hence, their total area $= \frac{\pi r^2}{4}$

Let the radius of the circle with centre S be x.

Thus, OS $= (r - x)$, OR $=$ and RS $= \left(\frac{r}{2} + x\right)$.

Applying Pythagoras theorem, we get

Solving this, we get $x = \frac{r}{3}$

Thus the area of the circle with centre S $= \frac{\pi r^2}{9}$

The total area that can be grazed

$= \pi r^2\left(\frac{1}{4} + \frac{1}{9}\right) = \frac{13\pi r^2}{36}$

Thus the fraction of the field that can be grazed

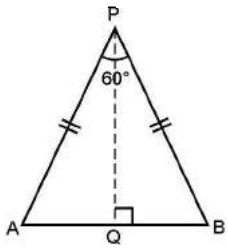
$= \frac{26}{36}\left(\frac{\text{Area that can be grazed}}{\text{Area of the field}}\right)$

The fraction that can not be grazed $= \frac{10}{36} = 28\%$

133. (a) It is very clear that a regular hexagon can be divided into six equilateral triangles. And triangle AOF is half of an equilateral triangle. Hence, the required ratio = 1 : 12

134. (c) If $y = 2$ (it can not be 0 or 1), then x can take 1 value and z can take 2 values. Thus with $y = 2$, a total of $1 \times 2 = 2$ numbers can be formed. With $y = 3$, $2 \times 3 = 6$ numbers can be formed. Similarly checking for all values of y from 2 to 9 and adding up we get the answer as 240.

135. (b)



Given $\angle APB = 60°$ and AB = b.

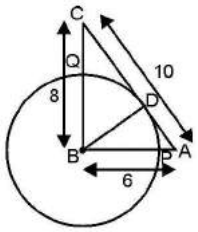
$\therefore$ PQ $= \frac{b}{2} \times \sqrt{3}$

Next, $\frac{b}{2}$, h and PQ from a right angle triangle.

$\therefore \frac{b^2}{4} + h^2 = \frac{3b^2}{4}$

$\therefore 2h^2 = b^2$

136. (d)



Triangle ABC is a right angled triangle.

Thus $\frac{1}{2} \times CB \times AB = \frac{1}{2} \times BD \times AC$

Or $6 \times 8 = BD \times 10$. Thus BD = 4.83.

Therefore, BP = BQ

So, AP = AB – BP = 6 – 4.8 = 1.2 and CQ = BC – BQ = 8 – 4.8 = 3.2

Thus, AP : CQ = 1.2 : 3.2 = 3:8

137. (b) Using the Basic Proportionality Theorem, $\frac{AB}{PQ} = \frac{BQ}{QD}$ and $\frac{PQ}{CD} = \frac{BQ}{BD}$

Multipying the two we get, $\frac{AB}{CD} = \frac{BQ}{QD} = 3:1.$

Thus CD : PQ = BD:BQ = 4:3 = 1:0.75

138. (c) Assume the number of horizontal layers in the pile be n.

So $\sum \frac{n(n+1)}{2} = 8436$

$\Rightarrow \frac{1}{2}\left[\sum n^2 + \sum n\right] = 8436$

$\Rightarrow n(n+1)\left[\frac{2n+4}{12}\right] = 8436$

$\Rightarrow n(n + 1)\,(n + 2) = 36 \times 37 \times 38$

So, $n = 36$

139. (c) The best way to do this is to take some value and verify.

e.g., 2, $\frac{1}{2}$ and 1. Thus, $n = 3$ and the sum of the three numbers = 3.5.

Thus options 1, 2 and 4 get eliminated.

Alternative method

Let the n positive numbers be $a1, a2, a3 \ldots a_n$

We know that AM ≥ GM

Hence, $\frac{1}{n}(a_1+a_2+a_3+\ldots a_n) \geq (a_1 a_2 \ldots an)^{1/n}$

or $(a_1+a_2+a_3+\ldots a_n) \geq n$

140. (c) Using $\log a - \log b = \log\left(\frac{a}{b}\right), \frac{2}{y-5}$

$= \frac{y-5}{y-3.5}$, where, $y = 2^x$

Solving we get $y = 4$ or 8 i.e., $x = 2$ or 3. It cannot be 2 as log of negative number is not defined (see the second expression).

141. (a)

∠BOC = 10° (Opposite equal sides)

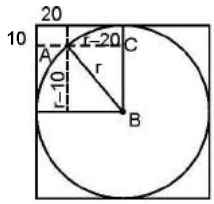
∠OBA = 20° (External angle of ABOC)

∠OAB = 20° (Opposite equal sides)

∠AOD = 30° (External angle of AAOC)

Thus, k = 3

142. (c)



Let the radius be r. Thus by Pythagoras' theorem for ΔABC we have

$(r - 10)^2 + (r - 20)^2$

i.e., $r^2 - 60r + 500 = 0$. Thus, r = 10 or 50.

It would be 10, if the corner of the rectangle had been lying on the inner circumference. But as per the given diagram, the radius of the circle should be 50 cm.

143. (b) u is always negative. Hence, for us to have a minimum vz value of $\frac{vz}{u}$, vz should be positive.

Also for the least u value, the numerator has to be the maximum positive value and the denominator has to be the smallest negative value. In other words, vz has to be 2 and u has to be −0.5.

Hence, the minimum value of $\frac{vz}{u} = \frac{2}{-0.5} = 4.$

For us to get the maximum value, vz has to be the smallest negative value and u has to be the highest negative value. Thus, vz has to be - 2 and u has to be −0.5.

Hence, the maximum value of $\frac{vz}{u} = \frac{-2}{-0.5} = 4$

144. (b) GRRRRR, RGRRRR, RRGRRR, RRRGRR, RRRRGR, RRRRRG

GGRRRR, RGGRRR, RRGGRR, RRRGGR, RRRRGG, GGGRRR, RGGGRR, RRGGGR, RRRGGG, GGGGRR, RGGGGR, RRGGGG

GGGGGR, RGGGGG, GGGGGG

Hence, 21 ways.

145. (d) When we substitute two values of x in the above curves, at $x = -2$, we get

$y = -8 + 4 + 5 = 1$

$y = 4 - 2 + 5 = 7$

Hence, at $x = -2$ the curves do not intersect.

At $x = 2, y_1 = 17$ and $y_2 = 11$

At $x = -1, y_1 = 5$ and 2 and $y_2 = 5$

When $x = 0, y_1 = 5$ and $y_2 = 5$

And at $x = 1$, $y_1 = 7$ and $y_2 = 7$

Therefore, the two curves meet thrice when $x = -1$, 0 and 1.

146. (a) Let us say there are only 3 questions. Thus there are $2^{3-1} = 4$ students who have done 1 or more questions wrongly, $2^{3-2} = 2$ students who have done 2 or more questions wrongly and $2^{3-3} = 1$ student who must have done all 3 wrongly. Thus the total number of wrong answers = 4 + 2 + 1 = $7 = 2^3 - 1 = 2^n - 1$.

In our question, the total number of wrong answers = $4095 = 2^{12} - 1$.

Thus $n = 12$.

147. (c) Here x, y, z are distinct positive real number

So, $\dfrac{x^2(y+z)+y^2(x+2)+z^2(x+y)}{xyz}$

$= \dfrac{x}{y}+\dfrac{x}{z}+\dfrac{y}{x}+\dfrac{y}{z}+\dfrac{z}{x}+\dfrac{z}{y}$ (we know that

$= \left(\dfrac{x}{y}+\dfrac{y}{x}\right)+\left(\dfrac{y}{z}+\dfrac{z}{y}\right)+\left(\dfrac{z}{x}+\dfrac{x}{z}\right)$ that $\dfrac{a}{b}+\dfrac{b}{a} > 2$ if

a and b are distinct numbers

$> 2 + 2 + 2$

> 6

148. (a) The least number of edges will be when one point is connected to each of the other 11 points, giving a total of 11 lines. One can move from any point to any other point via the common point. The maximum edges will be when a line exists between any two points. Two points can be selected from 12 points in $^{12}C_2$ i.e., 66 lines.

149. (b) From 12 to 40, there are 7 prime number, i.e., 13, 17, 19, 23, 29, 31, 37, which is not divisible by $(n - 1)!$

150. (d) $T_n = a + (n - 1)d$

$467 = 3 + (n - 1)\,8$

$n = 59$

Half of $n = 29$ terms

29th term is 227 and 30th term is 235 and when these two terms are added the sum is less than 470.

Hence, the maximum possible values the set S can have, is 30.

RETEST PAPER

Direction for questions 51 to 53: Answer the questions on the basis of the information given below.

The seven basic symbols in a certain numeral system and their respective values are as follows:

I = 1, V = 5, X = 10, L = 50, C = 100, D = 500 and M = 1000

In general, the symbols in the numeral system are (read from left to right), starting with the symbol representing the largest value; the same symbol cannot occur continuously more than three times: the value of the numeral is the sum of the values of the symbols. For example, XXVII = 10 + 10 + 5 + 1 + 1 = 27. An exception to the left-to-right reading occurs when a symbol is' followed immediately by a symbol of greater value; then the smaller value is subtracted from the larger.

For example, XLVI (50 — 10) + 5 + 1 = 46.

Q.51. The value of the numeral MDCCLXXXVII is

(a) 1687 (b) 1787

(c) 1887 (d) 1987

Q.52. The value of the numeral MCMXC IX is

(a) 1999 (b) 1899

(c) 1989 (d) 1889

Q.53. Which of the following represent the numeral for 1995?

I. MCMLXXV II. MCMXCV

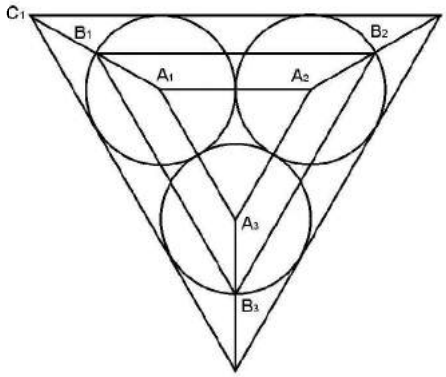
III. MVD IV. MVM

(a) Only I and II (b) Only Ill and IV

(c) Only II and IV (d) Only IV

Direction for questions 54 to 56: Answer the questions on the basis of the information given below.

Consider three circular parks of equal size with centres at A_1, A_2 and A_3 respectively. The parks touch each other at the edge as shown in the figure (not drawn to scale). There are three paths formed by the triangles $A_1A_2A_3$, $B_1B_2B_3$ and $C_1C_2C_3$ as shown. Three sprinters A, B, and C begin running from points A_1, B_1 and C_1 respectively. Each sprinter traverses her respective triangular path clockwise and returns to her starting point.



Q.54. Let the radius of each circular park be r, and the distances to be traversed by the sprinters A, B and C be *a*, *b* and *c* respectively. Which of the following is true?

(a) $b-a=c-b=3\sqrt{3}r$

(b) $b-a=c-b=\sqrt{3}r$

(c) $b=\frac{a+c}{2}=2\left(1+\sqrt{3}\right)r$

(d) $c=2b-a=\left(2+\sqrt{3}\right)r$

Q.55. Sprinter A traverses distance A_1A_2, A_2A_3 and A_3A_1 at average speeds of 20, 30 and 15 respectively. B traverses her entire path at a uniform speed of $\left(10+\sqrt{3}+20\right)$. C traverses distances C_1C_2, C_2C_3 and C_3C_1 at an average speed of $\frac{40}{3}\left(\sqrt{3}+1\right), \frac{40}{3}\left(\sqrt{3}+1\right)$ and 120 respectively. All speeds are in the same unit. Where would B and C be respectively when A finishes her sprint?

(a) B_1, C_1

(b) B_3, C_3

(c) B_1, C_3

(d) B_1, somewhere between C_3 and C_1

Q.56. Sprinters A, B and C traverse their respective paths at uniform speeds of u, v and w respectively. It is known that $u^2:v^2:w^2$ is equal to Area A:Area B: Area C, where Area A, Area B and Area C are the areas of triangles $A_1A_2A_3$, $B_1B_2B_3$, and $C_1C_2C_3$ respectively. Where would A and C be when B reaches point B_3?

(a) A_2, C_3

(b) A_3, C_3

(c) A_3, C_2

(d) Some where between A_2 and A_3, somewhere between C_3 and C_1

Direction for questions 57 to 59: Answer the questions on the basis of the information given below.

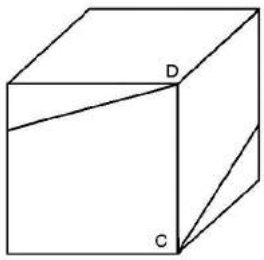
Consider a cylinder of height h cm and radius r = $\frac{2}{\pi}$cm as shown in the figure (not drawn to scale). A string of a certain length, when wound on its cylindrical surface, starting at point A and ending at point B, gives a maximum of n turns (in other words, the string's length is the minimum length required to wind n turns).

Q.57. What is the vertical spacing between the two consecutive turns?

(a) $\frac{h}{n}$cm (b) $\frac{h}{\sqrt{n}}$cm

(c) $\frac{h}{n^2}$cm (d) Cannot be determined

Q.58. The same string, when wound on the exterior four walls of a cube of side n cm, starting at point C and ending at point D, can give exactly one turn (see figure, not drawn to scale). The length of the string is



(a) $\sqrt{2}$ n cm (b) $\sqrt{17}$ n cm

(c) n cm (d) $\sqrt{13}$ n cm

Q.59. In the set-up of the previous two questions, how is h related to n?

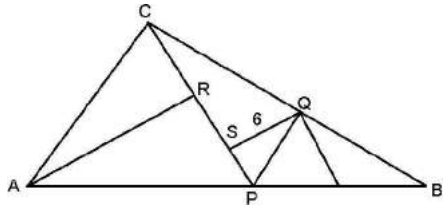
(a) $h=\sqrt{2}$ n (b) $h=\sqrt{17}$ n

(c) h = n (d) $h=\sqrt{13}$ n

Q.60. There are 12 towns grouped into four zones with three towns per zone. It is intended to connect the towns with a telephone lines such that every two towns are connected with three direct lines if they belong to the same zone, and with only one direct line otherwise. How many direct telephone lines are required?

(a) 72 (b) 90 (c) 96 (d) 144

Q.61. In the figure (not drawn to scale) given below, P is a point on AB such that AP: PB = 4:3. PQ is parallel to AC and QD is parallel to CP. In ΔARC, ∠ARC = 90° and in ΔPQS, ∠PSQ = 90°. The length of QS is 6 cm. What is the ratio of AP:PD?



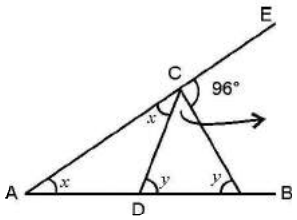
(a) 10:3 (b) 2:1 (c) 7:3 (d) 8:3

Q.62. A car is being driven, in a straight line and at a uniform speed, towards the base of a vertical tower. The top of the tower is observed from the car and, in the process, it takes 10 min for the angle of elevation to change from 45° to 60°. After how much more time will this car reach the base of the tower?

(a) $5\left(\sqrt{3}+1\right)$ (b) $6\left(\sqrt{3}+\sqrt{2}\right)$

(c) $7\left(\sqrt{3}-1\right)$ (d) $8\left(\sqrt{3}-2\right)$

Q.63. In the figure (not drawn to scale) given below. If AD = CD= BC and ∠BCE = 96°, how much is the value of ∠DBC?



(a) 32° (b) 84°

(c) 64° (d) Cannot be determined

Q.64. If both a and b belong to the set {1, 2, 3, 4}, then the number of equations of the of $ax^2 + bx + 1 = 0$ having real roots is

(a) 10 (b) 7

(c) 6 (d 12

Q.65. If $\log_{10}x - \log_{10}\sqrt{x} = 2\log_x 10$, then the possible value of x is given by

(a) 10 (b) $\frac{1}{100}$

(c) $\frac{1}{1000}$ (d) None of the

Q.66. What is the sum of all two-digit numbers that give a remainder of 3 when they are divided by 7?

(a) 666 (b) 676 (c) 683 (d) 777

Q.67. An intelligence agency forms a code of two distinct digits selected from 0, 1, 2,… , 9 such that the first digit of the code is non-zero. The code, handwritten on a slip, can however potentially create confusion when read upside down, for example, the code 91 may appear as 16. How many code are there for which no such confusion can arise?

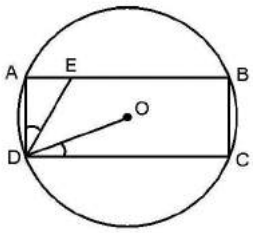
(a) 80 (b) 78 (c) 71 (d) 69

Q.68. Consider two different cloth-cutting processes. In the first one, n circular cloth pieces are cut from square cloth piece of side a in the following steps: the original square of side a is divided into smaller squares, not necessarily of the same size, then a circle of maximum possible area is from each of the smaller squares. In the second process, only one circle of maximum possible area is cut from the square of side a and the process ends there. The cloth pieces remaining after cutting the circles are scrapped in both the processes: The ratio of the total area of scrap cloth generated in the former to that in the latter, is

(a) 1:1 (b) $\sqrt{2}:1$

(c) $\frac{n(4-\pi)}{4n-\pi}$ (d) $\frac{4n-\pi}{n(4-\pi)}$

Q.69. In the figure below (not drawn to scale), rectangle ABCD is inscribed in the circle with centre at O.The length of side AB is greater than side BC. The ratio of the area of the circle to the area of the rectangle ABCD is $\pi:\sqrt{3}$. The line segment DE intersects AB at E such that ∠ODC = ∠ADE. Find ratio AE : AD.



(a) $1:\sqrt{3}$ (b) $1:\sqrt{2}$

(c) $1:2\sqrt{3}$ (d) 1:2

Q.70. If $\frac{1}{3}\log, M + 3\log_3 N = 1 + \log_{0.008} 5$, then

(a) $M^9 = \frac{9}{N}$ (b) $N^9 = \frac{9}{M}$

(c) $M^3 = \frac{3}{N}$ (d) $N^9 = \frac{3}{M}$

Q.71. Using only 2, 5, 10, 25, and 50 paisa coins, what will be the minimum number of coins required to pay exactly 78 paise, 69 paise and Rs 1.01 to three different persons?

(a) 19 (b) 20 (c) 17 (d) 18

Q.72. The length of the circumference of a circle equals the perimeter of a triangle of equal sides, and also the perimeter of a square. The areas covered by the circle, triangle, and square are c, t and s, respectively. Then,

(a) s > t > c (b) c > t > s

(c) c > s > t (d) s > c > t

Q.73. What is the remainder when 4^{96} is divided by 6?

(a) 0 (b) 2 (c) 3 (d) 4

Q.74. If x and y are integers, then the equation $5x + 1\ 9y = 64$ has

(a) No solution for $x < 300$ and $y < 0$

(b) No solution for $x > 250$ and $y > -100$

(c) A solution for $250 < x < 300$

(d) A solution for $-59 < y < -56$

Q.75. What is the sum of 'n' terms in the series

$$\log m + \log\left(\frac{m^2}{n}\right) + \log\left(\frac{m^3}{n^2}\right) + \log\left(\frac{m^4}{n^3}\right) + \ldots ?$$

(a) $\log\left[\frac{n^{(n-1)}}{m^{(n+1)}}\right]^{\frac{n}{2}}$ (b) $\log\left[\frac{m^m}{n^n}\right]^{\frac{n}{2}}$

(c) $\log\left[\frac{m^{(1-n)}}{n^{(1-m)}}\right]^{\frac{n}{2}}$ (d) $\log\left[\frac{m^{(n+1)}}{n^{(n-1)}}\right]^{\frac{n}{2}}$

Q.76. Let S_1 be a square of side a. Another square S_2 is formed by joining the mid-points of the sides of S_1. The same process is applied to S_2 to form yet another square S_3 and so on. If $A_1, A_2 A_3 \ldots$ be the areas and $P_1, P_2, P_3, \ldots$ be the perimeters of S_1, S_2, S_3 respectively, then the ratio $\frac{P_1 + P_2 + P_3 + \ldots}{A_1 + A_2 + A_3 + \ldots}$ equals

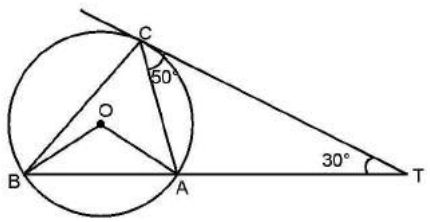
(a) $\frac{2\left(1+2\sqrt{2}\right)}{a}$ (b) $\frac{2\left(2-2\sqrt{2}\right)}{a}$

(c) $\frac{2\left(2+2\sqrt{2}\right)}{a}$ (d) $\frac{2\left(1+2\sqrt{2}\right)}{a}$

Q.77. If three positive real numbers x, y and z satisfy $y - x = z - y$ and $xyz = 4$, then what is the minimum possible value of y?

(a) $21^{1/3}$ (b) $2^{2/3}$ (c) $2^{1/4}$ (d) $2^{3/4}$

Q.78. In the figure given below (not drawn to scale), A, B and C are three points on a circle with centre O. The chord BA is extended to a point T such that CT becomes a tangent to the circle at point C. If $\angle ATC = 30°$ and $\angle ACT = 50°$, then the angle $\angle BOA$ is



(a) 100°
(b) 150°
(c) 80°
(d) Not possible to determine

Q.79. The infinite sum $1 + \frac{4}{7} + \frac{9}{7^2} + \frac{16}{7^3} + \frac{25}{7^4} + \ldots..$ equals

(a) $\frac{27}{14}$ (b) $\frac{21}{13}$ (c) $\frac{49}{27}$ (d) $\frac{256}{147}$

Q.80. Consider the sets $T_n = \{n, n+1, n+2, n+3, n+4\}$, where $n = 1, 2, 3 \ldots 96$. How many of these sets contain 6 or any integral multiple thereof (i.e. any one of the numbers 6, 12, 18, ...)?

(a) 80 (b) 81 (c) 82 (d) 83

Q.81. Let ABCDEF be a regular hexagon. What is the ratio of the area of the DACE to that of the hexagon ABCDEF?

(a) $\frac{1}{3}$ (b) $\frac{1}{2}$ (c) $\frac{2}{3}$ (d) $\frac{5}{6}$

Q.82. The number of roots common between the two equations $x^3 + 3x^2 + 4x + 5 = 0$ and $x^3 + 2x^2 + 7x + 3 = 0$ is

(a) 0 (b) 1 (c) 2 (d) 3

Q.83. A real number x satisfying $1 - \frac{1}{n} < x \le 3 + \frac{1}{n}$, for every positive integer n, is best described by

(a) $1 < x < 4$ (b) $1 < x \le 3$
(c) $0 < x \le 4$ (d) $1 \le x \le 3$

Q.84. If n is such that $36 \le n \le 72$, then $x = \frac{n^2 + 2\sqrt{n}(n+4) + 16}{n + 4\sqrt{n} + 4}$ satisfies

(a) $20 < x < 54$ (b) $23 < x < 58$
(c) $25 < x < 64$ (d) $28 < x < 60$

Q.85. If $13x + 1 < 2z$ and $z + 3 = 5y^2$, then,

(a) x is necessarily less than y
(b) x is necessarily greater than y
(c) x is necessarily equal to y
(d) None of the above is necessarily true

Q.86. Let n (>1) be a composite integer such that $\sqrt{n}$ is not an integer. Consider the following statements:

(A) n has a perfect integer-valued divisor which is greater than 1 and less than $\sqrt{n}$

(B) n has a perfect integer-valued divisor which is greater than $\sqrt{n}$ but less than n

(a) Both A and B are false
(b) A is true but B is false
(c) A is false but B is true
(d) Both A and B are true

Q.87. If $|b| \ge 1$ and $x = -|a|b$, then which one of the following is necessarily true?

(a) $a - xb < 0$ (b) $a - xb \ge 0$
(c) $a - xb > 0$ (d) $a - xb \le 0$

Q.88. A piece of paper is in the shape of a right-angled triangle and is cut along a line that is parallel to the hypotonuse, leaving a smaller triangle. There was 35% reduction in the length of the hypotonuse of the triangle. If the area of the original triangle was 34 square inches before the cut, what is the area (in square inches) of the smaller triangle?

(a) 16.665 (b) 16.565
(c) 15.465 (d) 14.365

Q.89. Two straight roads R_1 and R_2 diverge from a point A at an angle of 120°. Ram starts walking from point A along R_1 at a uniform speed of 3 km/h. Shyam starts walking at the same time from A along R_2 at a uniform speed of 2 km/h. They continue walking for 4 h along their respective roads and reach points B and C on R_1 and R_2 respectively. There is a straight-line path connecting B and C. Then Ram returns to point A after walking along the line segments BC and CA. Shyam also returns to A after walking along line segments BC and CA. Their speeds remain unchanged. The time interval (in h) between Ram's and Shyam's return to the point A is

(a) $\frac{10\sqrt{19}+26}{3}$ (b) $\frac{2\sqrt{19}+10}{3}$
(c) $\frac{\sqrt{19}+26}{3}$ (d) $\frac{\sqrt{19}+10}{3}$

Q.90. A square tin sheet of side 12 inches is converted into a box with open top in the following steps. The sheet is placed horizontally. Then, equal-sized squares, each of side x inches, are cut from the four corners of the sheet. Finally, the four resulting sides are bent vertically upwards in the shape of a box. If x is an integer, then what value of x maximizes the volume of the box?
(a) 3 (b) 4
(c) 1 (d) 2

Q.91. If a, $a+2$ and $a+4$ are prime numbers, then the number of possible solutions for a is
(a) One (b) Two
(c) Third (d) More than three

Q.92. Let a, b, c, d and e be integers such that $a = 6b = 12c$, and $2b = 9d = 12e$. Then which of the following pairs contains a number that is not an integer?

(a) $\left(\frac{a}{27}, \frac{b}{e}\right)$ (b) $\left(\frac{a}{36}, \frac{c}{e}\right)$
(c) $\left(\frac{a}{12}, \frac{bd}{18}\right)$ (d) $\left(\frac{a}{6}, \frac{c}{d}\right)$

Q.93. In a coastal village, every year floods destroy exactly half of the huts. After the flood water recedes, twice the number of huts destroyed are rebuilt. The floods occurred consecutively in the last three years—2001, 2002 and 2003. If the floods are expected again in 2004, the number of huts expected to be destroyed is
(a) Less than the number of huts existing at the beginning of 2001
(b) Less than the total number of huts destroyed by the floods in 2001 and 2003
(c) Less than the total number of huts destroyed by the floods in 2002 and 2003
(d) More than the total number of huts built in 2001 and 2002

Direction for questions 94 to 96: Answer the questions on the basis of the tables given below.

Two binary operations ⊕ and * are defined over the set {a, e, f, g, h} as per the following tables:

⊕	a	e	f	g	h
a	a	e	f	g	h
e	e	f	g	h	a
f	f	g	h	a	e
g	g	h	a	e	f
h	h	a	e	f	g

⊕	a	e	f	g	h
a	a	a	a	a	a
e	a	e	f	g	h
f	a	f	h	e	g
g	a	g	e	h	f
h	a	h	g	f	e

Thus, according to the first table f ⊕ g = a, while according to the second table g * h = f, and so on. Also, let $f^2 = f * f$, $g^3 = g*g*g$, and so on.

Q.94. What is the smallest positive integer n such that $g^n = e$?
(a) 4 (b) 5 (c) 2 (d) 3

Q.95. Upon simplification, f ⊕ [f * {f ⊕ (f * f)}] equals
(a) e (b) f (c) g (d) h

Q.96. Upon simplification, $\{a^{10} * (f^{10} \oplus g^g)\} \oplus e^8$ equals
(a) e (b) f (c) g (d) h

Direction for questions 97 and 98: Answer the questions on the basis of the information given below.

A string of three English letters is formed as per the following rules:

I. The first letter is any vowel.
II. The second letter is m, n or p.
III. If the second letter is m, then the third letter is any vowel which is different from the first letter.
IV. If the second letter is n, then the third letter is e or u.
V. If the second letter is p, then the third letter is the same as the first letter.

Q.97. How many strings of letters can possibly be formed using the above rules?
(a) 40 (b) 45 (c) 30 (d) 35

Q.98. How many strings of letters can possibly be formed using the above rules such that the third letter of the string is e?
(a) 8 (b) 9 (c) 10 (d) 11

Q.99. Let x and y be positive integers such that x is a prime and y is a composite. Then,
(a) $y - x$ cannot be an even integer
(b) xy cannot be an even integer
(c) $\frac{(x+y)}{x}$ cannot be an even integer
(d) None of these

Q.100. A survey on a sample of 25 new cars being sold at a local auto dealer was conducted to see which of the three popular options —— air conditioning, radio and power windows were already installed.

Following were the observation of the survey:

I. 15 had air conditioning
II. 2 had air conditioning and power windows but no radios
III. 12 had radio
IV. 6 had air conditioning and radio but no power windows
V. 11 had power windows
VI. 4 had radio and power windows
VII. 3 had all three options

What is the number of cars that had none of the options?
(a) 4 (b) 3 (c) 1 (d) 2

ANSWERS

■■■ CAT 2003

Q. No.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
Answer	(b)	(a)	(c)	(a)	(c)	(b)	(a)	(b)	(c)	(b)	(c)	(a)	(c)	(b)	(b)	(b)	(c)	(a)	(a)	(b)
Q. No.	71	72	73	74	75	76	77	78	80	81	82	83	84	85	86	87	88	90	91	92
Answer	(a)	(c)	(d)	(c)	(d)	(c)	(b)	(a)	(a)	(b)	(a)	(c)	(d)	(d)	(d)	(b)	(d)	(d)	(a)	(d)
Q. No.	93	94	95	96	97	98	99	100												
Answer	(c)	(a)	(d)	(a)	(d)	(c)	(d)	(d)												

HINTS AND SOLUTIONS

51. (b) MDCCLXXXVII = 1000 + 500 + 100 + 100 + 50 + 10 + 10 + 10 + 5 + 1 + 1 = 1787

52. (a) MCMXCIX = 1000 + (1000 – 100) + (100 – 10) + (10 – 1) = 1000 + 900 + 90 + 9 = 1999

53. (c) (I) MCMLXXV = 1000 + (1000 – 100) + 50 + 10 + 10 + 5 = 1975
(II) MCMXCV = 1000 + (1000 – 100) + (100 – 10) + 5 = 1995
(III) MVD = 1000 + (500 – 5) = 1495
(IV) MVM = 1000 + (1000 – 5) = 1995
Therefore, the answer is (II) and (IV), i.e. option (c).

Questions 54 to 56

$A_1A_2 = 2r$, $B_1B_2 = 2r + r\sqrt{3}$, $C_1C_2 = 2r + 2r\sqrt{3}$ r.
Hence, $a = 3 \times 2r$
$b = 3 \times (2r + r\sqrt{3})$
$c = 3 \times (2r + 2r\sqrt{3})$

54. (a) Difference between (1) and (2) is $3\sqrt{3}$ r and that between (2) and (3) is $3\sqrt{3}$ r. Hence, (1) is the correct choice.

55. (c) Time taken by A $= \frac{2r}{20} + \frac{2r}{30} + \frac{2r}{15} = \left(\frac{2r \times 9}{60}\right) = \frac{3}{10}r$

Therefore, B and C will also travel for time $\frac{3}{10}$r

Now the speed of B = $\left(10\sqrt{3} + 20\right)$

Therefore the distance covered = $\left(10\sqrt{3} + 20\right) \times \frac{3}{10}r = \left(\sqrt{3} + 2\right) \times 10 \times \frac{3}{10}$r

$= \left(2r + \sqrt{3}r\right) \times 3 = B_1B_2 + B_2B_3 + B_3B_1$

$\therefore$ B will be at B_1.

Now the time taken by C for each distance is

$$\frac{C_1C_2}{\frac{40}{3}\left(\sqrt{3}+1\right)}, \frac{C_2C_3}{\frac{40}{3}\left(\sqrt{3}+1\right)}, \frac{C_3C_1}{120}$$

$$\frac{\left(2+2\sqrt{3}\right)r}{\left(\sqrt{3}+1\right)}, \frac{3}{40} \times \frac{\left(2+2\sqrt{3}\right)r}{\left(\sqrt{3}+1\right)}, \frac{\left(2+2\sqrt{3}\right)}{120}$$

i.e., $\frac{3}{40} \times 2r, \frac{3}{40} \times 2r, \frac{\left(1+\sqrt{3}\right)}{60}r$

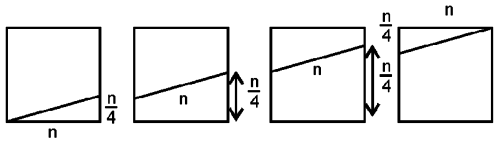
i.e., $\frac{3}{20}r, \frac{3}{20}r, \frac{\left(1+\sqrt{3}\right)}{60}r$

We can observe that time taken for C_1C_2 and C_2C_3 combined is $\frac{3}{20}r + \frac{3}{20}r = \frac{3}{10}r$, which is same as the time taken by A. Therefore, C will be at C_3.

56. (b) In similar triangles, the ratio of the area = ratio of the squares of the corresponding sides.
Hence, A and C reach A_3 and C_3 respectively.

57. (a) The whole height h will be divided into n equal parts. Therefore, the spacing between two consecutive turns is $\frac{h}{n}$.

58. (b) The four faces through which a string is passing can be shown as



Therefore, the length of the string in each face

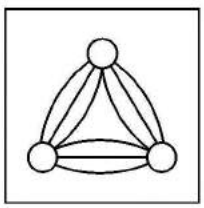
$$= \sqrt{n^2 + \left(\frac{n}{4}\right)^2}$$

$$= \sqrt{n^2 + \frac{n^2}{16}} = \frac{\sqrt{17}n}{4}$$

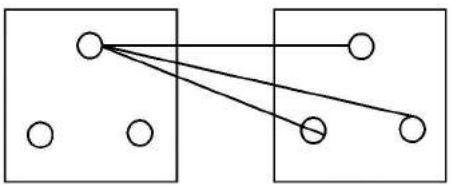
Therefore, the length of the string through four faces = $\frac{\sqrt{17}n}{4} \times 4 = \sqrt{17}n$

59. (c) As h/n = Number of turns = 1 (as given). Hence h = n.

60. (b) Consider the first zone. The number of telephone lines can be shown a follows.



Therefore, the total number of lines required for internal connections in each zone = 9 × 4 = 36 lines.
Now, consider the connection between any two zones.



Each town in the first zone can be connected to the three towns in the second zone.
Therefore, the lines required = 3 × 3 = 9
Therefore, the total number of lines required for connecting the towns of different zones = $^4C_2 \times 9 = 6 \times 9 = 54$.
Therefore, the total number of lines in all = 54 + 36 = 90

61. (c) PQ || AC

$\therefore \frac{CQ}{QB} = \frac{AB}{PB} = \frac{4}{3}$

QD||PC

$$\therefore \frac{PD}{DB} = \frac{CQ}{QB} = \frac{4}{3}$$

As $\frac{PD}{DB} = \frac{4}{3}$

$\therefore$ PD $= \frac{4}{7}PB$

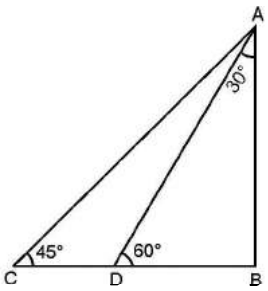
$$\therefore \frac{AP}{PD} = \frac{AR}{\frac{4}{7}PB}$$

$$= \frac{7}{4} \times \frac{AP}{PB}$$

$$= \frac{7}{4} \times \frac{4}{3}$$

$= 7:3$

62. (a)



Let AB = 1

Therefore, BC = 1

$\therefore \tan 60 = \frac{AB}{BD} \therefore \sqrt{3} = \frac{1}{BD}$

$$\therefore \text{BD} = \frac{1}{\sqrt{3}}$$

$\therefore$ CD = BC – BD

$$= 1 - \frac{1}{\sqrt{3}}$$

As time for travelling CD, i.e., $1 - \frac{1}{\sqrt{3}}$ is 10 min.

$\therefore$ Time required for travelling BD

$$= \frac{\frac{1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} \times 10 \times 10$$

$$= \frac{1}{\sqrt{3} - 1} \times 10$$

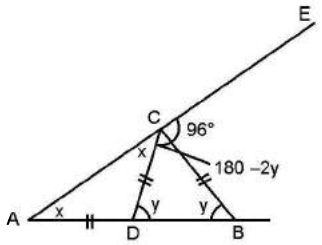
$$= \frac{10}{\sqrt{3} - 1}$$

$$= \frac{10}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{10\left(\sqrt{3} + 1\right)}{2}$$

$$= 5\left(\sqrt{3} + 1\right)$$

63. (c)



Using exterior angle theorem

$\angle A + \angle B = 96$

i.e., $x + y = 96$ … (i)

Also $x + (108 - 2y) + 96 = 180°$

$\therefore x - 2y + 96 = 0$

$\therefore x - 2y = -96$

Solving (i) and (ii)

$y = 64°$ and $x = 32°$

$\therefore \angle DBC = y = 64°$

64. (b) $ax^2 + bx + 1 = 0$

For real roots

$b^2 - 4ac \geq 0$

$\therefore b^2 - 4a(1) \geq 0$

$\therefore b^2 \geq 4a$

For $a = 1$, $4a = 4$, $\therefore b = 2, 3, 4$

$a = 2$, $4a = 8$, $\therefore b = 3, 4$

$a = 3$, $4a = 12$, $\therefore b = 4$

$a = 4$, $4a = 16$, $\therefore b = 4$

65. (b) $\log_{10} x - \log_{10} \sqrt{x} = 2\log_x 10$

$$\log_{10}\left[\frac{x}{\sqrt{n}}\right] = \log_x 100$$

$$\therefore \log_{10} \sqrt{x} = \frac{\log_{10} 100}{\log_{10} x}$$

$$\therefore \frac{1}{2}\log_{10} x = \frac{2}{\log_{10} x}$$

$\therefore (\log_{10} x)^2 = 4$

$\therefore \log_{10} x = \pm 2$

$\therefore \log_{10} x = 2$ or $\log_{10} x = -2$

$\therefore 10^2 = x$ or $10^{-2} = x$

$$\therefore x = 100 \text{ or } x = \frac{1}{100}$$

66. (b) Such numbers are 10, 17, ... 94.
These numbers are in an AP. There are 13 numbers.

$\therefore$ Sum $= \frac{10+94}{2} \times 13$

$= 52 \times 13 = 676$

67. (c) The total codes which can be formed = $9 \times 9 = 81$. (Distinct digit codes)
The digits which can be confusing are 1, 6, 8 and 9 from these digit we can form the codes $= 4 \times 3 = 12$
Out of these 12 codes two numbers 69 and 96 will not create confusion
Therefore, (12 – 2) = 10 codes will create a confusion.
Therefore the total codes without confusion
= 81 – 10 = 71.

68. (a) Consider a square of side x.
Therefore, its area = x^2

Therefore, the area of the largest circle = $\pi \left(\frac{x}{2}\right)^2$

Which can be cut from the square = $\frac{\pi x^2}{4}$

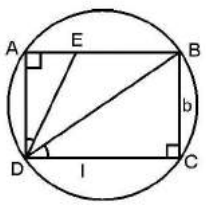
Therefore, the area scrapped

$= x^2 - \frac{\pi}{4}x^2 = x^2\left(1-\frac{\pi}{4}\right)$

$\therefore \frac{\text{Area Scrapped}}{\text{Area Scrapped}} = \frac{x^2\left(1-\frac{\pi}{4}\right)}{x^2} = 1-\frac{\pi}{4} = \text{Constant}$

As this ratio is constant whether we cut a circle from the smaller square or the larger square, the scrapped area will be a fixed percentage of the square. Therefore, in this problem, as the two squares are of the same size, the ratio will be 1 : 1

69. (a)



BD = 2r

$\frac{\text{Area of circle}}{\text{Area rectangle}} = \frac{\pi r^r}{lb} = \frac{\pi}{\sqrt{3}}$

$\frac{\frac{d^2}{4}}{lb} = \frac{1}{\sqrt{3}}$

$\therefore \frac{d^2}{4lb} = \frac{1}{\sqrt{3}}$

$\therefore \frac{l^2+b^2}{lb} = \frac{1}{\sqrt{3}}$

$\therefore \frac{l^2+b^2}{lb} = \frac{4}{\sqrt{3}}$

$\therefore \frac{l}{b}+\frac{b}{l} = \frac{4}{\sqrt{3}}$ ... (i)

Now $\Delta AEB \sim \Delta CBD$

$\therefore \frac{AE}{CB} = \frac{AD}{DC}$

$\therefore \frac{AE}{AD} = \frac{BC}{DC}$

$\therefore$ we have to find $\frac{AE}{AD}$, i.e., $\frac{b}{l}$

Let $\frac{b}{l} = x$

Therefore, from (i) we get

$\frac{1}{x} + x = \frac{4}{\sqrt{3}}$

$\frac{1+x^2}{x} = \frac{4}{\sqrt{3}}$

$\sqrt{3} + \sqrt{3}x^2 = 4$

$\therefore \sqrt{3}x^2 - 4x + \sqrt{3} = 0$

$\therefore x = \frac{-(-4) \pm \sqrt{16 - 4\left(\sqrt{3}\right)\sqrt{3}}}{2\sqrt{3}}$

$= \frac{4 \pm \sqrt{16-12}}{2\sqrt{3}}$

$= \frac{4 \pm 2}{2\sqrt{3}}$

$= \frac{6}{2\sqrt{3}}$

Or $\frac{2}{2\sqrt{3}}$

$\frac{\sqrt{3}}{1}$ OR $\frac{1}{\sqrt{3}}$

From options, the answer is $\frac{1}{\sqrt{3}}$ i.e., $1 : \sqrt{3}$.

70. (b) $\frac{1}{3}\log_3 M + 3\log_3 N = 1 + \log_{0.008} 5$

$\frac{1}{3}\log_3\left(M^{1/3}N\right) = 1 + \frac{(\log 10 - \log 2)}{\log 8 - \log 1000}$

$$\frac{1}{3}\log_3\left(M^{1/3}N\right)=1-\frac{(1-\log 2)}{3(1-\log 2)}$$

$$\Rightarrow \frac{1}{3}\log_3\left(M^{1/3}N\right)=1-\frac{1}{3}=\frac{2}{3}$$

Or $M^{1/3}N = 3^{2/3}$

Or $MN^9 = 3^2$

Or $N^9 = 9/M$

71. (a) Let us make the given sum by using minimum number of coins as

Value of coin	*Number of coins*	*Number of coins*	*Number of coins*	*Total number of coins*
50	1	1	1	3
25	–	–	1	1
10	1	2	2	5
5	1	–	–	1
2	2	4	3	9
Total Amount	69	78	101	19

72. (c) It is the standard property among circle, square and triangle. For a given parameter, area of a circle is the highest and area of the triangle is the least whereas area of the square is in-between i.e., c > s > t.

73. (d) Remainder when $\frac{4^{96}}{6}$

Let us come down to basic property of dividing the power of 4 by 6 i.e.,

$\frac{4^1}{6}=4$

$\frac{4^2}{6}=4$

$\frac{4^3}{6}=4$

$\frac{4^4}{6}=4$

Hence, any power of 4 when divided by 6 leaves a remainder of 4.

74. (c) $5x + 19y = 64$

We see that if $y = 1$, we get an integer solution for $x = 9$, now if y changes (increases or decreases by $5x$, will the change decrease or increase) by 19. Looking at the options, if $x = 256$, we get $y = 64$. Using these values we see options (a), (b) and (d) are eliminated and also that there exists a solution for $250 < x \le 300$.

75. (d) Sum of $\log m + \log\left(\frac{m^2}{n}\right)+\log\left(\frac{m^3}{n^2}\right)+\ldots n$ terms, such a problem must be solved by taking the value of number of the terms. Let us say 2 and check the given option. If we look at the sum of the 2 terms of the given series it comes out to be $\log m + \log \frac{m^2}{n} \Rightarrow \log \frac{m\times m^2}{n}=\log\left(\frac{m^3}{n}\right)$. Now look at the option and put number of terms as 2, only option (d) validates the above mentioned answer.

As $\log\left[\frac{m^{(n+1)}}{n^{(n-1)}}\right]^{\frac{n}{2}} \Rightarrow \log\left[\frac{m^3}{n}\right]^1 \Rightarrow \log\left(\frac{m^3}{n}\right)$

76. (c) $$\frac{P+\frac{P}{\sqrt{2}}+\ldots\infty}{A+\frac{A}{2}+\ldots\infty}=\frac{\frac{P}{1-\frac{1}{\sqrt{2}}}}{2A}=\frac{p\sqrt{2}}{\left(\sqrt{2}-1\right)}\times\frac{1}{2A}$$

$$=\frac{\sqrt{2}P\left(\sqrt{2}+1\right)}{2A}=\frac{\sqrt{2}\times 4a\left(\sqrt{2}+1\right)}{2\times a^2}$$

$$=\frac{\sqrt{2}\times 2\left(\sqrt{2}+1\right)}{a}=\frac{2\left(2+\sqrt{2}\right)}{a}$$

77. (b) $xyz = 4$

$y - x = z - y$

$2y = x + z$

y is the AM of x, y, z.

Also $\sqrt[3]{xyz}=4^{\frac{2}{3}} \Rightarrow \sqrt[3]{xyz}=2^{\frac{1}{3}}$

AM ≥ GM

$Y \ge 2^{\frac{2}{3}}$

Therefore, the minimum value of y is $2^{\frac{2}{3}}$

78. (a) $\angle BAC = \angle ACT + \angle ATC = 50 + 30 = 80°$

And $\angle ACT = \angle ABC$ (Angle in alternate segment)

So $\angle ABC = 50°$

$\angle BCA = 1870 - (\angle ABC + \angle BAC)$

$= 180 - (50 + 80) = 50°$

Since $\angle BOA = 2\angle BCA = 2 \times 50 = 100°$

Alternative Method:

Join OC

$\angle OCT = 90°$ (TC is tangent to OC)

$\angle OCA = 90° - 50° = 40°$

$\angle OAC = 40°$ (OA = OC being the radius)

$\angle BAC = 50° + 30° = 80°$

$\angle OAB = 80° - 40° = 40° = \angle OBA$ (OA = OB being the radius)

$\angle BOA = 180° - (\angle OBA + \angle OAB) = 100°$

79. Let $S = 1+\frac{4}{7}+\frac{9}{7^2}+\frac{16}{7^3}+\frac{25}{7^4}$ ... (i)

$\therefore \frac{1}{7}S = \frac{1}{7}+\frac{4}{7^2}+\frac{9}{7^3}+\frac{16}{7^4}+\frac{25}{7^5}$ ...(ii)

(i) – (ii) gives

$S\left(1-\frac{1}{7}\right) = 1+\frac{3}{7}+\frac{5}{7^2}+\frac{7}{7^3}+\frac{9}{7^4}$ ...(iii)

$\frac{1}{7}\times S\left(1-\frac{1}{7}\right) = \frac{1}{7}+\frac{3}{7^2}+\frac{5}{7^3}+\frac{7}{7^4}$ ...(iv)

(iii) – (iv) gives

$S\left(1-\frac{1}{7}\right)-\frac{1}{7}S\left(1-\frac{1}{7}\right) = 1+\frac{2}{7}+\frac{2}{7^2}+\frac{2}{7^3}+\frac{2}{7^4}$

$\therefore S\left(1-\frac{1}{7}\right)\left(1-\frac{1}{7}\right) = 1+\frac{2}{7}\left[1+\frac{1}{7}+\frac{1}{7^2}+\ldots\infty\right]$

$\therefore S\left(1-\frac{1}{7}\right)^2 = 1+\frac{2}{7}\times\frac{1}{1-\frac{1}{7}}$

$\therefore S\left(\frac{6}{7}\right)^2 = 1+\frac{2}{7}\times\frac{7}{6}$

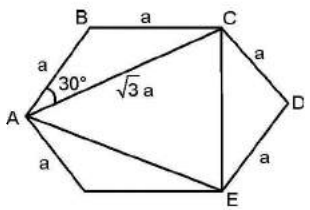
$\therefore S\times\frac{36}{49} = 1+\frac{1}{3}$

$\therefore S = \frac{49}{36}\times\frac{4}{3}$

$S = \frac{49}{27}$

80. (a) By observing, we see 6 will appear in 5 sets T_2, T_3, T_4, T_5 and T_6. Similarly, 12 will also appear in 5 sets and these sets will be distinct from the sets in which 6 appears, i.e. T_8, T_9, T_{10}, T_{11} and T_{12}. Thus, each multiple of 6 will be appear in 5 distinct sets. Till T_{96} there will be 16 multiples which will appear in $16 \times 5 = 80$ sets.

81. (b)



ΔACE is an equilateral triangle with side $\sqrt{3}a$

Area of hexagon = $\frac{\sqrt{3}}{4}a^2\times 6$

Area as $\Delta ACE = \frac{\sqrt{3}}{4}a^2\times\left(\sqrt{3}a\right)^2$

Therefore, the ratio = $\frac{1}{2}$

82. (a) Let a be the common root.

$\alpha^3 + 3\alpha^2 + 4\alpha + 5 = 0$

$\alpha^3 + 2\alpha^2 + 7\alpha + 3 = 0$

$\alpha^2 - 3\alpha + 2 = 0$

$\alpha = 2, \alpha = 1$

83. (c) $1-\frac{1}{n} < x \le 3+\frac{1}{n}$

Put $n = 1$

$\therefore 0 < x \le 4$

84. (d) $36 \le n \le 72$

$x = \frac{n^2+2\sqrt{n}(n+4)+16}{n+4\sqrt{n}+4}$

Put $x = 36$.

$\therefore x = \frac{(36)^2+2\times 6\times 40+16}{36+24+4}$

85. (d) $13x + 1 < 2z$ and $z + 3 = 5y^2$

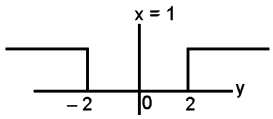
$13x + 1 < 2(5y^2 - 3)$

$13x + 1 < 10y^2 - 6$

$13x + 7 < 10y^2$ put $x = 1$

$20 < 10y^2$ $y^2 > 2$

$y^2 > \sqrt{2}$ $(y^2 - 2) > 0$



86. (d) Let $n = 6$

Therefore, $\sqrt{n} = \sqrt{6} = 2.4$

Therefore, divisors of 6 are 1, 2, 3.

If we take 2 as divisor, then $\sqrt{n} > 2 > 1$

Statement A is true

If we take 3 as divisor, then $6 > 3 > 2.4$, i.e., $n > 3 > \sqrt{n}$

Therefore, statement B is true.

87. (b) $x = -|a|b$

Now $a - xb = a - (-|a|b)b$

$= a + |a|b^2$

$\therefore a - xb = a + ab^2 \ldots a \ge 0$ or $a - xb$

$= a - ab^2 \ldots a < 0$

$= a(1+b^2) = a(1-b^2)$

Consider the first case:

As $a \geq 0$ and $|b| \geq 1$, therefore $(1+b^2)$ is positive.

$\therefore a(1+b^2) \geq 0$

$\therefore a - xb \geq 0$

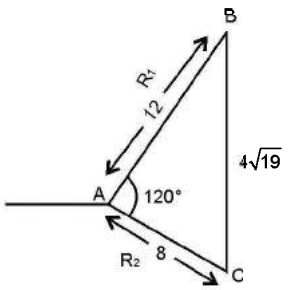
Consider the second case.

As $a < 0$ and $\}b\} \geq 1$, therefore $(1-b^2) \leq 0$

$\therefore a(1-b^2) \geq 0$ (since –ve ax –ve = +ve and $1-b^2$ can be zero also), i.e., $a - xb \geq 0$.

Therefore, in both cases $a - xb \geq 0$.

88. (d) The required answer is $34 \times 0.65 \times 0.65 = 14.365$. Because we get two similar triangles and the area is proportional to the square of its side.



$BC^2 = (12)^2 + 8^2 - 2 \times 12 \times \cos 120°$

$\therefore BC = 4\sqrt{19}$

$$t_1 = \frac{4\sqrt{19}+8}{3}$$

$$t_2 = \frac{4\sqrt{19}+12}{2}$$

(Where t_1 and t_2 are the time taken by Ram and Shyam to reach the starting point.)

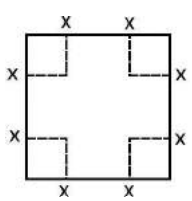
$$t_2 - t_1 = \frac{4\sqrt{19}+12}{2} - \frac{4\sqrt{19}+8}{3}$$

$$t_2 - t_1 = \frac{12\sqrt{19}+36-8\sqrt{19}+10}{6}$$

$$= \frac{4\sqrt{19}+20}{6} = \frac{2\sqrt{19}+10}{3}$$

Therefore, choice (b).

90. (d)



Volume = $l \times b \times h$

$V = (12-2x)(12-2x) \times x$

$V' = (12-2x)(12-2x)\,4x$

Where $V' = 4V$

Now sum = $12 - 2x + 12 - 2x + 4x = 24$ (constant)

As sum is constant for the maximum product $12 - 2x = 12 - 2x = 4x$

Therefore, $x = 2$.

91. (a) As any prime number greater than 3 can be expressed in the form $6n \pm 1$, the minimum difference between the three consecutive prime numbers will be 2 and 4. The values that satisfy the given conditions are only 3, 5 and 7 i.e., only one set is possible.

92. (d) $a = 6b = 12c$ and $2b = 9d = 12e$.

Dividing the first equations by 12 and second by 36.

We get, $\frac{a}{12} = \frac{b}{2} = \frac{c}{1}$ and $\frac{b}{18} = \frac{d}{4} = \frac{e}{3}$

$$\therefore \frac{a}{108} = \frac{b}{18} = \frac{c}{9} = \frac{d}{4} = \frac{e}{3}$$

$\therefore a:b:c:d:e = 108:18:19:4:3$

$\therefore \frac{c}{d} = \frac{9}{4}$ is not an integer

93. (c) Let the number of huts before the floods in 2001 $be = 32$.

$\therefore$ Huts destroyed $= \frac{32}{2} = 16$ and rebuilt $= 32$.

$\therefore$ Total huts after floods in 2001 = 16 + 32 = 48

Similarly, in 2002, the destroyed huts = 24

Rebuilt = 48

Total = 24 + 48 = 72

Similarly, in 2003, the destroyed huts = 36

Rebuilt = 72

Total = 36 + 72 = 108.

$\therefore$ Checking with choices the answer is (c).

94. (a) $g^2 = g * g = h$

$g^3 = g^2 * g = h * g = f$

$g^4 = g^3 * g = f * g = e$

$\therefore n = 4$.

95. (d) $f \oplus [f * \{f \oplus (f * F)\}]$

$= f \oplus [f * \{f \oplus h\}]$

$= f \oplus [f * e]$

$= f \oplus [f] = h$

96. (a) $e^8 = e^2 * e^2 * e^2$

$= e * e * e$

$= e$

If we observe a * anything = a

$\therefore a^{10} = a$

$\therefore \{a^{10} * (f^{10} \oplus g^9)\} \oplus e^8$

$= a \oplus e$

$= e$

97. (d) **Case I:** ___m___

First place can be selected in five ways and hence the third in four ways

$\therefore 5 \times 4 = 20$ was

Case II: ___n___

First place can be selected in 5 ways and the third in 2 ways

$\therefore 5 \times 2 = 10$ ways

Case III: ___p___

First place can be selected in 5 ways and the last letter will be the same, i.e., one way

$\therefore 5 \times 1 = 5$ way

Total ways = 20 + 10 + 5 = 35 ways

98. (c) As the third letter is e that can be selected in only one way

Case I: 4 m e = 4 ways

Case II: 5 n e = 5 ways

Case III: e p e = 1 ways

= 10 ways

99. (d) It will go by elimination

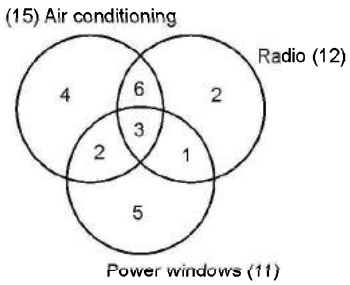
9 – 7 = 2 is even, therefore option (a) is not possible.

2 × 9 = 18 is even, therefore option (c) is not possible.

$\frac{3+9}{3} = \frac{12}{3} = 4$ is even, therefore option (c) is not possible.

$\therefore$ The correct option is (d).

100. (d)



Total = 4 + 6 + 2 + 2 + 3 +1 + 5 = 23

$\therefore$ Cars having none of the option = 25 – 23 = 2.

CAT 2004

This paper contains a total of 35 questions divided into two parts—Part A and Part—B. Part—A contains 20 questions and each question carry one mark each. Part—B contains 15 questions and each question carry two marks each.

There is a negative marking.

PART—A

Q.39. Two boats, travelling at 5 and 10 km/h, head directly towards each other. They begin at a distance of 20 km from each other. How far apart are they (in km) one min before they collide?

(a) $\frac{1}{12}$ (b) $\frac{1}{6}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

Q.40. A rectangular sheet of paper, when halved by folding it at the mid point of its longer side, results in a rectangle, whose longer and shorter sides are in the same proportion as the longer and shorter sides of the original rectangle. If the shorter side of the original rectangle is 2, what is the area of the smaller rectangle?

(a) $4\sqrt{2}$ (b) $2\sqrt{2}$

(c) $\sqrt{2}$ (d) None of these

Q.41. If the sum of the first 11 terms of an arithmetic progression equals that of the first 19 terms, then what is the sum of the first 30 terms?

(a) 0 (b) -1

(c) 1 (d) Not unique

Q.42. If a man cycles at 10 km/h, then he arrives at a certain place at 1 pm If he cycles at 15 km/h, he will arrive at the same place at 11 am At what speed must he cycle to get there at noon?

(a) 11 km/h (b) 12 km/h

(c) 13 km/h (d) 14 km/h

Q.43. On 1 January 2004, two new societies S_1 and S_2 are formed, each having n members. On the first day of each subsequent month, S_1 adds b members while S_2 multiples its current numbers by a constant factor r. Both the societies have the same number of members on 2 July 2004. If $b = 1\ 0.5n$, what is the value of r?

(a) 2.0 (b) 1.9 (c) 1.8 (d) 1.7

Q.44. If $f(x) = x^3 - 4x + p$, and $f(0)$ and $f(1)$ are of opposite sign, then which of the following is necessarily true?

(a) $-1 < p < 2$ (b) $0 < p < 3$

(c) $-2 < p < 1$ (d) $-3 < p < 0$

Q.45. Suppose n is an integer such that the sum of digits on n is 2, and $10^{10} < n < 10^{11}$. The number of different values of n is

(a) 11 (b) 10 (c) 9 (d) 8

Q.46. A milkman mixes 20 litres of water with 80 litres of milk. After selling one fourth of this mixture he adds water to replenish the quantity that he had sold. What is the current proportion of water to milk?

(a) 2:3 (b) 1:2 (c) 1:3 (d) 3:4

Q.47. If $\frac{a}{b+c} = \frac{b}{c+a} = \frac{c}{a+b} = r$ then r cannot take any value except

(a) 1/2

(b) -1

(c) 1/2 or -1

(d) Cannot be determined

Q.48. Let $y =$ $y = \cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{2+\cfrac{1}{3+\ldots}}}}$

What is the value of y?

(a) $\frac{\sqrt{11}+3}{2}$ (b) $\frac{\sqrt{11}-3}{2}$

(c) $\frac{\sqrt{15}+3}{2}$ (d) $\frac{\sqrt{15}-3}{2}$

Q.49. Karan and Arjun run a 100 m race, where Karan beats Arjun by10 m. To do a favour to Arjun, Karan starts 10 m behind the starting line in a second 100 m race. They both run at their earlier speeds. Which of the following is true in connection with the second race?

(a) Karan and Arjun reach the finishing line simultaneously
(b) Arjun beats Karan by 1 m
(c) Arjun beats Karan by 2 m
(d) Karan beats Arjun by 1 m

Q.50. N persons stand on the circumference of a circle at distinct points. Each possible pair of persons, not standing next to each other, sings a two-min song, one pair after the other. If the total time taken for singing is 28 min, what is N?

(a) 5 (b) 7
(c) 9 (d) None of the above

Q.51. In Nuts and Bolts factory, one machine produces only nuts at the rate of 100 nuts per min and needs to be cleaned for 5 mins after production of every 1,000 nuts. Another machine produces only bolts at the rate of 75 bolts per min and needs to cleaned for 10 mins after production of every 1,500 bolts. If both the machines start production at the same time, what is the minimum duration required for producing 9,000 pairs of nuts and bolts?

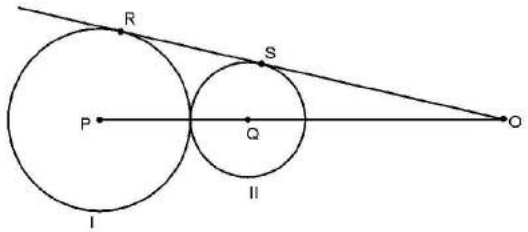
(a) 130 min (b) 135 min
(c) 170 min (d) 180 min

Q.52. A father and his son are waiting at a bus stop in the evening. There is a lamp post behind them. The lamp post, the father and his son stand on the same straight line. The father observes that the shadows of his head and his son's head are incident at the same point on the ground. If the heights of the lamp post, the father and his son are 6 m, 1.8 m and 0.9 m respectively, and the father is standing 2.1 m away from the post then how far (in m) is the son standing from his father?

(a) 0.9 (b) 0.75 (c) 0.6 (d) 0.45

Direction for questions 53 to 55: Answer the questions on the basis of the information given below.

In the adjoining figure, I and II are circles with P and Q as centres respectively. The two circles touch each other and have a common tangent that touches them at points R and S respectively. This common tangent meets the line joining P and Q at O. The diameters of I and II are in the ratio of 4 : 3. It is also known that the length of PQ is 28 cm.



Q.53. What is the ratio of the length of PQ to that of QO?

(a) 1 : 4 (b) 1 : 3 (c) 3 : 8 (d) 3 : 4

Q.54. What is the radius of the circle II?

(a) 2 cm (b) 3 cm (c) 4 cm (d) cm

Q.55. The length of SO is

(a) $8\sqrt{3}$ cm (b) $10\sqrt{3}$ cm

(c) $12\sqrt{3}$ cm (d) $14\sqrt{3}$ cm

Q.56. Let $f(x) = ax^2 - b|x|$, where a and b are constants. Then at $x = 0$, $f(x)$ is

(a) maximized whenever $a > 0, b > 0$
(b) maximized whenever $a > 0, b < 0$
(c) minimized whenever $a > 0, b > 0$
(d) minimized whenever $a > 0, b < 0$

Q.57. Each family in a locality has at most two adults, and no family has fewer than 3 children. Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families, Then the minimum possible number of families in the locality is

(a) 4 (b) 5 (c) 2 (d) 3

Q.58. The total number of integers pairs (x, y) satisfying the equation $x + y = xy$ is

(a) 0
(b) 1
(c) 2
(d) None of the above

PART B—NUMBER OF QUESTIONS = 15

Note: Questions 59 to 73 carry two marks each.

Q.59. Let C be a circle with centre P_0 and AR be a diameter of C. Suppose P_1 is the mid point of the line segment P_0B, P_2 is the mid point of the line segment P_1B and so on. Let C_1 C_2 C_3 be circles with diameters P_0P_1, P_1P_2, P_2P_3, respectively. Suppose the circles C_1 C_2 C_3 are all shaded. The ratio of the area of the unshaded portion of C to that of the original circle is
(a) 8 : 9 (b) 9 : 10 (c) 10 : 11 (d) 11 : 12

Q.60. Consider the sequence of numbers a_1, a_2, a_3 ... to infinity where $a_1 = 81.33$ and $a_2 = -19$ and $a_j = a_{j-1} - a_{j-2}$ for $j \geq 3$. What is the sum of the first 6002 terms of this sequence?
(a) −100.33 (b) −30.00
(c) 0 62.33 (d) 119.33

Q.61. A sprinter starts running on a circular path of radius r metres. Her average speed (in m/min) is πr during the first 30 s, $\frac{\pi r}{2}$ during next one min, $\frac{\pi r}{4}$ during next 2 mins, $\frac{\pi r}{8}$ during next 4 mins, and so on. What is the ratio of the time taken for the nth round to that for the previous round?
(a) 4 (b) 8 (c) 16 (d) 32

Q.62. Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$, has
(a) No solution for x
(b) Exactly one solution for x
(c) Exactly two distinct solutions for x
(d) Exactly three distinct solutions for x

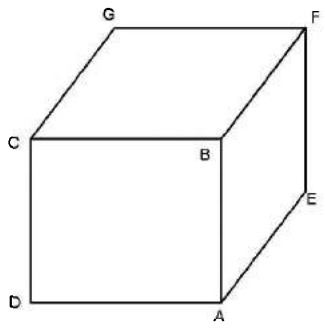
Direction for questions 63 and 64: Answer the questions on the basis of the information given below.

$f_1(x) = x \quad 0 \leq x \leq 1$
$= 1 \quad x \geq 1$
$= 0 \quad$ Otherwise
$f_2(x) = f_1(-x)$ for all x
$f_3(x) = -f_2(x)$ for all x
$f_3(x) = f_3(-x)$ for all x

Q.63. How many of the following products are necessarily zero for every x?
$f_1(x)f_2(x)$, $f_2(x)f_3(x)$, $f_2(x)f_4(x)$?
(a) 0 (b) 1 (c) 2 (d) 3

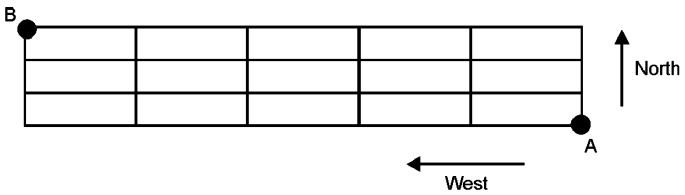
Q.64. Which of the following is necessarily true?
(a) $f_4(x) = f_1(x)$ for all x
(b) $f_1(x) = -f_3(-x)$ for all x
(c) $f_2(-x) = f_4(x)$ for all x
(d) $f_1(x) = f_3(x) = 0$ for all x

Q.65. If the lengths of diagonals DF, AG and CE of the cube shown in the adjoining figure are equal to the three sides of a triangle, then the radius of the circle circumscribing that triangle will be



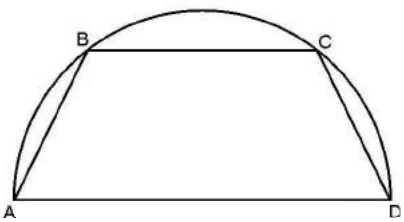
(a) Equal to the side of cube
(b) Times the side of the cube
(c) Times the side of the cube
(d) Impossible to find from the given information.

Q.66. In the adjoining figure shown below, the lines represent one-way roads allowing travel only northwards or only westwards. Along how many distinct routes can a car reach point B from point A?



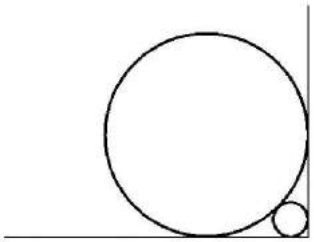
(a) 15 (b) 56 (c) 120 (d) 336

Q.67. On a semicircle with diameter AD, chord BC is parallel to the diameter. Further, each of the chords AB and CD have length 2, while AD has length 8. What is the length of BC?



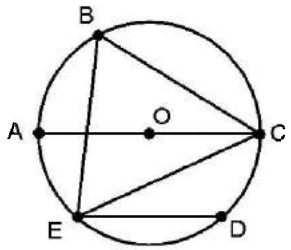
(a) 7.5 (b) 7
(c) 7.75 (d) None of these v

Q.68. A circle with radius 2 is placed against a right angle. Another smaller circle is also placed as shown in the adjoining figure. What is the radius of the smaller circle?



(a) $3-2\sqrt{2}$ (b) $4-2\sqrt{2}$
(c) $7-4\sqrt{2}$ (d) $6-4\sqrt{2}$

Q.69. In the figure shown below, chord ED is parallel to the diameter AC of the circle. If ∠CBE = 65°, then what is the value of ∠DEC?



(a) 35° (b) 55° (c) 45° (d) 25°

Direction for questions 70 and 71: Answer the questions on the basis of the information given below.

In an examination, there are 100 questions divided into three groups A, B and C, such that each group contains at least one question. Each question in group A carries 1 mark, each question in group B carries 2 marks and each question in group C carries 3 marks. It is known that the questions in group A together carry at least 60% of the total marks.

Q.70. If group B contains 23 questions, then how many questions are there in group C?
(a) 1 (b) 2
(c) 3 (d) Cannot be determined

Q.71. If group C contains 8 questions and group B carries at least 20% of the total marks, which of the following best describes the number of questions in group B?
(a) 11 or 12 (b) 12 or 13
(c) 13 or 14 (d) 14 or 15

Q.72. The reminder, when $(15^{23} + 23^{23})$ is divided by 19, is
(a) 4 (b) 15 (c) 0 (d) 18

Q.73. A new flag is to be designed with six vertical stripes using some or all of the colours yellow, green, blue and red. Then, the number of ways this can be done so that no two adjacent stripes have the same colour, is
(a) 12×81 (b) 16×192
(c) 20×125 (d) 24×216

ANSWERS

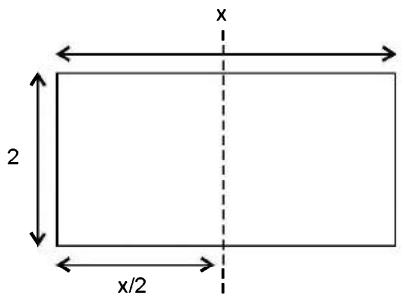
■■■ CAT 2004

Q. No.	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58
Answer	(c)	(b)	(a)	(b)	(a)	(b)	(a)	(a)	(c)	(d)	(d)	(b)	(c)	(d)	(b)	(b)	(c)	(d)	(d)	(c)
Q. No.	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73					
Answer	(d)	(c)	(c)	(b)	(c)	(b)	(a)	(b)	(b)	(d)	(d)	(a)	(c)	(c)	(a)					

HINTS AND SOLUTIONS

39. (c) The boats will be colliding after a time which is given by;
After this time of 80 min, boat (1) has covered $80\times\frac{5}{60}\text{km} = \frac{20}{3}\text{km}$ whereas boat (2) has covered $80\times\frac{10}{60}\text{km} = \frac{40}{3}$
After 79 min, distance covered by the first boat $= d_1 = \left(\frac{20}{3}-\frac{5}{60}\right)$
After 79 min, distance covered by the second boat $= d_2 = \left(\frac{40}{3}-\frac{10}{60}\right)$
So the separation between the two boats = $20 - (d_1 + d_2) = \frac{1}{4}\text{km}$
Alternative method
Relative speed of two boats = 5 + 10 = 15 km/h
i.e., in 60 min they cover (together) = 15 km
$\therefore$ In 1 min they will cover (together) $\frac{15}{60}=\frac{1}{4}\text{km}$

40. (b)



In original rectangle ratio = $\frac{x}{2}$
In smaller rectangle ratio = $\frac{2}{\left(\frac{x}{2}\right)}$
Given $\frac{x}{2}=\frac{2}{\frac{x}{2}}\Rightarrow\times 2\sqrt{2}$
Area of smaller rectangle $=\frac{x}{2}\times 2 = x = 2\sqrt{2}$

41. (a) Given
$t_1 + t_2 + \ldots + t_{11} = t_1 + t_2 + \ldots + t_{19}$ (for an AP)
$\Rightarrow \frac{11}{2}\left[2a+(11-1)d\right]=\frac{19}{2}\left[2a+(19-1)d\right]$
$22a + 110d = 28a + 342d$
$16a + 232d = 0$
$\Rightarrow \frac{30}{2}\left[2a+(30-1)d\right]=0$
$\Rightarrow S_{30\text{ terms}} = 0$

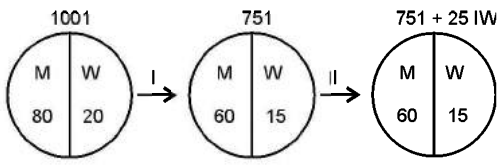
42. (b) When speed of the man = 10 km/h = $\frac{d}{t}$ and
When speed of the man = 15 km/h = $\frac{d}{t-2}$
Equating the value of d: $10 \times t = 15 \times (t - 2) \Rightarrow t = 6$ h
Finally desired speed $\frac{d}{t-1}=\frac{10t}{t-1}=\frac{10\times 6}{5} = 12$ km/h.

43. (a) There will be an increase of 6 times. Number of members s, will be in AP. On 2 July 2004, S_1 will have $n + 6b$ members = $n + 6 \times 10.5n = 64n$
Number of members in s_2 will be in GP
On 2 July 2004, number of members in $s_2 = nr6$
They are equal, hence $64n = nr^6$
$\Rightarrow 64 = r^6 \Rightarrow r = 2$

44. (b) We have
$f(0) = 0^3 - 4(0) + p = p$
$f(1) = 1^3 - 4(1) + p = p - 3$
If P and $p - 3$ are of opposite signs then $p(p - 3) < 0$
Hence, $0 < p < 3$.

45. (a) We have
(1) $10^{10} < n < 10^{11}$
(2) Sum of the digits for 'n' = 2 clearly
(n) min = 10000000001 (1 followed by 9 zeros and finally 1)
Obviously, we can form 10 such numbers by shifting '1' by one place from right to left again and again. Again, there is another possibility for n
n = 20000000000
So finally: Number of different values for n = 10 + 1 = 11 answer.

46. (a)



The diagram is self explanatory. Removal of 25 litres at stage I will result in the volume of milk being reduced by 80% of 25 litres i.e., 20 litres and the volume of water being reduced by the remaining 5 l. So M = 60 litres and W = 15 litres. Addition of 25 litres water will finally given M = 60 litres and W = 40 m. Hence the ratio of W and M = 40:60 = 2:3.

47. (c) If $\frac{a}{b+c}=\frac{b}{c+a}=\frac{c}{a+b}=r$

Then there are only two possibilities.

(i)

If $a+b+c\neq 0$, then

$$\frac{a}{b+c}=\frac{b}{c+a}=\frac{c}{a+b}=\frac{a+b+c}{(b+c)+(c+a)+(a+b)}$$

$$=\frac{a+b+c}{2(a+b+c)}=\frac{1}{2}$$

(ii)

If $a+b+c=0$, then

$b+c=-a$

$c+a=-b$

$a+b=-c$

Hence $\frac{a}{b+c}=\frac{a}{(-a)}=-1$

Similarly, $\frac{b}{c+a}=\frac{c}{a+b}=-1$

Therefore option (c) is the correct one $\frac{1}{2}$ or -1

48. (d) $y=\frac{1}{2+\frac{1}{3+y}}$

$\Rightarrow y=\frac{3+y}{7+2y}$

$\Rightarrow 2y^2+6y-3=0$

$\Rightarrow y=\frac{-6\pm\sqrt{36+24}}{4}$

$=\frac{-6\pm\sqrt{60}}{4}=\frac{-3\pm\sqrt{15}}{2}$

Since 'y' is a +ve number, therefore

$y=\frac{\sqrt{15}-3}{2}$

49. (d) Situation (I):

In whatever time Karan covers a distance of 100 m, Arjun covers 90 m in the same time.

Situation (II):

Now Karan is 10 m behind the starting point. Once again to cover 100 m from this new point Karan will be taking the same time as before. In this time Ariun will cover 90 metres only. This means that now both of them will be at the same point, which will be 10 metres away from the finish point. Since both of them are required to cover the same distance of 10 m now and Karan has a higher speed, he will beat Ariun. No need for calculations as option (d) is the only such option.

50. (b) Each person will form a pair with all other persons except the two beside him. Hence, he will form $(n-3)$ pairs.

If we consider each person, total pairs = $n(n-3)$ but here each pair is counted twice.

Hence, the actual number of pairs = $\frac{n(n-3)}{2}$

They will sing for $\frac{n(n-3)}{2}\times 2=n(n-3)$ min

Hence, $n(n-3)=28$

$\Rightarrow n^2-3n-28=0$

$\Rightarrow n=7$ or -4

Discarding the –ve value: $n=7$

51. (c) **Machine I:**

Number of nuts produced in one min = 100

To produce 1000 nuts the time required = 10 min

Cleaning time for nuts = 5 min

Over all time to produce 1000 nuts = 15 min

Over all time to produce 9000 nuts = 138 min – 5 min = 133 min ... (1)

Machine II:

To produce 75 bolts time required = 1 min

To produce 1500 bolts time required = 20 min

Cleaning time for bolts = 10 min

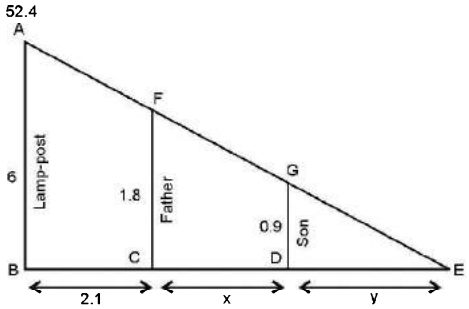
Effective time to produce 1500 bolts = 30 min

Effective time to produce 9000 bolts = $30\times 6-10$ = 170 min ... (2)

From (1) and (2)

Minimum time = 170 min

52. (d)



$\Delta ABE \sim \Delta FCE$

$\therefore \frac{6}{1.8}=\frac{2.1+x+y}{x+y}$ ... (i)

Also $\Delta ABE \sim \Delta GDE$

$\therefore \frac{6}{0.9}=\frac{2.1+x+y}{x+y}$ ... (ii)

From (i) and (ii) $x=0.45$.

53. (b) $\frac{OP}{OQ}=\frac{PR}{QS}=\frac{4}{3}$

OP = 28

OQ = 21

PQ = OP – OQ = 7

$\frac{PQ}{OQ} = \frac{7}{21} = \frac{1}{3}$

54. (b) PR + QS = PQ = 7

$= \frac{RP}{QS} = \frac{4}{3}$

$\Rightarrow$ QS = 3

55. (c) $SO = \sqrt{OQ^2 - QS^2} = \sqrt{21^2 - 3^2}$

$= \sqrt{24 \times 18} = 12\sqrt{3}$

56. (d) When $a > 0$, $b < 0$,
ax^2 and $-b|x|$ are non negative for all x,
i.e., $ax^2 - b|x| \geq 0$ $ax^2 - b|x|$ is minimum at $x = O$ when $a > 0$, $b < 0$.

57. (d)

Family	*Adults*	*Children*
I	0, 1, 2	3, 4, 5...
II	0, 1, 2	3, 4, 5...
III	0, 1, 2	3, 4, 5...

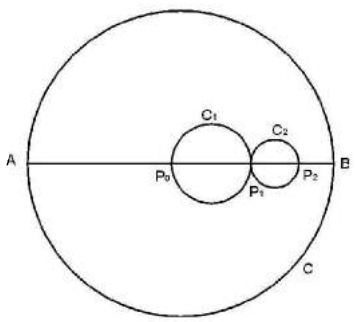
As per the question, we need to satisfy three conditions, namely:

1. Adults (A) > Boys (B)
2. Boys (B) > Girls (G)
3. Girls (G) > Families (F)

Clearly, if the number of families is 2, maximum number of adults can only be 4. Now, for the second condition to be satisfied, every family should have at least two boys and one girl each. This will result in noncompliance with the first condition because adults will be equal to boys. If we consider the same conditions for 3 families, then all three conditions will be satisfied.

58. (c) Given equation is $x + y = xy$
$\Rightarrow xy - x - y + 1 = 1$
$\Rightarrow (x - 1)(y - 1) = 1$
$x - 1 = 1$ & $y - 1 = 1$ or $x - 1 = -1$ & $y - 1 = -1$
Clearly (0, 0) and (2, 2) are the only pairs that will satisfy the equation.

59. (d)



Circle C	Radius r
C_1	$\frac{r}{4}$
C_2	$\frac{r}{8}$
C_3	$\frac{r}{16}$
:	:

$\Rightarrow$ Either $\frac{\text{Area of unshaded portion of } c}{\text{Area of } c}$

$= 1 - \frac{\text{Area of unshaded portion of } c}{\text{Area of } c}$

$= 1 - \frac{\pi\left(\left(\frac{r}{4}\right)^2 + \left(\frac{r}{8}\right)^2 + \ldots\right)}{\pi r^2}$

$= 1 - 1 - \left(\frac{1}{4^2} + \frac{1}{8^2} + \ldots\right) = 1 - \frac{\frac{1}{16}}{1 - \frac{1}{4}}$

$= \frac{11}{12}$

60. (c) Given $a_1 = 81.33$; $a_2 = -19$
Also:
$aj = a_{j-1} - a_{j-2}$, for $j \geq 3$
$\Rightarrow a_3 = a_2 - a_1 = -100.33$
$a_4 = a_3 - a_2 = -81.33$
$a_5 = a_4 - a_3 = 19$
$a_6 = a_5 - a_4 = +100.33$
$a_7 = a_6 - a_5 = +81.33$
$a_8 = a_7 - a_6 = -19$
Clearly onwards there is a cycle of 6 and the sum of terms in every such cycle = 0. Therefore, when we add a_1, a_2, a_3... upto '2' we will eventually be left with $a_1 + a_2$ only i.e., 81.33 – 19 = 62. 33.

61. (c) As options are independent of n let $n = 2$
Time taken for the first round $\frac{1}{2} + 1 + 2 + 4 =$ 7.5 min
Time taken for the second round = 8 + 16 + 32 + 64 = 120 min
Ratio $= \frac{120}{7.5} = 16$

62. (b) $u = (\log_2 x)^2 - 6\log_2 x + 12$
$x^u = 256$
Let $\log_2 x = y \Rightarrow x = 2^y$
$x^u = 2^8 \Rightarrow uy = 8 \Rightarrow u = \frac{8}{y}$
$\frac{8}{y} = t^2 - 6y + 12 \Rightarrow y^3 - 6y^2 - = 0$
$\Rightarrow (y - 2)^3 = 0 \Rightarrow y = 2$
$\Rightarrow x = 4$, $u = 4$

63. (c) $f_1 f_2 = f_1(x) f_1(-x)$

$$f_1(-x) = \begin{cases} -x & -1 \le -x \le 1 \\ 1 & -x \ge 1 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} -x & -1 \le -x \le 1 \\ 1 & -x \ge 1 \\ 0 & \text{other wise} \end{cases}$$

$f_1 f_1(-x) = 0 \ \forall \ x$

64. (b) Check with options

Option (2)

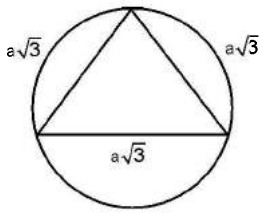
$f_3(-x) = -f_2(-x) = -f_1(x)$

$= f_1(x) = -f_3(-x) \ \forall \ x$

65. (a) DF, AG and ~ E are body diagonals of cube.

Let the side of cube = a

Therefore, body diagonal is a $\sqrt{3}$



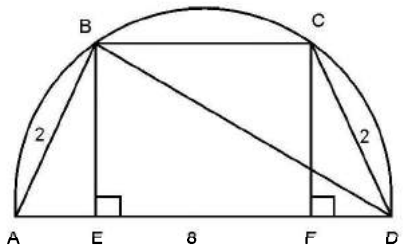
Circum radius for equilateral triangle side = $\frac{side}{\sqrt{3}}$

Therefore, $\frac{a\sqrt{3}}{\sqrt{3}} = a$

66. (b) From A to B, there are 8 on-way roads out of which 3 roads are in northwards and 5 roads are westwards.

Therefore, number of distinct routes is = $\frac{8!}{513!} = 56$

67. (b)



$$\frac{1}{2} \times AB \times BD = \frac{1}{2} \times AD \times BE$$

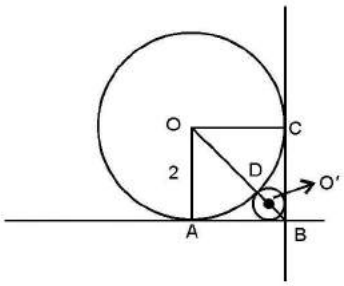
$$2\sqrt{8^2 - 2^2} = 8 \times BE$$

$$BE = \frac{\sqrt{60}}{4} = \frac{\sqrt{15}}{2}$$

$$AE = AE = \sqrt{2^2 - \left(\frac{\sqrt{15}}{2}\right)^2} = \sqrt{4 - \frac{15}{4}} = \frac{1}{2}$$

$$BC = EF = 8 - \left(\frac{1}{2} + \frac{1}{2}\right) = 7$$

68. (d)



Let the radius of smaller circle = r

$\therefore$ O′B = r $\sqrt{2}$

$\therefore$ OB = O′B + O′D + OD

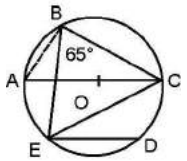
= r $\sqrt{2}$ + r + 2

Also OB = 2 $\sqrt{2}$

= r $\sqrt{2}$ + r + 2 = 2 $\sqrt{2}$

$\Rightarrow$ r = 6 − 4 $\sqrt{2}$

69. (d)



In ΔABC

∠B = 90° (Angles in semicircle)

Therefore ∠ABE = 90 − 65 = 25°

Also ∠ABE = ∠ACE (angel subtended by same arc AE)

Also ∠ACE = ∠CED [AC ||ED]

Therefore ∠CED = 25°

70. (a) Since Group (B) contains 23 questions, the marks associated with this group are 46. Now check for option (1). If Group (C) has one question, then marks associated with this group will be 3. This means that the cumulative marks for these two groups taken together will be 49. Since total number of questions are 100, Group (A) will have 76 questions, the corresponding weightage being 76 marks. This satisfies all conditions and hence is the correct option. It can be easily observed that no other option will fit the bill.

71. (c) Since Group (C) contains 8 questions, the corresponding weightage will be 24 marks. This figure should be less than or equal to 20% of the total marks. Check from the options . Option (c) provides 13 or 14 questions in Group (B), with, a corresponding weightage of 26 or 28 marks. This means that number of questions in Group (A) will either be 79 or 78 and will satisfy the desired requirement.

72. (c) $15^{23} = (19-4)^{23} - 19x + (-4)^{23}$ where x is a natural number.
$23^{23} - (19+4)^{23} = 19y + (4)^{23}$ where y is a natural number.
$15^{23} + 23^{23} - 19(x+y) + 4^{23} + (-4)^{23}$
$= 19\,(x+y)$

73. (a) The first strip can be of any of the four colours. The 2nd can be of any colour except that of the first (i.e., 3). Similarly, each subsequent strip can be of any colour except that of the preceding strip (= 3)
Hence, the number of ways = $4 \times 35 = 12 \times 81$

CAT 2005

This paper contains a total of 30 questions divided into two parts—Part A and Part B. Part A contains 10 questions and each question carry one mark each. Part B contains 10 questions and each question carry two marks each.

There is a negative marking of 1/3 of the marks associated with the particular question.

PART—A

Q.1. If $x = (16^3 + 17^{13} + 18^3 + 19^3)$, then x divided by 70 leaves a remainder of
(a) 0 (b) 1 (c) 69 (a) 35

Q.2. A chemical plant has four tanks (A, B, C and D), each containing 1000 litre of a chemical. The chemical is being pumped from one tank to another as follows:
From A to B @ 20 l/min
From C to A @ 90 l/min
From A to D @ 10 l/min
From C to D @ 50 l/min
From B to C @ 100 l/min
From D to B @ 110 l/min
Which tank gets emptied first, and how long does it take (in min) to get empty, after the pumping starts?
(a) A, 16.66 (b) C, 20
(c) D, 20 (d) D, 25

Q.3. Two identical circles intersect so that their centers, and the points at which they intersect, form a square of side 1 cm. The area in cm^2 of the portion that is common to the two circles is

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}-1$ (c) $\frac{\pi}{5}$ (d) $\sqrt{2}-1$

Q.4. A jogging park has two identical circular tracks touching each other, and a rectangular track enclosing the two circles. The edges of the rectangles are tangential to the circles. Two friends, A and B, start jogging simultaneously from the point where one of the circular tracks touches the smaller side of the rectangular track. A jogs along the rectangular track while B jogs along the two circular tracks in a figure of eight. Approximately, how much faster than A does B have to run, so that they take the same time to return to their starting point?
(a) 3.88% (b) 4.22% (c) 4.44% (d) 4.72%

Q.5. In a chess competition involving some boys and girls of a school, every student had to play exactly one game with every other student. It was found that in 45 games both the players were girls, and in 190 games both were boys. The number of games in which one player was a boy and the other was a girl is
(a) 200 (b) 216 (c) 235 (d) 256

Direction for questions 6 and 7: Answer the questions on the basis of the information given below.

Ram and Shyam run a race between points A and B 5 km apart. Ram starts at 9 am from A at a speed of 5 km/h, reaches B, and returns to A at the same speed. Shyam starts at 9.45 am from A at a speed of 10 km/h, reaches B, and comes back to A at the same speed.

Q.6. At what time do Ram and Shyam first meet each other?
(a) 10 am (b) 10.10 am
(c) 10.20 am (d) 10.30 am

Q.7. At what time does Shyam over take Ram?
(a) 10.20 am (b) 10.30 am
(c) 10.40 am (d) 10.30 am

Q.8. If $R = \frac{30^{65} - 29^{65}}{30^{64} + 29^{64}}$, then
(a) $0 < R \leq 0.1$ (b) $0.1 < R \leq 0.5$
(c) $0.5 < R \leq 1.0$ (d) $R > 1$

Q.9. What is the distance in cm between two parallel chords of lengths 32 cm and 24 cm in a circle of radius 20 cm?
(a) 1 or 7 (b) 2 or 14
(c) 3 or 21 (d) 4 or 28

Q.10. For which value of k does the following pair of equations yield a unique solution of x such that the solution is positive?
$x^2 - y^2 = 0$ and $(x - k)^2 + y^2 = 1$
(a) 2 (b) 0
(c) $\sqrt{2}$ (d) -2

PART—B

Note: Questions 11 to 30 carry two marks each.

Q.11. Let $n! = 1 \times 2 \times 3 \times \ldots \times n$ for integer $n \geq 1$. If $p = 1! + (2 \times 2!) + (3 \times 3!) + \ldots + (10 \times 10!)$, then $p + 2$ when divided by 11! leaves a remainder of
(a) 10 (b) 0 (c) 7 (d) 1

Q.12. Consider a triangle drawn on the X-Y plane with its three vertices of (41, 0), (0, 41) and (0, 0), each vertex being represented by its (X, Y) coordinates. The number of points with integer coordinates inside the triangle (excluding all the points on the boundary) is
(a) 780 (b) 800 (c) 820 (d) 741

Q.13. The digits of a three-digit number A are written in the reverse order to form another three-digit number B. If B> A and B-A is perfectly divisible by 7, then which of the following is necessarily true?
(a) 100<A<299 (b) 106<A<305
(c) 112<A<311 (d) 118<A<317

Q.14. If $a_1 = 1$ and $a_{n+1} - 3a_n + 2 = 4n$ for every positive integer n, then a_{100} equals
(a) $3^{99} - 200$ (b) $3^{99} + 200$
(c) $3^{100} - 200$ (d) $3^{100} + 200$

Q.15. Let S be the set of five-digit numbers formed by digits 1, 2, 3, 4 and 5, using each digit exactly once such that exactly two odd position are occupied by odd digits. What is the sum of the digits in the rightmost position of the numbers in S?
(a) 228 (b) 216 (c) 294 (d) 192

Q.16. The right most non-zero digits of the number 30^{2720} is
(a) 1 (b) 3 (c) 7 (d) 9

Q.17. Four points A, B, C and D lie on a straight line in the X-Y plane, such that AB = BC = CD, and the length of AB is 1 m. An ant at A wants to reach a sugar particle at D. But there are insect repellents kept at points B and C. The ant would not go within one m of any insect repellent. The minimum distance in m the ant must traverse to reach the sugar particle is
(a) $3\sqrt{2}$ (b) $1 + \pi$
(c) $\frac{4\pi}{3}$ (d) 5

Q.18. If $x \geq y$ and $y > 1$, then the value of the expression $\log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$ can never be
(a) -1 (b) -0.5
(c) 0 (d) 1

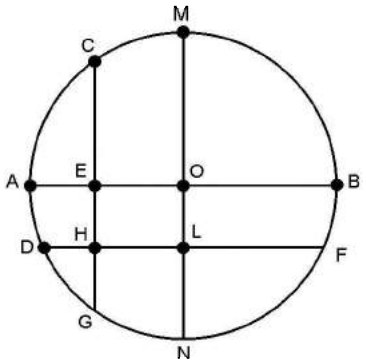
Q.19. For a positive integer n, let p_n, denote the product of the digits of n and S_n denote the sum of the digits of n. The number of integers between 10 and 1000 for which $p_n + s_n = n$ is
(a) 81 (b) 16 (c) 18 (d) 9

Q.20. Rectangular tiles, each of the size 70 cm by 30 cm must be laid horizontally on a rectangular floor of size 110 cm by 130 cm, such that the tiles do not overlap. A tile can be placed in any orientation so long as its edges are parallel to the edges of the floor. No tile should overshoot any edge of the floor. The maximum number of tiles that can be accommodated on the floor is
(a) 4 (b) 5 (c) 6 (d) 7

Q.21. In the X-Y plane, the area of the region bounded by the graph $|x + y| + |x - y| = 4$ is
(a) 8 (b) 12 (c) 16 (d) 20

Q.22. In the following figure, the diameter of the circle is 3 cm. AB and MN are two diameters such that MN is perpendicular to AB. In addition, CG is perpendicular to AB such that AE : EB = 1 : 2, and DF is perpen-

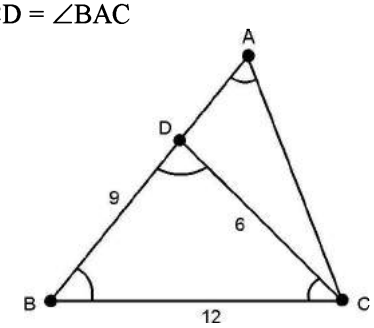
dicular to MN such that NL : LM = 1 : 2. The length of DH in cm is



(a) $2\sqrt{2}-1$ (b) $\frac{\left(2\sqrt{2}-1\right)}{2}$

(c) $\frac{\left(3\sqrt{2}-1\right)}{2}$ (d) $\frac{\left(2\sqrt{2}-1\right)}{2}$

Q.23. Consider the triangle ABC shown in the following figure where BC = 12 cm, DB = 9 cm, CD = 6 cm and ∠BCD = ∠BAC



What is the ratio of the perimeter of ΔADC to that of the ΔBDC?

Q.24. P, Q, S and R are points on the circumference of a circle of radius r, such that PQR is an equilateral triangle and PS is a diameter of the circle. What is the perimeter of the quadrilateral PQSR?

(a) $2r\left(1+\sqrt{3}\right)$ (b) $2r\left(2+\sqrt{3}\right)$

(c) $r\left(1+\sqrt{5}\right)$ (d) $2r+\sqrt{3}$

Q.25. Let S be a set of positive integers such that every element n of S satisfies the conditions:

I. $1000 \le n < 1200$

II. Every digit in n is odd

Then how many elements of S are divisible by 3?

(a) 9 (b) 10 (c) 11 (d) 12

Q.26. Let $x=\sqrt{4+\sqrt{4-\sqrt{4+\sqrt{4-....\infty}}}}$, then x equals

(a) 3

(b) $\left(\frac{\sqrt{13}-1}{2}\right)$

(c) $\left(\frac{\sqrt{13}+1}{2}\right)$

(d) $\sqrt{13}$

Q.27. Let $g(x)$ be a function such that $g(x+1)+g(x-1) = g(x)$ for every real x. Then for what value of p is the relation $g(x+p) = g(x)$ necessarily true for every real x?

(a) 3 (b) 3 (c) 2 (d) 6

Q.28. A telecom service provider engages male and female operators for answering 1000 calls per day. A male operator can handle 40 calls per day whereas a female operator can handle 50 calls per day. The male and the female operators get a fixed wage of Rs 250 and Rs 300 per day respectively. In addition, a male operator gets Rs 15 per call he answers and female operator gets Rs 10 per call she answers. To minimize the total cost, how many male operators should the service provider employ assuming he has to employ more than 7 of the 12 female operators available for the job?

(a) 15 (b) 14 (c) 12 (d) 10

Q.29. Three Englishmen and three Frenchmen work for the same company. Each of them knows a secret not known to others. They need to exchange these secrets over person-to-person phone calls so that eventually each person knows all six secrets. None of the Frenchmen knows English, and only one Englishman knows French. What is the minimum number of phone calls needed for the above purpose?

(a) 5 (b) 10 (c) 9 (d) 15

Q.30. A rectangular floor is fully covered with square tiles of identical size. The tiles on the edges are white and the tiles in the interior are red. The number of white tiles is the same as the number of red tiles. A possible value of the number of tiles along one edge of the floor is

(a) 10 (b) 12 (c) 14 (d) 16

ANSWERS

■■■ CAT 2005

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(a)	(c)	(b)	(d)	(a)	(b)	(b)	(d)	(d)	(c)	(d)	(a)	(b)	(c)	(b)	(a)	(b)	(d)	(d)	(c)
Q. No.	21	22	23	24	25	26	27	28	29	30										
Answer	(c)	(b)	(a)	(a)	(a)	(c)	(d)	(d)	(c)	(b)										

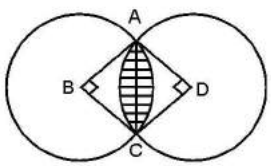
HINTS AND SOLUTIONS

1. (a) $\frac{a^n + b^n + c^n + \ldots}{a + b + c + \ldots}$ is divisible for all the odd values of n.

2. (c)

A	*B*	*C*	*D*
– 20	20		
90		– 90	
– 10			10
		– 50	50
	– 100	100	
	110		– 110
Total + 60	30	– 40	– 50

3. (b)

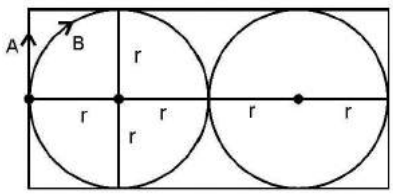


Shaded area = 2 × (area of sector ADC – area of ΔADC)

$= 2\times\left(\frac{\pi}{4}\times 1^2 - \frac{1}{2}\times 1\times 1\right)$

$= \left(\frac{\pi}{2} - 1\right)$

4. (d)



A covers 2r + 2r + 4r + 4r = 12r
B Covers 2πr + 2πr = 4 πr distance

$\frac{4\pi r}{S_B} = \frac{12r}{S_A} \Rightarrow S_B = \frac{\pi}{3} S_A$

$\frac{S_B - S_A}{S_A}\times 100 = \frac{\pi - 3}{3}\times 100 = 4.72\%$

5. (a) Let there be m boys and n girls

${}^nC_2 = 45 = \Rightarrow n\,(n-1) = 90 \Rightarrow n = 10$

${}^mC_2 = 190 \Rightarrow \frac{m(m-1)}{2} = 190 \Rightarrow m\,(m-1) = 380 \Rightarrow m = 20$

The number of games between one boy and one girl

$= {}^{10}C_1 \times {}^{20}C_1 = 10 \times 20 = 200$

6. (b) Obviously, Ram and Shyam shall meet each other between C and B, sometime after 10 am. At 10 am, they are moving as shown below:
From now, the time taken to meet

$= \frac{2.5}{(10+5)}\times 60\,\text{min} = 10\,\text{min}$

So, they meet each other at 10.10 am.

7. (b) Obvious from the diagram that at 10.30; Shyam overtakes Ram
Alternate: At 10.15 the situation is as shown:

Time taken for Shyam to overtake Ram = $\frac{2.5}{(10+5)} \times 60\,\text{min} = 15\,\text{min}$

Shyam overtakes Ram at 10.30 a.m.

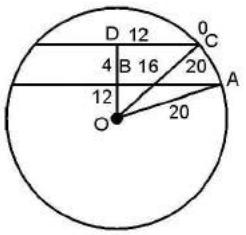
8. (d) $$\frac{30^{65}-(30-1)^{65}}{30^{64}+(30-1)^{64}} = \frac{30^{65}-30^{65}\left(1-\frac{1}{30}\right)^{65}}{30^{64}-30^{64}\left(1-\frac{1}{30}\right)^{64}}$$

$$\text{Or R} = 30\left\{\frac{1-\left(1-\frac{1}{30}\right)^{65}}{1+\left(1-\frac{1}{30}\right)^{64}}\right\}$$

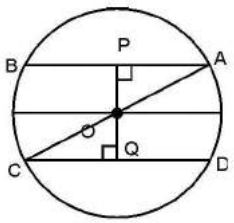
$$\text{R} = 30\left\{\frac{1-(0.96)^{65}}{1+(0.96)^{64}}\right\}$$

The numerator is only slightly less then 1 and denominator is only slightly more than 1. R is slightly less than 1.

9. (d) Case I: chords on same side of the centre.



$OB^2 = OA^2 - AB^2 = 20^2 - 16^2 = 144$
$OB = 12$
$OD^2 = 20^2 - 12^2 = 400 - 144 = 256$
$OD = 16$
$BD = 4$ cm
Case II: Chords on opposite side of the centre

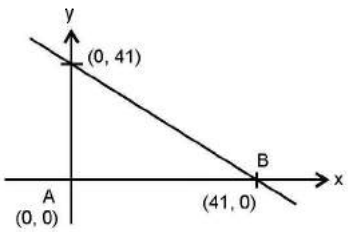


$AB = 32$ cm
$CD = 24$ cm
$OP = \sqrt{AO^2 - AP^2}$
$= \sqrt{(20)^2 - (16)^2}$
$OP = 12$ cm
and $OQ = \sqrt{(OC)^2 - (CQ)^2}$
$= \sqrt{(20)^2 - (12)^2}$
$OQ = 16$ cm
Distance = PQ = 12 + 16 = 28 cm.

10. (c) $y^2 = x^2$
$2x^2 - 2kx + k^2 - 1 = 0$
$D = 0$
$\Rightarrow 4k^2 = 8k^2 - 6$
$\Rightarrow 4k^2 = 8$
$k^2 = 2 \Rightarrow 8$
$k^2 = 2 \Rightarrow k = \pm\sqrt{2}$ with $k = \pm\sqrt{2}$
the equation is $2x^2 + 2\sqrt{2}x + 1 = 0$ roots is: $\frac{-b}{2a} = +\frac{1}{\sqrt{2}}$ but with
$k = -\sqrt{2}$, the equation is $-\frac{1}{\sqrt{2}}$ as this root is a –ve it will reject $k = -\sqrt{2}$, only answer is $\Rightarrow k = +\sqrt{2}$ only.

11. (d) If $p = 1! = 1$
Then $p + 2 = 3$ when divided by 2! remainder will be 1.
If $p = 1! + 2 \times 2! = 5$
Then $p + 2 = 7$ when divided by 3! remainder is still 1.
Hence, $p = 1! + (2\times2!) + (3\times3!) + \ldots + (10\times10!)$
When divided by 11! leaves remainder 1.
Hence, the remainder is 1.

12. (a)



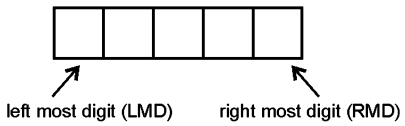
Equation of line = $x + y = 41$. If the (x, y) co ordinates of the points are integer, their sum shall also be integers so that $x + y = k$ (l, a variable) as we have to exclude points lying on the boundary of the triangle; k can take all values from 1 to 40 only. K = 0 is also rejected as at $k = 0$ will give the point A; which can not be taken. Now, $x + y = k$ ($k = 1, 2, 3, \ldots 40$)
With $k = 40$; $x + y = 40$; taking integral solutions. We get points (1, 39), (2, 38); (3, 37)... (39, 1) i.e., 39 points

1) $x + y = 40$ will be satisfied by 39 point is similarly $x + y = 39$ is satisfied by 38 points.
$x + y = 38$ by 37 point.
$x + y = 3$ by 2 point.
$x + y = 2$ is satisfied by 1 point
$x + y = 1$ by no points.
So, the total number of all such points is:
$39 + 38 + 37 + 36 + \ldots 3 + 2 + 1$
$= \frac{39 \times 40}{2} = 780$ points

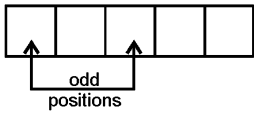
13. (b) Let A = abc, then B = cba
Given, B > A which implies $c > a$ ... (1)
As $B - A = (100c + 10b + a) - (100a + 10b + 1)$
$B - A = 100(c - a) + (a - c)$
$B - A = 99(c - a)$ and (B–A) is divisible by 7; and 99 is not divisible by 7 (no factor like 7 or 7^2) therefore $(c - a)$ must be divisible by 7 {i.e., $(c - a)$ must be 7, 7^2 etc) as c and a are single digits. $(c - a)$ must be 7 only, the possible values (c, a) {with $c > a$} are (2) and (8,1) with this we can write A as
A : $abc = 1b8$ or $2b9$
As b can take values from 0 to 9, the smallest and the largest possible values are:
$A_{min} = 108$
and $A_{max} = 299$
Only (b) satisfies so (b) is the answer.

14. (c) $a_1 = 1$, $a_{n+1} - 3a_n + 2 = 4n$
$A_{n+1} = 3a_n + 4n - 2$
When $n = 2$, then $a^2 = 3 + 4 - 2 = 5$
When $n = 3$, then $a_3 = 3 \times 5 + 4 \times 2 - 2 = 21$
From the options, we get an idea that a_n can be expressed in a combination of some power of 3 and some multiple of 100.
(a) $3^{99} - 200$; tells us that a_n could be: $3^{n-1} - 2 \times n$; but it does not fit a_1 or a_2 or a_3
(b) $3^{99} + 200$; tells us that a_n could be: $3^{n-1} + 2 \times n$; again not valid for a_1, a_2 etc.
(c) $3^{100} + 200$; tells $3^n + 2n$: again not valid.
So, (c) is the correct answer.

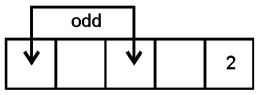
15. (b)



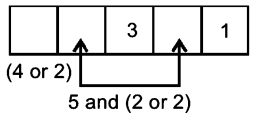
Odd positions can be counted in 2 ways.
(i) Counting form the LMD – end:



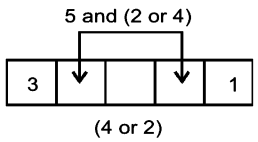
We have 1, 2, 3, 4, and 5 to be filled in these blocks. Odd numbers (1, 3, 5) to be be filled at odd positions. Other places are to be filled by even numbers (2 or 4). Let us count, how many such numbers are there with 2 at the unit's digit.



Odd numbers can be filled ${}^3P_2 = 6$ ways
The remaining two places are to be filled by 2 numbers (one odd number left out of 1, 3, 5 and one even i.e., 4) in = 2 ways
S, there are $6 \times 2 = 12$ number with 2 at the right most place. Similarly there are 12 such numbers with 4 at the rightmost digits.
The sum of the rightmost digits in all such number $= 12(2 + 4) = 72$
(ii) Now counting from the RMD-end.
Let us place 1 at the unit's place and check, how many numbers are possible with (1, 3) at the odd positions:



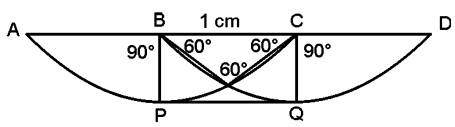
Number of such cases $= 2 \times 2 = 4$ ways



Here again number of ways $= 2 \times 2 = 4$ ways
So, there are $4 + 4 = 8$ numbers in which (1, 3) are at odd positions. Similarly there are 8 numbers in which (1, 5) are at odd positions. So, in all there are 16 numbers, where 1 is at the unit's place. Similarly there are 16 numbers with 3 at unit's place and 16 more with 5 at the unit's place.
Summing up all the odd unit's digits $= 16(1 + 3 + 5) = 144$
From (i) and (ii) we can now sum all (even or odd) numbers at unit place $= 72 + 144 = 216$
Hence, answer is (b)

16. (a) $((30)^4)^{680} = (8100)^{680}$. Hence, the right most non-zero digit is 1.

17. (b)



Draw figure since it could not go within distance of 1 m, so it will go along APQD, which is the path of minimum distance.
$AP = \frac{90}{360} \times 2\pi \times 1 = \frac{\pi}{2}$

Also AP = QD = $\frac{\pi}{2}$

So the minimum distance = AP + PQ + QD

$= \frac{\pi}{2} + 1 + \frac{\pi}{2} = 1 + p$

18. (d) $P = \log_x\left(\frac{x}{y}\right) + \log_y\left(\frac{y}{x}\right)$

$= \log_x x - \log = x_y + \log_y y - \log_y x$

$= 2 - \log_x y - \log_y x$

Let, $t = \log_x y$

$\Rightarrow p = 2 - \frac{1}{t} - t = -\left[\sqrt{t} - \frac{1}{\sqrt{t}}\right]^2$

Which can never be positive, out of the given option it cannot assume a value of +1. So (d) is the answer.

19. (d) $10 < n < 1000$

Let n be two digit number

$n = 10a + b \Rightarrow p_n = ab, s_n = a + b$

Then $ab + a + b = 10a + b$

$\Rightarrow ab = 9a \Rightarrow b = 9$

There are 9 such numbers 19, 29, 33, ... 99

Then let n be three digit number

$\Rightarrow n = 100a + 10b + c \Rightarrow P_n = abc, s_n = a + b + c$

Then $ab + a + b + c = 100a + 10b + c$

$\Rightarrow abc = 99a + 9b$

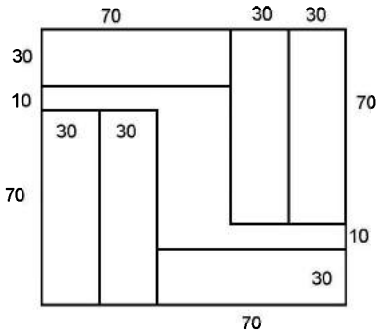
$\Rightarrow bc = 99 + 9\ \frac{b}{a}$

But the minimum value for bc = 81 (i.e., when both b and c are 9)

And RHS is more than 99. Hence, no such number is possible.

Hence, option (d).

20. (c)



21. (c) $|x + y| + |x - y| = 4$

Replace "$+x$" by "$-x$" & "$+y$" by "y" everywhere in the curve: we again get the same equation.

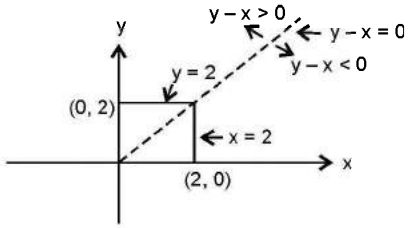
$\Rightarrow$ Curve is symmetric in the 4-quadrants of X–Y plane.

In 1-quadrant $(x, y > 0)$

$|x + y| + |x - y| = 4$

$= \begin{cases} (x+y)+(y-x) = 4; y > x \\ (x+y)-(y-x) = 4; y < x \end{cases}$

Or $\begin{cases} y = 2; y > x \\ x = 2; y < x \end{cases}$



We can now plot the graph:

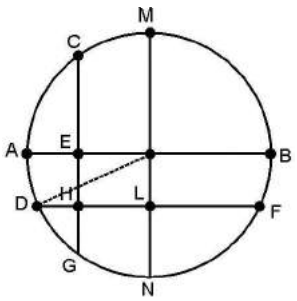
Area in L-quadrant = $(2)^2 = 4$ unit^2

Total area of $|x + y| + |x - y| = 4$ is: 4 x (area of I-quadrant)

$= 4 \times 4$

= 16 sq. unit.

22. (b)



AE = 1 cm

BE = 2 cm and NL = 1 cm

Ml = 2 cm

$HL = OE = \frac{1}{2}$

$DL = DH + HL$

$DL = DH + \frac{1}{2}$

OB = AO = radius = 1.5

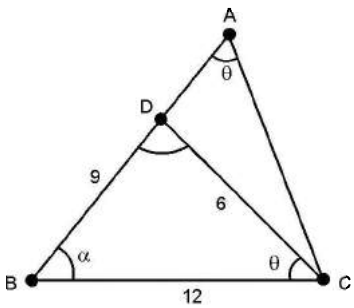
$DO^2 = OL^2 + Dl^2$

$\left(\frac{3}{2}\right)^2 = \left(\frac{1}{2}\right)^2 + \left(DH + \frac{1}{2}\right)^2$

$= \left(DH + \frac{1}{2}\right)^2 = 2 \Rightarrow DH\sqrt{2} - \frac{1}{2}$

Hence, option (b)

23. (a)



Hence, $\angle ACB = \theta + [180 - (20 + a] = 180 - (\theta + \alpha)$

So, here we can say that the triangle BCD and triangle ABC will be similar $\Delta BCD \sim \Delta BAC$.

Hence, from the property of similarity

$\frac{AB}{12} = \frac{12}{9}$ Hence, AB = 16

$\frac{AC}{6} = \frac{12}{9}$ Hence, AC = 8

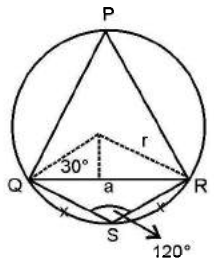
Hence, AD = 7, AC = 8

$S_{ADC} = 8 + 7 + 6 = 21$

$S_{BDC} = 27$

Hence, $r = \frac{21}{27} = \frac{7}{9}$

24. (a)



Here, $\cos 30° = \frac{a}{2r}$

$a = r\sqrt{3}$

From the diagram $\cos 120° = \frac{x^2 + x^2 - a^2}{2x^2}$

$a^2 = 3x^2$

$x = r$

Hence, the circumference will be $2r\left(1+\sqrt{3}\right)$

Hence, answer is (a).

25. (a) The 100th and 1000th position value will be only 1. Now the possibility of unit and tens digits are (1, 3), (1, 9), (3, 1), (5, 5), (7, 3), (7, 9), (9, 1), (9, 7).

26. (c) $x = \sqrt{4+\sqrt{4-x}} \Rightarrow x^2 = 4 + \sqrt{4-x}$

$(X^2 - 4) = \sqrt{4-x}$

Now put the values from the options. Only the 3rd Option satisfies the condition.

27. (d) $g(x + 1) + g(x - 1) = g(x)$

$g(x + 2) + g(x) = g(x + 1)$

Adding these two equations we get

$g(x + 2) + g(x - 1) = 0$

$\Rightarrow g(x + 3) + g(x) = 0$

$\Rightarrow g(x + 4) + g(x + 1) = 0$

$\Rightarrow g(x + 5) + g(x + 2) = 0$

$\Rightarrow g(x + 6) + g(x + 3) = 0$

$\Rightarrow g(x + 6) - g(x) = 0$

28. (d) Let us form both the equations first:

$40 m + 50f = 1000$

$250 m + 300 f + 40 \times 15 m + 50 \times 10 \times f = A$

$850 m + 8000 f = A$

Where m and f are the number of males and females and A is the amount paid by the service provider.

Then the possible values for f are 8, 9, 10, 11 and 12

If $f = 8$, then m = 15

If $f = 9$, 10, 11 then m will not be an integer while $f = 12$ then m will be 10.

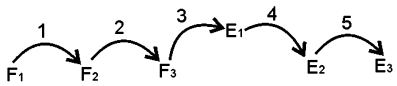
By putting f = 8 and m = 15, A = 18800. When $f = 12$ and m = 10 then A = 18100.

Hence, the number of males will be 10.

29. (c) Frenchmen: F_1, F_2, F_3

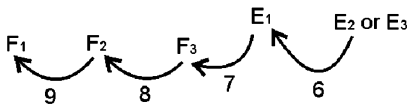
Englishmen: E_1, E_2, E_3

Let E_1 knows fresh 1 round of calls:



Persons	Secrets know after I-round
F_1	F_1, F_2,
F_2,	F_1, F_2, F_3
F_3,	F_1, F_2, F_3, F_4
E_1,	F_1, F_2, F_3, F_2
E_2,	$F_1, F_2, F_3, E_1, E_2, E_3$) all known
E_3,	$F_1, F_2, F_3, E_1, E_2, E_3$) all known

In the 6th call E_1 knows all the secrets. Similarly, after the 9th Call, everybody knows all the secrets.



30. (b) Let the rectangle has *m* and *n* tiles along its length and breadth respectively.

The number of white tiles

$W = 2m + 2(n-2) = 2(m + n - 2)$

And the number of red tiles = = $R = mn - 2(m + n - 2)$ Give $W = R \Rightarrow 4(m + n - 2) = mn$

$\Rightarrow mn - 4m - 4n = -8$

$\Rightarrow (m - 4)(n - 4) = 8$

As *m* and *n* are integers so $(m - 4)$ & $(n - 4)$ are both integers. The possibilities are $(m - 4, n - 4) = (1, 8)$ or $(2, 4)$ giving, (m, n) as (5, 12) or (6, 8) so the edges can have 5, 12, 6 or 8 tiles. Answer is (b) only.

CAT 2006

SECTION III

Each question carries 4 marks with a negative marking of 25%.

Q.51. If $a/b = 1/3$, $b/c = 2$, $c/d = 1/2$, $d/e = 3$ and $e/f = 1/4$, then what is the value of abc/def?
(a) 3/8 (b) 27/8 (c) 3/4 (d) 27/4
(e) 1/4

Q.52. If $x = -0.5$, then which of the following has the smallest value?
(a) $21/x$ (b) $1/x$ (c) $1/x2$
(d) $2X$ (e) $1/\sqrt{-x}$

Q.53. Consider a sequence where the nth term,
$tn = n/(n + 2)$, $n = 1, 2, \ldots$.
The value of $t_3 \times t_4 \times t_5 \times \ldots \times t_{53}$ equals:
(a) 2/495 (b) 2/477
(c) 12/55 (d) 1/1485
(e) 1/2970

Q.54. Which among 21/2, 31/3, 41/4, 61/6 and 121/12 is the largest?
(a) 21/2 (b) 31/3
(c) 41/4 (4) 61/6
(e) 21/12

Q.55. The length, breadth and height of a room are in the ratio 3 : 2 : 1. If the breadth and height are halved and the length is doubled, then the total area of the four walls of the room will
(a) remain the same (b) decrease by 13.64%
(c) decrease by 15% (d) decrease by 18.75%
(e) decrease by 30%

Q.56. A survey of a hundred people was conducted to find out whether they had read the recent issues of *Golmal*, a monthly magazine. A summary of the information regarding the readership in 3 months is given below:

Only September: 18; September but not August: 23; September and July: 28;

September: 28; July: 48; July and August: 10;

None of the three months: 24.

What is the number of surveyed people who have read exactly two consecutive issues (out of the three)?
(a) 7 (b) 9 (c) 12 (d) 14
(e) 17

Q.57. A semi-circle is drawn with AB as its diameter. From C, a point on AB, a line perpendicular to AB is drawn meeting the circumference of the semi-circle at D. Given that AC = 2 cm and CD = 6 cm, the area of the semi-circle (in cm^2) will be:
(a) 32 π (b) 50 π (c) 40.5 π (d) 81 π
(e) undeterminable

Direction for questions 58 and 59: Answer Questions 58 and 59 on the basis of the information given below:

An airline has a certain free luggage allowance and it charges for excess luggage at a fixed rate per kg. Two passengers, Raja and Praja have 60 kg of luggage between them, and are charged Rs 1200 and Rs 2400 respectively for excess luggage. If the entire luggage belonged to one of them, the excess luggage charge would have been Rs 5400.

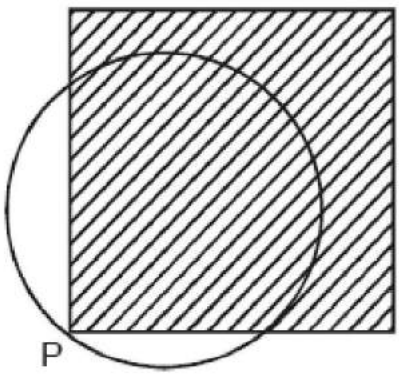
Q.58. What is the weight of Praja's luggage?
(a) 20kg (b) 25 kg (c) 30 kg (d) 35 kg
(e) 40 kg

Q.59. What is the free luggage allowance?
(a) 10kg (b) 5 kg (d) 20 kg (d) 25 kg
(e) 30 kg

Q.60. A group of 630 children is arranged in rows for a group photograph session. Each row contains three fewer children than the row in front of it. Which of the following number of rows is not possible?
(a) 3 (b) 4 (c) 5 (d) 6
(e) 7

Direction for questions 61 to 62: Answer Questions 61 and 62 on the basis of the information given below:

A punching machine is used to punch a circular hole of diameter two units from a square sheet of aluminium of width 2 units, as shown below. The hole is punched such that the circular hole touches one corner P of the square sheet and the diameter of the hole originating at P is in line with a diagonal of the square.



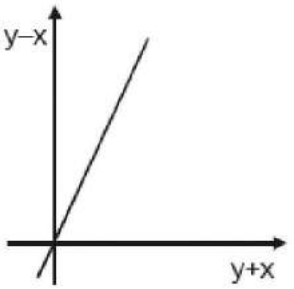
Q.61. The proportion of the sheet area that remains after punching is
(a) $(\pi + 2)/8$ (b) $(6 - \pi)/8$
(c) $(4 - \pi)/4$ (d) $(\pi - 2)/4$
(e) $(14 - 3\pi)/6$

Q.62. Find the area of the part of the circle (round punch) falling outside the square sheet.
(a) $\pi/4$ (b) $(\pi - 1)/2$
(c) $(\pi - 1)/4$ (d) $(\pi - 2)/2$
(e) $(\pi - 2)/4$

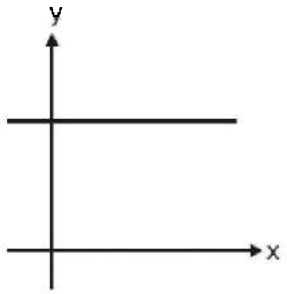
Q.63. What values of x satisfy $x^{2/3} + x^{1/3} - 2 \leq 0$?
(a) $-8 \leq x \leq 1$ (b) $-1 \leq x \leq 8$
(c) $1 < x < 8$ (d) $1 \leq x \leq 8$
(e) $-8 \leq x \leq 8$

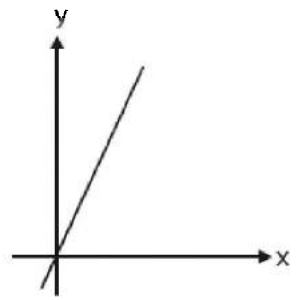
Q.64. Consider the set S = {1, 2, 3, . . ., 1000}. How many arithmetic progressions can be formed from the elements of S that start with 1 and end with 1000 and have at least 3 elements?
(a) 3 (b) 4 (c) 6 (d) 7
(e) 8

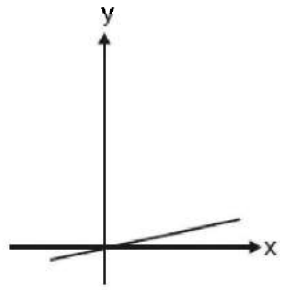
Q.65. The graph of $y - x$ against $y + x$ is as shown below. (All graphs in this question are drawn to scale and the same scale has been used on each axis.)

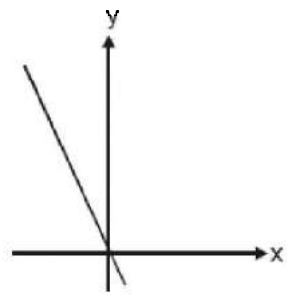


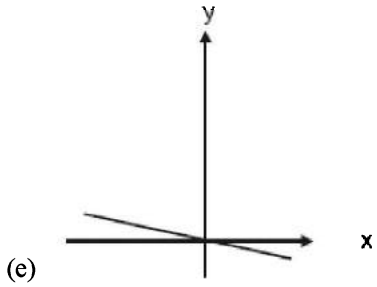
Which of the following shows the graph of y against x?

(a)


(b)


(c)


(d)


(e) 

Q.66. The sum of four consecutive two-digit odd numbers, when divided by 10, becomes a perfect square. Which of the following can possibly be one of these four numbers?
(a) 21 (b) 25 (c) 41 (d) 67
(e) 73

Q.67. The number of solutions of the equation $2x + y = 40$ where both x and y are positive integers and $x \leq y$ is
(a) 7 (b) 13 (c) 14 (d) 18
(e) 20

Q.68. The number of employees in Obelix Menhir Co. is a prime number and is less than 300. The ratio of the number of employees who are graduates and above, to that of employees who are not, can possibly be
(a) 101 : 88 (b) 87 : 100
(c) 110 : 111 (d) 85 : 98
(e) 97 : 84

Q.69. There are 6 tasks and 6 persons. Task 1 cannot be assigned either to person 1 or to person 2, task 2 must be assigned to either person 3 or person 4. Every person is to be assigned one task. In how many ways can the assignment be done?
(a) 144 (b) 180 (c) 192 (d) 360
(e) 716

Q.70. If $\log yx = (a \,.\, \log zy) = (b \,.\, \log xz) = ab$, then which of the following pairs of values for (a, b) is not possible?
(a) (–2, 1/2) (b) (1, 1)
(c) (0.4, 2.5) (d) $(\pi, 1/\pi)$
(e) (2, 2)

Q.71. What are the values of x and y that satisfy both the equations?
$2^{0.6x}.3^{1.25y} = 8\sqrt{6}/27$
$4^{0.3x}.9^{0.2y} = 8.(81)^{1/5}$
(a) $x = 2, y = 5$ (b) $x = 2.5, y = 6$
(c) $x = 3, y = 5$ (d) $x = 3, y = 4$
(e) $x = 5, y = 2$

Q.72. Let $f(x) = \max(2x + 1, 3 - 4x)$, where x is any real number. Then the minimum possible value of $f(x)$ is
(a) 1/3 (b) 1/2 (c) 2/3 (d) 4/3
(e) 5/3

Q.73. When you reverse the digits of the number 13, the number increases by 18. How many other two-digit numbers increase by 18 when their digits are reversed?
(a) 5 (b) 6 (c) 7 (d) 8
(e) 10

Q.74. An equilateral triangle BPC is drawn inside a square ABCD. What is the value of the angle APD in degrees?
(a) 75 (b) 90 (c) 120 (d) 135
(e) 150

Q.75. Arun, Barun and Kiranmala start from the same place and travel in the same direction at speeds of 30, 40 and 60 km per hour respectively. Barun starts two hours after Arun. If Barun and Kiranmala overtake Arun at the same instant, how many hours after Arun did Kiranmala start?
(a) 3 (b) 3.5 (c) 4 (d) 4.5
(e) 5

ANSWERS

■■■ CAT 2006

Q. No.	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70
Answer	(a)	(b)	(a)	(b)	(e)	(b)	(b)	(d)	(*)	(d)	(b)	(d)	(a)	(d)	(d)	(c)	(b)	(e)	(a)	(e)
Q. No.	71	72	73	74	75															
Answer	(e)	(e)	(b)	(e)	(c)															

* None of the option was correct

HINTS AND SOLUTIONS

51. (a) – $a : b : c = 2 : 6 : 3$
$a : b : c : d : e : f = 6 : 18 : 9 : 18 : 6 : 24$
$abc/def = 3/8$

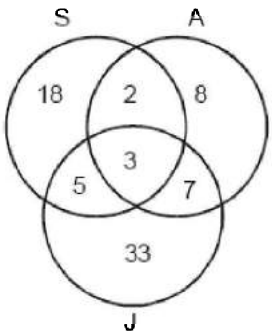
52. (b) Going through options,
Obviously options (a) and (d) are not possible (Since any power positive or negative raised to any positive number will be positive).
Options (c) and (d) will be positive. Only option (b) is negative.

53. (a) – $t_3 = 3/5$, $t_4 = 4/6$, t_5
$= 5/7$, ……,$t_{53} = 53/55$.
$t_3 \times t_4 \times t_5 \times$ ……$\times t_{53}$
$= 3/5 \times 4/6 \times 5/7 \times$……$\times 53/55$
$= (3\times4)/(54\times55) = 2/495$.

54. (b) Take LCM of 2, 3, 4, 6, 12 = 12
$2^{1/2} = (2^6)^{1/12} = (64)^{1/12}$
$3^{1/3} = (3^4)^{1/12} = (81)^{1/12}$
$4^{1/4} = (4^3)^{1/12} = (64)^{1/12}$
$6^{1/6} = (6^2)^{1/12} = (36)^{1/12}$
$12^{1/12} = (12)^{1/12} = (12)^{1/12}$

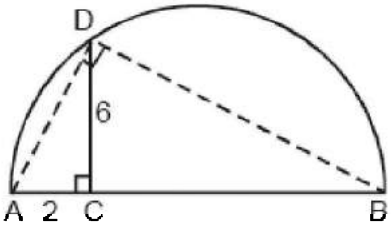
55. (e) Area of 4 walls = 2 H(L + B)
Assume that the length, breadth and height of the room is 30 units, 20 units and 10 units.
Initial total area of the four walls = 2 × 10 (50) = 1000 sq. units
After the changes, new length = 60 units, new breadth = 10 units and height = 5 units
So, new total area of the four walls = 700 sq. units
So, area decreases by 30%.

56. (b)



As given in the above diagram, total persons reading in consecutive months i.e., July and August and August and September is 2 + 7 = 9 people.

57. (b)

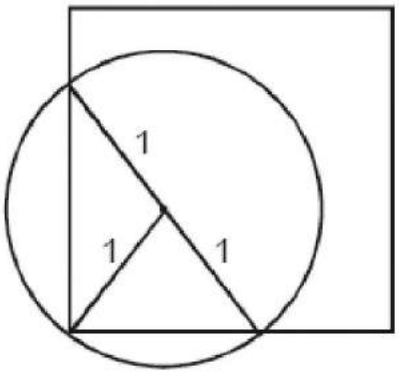


$\angle\angle ADC = 90^\circ$ (Anglein a semicircle)
$CD^2 = AC \times CB$
$(6)^2 = 2 \times CB$. So, CB = 18
Hence AB = AC + CB = 20
Area of semicircle = 50 π sq. units

58 and 59.
Let us assume that total allowed luggage for Raja is A and excess luggage is E
Hence, total luggage for Praja should be A + 2E.
E corresponds to Rs 1200.
Hence, A must correspond to (5400 – 3600)
= Rs 1800
If E = 2N; A = 3N
So, total weight = 2(A) + 3E = 12N
Or N = 5
Hence, Praja's luggage weight = 7N = 35 kg
It should be noted here that none of the options given in Question no. 59 was correct.

60. (d) Let the number of students in front row be x.
So, the number of students in next rows be $x – 3$, $x – 6$, $x – 9$…. so on.
If n i.e. number of rows be 3 then no. of students
$x + (x – 3) + (x – 6) = 630$, or, $3x = 639$, so, $x = 213$. Hence, we can conclude that total number of rows can be equal to 3.
Similarly, when n = 4, then
$x + (x –3) + (x – 6) + (x – 9) = 630$, or, $4x = 648$. So number of rows = 4 is also possible.
Similarly, when $n = 5$, then
$x + (x –3) + (x – 6) + (x – 9) + (x – 12) = 630$, or, $5x = 660$. So number of rows = 5 is also possible.
Similarly, when n = 6, then
$x + (x – 3) + (x – 6) + (x – 9) + (x – 12) + (x – 15) = 630$, or, $6x = 675$.
So, number of rows = 6 is not possible.

61. (b)

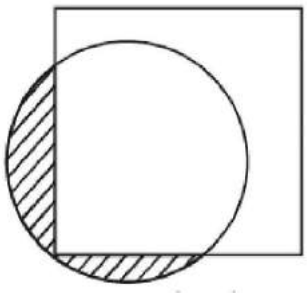


Remaining Area = $(6 - \pi)/2$

So, remaining proportion = $(6 - \pi)/8$

Hence, option (b) is the answer.

62. (d)



Area of the required area = $\pi(1)^2 - [(\pi/2) + 1] = (\pi - 2)/2$

63. (a) $x^{2/3} + x^{1/3} - 2) \leq 0$

Or, $(x^{1/3} - 1)(x^{1/3} + 2) \leq 0$

Or, $-8 \leq x \leq 1$

64. (d) Let number of terms in progression be n, then using $t_n = a + (n - 1)d$

$1000 = 1 + (n - 1)d$, or, $(n - 1)d = 999$

$= 3^3 \times 37$.

Now, making a table of values of n and d gives us the following values of (n-1):

$(n - 1) = 3$ or 9 or 27 or 37 or 111 or 333 or 999

Hence, 7 progressions are possible.

65. (d) From the graph of $(y - x)$ Vs. $(y + x)$, we can conclude

Slope of line = $(y - x) / (y + x) = p$, $0 < p < 1$.

Or, $(y - x)p = (y + x)$

Or, $y / x = (1 + p) / (1 - p)$ = slope of the line that shows the graph of y vs. x.

And we know $0 < p < 1$.

$(1 + p) / (1 - p)$ is –ve and is also greater than 1.

The slope of the graph y Vs. x must be negative and greater than 1. Accordingly, only option (d) satisfies.

Option (e) negative slope but absolute value is less than 1.

66. (c) Odd digits are 1, 3, 5, 7 and 9. If sum of four consecutive odd numbers is divisible by 10, then the digits should be 7, 9, 1 and 3.

Now using the option (c) – the numbers are 37, 39, 41 and 43.

67. (b) $-2x + y = 40$, $x \leq y$.

Or, $y = 40 - 2x$

Values of x and y that satisfy the equation can be calculated like:

x	y
1	38
2	36
..	
. ..	.
. ..	.
. ..	.
13 ..	14

These are 13 values.

68. (e) The sum of the numerator and denominator of the ratio should be a prime number.

Note that the number of employees is less then 300.

Consider options.

(a) 101 : 88

101 + 88 = 189

$189 \times 2 > 300 \Rightarrow$ Number of employees who are graduates and above = 101 and the others are 88 in number 189 is not a prime number. Option (a) is eliminated.

(b) 87 : 100

87 + 100 = 187, $187 \times 2 > 300$

187 is not a prime number.

(c) 110 : 111

110 + 111 = 221, $221 \times 2 > 300$

221 is not a prime number.

(d) 85 : 98

85 + 98 = 183, $183 \times 2 > 300$

183 is not a prime number.

69. (a) Task 2 can be given to two persons only viz. (3 and 4), so, Number of ways = 2 ways.

First task can be done in 3 ways by 3 persons.

And finally third task can be done by 4 persons.

So, 4 ways similarly for fourth, five and six tasks, total number of ways of doing is 3, 2 and 1 respectively.

So, total number of ways = $2 \times 3 \times 4 \times 3 \times 2 \times 1 = 144$ ways

70. (e) $\log_v^x = a.\log_x^z = a \times b$

$$\Rightarrow a \times b = \frac{\log_y^x}{\log_z^y} \times \left(\frac{\log_y^x}{\log_z^y}\right)$$

$$\left(\frac{\log_k^x}{\log_k^y}\right)^3 = (\log_y^x)^3 = (ab)^3$$

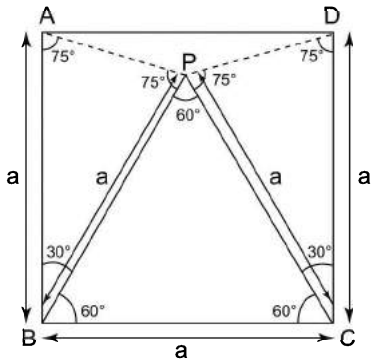
Or, $ab - a^3b^3 = 0$, hence $ab = \pm 1$.
Only option (e) does not satisfy.

71. (e) Equation (ii) can be written as
$4^{0.3} \times 9^{0.2y} = 8 \times (81)^{1/5}$
$0.6x = 3$, so, $x = 5$ and $0.4y = 4/5$, hence $y = 2$.
If we put the values of x and y in first equation these values satisfy the first equation also.
So the answer is $x = 5$, $y = 2$

72. (e) To find out min of f(x) = max $(2x + 1, 3 - 4x)$, we should be taking the point of intersection of $(2x + 1)$ and $(3 - 4x)$. [Since one of these equations is increasing and other one is decreasing]
$2x + 1 = 3 - 4x$, or, $x = 1/3$
$2x + 1 = 3 - 4x = 5/3$
Hence option (e) is the answer.

73. (b) We know that $xy - yx = 9(x - y)$, where xy is a two digit number and $xy \geq yx$.
So, $x - y = 2$
The possible pairs of (x, y) is (3, 1) (4, 2) (5, 3) (6, 4), (7, 5) (8, 6) (9, 7)
But we want the number other than 13 so, there are 6 possible numbers are there i.e., 24, 35, 46, 57,68, 79.
So, total possible numbers are 6.

74. (e)



$\angle PBC = \angle CPB = \angle BPC$ (equilateral triangle) and PC = CD = a.
$\angle CPD = \angle PDC = 75^\circ[(180^\circ - 30^\circ)/2]$
Similarly, $\angle BAP = \angle BPA = 75^\circ$
Hence $\angle APD = 360^\circ - (75^\circ + 75^\circ + 60^\circ) = 150^\circ$

75. (c) Generate the equations to solve this question. Alternatively, options can also be used to solve this question.

CAT 2007

There are three sections in this paper. Each section has 25 questions. Each section carries 100 marks. All questions carry four marks each. Each wrong answer will attract a penalty of one mark.

PART—A

Q.1. How many pairs of positive integers m, n satisfy:

$$\frac{1}{m}+\frac{4}{n}=\frac{1}{12}$$

Where n is an odd integer less than 60?

(a) 7 (b) 5 (c) 3 (d) 6
(e) 4

Q.2. A confused bank teller transposed the rupees and paise when he cashed a cheque for Shailaja, giving her rupees instead of paise instead of rupees. After buying a toffee for 50 paise, Shailaja noticed that she was left with exactly three times as much as the amount on the cheque. Which of the following is a valid statement about the cheque amount?

1. Over rupees 22 but less than rupees 23
2. Over rupees 18 but less than rupees 19
3. Over rupees 4 but less than rupees 5
4. Over rupees 13 but less than rupees 14
5. Over rupees 7 but less than rupees 8

Q.3. Suppose you have a currency, named Miso, in three denominations: 1 Miso, 10 Misos and 50 Misos. In how many ways can you pay a bill of 107 Misos?

(a) 18 (b) 15 (c) 19 (d) 17
(e) 16

Direction for questions 4 and 5: Answer the questions on the basis of the information given below.

Mr David manufactures and sells a single product at a fixed price in a niche market. The selling price of each unit is Rs 30. On the other hand, the cost, in rupees, or producing x units is $240 + bx + cx^2$, where b and c are some constants. Mr David noticed that doubling the daily production from 20 to 40 units increases the daily production cost by $66\frac{2}{3}\%$. However, an increase in daily production from 40 to 60 units results in an increase of only 50% in the daily production cost. Assume that demand is unlimited and that Mr David can sell as such as he can produce. His objective is to maximize the profit.

Q.4. How many units should Mr David produce daily?

(a) 70 (b) 150 (c) 130 (d) 100
(e) Cannot be determined

Q.5. What is the maximum daily profit, in rupees, that Mr David can realize from his business?

(a) 840 (b) 760 (c) 620 (d) 920
(e) Cannot be determined

Direction for questions 6 and 7: Answer the questions on the basis of the information given below.

Let $a_1 = p$ and $b_1 = q$, where p and q are positive quantities. Define

$a_n = pb_{n-1}$, $b_n = qb_{n-1}$, for even $n > 1$.
and $a_n = pa_{n-1}$, $b_n = qa_{n-1}$, for even $n > 1$.

Q.6. Which or the following best describes $a_n + b_n$ for even n?

(a) $q^{\frac{1}{2}n}(p+q)$ (b) $q^{\frac{1}{2}n}(p+q)^{\frac{1}{2}n}$
(c) $q(qp)^{\frac{1}{2}n-1}(p+q)^{\frac{1}{2}n}$ (d) $q(oq)^{\frac{1}{2}n-1}(p+q)$
(e) $qp^{\frac{1}{2}n-1}(p+q)$

Q.7. If $p = \frac{1}{3}$ and $q = \frac{2}{3}$, then what is the smallest odd n such that $a_n + b_n < 0.01$?

(a) 11 (b) 9 (c) 15 (d) 7
(e) 13

Q.8. In a tournament, there are n teams $T_1, T_2, ..., T_n$, with $n > 5$. Each team consists of k players, $k > 3$. The following pairs of teams have one player in common: T_1 and T_2, T_2 and T_3, ..., T_{n-1} and T_n and T_n and T_1. No other pair of teams has any player in common. How many players are participating in the tournament, considering all the n teams together?

(a) $n(k-2)$ (b) $k(n-2)$
(c) $(n-1)(k-1)$ (d) $n(k-1)$
(e) $k(n-1)$

Q.9. Consider four digit numbers for which the first two digits are equal and also equal. How many such numbers are perfect squares?

(a) 4 (b) 0 (c) 1 (d) 3
(e) 2

Direction for questions 10 and 11: Answer the questions on the basis of the information given below.

Shabnam is considering three alternatives to invest her surplus cash for a week. She wishes to guarantee maximum returns on her investment. She has three options, each of which can be utilized fully or partially in conjunction with others.

Option A: Invest in a public sector bank. It promises a return of +0.10%

Option B: Invest in mutual funds of ABC Ltd. A rise in the stock market will result in a return of +5%, while a fall will entail a return of –3%.

Option C: Invest in mutual funds of CBA Ltd. A rise in the stock market will result in a return of –2.5%, while a fall will entail of +2%.

Q.10. The maximum guaranteed return to Shabnam is

(a) 0.20% (b) 0.15%
(c) 0.30% (d) 0.25%
(e) 0.10%

Q.11. What strategy will maximize the guaranteed return to Shabnam?

(a) 64% in option B and 36% in option C
(b) 1/3 in each of the three options
(c) 30% in option A, 32% in option B and 38% in option C
(d) 100% in option A
(e) 36% in option B and 64% in option C

Direction for questions 12 and 13: Answer the questions on the basis of the information given below.

Let S be the set of all pairs (I, j) where $1 \le i < j \le n$, and $n \ge 4$. Any two distinct members of S are called "friends" if they have one constituent of the pairs in common and "enemies" otherwise. For example, if $n = 4$, then S = {(1, 2), (1, 3), (1, 4), (2, 3), (2, 4), (3, 4)}. Here, (1, 2) and (1, 3) are friends, (1, 2) and (2, 3) are also friends, but (1, 4) and (2, 3) are enemies.

Q.12. For general n, how many enemies will each members of S have?

(a) $2n - 7$ (b) $5 - \frac{5k}{2}$ or $2k - 3$
(c) $\frac{1}{2}(n^2 - 7n + 16)$ (d) $n - 3$
(e) $\frac{1}{2}(n^2 - 3n - 2)$

Q.13. For general n, consider any two members of S that are friends. How many other members of S will be common friends of both these members?

(a) $\frac{191}{8} \approx 24$ (b) $n - 2$
(c) $\frac{1}{2}(n^2 - 7n + 16)$ (d) $\frac{1}{2}(n^2 - 5n + 8)$
(e) $2n - 6$

Q.14. Two circles with centres P and Q cut each other at two distinct points A and B. The circles have the same radii and neither P nor Q falls within the intersection of the circles. What is the smallest range that includes all possible values of the angle AQP in degrees?

(a) Between 0 and 60 (b) Between 0 and 75
(c) Between 0 and 45 (d) Between 0 and 90
(e) Between 0 and 30

Q.15. The price of Darjeeling tea (in rupees per kilogram) is $100 + 0.10n$, on the nth day of 2007 (n = 1, 2, ... 100). And then remains constant. On the other hand, the price of Ooty tea (in rupees per kilogram) is $89 + 0.15n$, on the nth day of 2007 (n = 1, 2, ...,, 365). On which date in 2007 will the prices of these two varieties of tea be equal?

(a) May 20 (b) April 10
(c) June 30 (d) May 21
(d) April 11

Q.16. A quadratic function f(x) attains a maximum of 3 at x = 1. The value of the function at x = 0 is 1. What is the value of f (x) = 10?

(a) –110 (b) –180 (c) –105 (d) – 119
(e) – 159

Direction for questions 17 and 18: Answer the questions on the basis of the information given below.

Cities A and B are in different time zones. A is located 3000 km east of B. The table below describes the schedules of an airline operating non-stop flights between A and B. All the times indicated are local and on the same day.

Departure		Arrival	
City	Time	City	Time
B	8:00 am	A	3:00 pm
A	4:00 pm	B	8:00 pm

Assume that planes cruise at the same speed in both directions. However, the effective speed is influenced by a steady wind blowing from east to west at 50 km per hour.

Q.17. What is the time difference between A and B?
(a) 2 hours and 30 minutes
(b) 1 hour
(c) 1 hour and 30 minutes
(d) 2 hours
(e) Cannot be determined

Q.18. What is the plane's cruising speed in km per hour?
(a) 600 (b) 500 (c) 700 (d) 550
(e) Cannot be determined

Q.19. A function $f(x)$ satisfies $f(1) = 3600$, and $f(1) + f(2) + f(3) + \ldots\ldots + f(n) = n^2 f(n)$, for all positive integers $n > 1$. What is the value of $f(9)$?
(a) 200 (b) 100 (d) 120 (d) 80
(e) 240

Q.20. Consider the set S = {2, 3, 4, ..., 2n + 1}, where n is a positive integer larger than 2007. Define X as the average of the odd integers in S and Y as the average of the even integers in S. What is the value of X – Y?
(a) $\frac{1}{2}n$ (b) $\frac{n+1}{2n}$
(c) 2008 (d) 0
(e) 1

Q.21. Ten years ago, the ages of the members of a joint family of eight people added up to 231 years. Three years later, one member died at the age of 60 years and a child was born during the same year. After another three years, one more member died, again at 60, and a child was born during the same year. The current average age of this eight-member joint family is nearest to
(a) 21 years (b) 25 years
(c) 24 years (d) 23 years
(e) 22 years

Direction for questions 22 and 25: Answer the questions on the basis of the information given below.

Each question is followed by two statements A and B. Indicate your responses based on the following directives:

Mark (a): If the question can be answered using A alone but not using B alone.

Mark (b): If the question can be answered using B alone but not using A alone.

Mark (c): If the question can be answered using A and B together, but not using either A or B alone.

Mark (d): If the question cannot be answered even using A and B together.

Q.22. ABC Corporation is required to maintain at least 400 Kilolitres of water at all times in its factory, in order to meet safety and regulatory requirements. ABC is considering the suitability of a spherical tank with uniform wall thickness for the purpose. The outer diameter of the tank is 10 meters. Is the tank capacity adequate to meet ABC's requirements?
A: The inner diameter of the tank is at least 8 meters.
B: The tank weighs, 30,000 kg when empty, and is made of a material with density of 3 gm/cc.

Q.23. Consider integers x, y and z. What is the minimum possible value of $x^2 + y^2 + z^2$?
A: $x + y + z = 89$
B: Among x, y, z two are equal

Q.24. The average weight of a class of 100 students is 45 kg. The class consists of two sections, I and II, each with 50 students. The average weight, W_I, of Section I is smaller than the average weight, W_{II}, of Section II. If the heaviest student, say Deepak, of Section II is moved to Section I, and the lightest student, say Poonam, of section I is moved to section II then the average weights of the two sections are switched, i.e., the average weight of Section I becomes W_{II} and that of Section II becomes W_I. What is the weight of Poonam?
A: $W_{II} - W_I = 1.0$
B: Moving Deepak from Section II to I (without any move from I to II) makes the average weights of the two sections equal.

Q.25. Rahim plans to draw a square JKLM with a point O on the side JK but is not successful. Why is Rahim unable to draw the square?
A: The length of OM is twice that of OL.
B: The length of OM is 4 cm.

ANSWERS

■■■ CAT 2007

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(b)	(a)	(d)	(d)	(d)	(b)	(d)	(c)	(a)	(a)	(b)	(b)	(a)	(a)	(e)	(b)	(d)	(d)	(e)
Q. No.	21	22	23	24	25															
Answer	(c)	(b)	(a)	(c)	(a)															

HINTS AND SOLUTIONS

1. (c) $\frac{1}{m}=\frac{1}{12}-\frac{4}{n}$

$\Rightarrow m=\frac{12n}{n-48}$

Since m is positive, n must be greater than 48. Possible odd values of n such that $48 < n < 60$ are 49, 51, 53, 55, 57 and 59 and 49, 51 and 57 give integral values of m. Hence (c).

2. (b) Let the original amount be Rs x and y paise. Then interchanged amount = Rs y and x paise. From the given condition:

$3(100x + y) = 100y + x - 50$

$\Rightarrow 300x + 3y = 10y + x - 50$

$\Rightarrow 299x = 97y - 50$

$\Rightarrow y=\frac{299x+50}{97}$

Considering the options, only for $x = 18$, we get an integral value for y i.e., $y = 56$. Hence, (b)

3. (a) Possible combinations

Case 1: Bill is paid with only two 50's misos

$2 \times 50 + 1 \times 10 + 7 \times 1$ ... 1 way

Case 2: Bill is paid with only one 50 miso

$1\times50+5\times10+7\times1$
$1\times50+4\times10+17\times1$
$1\times50+3\times10+27\times1$
. . .
. . .
. . .
$1\times50+0\times10+57\times1$
} i.e., 6 ways

Case 3: Bill is paid with only 50 misos

$10\times10+7\times1$
$9\times10+17\times1$
. .
. .
. .
$1\times10+97\times1$
} i.e., 10 ways

Case 4: Bill is paid with only 10 misos and only 50 misos

1×107 ... 1 way

Therefore, total number of ways = 1 + 6 + 10 + 1 = 18 ways

4. (d)

Quantity produced	CP	SP	Profit
x	$240+bx+cx^2$	$30x$	$30x-240-bx-cx^2$
20	$240+20b+400c$	600	$600-240-20b-400c$
40	$240+40b+1600c$	1200	$1200-240-40b-600c$
60	$240+60b+3600c$	1800	$1800-240-b+3600c$

From the given conditions,

$$(240+40b+1600c)=\frac{5}{3}(240+20b+400c)\left(\frac{\pi}{2}-\theta\right) \quad \text{(i)}$$

Also

$$240+60b+3600c=\frac{3}{2}(240+40b+1600b)$$

$$=\frac{5}{2}(240+20b+400c) \quad \text{(ii)}$$

From (i)

$2800c + 20b - 480 = 0$ (iii)

$200c + 20b - 720 = 0$ (iv)

$2400c = 240$

$c = \frac{1}{10} \Rightarrow b = 10$

Profit on x units $= f(x) = 30x - 240 - 10x - \frac{x^2}{10}$

i.e., $f(x) = -\frac{x^2}{10} + 20x - 240$

$f(x)$ is maximum at x if $f'(x) = 0$

i.e., $-\frac{2x}{10} + 20 = 0$

$2x = 200$, or, $x = 100$

5. (d) Maximum daily profit $= f(100)$

$= -1000 + 2000 - 240$

= Rs 760

Answers to 6 and 7:

$a_1 = p, b_1 = q$

$n = 2$: $a_2 = p\,b_1 = pq$; $b_2 = q\ \ b_1 = q^2$

$n = 3$: $a_3 = pq_2 = p^2q$; $b_3 = qa_2 = pq^2$

$n = 4$: $a_4 = pb_3 = p^2q^2$; $b_4 = qb_3 = pq^3$

$n = 5$: $a_5 = pa_4 = p^3q^2$; $b_5 = qa_4 = p^2q^3$

$n = 6$: $a_6 = pb_5 = p^3q^3$; $b_6 = qb_2 = p^2q^4$

$n = 7$: $a_7 = pa_6 = p^4q^3$; $b_7 = qa_6 = p^3q^4$

6. (d) $a_2 + b_2 = pq + q^2 = q(p+q)$

$a_4 + b_4 = p^2q^2 + pq^3 = pq^2(p+q) = q(pq)(p+q)$

$a_6 + b_6 = p^3q^3 + p^2q^4 = p^2q^3\,(p+q) = q(pq)^2(p+q)$

Hence in general, $a_n + b_n = q(pq)^{\frac{1}{2}n-1}(p+q)$

7. (b) $p = \frac{1}{3}, q = \frac{2}{3} \Rightarrow p + q = 1 \text{ and } pq = \frac{2}{9}$

i.e., $a_1 + b_1 = 1$

Now, $a_3 + b_3 = pq(p+q) = pq$

$a_5 + b_5 = (pq)^2(p+q) = (pq)^2$

$a_7 + b_7 = (pq)^3(p+q) = (pq)^3$

In general, for odd 'n' and $p = \frac{1}{3}, \quad q = \frac{2}{3}$

$a_n + b_n = (pq)^{\frac{(n-1)}{2}} = \left(\frac{2}{9}\right)^{\left(\frac{n-1}{2}\right)}$

Starting from the smallest option

$a_7 + b_7 = \left(\frac{2}{9}\right)^3 = 0.01$

$a_9 + b_9 = \left(\frac{2}{9}\right)^4 = 0.002 < 0.01$

8. (d) Total number of teams = n and

Number of players in each team = k

Number of players common to two teams = number of teams = n

Hence, total number of players participating in the tournament

$= nk - n = n\,(k-1)$

9. (c) Let the four-digit number be

$1000a + 100a + 10b + b = 1100a + 11b$

This number will be a perfect square if

$1100a + 11b = k^2$; where k is an integer

$\Rightarrow 11(100a + b) = k^2$

$\Rightarrow 100a + b = \frac{k^2}{11}$

Therefore, k should be a multiple of 11 such that $100a + b$ is a 3-digit number k = 44, 55, 66, ...99. Corresponding values of $100a + b$ will be 176, 275,396,539,704,891.

Therefore, $a, b < 9$ only '704' satisfies this.

Therefore, $1100a + 11b = 7700 + 44 = 7744 = 88^2$

Alternatively, we can solve this question by using elimination and fundas of number system.

Unit digit of a perfect square can be – 0, 1, 4, 5, 6, 9

Only these digits will get repeated at the last two places of the said square in this question.

So, the last two digits can only be – 00, 11, 44, 55, 66, 99

A perfect square cannot have 11, 55, 66, 99 as its last two digits.

(Reason – A perfect square, when divided by 4, should give either of 0 or 1 as the remainder).

So, now we are left with = 00 and 44.

Perfect square lies in between 1000 and 9999 (both including). If the last two digits are 00, then the unit digit should be '0' in the number. Then it is not possible to have other two digits (at the thousands and hundreds place) same.

So the only option left, if possible = 44

Taking a cue from $12^2 = 144$, $88^2 = 7744$

This is the only number satisfying the condition given in the question.

Note:

1. The same question with a different format was asked in CAT 99. The question was – Given $(aa)^2 = bbcc$, where a, b and c are non-zero digits. Find the value of b.
2. $38^2 = 1444$ is another such interesting number. "444" occurring at the end of this square is the maximum number of times a non-zero digit is occurring at the end of any perfect square.

10. (a) Let the amount invested in option B and C be in the ratio $1 : K$

So, depending on whether there is a rise or fall in the stock market, The amount earned will be $5-\frac{5k}{2}$ or $2k-3$.

Therefore, guaranteed return

$$= \min\left\{\frac{5-5k}{2}, 2k-3\right\}$$

Therefore, the maximum guaranteed will be earned when $\frac{5-5k}{2} = 2k-3$ i.e., $9k = 16$

i.e., $k = \frac{16}{9}$.

Therefore, the maximum guaranteed return is when, the amounts invested are in the ratio 9 : 16 i.e., 36% and 64% respectively. Now, the guaranteed return for this distribution is 0.2% (see Q.11). Since option A gives a return of 0.1% which is lesser than this, no amount should be invested in option A.

Hence, maximum guaranteed return = 0.20%.

11. (a) Let us assume that Shabnam has Rs 100

We calculate her guaranteed return in case of each of the given options:

Option (d): 0.1% of 100 = 0.1

Option (e) : If there is a rise in the stock market, earning = 5% of 36 – 2.5% of 64

$$\frac{-108+128}{100} = 0.2$$

If there is a fall in the stock market, earning = –3% of 36 + 2% of +64 $= \frac{-108+128}{100} = 0.2$

Therefore guaranteed return = 0.2

Option (a): Rise in market gives earning = 5% if 64 + 2.5% of 36 = 2.4

Fall in market gives earning = –3% of 64+ 2% of 36 = –1.2 which is negative. Similarly, the guaranteed returns for options (b) and (c) are also negative.

Therefore option (e) offers the highest guaranteed return.

12. (b) A member (a, b) will have an enemy of the form (c, d) where c, d, a, b are all distinct. Therefore c and d can be chosen in ${}^{n-2}C_2$ ways.

So, number of enemies

$$= \frac{(n-2)(n-3)}{2} = \frac{1}{2}(n^2-5n+6)$$

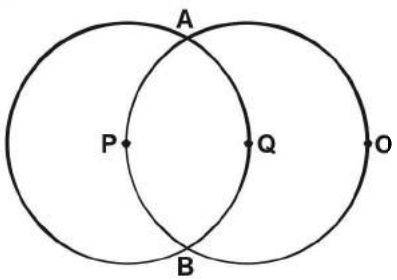
13. (b) Consider friends (a, b) and (a, c).

Their common friend can be either (b, c) or a member of the form (a, d) or (d, a) where d is different from a, b, c.

d can be chosen in $(n-3)$ ways.

So, number of common friends = $(n-3) + 1 = n-2$.

14. (a) P and Q do not lie within intersection of the circles. The extreme case can be that they lie on the circumference of the other circle as shown in the figure.



In this case ΔAPQ will be an equilateral triangle.

Therefore, m $\angle AQP$ is more than 60°, then P and Q will lie within the intersection of the circles.

15. (a) From $100 + 0.10n = 89 + 0.15n$, n will be definitely greater than 100

Price of Darjeeling tea on 100th day and onwards = 100 + 0.10 (100) = Rs 110

Now, price of Ooty tea will be Rs 110 When $89 + 0.15n = 110$

$\Rightarrow 0.15n = 21 \Rightarrow n = 140$

therefore prices will be equal on 140th day i.e., May 20.

16. (e) Let $f(x) = ax^2 + bx + c$

f attains a maximum at $x = 1$

$f'(x) = 0$

$\Rightarrow 2ax + b = 0$

$\Rightarrow x = \frac{-b}{2a} = 1$

$\Rightarrow -b = 2a$

Also max $f(x) = 3$

$\Rightarrow a + b + c = 3$

$\Rightarrow a - 2a + c = 3$

$\Rightarrow c - a = 3$

$f(0) = 1, \Rightarrow c = 1$ and $a = -2$

Therefore $f(x) = -2x^2 + 4x + 1$

$F(10) = -2(100) + 4(10) + 1 = -200 + 41 = -159$

Answers to questions 17 and 18:

Let the speed of the plane be x km/h

Then its speed from B to A = $(x + 50)$ km/ h

and its speed from A to B = $(x - 50)$ km/ h

The fight starts from city B (8.00 am) and arrives at city B (8.00 am) after taking a halt for 1 hour in city A

Therefore Total time taken = 11 hours

i.e., $\frac{3000}{x+50} + \frac{3000}{x-50} = 11$

$\Rightarrow \frac{2x}{x^2 - 2500} = \frac{11}{3000}$

$\Rightarrow 11x^2 - 6000x - 27500 = 0$

Solving the quadratic equation, $x = 550$

Therefore, time taken from B to A = $\frac{3000}{500} = 6$

Flight reaches A when local time at B is 2.00 p.m. which is same as local time 3.00 p.m. at A. Therefore, required time difference = 1 hour

17. (b)

18. (d)

19. (d) $f(n) = \frac{f(1) + f(2) + \ldots + f(n-1)}{n^2 - 1}$

$f(1) = 3600$

$f(2) = \frac{f(1)}{3} = \frac{3600}{3} = 1200$

$f(3) = \frac{f(1) + f(2)}{8} = \frac{4800}{8} = 600$

$f(4) = \frac{f(1) + f(2) + f(3)}{15} = \frac{5400}{15} = 360$

$f(5) = \frac{f(1) + \ldots + f(4)}{24} = \frac{5760}{24} = 240$

$f(6) = \frac{f(1) + \ldots + f(5)}{35} = \frac{6000}{35} = \frac{1200}{7}$

$f(7) = \frac{f(1) + \ldots + f(6)}{48} = \frac{6000}{48} + \frac{1200}{7 \times 48}$

$125 + 25/7$

$f(8) = \frac{f(1) + \ldots f(7)}{63} = \frac{6125}{63} + \frac{1225}{7 \times 63} = 100$

Similarly $f(9) = 80$

20. (e) S = { 2,3,4,… $2n + 1$}

Total number of elements in S = $2n$

$X = \frac{3 + 5 + \ldots + (2n+1)}{n}$

$Y = \frac{2 + 4 + 6 \ldots + 2n}{n}$

Therefore X – Y = $\frac{1 + 1 + \ldots + 1(n \text{ times})}{n} = \frac{n}{n} = 1$

21. (c) 10 years ago, total age of 8 members = 231

Three years later, sum of the ages

$= 231 + 8 \times 3 - 60 = 195$

Three more years later, sum of the ages

$= 195 + 8 \times 3 - 60 = 159$

Sum of current ages = $159 + 8 \times 4 = 191$

Now, current average age = $\frac{191}{8} \approx 24$ years

22. (b) Using statement A:

For minimum diameter i.e., 8 m, the capacity of the tank = $\frac{4}{3} \times \frac{22}{7} \times 4^3 \text{ m}^3 = 268.19 < 400$

For diameter greater than 8m, for example radius = 4.95 m, capacity of the tank = 508.25.71 > 400

Hence, statement A alone cannot be used to answer the question.

Using statement B:

Volume of material used = $\frac{\text{Mass}}{\text{Density}}$ = V (say), which is given

Therefore Outer volume – Inner Volume = V

Therefore, we can find the inner volume and question can be answered.

23. (a) Using statement A:

$x + y + z = 89$

For $x^2 + y^2 + z^2$ to be minimum, each of x, y, z must take integral value nearest to $\frac{89}{3}$

Let $x = 30$, $y = 30$, $z = 29$

Therefore, minimum value of

$x^2 + y^2 + z^2 = (30)^2 + (30)^2 + (29)^2 = 2641$

Thus, statement A alone is sufficient to answer the question.

24. (c) W_I = Average weight of Section I

W_{II} = Average weight of section II

$W_I + W_{II} = 90$ where $W_1 < W_{II}$

Let weight of Deepak and Poonam be D and P kgs respectively

$$\frac{50 \times W_{II} - D + P}{50} = W_I \text{ and } \frac{50 \times W_{II} - D + P}{50} = W_I$$

$$\Rightarrow \quad 50 (W_{II} - W_I) = D - P$$

Using Statement A alone:

$50 \times 1 = D - P$

Thus, D and P can take various values.

So, statement A alone

$50 \times 1 = D - P$ (i)

Thus, D and P can take various values.

Hence, Stat. A alone is not sufficient.

Using Stat. B alone:

$$\frac{50 \times W_I + D}{51} = \frac{50\ W_{II} - D}{49} \quad \text{(ii)}$$

Since values of W_I and W_{II} are not known

We cannot find the value of D.

Combining both the statements,

Values of W_I and W_{II} can be found and hence value of D and P can be found, using (i) and (ii).

25. (a) The farthest point from point M which is on the square is the diagonally opposite point.

Length OM $\leq \sqrt{2} \times$ side of the square $\Rightarrow$ Length OM $\leq \sqrt{2} \times$ length of LK (i)

And length of OL $\geq$ length of LK (ii)

From (i) and (ii) we can conclude that such a point can not be drawn.

CAT 2008

SECTION I

This section contains 25 questions.

Q.1. The integers 1, 2, ..., 40 are written on a blackboard. The following operation is then repeated 39 times: In each repetition, any two numbers, say a and b, currently on the blackboard erased and a new number a + b – 1 is written. What will be the number left on the board at the end?
(a) 820 (b) 821
(c) 781 (d) 819
(e) 780

Q.2. What are the last two digits of 7^{2008}?
(a) 21 (b) 61
(c) 01 (d) 41
(e) 81

Q.3. If the roots of the equation $x^3 - ax^2 + bx - c = 0$ are three consecutive integers, then what is the smallest possible value of b?
(a) $-\frac{1}{\sqrt{3}}$ (b) -1
(c) 0 (d) 1
(e) $\frac{1}{\sqrt{3}}$

Q.4. A shop stores x kg of rice. The first customer buys half this amount plus half a kg of rice. The second customer buys half the remaining amount plus half a kg of rice. Then the third customer also buys half the remaining amount plus half a kg of rice. Thereafter, no rice is left in the shop. Which of the following best describes the value of x?
(a) $2 \le x \le 6$ (b) $5 \le x \le 8$
(c) $9 \le x \le 12$ (d) $11 \le x \le 14$
(e) $13 \le x \le 18$

Direction for questions 5 and 6: Answer the questions on the basis of the information given below.

Let $f(x) = ax^2 + bx + c$, where a, b and c are certain constants and $a \neq 0$. It is known that $f(5) = -3f(2)$ and that 3 is a root of $f(x) = 0$.

Q.5. What is the other root of $f(x) = 0$?
(a) –7 (b) – 4
(c) 2 (d) 6
(e) cannot be determined

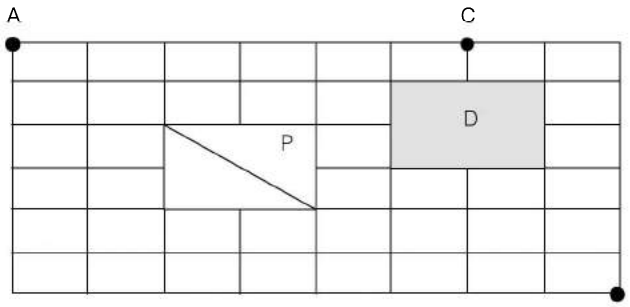
Q.6. What is the value of a + b + c?
(a) 9 (b) 14
(c) 13 (d) 37
(e) cannot be determined

Q.7. The number of common terms in the two sequences 17, 21, 25, ..., 417 and 16, 21, 26, ..., 466 is
(a) 78 (b) 19
(c) 20 (d) 77
(e) 22

Q.8. How many integers, greater than 999 but not greater than 4000, can be formed with the digits 0, 1, 2, 3 and 4, if repetition of digits is allowed?
(a) 499 (b) 500
(c) 375 (d) 376
(e) 501

Direction for questions 9 and 10: Answer the questions on the basis of the information given below.

The figure below shows the plan of a town. The streets are at tight angles to each other. A rectangular park (P) is situated inside the town with a diagonal road running through it. There is also a prohibited region (D) in the town.



Q.9. Neelam rides her bicycle from her house at A to her office at B, taking the shortest path. Then the number of possible shortest paths that she can choose is
(a) 60 (b) 75
(c) 45 (d) 90
(e) 72

Q.10. Neelam rides her bicycle from her house at A to her club at C, via B taking the shortest path. Then the number of possible shortest paths that she can choose is
(a) 1170 (b) 630
(c) 792 (d) 1200
(e) 936

Q.11. Let $f(x)$ be a function satisfying $f(x)f(y) = f(xy)$ for all real x, y. If $f(2) = 4$ then what is the value of $f(1/2)$?
(a) 0 (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) 1
(e) cannot be determined

Q.12. Suppose, the seed of any positive integer i is defined as follows:

seed(n) = n, if n < 10
= seed(s(n)), otherwise

where s(n) indicates the sum of digits of n. For example, seed (7) = 7, seed (248) = seed (2 + 4 + 8) = seed (14)= (1 + 4) = seed = 5 etc. How many positive integers n, such that n < 500, will have seed(n) = 9?
(a) 39 (b) 72
(c) 81 (d) 108
(e) 55

Q.13. In a triangle ABC, the lengths of the sides AB and AC equal 17.5 cm and 9 cm respectively. Let D be a point on the line segment BC such that AD is perpendicular to BC. If AD=3 cm, then what is the radius (in cm) of the circle circumscribing the triangle ABC?
(a) 17.05 (b) 27.85
(c) 22.45 (d) 32.25
(e) 26.25

Q.14. Consider obtuse-angled triangles with sides 8 cm, 15 cm and x cm. If x is an integer, then how many such triangles exist?
(a) 5 (b) 21
(c) 10 (d) 15
(e) 14

Q.15. Consider a square ABCD with mid points E, F, G, H of AB, BC, CD and DA respectively. Let L denote the line passing through F and H. Consider points P and Q, on L and inside ABCD, such that the angles APD and BQC both equal 120°. What is the ratio of the area of ABQCDP to the remaining area inside ABCD?
(a) $\frac{4\sqrt{2}}{3}$ (b) $2+\sqrt{3}$
(c) $\frac{10-3\sqrt{3}}{9}$ (d) $1+\frac{1}{\sqrt{3}}$
(e) $2\sqrt{3}-1$

Q.16. What is the number of distinct terms in the expansion of $(a + b + c)^{20}$?
(a) 231 (b) 253
(c) 242 (d) 210
(e) 228

Direction for questions 17 and 18: Answer the questions on the basis of the information given below.

Five horses, Red, White, Grey, Black and Spotted participated in a race. As per the rules of the race, the persons betting on the winning horse get four times the bet amount and those betting on the horse that came in second get thrice the bet amount. Moreover, the bet amount is returned to those betting on the horse that came in third, and the rest lose the bet amount. Raju bets Rs.3000, Rs.2000 and Rs.1000 on Red, White and Black horses respectively and ends up with no profit and no loss.

Q.17. Which of the following cannot be true?
(a) At least two horses finished before Spotted
(b) Red finished last
(c) There were three hones between Black and Spotted
(d) There were three horses between White and Red
(e) Grey came in second

Q.18. Suppose, in addition, it is known that Grey came in fourth. Then which of the following cannot be true?
(a) Spotted came in first
(b) Red finished last
(c) White came in second
(d) Black came in second
(e) There was one horse between Black and White

Direction for questions 19 and 20: Answer the questions on the basis of the information given below.

Mark (a): If Question can be answered 1mm A alone but not from B alone.

Mark (b): If Question can be answered from B alone but not from A alone.

Mark (c): If Question can be answered from either of A or B alone.

Mark (d): If Question can be answered from A and B together but not from any of them alone.

Mark (e): If Question cannot be answered even from A and B together.

In a single elimination tournament, any player is eliminated with a single loss. The tournament is played in multiple rounds subject to the following rules:

(a) If the number of players, say n, in any round is even, then the players are grouped in to n/2 pairs. The players in each pair play a match against each other and the winner moves on to the next round.

(b) If the number of players, say it, in any round is odd, then one of them is given a bye, that is, he automatically moves on to the next round. The remaining (n-1) players are grouped into (n-l)/2 pairs. The players in each pair play a match against each other and the winner moves on to the next round. No player gets more than one bye in the entire tournament.

Thus, if n is even, then n/2 players move on to the next round while if n is odd, then (n+l)/2 players move on to the next round. The process is continued till the final round, which obviously is played between two players. The winner in the final round is the champion of the tournament.

Q.19. Question: What is the number of matches played by the champion?
A: The entry list for the tournament consists of 83 players.
B: The champion received one bye.

Q.20. Question: If the number of players, say n, in the first round was between 65 and 128, then what is the exact value of n?
A: Exactly one player received a bye in the entire tournament.
B: One player received a bye while moving on to the fourth round from the third round

Q.21. Two circles, both of radii 1 cm, intersect such that the circumference of each one passes through the centre of the other. What is the area (in sq cm) of the intersecting region?

(a) $\frac{\pi}{3}-\frac{\sqrt{3}}{4}$ (b) $\frac{2\pi}{3}+\frac{\sqrt{3}}{2}$
(c) $\frac{4\pi}{3}-\frac{\sqrt{3}}{2}$ (d) $\frac{4\pi}{3}+\frac{\sqrt{3}}{2}$
(e) $\frac{2\pi}{3}-\frac{\sqrt{3}}{2}$

Q.22. Rahim plans to drive from city A to station C, at the speed of 70 km per hour, to catch a train arriving there from B. He must reach C at least 15 minutes before the arrival of the train. The train leaves B, located 500 km south of A, at 8:00 am and travels at a speed of 50 km per hour. It is known that C is located between west and northwest of B, with BC at 60° to AB. Also, C is located between south and southwest of A with AC at 30° to AB. The latest time by which Rahim must leave A and still catch the train is closest to
(a) 6:15am (b) 6:30 am
(c) 6:45am (d) 7:00am
(e) 7:15am

Q.23. Three consecutive positive integers are raised to the first, second and third powers respectively and then added. The sum so obtained is a perfect square whose square root equals the total of the three original integers. Which of the following best describes the minimum, say m, of these three integers?
(a) $1 \le m \le 3$ (b) $4 \le m \le 6$
(c) $7 \le m \le 9$ (d) $10 \le m \le 12$
(e) $13 \le m \le 15$

Q.24. Find the
$$\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}}+\sqrt{1+\frac{1}{2^2}+\frac{1}{3^2}}+.....+\sqrt{1+\frac{1}{2007^2}+\frac{1}{2008^2}}$$
(a) $2008-\frac{1}{2008}$ (b) $2007-\frac{1}{2007}$
(c) $2007-\frac{1}{2008}$ (d) $2008-\frac{1}{2007}$
(e) $2008-\frac{1}{2009}$

Q.25. Consider a right circular cone of base radius 4 cm and height 10 cm. A cylinder is to be placed inside the cone with one of the flat surfaces resting on the base of the cone. Find the largest possible total surface area of the cylinder (in sq. cm).
(a) $100\pi/3$ (b) $80\pi/3$
(c) $120\pi/7$ (d) $130\pi/9$
(e) $110\pi/7$

ANSWERS

■■■ CAT 2008

Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Answer	(c)	(c)	(b)	(b)	(b)	(e)	(c)	(d)	(d)	(a)	(b)	(e)	(e)	(c)	(e)	(a)	(d)	(c)	(d)	(d)
Q. No.	21	22	23	24	25															
Answer	(e)	(b)	(a)	(a)	(a)															

HINTS AND SOLUTIONS

1. Let S = 1 + 2 + ... + 40. So 1 + 2 + a + b +... + 40 = S
After 1st step:
1 + 2 + ... + (a + b — 1) + ... + 40 = S — 1
So, after each step the total reduces by 1. Continuing this way after 39 steps numbers left S — 39 = $\frac{40\times 41}{2} - 39 = 781$
Hence, (c)

2. $7^{2008} = (7^4)502 = (2401)^{502}$
To find tens digit, we need to take care only of last two digits of this number.
Last two digits of $(2401)^{502}$ = Last two digits of $(01)^{502}$
Thus, 7^{2008} end with 01. Hence, [3].

3. Assume that the consecutive roots are $(x - 1)$, x and $(x + 1)$
Now, $b = x(x-1) + x(x+1) + (x-1)(x+1) = 3x^2 - 1$
Now as $3x^2 \geq 0$ minimum value of $b = 3 \times 0^2 - 1 = -1$
Hence, (b)

4. Let us make a table from the given information:

Quantity of Rice in the shop	Quantity of Rice bought	Quantity Left
x	$\left(\frac{x}{2}+\frac{1}{2}\right)$	$\frac{x}{2}-\frac{1}{2}$
$\frac{x}{2}-\frac{1}{2}$	$\left(\frac{x}{2}-\frac{1}{2}\right)+\frac{1}{2}$	$\frac{x}{4}-\frac{3}{4}$
$\frac{x}{4}-\frac{3}{4}$	$\left(\frac{x}{8}-\frac{3}{8}\right)+\frac{1}{2}$	$\frac{x}{8}-\frac{7}{8}$

$\frac{x}{8}-\frac{7}{8} = 0 \Rightarrow x = 7$

Now, $5 \leq x \leq 8$

Hence, (b)

5. Given $f(x) = ax^2 + bx + c$ $(a \neq 0)$.
Since 3 is a root of $f(x)$, so, $9a + 3b + c = 0$ .....(i)
Also, $f(5) = 3f(2)$.
So, $25a + 5b + c = -3(4a + 2b + c) = -12a - 6b - 3c$
$\Rightarrow 37a + 11b + 4c = 0$ .....(ii)
From (i) and (ii), $a - b = 0 \Rightarrow a = b$
Thus we get $f(x) = ax^2 + ax + c$
Dividing $f(x)$ by $x - 3$, we get $c = -12a$
$\Rightarrow f(x) = ax^2 + ax - 12a \Rightarrow f(x) = 0 \Rightarrow -4$ is second root.
Hence, (b).

6. $a + b + c = a + a - 12a = -10a$
Value of a is not given, so cannot be determined.
Hence, (e).

7. In 1st series, terms of are in the form $4a + 1$ $(4 < a < 104)$
In 2nd series, terms of are in the form $5b + 1$ $(3 < b < 93)$
In order to have same terms, we should get $4a = 5b$.
This happens only 20 times.
Thus, we get 21, 41, 61... i.e., 20 common terms.
Hence, (c).

8. Numbers are of 4 digits, and less than or equal to 4000.
For numbers less than 4000, following is the box diagram:
$\underline{3} \times \underline{5} \times \underline{5} \times \underline{5} = 375$
Including 4000, total number of numbers = 376
Hence, (d)

9. Neelam has to take path XY
A to X = ${}^4C_2$ = 6 possibilities
Y to B = ${}^6C_2$ = 15 possibilities
So total possibilities = 6 × 15 = 90 possibilities
Hence, (d)

10. From A to B = 90 paths
From B to C via N = 6 (and not via M)
From B to C via M = 7 paths
In all 90 V (6 + 7) = 1170 paths
Hence, (a)

11. We have $f(x)f(y) = f(xy)$
So, $f(1)f(1) = f(1 \times 1) = f(1)$
$\Rightarrow f(1)^2 = f(1) \Rightarrow f(1)^2 - f(1) = 0 \Rightarrow f(1)[f(1) - 1] = 0$
Hence $f(1) = 0$ or $f(1) = 1$
If $f(1) = 0$ then $f(x) = 0$ for any x since $x = x \times 1$
$\Rightarrow f(1) = 1$
Now, $f(2) = 4$
So, $1 = f(1) = f\left(\frac{1}{2} \times 2\right) = f\left(\frac{1}{2}\right) f(2) = f\left(\frac{1}{2}\right) \times 4 \Rightarrow f\left(\frac{1}{2}\right) = \frac{1}{4}$
Hence, (b).

12. Our answer would be the number of integers between 1 and 500, which are divisible by 9.
The smallest is 9 and the largest is 495.
In the first 499 natural numbers, we have 495 as the last multiple of 9, and this is 55th multiple of 9.
Hence, (e).

13. Let the sides be a = AB, b = AC, e = BC
$$A(\Delta ABC) = \frac{abc}{4R} \Rightarrow A(\Delta ABC) = \frac{1}{2}$$
$$\text{So, } \frac{abc}{4R} = \frac{1}{2} \times c \times 3$$
$$R = \frac{a \times b}{2 \times 3} = \frac{17.5 \times 9}{2 \times 3} = 26.25 \text{ cm}^2$$
Hence, (e).

14. Let the three sides be a = 8, b = 15 and c = x
Case I:
c is the longest side.
We know that 15 < c < 23. (8 + 15 = 23). For c = 17 we get a right angled triangle. So, c > 17 for an obtuse angled triangle.
Therefore c is 18, 19, 20, 21, 22.
Case II
b is the longest side. In this case 7 < c < 15 (15 — 8 = 7)
For a right angled triangle, $c < \sqrt{15^2 - 8^2} \Rightarrow c < \sqrt{161} \cong 12.7$
So, c can be 8, 9, 10, 11 or 12.
Hence in total there are 10 possibilities.

15. (e).

16. $(a + b + c)^{20}$ has each term of the form $a^x b^y c^z$ where $x + y + z = 20$. This has ${}^{20+3-1}C_{3-1} = 231$ solutions.
(This question is same as distributing 20 apples among three children where any child can get any number of apples. Formula for this is – ${}^{n+r-1}C_{r-1}$).
Hence, (a).

17. (d) There are 3 possibilities

Horse	Rank		
	Case 1	Case 2	Case 3
Red	4/5	4/5	3
White	2	3	4/5
Black	5/4	1	2
Grey	1/3	2/4/5	1/4/5
Spotted	3/1	2/4/5	1/4/5

18. (c)

19. Using statement A alone, if there are 83 if there are 83 players, the number of players in each subsequent round will be as follows:

Round	Players	
1	83	
2	42	
3	21	
4	11	
5	6	
6	3	
7	2	→ Final

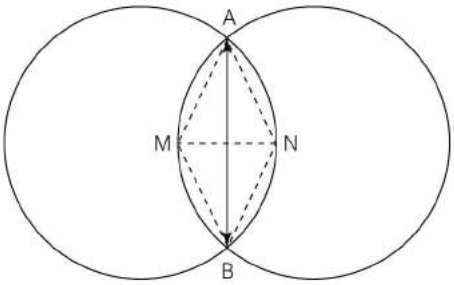
The champion plays in the final and so can play either 6 or 7 matches (depending on whether he gets a bye or not). Hence no unique answer.
Using statement B alone, we can draw no conclusion without knowing the number of rounds. With both the statement together, we can say that the champion plays 6 matches. Hence, (d).

20. Using statement A alone we get various possible cases. We can see that 7 rounds have to be played. The bye could have occurred in any of the first 6. Let us look at the table below now:

Round	Bye in					
	Round 1	Round 2	Round 3	Round 4	Round 5	Round 6
1	127	126	124	120	112	96
2	64	63	62	60	56	47
3	32	32	31	30	28	24
4	16	16	16	15	14	12
5	8	8	8	8	7	6
6	4	4	4	4	4	3
7	2	2	2	2	2	2

Using Statement B alone, we cannot say anything as other players too might have received byes. Using both together we can say that the number of players is 124. Hence, (d).

21. Area required region = 2 × [(Area of sector – AMB) – A (ΔANB)]



$= 2 \times \frac{120^\circ}{360^\circ} \times \pi r^2 - 2 \times \text{Area}(\Delta ANB)$

$= \frac{2}{3}\pi - 2 \times \frac{\sqrt{3}}{4} \times 1 = \frac{2}{3}\pi - \frac{\sqrt{3}}{2}$ sq. cm.

Hence (e).

22. (b).

23. $3^1 + 4^2 + 5^3 = 144$

$\sqrt{144} = 3 + 4 + 5$. So, the numbers are 3, 4 and 5.

Hence, (a).

Alternatively, assume the numbers to be n, n+1 and n+2.

So, $n^1 + (n+1)^2 + (n+2)^3 = (n+n+1+n+2)^2 = (3n+3)^2$

Expanding it, we get $n(n^2 - 2n - 3) = 0$

Since $n \neq 0$, so, $(n^2 - 2n - 3) = 0 \Rightarrow (n-3)(n+1) = 0$

Hence $n = 3$

24. Go through the method of induction,

$$\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{\frac{9}{4}} = \frac{3}{2} = 2 - \frac{1}{2}$$

$$\sqrt{1+\frac{1}{1^2}+\frac{1}{2^2}} + \sqrt{1+\frac{1}{2^2}+\frac{1}{3^3}}$$

$$= \frac{3}{2} + \frac{7}{6} = \frac{8}{3} = 3 - \frac{1}{3}$$

Similarly,

$$\sqrt{1=\frac{1}{2^2}+\frac{1}{2^2}} + \ldots\ldots\sqrt{1+\frac{1}{2007^2}+\frac{1}{2008^2}} = 2008 - \frac{1}{2008}$$

Hence, (a).

25. Consider the figures given below:

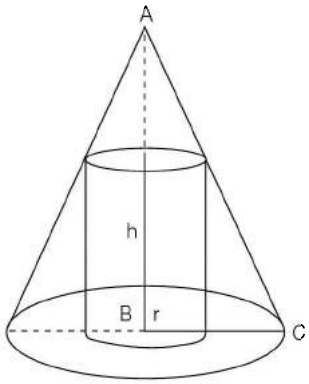
$$\frac{AP}{PB} = \frac{AQ}{QC} = \frac{10-h}{h}$$

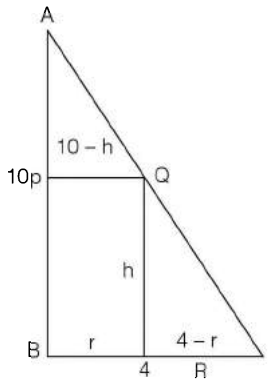
$$\frac{BR}{RC} = \frac{AQ}{QC} = \frac{R}{4-r}$$

$$\Rightarrow \frac{10-h}{h} = \frac{r}{4-r} \Rightarrow 4h = 40 - 10r \Rightarrow 2h = 20 - 5\,r$$

Surface area of cylinder = $2\pi r^2 + 2\pi rh$

$= 2\pi r^2 + \pi r\,(20 - 5r)$





$= 20\,\pi r - 2\,\pi r^2 = \pi\,(20r - 3r^2)$

So, we have to maximize $f(r) = 20r - 3r^2$

We know that that maximum of quadratic equation $ax^2 + bx + c = 0$ arises for $x = \frac{-b}{2a}$ and the maximum value $= \frac{-D}{4a}$

In this case $\frac{-D}{4a} = 100/3$

So the maximum surface area = $\frac{100\pi}{3}$

Hence, (a).

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PART 4

Section Tests

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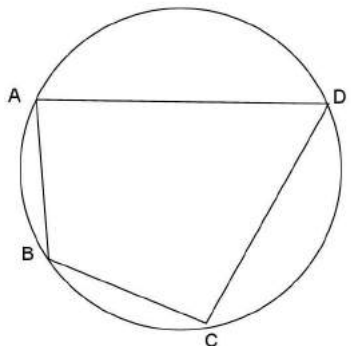
Section Test 1

Q.1. For how many natural number values of N, $N^4 + 4$ will be a prime number?
(a) 0 (b) 1
(c) 2 (d) None of these

Q.2. In the figure given, ABCD is a cyclic quadrilateral and AB = 25 cm, BC = 39 cm, CD = 52 cm and AD = 60 cm. What is the diameter of the circle?



(a) 60 cm (b) 65 cm
(d) 72 cm (d) 78 cm

Q.3. How many natural numbers are there which give a remainder of 41 after dividing 1997?
(a) 2 (b) 4
(c) 6 (d) None of these

Q.4. You are selecting 10 numbers randomly out of the first 100 odd numbers. Sum of these 10 odd numbers is N. How many different values of N are possible?
(a) 900 (b) 1801
(c) 1800 (d) 901

Q.5. How many values of x will satisfy the following equation:

$$\sqrt{x+\sqrt{x+\sqrt{x}.....}.....\infty} = \sqrt{x.\sqrt{x.\sqrt{x}.....}.....\infty}$$

(a) 0 (b) 1
(c) 2 (d) None of these

Q.6. ABC is an equilateral triangle with side length 1 unit. P is any point on the circumcircle of this triangle. What is the value of $AP^2 + BP^2 + CP^2$?
(a) $\sqrt{2}$ (b) 2
(c) $2\sqrt{2}$ (d) 3

Q.7. How many integral points are contained inside a triangle with the vertices (0, 0), (21, 0) and (0, 21)?
(a) 190 (b) 231
(c) 210 (d) 171

Q.8. In a race on a circular track, A, B and C start from the same point N in the same direction. Their speeds are n m/s, (n+1) m/s and (n+2) m/s respectively where n is a natural number. How many times will they meet before they meet for the first time at point N(n>0, N>0)?
(a) 1
(b) 1
(c) 2
(d) None of these

Q.9. There are a certain number of students in a class and each of them have at most 4 books. The number of students who have at most 1, 2, 3 and 4 books is 11, 18, 19 and 21 respectively. What is the number of students with at least 3 books?
(a) 0 (b) 1
(c) 2 (d) 3

Direction for questions 10 to 12: Go through the information given below and solve the questions based on it.

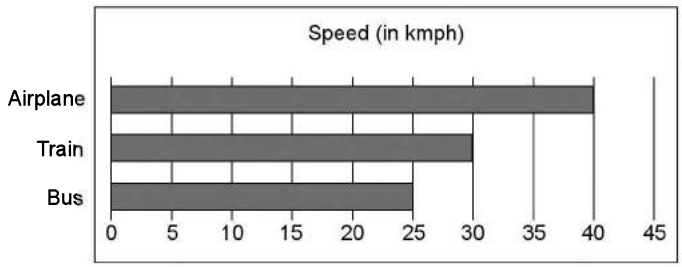
Table 1 – Mode of travel between two cities

	Patna	*Mumbai*	*Calcutta*	*Chennai*	*Bangalore*
Patna	-	Bus	Train	Air plane	Train
Mumbai	Bus	-	Train	Train	Air plane
Calcutta	Train	Train	-	Air plane	Bus

	Patna	*Mumbai*	*Calcutta*	*Chennai*	*Bangalore*
Chennai	Air plane	Train	Air plane	-	Bus
Bangalore	Train	Air plane	Bus	Bus	-

Table 2 - Distance between two cities (In kms)

	Patna	*Mumbai*	*Calcutta*	*Chennai*	*Bangalore*
Patna	-	100	150	150	300
Mumbai	100	-	200	300	500
Calcutta	150	200	-	500	300
Chennai	150	300	500	-	100
Bangalore	300	500	300	100	-



Q.10. Which city can be reached from Patna in the least time?
(a) Mumbai
(b) Calcutta
(c) Chennai
(d) Bangalore

Q.11. For which of the following options, travel time is the least?
(a) Mumbai-Chennai
(b) Patna-Bangalore
(c) Bangalore-Chennai
(d) Mumbai - Calcutta

Q.12. If one can travel by bus or train only, then which two cities will have the least travel time?
(a) Patna-Mumbai
(b) Bangalore-Calcutta
(c) Chennai - Mumbai
(d) None of these

Q.13. In a town 25% families own a phone and 15% own a car. 65% families own neither a phone nor a car. 2000 families own both a phone and a car. Consider the following statements in this regard:
1. 10% families own both a car and a phone
2. 35% families own either a car or a phone
3. 40,000 families live in the town

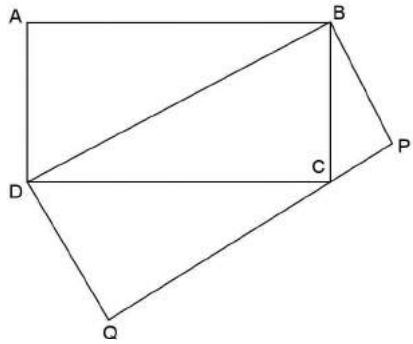
Which of the above statements are correct?
(a) 1 and 2 (b) 1 and 3
(d) 2 and 3 (d) 1, 2 and 3

Q.14. The function $f(x) = |x-2| + |2.5-x| + |3.6-x|$, where x is any real number, attains a minimum value at
(a) x = 2.3 (b) x = 2.5
(d) x = 2.7 (d) None of these

Q.15. Find the number of natural numbers which are smaller than 2.10^8 written by means of the digits 1 and 2.
(a) 760 (b) 766
(c) 754 (d) 782

Q.16. The quadratic equation $g(x) = (px^2 + qx + r)$, $p \neq 0$, attains its maximum value at x = 7/2. Product of the roots of the equation g(x) = 0 is equal to 10. What is the value of p×q×r?
(a) 70 (b) –70
(c) 0 (d) Cannot be determined

Q.17. In the given figure, ABCD and BDQP are two rectangles, in which AB = 12 units and AD = 5 units. Find the area of the rectangle BDQP.



(a) 78 sq. units (b) 60 sq. units
(c) 65 sq. units (d) 52 sq. units

Direction for questions 18 to 20: Read the following instructions and solve the questions based on it.

Choose (a): If the question can be answered by one of the statements alone and not by the other.

Choose (b): If the question can be answered by using either statement alone.

Choose (c): If the question can be answered by using both the statements together, but cannot be using either statement alone.

Choose (d): If the question cannot be answered even by using both the statements together.

Q.18. Is x > 2?
Statement A: $|2x-19| < 7$
Statement B: $x^2 - 4x = 0$

Q.19. Is x > y?
Statement A: 3x + 5y = 11 where x and y are integers
Statement B: $x^5 > y^5$

Q.20. If both m and n are positive integers less than 100 and greater than 10, is the sum m + n a multiple of 11?
Statement A: m – n is a multiple of 22
Statement B: The tens digit and the units digit of m are the same; the tens digit and the units digit of n are the same

Q.21. Rs 180 is divided among 66 boys and girls of a class. The sum of the boy's share and girl's share are in ratio 5 : 4 but their individual shares are 3 : 2 respectively. The number of boys in the class is:
(a) 36 (b) 30
(c) 5 (d) 40

Q.22. If the sum of five consecutive positive integers is A, then the sum of the next five consecutive integers in terms of A is:
(a) A+1 (b) A+5
(c) A+25 (d) 2A

Direction for questions 23 to 26. Go through the table given below and solve the questions based on it.

The following table gives the tariff (in paisa per kilo–watt–hour) levied by the NTPC in 2004–05, in the four sectors and the regions within them. The table also gives the percentage change in the tariff as compared to 2001–02.

	Region 1		*Region 2*		*Region 3*		*Region 4*		*Region 5*	
Sector 1	425	+15	472	+5	420	–4	415	+8	440	+10
Sector 2	430	+12	468	+8	448	+7	423	–3	427	+11
Sector 3	428	+8	478	–4	432	+6	441	+10	439	+8
Sector 4	434	–5	470	+15	456	+10	451	+12	446	–12

Q.23. If the amount of power consumed by the various Regions in Sector 1 is the same, then, as compared to 2001–02, the net tariff in 2004–05...
(a) Increased by 9.5%
(b) Decrease by 1.5%
(c) Decrease by 13.3%
(d) None of these

Q.24. What approximately was the average tariff per sector in Region 3 in 2001–02?
(a) 402 (b) 424
(c) 433 (d) None of these

Q.25. Which of the following has got a maximum percentage increase for the given period?
(a) Region 1 – Sector 1
(b) Region 1 – Sector 2
(c) Region 5 – Sector 2
(d) None of these

Q.26. There are a total of 20 units of NTPC divided into regions and then subdivided into sectors, as given in the above table. For how many units of NTPC in 04 – 05, there is a net increase in the tariff?
(a) 15 (b) 5
(c) 8 (d) None of these

Direction for questions 27 to 30: Go through the table given below and solve the questions based on it.

Table given below gives the Estimated and Actual Sales revenue and cost figures for three years.
Answer the questions based on it.

Sales Revenue	2007	2008	2009	Cost	2007	2008	2009
Estimated	1281	1452	1228	Estimated	986	1051	1024
Actual	1326	1286	1452	Actual	988	1076	1068

Estimated Profit = Estimated Revenue – Estimated Cost
Actual Profit = Actual Revenue – Actual Cost

Q.27. For how many years, actual profit is more than estimated profit?
(a) 0 (b) 1
(c) 2 (d) 3

Q.28. A measure of forecasting done by marketing team is the ratio of Estimated Sales to actual sales. Lower is this value, better is the forecasting. Which year's forecasting is best?
(a) 2007
(b) 2008
(c) 2009
(d) cannot be determined

Q.29. Company plans to use a different mechanism for forecasting of its maximum possible profit = Max(Estimate, Actual(Sales Revenue)) – Min(Estimate, Actual(Cost)). What is the maximum possible profit derived this way?
(a) 340 (b) 401
(c) 428 (d) None of these

Q.30. In a geometric progression, product of 11th term, 18th term and 20th term is equal to the product of 28th term and 5th term. Which of the following is necessarily true about that GP?
(a) One of terms of this GP has to be equal to zero.
(b) One of the terms of this GP has to be equal to one.
(c) This GP will have infinite terms.
(d) Common ratio of this GP will be less than 1.

My performance in this Test

Attempts	Right	Wrong	Net Marks	% Accuracy